

# GENERAL THETA FOUNDATIONS I: CONTINUATION, TESTING, AND MICROSCOPIC TRANSPORT

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ABSTRACT. We study marked causal experiments with a fixed persistent register and separately charged hidden or visible randomization. We characterize finite registers by compatible continuation polytopes and determine an exact memory–error law at an erasure cut. On the original nonconvex behavior image, a finite reset-audit game has an attained minimax representation and error at most  $\sqrt{D/n}$ , where  $D$  counts independent observable probabilities. A binary family gives a matching uniform sampling exponent. We then prove transport of these certificates together with likelihoods, posterior processes, and bounded entropic backward equations under Gaussian observation of changing microscopic trajectories. For two hard spheres this yields one explicit finite-data certificate, uniform on a contact-diameter interval, separating private and visible width one from a hidden two-valued selector. The collision creates the observed signal from a prepared impact variable; removing the collision makes all three deficiencies zero. Varying the contact diameter changes the collision time and the microscopic path, while the certified resource gap and the conditional-law consequences persist. The effective constructions retain the original resource constraints.

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## 1. INTRODUCTION

A causal reduction consumes observations, emits reports, and carries some information into the future. The size and visibility of that carried information are mathematical data. In particular, a persistent hidden selector is not equivalent to fresh private randomization, and revealing the selector can change a comparison even when the retained register is unchanged. A theory of reduction must therefore compare marked laws before convexifying its feasible set.

There are two further requirements. A separating test should specify what it observes about a proposed reduction; an exact probability law is a much stronger observation than finitely many independent runs. Moreover, a certificate obtained in a microscopic model should survive an approximation of the dynamics, not only an approximation of a formula written after the dynamics have been fixed. This paper joins these requirements.

Our basic object is the resource spectrum of a marked causal experiment. A point of that spectrum records a register-width profile, an error, and, where applicable, a hidden or visible selector alphabet. The mark is an actual system variable. It is not available to the online encoder. The comparison tests include its terminal value. Theorem 2.2 gives the elementary compactness, factorization and costed composition properties. The subsequent argument does not replace the private spectrum by its convex hull.

**1.1. The principal mechanism.** Let  $B$  be the compact behavior image of a fixed resource signature. A behavior  $b = (b_j)_{j=1}^J$  lists the laws of finitely many marked causal transcripts. Fix target laws  $Q_j$  and put

$$F(b) = \max_{j,A} \{Q_j(A) - b_j(A)\}, \quad \delta(B, Q) = \min_{b \in B} F(b).$$

Complements of events supply both signs. An auditor is allowed  $n$  independent, reset runs of each row, followed by independent validation. The candidate is held fixed during the whole audit, but its realized private randomness is reset between runs. The auditor sees completed marked transcripts, not the candidate's probabilities or its retained states. This information pattern is different both from an ordinary single-run comparison and from a revealed-law verification game.

**Theorem 1.1** (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law  $\pi$ , the optimal error with  $K$  retained states is*

$$1 - \sum_{i=1}^{\min(K,N)} \pi(i), \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

*Independent hidden or visible selectors do not improve this cut value.*

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original  $B$ , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

*The exponent  $1/2$  cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.*

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals  $H, H'$  on  $[0, T]$ , and the same positive Gaussian noise level  $\sigma$ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

*Their joint marked observation laws differ by at most  $\varepsilon$ . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum  $L^2$  norm under the common Wiener reference when  $\eta \rightarrow 0$ . For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.*

(iv) *In the two-sphere model of Theorem 11.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than  $2/5$  against every private width-one simulator, uniformly for  $9/10 \leq d \leq 11/10$  and  $0 < \sigma \leq 1/24$ . A hidden two-valued selector has error less than  $13/100$ ; every visible-selector width-one error exceeds  $9/20$ . For  $d_n \rightarrow d$  the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For  $d \leq 2/5$  all three deficiencies vanish.*

Part (iv) is a composition, not a juxtaposition: the scattering calculation provides a marked gap, the reset-audit theorem turns it into one finite-data certificate on the nonconvex private image, and microscopic transport carries this certificate together with a conditional-law consumer through a changing collision geometry. The proofs are given in Theorems 3.2, 3.3, 4.1, 4.3, 10.1, 10.3, and 11.2.

**1.2. What the information restrictions do.** The continuation criterion is a geometric realization criterion, not a claim that ordinary Hankel rank equals nonnegative state complexity. Its generators must be stable under all one-step input–output shifts. The cut theorem gives a sharp obstruction even when all other checkpoints have unrestricted memory. In particular,  $m$  erased unbiased bits require at least  $(1 - \epsilon)2^m$  retained states to attain error  $\epsilon$ .

The reset-audit dual randomizes over a candidate chosen once for an entire audit. Its moment measure cannot be replaced by the barycenter behavior: the replicated samples are conditionally independent given that candidate, not after forgetting it. This is why minimax on the auditor’s finite coefficient simplex does not supply free persistent randomness to the simulator. The auditor’s

count register, within-run buffer, validation choice, and reset interface are all explicit. No sample-complexity statement here is a polynomial-time statement about minimizing on a nonconvex image.

We also retain revealed-behavior testing and its behavior-dimensional Bernstein bound. The infinite-dimensional representation is a general partition-of-unity identity. Its finite polynomial hierarchy and its small exact nonlinear witness remain useful, but they have different information requirements from the reset audit. Proposition 7.1 gives an exact input–output algorithm for the rational observable quotient and chart used by that hierarchy.

**1.3. Relation to earlier comparison and realization theory.** Blackwell and Le Cam supply the classical comparison viewpoint [1, 2, 13]. Filtered comparison has a substantial earlier literature, including Norberg and Weisshaupt [3, 19]. We do not assert that inserting a filtration into an experiment is new. The extra data here are fixed persistent width, the visibility of separately charged selectors, and the finite reset interface of the testing problem. An original-text proof-by-proof crosswalk with Norberg is not established in this revision; no claim of priority over an uninspected theorem is used in a proof.

Positive realization by stable finitely generated cones is classical. Fliess’s Theorem 3.1 and the stochastic-process specialization in Theorem 5.4 and its second corollary make this particularly clear [23]. Our continuation statement is a finite-horizon, input-indexed normalization of that mechanism, with a width at each cut. The new use is its connection to the marked resource spectrum and to an exact cut obstruction; the existence of a positive-cone realization is not claimed as a discovery.

The minimax exchange is classical [18]. The statistical lower bound is a two-point testing argument, and the empirical upper bound uses multinomial second moments. Bernstein and positivity methods are also classical [20, 21, 22]. What is proved here is the compatibility of these constructions with the unconvexified resource object, including their information costs and their transport through an interacting microscopic limit. Bayes’ formula, Gaussian change of measure, martingale representation, and maximal inequalities are the analytic inputs to the transport step; the moving-collision estimate verifies that step without replacing the microscopic trajectory by a fixed flow.

**1.4. Scope and organization.** The physical theorem is at fixed particle number and fixed positive noise. The  $B$ – $W$  correlation belongs to the preparation. Collision transduces  $B$  into the measured transverse signal; it does not create that correlation. The feedback is an acquisition gate, not a mechanical force. Changing the contact diameter changes a genuine microscopic path, but is not a Boltzmann–Grad or particle-number limit.

The foundations developed here remain applicable to the larger dynamical program. A concrete downstream consequence is the bounded Gaussian changing-path likelihood, posterior and entropic-value theorem proved in Section 10; it replaces a finite-dimensional-to-tightness shortcut by uniform estimates. Algebraic geometry, kinetic action compactness and particle-uniform hierarchy recovery retain their own hypotheses. No independence of an algebraic-geometric theorem is converted into a dependency merely by the order of papers.

The canonical proof chain follows the diagram in Appendix A. The full preceding comparison, physical-domain, task, and historical development is preserved in the companion volume. It is not necessary to reconstruct that archive to read the proofs in this article.

## 2. INTRINSIC CAUSAL DEFICIENCY

**2.1. Experiments and executable signatures.** Fix a finite parameter set  $\Theta$ , a horizon  $T$ , and finite control alphabets  $A_0, \dots, A_{T-1}$ . An experiment has a report at checkpoints  $0, \dots, T$  and an unobserved terminal mark  $W$  in a common finite set. It is specified by finite causal instruments:

for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write  $Y_t$  for the source report and  $Z_t$  for the target report. A private simulator of width  $K$  has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m &\in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' &\in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in  $m'$ . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted  $\mathcal{K}_K$ .

A deterministic feedback tree  $g$  chooses  $a_t$  from  $z_{0:t}$  and its own previous controls. It is the same controller on both sides of a comparison. Let  $\Omega_g$  be the finite marked transcript space, including reports, controls, and  $W$ . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here  $W$  on the left is the source's actual mark. The total variation convention is  $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$ . Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector*  $U \in [r]$ , independent of the source, chooses one simulator  $q^u$ . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions  $\lambda \in \Delta_r$  and  $q^u \in \mathcal{K}_K$ . The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining  $(u, m)$ , at width  $rK_t$ . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

**Definition 2.1** (Deficiency spectrum). The private resource spectrum of  $E$  relative to  $F$  is the upward-closed set of pairs  $(K, \varepsilon)$  satisfying  $\delta_K^p(E, F) \leq \varepsilon$ . The hidden spectrum consists of triples  $(K, r, \varepsilon)$  satisfying  $\delta_{K,r}^h(E, F) \leq \varepsilon$ , and the visible spectrum uses  $\delta_{K,r}^v$ . Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is  $\delta_{K,r}^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$ .

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize  $\max_t K_t$ , with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

**2.2. The finite comparison theorem.** Concatenate all probabilities  $P_{\theta,g}^q(\omega)$  into  $b(q) \in \mathbb{R}^N$ , retaining their row indices; put  $b_F = (Q_{\theta,g}(\omega))$ . Let  $\mathcal{D}_K$  be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For  $i = (\theta, g, B)$  and  $d \in \mathcal{D}_K$ , define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

**Theorem 2.2** (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- $r$  deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

*In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At  $K_t = 1$  for every  $t$ , this is precisely the same factorization with rows  $q_t(z \mid a, y)$  and  $q_0(z \mid y)$ .*

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

*An optimal simulation uses at most  $a_K + 1$  selector values. Thus every weighted causal test has optimal discrepancy at most  $\varepsilon$  if and only if a hidden mixture of  $K$ -state simulators has error at most  $\varepsilon$ ; it has a private realization of width at most  $(a_K + 1)K$ . More specifically,*

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put  $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$ . Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

*Writing  $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$ , the infimum over all finite visible selectors is attained using  $a_K^v + 1 \leq |\Theta| + 1$  values and equals*

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

*For every  $r \geq 1$ , and also after optimizing over all  $r$ ,*

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

*No corresponding equivalence with  $\delta_K^{\text{mix}} = 0$  holds in general.*

(iv) *For compatible experiments  $E, F, G$  and profiles  $K, L$ ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

*Here  $KL$  is the pointwise product, and the second line uses independent selectors. If an  $F$ -consumer has widths  $S_t$ , composing it with a private simulator gives widths  $K_t S_t$  and changes every  $[0, 1]$  marked risk by at most  $\delta_K^p(E, F)$ . A hidden selector is stored at the additional price  $r$ , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.*

*Proof.* Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes  $\lambda$  and the  $r$  policies in (2.3). For the visible problem there are only finitely many collections  $(g_u)$  and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting  $\varepsilon = 0$  forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of  $q$ , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability  $\Lambda_q$  on  $\mathcal{D}_K$ . On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every  $\theta, g$ , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies  $\text{conv } b(\mathcal{K}_K) = \text{conv} \{b(d) : d \in \mathcal{D}_K\}$ . The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most  $a_K + 1$  of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients  $c_d$  with  $\sum c_d = 0$  and  $\sum c_d b(d) = 0$ . Subtract the largest permissible multiple of  $c$  from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every  $w$  is equivalent to the maximum over  $w$  being at most  $\varepsilon$ , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by  $\lambda_u$ . For a fixed  $\theta$ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is  $\sum_u \lambda_u e_\theta(q^u)$ , proving (2.9). The map  $q \mapsto e(q)$  is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in  $\mathbb{R}^{|\Theta|}$  and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic  $q$ , not over deterministic tables: the nonlinear function  $e_\theta$  does not commute with (2.13). If the attained visible optimum is zero, then for every  $u$  of positive weight and every  $\theta$  one has  $e_\theta(q^u) = 0$ , since every summand is nonnegative. Any such  $q^u$  is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a  $K$ -state  $E$ -to- $F$  transducer and an  $L$ -state  $F$ -to- $G$  transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most  $K_t L_t$  values. For any terminal tester on  $G$ , the second transducer together with that tester is an admissible randomized feedback system on  $F$ . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine  $F$  system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product

with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any  $[0, 1]$  marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities.  $\square$

*Remark 2.3* (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

**2.3. A delayed channel and a strict signature separation.** Let the source reveal  $i \in [n]$  at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws  $j \in [m]$  from a specified row-stochastic matrix  $V = (V_{ij})$ . The unknown parameter is  $i$ . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size  $K$  matters.

**Proposition 2.4** (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where  $A$  is an  $n$ -by- $K$  row-stochastic matrix and  $B$  is a  $K$ -by- $m$  row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For  $1 \leq K \leq n$  the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance  $0 \leq \varepsilon < 1$ , the least private width is  $\lceil n/(1 - \varepsilon)^{-1} \rceil$ , whereas the least within-mode width with optimized hidden selection is  $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$ . The latter still pays for its selector. In particular, for  $n = m = 3$  and  $K = 2$ ,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For  $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$ ,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

*Proof.* The encoder maps the consumed  $i$  into its retained label by  $A$ ; the terminal decoder maps that label into  $j$  by  $B$ . This proves (2.14) including its converse implementation. If  $V = CD$  is any nonnegative factorization, let  $s_k = \sum_j D_{kj}$ . Delete zero rows of  $D$ , define  $B_{kj} = D_{kj}/s_k$  and  $A_{ik} = C_{ik}s_k$ . Since  $V$  is stochastic, the rows of  $A$  sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For  $V = I_n$ , row error is  $1 - (AB)_{ii}$ . If the least correct probability is  $c$ , for each  $i$  select a row  $k(i)$  of  $B$  with  $B_{k(i),i} \geq c$ . Some row is selected at least  $\lceil n/K \rceil$  times, so stochasticity gives  $c \leq 1/\lceil n/K \rceil$ . A partition into  $K$  groups of sizes at most  $\lceil n/K \rceil$ , each decoded uniformly, attains that bound. Also  $\sum_i (AB)_{ii} \leq K$ , by the calculation below with  $K$  in place of two. It remains valid under mixing, giving hidden error at least  $1 - K/n$ . Start with a deterministic channel that reproduces  $K$  input symbols exactly and sends all others to those  $K$  outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then  $K/n$ , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For  $V = I_3$ , total variation in row  $i$  is  $1 - (AB)_{ii}$ . If all three diagonal entries of  $AB$  were greater than  $1/2$ , then for each  $i$  some row of  $B$  would have its  $i$ th entry greater than  $1/2$ . A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound  $1/2$ . Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least  $1/3$ . Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are  $2/3$  and the off-diagonal entries are  $1/6$ . This is exactly  $V_*$  and has error  $1/3$  from  $I_3$ . It proves the hidden claims in both displays.

The smallest singular value of  $V_*$  is  $1/2$ . Any  $AB$  with two intermediate states has a unit vector  $x$  in its right null space, so  $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$ . If the maximal row total variation error is  $\varepsilon$ , each row has Euclidean error at most  $2\varepsilon$ , and hence the Frobenius error is at most  $2\sqrt{3}\varepsilon$ . This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity.  $\square$

**2.4. Optimal resource sets and decision certificates.** For a class  $\mathcal{C}_{S,q}(F)$  of common consumers, with all other information conventions fixed, let  $\mathcal{R}_{S,q}(F)$  be its vector of attainable  $[0, 1]$  risks. For a nonnegative row weight  $w$  summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If  $E$  privately simulates  $F$  with cost  $K$  and error  $\varepsilon$ , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance  $\varepsilon$ ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by  $r$ ; visible selectors multiply the separate public alphabet by  $r$ . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most  $\varepsilon$ , paired risk sets and their support values at unchanged width are within  $\varepsilon$ . If each is centered by a paired unrestricted benchmark whose error is also at most  $\varepsilon$ , the centered values are within  $2\varepsilon$ . Therefore a gap  $g > 0$  between zero excess and infeasible widths is stable under  $\varepsilon < g/4$ : a value below  $g/2$  in the approximation certifies feasibility in the exact finite problem, and a value above  $g/2$  certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

**Proposition 2.5** (Intrinsic finite-dimensional selector approximation). *Let  $\mathcal{R}$  be a compact risk image in  $C(J)$ , where  $J$  is compact. Suppose a bounded linear operator  $L : C(J) \rightarrow C(J)$  has rank at most  $m$  and  $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$ . Every barycenter  $\bar{r}$  of a probability on  $\mathcal{R}$  has a barycenter  $r_*$  of at most  $m + 1$  members of  $\mathcal{R}$  with  $\|\bar{r} - r_*\|_\infty \leq 2\eta$ . If  $\mathcal{R}$  lies in a finite affine space of dimension  $m$ , the error is zero with at most  $m + 1$  members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

*Proof.* The compact set  $L\mathcal{R}$  lies in an  $m$ -dimensional space. Its convex hull is compact, and  $L\bar{r}$  belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives  $r_1, \dots, r_{m+1} \in \mathcal{R}$  and weights  $\lambda$  such that  $L\bar{r} = \sum_i \lambda_i Lr_i$ . Put  $r_* = \sum_i \lambda_i r_i$ . The two residual

terms  $\bar{r} - L\bar{r}$  and  $r_* - Lr_*$  have norms at most  $\eta$ , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information.  $\square$

*Remark 2.6* (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- $r$  formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

### 3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

**3.1. Normalized continuation laws.** We first separate the transducer from the source to which it is attached. There are  $L$  epochs with finite input alphabets  $X_t$  and output alphabets  $Z_t$ . A causal kernel  $p(z_{1:L} \mid x_{1:L})$  is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let  $\mathcal{C}_t$ ,  $0 \leq t \leq L$ , be the finite-dimensional polytope of normalized causal kernels from  $x_{t+1:L}$  to  $z_{t+1:L}$ ; put  $\mathcal{C}_L = \{1\}$ . For  $v \in \mathcal{C}_t$  define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass  $v(z \mid x)$  is independent of the future input word. A zero shift is allowed.

**Definition 3.1** (Compatible continuation polytopes). A continuation cover of width  $K = (K_0, \dots, K_L)$  consists of lists  $V_t = (v_{t,m})_{m \leq K_t}$  in  $\mathcal{C}_t$  such that, for every  $m, x, z$ ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers  $p$  when  $p \in \text{conv } V_0$ . Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

**Theorem 3.2** (Continuation realization with a width profile). *A causal kernel  $p$  has a stochastic transducer realization with an initial  $K_0$ -valued register and  $K_t$  retained states after epoch  $t$  if and only if it has a compatible continuation cover of width  $K$ . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is  $p = v_{0,1}$  and  $K_0 = 1$ .*

*Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut  $t$  has nonnegative rank at most  $K_t$ . Individual rank bounds do not replace the simultaneous shift conditions (3.1).*

*Proof.* Given a transducer, let  $v_{t,m}$  be the suffix response when its register at time  $t$  equals  $m$ . The remaining rows define this response even for an unreachable state. If the next row is  $q_{t+1}(z, m' \mid m, x)$ , conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses  $p$  as a convex combination of  $V_0$ .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future

input word, the sum of the coefficients for a fixed  $m, x, z$  is  $v_{t,m}(z \mid x)$ . Summing over  $z$  gives one. The coefficients are therefore stochastic rows  $q_{t+1}(z, m' \mid m, x)$ . Choose an initial distribution representing  $p$  in  $\text{conv } V_0$ . Backward induction from  $\mathcal{C}_L = \{1\}$  proves that the constructed machine has exactly the response  $v_{t,m}$  from each register state. In particular its initial response is  $p$ . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of  $v_{t,m}$  with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through  $K_t$  generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance.  $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

**3.2. An erasure cut.** A simple family makes the width obstruction exact, including its error. Let  $X$  take values in  $[N]$  with law  $\pi$ . Before a specified cut the source reveals  $X$ . After the cut it has no reports correlated with  $X$ . The controller has no feedback channel carrying  $X$  across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its  $K$ -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual  $W = X$ . The target’s terminal report is  $X$ . All memory at other checkpoints is unrestricted.

**Theorem 3.3** (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform  $X$ ,  $\delta_K = (1 - K/N)_+$ . Thus an erasure cut for  $m$  unbiased bits requires  $K \geq (1 - \epsilon)2^m$  whenever its error is at most  $\epsilon$ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every  $\pi_x > 0$ , exact transmission has nonnegative cut rank  $N$ .

*Proof.* With all post-cut randomness averaged into the decoder, any private machine gives kernels  $e(m \mid x)$  and  $d(z \mid m)$ . Its joint terminal law is  $\pi_x \sum_m e(m \mid x) d(z \mid m)$ . Since the target law is  $\pi_x \mathbf{1}_{z=x}$ , total variation is exactly one minus the correct-decoding probability. Put  $u_x = \max_m d(x \mid m)$ . Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in  $u$  from a smaller weight to a larger one until the  $K$  largest weights are filled. To attain it, retain distinct labels for those letters and decode them

deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of  $X$ , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its  $N$  positive diagonal entries  $\pi_x$ . Its ordinary rank is  $N$ , so its nonnegative rank is at least  $N$  and its diagonal factorization attains  $N$ .  $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing  $X$  and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with  $X$  is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

#### 4. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

**4.1. The reset interface.** Let  $B \subset \prod_{j=1}^J \Delta(\Omega_j)$  be compact, and let  $Q = (Q_j)$  be fixed. Set  $d_j = |\Omega_j|$ ,  $D = \sum_j (d_j - 1)$ ,  $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ , and  $f_{j,A}(b) = Q_j(A) - b_j(A)$ . Include the empty event. No convexity of  $B$  is assumed.

A candidate  $b$  is fixed for the whole audit. The auditor obtains  $n$  independent samples of each  $b_j$ , with independence between rows. In a causal experiment, row  $j$  means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write  $C_{j,w}$  for the counts, and  $\mathcal{N}_n$  for all count arrays with  $\sum_w C_{j,w} = n$ . Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array  $c = (c_{C,i})$  with  $c_C \in \Delta(\mathcal{I})$ . After training it chooses  $i = (j, A)$  according to  $c_C$ . Independent candidate and target validation runs give the signed score  $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$ . Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (4.1)$$

Thus every audit payoff is polynomial of total degree at most  $Jn + 1$  in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

**Theorem 4.1** (Reset-audit duality and finite-sample approximation). *Let  $\delta = \min_{b \in B} \max_i f_i(b)$ . Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (4.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (4.3)$$

*The values are nondecreasing in  $n$ . If  $a = \dim \text{aff } B$ , a minimizing measure in (4.2) can be supported on at most  $\binom{a+Jn+1}{a}$  points.*

*An explicit feasible audit attaining the lower bound  $\delta - \sqrt{D/n}$  chooses an event maximizing  $Q_j(A) - C_j(A)/n$ , with a fixed deterministic tie rule. If the rational target probabilities are given,*

this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1)^{\sum_j d_j},$$

where  $H$  is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

*Proof.* The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in  $b$  and affine in  $c$ . The probability measures on compact  $B$  form a compact convex set for weak convergence. The minimum over  $b$  equals the minimum over such measures for fixed  $c$ . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (4.2). Continuity and compactness give attainment. This step uses measures on the original  $B$ , not feasible private realizations of their barycenters.

For any audit,  $g_c(b) \leq F(b) = \max_i f_i(b)$ . At a minimizer of  $F$  this gives  $v_n^{\text{aud}} \leq \delta$ . The zero event gives  $v_n^{\text{aud}} \geq 0$ . For the stated explicit rule put  $\hat{b}_j = C_j/n$  and  $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$ . Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E}(\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently  $\mathbb{E}\rho \leq (\sum_j \mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$ . Averaging the regret uniformly over  $B$  proves (4.3).

From  $n + 1$  unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the  $n$ -sample law, independently across rows. This realizes every  $n$ -sample audit as a randomized  $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on  $\text{aff } B$ . All the functions  $p_C f_i$  lie in the span of monomials of total degree at most  $Jn + 1$ , a space of dimension at most  $s = \binom{a + Jn + 1}{a}$  including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most  $s - 1$ . Eliminating affine dependencies gives a representation by at most  $s$  points. It preserves every integral in (4.2), and hence preserves the objective.

Finally, each count is in  $\{0, \dots, n\}$ . The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation.  $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by  $\bar{b} = \int b d\mu$  would replace  $\int p_C(b) f_i(b) d\mu$  by  $p_C(\bar{b}) f_i(\bar{b})$ , generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

**Proposition 4.2** (Both sides sampled). *Suppose the target reference  $Q^b$  varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both  $b$  and  $Q^b$ , and use their empirical difference. If their row dimensions are  $D_P, D_Q$ , the same construction has regret at most  $\sqrt{D_P/n} + \sqrt{D_Q/n}$ . Minimax and attainment hold on the*

compact image of pairs  $(b, Q^b)$ . In particular a variable reference law must not be treated as known fixed input to Theorem 4.1.

*Proof.* The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature.  $\square$

#### 4.2. A matching uniform sampling obstruction.

**Theorem 4.3** (Binary reset lower bound). *Take one binary row with target  $\text{Ber}(1/2)$  and candidate class  $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$ , where  $0 < h < 1/2$ . Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (4.4)$$

For every  $n \geq 1$ , choosing  $h = 1/(8\sqrt{n})$  gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (4.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

*Proof.* Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic  $d$  in  $[-1, 1]$ . The two expected payoffs are  $h\mathbb{E}_+d$  and  $-h\mathbb{E}_-d$ . Their minimum is at most half their sum, which is at most  $h\|P_+^n - P_-^n\|_{\text{TV}}$ . Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges  $P_+^n$  and  $P_-^n$  and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective  $h$ , whereas for finite  $n$  the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (4.4).

For  $h \leq 1/8$ , elementary integration of  $(1 - u^2)^{-1}$  gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most  $16nh^2$ . Pinsker's inequality bounds its total variation by  $\sqrt{8nh^2}$ . At the specified  $h$  this is less than  $1/2$ . Substituting in (4.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game.  $\square$

#### 4.3. Transport of one fixed audit.

**Proposition 4.4** (Reset cost under model perturbation). *Let  $q$  range over the same compact set of executable row designs in two models. Suppose uniformly in  $q, j$  that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule  $c$ , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (4.5)$$

The deficiencies themselves differ by at most  $\alpha + \beta$ , at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most  $\beta$ , independently of its reset count.

*Proof.* A product coupling of the  $Jn$  training runs fails with probability at most  $Jn\alpha$ . Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in  $[-1, 1]$ , so the change in its expectation from this training-law perturbation is at most  $2Jn\alpha$ . At each event the discrepancy changes by at most  $\alpha + \beta$ . This proves (4.5). For deficiency, every row variation changes by at most  $\alpha + \beta$ ; taking the maximum and then the minimum preserves that bound. No new simulator is inserted.  $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of  $g_c(q)$  above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

## 5. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 6.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as  $F(q) = \max_{1 \leq i \leq m} f_i(q)$ , where  $f_i$  are all its marked event discrepancies, including the empty event  $f_1 = 0$ . For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in  $f_i$ . Thus  $-1 \leq f_i \leq 1$  for all three signatures; the polynomials may be different.

Parameterize a row with  $d$  outcomes by  $d - 1$  variables in  $[0, 1]$ :

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map  $q(x)$  from  $[0, 1]^D$  to the feasible row product. Put  $h_i(x) = f_i(q(x))$  and  $g_i = (1 + h_i)/2$ . All  $h_i$  are rational polynomials,  $0 \leq g_i \leq 1$ , and  $g_1 = 1/2$ . The case  $D = 0$  is a single feasible table and is interpreted directly.

**Theorem 5.1** (Strict polynomial lower-certificate hierarchy). *For  $p \geq 1$  define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

*Then the denominators are positive,  $a_p$  is nondecreasing, and*

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (5.1)$$

*For rational  $L$  let*

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (5.2)$$

*A nonnegative tensor Bernstein expansion of  $P_{p,L}$  on  $[0, 1]^D$  certifies  $\delta \geq L$ . If  $L < \delta$ , some  $p$  and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies  $L < \delta$ . These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.*

*Proof.* Fix  $x$  and put  $b = \max_i g_i(x) \geq 1/2$  and  $r_i = g_i(x)/b$ . At least one  $r_i$  is one. Write  $S_j = \sum_i r_i^j$ . Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently  $S_{p+1}/S_p \geq m^{-1/(p+1)}$ . Since  $A_p = 2bS_{p+1}/S_p - 1$  and  $\max_i h_i = 2b - 1$ ,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy–Schwarz gives  $S_{p+2}S_p \geq S_{p+1}^2$ , so  $A_p$  is nondecreasing in  $p$ . Its denominator is at least  $2^{-p}$  and it is continuous on the cube. Taking minima proves (5.1). The bound is also at most  $2 \log m/(p+1)$ , by  $1 - e^{-u} \leq u$ .

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply  $P_{p,L} \geq 0$ . Dividing by the positive denominator gives  $F(q(x)) \geq A_p(x) \geq L$  for every  $x$ , which proves soundness. For completeness choose  $p$  so that the error in (5.1) is less than  $\delta - L$ . Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 5.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most  $m$ , so they give a strict positive uniform gap above  $L$ .  $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

**Lemma 5.2** (An explicit positivity degree). *Let  $P(x) = \sum_k c_k x^k$  be a rational polynomial in  $D$  variables and  $n \geq \max_{k,j} k_j$ . In its tensor Bernstein expansion of degree  $n$  in each coordinate the coefficient at  $\alpha \in \{0, \dots, n\}^D$  is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (5.3)$$

*In particular,  $\min P > 0$  implies positivity of every coefficient whenever  $n > C(P)/\min P$  and the degree restriction holds.*

*Proof.* For a binomial random variable  $B$  with parameters  $(n, x)$ ,  $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$ . Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is  $\binom{a}{k} = 0$  for  $a < k$ . To compare  $\binom{a}{k}/\binom{n}{k}$  with  $(a/n)^k$ , draw  $k$  indices uniformly from  $\{1, \dots, n\}$ . Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most  $\binom{k}{2}/n$ . Apply this total variation bound to the event that all drawn indices lie in a prescribed  $a$ -element subset. Hence the difference of the two factors is at most  $\binom{k}{2}/n$ . All factors lie in  $[0, 1]$ . A product telescoping bound followed by the triangle inequality proves (5.3). The final assertion follows by taking the minimum of  $P$ .  $\square$

**Corollary 5.3** (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level  $L$  is equivalent to the existence of the following finite rational certificate: an integer  $p$ , an integer  $n$ , and a strictly positive coefficient array for (5.2) whose identity is checked by (5.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to  $\delta$ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

*Proof.* Soundness and completeness are Theorem 5.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all  $(p, n)$  and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed  $(p, n)$  test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument.  $\square$

*Remark 5.4* (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions  $g_i^p / \sum_j g_j^p$  are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count  $(n + 1)^D$ , the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 6.3; its verification is a different cost.

**5.1. A sharp two-time nonconvex example.** The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits  $(0, 0)$  or  $(1, 1)$  with probability  $1/2$  each. At width one, a private simulator emits two independent bits with probabilities  $p, q$ . It cannot reread its first emitted bit.

**Proposition 5.5** (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

*The private objective is the maximum of the four polynomials*

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1 - p)(1 - q), \quad p + q - 2pq.$$

*Every constant mixture of these tests has minimum at most zero, whereas Corollary 5.3 certifies every rational  $L < \sqrt{2} - 1$  at private width one.*

*Proof.* Put  $u = pq$  and  $v = (1 - p)(1 - q)$ . Direct total variation calculation gives  $F = 1 - \min(u, 1/2) - \min(v, 1/2)$  and the displayed maximum. By Cauchy–Schwarz,  $\sqrt{u} + \sqrt{v} \leq 1$ . If  $u \geq 1/2$ , then  $v \leq (1 - 1/\sqrt{2})^2$ , so the sum of the two capped terms is at most  $2 - \sqrt{2}$ ; the case  $v \geq 1/2$  is symmetric. Otherwise set  $a = \sqrt{u}$ ,  $b = \sqrt{v}$ . On  $0 \leq a, b \leq 1/\sqrt{2}$ ,  $a + b \leq 1$ , the maximum of  $a^2 + b^2$  is at a vertex of this polygon and equals  $2 - \sqrt{2}$ . Equality is attained at  $p = q = 1/\sqrt{2}$ . A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation.  $\square$

## 6. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

**6.1. The testing game.** Let  $B$  be a nonempty compact subset of a finite-dimensional affine space, and let  $f_1, \dots, f_m$  be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2,  $b$  is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts;  $f_i$  ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set  $B$ . No convex hull is taken.

An *adaptive test* is a continuous map  $\kappa : B \rightarrow \Delta_m$ , chosen before a candidate  $b$ . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function  $\kappa$ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

**Theorem 6.1** (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{6.1}$$

*The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.*

*Proof.* Fix  $\eta > 0$ . The sets  $U_i = \{b : f_i(b) > F(b) - \eta\}$  form an open cover of the compact metric space  $B$ . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has  $F(b) - \eta < G(b, \kappa) \leq F(b)$  for all  $b$ . Consequently the first supremum equals  $\delta$ . The same uniform inequality shows, for every probability  $\mu$ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is  $\delta$ , attained by a point mass at a minimizer of  $F$ . This also proves the displayed minimax exchange without an assumption that  $B$  is convex. Equivalently one can apply the usual compact-convex minimax theorem to  $\mathcal{P}(B)$  and the convex class of continuous tests, with the pairing  $\int G d\mu$ . Finally choose the smallest index attaining  $F(b)$ . Its level sets are Borel, and its indicator vector is a Borel test with score exactly  $F$ .  $\square$

The probability  $\mu$  here describes a lottery over *revealed candidate laws*. The score is  $\int G(b, \kappa) d\mu(b)$ , not  $G(\int b d\mu, \kappa)$ . Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

**6.2. A size bound in behavior dimension.** Factor out directions annihilated by every  $f_i - f_i(b_0)$ , so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by  $a$  and let  $R = \text{diam}_*(B)$ . All conclusions below are unaffected by this observable quotient. The case  $a = 0$  is a single constant comparison and requires one test. Suppose henceforth that  $a \geq 1$ , so  $R > 0$ .

**Lemma 6.2** (A bounded behavior chart). *There exist  $b_0 \in B$  and vectors  $v_j \in B - b_0$ ,  $1 \leq j \leq a$ , forming a basis, such that  $\|v_j\|_* \leq R$  and every  $b \in B$  has coordinates  $b = b_0 + \sum_j t_j v_j$  with  $|t_j| \leq 2$ . If  $B$  is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

*Proof.* Fix  $b_0 \in B$ . In any fixed coordinates on the difference space, let  $D_0 > 0$  be the maximum absolute determinant of  $a$  vectors in  $B - b_0$ . Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than  $D_0/2$ . Replacing its  $j$ th vector by  $b - b_0$  multiplies the determinant by  $t_j$ ; maximality gives  $|t_j| < 2$ . The norm bound is the definition of  $R$ . For a rational polynomial image, take  $b_0$  at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart.  $\square$

Set  $b(x) = b_0 + \sum_j (4x_j - 2)v_j$  and  $C = \{x \in [0, 1]^a : b(x) \in B\}$ . The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write  $\ell_i(x) = f_i(b(x))$ . Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (6.2)$$

For  $n \geq 0$  let  $\beta_{\alpha,n}$  be the tensor Bernstein basis of coordinatewise degree  $n$ . A feasible coefficient array has  $c_{\alpha i} \geq 0$  and  $\sum_i c_{\alpha i} = 1$  for every  $\alpha \in \{0, \dots, n\}^a$ . It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (6.3)$$

**Theorem 6.3** (Intrinsic adaptive degree and support bounds). *The maxima in (6.3) are attained and  $d_n \leq d_{n+1} \leq \delta$ . For  $n \geq 1$ ,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (6.4)$$

*There is an array with at most  $(n+1)^a$  nonzero coefficients, all equal to one, whose minimum score is at least  $\delta - 4Ra/\sqrt{n}$ . Its test has coordinatewise degree  $n$ , and its score degree at most  $n+1$ . Furthermore*

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (6.5)$$

*and a minimizing measure can be supported on at most  $(n+2)^a$  points. In particular a strict level  $L < \delta$  admits an adaptive test of positive uniform margin with  $n > (8Ra/(\delta - L))^2$  and at most  $(n+1)^a$  occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.*

*Proof.* The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact  $C$  is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the  $\ell_i(x)$ , proving the upper bound by  $\delta$ .

At each grid point  $\alpha/n$  choose an index  $i_{\alpha}$  maximizing  $\ell_i$  there. The choice is made even at infeasible grid points; only the final score is minimized over  $C$ . Give that index coefficient one and

the others zero. For fixed  $x$  let  $A_j$  be independent  $\text{Bin}(n, x_j)$  variables. If  $i_*$  maximizes at  $x$ ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses  $\text{Var}(A_j/n) \leq 1/(4n)$ . This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over  $C$  by a minimum over  $\mathcal{P}(C)$ . The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (6.5). The integrands are polynomials of coordinatewise degree at most  $n+1$ . Preserve all monomial moments of that degree by Carathéodory's theorem. There are  $(n+2)^a$  monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most  $(n+2)^a - 1$ . A measure with at most  $(n+2)^a$  atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than  $(\delta - L)/2$ .  $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (6.5) cannot be replaced by its behavior barycenter. For  $n=0$  it can:  $d_0$  is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 5.5, the target belongs to the convex hull of private behaviors, so  $d_0 = 0$ , whereas  $\lim_n d_n = \sqrt{2} - 1$ . Thus the hierarchy recovers a gap which every constant test misses.

*Remark 6.4* (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set  $C$ , the bit lengths of a rational chart, or the size of a description of  $C$ . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use  $R \leq 1$  with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound  $R \leq 2$  is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

### 6.3. A three-weight certificate invisible to constant tests.

**Proposition 6.5** (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities  $p, q$ , for the target which assigns probability  $1/2$  to each of 00, 11. Put  $u = pq$ ,  $v = (1-p)(1-q)$  and  $J(t) = 3t^2 - 2t^3$ . The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

*are nonnegative and sum to one. Against the event discrepancies  $1/2 - u, 1/2 - v, 1 - u - v$ , respectively, their score  $G$  satisfies*

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (6.6)$$

*The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.*

*Proof.* We have  $u + v \leq 1$ . Since  $J$  is increasing on  $[0, 1]$  and  $J(1-t) = 1 - J(t)$ , the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set  $a = (p + q - 1)^2$  and  $b = (p - q)^2$ . Then  $a, b \geq 0$  and  $a + b \leq 1$ . Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let  $c = 1 - a - b$ . Another polynomial identity, valid when  $a + b + c = 1$ , is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus  $G(a, b) \geq G(a, 0)$ . The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients  $z_k/349440$ , where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is  $351/349440 = 9/8960$ , proving (6.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero.  $\square$

For comparison, the affine weights  $v, u, 1 - u - v$  already give the smaller bound  $3/8$ , with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous  $f_i$  on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of  $F$ . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 4 imposes an explicit finite-data restriction.

**6.4. Strict certificates and effective physical input.** Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at  $L = \delta$ .

**Proposition 6.6** (Uniform effective presentation input). *Suppose a single algorithm, on input  $j$ , returns rational finite instruments  $E_j, F_j$  and nonnegative rational error enclosures  $a_j, c_j$  for two-sided stateless marked feedback presentations of  $E, F$ . Suppose  $a_j + c_j \rightarrow 0$ . All alphabets, widths, selector laws and finite subroutines must be computable uniformly from  $j$  and the requested*

signature. Then, for every positive rational  $\eta$ , an algorithm returns a rational interval of width below  $\eta$  containing  $\delta_b(E, F)$  at the same signature.

*Proof.* Enumerate  $j$  until  $a_j + c_j < \eta/8$ . This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below  $\eta/2$ . Enlarge it by  $a_j + c_j$  at both ends using stateless comparison. The enlarged width is below  $3\eta/4$ . Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition.  $\square$

## 7. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 6.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope  $\Omega \subset \mathbb{R}^r$ . Zero-dimensional factors are omitted.

**Proposition 7.1** (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map  $b : \Omega \rightarrow \mathbb{R}^N$ , and finitely many rational affine functionals  $f_i$  on  $\mathbb{R}^N$ . There is a terminating algorithm which returns its observable affine dimension  $a$ , a rational basis and coordinate map for the observable quotient, and, when  $a > 0$ , rational parameter points  $q_0, q_1, \dots, q_a$  with the following properties. With*

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

*the vectors  $v_j$  are a basis of the observable difference space, and all coordinates  $t(q)$  in that basis satisfy  $|t_j(q)| < 2$ . The output also contains the rational polynomial cube map  $x(q) = (t(q) + 2)/4$ , a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in  $x$ . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.*

*Proof.* Choose a rational relative interior point  $q_0$  of the input simplexes. Expand the vector polynomial  $z(q)$  in the free row coordinates. Let  $V$  be the rational span of all its coefficient vectors, including its constant coefficient. This equals  $\text{span } z(\Omega)$ : a linear functional vanishes on  $z(\Omega)$  exactly when its polynomial vanishes on the nonempty interior of  $\Omega$ , hence identically, hence on every coefficient. Gaussian elimination therefore gives  $a = \dim V$  and a rational basis matrix  $U$ . Choosing independent rows gives a rational left inverse  $L$  with  $LU = I_a$ . Thus  $y(q) = Lz(q)$  provides exact quotient coordinates. If  $a = 0$ , all discrepancies are constant and no chart is required.

For  $a > 0$  let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives  $D_0 > 0$ . Enumerate rational tuples  $(q_1, \dots, q_a)$  in  $\Omega^a$ . Membership and their determinant  $d$  are decided by rational arithmetic. For  $d \neq 0$ , decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (7.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has  $|d| > D_0/2$ . For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns  $y(q_j)$  to obtain  $t(q)$ . Cramer's rule and (7.1) give  $|t_j(q)| < 2$  for every  $q$ . In the observable norm  $\|z\|_* = \max_i |z_i|$ , each  $v_j$  has norm at most the diameter of  $z(\Omega)$ . An upper bound for that diameter can be produced, for example, by bounding each free coordinate in  $[0, 1]$  and summing the absolute polynomial coefficients for  $f_i(b(q)) - f_i(b(q'))$ . This bound is rational and computable. Rational matrix multiplication and substitution give  $x(q)$  and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations.  $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

## 8. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

**8.1. The exact marked two-report experiment.** Let  $B, W \in \{0, 1\}$  have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (8.1)$$

The blind source  $E_*$  has two constant reports and actual mark  $W$ . The target  $F_*$  reports  $Y_1 = B$ , receives  $A \in \{0, 1\}$  after that report, and reports  $Y_2 = AB$ . The feedback tester may choose  $A$  as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate  $A$ . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of  $(B, W)$ . Put

$$d_* = \sqrt{5/2} - 9/8.$$

**Theorem 8.1** (An exact marked causal separation). *At the width profile  $(1, 1)$ ,*

$$\delta^p(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (8.2)$$

*The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter  $\sqrt{5/8}$ , followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.*

*Proof.* First choose the constant gate  $A = 1$ . The two private emissions are independent Bernoulli variables with parameters  $p, q$  and are independent of  $W$ . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is  $H(u) + H(v)$ , where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two  $W$ -masses at each equal pair are  $3/16$  and  $5/16$ . Total variation is one minus this overlap.

Cauchy–Schwarz gives  $\sqrt{u} + \sqrt{v} \leq 1$ . By symmetry suppose  $u \geq v$ . If  $u \leq 1/4$  the overlap is at most  $1/2$ . If  $u \geq 1/4$ , then  $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$ , so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of  $[1/4, a]$  and  $[a, b]$  and nonincreasing on  $[b, 1]$ . Its maximum is therefore attained at  $1/4$ ,  $a$  or  $b$ . The values at  $a$  and  $b$  are  $1 + 2a - 2\sqrt{a}$  and  $3/2 + b - 2\sqrt{b}$  respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using  $\sqrt{a} > 3/5$ ,  $\sqrt{b} > 3/4$ . It also exceeds  $1/2$ . Taking  $p = q = \sqrt{b}$  attains this maximum, and hence the minimum error is  $1 - (3/2 + b - 2\sqrt{b}) = d_*$ .

For any hidden selector, the all-on report pair is still independent of  $W$ . If its probabilities of 11 and 00 are  $u, v$ , then  $u + v \leq 1$ . Concavity and monotonicity of  $H$  give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit  $J$ , emitting  $J$  twice, attains this overlap. This requires precisely two deterministic selector tables and gives error  $1/8$ . In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair  $(X_1, X_2)$ , emit  $X_1$ , and after receiving  $A$  emit  $AX_2$ . For a deterministic feedback  $A = g(X_1)$  the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit  $J$  is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of  $(B, W)$ . The all-on gate in every fiber has error at least  $d_*$  in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains  $d_*$ .  $\square$

**8.2. A full-dimensional collision preparation.** Consider two labeled equal-mass hard spheres in  $\mathbb{R}^3$  with contact distance  $d$ . The center of mass  $Q = (q_1 + q_2)/2$  ranges over any fixed rational position box. Write  $r = q_1 - q_2$ ,  $g = v_1 - v_2$ , and  $V = (v_1 + v_2)/2$ . Conditional on  $(B, W)$  in (8.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (8.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is  $W = \mathbf{1}_{\{V_z > 0\}}$ ; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

**Lemma 8.2** (Uniform transverse collision signal). *For every preparation in (8.3) and every  $d \in [9/10, 11/10]$ , exactly one collision occurs before time 3. Its time  $t_c$  lies in  $(4/5, 3/2)$ , and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (8.4)$$

*For  $0 < d \leq 2/5$  no collision occurs before time 3, and the detector is zero throughout that interval.*

*Proof.* Up to a collision  $r(t) = r + tg$ . For  $t \leq 4/5$ , its  $x$  coordinate is at most  $-29/10 + (4/5)(21/10) = -61/50$ , so its norm exceeds  $11/10$ . At time  $3/2$  the absolute coordinate bounds are  $1/4$ ,  $113/200$  and  $1/40$ , whose squared sum is less than  $(9/10)^2$ . Thus a first contact occurs in the stated interval. At the free zero of the  $x$  coordinate, whose time is at most  $31/19$ , the transverse norm is less than  $9/10$  as well. The first contact therefore has  $n_x < 0$ , where  $n = r(t_c)/d$ . The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that  $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$ . The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing,  $r(t_c) \cdot g' > 0$ , so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (8.4). For  $d \leq 2/5$  and  $0 \leq t \leq 3$ ,  $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$ . Thus there is no collision. The incoming  $|v_{1,y}| \leq 3/200$  then makes  $h$  zero.  $\square$

At time 2 the physical target  $F_{d,\sigma}$  reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate  $A \in \{0, 1\}$ , and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the  $\xi_i$  are independent standard normal variables, independent of the preparation. The source  $E_d$  has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking  $A = 0$  suppresses later information and taking  $A = 1$  exposes another noisy observation of a collision-generated quantity.

**Theorem 8.3** (Collision-sensitive physical resource separation). *Let  $d \in [9/10, 11/10]$  and  $0 < \sigma \leq 1/12$ . At retained width  $(1, 1)$  the physical deficiencies obey, uniformly in  $d$ ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (8.5)$$

*In particular private and visible error exceed  $9/20$ , while hidden two-selector error is less than  $13/100$ . More generally the errors in these three comparisons are at most  $2\Phi(-1/(4\sigma))$ , which tends to zero as  $\sigma \downarrow 0$ . For  $0 < d \leq 2/5$  all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.*

*Proof.* Couple the ideal and physical target by the same  $(B, W)$  and the same feedback random table. Lemma 8.2 bounds each potential bit error by  $\Phi(-1/(4\sigma))$ . With probability at least  $1 - 2\Phi(-1/(4\sigma))$  both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly  $E_*$ . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound  $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$  at  $x = 3$  gives  $\Phi(-3) < 1/600$ : use  $e^{9/2} > 80$  and  $\sqrt{2\pi} > 5/2$ . Hence the total error is below  $1/300$ . The inequalities  $d_* - 1/300 > 9/20$  and  $1/8 + 1/300 < 13/100$  follow by squaring positive rational bounds.

When  $d \leq 2/5$ , both potential reports are independent fair noise bits and are independent of  $W$ . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under  $(q, d, t) \mapsto (sq, sd, st)$  with  $v$  unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit.  $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

**8.3. Uniformly effective finite presentations.** The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

**Proposition 8.4** (Collision-chart rational presentations). *For computable  $d \in [9/10, 11/10]$  and  $\sigma > 0$ , one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for  $(E_d, F_{d,\sigma})$ . On a compact computable parameter interval with  $\sigma \geq \sigma_0 > 0$ , the errors and the construction are uniform. Thus Proposition 6.6 applies at every fixed finite state and selector signature, not only at  $(1, 1)$ .*

*Proof.* On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals  $-d(g \cdot n) > (9/10)(1443/1000)$ . All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound  $C_{\text{geo}}$  for  $v'_{1,y}$  in the packet variables and  $d$ . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability  $p = \Phi(h(v'_{1,y})/\sigma)$ . Their four probabilities are  $p^2$ ,  $p(1-p)$ ,  $p(1-p)$  and  $(1-p)^2$ . The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most  $2C_{\text{geo}}/\sigma_0$  for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh  $h$  therefore have explicit error at most a computable constant times  $h$ , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (8.1). For each  $W$ , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep  $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$  exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent

and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence.  $\square$

## 9. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let  $Z$  take values in a standard Borel space and let  $Y$  take values in a standard Borel canonical path space on  $[0, T]$ , with a right-continuous observation filtration. Consider equivalent joint probability measures  $P, Q$  on  $(Z, Y)$ , with the same  $Z$ -marginal  $\nu$ . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable  $f(Z)$  write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include  $T$  and can be evaluated on a countable dense time set.

**Theorem 9.1** (Marked variation controls the whole posterior process). *Let  $\varepsilon = \|P - Q\|_{\text{TV}}$  and  $|f| \leq 1$ . When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (9.1)$$

*There is a coupling of  $(Z, Y^P)$  and  $(Z, Y^Q)$  which retains exactly the same latent variable and satisfies*

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (9.2)$$

*The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.*

*Proof.* Let  $D = dP/dQ$ ,  $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$ , and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since  $\mathbb{E}_Q|D - 1| = 2\varepsilon$ , both martingales  $A, B$  have terminal absolute expectation at most  $2\varepsilon$ . Doob's weak  $L^1$  inequality gives, for  $0 < u \leq 2$ ,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating  $2u$  times this bound on  $(0, 2)$  gives  $16\varepsilon + 64\varepsilon = 80\varepsilon$ , proving (9.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on  $Z$ , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the

event of identical full output paths use (9.1) with the  $Q$ -marginal; on its complement the squared difference is at most four. This proves (9.2). The maximal coupling may depend on the full output path; it has no asserted online implementation.  $\square$

**Corollary 9.2** (Changing-filtration convergence from marked comparison). *Let  $P_n$  and  $P$  be equivalent joint laws with the same latent marginal  $\nu$ , and  $\|P_n - P\|_{\text{TV}} \rightarrow 0$ . They admit a common probability-space realization  $(Z, Y, Y^1, Y^2, \dots)$  such that, for every  $f \in L^2(\nu)$ ,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

*The same realization gives convergence uniformly in time in probability for  $f \in L^1(\nu)$ . It also permits  $f_n \rightarrow f$  in  $L^2(\nu)$ , respectively in  $L^1(\nu)$ , in the first conditional expectation.*

*Proof.* Glue the conditional maximal couplings of  $(Z, Y^n)$  with the fixed reference  $(Z, Y)$  by taking their conditional kernels independently on a countable product. The coupling does not depend on  $f$ . For the truncation  $f^{(K)} = (-K) \vee (f \wedge K)$ , Theorem 9.1 bounds the middle difference in the  $L^2$  norm of the supremum by  $K\sqrt{84\varepsilon_n}$ . Doob's  $L^2$  inequality bounds each truncation tail by  $2\|f - f^{(K)}\|_{L^2(\nu)}$ . The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let  $n \rightarrow \infty$ , then  $K \rightarrow \infty$ . For  $L^1$  use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference  $f_n - f$  is handled by the same maximal inequalities under  $P_n$ , whose latent marginal remains  $\nu$ .  $\square$

**9.1. A microscopic sufficient condition.** Let  $(\Phi_t)_{t \in \mathbb{R}}$  be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability  $\mu$ . Fix a prepared law  $\nu$  with  $d\nu/d\mu \leq C$ , and bounded real observables  $h, h_n$ . These may be collision-sensitive tagged observables. Put  $\eta_n = \|h_n - h\|_{L^2(\mu)}$ . With a common latent initial state  $Z \sim \nu$ , consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of  $Z$  in each model. The notation does not prescribe a synchronous coupling.

**Theorem 9.3** (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (9.3)$$

*For every  $|f(Z)| \leq 1$  they admit a latent-preserving coupling with*

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (9.4)$$

*In particular  $\eta_n \rightarrow 0$  supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when  $h_n$  are orbit-mollified graph-core approximants to any bounded  $h \in L^2(\mu)$ .*

*Proof.* Conditional on  $Z$ , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses  $\nu \leq C\mu$  and invariance of  $\mu$ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (9.3). Applying Theorem 9.1 proves (9.4).

For the last assertion choose a nonnegative smooth approximate identity  $\rho_n$  on the time line and set  $h_n = \int \rho_n(s) h \circ \Phi_s ds$ . Strong continuity of the Koopman group in  $L^2(\mu)$  gives  $\eta_n \rightarrow 0$  and  $\|h_n\|_\infty \leq \|h\|_\infty$ . Repeated integration by parts puts  $h_n$  in every power domain of the generator, with derivatives obtained from  $\rho_n^{(k)}$ . This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate.  $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (9.3). That is the resource-comparison consequence. The latent-preserving coupling in (9.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

## 10. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

**10.1. A common preparation, not a common flow.** Let  $(\mathcal{Z}, \nu)$  be a standard Borel probability space, and let  $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$  be jointly measurable, with  $|H|, |H_n| \leq M$ . They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from  $\mathcal{Z}$ . No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On  $\mathcal{Z} \times C([0, T])$  let  $R$  be the law of independent  $Z \sim \nu$  and  $Y = \sigma B$ , where  $\sigma > 0$ . Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (10.1)$$

$$dP_n = L_T^n dR. \quad (10.2)$$

The reference model  $P, L$  uses  $H$ . Conditional on  $Z$ , these are Gaussian shifts, so under  $P_n$  the observed path has law  $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$ . Boundedness gives true strictly positive density martingales. Write  $\mathcal{F}_t$  for the usual canonical observation filtration and  $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$  for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

**Theorem 10.1** (Likelihood and posterior transport for changing paths). *The measures  $P_n, P, R$  are equivalent and have common  $Z$ -marginal  $\nu$ . With  $A = M^2 T / \sigma^2$  and  $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$ ,*

one has

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (10.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (10.4)$$

For  $|f(Z)| \leq K$  put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (10.5)$$

In particular this applies to the projected likelihood  $\ell^n$ . For every  $|f| \leq 1$ , the posterior functions satisfy the same-path bound  $80\varepsilon_n$  of Theorem 9.1, and there is one latent-preserving coupling, independent of  $f$ , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (10.6)$$

If  $\eta_n \rightarrow 0$  and deterministic microscopic paths  $X_n(Z) \rightarrow X(Z)$  in a specified Polish path metric in  $\nu$ -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

*Proof.* Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (10.3).

For the quantitative density estimate, condition first on  $Z = z$  and write  $D_t = L_t^n - L_t$ . The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of  $L_t$  is at most  $e^A$ . Itô's isometry and  $(u + v)^2 \leq 2u^2 + 2v^2$  therefore imply

$$\mathbb{E}_R[D_t^2 \mid z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 \mid z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in  $z$ , gives  $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$ . The difference is a square-integrable  $\mathcal{H}_t$ -martingale. Doob's  $L^2$  inequality proves (10.4). Applying the same maximal inequality to the  $\mathcal{F}_t$ -martingale  $\mathbb{E}_R[(L_T^n - L_T)f \mid \mathcal{F}_t]$  proves (10.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 9.1 and its countable conditional-coupling construction. For the last claim, on the event  $Y^n = Y$  the two likelihood functions are evaluated on the same  $(Z, Y)$ . Their uniform difference tends to zero in  $R$ -probability by the estimates just proved, and therefore in  $P$ -probability by absolute continuity of the fixed  $P$  with respect to  $R$ . The complementary event has probability at most  $\varepsilon_n$ . The same argument applies to  $\ell^n$ , and (10.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same  $Z$ . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence.  $\square$

For  $f \in L^2(\nu)$  the posterior conclusion follows by truncation exactly as in Corollary 9.2; it also permits  $f_n \rightarrow f$  in  $L^2(\nu)$ . Boundedness of  $H_n$  is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

**10.2. The projected stochastic exponential.** Let  $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t]$ , using a predictable version for Lebesgue-almost every  $t$ . Put  $\theta_t^n = \hat{H}_t^n / \sigma$ .

**Proposition 10.2** (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (10.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (10.8)$$

Here  $I^n$  is a  $P_n$ -Brownian motion in  $\mathcal{F}_t$ . In particular the limit of the projected likelihoods in (10.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

*Proof.* Project  $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$  onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) \mid \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since  $|\theta_t^n| \leq M/\sigma$ , the solution is the positive stochastic exponential (10.7); Girsanov gives the stated innovation. The same argument with  $H$  identifies the limiting exponential. Finally  $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$  for bounded signed  $F$ . There is no interchange based only on a formal limit of logarithms.  $\square$

**10.3. A bounded entropic backward-equation consumer.** Fix  $\gamma > 0$  and a latent payoff  $f$  with  $|f| \leq 1$ . Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n, e^{\gamma f}}}{\ell_t^n}. \quad (10.9)$$

The terminal condition here is the observable conditional certainty equivalent  $U_T^n$ , not the generally unobserved  $f(Z)$  itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

**Theorem 10.3** (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value  $U_T^n$ ; for a predictable integrand  $Z^n$  it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (10.10)$$

On the coupling of Theorem 10.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left( \frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (10.11)$$

For each  $r > e^\gamma$  let  $\tau_r^n$  be the first exit of any of  $\ell^n, \ell, N^{n, e^{\gamma f}}, N^{e^{\gamma f}}$  from  $(1/r, r)$ , capped at  $T$ . If  $\eta_n \rightarrow 0$ , then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (10.12)$$

Furthermore  $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$ . Thus the localizations exhaust the observation experiment in probability.

*Proof.* For a positive bounded latent  $g$ , write  $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$  and  $dN_t^{n,g} = a_t^{n,g} dB_t$ . Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with  $g = 1$  gives  $d\ell_t^n = b_t^n dB_t$  and  $b_t^n/\ell_t^n = \theta_t^n$ . Put  $N^n = N^{n,e^{\gamma f}}$  and  $a^n = a^{n,e^{\gamma f}}$ . Itô's formula in (10.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n/N_t^n - b_t^n/\ell_t^n).$$

Replacing  $dB_t$  by  $dI_t^n + \theta_t^n dt$  reduces its drift to  $-\gamma|Z_t^n|^2/2$ , which proves (10.10). The ratios are bounded by  $M/\sigma$ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (10.10) produces a bounded  $P_n$ -martingale with terminal value  $e^{\gamma U_T^n}$ . Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 9.1 to  $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^\gamma - e^{-\gamma}) \in [0, 1]$ . The map taking its conditional expectation to  $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$  has Lipschitz constant at most  $(e^{2\gamma} - 1)/\gamma$ . This proves (10.11).

The initial values of  $N^n$  and  $N$  agree, since  $Z$  has the same marginal. Itô's isometry and the terminal version of (10.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before  $\tau_r^n$ , all denominators lie in  $[1/r, r]$ , and the limiting numerators  $a, b$  are bounded in absolute value by  $(M/\sigma)r$ . For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r|a^n - a| + (M/\sigma)r^3|N^n - N|.$$

The supremum estimates (10.5), followed by integration in time, prove (10.12). The same inequality applies to  $b^n/\ell^n$ .

Finally  $e^{-\gamma} \ell^n \leq N^n \leq e^\gamma \ell^n$ . Each  $\ell^n$  is a stochastic exponential with uniformly bounded integrand  $\theta^n$ . Its positive and negative moments of the supremum are bounded uniformly in  $n$ : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of  $N^n$  with  $\ell^n$ , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers.  $\square$

**Corollary 10.4** (A changing-filtration stopping consumer). *Let  $\psi(t, x) \in [0, 1]$  be measurable and  $L_\psi$ -Lipschitz in  $x$ , uniformly in  $t$ . For a bounded latent  $f$  with  $|f| \leq 1$ , put  $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$ . Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

*The same statement for the entropic value  $U$  has the additional Lipschitz factor  $(e^{2\gamma} - 1)/\gamma$  on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.*

*Proof.* For a fixed stopping time, first change the measure while keeping its  $[0, 1]$  reward function, at cost at most  $\varepsilon_n$ . Then compare the two reward functions on the same canonical path under  $P$ , using Theorem 9.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem.  $\square$

**10.4. Resource transport is not the posterior coupling.** Apply any common finite causal acquisition map to  $Y$  and retain a fixed finite mark  $w(Z)$ . Data processing makes its marked row variations at most  $\varepsilon_n$ , uniformly over a declared finite feedback family. If the source and target have respective bounds  $\alpha_n, \beta_n$ , Theorem 2.2 gives same-budget deficiency perturbation at most  $\alpha_n + \beta_n$ , and Proposition 4.4 transports a fixed reset audit. A subsequent consumer of width  $S$  composed with a simulator of width  $K$  has width  $KS$ , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

**Example 10.5** (Why an amplitude bound is not a tightness argument). Let  $\xi_1, \xi_2$  be independent symmetric signs, and, for  $n \geq 5$ , let  $\mathcal{G}_t^n$  reveal  $\xi_1$  at  $1/2 - 2/n$  and  $\xi_2$  at  $1/2 - 1/n$ . On  $[0, 1]$  the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale  $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$ . Nevertheless its laws are not tight in the Skorokhod  $J_1$  topology. On the event  $\xi_2 = -\xi_1$ , of probability  $1/2$ , its path has an excursion of height one on an interval of length  $1/n$  and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than  $2/n$ . This violates the necessary  $J_1$  tightness modulus criterion. All maximal moments are uniformly bounded by  $2^p$ .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 10.1 remove the need for that inference in the observation class treated here.

## 11. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 8.1 and Lemma 8.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let  $z$  denote a preparation in the boxes of (8.3), and let  $\nu$  be that prepared law. The sign of the impact orientation is  $2B - 1$ , and the conserved center-of-mass longitudinal sign is the actual mark  $W$ . Their correlation is prepared:  $\mathbb{P}(B = b, W = w) = 5/16$  for  $b = w$  and  $3/16$  otherwise. The contact diameter  $d$  ranges over  $I = [9/10, 11/10]$ . Let  $X_d(t, z)$  be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (11.1)$$

The acquisition gate  $A \in \{0, 1\}$  is chosen after  $Z_1^d$ ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

**Lemma 11.1** (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time  $t_c(d)$  and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing  $\Delta = |d' - d|$ , one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (11.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod  $J_1$  metric as  $d' \rightarrow d$ . More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives  $J_1$  distance at most  $100\Delta$ .

*Proof.* Write  $r$  and  $g$  for the incoming relative position and velocity, and  $n(d) = (r + t_c(d)g)/d$  for the contact normal. The collision root satisfies  $|r + t_cg| = d$ . The nongrazing estimate from Lemma 8.2 is  $-g \cdot n > 1443/1000$ . Differentiating the root with respect to  $d$  gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is  $|g| < 3$ , so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since  $v_1^+ = V + g/2 - (g \cdot n)n$  and  $v_2^+ = V - g/2 + (g \cdot n)n$ , differentiation bounds either velocity derivative by  $2|g| |n'| < 30$ .

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most  $30\Delta$ . The intervening interval has length at most  $\Delta$ , and the whole observation lasts three units. This proves (11.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending  $t_c(d)$  to  $t_c(d')$ . Its uniform displacement is at most  $\Delta$ . Aligned velocities differ by at most  $30\Delta$ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by  $3\Delta + 3(30\Delta) = 93\Delta$ , by integrating the velocity difference before and after the two jump times. Aligning time adds at most  $4\Delta$ . The asserted bound follows, with a harmless rounding to  $100\Delta$ .  $\square$

**Theorem 11.2** (Microscopic-to-audit and conditional-law composition). *For the single family (11.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

*uniformly for  $d \in I$  and  $0 < \sigma \leq 1/24$ . When  $d \leq 2/5$  the detector is zero on  $[0, 3]$ , and all three finite-experiment deficiencies are zero.*

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and  $n = 2800$  training runs, whose expected validation score exceeds  $2/5$  against every private width-one simulator throughout that same parameter region. In detail, for counts  $C$  choose an event  $A_C \subseteq \{0, 1\}^3$  maximizing*

$$Q_*(A_C) - C(A_C)/2800, \quad (11.3)$$

*with lexicographic tie breaking. Here  $Q_*$  is the rational marked reference law of  $(B, B, W)$ . The independent validation uses the actual physical target, not  $Q_*$ . This finite rule has at most  $\binom{2807}{7}$  training-count states at any scheduled epoch; a conservative complete auditor bound is*

$8(256 + 1)\binom{2807}{7}$ . The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.

(c) Fix  $\sigma > 0$  and let  $d_n, d \in I$  with  $d_n \rightarrow d$ . Set

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \quad (11.4)$$

The joint preparation–continuous-observation laws differ by at most  $\epsilon_n$ . Every finite resource deficiency above changes by at most  $\epsilon_n$ , and a fixed audit changes by at most  $\epsilon_n$  because its source behavior is unchanged. On a latent-preserving coupling,  $X_{d_n} \rightarrow X_d$  in  $J_1$  and each bounded posterior converges in supremum  $L^2$  with bound  $84\epsilon_n$ . The full and projected likelihoods converge in reference supremum  $L^2$ , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 10.2, Theorem 10.3, and Corollary 10.4.

(d) On compact parameter sets with  $\sigma \geq \sigma_0 > 0$ , the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 7.1 and 6.6.

*Proof.* Every collision occurs before time  $3/2$ . Conditional on preparation, the two normalized window increments in (11.1) are independent Gaussians with mean  $h(v_{1,y}^+)$  and standard deviation  $\sqrt{2}\sigma \leq 1/12$ . The outgoing detector has sign  $2B - 1$  and absolute value greater than  $1/4$ . Coupling each noisy report to  $B$  gives total marked variation less than  $1/300$  from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 8.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed  $d_* - 1/300 > 9/20$ , and the hidden value is less than  $1/8 + 1/300 < 13/100$ . The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of  $W$ . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities  $p, q$ , independently of the actual  $W$ . The calculation in Theorem 8.1 already gives minimum variation  $d_*$  for this row. Its alphabet  $(z_1, z_2, W)$  has eight elements, hence  $D = 7$ . The explicit empirical audit in Theorem 4.1, trained using  $Q_*$ , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than  $1/300$ . Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (11.5)$$

For completeness, the last strict inequality is the positive-root comparison  $\sqrt{5/2} > 947/600$ ; its squared margin is  $5/2 - (947/600)^2 = 3191/360000 > 0$ . All probabilities in  $Q_*$  are 0,  $3/16$  or  $5/16$ , so (11.3) is a finite rational rule independent of  $d, \sigma, p, q$ . At training epoch  $k$  the number of eight-category count vectors is  $\binom{k+7}{7} \leq \binom{2807}{7}$ . A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 11.1 gives the signal-distance hypothesis of Theorem 10.1, with  $T = 3$  and  $M = 1$ . It also supplies its microscopic  $J_1$  hypothesis. That theorem gives (11.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous observation, so marked variation contracts for every gate. The source is blind and independent of  $d$ . Consequently its executable candidate behaviors are identical across  $d$ , and Proposition 4.4 has  $\alpha = 0$  and  $\beta = \epsilon_n$ . The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance  $2\sigma^2$  bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 8.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as  $\sigma \downarrow 0$  is not asserted.  $\square$

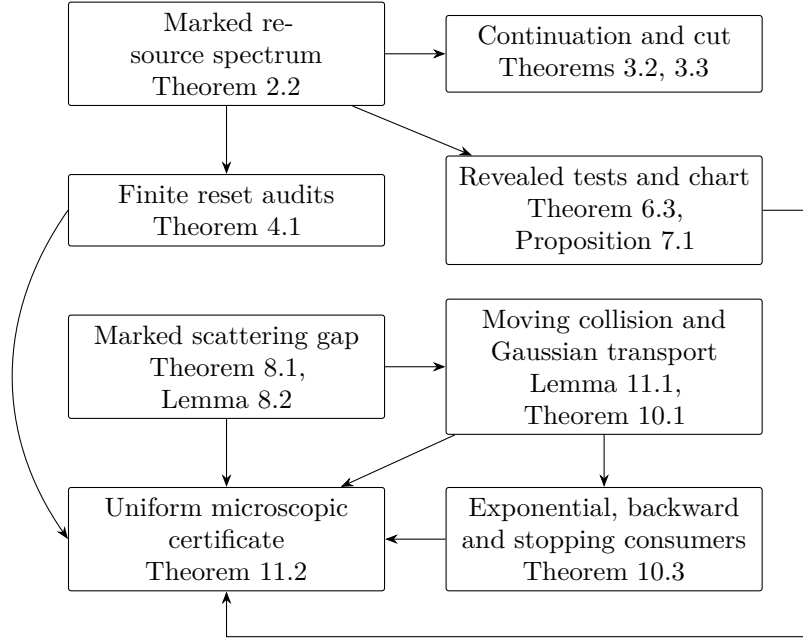
The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error  $\rho$ : its lower score decreases by at most  $\rho$  when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 11.2 and the preceding results prove Theorem 1.1. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

## APPENDIX A. THE CANONICAL THEOREM CHAIN

The diagram records dependencies within this article. A solid arrow means that the conclusion at its tail is used in a proof at its head. The continuation characterization identifies the original finite-resource object; the exact cut theorem supplies its independent sharp obstruction. The two testing constructions have different information conventions.



In detail, the proof of Theorem 11.2(b) uses the audit theorem and the exact marked gap, while part (c) uses the moving-collision lemma, the TV-to-posterior estimate, the likelihood theorem and its conditional consumers. Part (d) consumes the chart and physical effective-presentation constructions. No testing theorem is an input to the elementary elastic-scattering calculation.

The comparison theorem, strict polynomial testing, revealed-behavior hierarchy, exact marked gap, scattering bounds, and common-latent TV-to-posterior estimate are inherited mathematical inputs with complete proofs reproduced here. The continuation normalization, exact cut spectrum, reset-audit game and lower bound, formal chart algorithm, changing-trajectory likelihood estimates, entropic consumer, and uniform moving-collision composition are additions of the present revision. The positive-cone, minimax, two-point testing, polynomial positivity, and stochastic-calculus mechanisms are classical inputs, not priority claims.

The parameter and the actual mark remain unavailable to the online simulator. The reset auditor sees completed marked trials through its separate interface. A revealed-behavior test sees an exact proposed law. The maximal coupling of observation paths may use their entire future and is not an executable transducer. These three information privileges are never substituted for one another.

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# GENERAL THETA FOUNDATIONS I: ADAPTIVE TESTING AND COLLISION-SENSITIVE COMPARISON

QIAN QI

**ABSTRACT.** We study comparison of marked causal experiments with fixed persistent state and separately priced hidden or visible randomness. On the nonconvex observable behavior image we construct an independent adaptive testing game and polynomial test classes with attained finite-level duality. In behavior dimension  $a$  and diameter  $R$ , their error is at most  $4Ra/\sqrt{n}$ , using at most  $(n+1)^a$  occupied test coefficients; this does not charge redundant simulator rows. We then construct a two-sphere scattering experiment with an actual correlated mark and a gate chosen after the first noisy report. Its private and visible deficiencies exceed  $9/20$ , whereas a hidden two-valued selector gives error below  $13/100$ , at the same width-one register. Removing the collision makes every deficiency zero. Exact finite values, uniform collision bounds and computable rational presentations prove this separation without convexifying the private image. Finally, marked variation controls the entire optional-projection process; Gaussian observation of graph-core approximants yields a quantitative microscopic changing-filtration limit. The costed comparison, strict-certificate, task and domain theory is developed with its separate information and analytic hypotheses.

## COMPLETE DEVELOPMENT

The canonical article and the preserved historical mathematical development.

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## 1. INTRODUCTION

Comparison of experiments asks whether the information supplied by one observation system can replace that supplied by another. A causal comparison must also respect the order in which

observations and actions become available. A resource-constrained comparison asks a further question: how much persistent state and shared randomness does the replacement require? These questions differ even for two binary reports. A simulator that cannot retain or reread its first emission cannot in general reproduce a correlated second emission. Revealing a persistent random selector to the tester is also different from hiding it. The distinction concerns the joint laws being compared, not merely the implementation of a convex combination.

We study marked experiments. A mark is an actual latent quantity of the source or target system, not a fresh variable sampled after a comparison. The comparison tests the joint law of the mark and the controlled report history. Private simulation uses finite retained registers and fresh private randomization. A hidden selector is an additional priced persistent variable; a visible selector is exposed and must occur with the same independent reference law on the target side. With these conventions the attainable behavior image is compact but usually nonconvex. The central issue is to compare this original image rather than a convex relaxation with a different information pattern.

**1.1. Main results.** For fixed finite instruments and a resource signature, let  $B$  denote the observable discrepancy image and  $F(b) = \max_i f_i(b)$  its marked feedback error. The  $f_i$  are affine event tests, including zero. The deficiency is  $\delta = \min_B F$ . An adaptive test is an independently chosen function  $\kappa : B \rightarrow \Delta_m$ , scored at  $b$  by  $\sum_i \kappa_i(b) f_i(b)$ . It observes a proposed law for purposes of verification; it is not an added physical information channel.

Theorems 3.1 and 3.3 give the independent variational identity

$$\delta = \sup_{\kappa} \min_{b \in B} \sum_i \kappa_i(b) f_i(b) = \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa} \int \sum_i \kappa_i(b) f_i(b) d\mu(b),$$

and a quantitative polynomial hierarchy. In a bounded chart of the observable affine space, of dimension  $a$  and metric diameter  $R$ , its attained finite-level values satisfy

$$0 \leq \delta - d_n \leq 4Ra/\sqrt{n}.$$

An approximating test needs at most  $(n+1)^a$  occupied Bernstein coefficients, and the corresponding moment dual needs at most  $(n+2)^a$  support points. The dimension counts behavior, not row variables. The nonlinear score prevents replacement of a lottery over revealed candidate laws by its barycenter. At degree zero such a replacement is possible, and the resulting convex relaxation can miss a strictly positive private deficiency.

Proposition 3.5 gives a concrete nonlinear witness with three cubic behavior weights, nine nonnegative derivative terms and seventeen positive rational coefficients. It certifies a private lower bound above  $2/5$ , while constant tests give zero.

The independent game and its size theorem complement the strict polynomial certificate hierarchy proved in Theorem 6.1. The latter remains an exact verifier on a chosen row cube; it is not itself called an independent dual. Strict lower levels are complete, and their verification complexity is kept separate from the behavior-coordinate degree bound. With uniformly effective stateless finite presentations this produces certified intervals at the original state and randomization budgets.

The second main result makes collision mechanics essential. We prepare two spheres in full-dimensional incoming phase-space boxes and observe a bounded function of the tagged transverse velocity after scattering. An actual binary mark, correlated with the incoming impact orientation, is retained but never revealed. A binary gate is chosen after the first noisy report. Theorem 4.1 first solves the limiting finite experiment exactly:

$$\delta^p = \delta_r^v = \sqrt{5/2} - 9/8, \quad \delta_2^h = 1/8 \quad \text{at width } (1, 1).$$

Uniform nongrazing collision estimates and Gaussian tails then give, for contact distance  $9/10 \leq d \leq 11/10$  and noise  $0 < \sigma \leq 1/12$ ,

$$\delta^p, \delta_r^v > 9/20, \quad \delta_2^h < 13/100.$$

The same detector and preparation have zero deficiency for  $0 < d \leq 2/5$ , when no collision reaches the detector during acquisition. Thus conserved collective motion cannot account for the gap. The estimates are uniform on an interval of contact distances and under geometric similarity. Explicit collision charts also produce rational marked presentations with certified errors tending to zero.

The third result strengthens the conditional-law content of marked comparison. For equivalent joint laws with a common latent marginal, Theorem 5.1 constructs a latent-preserving coupling under which the squared uniform distance between bounded posterior martingales is at most 84 times the joint total variation. The coupling is a convergence device, not an executable simulator. Gaussian observation of an invariant microscopic flow gives

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \|h_n - h\|_{L^2(\mu)},$$

so graph-core approximation yields convergence of entire optional projection processes. The maximum-over-time estimate, rather than a finite-dimensional argument, supplies the changing-filtration input. Finite-checkpoint risk transport still uses the stateless identity comparison and keeps its original budget.

**1.2. Relation to classical theory.** Classical statistical comparison and randomization originate with Blackwell and Le Cam [1, 47, 46]. Filtered comparison has important antecedents, including Norberg [48]. We do not claim priority for introducing causal deficiency. In the adapted setting Weisshaupt’s Definition 5.1 and Theorems 5.1–5.2 [72] use a compact convex closure of adapted stochastic operators, allowing finitely additive outputs. A fixed finite-state private image need not have that convexity. The two-report examples identify the precise failed step in inserting the state constraint into such a separation proof. Our testing game retains the nonconvex image by changing the verifier’s strategy space, not by asserting a new unrestricted randomization criterion. Its minimax ingredient is classical [71].

Bernstein positivity is likewise not new here. Theorem 1.4 of de Klerk–Laurent [73] gives constructive hypercube degree and error bounds through Bernstein and Handelman representations. Laurent–Slot [74], Theorem 2 and Corollary 3, obtain stronger Schmüdgen degree bounds using Jackson kernels. Theorem 9 and Proposition 2 of de Klerk–Vera [75] connect Chebyshev coefficient norms to effective hypercube certificates. Our bound concerns an independent simplex-valued adaptive test on a nonconvex behavior image; it is not an improvement to those general positivity bounds. Pulling that test back to row coordinates still incurs a separate positivity-verification cost.

The finite-state, common-information and automata antecedents [54, 53, 70, 58] remain relevant to the task and synthesis theory. The physical flow and collision rule are classical [63]. The contribution of the scattering construction is the marked, feedback-sensitive resource comparison with uniform quantitative certificates in that flow, not a new mechanics of elastic collision.

**1.3. Organization and analytic scope.** After the exact comparison conventions, Sections 3 and 4 prove the adaptive hierarchy and its collision-sensitive realization. Section 5 proves the posterior-process estimate and its microscopic sufficient condition. The remaining sections develop the complete finite certificate, task, regeneration, graph-domain and physical-presentation machinery used for broader acquisitions, including the explicitly solvable collective benchmark. Appendix A records the theorem dependencies in mathematical terms.

All physical conclusions have their stated finite-time and preparation hypotheses. A binary scattering chart and a fixed-particle optional-projection estimate are different from particle-uniform

recollision estimates or a nonlinear kinetic limit. In particular a large-deviation action, a hierarchy corrector uniform on action sublevels, and a limiting nonlinear resolvent require their own analytic inputs. The comparison theorem transports consequences of such inputs; it does not create them by naming a dependency.

## 2. INTRINSIC CAUSAL DEFICIENCY

**2.1. Experiments and executable signatures.** Fix a finite parameter set  $\Theta$ , a horizon  $T$ , and finite control alphabets  $A_0, \dots, A_{T-1}$ . An experiment has a report at checkpoints  $0, \dots, T$  and an unobserved terminal mark  $W$  in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write  $Y_t$  for the source report and  $Z_t$  for the target report. A private simulator of width  $K$  has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in  $m'$ . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted  $\mathcal{K}_K$ .

A deterministic feedback tree  $g$  chooses  $a_t$  from  $z_{0:t}$  and its own previous controls. It is the same controller on both sides of a comparison. Let  $\Omega_g$  be the finite marked transcript space, including reports, controls, and  $W$ . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here  $W$  on the left is the source's actual mark. The total variation convention is  $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$ . Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector*  $U \in [r]$ , independent of the source, chooses one simulator  $q^u$ . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta,g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta,g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions  $\lambda \in \Delta_r$  and  $q^u \in \mathcal{K}_K$ . The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining  $(u, m)$ , at width  $rK_t$ . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

**Definition 2.1** (Deficiency spectrum). The private resource spectrum of  $E$  relative to  $F$  is the upward-closed set of pairs  $(K, \varepsilon)$  satisfying  $\delta_K^p(E, F) \leq \varepsilon$ . The hidden spectrum consists of triples  $(K, r, \varepsilon)$  satisfying  $\delta_{K,r}^h(E, F) \leq \varepsilon$ , and the visible spectrum uses  $\delta_{K,r}^v$ . Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is  $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$ .

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize  $\max_t K_t$ , with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

**2.2. The finite comparison theorem.** Concatenate all probabilities  $P_{\theta,g}^q(\omega)$  into  $b(q) \in \mathbb{R}^N$ , retaining their row indices; put  $b_F = (Q_{\theta,g}(\omega))$ . Let  $\mathcal{D}_K$  be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For  $i = (\theta, g, B)$  and  $d \in \mathcal{D}_K$ , define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

**Theorem 2.2** (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- $r$  deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

*In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At  $K_t = 1$  for every  $t$ , this is precisely the same factorization with rows  $q_t(z \mid a, y)$  and  $q_0(z \mid y)$ .*

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

*An optimal simulation uses at most  $a_K + 1$  selector values. Thus every weighted causal test has optimal discrepancy at most  $\varepsilon$  if and only if a hidden mixture of  $K$ -state simulators has error at most  $\varepsilon$ ; it has a private realization of width at most  $(a_K + 1)K$ . More specifically,*

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put  $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$ . Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

*Writing  $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$ , the infimum over all finite visible selectors is attained using  $a_K^v + 1 \leq |\Theta| + 1$  values and equals*

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

*For every  $r \geq 1$ , and also after optimizing over all  $r$ ,*

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

*No corresponding equivalence with  $\delta_K^{\text{mix}} = 0$  holds in general.*

(iv) For compatible experiments  $E, F, G$  and profiles  $K, L$ ,

$$\begin{aligned}\delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}.\end{aligned}\tag{2.12}$$

Here  $KL$  is the pointwise product, and the second line uses independent selectors. If an  $F$ -consumer has widths  $S_t$ , composing it with a private simulator gives widths  $K_t S_t$  and changes every  $[0, 1]$  marked risk by at most  $\delta_K^p(E, F)$ . A hidden selector is stored at the additional price  $r$ , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

*Proof.* Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes  $\lambda$  and the  $r$  policies in (2.3). For the visible problem there are only finitely many collections  $(g_u)$  and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting  $\varepsilon = 0$  forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of  $q$ , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability  $\Lambda_q$  on  $\mathcal{D}_K$ . On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every  $\theta, g$ , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d).\tag{2.13}$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies  $\text{conv } b(\mathcal{K}_K) = \text{conv } \{b(d) : d \in \mathcal{D}_K\}$ . The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most  $a_K + 1$  of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients  $c_d$  with  $\sum c_d = 0$  and  $\sum c_d b(d) = 0$ . Subtract the largest permissible multiple of  $c$  from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every  $w$  is equivalent to the maximum over  $w$  being at most  $\varepsilon$ , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by  $\lambda_u$ . For a fixed  $\theta$ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is  $\sum_u \lambda_u e_\theta(q^u)$ , proving (2.9). The map  $q \mapsto e(q)$  is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in  $\mathbb{R}^{|\Theta|}$  and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic  $q$ , not over deterministic tables: the nonlinear function  $e_\theta$  does not commute with (2.13). If the attained visible optimum is zero, then for every  $u$  of positive weight and every  $\theta$  one has  $e_\theta(q^u) = 0$ , since every summand is nonnegative. Any such  $q^u$  is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a  $K$ -state  $E$ -to- $F$  transducer and an  $L$ -state  $F$ -to- $G$  transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most  $K_t L_t$  values.

For any terminal tester on  $G$ , the second transducer together with that tester is an admissible randomized feedback system on  $F$ . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine  $F$  system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any  $[0, 1]$  marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities.  $\square$

*Remark 2.3* (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

**2.3. A delayed channel and a strict signature separation.** Let the source reveal  $i \in [n]$  at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws  $j \in [m]$  from a specified row-stochastic matrix  $V = (V_{ij})$ . The unknown parameter is  $i$ . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size  $K$  matters.

**Proposition 2.4** (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where  $A$  is an  $n$ -by- $K$  row-stochastic matrix and  $B$  is a  $K$ -by- $m$  row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For  $1 \leq K \leq n$  the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance  $0 \leq \varepsilon < 1$ , the least private width is  $\lceil n/(1 - \varepsilon)^{-1} \rceil$ , whereas the least within-mode width with optimized hidden selection is  $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$ . The latter still pays for its selector. In particular, for  $n = m = 3$  and  $K = 2$ ,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For  $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$ ,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

*Proof.* The encoder maps the consumed  $i$  into its retained label by  $A$ ; the terminal decoder maps that label into  $j$  by  $B$ . This proves (2.14) including its converse implementation. If  $V = CD$  is any nonnegative factorization, let  $s_k = \sum_j D_{kj}$ . Delete zero rows of  $D$ , define  $B_{kj} = D_{kj}/s_k$  and

$A_{ik} = C_{ik}s_k$ . Since  $V$  is stochastic, the rows of  $A$  sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For  $V = I_n$ , row error is  $1 - (AB)_{ii}$ . If the least correct probability is  $c$ , for each  $i$  select a row  $k(i)$  of  $B$  with  $B_{k(i),i} \geq c$ . Some row is selected at least  $\lceil n/K \rceil$  times, so stochasticity gives  $c \leq 1/\lceil n/K \rceil$ . A partition into  $K$  groups of sizes at most  $\lceil n/K \rceil$ , each decoded uniformly, attains that bound. Also  $\sum_i (AB)_{ii} \leq K$ , by the calculation below with  $K$  in place of two. It remains valid under mixing, giving hidden error at least  $1 - K/n$ . Start with a deterministic channel that reproduces  $K$  input symbols exactly and sends all others to those  $K$  outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then  $K/n$ , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For  $V = I_3$ , total variation in row  $i$  is  $1 - (AB)_{ii}$ . If all three diagonal entries of  $AB$  were greater than  $1/2$ , then for each  $i$  some row of  $B$  would have its  $i$ th entry greater than  $1/2$ . A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound  $1/2$ . Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least  $1/3$ . Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are  $2/3$  and the off-diagonal entries are  $1/6$ . This is exactly  $V_*$  and has error  $1/3$  from  $I_3$ . It proves the hidden claims in both displays.

The smallest singular value of  $V_*$  is  $1/2$ . Any  $AB$  with two intermediate states has a unit vector  $x$  in its right null space, so  $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$ . If the maximal row total variation error is  $\varepsilon$ , each row has Euclidean error at most  $2\varepsilon$ , and hence the Frobenius error is at most  $2\sqrt{3}\varepsilon$ . This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity.  $\square$

**2.4. Optimal resource sets and decision certificates.** For a class  $\mathcal{C}_{S,q}(F)$  of common consumers, with all other information conventions fixed, let  $\mathcal{R}_{S,q}(F)$  be its vector of attainable  $[0, 1]$  risks. For a nonnegative row weight  $w$  summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If  $E$  privately simulates  $F$  with cost  $K$  and error  $\varepsilon$ , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance  $\varepsilon$ ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by  $r$ ; visible selectors multiply the separate public alphabet by  $r$ . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most  $\varepsilon$ , paired risk sets and their support values at unchanged width are within  $\varepsilon$ . If each is centered by a paired unrestricted benchmark whose error is also at most  $\varepsilon$ , the centered values are within  $2\varepsilon$ . Therefore a gap  $g > 0$  between zero excess and infeasible widths is stable under  $\varepsilon < g/4$ : a value below  $g/2$  in the approximation certifies feasibility in the exact finite problem, and a value above  $g/2$  certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

**Proposition 2.5** (Intrinsic finite-dimensional selector approximation). *Let  $\mathcal{R}$  be a compact risk image in  $C(J)$ , where  $J$  is compact. Suppose a bounded linear operator  $L : C(J) \rightarrow C(J)$  has rank at most  $m$  and  $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$ . Every barycenter  $\bar{r}$  of a probability on  $\mathcal{R}$  has a barycenter  $r_*$  of at most  $m + 1$  members of  $\mathcal{R}$  with  $\|\bar{r} - r_*\|_\infty \leq 2\eta$ . If  $\mathcal{R}$  lies in a finite affine space of dimension  $m$ , the error is zero with at most  $m + 1$  members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

*Proof.* The compact set  $L\mathcal{R}$  lies in an  $m$ -dimensional space. Its convex hull is compact, and  $L\bar{r}$  belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives  $r_1, \dots, r_{m+1} \in \mathcal{R}$  and weights  $\lambda$  such that  $L\bar{r} = \sum_i \lambda_i Lr_i$ . Put  $r_* = \sum_i \lambda_i r_i$ . The two residual terms  $\bar{r} - L\bar{r}$  and  $r_* - Lr_*$  have norms at most  $\eta$ , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information.  $\square$

*Remark 2.6* (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- $r$  formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

**2.5. Local testing at the prescribed private budget.** The convex converse in Theorem 2.2 has an explicit selector price. We now give a different converse that leaves the private register unchanged. Its witnesses are finite covers of the policy space by local testing inequalities, rather than one separating functional on a convex hull. Throughout this subsection the experiments, horizon, and alphabets are finite; for effective statements their instrument probabilities are rational. The prescribed widths and, when present, selector cardinality are fixed.

Let  $\mathcal{P}$  be the product of the row simplexes of the private transducer. For a hidden or visible selector with at most  $r$  values, include its simplex and  $r$  copies of the transducer rows in  $\mathcal{P}$ . Zero selector weights are permitted. The relevant deficiency has the form

$$\delta = \min_{q \in \mathcal{P}} F(q), \quad F(q) = \max_{i \in I} f_i(q), \quad 0 \leq F \leq 1. \quad (2.19)$$

Here  $I$  consists of all parameter, feedback-tree, and marked-event tests. The empty event is included. For a visible selector a test specifies a feedback tree and event in each selector fiber. Thus

$$f_i(\lambda, q) = \sum_u \lambda_u \{Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)\}.$$

For a hidden selector the target term is independent of  $\lambda$ . Every  $f_i$  is affine in each row separately, with rational coefficients. This follows by expanding complete controlled paths: a time-indexed row is used at most once along each path. It holds simultaneously for all feedback tests, not just for open-loop words.

For an integer  $M \geq 1$ , intersect a row simplex  $\Delta_{d-1}$  with each closed coordinate box

$$\prod_{j=1}^d [b_j/M, (b_j + 1)/M], \quad b_j \in \{0, \dots, M - 1\}.$$

Discard empty intersections. Products of these row polytopes form a finite closed cover  $\mathcal{C}_M$  of  $\mathcal{P}$ . Overlapping faces cause no ambiguity. Write  $V(C)$  for the products of row vertices of a cell  $C$ . These are exactly its vertices. They are rational and are obtained by setting all but at most one

coordinate of each row to a box endpoint and imposing that the coordinates sum to one. This prescription also includes lower-dimensional cells and one-point rows. Define

$$\ell(C) = \max_{i \in I} \min_{v \in V(C)} f_i(v), \quad u(C) = \min_{v \in V(C)} F(v), \quad (2.20)$$

$$L_M = \min_{C \in \mathcal{C}_M} \ell(C), \quad U_M = \min_{C \in \mathcal{C}_M} u(C). \quad (2.21)$$

For time  $t$ , let  $d_t$  be the largest row alphabet cardinality, including both emitted symbol and next retained label. Put

$$\rho_t(M) = \begin{cases} 0, & d_t = 1, \\ \min\{1, d_t/(2M)\}, & d_t > 1, \end{cases} \quad b_M = 1 - \prod_{t=0}^T (1 - \rho_t(M)).$$

For a selector row set  $s_M = 0$  if  $r = 1$ , and  $s_M = \min\{1, r/(2M)\}$  otherwise. The three bounds used below are

$$\omega_M^p = b_M, \quad \omega_M^h = s_M + (1 - s_M)b_M, \quad \omega_M^v = \min\{1, 2s_M + (1 - s_M)b_M\}. \quad (2.22)$$

They do not count the number of histories or the number of policy rows. The visible-selector bound includes a second selector term because the reference joint experiment changes when its selector law changes.

**Theorem 2.7** (Complete local testing certificates). *For any of the three fixed resource signatures in (2.19), the rational quantities (2.21) satisfy*

$$0 \leq L_M \leq \delta \leq U_M \leq 1, \quad U_M - L_M \leq \omega_M. \quad (2.23)$$

*The upper witness is an executable rational transducer at the prescribed widths and selector cardinality. A lower witness consists of one test  $i_C$  for each cell and its inequalities  $f_{i_C}(v) \geq L_M$  at all  $v \in V(C)$ .*

*For  $M = 2^n$ , the lower bounds increase and the upper bounds decrease to  $\delta$ . In particular,*

$$\delta = \sup_n \max_{(i_C)_{C \in \mathcal{C}_{2^n}}} \min_{C \in \mathcal{C}_{2^n}} \min_{v \in V(C)} f_{i_C}(v) = \inf_n U_{2^n}. \quad (2.24)$$

*If  $\delta > \varepsilon$ , a finite local testing certificate eventually proves this strict inequality. If  $\delta < \varepsilon$ , a rational original-budget simulator eventually proves feasibility below  $\varepsilon$ . For rational  $\varepsilon$ , non-strict feasibility, including equality, is decidable by real quantifier elimination. The optimum is algebraic, and an optimal row table can be chosen with real algebraic entries.*

*Proof.* Express each row of  $q \in C$  as a convex combination of its row vertices. Expand a separately affine  $f_i$  in those combinations, one row at a time. The resulting weights on  $V(C)$  are nonnegative and sum to one. Consequently

$$\min_{q \in C} f_i(q) = \min_{v \in V(C)} f_i(v). \quad (2.25)$$

It follows that  $\ell(C) \leq \min_C F \leq u(C)$  and hence  $L_M \leq \delta \leq U_M$ . No assertion that  $F$  itself is separately affine has been made. This distinction is essential at a randomized optimum. The empty event gives  $\ell(C) \geq 0$ .

We next prove the oscillation estimate used for the gap. Two probability rows in the same  $M$ -cell differ in total variation by at most  $\rho_t(M)$ . Run the two transducers against the same tester and source. As long as all earlier emitted symbols and retained labels coincide, the controls and source laws agree. Maximally couple the next joint output–state row. The conditional chance of a mismatch is at most  $\rho_t(M)$ . Thus the chance of any mismatch through time  $T$  is at most  $b_M$ . On the event of no mismatch the source can be coupled identically, including its actual mark. This bounds the difference of the probability of every marked feedback event by  $b_M$ , uniformly in the parameter and tester. A finite causal instrument provides precisely the conditional kernels needed for this coupling.

For a hidden selector first maximally couple its two laws. The selector mismatch has probability at most  $s_M$ ; on matched fibers the preceding bound is  $b_M$ . The difference of any fixed test is at most  $s_M + (1 - s_M)b_M$ . There is only one executed fiber, so this estimate has no extra factor  $r$ . For a visible selector write its fixed test as the expectation of  $a_u(q) = Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)$ . These numbers lie in  $[-1, 1]$ . On mismatched selector fibers their difference is at most two; on matched fibers it is at most  $b_M$ . Thus its oscillation is at most  $2s_M + (1 - s_M)b_M$ . This estimate does not forget the selector in either compared experiment.

Choose a vertex  $v_*$  minimizing  $F$  on  $V(C)$  and an active test  $i_*$  there. For every other vertex,

$$f_{i_*}(v) \geq F(v_*) - \omega'_M = u(C) - \omega'_M,$$

where  $\omega'_M$  is the untruncated corresponding oscillation bound. It follows that  $u(C) - \ell(C) \leq \omega'_M$ . Since both bounds lie in  $[0, 1]$ , truncation by one is valid. Choose a cell attaining  $L_M$  to obtain  $U_M - L_M \leq \omega_M$ . A vertex attaining  $U_M$  is an original row table; it is not a mixture of different designs.

Dyadic refinement only replaces each cell by subsets. By (2.25), its descendants cannot lower the minimum of a fixed test and hence cannot lower  $L_M$ . Every coarse vertex belongs to a finer cell and remains an extreme point of that subset; therefore the set of available vertices can only increase, and  $U_M$  cannot increase. The bounds in (2.22) tend to zero. Independent choices of the maximizing test in each of the finitely many cells give (2.24) and the two strict-decision assertions.

Finally  $q \in \mathcal{P}$  and  $f_i(q) \leq \varepsilon$  are finitely many polynomial conditions over the rationals. Quantifier elimination over real closed fields decides their existential conjunction; this is the classical effective Tarski–Seidenberg theorem, not a new elimination algorithm [69, §2.1, Theorem 2.1]. The set of feasible objective values is a semialgebraic ray with included endpoint  $\delta$ . Its endpoint is algebraic over  $\mathbb{Q}$ . The nonempty optimum fiber has a point over the real algebraic numbers by the same real-closed-field transfer, proving the final assertion.  $\square$

**Corollary 2.8** (Private feasibility is locally test complete). *At fixed private widths,  $\delta_K^P(E, F) \leq \varepsilon$  if and only if no finite dyadic local testing certificate has value greater than  $\varepsilon$ . For any finite prescribed list of resource profiles, all strictly feasible and strictly infeasible profiles can be separated by rational original-budget witnesses and finite local certificates. With rational tolerance, exact feasibility and the minimal elements in that finite list are decidable. These assertions require neither a free public selector nor a convex private behavior set.*

*Proof.* The first claim is (2.24). Apply the strict parts of Theorem 2.7 to each profile in the finite list and its quantifier-elimination clause to profiles on the boundary. The partial order is coordinatewise state and selector cost. Enumeration of feasible profiles then determines the minimal elements. It need not produce a single least profile, and no polynomial running time is asserted.  $\square$

For the three-symbol channel  $V_*$  in Proposition 2.4,  $\delta_2^{\text{mix}} = 0$  whereas  $\delta_2^P \geq 1/(4\sqrt{3}) > 1/8$ . Theorem 2.7 therefore eventually gives a finite local certificate exceeding  $1/8$  at private width two. A single convex testing separator cannot have that conclusion, since the hidden convex hull already contains the target. This example identifies the logical work done by localization; it does not claim a polynomial bound on the size of the certificate.

The exact use of row vertices and real algebraic decision procedures is classical. So is the unrestricted filtered randomization criterion: The precise filtered antecedent is [68, Lemma 1 and Theorem 5], discussed in the introduction. The conclusion here is instead a complete system of local tests with executable witnesses at that fixed register, together with an explicit feedback modulus. These are different conclusions even on the finite delayed-channel slice. The original-source audit records separately the still-unverified comparison with Norberg’s full proof and with the original incomplete-machine minimization papers.

**2.6. Transferring intrinsic certificates across presentations.** For compatible marked experiments write  $E \simeq_a \tilde{E}$  when two stateless, seed-free, marked causal channels, one in each direction, have feedback error at most  $a$ . Their report timing and action interfaces are the same. On a visible signature they compare the complete joint interface with the independent selector. These are assumptions about executable channels, not about unseeded marginal distributions.

**Theorem 2.9** (Original-budget stability of intrinsic certificates). *Suppose  $E \simeq_a \tilde{E}$  and  $F \simeq_b \tilde{F}$ . For private, fixed hidden-selector, and fixed visible-selector deficiencies at the same budget,*

$$|\delta_K(E, F) - \delta_K(\tilde{E}, \tilde{F})| \leq a + b. \quad (2.26)$$

*If the approximating experiments are finite and rational, their local certificates consequently give the rigorous interval*

$$\max\{0, L_M - a - b\} \leq \delta_K(E, F) \leq \min\{1, U_M + a + b\}. \quad (2.27)$$

*An upper witness consists of the rational transducer and the two stateless bridge channels and has the original persistent resource budget. A lower certificate excludes every simulator at that budget for the original experiments whenever  $L_M > a + b + \varepsilon$ .*

*Proof.* Compose  $E \rightarrow \tilde{E}$ , a near-optimal  $\tilde{E} \rightarrow \tilde{F}$  transducer, and  $\tilde{F} \rightarrow F$ . The optimized composition inequality in Theorem 2.2, or its identical channel proof on measurable carriers, adds the three errors and multiplies the widths by  $1, K, 1$ . The bridges introduce no selector, so a prescribed hidden or visible cardinality remains unchanged. Reverse the bridges for the opposite inequality and take the infima. Equation (2.27) follows from Theorem 2.7. For the last assertion any original simulator would, after reverse bridging, contradict the finite lower certificate. Thus the lower statement does not assume the existence of an optimal simulator on a nonfinite carrier.  $\square$

**2.7. Witness size in the intrinsic behavior dimension.** The local theorem can be complemented by a bound on how many different tests are needed. This bound uses the dimension of attainable behaviors, not the number of stochastic-row coordinates in a chosen realization. It does not confuse the number of tests with the cost of verifying that their violation regions cover a nonconvex set.

Let  $\mathcal{B}_K = \{b(q) : q \in \mathcal{K}_K\}$  be the private behavior image from Theorem 2.2. Differences of behaviors have zero mass in each parameter–feedback row. On their linear span put

$$\|b - b'\|_* = \max_{\theta, g} \frac{1}{2} \sum_{\omega \in \Omega_g} |b_{\theta, g}(\omega) - b'_{\theta, g}(\omega)|.$$

This is a norm,  $\text{diam } \mathcal{B}_K \leq 1$ , and every event discrepancy  $f_i(b) = Q_{\theta, g}(B) - b_{\theta, g}(B)$  is 1-Lipschitz. Let  $a = \dim \text{aff } \mathcal{B}_K$ ; redundant realization coordinates do not change  $a$ .

**Theorem 2.10** (Intrinsic-dimension testing witnesses). *Suppose  $\delta_K^p(E, F) \geq L + \gamma$  with  $\gamma > 0$ . There is a subfamily  $\mathcal{I}_0$  of the original marked feedback tests, of size at most*

$$|\mathcal{I}_0| \leq \lceil (1 + 4/\gamma)^a \rceil, \quad (2.28)$$

*such that*

$$\min_{q \in \mathcal{K}_K} \max_{i \in \mathcal{I}_0} f_i(b(q)) \geq L + \gamma/2. \quad (2.29)$$

*Thus the same-budget private lower certificate can use this fixed subfamily on every policy cell. With rational finite input, rational  $L, \gamma$ , and classical real quantifier elimination, such a subfamily can be extracted effectively. No convex mixture of simulators, state increase or selector is introduced.*

*Proof.* In the  $a$ -dimensional affine behavior space choose a maximal set  $b_1, \dots, b_s \in \mathcal{B}_K$  with pairwise distances strictly greater than  $r = \gamma/2$ . It is an  $r$ -net by maximality. Balls of radius  $r/2$  about these points have disjoint interiors and lie in a ball of radius  $1 + r/2$  about  $b_1$ . Lebesgue

volume in this affine space, and its homogeneity under dilation, give  $s \leq (1 + 2/r)^a$ . For  $a = 0$  one point suffices. At each center choose a test  $i_j$  attaining its maximal discrepancy. This discrepancy is at least  $L + \gamma$ . Every behavior within  $r$  of that center therefore satisfies  $f_{i_j} \geq L + \gamma/2$ . Take the union of these tests and remove repetitions. This proves both assertions.

For effective extraction use rational row tables as centers. Their behavior images are dense, since rational points are dense in products of simplexes and  $b$  is continuous. Given finitely many centers, the assertion that some feasible  $q$  lies at distance strictly greater than  $r$  from all centers is a finite existential semialgebraic assertion: absolute values and row maxima can be expressed by finitely many polynomial inequalities and disjunctions. Real quantifier elimination decides it [69]. When true, rational row enumeration finds such a point because the strict inequality defines a relatively open nonempty subset of the row simplexes. When false, the chosen points form an  $r$ -net. The volume bound forces termination. All discrepancies at rational centers are rational and can be compared exactly.

Finally apply Theorem 2.7 to the reduced finite family. Its maximal discrepancy has the strict gap in (2.29); sufficiently fine policy cells certify  $L$  using only these tests. The extraction bound is on the number of distinct tests, not on the number of cells or the complexity of real quantifier elimination.  $\square$

**Proposition 2.11** (Dimension dependence cannot be discarded). *For the delayed identity experiment on  $n \geq 2$  symbols at private width one,  $a = n - 1$ . If*

$$0 < \gamma < \frac{1}{n(n-1)}, \quad L = 1 - \frac{1}{n} - \gamma,$$

*any subfamily of original event tests that proves  $\min_q \max_i f_i(q) > L$  contains at least  $n = a + 1$  tests.*

*Proof.* A private one-state behavior is a probability vector  $p$  independent of the parameter. A positive discrepancy in parameter row  $i$  has form  $1 - p(B)$  with  $i \in B$ , and is bounded above by  $1 - p_i$ . If fewer than  $n$  tests are used, let  $S$  be the parameter indices of the positive-discrepancy tests. It has size at most  $n - 1$ . When it is nonempty choose  $p$  uniform on  $S$ . Every chosen positive discrepancy is at most  $1 - 1/|S| \leq 1 - 1/(n - 1) < L$ ; tests with target probability zero are nonpositive. If  $S$  is empty all tests are nonpositive. Hence no such subfamily certifies the stated bound. The full behavior set is the  $(n - 1)$ -simplex, giving its dimension.  $\square$

The volumetric packing step and real-algebraic elimination are classical. The conclusion here is a structural bound for the nonconvex original resource image, invariant under the addition of unobservable row coordinates. It supplies an alternative to retaining every possible event at every policy cell. It is neither a polynomial-time algorithm nor a claim that one constant convex separator characterizes the private feasible set.

### 3. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

**3.1. The testing game.** Let  $B$  be a nonempty compact subset of a finite-dimensional affine space, and let  $f_1, \dots, f_m$  be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2,  $b$  is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts;  $f_i$  ranges over their

signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set  $B$ . No convex hull is taken.

An *adaptive test* is a continuous map  $\kappa : B \rightarrow \Delta_m$ , chosen before a candidate  $b$ . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function  $\kappa$ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

**Theorem 3.1** (Independent adaptive testing duality). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{3.1}$$

*The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.*

*Proof.* Fix  $\eta > 0$ . The sets  $U_i = \{b : f_i(b) > F(b) - \eta\}$  form an open cover of the compact metric space  $B$ . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has  $F(b) - \eta < G(b, \kappa) \leq F(b)$  for all  $b$ . Consequently the first supremum equals  $\delta$ . The same uniform inequality shows, for every probability  $\mu$ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is  $\delta$ , attained by a point mass at a minimizer of  $F$ . This also proves the displayed minimax exchange without an assumption that  $B$  is convex. Equivalently one can apply the usual compact-convex minimax theorem to  $\mathcal{P}(B)$  and the convex class of continuous tests, with the pairing  $\int G d\mu$ . Finally choose the smallest index attaining  $F(b)$ . Its level sets are Borel, and its indicator vector is a Borel test with score exactly  $F$ .  $\square$

The probability  $\mu$  here describes a lottery over *revealed candidate laws*. The score is  $\int G(b, \kappa) d\mu(b)$ , not  $G(\int b d\mu, \kappa)$ . Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

**3.2. A size bound in behavior dimension.** Factor out directions annihilated by every  $f_i - f_i(b_0)$ , so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by  $a$  and let  $R = \text{diam}_*(B)$ . All conclusions below are unaffected by this observable quotient. The case  $a = 0$  is a single constant comparison and requires one test. Suppose henceforth that  $a \geq 1$ , so  $R > 0$ .

**Lemma 3.2** (A bounded behavior chart). *There exist  $b_0 \in B$  and vectors  $v_j \in B - b_0$ ,  $1 \leq j \leq a$ , forming a basis, such that  $\|v_j\|_* \leq R$  and every  $b \in B$  has coordinates  $b = b_0 + \sum_j t_j v_j$  with  $|t_j| \leq 2$ . If  $B$  is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

*Proof.* Fix  $b_0 \in B$ . In any fixed coordinates on the difference space, let  $D_0 > 0$  be the maximum absolute determinant of  $a$  vectors in  $B - b_0$ . Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than  $D_0/2$ . Replacing its  $j$ th vector by  $b - b_0$  multiplies the determinant by  $t_j$ ; maximality gives  $|t_j| < 2$ . The norm bound is the definition of  $R$ . For a rational polynomial image, take  $b_0$  at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart.  $\square$

Set  $b(x) = b_0 + \sum_j (4x_j - 2)v_j$  and  $C = \{x \in [0, 1]^a : b(x) \in B\}$ . The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write  $\ell_i(x) = f_i(b(x))$ . Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (3.2)$$

For  $n \geq 0$  let  $\beta_{\alpha,n}$  be the tensor Bernstein basis of coordinatewise degree  $n$ . A feasible coefficient array has  $c_{\alpha i} \geq 0$  and  $\sum_i c_{\alpha i} = 1$  for every  $\alpha \in \{0, \dots, n\}^a$ . It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (3.3)$$

**Theorem 3.3** (Intrinsic adaptive degree and support bounds). *The maxima in (3.3) are attained and  $d_n \leq d_{n+1} \leq \delta$ . For  $n \geq 1$ ,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (3.4)$$

*There is an array with at most  $(n+1)^a$  nonzero coefficients, all equal to one, whose minimum score is at least  $\delta - 4Ra/\sqrt{n}$ . Its test has coordinatewise degree  $n$ , and its score degree at most  $n+1$ . Furthermore*

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (3.5)$$

*and a minimizing measure can be supported on at most  $(n+2)^a$  points. In particular a strict level  $L < \delta$  admits an adaptive test of positive uniform margin with  $n > (8Ra/(\delta - L))^2$  and at most  $(n+1)^a$  occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.*

*Proof.* The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact  $C$  is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the  $\ell_i(x)$ , proving the upper bound by  $\delta$ .

At each grid point  $\alpha/n$  choose an index  $i_{\alpha}$  maximizing  $\ell_i$  there. The choice is made even at infeasible grid points; only the final score is minimized over  $C$ . Give that index coefficient one and the others zero. For fixed  $x$  let  $A_j$  be independent  $\text{Bin}(n, x_j)$  variables. If  $i_*$  maximizes at  $x$ ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses  $\text{Var}(A_j/n) \leq 1/(4n)$ . This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over  $C$  by a minimum over  $\mathcal{P}(C)$ . The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [71] exchanges the extrema. Optimizing each coefficient simplex gives (3.5). The integrands are polynomials of coordinatewise degree at most  $n + 1$ . Preserve all monomial moments of that degree by Carathéodory's theorem. There are  $(n + 2)^a$  monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most  $(n + 2)^a - 1$ . A measure with at most  $(n + 2)^a$  atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than  $(\delta - L)/2$ .  $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (3.5) cannot be replaced by its behavior barycenter. For  $n = 0$  it can:  $d_0$  is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 6.5, the target belongs to the convex hull of private behaviors, so  $d_0 = 0$ , whereas  $\lim_n d_n = \sqrt{2} - 1$ . Thus the hierarchy recovers a gap which every constant test misses.

*Remark 3.4* (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set  $C$ , the bit lengths of a rational chart, or the size of a description of  $C$ . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use  $R \leq 1$  with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound  $R \leq 2$  is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

### 3.3. A three-weight certificate invisible to constant tests.

**Proposition 3.5** (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities  $p, q$ , for the target which assigns probability  $1/2$  to each of 00, 11. Put  $u = pq$ ,  $v = (1 - p)(1 - q)$  and  $J(t) = 3t^2 - 2t^3$ . The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

*are nonnegative and sum to one. Against the event discrepancies  $1/2 - u, 1/2 - v, 1 - u - v$ , respectively, their score  $G$  satisfies*

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (3.6)$$

*The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.*

*Proof.* We have  $u + v \leq 1$ . Since  $J$  is increasing on  $[0, 1]$  and  $J(1 - t) = 1 - J(t)$ , the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set  $a = (p + q - 1)^2$  and  $b = (p - q)^2$ . Then  $a, b \geq 0$  and  $a + b \leq 1$ . Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let  $c = 1 - a - b$ . Another polynomial identity, valid when  $a + b + c = 1$ , is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus  $G(a, b) \geq G(a, 0)$ . The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients  $z_k/349440$ , where

$$(z_0, \dots, z_{16}) = (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944).$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is  $351/349440 = 9/8960$ , proving (3.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero.  $\square$

For comparison, the affine weights  $v, u, 1 - u - v$  already give the smaller bound  $3/8$ , with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

**3.4. Strict certificates and effective physical input.** Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at  $L = \delta$ .

**Proposition 3.6** (Uniform effective presentation input). *Suppose a single algorithm, on input  $j$ , returns rational finite instruments  $E_j, F_j$  and nonnegative rational error enclosures  $a_j, c_j$  for two-sided stateless marked feedback presentations of  $E, F$ . Suppose  $a_j + c_j \rightarrow 0$ . All alphabets, widths, selector laws and finite subroutines must be computable uniformly from  $j$  and the requested signature. Then, for every positive rational  $\eta$ , an algorithm returns a rational interval of width below  $\eta$  containing  $\delta_b(E, F)$  at the same signature.*

*Proof.* Enumerate  $j$  until  $a_j + c_j < \eta/8$ . This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below  $\eta/2$ . Enlarge it by  $a_j + c_j$  at both ends using stateless comparison. The enlarged width is below  $3\eta/4$ . Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition.  $\square$

#### 4. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

**4.1. The exact marked two-report experiment.** Let  $B, W \in \{0, 1\}$  have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (4.1)$$

The blind source  $E_*$  has two constant reports and actual mark  $W$ . The target  $F_*$  reports  $Y_1 = B$ , receives  $A \in \{0, 1\}$  after that report, and reports  $Y_2 = AB$ . The feedback tester may choose  $A$  as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate  $A$ . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of  $(B, W)$ . Put

$$d_* = \sqrt{5/2} - 9/8.$$

**Theorem 4.1** (An exact marked causal separation). *At the width profile  $(1, 1)$ ,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (4.2)$$

*The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter  $\sqrt{5/8}$ , followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.*

*Proof.* First choose the constant gate  $A = 1$ . The two private emissions are independent Bernoulli variables with parameters  $p, q$  and are independent of  $W$ . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is  $H(u) + H(v)$ , where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two  $W$ -masses at each equal pair are  $3/16$  and  $5/16$ . Total variation is one minus this overlap.

Cauchy–Schwarz gives  $\sqrt{u} + \sqrt{v} \leq 1$ . By symmetry suppose  $u \geq v$ . If  $u \leq 1/4$  the overlap is at most  $1/2$ . If  $u \geq 1/4$ , then  $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$ , so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of  $[1/4, a]$  and  $[a, b]$  and nonincreasing on  $[b, 1]$ . Its maximum is therefore attained at  $1/4$ ,  $a$  or  $b$ . The values at  $a$  and  $b$  are  $1 + 2a - 2\sqrt{a}$  and  $3/2 + b - 2\sqrt{b}$  respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using  $\sqrt{a} > 3/5$ ,  $\sqrt{b} > 3/4$ . It also exceeds  $1/2$ . Taking  $p = q = \sqrt{b}$  attains this maximum, and hence the minimum error is  $1 - (3/2 + b - 2\sqrt{b}) = d_*$ .

For any hidden selector, the all-on report pair is still independent of  $W$ . If its probabilities of 11 and 00 are  $u, v$ , then  $u + v \leq 1$ . Concavity and monotonicity of  $H$  give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit  $J$ , emitting  $J$  twice, attains this overlap. This requires precisely two deterministic selector tables and gives error  $1/8$ . In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair  $(X_1, X_2)$ , emit  $X_1$ , and after receiving  $A$  emit  $AX_2$ . For a deterministic feedback  $A = g(X_1)$  the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit  $J$  is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of  $(B, W)$ . The all-on gate in every fiber has error at least  $d_*$  in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains  $d_*$ .  $\square$

**4.2. A full-dimensional collision preparation.** Consider two labeled equal-mass hard spheres in  $\mathbb{R}^3$  with contact distance  $d$ . The center of mass  $Q = (q_1 + q_2)/2$  ranges over any fixed rational position box. Write  $r = q_1 - q_2$ ,  $g = v_1 - v_2$ , and  $V = (v_1 + v_2)/2$ . Conditional on  $(B, W)$  in (4.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (4.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is  $W = \mathbf{1}_{\{V_z > 0\}}$ ; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

**Lemma 4.2** (Uniform transverse collision signal). *For every preparation in (4.3) and every  $d \in [9/10, 11/10]$ , exactly one collision occurs before time 3. Its time  $t_c$  lies in  $(4/5, 3/2)$ , and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (4.4)$$

For  $0 < d \leq 2/5$  no collision occurs before time 3, and the detector is zero throughout that interval.

*Proof.* Up to a collision  $r(t) = r + tg$ . For  $t \leq 4/5$ , its  $x$  coordinate is at most  $-29/10 + (4/5)(21/10) = -61/50$ , so its norm exceeds  $11/10$ . At time  $3/2$  the absolute coordinate bounds are  $1/4$ ,  $113/200$  and  $1/40$ , whose squared sum is less than  $(9/10)^2$ . Thus a first contact occurs in the stated interval. At the free zero of the  $x$  coordinate, whose time is at most  $31/19$ , the transverse norm is less than  $9/10$  as well. The first contact therefore has  $n_x < 0$ , where  $n = r(t_c)/d$ . The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that  $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$ . The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing,  $r(t_c) \cdot g' > 0$ , so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (4.4). For  $d \leq 2/5$  and  $0 \leq t \leq 3$ ,  $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$ . Thus there is no collision. The incoming  $|v_{1,y}| \leq 3/200$  then makes  $h$  zero.  $\square$

At time 2 the physical target  $F_{d,\sigma}$  reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate  $A \in \{0, 1\}$ , and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the  $\xi_i$  are independent standard normal variables, independent of the preparation. The source  $E_d$  has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking  $A = 0$  suppresses later information and taking  $A = 1$  exposes another noisy observation of a collision-generated quantity.

**Theorem 4.3** (Collision-sensitive physical resource separation). *Let  $d \in [9/10, 11/10]$  and  $0 < \sigma \leq 1/12$ . At retained width  $(1, 1)$  the physical deficiencies obey, uniformly in  $d$ ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^v(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (4.5)$$

*In particular private and visible error exceed  $9/20$ , while hidden two-selector error is less than  $13/100$ . More generally the errors in these three comparisons are at most  $2\Phi(-1/(4\sigma))$ , which tends to zero as  $\sigma \downarrow 0$ . For  $0 < d \leq 2/5$  all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.*

*Proof.* Couple the ideal and physical target by the same  $(B, W)$  and the same feedback random table. Lemma 4.2 bounds each potential bit error by  $\Phi(-1/(4\sigma))$ . With probability at least  $1 - 2\Phi(-1/(4\sigma))$  both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly  $E_*$ . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound  $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$  at  $x = 3$  gives  $\Phi(-3) < 1/600$ : use  $e^{9/2} > 80$  and  $\sqrt{2\pi} > 5/2$ . Hence the total error is below  $1/300$ . The inequalities  $d_* - 1/300 > 9/20$  and  $1/8 + 1/300 < 13/100$  follow by squaring positive rational bounds.

When  $d \leq 2/5$ , both potential reports are independent fair noise bits and are independent of  $W$ . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under  $(q, d, t) \mapsto (sq, sd, st)$  with  $v$  unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit.  $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

**4.3. Uniformly effective finite presentations.** The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

**Proposition 4.4** (Collision-chart rational presentations). *For computable  $d \in [9/10, 11/10]$  and  $\sigma > 0$ , one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for  $(E_d, F_{d,\sigma})$ . On a compact computable parameter interval with  $\sigma \geq \sigma_0 > 0$ , the errors and the construction are uniform. Thus Proposition 3.6 applies at every fixed finite state and selector signature, not only at  $(1, 1)$ .*

*Proof.* On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals  $-d(g \cdot n) > (9/10)(1443/1000)$ . All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound  $C_{\text{geo}}$  for  $v'_{1,y}$  in the packet variables and  $d$ . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability  $p = \Phi(h(v'_{1,y})/\sigma)$ . Their four probabilities are  $p^2$ ,  $p(1-p)$ ,  $p(1-p)$  and  $(1-p)^2$ . The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most  $2C_{\text{geo}}/\sigma_0$  for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh  $h$  therefore have explicit error at most a computable constant times  $h$ , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (4.1). For each  $W$ , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep  $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$  exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence.  $\square$

## 5. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let  $Z$  take values in a standard Borel space and let  $Y$  take values in a standard Borel canonical path space on  $[0, T]$ , with a right-continuous observation filtration. Consider equivalent joint probability measures  $P, Q$  on  $(Z, Y)$ , with the same  $Z$ -marginal  $\nu$ . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable  $f(Z)$  write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include  $T$  and can be evaluated on a countable dense time set.

**Theorem 5.1** (Marked variation controls the whole posterior process). *Let  $\varepsilon = \|P - Q\|_{\text{TV}}$  and  $|f| \leq 1$ . When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (5.1)$$

*There is a coupling of  $(Z, Y^P)$  and  $(Z, Y^Q)$  which retains exactly the same latent variable and satisfies*

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (5.2)$$

*The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.*

*Proof.* Let  $D = dP/dQ$ ,  $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$ , and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since  $\mathbb{E}_Q|D - 1| = 2\varepsilon$ , both martingales  $A, B$  have terminal absolute expectation at most  $2\varepsilon$ . Doob's weak  $L^1$  inequality gives, for  $0 < u \leq 2$ ,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating  $2u$  times this bound on  $(0, 2)$  gives  $16\varepsilon + 64\varepsilon = 80\varepsilon$ , proving (5.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on  $Z$ , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (5.1) with the  $Q$ -marginal; on its complement the squared difference is at most four. This proves (5.2). The maximal coupling may depend on the full output path; it has no asserted online implementation.  $\square$

**Corollary 5.2** (Changing-filtration convergence from marked comparison). *Let  $P_n$  and  $P$  be equivalent joint laws with the same latent marginal  $\nu$ , and  $\|P_n - P\|_{\text{TV}} \rightarrow 0$ . They admit a common probability-space realization  $(Z, Y, Y^1, Y^2, \dots)$  such that, for every  $f \in L^2(\nu)$ ,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for  $f \in L^1(\nu)$ . It also permits  $f_n \rightarrow f$  in  $L^2(\nu)$ , respectively in  $L^1(\nu)$ , in the first conditional expectation.

*Proof.* Glue the conditional maximal couplings of  $(Z, Y^n)$  with the fixed reference  $(Z, Y)$  by taking their conditional kernels independently on a countable product. The coupling does not depend on  $f$ . For the truncation  $f^{(K)} = (-K) \vee (f \wedge K)$ , Theorem 5.1 bounds the middle difference in the  $L^2$  norm of the supremum by  $K\sqrt{84\epsilon_n}$ . Doob's  $L^2$  inequality bounds each truncation tail by  $2\|f - f^{(K)}\|_{L^2(\nu)}$ . The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\epsilon_n}.$$

First let  $n \rightarrow \infty$ , then  $K \rightarrow \infty$ . For  $L^1$  use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference  $f_n - f$  is handled by the same maximal inequalities under  $P_n$ , whose latent marginal remains  $\nu$ .  $\square$

**5.1. A microscopic sufficient condition.** Let  $(\Phi_t)_{t \in \mathbb{R}}$  be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability  $\mu$ . Fix a prepared law  $\nu$  with  $d\nu/d\mu \leq C$ , and bounded real observables  $h, h_n$ . These may be collision-sensitive tagged observables. Put  $\eta_n = \|h_n - h\|_{L^2(\mu)}$ . With a common latent initial state  $Z \sim \nu$ , consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of  $Z$  in each model. The notation does not prescribe a synchronous coupling.

**Theorem 5.3** (Collision-observation optional-projection limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (5.3)$$

For every  $|f(Z)| \leq 1$  they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (5.4)$$

In particular  $\eta_n \rightarrow 0$  supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when  $h_n$  are orbit-mollified graph-core approximants to any bounded  $h \in L^2(\mu)$ .

*Proof.* Conditional on  $Z$ , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses  $\nu \leq C\mu$  and invariance of  $\mu$ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (5.3). Applying Theorem 5.1 proves (5.4).

For the last assertion choose a nonnegative smooth approximate identity  $\rho_n$  on the time line and set  $h_n = \int \rho_n(s) h \circ \Phi_s ds$ . Strong continuity of the Koopman group in  $L^2(\mu)$  gives  $\eta_n \rightarrow 0$  and  $\|h_n\|_\infty \leq \|h\|_\infty$ . Repeated integration by parts puts  $h_n$  in every power domain of the generator,

with derivatives obtained from  $\rho_n^{(k)}$ . This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate.  $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (5.3). That is the resource-comparison consequence. The latent-preserving coupling in (5.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

## 6. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 3.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as  $F(q) = \max_{1 \leq i \leq m} f_i(q)$ , where  $f_i$  are all its marked event discrepancies, including the empty event  $f_1 = 0$ . For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in  $f_i$ . Thus  $-1 \leq f_i \leq 1$  for all three signatures; the polynomials may be different.

Parameterize a row with  $d$  outcomes by  $d - 1$  variables in  $[0, 1]$ :

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map  $q(x)$  from  $[0, 1]^D$  to the feasible row product. Put  $h_i(x) = f_i(q(x))$  and  $g_i = (1 + h_i)/2$ . All  $h_i$  are rational polynomials,  $0 \leq g_i \leq 1$ , and  $g_1 = 1/2$ . The case  $D = 0$  is a single feasible table and is interpreted directly.

**Theorem 6.1** (Strict polynomial lower-certificate hierarchy). *For  $p \geq 1$  define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

*Then the denominators are positive,  $a_p$  is nondecreasing, and*

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (6.1)$$

*For rational  $L$  let*

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (6.2)$$

*A nonnegative tensor Bernstein expansion of  $P_{p,L}$  on  $[0, 1]^D$  certifies  $\delta \geq L$ . If  $L < \delta$ , some  $p$  and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such*

expansion certifies  $L < \delta$ . These are lower certificates at the original state and selector budgets. No convergence of the feasible behavior image occurs.

*Proof.* Fix  $x$  and put  $b = \max_i g_i(x) \geq 1/2$  and  $r_i = g_i(x)/b$ . At least one  $r_i$  is one. Write  $S_j = \sum_i r_i^j$ . Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently  $S_{p+1}/S_p \geq m^{-1/(p+1)}$ . Since  $A_p = 2bS_{p+1}/S_p - 1$  and  $\max_i h_i = 2b - 1$ ,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy–Schwarz gives  $S_{p+2}S_p \geq S_{p+1}^2$ , so  $A_p$  is nondecreasing in  $p$ . Its denominator is at least  $2^{-p}$  and it is continuous on the cube. Taking minima proves (6.1). The bound is also at most  $2 \log m/(p+1)$ , by  $1 - e^{-u} \leq u$ .

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply  $P_{p,L} \geq 0$ . Dividing by the positive denominator gives  $F(q(x)) \geq A_p(x) \geq L$  for every  $x$ , which proves soundness. For completeness choose  $p$  so that the error in (6.1) is less than  $\delta - L$ . Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 6.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most  $m$ , so they give a strict positive uniform gap above  $L$ .  $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [73]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

**Lemma 6.2** (An explicit positivity degree). *Let  $P(x) = \sum_k c_k x^k$  be a rational polynomial in  $D$  variables and  $n \geq \max_{k,j} k_j$ . In its tensor Bernstein expansion of degree  $n$  in each coordinate the coefficient at  $\alpha \in \{0, \dots, n\}^D$  is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (6.3)$$

*In particular,  $\min P > 0$  implies positivity of every coefficient whenever  $n > C(P)/\min P$  and the degree restriction holds.*

*Proof.* For a binomial random variable  $B$  with parameters  $(n, x)$ ,  $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$ . Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is  $\binom{a}{k} = 0$  for  $a < k$ . To compare  $\binom{a}{k}/\binom{n}{k}$  with  $(a/n)^k$ , draw  $k$  indices uniformly from  $\{1, \dots, n\}$ . Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most  $\binom{k}{2}/n$ . Apply this total variation bound to the event that all drawn indices lie in a prescribed  $a$ -element subset. Hence the difference of the two factors is at most  $\binom{k}{2}/n$ . All factors lie in  $[0, 1]$ . A product telescoping bound followed by the triangle inequality proves (6.3). The final assertion follows by taking the minimum of  $P$ .  $\square$

**Corollary 6.3** (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level  $L$  is equivalent to the existence of the following finite rational certificate: an integer  $p$ , an integer  $n$ , and a strictly positive coefficient array for (6.2) whose identity is checked by (6.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to  $\delta$ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

*Proof.* Soundness and completeness are Theorem 6.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all  $(p, n)$  and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed  $(p, n)$  test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument.  $\square$

*Remark 6.4* (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions  $g_i^p / \sum_j g_j^p$  are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count  $(n + 1)^D$ , the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 3.3; its verification is a different cost.

**6.1. A sharp two-time nonconvex example.** The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits  $(0, 0)$  or  $(1, 1)$  with probability  $1/2$  each. At width one, a private simulator emits two independent bits with probabilities  $p, q$ . It cannot reread its first emitted bit.

**Proposition 6.5** (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

*The private objective is the maximum of the four polynomials*

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1 - p)(1 - q), \quad p + q - 2pq.$$

*Every constant mixture of these tests has minimum at most zero, whereas Corollary 6.3 certifies every rational  $L < \sqrt{2} - 1$  at private width one.*

*Proof.* Put  $u = pq$  and  $v = (1 - p)(1 - q)$ . Direct total variation calculation gives  $F = 1 - \min(u, 1/2) - \min(v, 1/2)$  and the displayed maximum. By Cauchy–Schwarz,  $\sqrt{u} + \sqrt{v} \leq 1$ . If  $u \geq 1/2$ , then  $v \leq (1 - 1/\sqrt{2})^2$ , so the sum of the two capped terms is at most  $2 - \sqrt{2}$ ; the case  $v \geq 1/2$  is symmetric. Otherwise set  $a = \sqrt{u}$ ,  $b = \sqrt{v}$ . On  $0 \leq a, b \leq 1/\sqrt{2}$ ,  $a + b \leq 1$ , the maximum of  $a^2 + b^2$  is at a vertex of this polygon and equals  $2 - \sqrt{2}$ . Equality is attained at  $p = q = 1/\sqrt{2}$ . A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation.  $\square$

## 7. ENDOGENOUS COMMON DECISIONS

**7.1. A shared encoder on a controlled tree.** Let  $\mathcal{H}_t$  be the finite supported observation–control histories of a known controlled experiment. The initial history has distribution  $\mu$ , positive on  $\mathcal{H}_0$ . For  $h \in \mathcal{H}_t$  and  $a \in A_t$ , write  $p_t(y \mid h, a)$  for the next-report law and  $hay$  for the successor

when this probability is positive. Zero-probability successors impose no condition. The finite task set is  $D$ . Task  $d$  has conditional expected stage costs  $c_{t,d}(h, a)$  and terminal cost  $c_{T,d}(h)$ . These may be obtained by conditioning bounded losses of the actual hidden mark; the task does not alter the physical transition law. Controls do alter that law through  $a$ .

The common source-consuming encoder is

$$q_0(m \mid h), \quad q_{t+1}(m' \mid m, a, y), \quad m \in [S_t], \quad m' \in [S_{t+1}].$$

Controller  $d$  chooses its action by  $\alpha_{t,d}(a \mid m)$ . It cannot query the discarded history. The clock is available. The encoder does not know  $d$ ; the update sees the actual selected action, as allowed by the interface. All kernels can be stochastic. The same encoder must serve every task, even though different tasks induce different laws on histories.

Define the full-history Bellman functions and regrets by

$$\begin{aligned} V_{T,d}(h) &= c_{T,d}(h), \\ V_{t,d}(h) &= \min_{a \in A_t} \left( c_{t,d}(h, a) + \sum_y p_t(y \mid h, a) V_{t+1,d}(hay) \right), \\ G_{t,d}(h, a) &= c_{t,d}(h, a) + \sum_y p_t(y \mid h, a) V_{t+1,d}(hay) - V_{t,d}(h) \geq 0. \end{aligned} \tag{7.1}$$

The benchmark for task  $d$  is  $v_d = \sum_h \mu(h) V_{0,d}(h)$ . It is an ordinary full-history control value, not a posterior quantity conditioned on a fixed open-loop preparation.

**Lemma 7.1** (Occupation regret). *For every admissible common encoder and task controller, the task excess satisfies the exact identity*

$$R_d - v_d = \mathbb{E}_d \sum_{t=0}^{T-1} G_{t,d}(H_t, A_t). \tag{7.2}$$

Moreover the entire vector of risks of a randomized finite-horizon design is a convex combination of risk vectors of deterministic designs with the same widths and one common encoder.

*Proof.* Take conditional expectations in (7.1) along the actual controlled path. Summing the resulting expressions for  $G_{t,d}$  telescopes the terms  $V_{t,d}(H_t)$ , leaving the sum of stage and terminal costs minus  $\mathbb{E} V_{0,d}(H_0)$ . This proves (7.2) without assuming the control is open loop.

Sample in advance a deterministic outcome for every time-indexed row of the common encoder and every time-indexed row of every task controller. Row choices are independent with their prescribed probabilities; the sampled encoder table is shared across tasks. Along each trajectory each time-indexed row is encountered at most once. The product of its row probabilities is therefore the probability of that path under the sampled-table mixture. This identity holds for all tasks simultaneously after summing paths and costs. An additional independent visible or hidden seed is first conditioned on and then integrated. The conclusion is a mixture identity; the mixing variable is not claimed to be executable privately without retaining its index.  $\square$

**Definition 7.2** (Endogenous compatible cover). A cover of profile  $S$  consists of an initial map  $i : \mathcal{H}_0 \rightarrow [S_0]$ , common maps  $\delta_t : [S_t] \times A_t \times Y_{t+1} \rightarrow [S_{t+1}]$ , task action maps  $a_{t,d} : [S_t] \rightarrow A_t$ , and cells  $C_{t,m}^d \subseteq \mathcal{H}_t$ . Empty cells and overlapping cells are allowed. The requirements are

$$\begin{aligned} h &\in C_{0,i(h)}^d & (h \in \mathcal{H}_0, d \in D), \\ G_{t,d}(h, a_{t,d}(m)) &= 0 & (h \in C_{t,m}^d), \\ h \in C_{t,m}^d, p_t(y \mid h, a_{t,d}(m)) > 0 &\implies ha_{t,d}(m)y \in C_{t+1,\delta_t(m,a_{t,d}(m),y)}^d. \end{aligned} \tag{7.3}$$

Thus the cells cover the subtree generated by the selected actions. They need not cover histories generated by actions the controller never selects. Crucially,  $i$  and  $\delta_t$  cannot depend on  $d$ .

**Theorem 7.3** (Exact endogenous compression). *For a prescribed finite width profile  $S$ , the following are equivalent: a deterministic common design attains  $R_d = v_d$  for every task; a privately randomized common design does so; a common design with an independent visible or hidden seed does so; and an endogenous compatible cover of profile  $S$  exists.*

Let  $\beta_d > 0$  sum to one, and put  $\mathcal{G} = \sum_d \beta_d (R_d - v_d)$ . If the profile is infeasible, then every such design satisfies

$$\mathcal{G} \geq g_S := \beta_* \mu_* p_*^T \gamma_* > 0, \quad (7.4)$$

where  $\beta_* = \min_d \beta_d$ ,  $\mu_*$  is the least positive initial mass,  $p_*$  is the least positive transition probability, and  $\gamma_*$  is the least positive Bellman regret among supported entries. If there are no positive regrets, every profile with positive widths is feasible. The convention  $p_* = 1$  applies when there are no transitions.

More generally, replacing the zero in (7.3) by the requirement  $G_{t,d}(h, a_{t,d}(m)) \leq \eta_{t,d}$  gives a common design with  $R_d - v_d \leq \sum_t \eta_{t,d}$ . The sharper value is exactly the occupation-weighted sum in (7.2).

*Proof.* A cover defines the deterministic encoder  $M_0 = i(H_0)$  and  $M_{t+1} = \delta_t(M_t, A_t, Y_{t+1})$ , and the task controller uses  $a_{t,d}(M_t)$ . Induction in (7.3) shows that every reached history lies in its indicated cell. Each incurred Bellman regret is zero, so Lemma 7.1 proves exactness. The same induction and identity prove the approximate-cover statement.

Conversely, an exact deterministic design defines  $C_{t,m}^d$  to be the set of histories reached with positive probability under task  $d$  and state  $m$ . Initial coverage and common successor closure follow from its encoder. Since the nonnegative sum in (7.2) has zero expectation, every positively reached history has zero regret for its selected action. This proves all cover requirements. It also explains why adding closure for every unused action would impose a strictly different problem.

For a randomized design, Lemma 7.1 expresses the vector of excesses as an average of deterministic excess vectors. Each coordinate is nonnegative. If all averaged coordinates are zero, then, for almost every sampled design, all coordinates are zero, since the task set is finite. Choose one such deterministic table. This applies after conditioning on an arbitrary independent seed, with recovery from the seed and retained state when the seed is visible. It does not use a seed correlated with the initial history. The deterministic cover already constructed then proves necessity.

If no exact deterministic design exists, each deterministic design has some task, stage, and positively reached history with positive selected regret. That history has probability at least  $\mu_* p_*^t$ , and its regret is at least  $\gamma_*$ . Since  $0 < p_* \leq 1$  and  $t \leq T$ , its contribution to  $\mathcal{G}$  is at least the right side of (7.4). Every randomized design is an average of these vectors, so the same bound follows. When there is no positive regret, every action is Bellman optimal wherever supported, and a one-state encoder with arbitrary task actions is exact.  $\square$

**Example 7.4** (Why unused controls cannot define the cover). At the first time there is no report. Action **rest** leads to a deterministic hidden bit zero and a constant report, at zero cost. Action **probe** costs one, draws a fair bit, and reports it. After this report is consumed there is one blank checkpoint, followed by a binary guess with zero-one loss. The full-history optimum chooses **rest** and guesses zero; one retained state is exact. A restart family that insists on optimal guessing after the unused **probe** branch needs two retained states. The endogenous criterion and the all-controls restart criterion are both meaningful, but they have different feasible sets. The cover in Definition 7.2 implements the former, not the latter.

**7.2. Lossy common design and intrinsic comparison.** Let  $\mathcal{D}_S^{\text{ctl}}$  be the finite set of deterministic common encoders and deterministic task controllers in this section. Put  $r_d(\pi) = R_d(\pi) - v_d$ . Then the hidden-selector lossy minimax value is

$$\min_{\lambda \in \Delta(\mathcal{D}_S^{\text{ctl}})} \max_d \sum_{\pi} \lambda_{\pi} r_d(\pi) = \max_{\beta \in \Delta(D)} \min_{\pi \in \mathcal{D}_S^{\text{ctl}}} \sum_d \beta_d r_d(\pi). \quad (7.5)$$

Here a selector chooses the entire design, including its task-controller tables; its mode is internal to that design. This follows from the simultaneous mixture identity in Lemma 7.1 and finite minimax. A mixture using at most  $|D| + 1$  designs suffices and can be stored with at most  $(|D| + 1)S_t$  private labels. Private finite-width optimization is instead the polynomial optimization in the actual encoder and controller rows. The exact zero sets agree by Theorem 7.3, but positive minimax values need not agree. Thus the cover theorem supplies a distinguished zero face inside a quantitative lossy theory, rather than purporting to solve every lossy design problem by equivalence classes.

**Corollary 7.5** (Intrinsic certification of endogenous widths). *Suppose the losses of this finite problem are normalized so that every total task loss lies in  $[0, 1]$ . Let an approximation  $F$  have the same control and task interfaces, with each task represented by the same bounded marked-transcript loss on both sides and the full-history benchmark included in the paired decision class. If*

$$\max\{\delta_1^p(E, F), \delta_1^p(F, E)\} \leq \varepsilon < g_S/4,$$

*then the optimized  $\beta$ -weighted excess at profile  $S$  in  $F$  is less than  $g_S/2$  exactly when the endogenous cover exists in  $E$ . If only a cost- $K$  simulation from  $E$  to  $F$  is known and an exact  $S$ -state  $F$  design transports with zero error and the same benchmark, then an endogenous cover of profile  $KS$  exists in  $E$ . A hidden selector multiplies that profile by its charged selector size.*

*Proof.* Full-history controllers also transport through a stateless simulator, so their optimized benchmarks differ by at most  $\varepsilon$ . Equation (2.18) bounds the optimized constrained values by  $\varepsilon$  in each direction. The optimized excesses therefore differ by at most  $2\varepsilon$ . In  $E$  the value is zero on the cover face and at least  $g_S$  otherwise, by Theorem 7.3. The strict threshold follows. For the one-sided statement, the product construction transports the exact design at the stated width and benchmark. The same theorem then converts exactness into a cover.  $\square$

*Remark 7.6* (Dependence of the finite witness). The number  $g_S$  depends on the least positive probability and regret of the specified finite model. It can tend to zero as a transition or a cost is perturbed. Neither (7.4) nor the certification corollary asserts a model-uniform robustness radius. Likewise an inverse-limit assertion that every finite prefix has a feasible cover would give existence of an infinite consistent design, not a computable bound on the first infeasible prefix. No such bound is used here.

**7.3. Task resolution, positive error, and the complete state.** We give a quantitative state law at a source-consuming restart. It complements the endogenous zero-face theorem; it does not replace the causal update constraint by a terminal partition.

Let  $\mathcal{H}$  be a finite set of supported histories at a fixed checkpoint. Choose  $d$  actual marked continuation-event prediction tasks, each including its declared open-loop continuation control word. For  $h \in \mathcal{H}$  let  $v(h) \in [0, 1]^d$  be the vector of their conditional probabilities. A source-consuming statistic has rows  $Q(m | h)$ ,  $m \in [K]$ . For every preparation  $\lambda$  on  $\mathcal{H}$ , the task is announced after storage and its forecast is scored by squared error against the actual event. Define

$$e(Q) = \sup_{\lambda} \max_{1 \leq j \leq d} \mathbb{E}_{\lambda} \text{Var}(v_j(H) | M), \quad e_K = \inf_Q e(Q).$$

The expression is precisely excess Brier risk over the full-history forecast: conditional on  $H$ , event variance is an unavoidable common term, and the law of total variance gives the displayed residual. Decoders may use the announced preparation and task, but the same statistic  $Q$  must work for all of them. Let  $N(\eta)$  be the maximum size of a subset of  $v(\mathcal{H})$  whose Euclidean pair distances are at least  $\eta > 0$ . Let  $C(r)$  be the minimum number of sets covering  $v(\mathcal{H})$  with coordinatewise diameter at most  $r$ .

**Theorem 7.7** (Positive-error state bounds). *For all positive  $K$  and  $\eta$ ,*

$$e_K \geq \frac{\eta^2}{2d} \left(1 - \frac{K}{N(\eta)}\right)_+. \quad (7.6)$$

*For  $0 \leq \varepsilon \leq 1/4$  there is a deterministic statistic of width  $C(2\sqrt{\varepsilon})$  with  $e(Q) \leq \varepsilon$ . Consequently, writing  $K(\varepsilon) = \min\{K : e_K \leq \varepsilon\}$ ,*

$$N(\eta) \left(1 - \frac{2d\varepsilon}{\eta^2}\right) \leq K(\varepsilon) \leq C(2\sqrt{\varepsilon}). \quad (7.7)$$

*At zero error the exact width is  $|v(\mathcal{H})|$ . It is at most the universal marked continuation width, and it equals that width when the event family separates all distinct marked continuation laws. The lower bound also holds when an independent exposed selector is available and each selector fiber has at most  $K$  private labels; in that case the conditional variance is taken given  $(U, M)$ .*

*Proof.* Choose  $n = N(\eta)$  separated histories and prepare them uniformly. For an output  $m$ , write  $\pi_h(m) = \mathbb{P}(H = h \mid M = m)$ . The variance identity gives

$$\begin{aligned} \sum_{j=1}^d \mathbb{E} \text{Var}(v_j(H) \mid M) &= \frac{1}{2} \sum_m \mathbb{P}(M = m) \sum_{h, h'} \pi_h(m) \pi_{h'}(m) \|v(h) - v(h')\|_2^2 \\ &\geq \frac{\eta^2}{2} \left(1 - \sum_m \mathbb{P}(M = m) \sum_h \pi_h(m)^2\right). \end{aligned}$$

Furthermore

$$\sum_m \mathbb{P}(M = m) \sum_h \pi_h(m)^2 \leq \sum_m \mathbb{P}(M = m) \max_h \pi_h(m) = \frac{1}{n} \sum_m \max_h Q(m \mid h) \leq \frac{K}{n}.$$

At least one coordinate variance is the average of these  $d$  terms or larger. Nonnegativity proves (7.6). Conditioning on an independent exposed selector gives the identical bound on the sum of coordinate variances at every fiber, with the same uniform preparation. Average this sum over the selector and only then choose a coordinate at least as large as its average. Thus the task need not depend on the selector, and the same lower bound follows.

For the upper bound assign each history to a covering set containing its feature vector and retain its set index. Every conditional coordinate range then has length at most  $2\sqrt{\varepsilon}$ . A real random variable in  $[a, b]$  has variance at most  $(b - a)^2/4$ :  $\mathbb{E}(X - a)(b - X) \geq 0$  implies  $\text{Var } X \leq (\mathbb{E}X - a)(b - \mathbb{E}X) \leq (b - a)^2/4$ . This proves the upper bound uniformly over preparations.

If two histories with different feature vectors share a retained symbol with positive probability, a preparation on just those two histories has a strictly positive posterior variance at that symbol in a differing coordinate. Therefore zero excess forces disjoint retained supports for distinct feature vectors. Conversely the feature vector itself is an exact statistic. This proves the zero law. Every feature is determined by the universal marked continuation law; a finite separating event family distinguishes all its finitely many classes. The final assertion follows.  $\square$

**Corollary 7.8** (A positive-error obstruction for marked simulation). *Suppose a prescribed source-consuming reverse simulator of width  $K$  compares the stored-state restart to the full-history restart, preserving the actual marked continuation experiment with error at most  $\delta$  uniformly over preparations and announced continuation tasks. Then*

$$\delta \geq e_K \geq \frac{\eta^2}{2d} \left(1 - \frac{K}{N(\eta)}\right)_+.$$

*Proof.* Compose the reverse simulator with the full-history optimal Brier forecast for each announced task. Squared error belongs to  $[0, 1]$ ; marked comparison changes its expectation by at

most  $\delta$ . The optimal stored-state forecast is no worse than this composed forecast. Hence its excess is at most  $\delta$  for every preparation and task. Apply Theorem 7.7.  $\square$

For several checkpoints these are necessary width bounds at every restart. Their simultaneous sufficiency additionally requires that the chosen covers admit causal updates on the reached support, exactly as in the preceding endogenous theorem. A feature cover alone does not construct a reverse channel preserving an arbitrary actual mark. For a full separating family the zero-error envelope recovers the universal continuation quotient; before completion, the quantitative bounds depend on the declared tasks rather than on irrelevant marks or unused future controls.

## 8. REGENERATIVE AVERAGE RISKS

**8.1. The reset signature.** Let the physical state space  $X$  be standard Borel, with any actual persistent hidden mark included in  $X$ . Reports and actions are finite. A stationary common finite-register design consists of stochastic encoder rows and task-controller rows, independent of time; their product of simplexes is a compact set  $\Pi_S$ . A fixed finite task set is part of the row index  $j \in J$ , where  $J$  is compact metrizable. For each design and row, the joint physical–register transition has form

$$P_j^\pi = \eta N_j + (1 - \eta) Q_j^\pi, \quad 0 < \eta \leq 1, \quad N_j(x, m; \cdot) = \nu_j \otimes \delta_{m_0}. \quad (8.1)$$

The independent reset flag is physically supplied; it refreshes the physical preparation and sets the within-policy memory to  $m_0$ . Thus  $N_j$  is a constant kernel independent of  $\pi$ . Actions between resets may change the physical state. No domination of  $Q_j^\pi$  by one reference measure is assumed.

Let  $c_j^\pi : X \times [S] \rightarrow [0, 1]$  be the expected one-stage marked cost. We assume the following uniform continuity conditions, using the supremum over starting states for kernel total variation:

$$(j, \pi) \mapsto Q_j^\pi \text{ is continuous in } \sup_{x, m} \|\cdot\|_{\text{TV}}, \quad j \mapsto \nu_j \text{ in TV}, \quad (j, \pi) \mapsto c_j^\pi \text{ in } \|\cdot\|_\infty. \quad (8.2)$$

For fixed physical kernels with finite action and report alphabets, continuity in  $\pi$  follows directly by expanding the finite sums of encoder and controller probabilities; the physical kernels can be singular. Row dependence must independently satisfy (8.2). In particular changing a deterministic flow with the row does not automatically satisfy a total-variation continuity hypothesis.

**Theorem 8.1** (Nonsummable average-risk duality). *Under (8.1)–(8.2), every  $(j, \pi)$  has a unique invariant probability  $\zeta_j^\pi$ . Its average risk*

$$r_\pi(j) = \zeta_j^\pi c_j^\pi$$

*is the limit of expected Cesàro average cost from every initial law. For  $N \geq 1$  and  $0 < \beta < 1$  the following errors are uniform in  $j, \pi$ , and the initial law  $\xi$ :*

$$\left| \frac{1}{N} \sum_{t=0}^{N-1} \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1}{N\eta}, \quad (8.3)$$

$$\left| (1 - \beta) \sum_{t=0}^{\infty} \beta^t \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1 - \beta}{1 - \beta(1 - \eta)}. \quad (8.4)$$

*The map  $\pi \mapsto r_\pi$  is continuous into  $C(J)$  and has compact image. The attainable average-risk body under independent selection of a stationary design is compact and convex in  $C(J)$ . For continuous  $b : J \rightarrow \mathbb{R}$  its common-design minimax value satisfies*

$$\min_{\Lambda \in \mathcal{P}(\Pi_S)} \max_{j \in J} \left( \int r_\pi(j) d\Lambda(\pi) - b(j) \right) = \max_{w \in \mathcal{P}(J)} \min_{\pi \in \Pi_S} \int_J (r_\pi(j) - b(j)) dw(j), \quad (8.5)$$

and all displayed extrema are attained. For finite  $J$  a selector of at most  $|J| + 1$  values realizes every required optimum. More generally Proposition 2.5 gives an intrinsic finite-dimensional selector approximation.

*Proof.* A probability kernel contracts total variation. Subtracting two probabilities in (8.1) cancels the common reset part, giving

$$\|\xi P_j^\pi - \xi' P_j^\pi\|_{\text{TV}} \leq (1 - \eta) \|\xi - \xi'\|_{\text{TV}}.$$

The explicit probability

$$\zeta_j^\pi = \eta \sum_{k=0}^{\infty} (1 - \eta)^k (\nu_j \otimes \delta_{m_0}) (Q_j^\pi)^k \quad (8.6)$$

is invariant by direct substitution, with convergence in total variation. Contraction proves uniqueness and  $\|\xi(P_j^\pi)^t - \zeta_j^\pi\|_{\text{TV}} \leq (1 - \eta)^t$ . Integrating the bounded cost and summing the geometric series proves (8.3) and (8.4), including their uniformity.

For two parameter-policy pairs let  $P, P'$  denote their kernels and  $\zeta, \zeta'$  their invariant laws. Invariance, the preceding contraction, and a one-step difference give

$$\|\zeta - \zeta'\|_{\text{TV}} \leq (1 - \eta) \|\zeta - \zeta'\|_{\text{TV}} + \sup_{x, m} \|P(x, m; \cdot) - P'(x, m; \cdot)\|_{\text{TV}}.$$

Thus the invariant-law difference is at most the last supremum divided by  $\eta$ . Together with the cost difference, this proves joint continuity of  $(j, \pi) \mapsto r_\pi(j)$ . Compactness of  $J \times \Pi_S$  upgrades it to continuity of  $\pi \mapsto r_\pi$  in the uniform norm. Its image is compact.

Probabilities on compact  $\Pi_S$  form a weakly compact set. The barycenter map into  $C(J)$  is norm continuous: approximate the compact risk image uniformly by finitely many continuous coordinates, or use uniform equicontinuity of  $r_\pi(j)$  and a finite net in  $J$ . The image is therefore compact and convex. The payoff  $\int (r_\pi - b) d\Lambda dw$  is continuous and affine in both probabilities on compact spaces. Convex minimax gives (8.5). The inner minimum over a probability on  $\Pi_S$  equals the minimum over  $\pi$ , and compactness attains it. A finite-dimensional barycenter is reduced by the affine-dependence argument already proved in Theorem 2.2. This proves the finite selector assertion.  $\square$

*Remark 8.2* (Which policies are optimized). The theorem optimizes stationary finite-register common policies with the reset signature (8.1). It does not assert that a stationary policy is optimal among arbitrary nonstationary policies without that signature. A selector choosing a stationary policy is sampled once; its mode persists through resets while that mode's within-policy register is reset. If the selector must be private, store the mode and the register, paying their product. Conditioned on the mode, (8.3)–(8.4) hold with the same constants, so they also hold after averaging. The joint chain including a persistent mode need not have a unique invariant law across modes, and no such claim is needed.

**8.2. Cycle comparison before optimization.** Consider two reset experiments with the same reset probability and reset flag. A comparison within a fresh cycle starts with the specified physical preparation and initial register. Suppose, for every feedback design and every independent visible side seed, the joint laws of the marked information needed to evaluate the age- $k$  stage cost differ by at most  $\varepsilon_k \leq 1$ . Here age zero means the first stage following a reset; its law includes any reports used at that stage. The same control law and encoder rows are used on both sides. This is a uniform channel comparison, not a comparison obtained after separately optimizing policies.

**Theorem 8.3** (Regenerative resource transport). *The resulting paired average risks satisfy, for every consumer width,*

$$\|r_E - r_F\|_\infty \leq \mathfrak{e} := \eta \sum_{k=0}^{\infty} (1 - \eta)^k \varepsilon_k. \quad (8.7)$$

*The bound holds with a priced simulator by replacing the consumer width  $S$  by its product with the simulator width and by charging the selector as in Theorem 2.2. For a sequence of stateless two-sided comparisons with  $\varepsilon_{n,k} \rightarrow 0$  for each fixed  $k$ ,  $\mathfrak{e}_n \rightarrow 0$ , uniformly over all consumer widths. At every width, the corresponding average-risk bodies and optimized lower support values have the same error bound. The minimax values in (8.5) with the same centering  $b$  differ by at most  $\mathfrak{e}_n$ .*

*Proof.* Formula (8.6) expresses a stationary average risk as the geometric mixture of costs at each cycle age. The expectation of a  $[0, 1]$  marked stage cost changes by at most  $\varepsilon_k$  in that age's joint law, yielding (8.7). The proof is uniform in the consumer because its error hypothesis is uniform before integration. For a priced transducer, include its retained state in the joint register and reset it to its specified initial state at the cycle boundary. The same product construction and conditioning on independent mode choices applies.

The geometric weights sum to one. Given  $\delta > 0$ , first choose  $L$  so their tail beyond  $L$  is less than  $\delta/2$ , and then choose  $n$  so the finite weighted sum through  $L$  is less than  $\delta/2$ . This proves  $\mathfrak{e}_n \rightarrow 0$  without a rate uniform in  $k$ . Pairing each design with the same design on the other side gives the risk-body comparison. Taking weighted infima or the supremum of centered row risks proves the optimized statements; taking mixtures preserves the same bound.  $\square$

**Example 8.4** (A nondominated physical kernel). On  $X = [0, 1]$ , let  $Q^a(x, \cdot) = \delta_{(x+a) \bmod 1}$  for finitely many rotations  $a$ , and let  $\nu$  be Lebesgue probability. The transition  $\eta\nu + (1 - \eta)Q^a(x, \cdot)$ , as  $x$  ranges over  $X$ , is not dominated by a common  $\sigma$ -finite measure when  $\eta < 1$ : such a measure would have to give positive mass to uncountably many singleton points. A  $\sigma$ -finite measure has at most countably many positive-mass atoms, by decomposing into finite-measure sets and positive lower bounds on atomic mass. Finite report partitions and any common finite encoder fit Theorem 8.1. This is a nonsummable controlled risk theorem with a genuinely nondominated physical transition family, not an application of full-path density compactness.

*Remark 8.5* (Why the reset is explicit). A hidden physical process may mix while a retained controller bit never changes. Physical mixing alone then gives no uniform contraction on the joint physical–register space. Equation (8.1) requires an actual clearing of that register. Conversely, without a cycle tail or another uniform long-time condition, finite-prefix comparison errors need not control average cost: a sequence of processes that agrees with a zero-cost process for its first  $n$  stages and incurs unit cost forever afterward has vanishing error at each fixed prefix but average cost one. The positive theorem uses the explicit geometric cycle law to control this precise obstruction.

## 9. AN ACTIVE MICROSCOPIC REALIZATION

**9.1. Interventions and the unbounded flow.** Fix a nonempty finite system of  $N \geq 2$  identical hard spheres of positive diameter on a flat torus, with positive configuration volume. Let  $\Gamma_N$  be the admissible phase space and let  $\mu$  be the normalized measure with density proportional to  $\mathbf{1}_{\Gamma_N}(q, v) \exp(-b \sum_i |v_i|^2)$ ,  $b > 0$ . We use the classical almost-everywhere hard-sphere flow  $\Phi_t$ , which is an invertible measure-preserving flow off a null exceptional set; see [63] and the detailed microscopic construction retained in the companion. No claim about particle-number uniformity is needed. On  $H = L^2(\mu)$ ,

$$U(t)f = f \circ \Phi_t$$

is a strongly continuous unitary group, with skew-adjoint generator  $L$ . Indeed invariance gives isometry and invertibility; for bounded continuous phase functions, almost-everywhere continuity of the orbit at zero and dominated convergence give strong continuity. Density and isometry extend it to  $H$ . The generator domain includes the collision identifications of the flow; it is not the unrestricted formal transport derivative on arbitrary boundary data.

Let  $A$  be finite and choose orthogonal matrices  $O_a$  on velocity space. The intervention

$$R_a(q_1, \dots, q_N, v_1, \dots, v_N) = (q_1, \dots, q_N, O_a v_1, \dots, O_a v_N)$$

is an invertible  $\mu$ -preserving map. Put  $C_a f = f \circ R_a$ . For a fixed step length  $\Delta > 0$ , the controlled physical step is  $F_a = \Phi_\Delta \circ R_a$ , and its Koopman operator is

$$T_a = C_a U(\Delta). \quad (9.1)$$

The order matters. If both  $O_a = I$  and  $O_a = -I$  are allowed, the two steps give different subsequent positions on a positive-measure set of configurations with no collision in the next  $\Delta$  units of time and nonzero sufficiently small velocities. Such a set exists inside any nonempty interior admissible configuration region by choosing small velocities and then a positive interval away from zero. The action therefore changes the dynamics, rather than merely choosing which observable to read. We require no assertion that  $C_a D(L)$  is contained in  $D(L)$ .

For all finite control words, the flow and intervention compositions are well defined on a common full-measure subset: the preimage of a null exceptional flow set under each finite measure-preserving composition is null, and there are countably many finite words. Every preparation absolutely continuous with respect to  $\mu$  is supported there, including under a feedback choice of the word.

**9.2. A graph-core approximation and controlled words.** Choose an increasing family of finite-dimensional real subspaces  $V_n \subset D(L)$  whose union is a graph core, and let  $P_n$  be the orthogonal projections. Such a family can be constructed without postulating smooth collision-boundary coordinates. Start with a countable dense family  $(f_j)$  in  $H$  and a smooth compactly supported approximate identity  $\chi_\varepsilon$  on  $\mathbb{R}$ . Define

$$f_{j,\varepsilon} = \int_{\mathbb{R}} \chi_\varepsilon(s) U(s) f_j ds.$$

A difference quotient followed by translation of  $\chi_\varepsilon$  shows that  $f_{j,\varepsilon} \in D(L^k)$  for every  $k$  and

$$L^k f_{j,\varepsilon} = (-1)^k \int \chi_\varepsilon^{(k)}(s) U(s) f_j ds, \quad \|L^k f_{j,\varepsilon}\|_2 \leq \|\chi_\varepsilon^{(k)}\|_1 \|f_j\|_2. \quad (9.2)$$

For  $f \in D(L)$  its smoothed versions converge to  $f$  in graph norm: commuting  $L$  on its domain reduces this to strong continuity for  $f$  and  $Lf$ . At each fixed smoothing scale, approximation of  $f$  by the dense family controls both the function and its first derivative by (9.2). A countable sequence of scales and finite linear spans consequently gives a graph core. Enumerate those vectors and let  $V_n$  be their successive spans. This also shows that their collision compatibility comes from the actual flow domain, not from an imposed incompatible trace.

Set  $L_n = P_n L P_n$  on  $V_n$ ,  $U_n(t) = \exp(tL_n)$  on  $V_n$ , and, as an operator on  $H$  extended by zero on  $V_n^\perp$ , define

$$T_{a,n} = P_n C_a P_n U_n(\Delta) P_n. \quad (9.3)$$

These operators are contractions. Their finite-dimensional matrices need not be Markov matrices: they approximate *observables*; the observation experiment constructed below is nevertheless a genuine probability experiment.

**Lemma 9.1** (Graph-core and controlled-word convergence). *For every  $f \in H$  and  $T_0 < \infty$ ,*

$$\sup_{0 \leq t \leq T_0} \|U_n(t) P_n f - U(t) f\|_2 \longrightarrow 0. \quad (9.4)$$

*For every fixed word length  $k$ ,*

$$\max_{(a_1, \dots, a_k) \in A^k} \|T_{a_1, n} \cdots T_{a_k, n} f - T_{a_1} \cdots T_{a_k} f\|_2 \longrightarrow 0. \quad (9.5)$$

*Neither statement needs domain invariance under the interventions.*

*Proof.* The matrix  $L_n$  is skew-adjoint on  $V_n$  because  $L$  is skew-adjoint and  $V_n \subset D(L)$ . Thus its group is unitary and  $\|(z - L_n)^{-1}\| \leq 1/\Re z$  when  $\Re z > 0$ . If  $u \in V_m$  and  $n \geq m$ , then

$$(z - L_n)^{-1} P_n (z - L) u = u.$$

The set  $(z - L) \bigcup_m V_m$  is dense in  $H$ , since the union is a graph core and  $z - L$  maps  $D(L)$  onto  $H$ . The uniform resolvent bounds and  $P_n \rightarrow I$  strongly therefore give

$$(z - L_n)^{-1} P_n f \rightarrow (z - L)^{-1} f \quad (f \in H).$$

This is the graph-core consistency and stable-resolvent setting of Trotter's theorem [67, Theorem 5.2]. We give the contraction argument to specify the hypotheses used here. For any contraction generator  $A$  and  $u \in D(A)$ , the Euler resolvent formula can be written as a Gamma integral:

$$(I - tA/m)^{-m} u = \int_0^\infty e^{sA} u d\gamma_{m,t}(s), \quad \mathbb{E}_{\gamma_{m,t}} s = t, \quad \text{Var}_{\gamma_{m,t}} s = t^2/m.$$

This follows by multiplying the Laplace resolvent integrals and convolving their exponential densities. Since  $\|e^{sA} u - e^{tA} u\| \leq |s - t| \|Au\|$ , its error from  $e^{tA} u$  is at most  $t \|Au\|/\sqrt{m}$ . For  $u \in V_l$  one has  $\|L_n u\| \leq \|Lu\|$  for all  $n \geq l$ . At each fixed  $t > 0$  and Euler order  $m$ , the resolvent convergence already proved gives convergence of its  $m$ th power. First letting  $n \rightarrow \infty$  and then  $m \rightarrow \infty$  proves orbit convergence on the core. The orbit Lipschitz bound with constant  $\|Lu\|$ , uniform in  $n$ , upgrades it to uniformity on  $[0, T_0]$  by a finite time mesh. Density and contraction extend it to all  $f \in H$ . No derivative of  $U(t)f$  for  $f \notin D(L)$  was taken.

Now  $P_n C_a P_n \rightarrow C_a$  strongly by boundedness of  $C_a$  and strong convergence of  $P_n$ . Equation (9.4) gives  $T_{a,n} \rightarrow T_a$  strongly. For a fixed finite word, subtract the two products one factor at a time. Each difference acts on a fixed vector formed from the limiting factors, while every factor to its left is a contraction. Each term tends to zero. There are finitely many words of length  $k$ , which proves the maximum in (9.5). At no step was an expression such as  $LC_a f$  required.  $\square$

**9.3. Actual marked acquisition under feedback.** Fix real observables  $f_k \in H$  and noises  $\sigma_k > 0$ . At step  $k$ , a controller chooses  $a_k$  from its previously available reports and retained state, the action is applied to the physical state, and the new report is

$$Y_k = (T_{a_1} \cdots T_{a_k} f_k)(x_0) + \sigma_k \xi_k, \quad \xi_k \text{ independent standard Gaussians.} \quad (9.6)$$

The initial law is  $p_\theta d\mu$ , with  $\sup_\theta \|p_\theta\|_2 \leq D < \infty$ . An unobserved initial physical mark  $W$  is drawn jointly with  $x_0$  through a fixed channel  $\gamma(dw | x_0)$ , independently of the acquisition noises. Losses may use that actual mark and the observed control-report transcript. This specification does not allow an unproved replacement of a controlled terminal physical quantity by an independent variable of the same marginal law.

The approximation replaces the product in (9.6) by  $T_{a_1,n} \cdots T_{a_k,n} f_k$ . It has a finite-dimensional physical realization. In a real orthonormal basis  $(\varphi_1, \dots, \varphi_{d_n})$  of  $V_n$ , write the matrix of  $T_{a,n}$  as  $M_{a,n}$  and the coefficient vector of  $P_n f_k$  as  $c_{k,n}$ . The initial row vector is  $v_0 = (\varphi_1(x_0), \dots, \varphi_{d_n}(x_0))$ , and

$$v_k = v_{k-1} M_{a_k,n}, \quad Y_k^{(n)} = v_k c_{k,n} + \sigma_k \xi_k.$$

Initialize  $(v_0, W)$  with the actual push-forward joint law of  $(x_0, W)$ . This is a valid controlled probability model with a real vector state; no positivity of  $M_{a,n}$  is needed. Its real dimension  $d_n$  is not counted as  $d_n$  finite retained labels. The consumer register remains a separate constrained object. Measurable representatives of the finitely many functions may be chosen once; their null ambiguities have zero probability under the specified preparations.

**Theorem 9.2** (Active marked deficiency bound). *For each fixed horizon  $T$ , define*

$$\epsilon_{n,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \sum_{a \in A^k} \|T_{a_1} \cdots T_{a_k} f_k - T_{a_1,n} \cdots T_{a_k,n} f_k\|_2. \quad (9.7)$$

*The exact and approximate acquisitions have two-sided stateless marked causal deficiency at most  $\epsilon_{n,T}$ , using the identity report channel. The bound includes every independent visible seed in the joint comparison, holds uniformly over every finite-register consumer and feedback design, and tends to zero for each fixed  $T$ . Consequently all paired  $[0, 1]$  task risks and optimized lower support values have error at most  $\epsilon_{n,T}$  at unchanged widths; paired centered values have error at most  $2\epsilon_{n,T}$  when their unrestricted benchmarks are also paired.*

*Proof.* Couple the two experiments with the same  $x_0, W$  and every independent visible seed. Couple their private controller randomness so that the same available history produces the same action. At each step before the first report mismatch, use a maximal coupling of the two Gaussian reports. Gaussian densities satisfy

$$\|N(u, \sigma^2) - N(v, \sigma^2)\|_{\text{TV}} = 2\Phi(|u - v|/(2\sigma)) - 1 \leq \frac{|u - v|}{\sigma\sqrt{2\pi}}.$$

The equality follows by separating the densities at their midpoint; the inequality follows by bounding the standard normal density by  $1/\sqrt{2\pi}$ . Maximal couplings can be chosen measurably from these densities by their common part and residual parts.

For a fixed  $x_0$ , the probability of a first mismatch at time  $k$  is bounded by the sum, over the possible matched action words  $a \in A^k$ , of their matched-history probabilities times the displayed Gaussian bound for that word. Discarding these probabilities gives an upper bound by the sum over all words. Integrate under the *original* preparation  $p_\theta d\mu$ . Cauchy–Schwarz bounds the integral of each absolute mean difference by  $D$  times its  $L^2$  norm. Summing first-mismatch probabilities yields (9.7). In particular no bound on a feedback-conditioned posterior density has been assumed. When no mismatch occurs the entire transcript and actual mark, including the visible seed, coincide. The coupling inequality gives their joint total variation bound.

The same argument is symmetric in the two report models; the stateless identity channel is executable in both directions, so it bounds the intrinsic infima. Lemma 9.1 makes every term in the fixed finite sum tend to zero. The comparison precedes any consumer optimization, hence its bound does not depend on the consumer’s number of labels. The remaining conclusions are the channel and support inequalities in Theorem 2.2 and (2.18), which remain valid on these standard Borel carriers.  $\square$

**9.4. Active regenerative average decisions.** Take a fixed observable  $f$ , fixed noise  $\sigma > 0$ , and a fixed finite measurable report partition applied to each Gaussian acquisition. At each cycle boundary the physical system is re-prepared from  $p_\theta d\mu$  with its actual mark channel  $\gamma$ , and the within-policy register is reset to  $m_0$ . Between boundaries the orthogonal interventions and hard-sphere flow continue as above. The independent reset probability is  $\eta > 0$ . In the approximate system reset the real vector and mark by the same joint push-forward preparation. A stationary consumer uses the partitioned reports, actions, and its finite register. Its  $[0, 1]$  one-stage losses depend on the actual cycle mark, control, and current finite report; reset stage costs are fixed and identically paired. A finite task index is available to the controller but not to the common encoder.

Assume  $j \mapsto p_j$  is continuous in  $L^1(\mu)$  on a compact row set, the  $L^2$  bounds are uniform, and the bounded row losses vary uniformly continuously. The flow, intervention maps, observable, noise law, report partition, and mark channel are fixed across rows. Then the exact and every approximate model satisfy (8.2): only the reset law and losses vary with the row, and the dependence on finite policy rows is a finite polynomial expansion in kernels of norm one. The underlying physical transitions are allowed to be deterministic between resets.

**Corollary 9.3** (Active average-risk realization). *Both models have the compact average-risk bodies and attained duality of Theorem 8.1. At every finite consumer width their paired average risks,*

lower supports, and equally centered minimax values differ by at most

$$\epsilon_n = \eta \sum_{k=0}^{\infty} (1 - \eta)^k \epsilon_{n,k+1} \longrightarrow 0. \quad (9.8)$$

The convergence is uniform in the consumer width and in any independent priced mixture of stationary designs. It does not require a controlled word approximation rate uniform in word length.

*Proof.* A deterministic report partition contracts total variation. The age- $k$  stage is evaluated after at most  $k + 1$  controlled acquisitions, so Theorem 9.2 supplies the cycle comparison with  $\epsilon_{n,k} = \epsilon_{n,k+1}$ . Apply Theorem 8.3. Its geometric tail proof gives the limit although the finite-word bound in (9.7) can grow with  $k$ . The reset and row-continuity assertions just verified give Theorem 8.1. Conditioning on each persistent mode and charging its index proves the mixture statement.  $\square$

*Remark 9.4* (Fixed physical scope). The interventions above are active changes to microscopic dynamics, and the average theorem uses actual repeated physical preparation. Neither observation control alone nor an assumed posterior  $L^2$  bound is substituted for them. The assertions are for fixed  $N$ , sphere diameter,  $A$ ,  $\Delta$ , and positive noise, and for the declared initial-mark losses. No estimate is asserted uniform in particle number, a growing action alphabet, vanishing noise, or a kinetic scaling. The historical nonlinear kinetic corrector and the Sinai left-strip spectral problem remain different analytic statements. They are preserved in the development with their original hypotheses, not inferred from the present operator approximation.

**9.5. A posteriori bounds from graph and intervention residuals.** Strong controlled-word convergence supplies an asymptotic comparison. For a fixed approximation space one can also bound its actual comparison error using finite Gram data. We retain the physical setup of Theorem 9.2: the particle number, interventions, noise, step length, and actual initial-mark channel are fixed. These statements do not require the interventions to map  $D(L)$  into itself.

Suppress  $n$  and let  $V \subset D(L)$  have orthonormal basis  $\phi_1, \dots, \phi_d$ . Identify  $V$  with  $\mathbb{R}^d$  by an isometry  $\iota$ ; complexification gives the same formulas with adjoints. Write

$$P = \iota^* \iota, \quad J = \iota^* L \iota, \quad B_a = \iota^* C_a \iota, \quad M_a = B_a e^{\Delta J}.$$

Because  $L$  is skew-adjoint and  $C_a$  is unitary,  $J$  is skew-adjoint,  $e^{sJ}$  is unitary and  $M_a$  is a contraction. The two residual Gram matrices are

$$R = (L \iota)^* L \iota - J^* J = ((I - P) L \iota)^* ((I - P) L \iota), \quad (9.9)$$

$$S_a = I - B_a^* B_a = ((I - P) C_a \iota)^* ((I - P) C_a \iota). \quad (9.10)$$

In particular both are positive semidefinite. For a coefficient vector  $c$ , set

$$d_a(c) = \int_0^\Delta \sqrt{(e^{sJ} c)^* R (e^{sJ} c)} ds + \sqrt{(e^{\Delta J} c)^* S_a (e^{\Delta J} c)}. \quad (9.11)$$

All quantities in this formula are finite-dimensional. For a real observable  $f \in H$  put

$$c_f = \iota^* f, \quad \tau_f = (\|f\|_H^2 - \|c_f\|^2)^{1/2}.$$

If  $w = (a_1, \dots, a_k)$  and  $c_j = M_{a_{j+1}} \cdots M_{a_k} c_f$  (empty product equal to the identity), define

$$E_w(f) = \tau_f + \sum_{j=1}^k d_{a_j}(c_j). \quad (9.12)$$

The suffix order agrees with the Koopman product for a physically chronological word; its reversal would approximate a different dynamics.

**Theorem 9.5** (Residual control of the active marked experiment). *For every such space  $V$  and every controlled word,*

$$\|T_{a_1} \cdots T_{a_k} f - \iota M_{a_1} \cdots M_{a_k} c_f\|_H \leq E_w(f). \quad (9.13)$$

*For the preparations  $p_\theta$  with  $\sup_\theta \|p_\theta\|_2 \leq D$  and Gaussian acquisitions of Theorem 9.2, the two stateless marked feedback deficiencies are bounded by*

$$\hat{\epsilon}_{V,T} = \min \left\{ 1, \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left( \sum_{w \in A^k} E_w(f_k)^2 \right)^{1/2} \right\}. \quad (9.14)$$

*The comparison includes every independent exposed seed and is uniform over all finite consumer widths. Thus it bounds raw paired risks and lower supports at unchanged widths. When unrestricted benchmarks are also paired, the centered risk bound is  $2\hat{\epsilon}_{V,T}$ .*

*Proof.* The orthogonal splitting of  $L\iota c$  into its  $V$  component  $\iota Jc$  and its orthogonal component proves (9.9). Unitarity of  $C_a$  similarly proves (9.10). For  $v = \iota c$ , the path  $s \mapsto \iota e^{sJ} c$  lies in the finite space  $V \subset D(L)$  and is differentiable in the graph norm. Hence variation of constants is legitimate and gives

$$U(\Delta)v - \iota e^{\Delta J} c = \int_0^\Delta U(\Delta - s)(L\iota - \iota J)e^{sJ} c ds.$$

Taking norms and applying (9.9) bounds this difference by the integral in (9.11). Apply the unitary  $C_a$  and then add the orthogonal projection error  $\|(I - P)C_a \iota e^{\Delta J} c\|$  to obtain

$$\|T_a \iota c - \iota M_a c\| \leq d_a(c).$$

No derivative of  $C_a \iota c$  has occurred. First remove  $(I - P)f$  at cost  $\tau_f$ . Telescope the difference of the two word products with exact prefixes and approximate suffixes. All prefixes are isometries and all approximate suffixes are contractions. The  $j$ th term is bounded by  $d_{a_j}(c_j)$ , proving (9.13).

For the probabilistic assertion use the common initial  $x_0$ , its actual conditional mark  $W$ , and the same visible seed, as in Theorem 9.2. Before the first report mismatch the selected actions agree. Conditional on  $x_0$ , the weights of the matched histories leading to the possible words of length  $k$  sum to at most one. If  $e_w(x)$  is the corresponding exact-minus-approximate mean, the first-mismatch contribution at time  $k$  is therefore at most

$$\frac{1}{\sigma_k \sqrt{2\pi}} \int p_\theta(x) \max_{w \in A^k} |e_w(x)| d\mu(x).$$

This retains the history weights until after taking their maximum, instead of replacing each by one. Cauchy–Schwarz and  $\max_w |e_w|^2 \leq \sum_w |e_w|^2$  bound the integral by  $D(\sum_w \|e_w\|_2^2)^{1/2}$ . Apply (9.13) and sum the first-mismatch probabilities. The coupling is symmetric and preserves the whole actual marked transcript on its matching event, so it proves both stateless inequalities in (9.14). It conditions only on the original preparation, not on a feedback posterior with a supposed  $L^2$  bound. Conditioning on an independent exposed seed gives the same uniform estimate. Risk and support statements follow by composition with any consumer and by taking the appropriate infima.  $\square$

**Corollary 9.6** (Finite-data average-risk and presentation certificates). *In the active regenerative model of Corollary 9.3, a finite age cutoff  $H$  gives an error bound for paired average risks and lower supports, uniform in consumer width,*

$$\hat{\epsilon}_{V,H} = \eta \sum_{k=0}^H (1 - \eta)^k \hat{\epsilon}_{V,k+1} + (1 - \eta)^{H+1}. \quad (9.15)$$

*The same bound applies to identically centered values; separately paired unrestricted benchmarks give twice the bound. Finite-horizon bounds (9.14), combined with any other verified stateless presentation bridges, can be inserted into (2.27) without changing the intrinsic resource budget.*

*Proof.* At age  $k$  there are at most  $k+1$  acquisitions. Use Theorem 9.5 at those horizons and bound each remaining age error by one. The remaining geometric mass is exactly  $(1-\eta)^{H+1}$ . The cycle formula in Theorem 8.3 proves (9.15). The final assertion is Theorem 2.9. A finite-dimensional vector model is not automatically a finite report experiment; a further finite-report bridge must actually be proved before invoking the rational local testing algorithm. No such bridge is implicit in a matrix representation.  $\square$

The data required in (9.9)–(9.12) are inner products of  $f, \phi_i, L\phi_i, C_a\phi_i$ . Once these data have rigorous enclosures, the formulas give a posteriori bounds without knowing the exact controlled orbit. For example, certified numbers  $r_0 \geq \|R\|$  and  $s_{a,0} \geq \|S_a\|$  imply the simpler bound

$$d_a(c) \leq (\Delta\sqrt{r_0} + \sqrt{s_{a,0}})\|c\|.$$

With matrix-entry enclosures, interval arithmetic and this inequality avoid an uncertified numerical integral. Floating-point Gram entries alone are not an analytic certificate. Nor is it asserted that these residual majorants converge for every graph-core sequence: cancellation may make the actual orbit error small while its residual bound is large. The convergence theorem remains Lemma 9.1; Theorem 9.5 is a complementary finite-space bound. This distinction separates a proved limiting approximation from a checkable error estimate for a particular space.

## 10. FINITE RATIONAL PRESENTATIONS OF GAUSSIAN ACQUISITIONS

We now join finite resource certification to the actual marked physical experiment. The intermediate model is allowed to have a nonfinite latent state. Its reports, not merely its observable coordinates, must eventually be finite. Throughout this section the parameter, action and mark alphabets are finite, the horizon is finite, and the mark is drawn with the initial state. An initial report, when present, is treated as one additional acquisition. All channels below are parameter blind and task blind.

**10.1. A stateless report channel.** Let  $(X, \mu)$  be a probability space, let the preparations be  $p_\theta\mu$  with  $\|p_\theta\|_2 \leq D$ , and let  $\gamma(dw | x)$  be the actual initial-mark channel. For each action word  $w \in A^k$ , let  $m_{k,w} \in L^2(\mu)$  be a real mean function. The report at time  $k$  is

$$Y_k = m_{k,a_{1:k}}(x) + \sigma_k \xi_k, \quad \sigma_k > 0,$$

where the noises are independent standard Gaussians, independent of  $(x, W)$ . Actions may be selected from all previously emitted reports. The finite-dimensional models in the preceding section have this form; so do the enriched approximations in Section 11. Put

$$B_k = \left( \sum_{w \in A^k} \|m_{k,w}\|_2^2 \right)^{1/2}.$$

Choose  $R > 0$  and a finite interval partition of  $[-R, R]$  with maximum cell length  $h$ . Adjoin the two tail cells. Let  $Q$  return the cell of a real report. Let  $H$  reconstruct an interior cell uniformly and reconstruct the tail cells at  $-R$  and  $R$ , respectively. The endpoints may be assigned arbitrarily to adjacent cells. These maps use no persistent memory, seed, parameter or mark. Fresh randomness in  $H$  is transient. Denote by  $F^{R,h}$  the physical acquisition with the report  $Q(Y_k)$ ; its latent state and actual mark are unchanged.

**Lemma 10.1** (Uniform marked report quantization). *The channel  $Q$  simulates  $F^{R,h}$  from  $F$  exactly. The reverse channel  $H$  has marked feedback error at most*

$$\beta(R, h) = 1 \wedge \sum_{k=1}^T \left\{ \frac{h}{\sigma_k \sqrt{2\pi}} + \frac{2DB_k}{R} + \frac{4\sigma_k^2}{R^2} \right\}. \quad (10.1)$$

*The comparison includes the joint law of every independent exposed selector and is uniform over every finite consumer width. In particular  $h \downarrow 0$  and  $R \uparrow \infty$  give two-sided stateless marked comparison with error tending to zero.*

*Proof.* For an absolutely continuous density  $g$  and an interval  $I$  of length at most  $h$ , write  $g_I = |I|^{-1} \int_I g$ . The fundamental theorem of calculus and Fubini give

$$\int_I |g - g_I| \leq |I|^{-1} \int_{I^2} |g(u) - g(v)| du dv \leq h \int_I |g'|.$$

Consequently the variation error of the interior histogram is at most  $\frac{h}{2} \int |g'|$ . For a Gaussian of variance  $\sigma^2$  the latter integral is  $\sqrt{2/\pi}/\sigma$ . Replacement of the tails costs at most their total probability, even though the reconstruction is atomic. For any mean  $m$ , this probability is at most

$$\mathbf{1}_{\{|m| > R/2\}} + \mathbb{P}\{|\sigma\xi| > R/2\} \leq \frac{2|m|}{R} + \frac{4\sigma^2}{R^2}.$$

Thus the conditional report error, given the original  $x$  and the selected action word, has the three terms in (10.1) before integration.

Couple the exact and reconstructed systems with the same original  $(x, W)$  and independent selector. Before their first emitted-report mismatch use the same controller randomness and maximally couple the next reports. Matched-history weights of different words at time  $k$  sum to at most one. Therefore the mean term is bounded by

$$\frac{2}{R} \int p_\theta(x) \max_{w \in A^k} |m_{k,w}(x)| d\mu(x) \leq \frac{2D}{R} \left( \sum_w \|m_{k,w}\|_2^2 \right)^{1/2}.$$

This uses the original preparation, not an assumed bound for a feedback-conditioned posterior. Sum the first-mismatch bounds. On the matching event the complete emitted transcript, controls, mark and selector agree. This proves the reverse comparison. Applying  $Q$  to each original report is the exact forward simulation, also under feedback. Conditioning on the exposed selector throughout proves the joint assertion. Composition never enlarges a retained register.  $\square$

The bound is deliberately an elementary certified bound; Gaussian tail integrals or higher moments may improve it. No uniform bound on the pointwise means is needed. The quantity  $B_k$  can grow with the number of action words. Uniformity in memory width is not uniformity in the size of the finite instrument being constructed.

**10.2. Rationalization without a small-prefix denominator.** Let  $G$  now be a finite-report acquisition of this form. For each  $\theta$ , write

$$\pi_\theta(w) = \mathbb{P}_\theta(W = w), \quad p_\theta(w, h; y | a) = \mathbb{P}_\theta(W = w, Y_{1:k-1} = h_y, Y_k = y | a_{1:k-1} = h_a, a_k = a)$$

for its joint marked prefix masses under the indicated open-loop control word. Here  $h = (h_y, h_a)$ , and the conditioning on controls means intervention, not conditioning on the probability of choosing a control. Causality and the initial-mark convention imply

$$\sum_y p_\theta(w, h; y | a) = p_\theta(w, h)$$

independently of the current action  $a$ . A parameterized latent history-tree instrument may use these masses; its latent dependence on  $\theta$  is part of the experiment, not information given to a simulator.

Suppose nonnegative rational arrays  $t_\theta(w)$  and  $t_\theta(w, h; y | a)$  approximate the indicated joint masses. Normalize  $t_\theta(w)$  to obtain a rational initial-mark law. At each node  $(w, h, a)$  normalize its vector of children to obtain a rational report kernel. If a normalizing sum is zero, use a fixed rational probability vector, for example the uniform vector. This defines a coherent finite rational

marked causal experiment  $G_{\mathbb{Q}}$ . Its prefix probabilities need not equal the originally rounded joint masses; they are the probabilities generated by these normalized kernels. That distinction is essential.

**Lemma 10.2** (Weighted normalization). *If  $u, v \in [0, \infty)^b$ ,  $s = \sum u_j$ , and  $\bar{u} = u/s$  when  $s > 0$ , then, for  $\bar{v} = v/\sum v_j$  with an arbitrary probability vector chosen when its denominator vanishes,*

$$s\|\bar{u} - \bar{v}\|_{\text{TV}} \leq \sum_j |u_j - v_j|. \quad (10.2)$$

For  $s = 0$  the left side is defined to be zero.

*Proof.* If both sums are positive, the triangle inequality gives

$$\sum_j |u_j - s\bar{v}_j| \leq \sum_j |u_j - v_j| + \left| \sum_j v_j - s \right| \leq 2 \sum_j |u_j - v_j|.$$

If  $\sum v_j = 0$ , the left side of (10.2) is at most  $s = \sum |u_j - v_j|$ . The remaining case is immediate.  $\square$

**Theorem 10.3** (Rational causal tree bridge). *The identity report channels compare  $G$  and  $G_{\mathbb{Q}}$  in both directions, including the actual marks, with feedback error at most*

$$\rho = 1 \wedge \max_{\theta} \left[ \sum_w |\pi_{\theta}(w) - t_{\theta}(w)| + \sum_{k=1}^T \sum_{w,h,a,y} |p_{\theta}(w, h; y | a) - t_{\theta}(w, h; y | a)| \right]. \quad (10.3)$$

Here the finite sum includes all open-loop histories at the specified time, not only the histories reached by a particular controller. The bound is independent of simulator and consumer memory. It remains valid with a visible selector and its joint reference law.

If every joint mass has computable rational enclosures of arbitrarily small width, the construction is effective and produces  $\rho \rightarrow 0$ . No lower bound for any nonzero prefix probability, and no decision whether a prefix has exactly zero probability, is required.

*Proof.* Couple the initial marks by their normalized laws. Lemma 10.2 bounds the initial failure probability by the first sum in (10.3). Use the same controller and selector on both sides. Conditional on a matched marked history, maximally couple the next reports. Under the original instrument, the probability of reaching a particular marked history under any deterministic controller is either zero or the corresponding open-loop prefix mass; for a randomized controller it is at most that mass. The probability of reaching it with the coupling still matched is smaller still. Thus its first-mismatch contribution is bounded by the original prefix mass times the conditional variation distance. Applying (10.2) at that node bounds this product by the sum of joint child-mass errors. Sum over all nodes and times. The coupling inequality gives the asserted joint variation distance. The estimate is symmetric as a comparison of the two emitted laws; identity channels require no state in either direction.

For effectiveness, take a nonnegative rational lower endpoint of a certified enclosure for every mass, clipping a negative lower endpoint at zero. If there are  $M$  mass entries for a parameter and every interval has width at most  $\eta/M$ , the displayed bound is at most  $\eta$ . Normalization uses only rational arithmetic. A zero rounded sum invokes the declared fallback. The weighted bound, rather than a bound on a ratio at an individual rare history, proves convergence. Conditioning on an independent visible selector leaves every bound unchanged. The mark is part of the finite tree and is never redrawn independently of its reports.  $\square$

For the Gaussian acquisition its required masses are explicitly

$$\int_X p_\theta(x) \gamma(w \mid x) \prod_{j=1}^k \mathbb{P}\{m_{j,a_{1,j}}(x) + \sigma_j \xi_j \in I_{y_j}\} d\mu(x). \quad (10.4)$$

This is an integration problem for a specified preparation and trial family. The theorem does not infer its effective solvability from the mere words “standard Borel”. Rigorous enclosures for (10.4) are part of the effective data specification in Theorem 13.1; explicit physical families are treated in Section 14.

## 11. ENRICHED DEFECT CERTIFICATES AND THEIR CONVERGENCE

A small error of a Galerkin orbit need not force a chosen residual majorant to be small. We replace that implication by an estimator whose own convergence is proved. The approximation is allowed to be a finite tree of trial observables rather than powers of one compressed matrix. This enrichment changes the numerical presentation, not the physical experiment or the charged consumer register.

Let  $H$  be a separable real Hilbert space,  $U(t)$  a strongly continuous orthogonal group with generator  $L$ , and  $C_a$  finitely many orthogonal operators. Write  $T_a = C_a U(\Delta)$ , with fixed rational  $\Delta > 0$ . Let  $\mathcal{D}$  be a countable rational vector space in  $D(L)$  whose real span is graph dense in  $D(L)$ , and suppose its rational span is also graph dense. It suffices to start with a countable graph-core family and all rational linear combinations.

**Definition 11.1** (Effective graph data). The data are effective if vectors of  $\mathcal{D}$  can be enumerated and rational intervals of any requested width can be computed for inner products among finitely many vectors of the forms

$$v, \quad Lv, \quad C_a v, \quad f_1, \dots, f_T, \quad v \in \mathcal{D}.$$

Only inner products actually used in the following norm expressions are necessary. The initial observables have specified  $H$  names. For physical probability certification we require, in addition, computable enclosures for (10.4) for finite trial families selected by this procedure and certified preparation bounds. These are finite analytic input operations; no deficiency optimizer or exact controlled orbit is an input operation.

For nodes  $v_0, \dots, v_l \in \mathcal{D}$  at rational times  $0 = s_0 < \dots < s_l = \Delta$ , let  $z(s)$  be their piecewise affine interpolant, and set

$$\begin{aligned} \mathcal{D}(z) &= \frac{1}{2} \sum_{j=0}^{l-1} (s_{j+1} - s_j) (\|d_j - Lv_j\| + \|d_j - Lv_{j+1}\|), \\ d_j &= (v_{j+1} - v_j) / (s_{j+1} - s_j). \end{aligned} \quad (11.1)$$

The interpolation lies in  $D(L)$  at every time and is absolutely continuous in its graph norm. For a trial input  $g \in \mathcal{D}$  and trial output  $v \in \mathcal{D}$ , define the one-step certificate

$$\mathcal{E}_a(g, z, v) = \|g - z(0)\| + \mathcal{D}(z) + \|C_a z(\Delta) - v\|. \quad (11.2)$$

No derivative of an intervened vector appears.

**Lemma 11.2** (Certified path defect). *For every such triple,*

$$\|T_a g - v\| \leq \mathcal{E}_a(g, z, v).$$

*Given  $g \in \mathcal{D}$ , an action  $a$ , and  $\eta > 0$ , there are rational nodes with  $z(0) = g$  and a rational-core output  $v$  such that  $\mathcal{E}_a(g, z, v) < \eta$ .*

*Proof.* On each interpolation interval, variation of constants gives

$$U(\Delta)z(0) - z(\Delta) = \int_0^\Delta U(\Delta - s)(Lz(s) - z'(s)) ds.$$

The identity follows first on each graph-differentiable interval and then by concatenation. The integrand's norm is bounded by the affine interpolation of its endpoint norms, since  $z'$  is constant on that interval and  $Lz$  is affine. Thus its integral is at most  $\mathcal{D}(z)$ . Orthogonality of  $C_a$  and the two endpoint errors give (11.2).

For existence put  $u(s) = U(s)g$ . It is continuously differentiable in  $H$ , continuous in the graph norm, and  $u' = Lu$ . Let a uniform rational mesh have size  $h$ . Its exact-node interpolant has derivative error bounded by the modulus of continuity of  $u'$  at scale  $h$ , and its  $L$ -image differs from  $Lu(s)$  by at most the modulus of continuity of  $Lu$  at that scale. Both moduli tend to zero. Approximate each noninitial node by a member of  $\mathcal{D}$  within  $h^2$  in graph norm; keep the initial node equal to  $g$ . The change in the interpolant derivative is at most  $2h$  and the change in its  $L$ -image at most  $h^2$ . The endpoint norms in (11.1) obey these same modulus bounds, so  $\mathcal{D}(z) \rightarrow 0$ , not only the orbit error. Finally use density of  $\mathcal{D}$  in  $H$  to approximate  $C_a z(\Delta)$  by  $v$ . No assertion that  $C_a z(\Delta) \in D(L)$  is needed.  $\square$

**Theorem 11.3** (Effective enriched controlled approximation). *Under the effective graph-data hypothesis there is an algorithm which, for a fixed finite horizon and rational tolerance  $\eta > 0$ , returns finite-core mean functions  $\hat{m}_{k,w}$  and rational upper bounds  $e_{k,w}$  such that*

$$\|T_{a_1} \cdots T_{a_k} f_k - \hat{m}_{k,w}\| \leq e_{k,w} < \eta, \quad w = (a_1, \dots, a_k), \quad 1 \leq k \leq T. \quad (11.3)$$

*Each bound is a sum of an initial approximation error and certified one-step path defects. The algorithm terminates; no rate or polynomial complexity is asserted. Its certified physical marked error is*

$$\alpha = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left( \sum_{w \in A^k} e_{k,w}^2 \right)^{1/2}, \quad (11.4)$$

*and tends to zero along requested tolerances tending to zero. The source and approximation are compared by stateless identity report channels, including all independent exposed selectors.*

*Proof.* Enumerate  $\mathcal{D}$  and increasingly accurate rational norm upper bounds until a trial vector for  $f_k$  has error strictly below a prescribed small tolerance. Density ensures termination. For each step enumerate all finite rational time partitions and all rational core nodes and outputs, with initial node the trial input. Dovetail these candidates and interval precisions. Lemma 11.2 provides a candidate whose true defect is strictly smaller than the requested tolerance, so eventually its rational upper enclosure is also smaller. This is a terminating semidecision search with a proved strict witness. It does not test an undecidable equality of real numbers.

For each final acquisition  $k$ , build a suffix tree starting at the trial  $f_k$ . At the next level approximate  $T_a$  applied to each current trial vector using the preceding search. After  $k$  levels, the chronological word  $a_1, \dots, a_k$  corresponds to applying  $a_k$  first to the observable and  $a_1$  last. This gives exactly  $T_{a_1} \cdots T_{a_k} f_k$ , not its reversed physical dynamics. Orthogonality propagates the accumulated error additively. Allocate less than  $\eta/(2(k+1))$  to the initial approximation and each step, and refine finite norm enclosures until their rational sum is below  $\eta$ . There are finitely many nodes and every search terminates.

All trial vectors used at finitely many nodes span a finite-dimensional subspace of  $H$ . Initialize its evaluation vector jointly with the actual mark and use the stored action-word table to read the appropriate linear functional at each time. This is a causal probability model. Its physical lookup table is not declared to be a finite consumer register, and its coefficient matrices are not called Markov matrices. The first-mismatch proof of Theorem 9.5, with the verified errors in (11.3), gives

(11.4). Every term tends to zero at fixed horizon. All numerical square roots and constants in an implementation are replaced by rational upper enclosures; their extra tolerances may also be made arbitrarily small.  $\square$

*Remark 11.4* (Existence, effectiveness, and the old Galerkin estimator). Graph-core density alone proves existence of arbitrarily small enriched defects. Effective graph data make the search an algorithm. Neither assertion says that the Galerkin majorants (9.12) converge along every graph-core sequence. Those estimates remain valid for every selected space and are kept in the article. The new estimator has a different convergence proof: its piecewise affine residual itself is made small. Effective integration of the resulting finite report probabilities is an additional operation, explicit in Definition 11.1, and is not inferred from Hilbert-space density.

## 12. EFFECTIVE GRAPH DATA FROM REGULAR COLLISION CHARTS

The effective-data hypothesis can be discharged for a class of genuine colliding systems; it need not remain an oracle for their exact flow. Here the particle number is fixed. No bound uniform in particle number, number of collisions, or kinetic scaling is used or concluded.

Fix  $N \geq 2$ , a flat square or cubic torus of positive rational side length, and a positive rational sphere diameter smaller than half that length. Assume the allowed configuration region has nonempty interior. Let  $\mu$  be the normalized equilibrium law with independent standard Gaussian velocity coordinates, and let the finite intervention family consist of algebraic orthogonal transformations applied identically to every particle velocity. Flight times are rational. Noise scales are computable positive reals with supplied rational positive lower bounds. We use the classical almost-everywhere regular hard-sphere flow: on each bounded time interval almost every phase point has a finite itinerary of simple, nongrazing binary collisions, and the flow preserves equilibrium measure [63]. The following effective statements are consequences of that flow theorem and the explicit collision formulas, rather than additional collision-count estimates.

A bounded observable or preparation is called elementary-computable here if it has a supplied rational bound and admits uniformly computable rational interval enclosures on compact rational boxes of continuity, with mesh errors tending to zero. We allow piecewise descriptions whose exceptional boundaries are a specified finite union of nonzero polynomial zero sets. Globally computable continuous functions with known local moduli, and bounded piecewise polynomial functions, are examples. Preparations are nonnegative normalized densities of this class. A finite actual-mark channel has elementary-computable probabilities  $\gamma(w \mid x)$  summing to one. These specifications exclude arbitrary noncomputable measurable inputs.

**Lemma 12.1** (Effective full-measure chart exhaustion). *For any fixed finite collection of controlled words and bounded time windows, regular collision charts can be enumerated with rational box certificates. Their union has full equilibrium measure. Integrals of bounded functions which have computable interval enclosures on these charts have rational enclosures of arbitrarily small width, provided a uniform bound for the integrand is supplied. The assertion also holds with finitely many compact time-integration variables.*

*Proof.* On a velocity-cutoff box and a bounded time window, energy conservation gives a computable speed bound. Only finitely many periodic images can participate in a collision; enumerate all images within that displacement bound. The candidate time for a pair is a root of

$$|q_i - q_j + n\ell + t(v_i - v_j)|^2 = d_0^2,$$

where  $\ell$  is the torus side and  $d_0$  the diameter. On a simple nongrazing itinerary, the relevant discriminant is positive, the selected future root is separated from every competing root, and the normal relative velocity does not vanish. Reflection of the velocities is an explicit algebraic function of the positions and velocities at that root. An already realized contact has its zero

root removed; the outgoing sign specifies the next free flight. In unfolded coordinates there is no coordinate-wrap event to treat as a physical collision. A pair with zero relative velocity is handled by a strict positive separation bound on its whole flight interval, not by division by a vanishing quadratic coefficient. The same no-contact interval test handles sufficiently small perturbations when no root is present.

Enumerate finite collision lists, rational velocity and position boxes, and rational strict margins. Interval arithmetic for algebraic operations and square roots verifies the selected roots, their order, nongrazing margins, and absence of another collision before the final time. On sufficiently small boxes about any regular itinerary with a noncollision final time, all these strict inequalities hold. This is because there are finitely many collisions and every relevant gap and nongrazing angle is positive there. Thus enumeration produces a neighborhood of every such point. Endpoint-collision sets have measure zero. Reverse time uses the same construction. Algebraic kicks have computable inverses and preserve  $\mu$ , so the finite intersections needed for controlled words and transformed chart inputs again have full measure. This proves effective enumerability and exhaustion; it does not assert a computable uniform collision-count bound.

The equilibrium measure itself is computable on boxes. The configuration normalizer is the volume of a bounded region defined by finitely many algebraic separation inequalities, with boundary of Lebesgue measure zero. Inside and outside rational-box approximations converge to its volume. A rational interior box provides a strictly positive lower bound for the normalizer; it can be found by enumeration because the interior is nonempty. Velocity-box masses are products of computable Gaussian interval probabilities. Gaussian tails provide effective cutoffs. These facts give rational upper and lower enclosures for  $\mu$  and for the mass of every box used in the construction.

Replace finitely many accepted boxes by a disjoint dyadic refinement. Continue enumeration and refinement until the certified lower mass of their union exceeds  $1 - \varepsilon$ . This search terminates since the countable union has full measure. On the accepted compact boxes use interval enclosures and subdivision to bound the integrand's integral; on the complement use its known absolute bound  $M$ , at cost at most  $M\varepsilon$ . Refining the mass and integration enclosures gives any specified absolute error. Exceptional input-function boundaries are also null and can be excluded by strict interval tests for their nonzero defining polynomials. Compact time variables are handled by product-box integration; their collision-time sets are null by the finite-itinerary property and Fubini's theorem. At no point is an uncertified sample estimate for the omitted set used.  $\square$

Let  $(\psi_j)$  be a countable uniformly described dense family of bounded, compact-velocity-supported continuous piecewise polynomial functions on phase space. For instance, use rational tent functions on successively finer coordinate boxes. Restrictions of this family are dense in  $L^2(\mu)$ . For rational  $\delta > 0$  set

$$\kappa_\delta(s) = \frac{15}{16\delta}(1 - s^2/\delta^2)^2 \mathbf{1}_{[-\delta, \delta]}(s), \quad v_{j,\delta} = \int \kappa_\delta(s) U(s) \psi_j ds.$$

Let  $\mathcal{D}$  be all rational linear combinations of these vectors.

**Theorem 12.2** (Constructive microscopic graph data). *For the fixed-particle equilibrium system just specified,  $\mathcal{D}$  is an enumerable graph core for  $L$ . Each vector has the exact representation*

$$Lv_{j,\delta} = - \int \kappa'_\delta(s) U(s) \psi_j ds. \quad (12.1)$$

*All inner products required in Definition 11.1, for bounded elementary-computable acquisition observables, have arbitrarily accurate rational enclosures. The marked Gaussian prefix masses of every finite trial family in  $\mathcal{D}$  also have such enclosures. Thus the complete effective-data hypothesis of Theorem 13.1 holds in this class, including its probability integration clause.*

*Proof.* The kernel is continuously differentiable, compactly supported, has integral one, and vanishes at its support endpoints. The strong group identity and integration by parts prove domain membership and (12.1). If  $u \in D(L)$ , its mollified orbit converges to  $u$  in graph norm as  $\delta \downarrow 0$ . At fixed  $\delta$ , approximating  $u$  in  $H$  by rational combinations of the  $\psi_j$  approximates its mollification in graph norm, since

$$\|A_\delta(u - \psi)\| + \|LA_\delta(u - \psi)\| \leq (\|\kappa_\delta\|_1 + \|\kappa'_\delta\|_1)\|u - \psi\|.$$

A diagonal choice proves graph density. No decision of linear independence between formal core vectors is needed by the enriched search.

For inner products insert the time-integral representations on each side and use Fubini. The resulting integrands are bounded products of known kernels and of observables evaluated along finitely many exact collision itineraries, possibly after an algebraic kick. Known sup-norm bounds for  $\psi_j$ ,  $\kappa_\delta$  and  $\kappa'_\delta$  bound these products. Lemma 12.1 therefore computes their integrals, including cross terms involving  $C_a v$  and the given bounded  $f_k$ . The signs of  $\kappa'$  present no difficulty because absolute bounds are available.

For completeness, the time-integral value of a core function itself admits uniform enclosures on each compact regular itinerary chart. Enclose all collision times there. Away from small strips containing those times, the algebraic flow formulas and input observables have uniform interval approximations. The strips contribute at most their total length times the known kernel and observable bounds. Strict itinerary margins allow these strips to shrink under subdivision. Thus both  $v(x)$  and  $Lv(x)$  are computable continuous functions on that chart, even if a velocity-dependent  $\psi_j$  jumps at a collision. The integral across its moving jump is controlled by the strip bound, not by pretending the jump is absent.

Every trial mean is a finite rational combination of such bounded core functions. Gaussian cell probabilities are computable continuous functions of that mean, including the two tails, and their derivatives are bounded by the positive noise scales. The integrand in (10.4), multiplied by a bounded preparation and actual-mark probability, consequently satisfies the hypotheses of Lemma 12.1. The lemma computes its mass enclosures. It also supplies  $L^2$  preparation and trial-norm enclosures. This verifies all finite operations required by Definition 11.1. The exhaustive searches may have no useful uniform rate, but each requested enclosure terminates by the full-measure exhaustion proved above.  $\square$

**Corollary 12.3** (Certified approximation for a colliding physical class). *For two systems in Theorem 12.2 with a common finite control interface, finite parameter and mark alphabets, bounded elementary-computable preparations and observables, rational positive noise variances and rational observation times, the procedure of Theorem 13.1 produces convergent same-budget intervals for their physical intrinsic deficiency. It uses certified collision charts rather than an exact-flow integration oracle. It does not require the acquisition observables to be conserved quantities.*

*Proof.* The theorem supplies the effective graph data and marked prefix integrals; Theorem 11.3 supplies convergent analytic certificates; the finite report and rational-tree bridges preserve the resource signature. Apply Theorem 13.1. One may work in the full labeled  $L^2$  space. For symmetric input data the construction also restricts to its symmetric subspace by symmetrizing the tent family; the flow and identical-particle kicks commute with that finite symmetrization.  $\square$

The use of a null exceptional set is constructive here because its regular complement has an effective increasing chart enumeration and because the probability of each accepted finite cover is certified. A bare assertion of almost-everywhere computability without those two properties would not justify the conclusion. Conversely, the argument is an exhaustion algorithm, not a practical complexity bound or a uniform estimate in the Boltzmann–Grad regime.

## 13. PHYSICAL INTRINSIC RESOURCE CERTIFICATION

We can now state a single conclusion joining the analytic and finite parts of the theory. The original experiments, not their approximations, are the objects whose intrinsic deficiency is bounded. Fix one of the private, hidden-selector, or visible-selector signatures, denoted  $b$ ; its state widths and, where relevant, selector cardinality are fixed. Write  $\delta_b(E, F)$  for the corresponding intrinsic deficiency, with source  $E$  simulating target  $F$ . The marked feedback comparison always uses the actual initial mark. For a visible selector its reference law is the joint law specified in Section 2.

**Theorem 13.1** (Same-budget physical certification). *Let  $E, F$  be finite-horizon active Gaussian acquisitions on Hilbert probability models of the form considered above, with finite parameter, action and actual-mark alphabets, positive noise variances, and certified  $L^2$  preparation bounds. For each model choose a finite trial presentation with rigorous marked error  $\alpha_E, \alpha_F$ , supplied either by Theorem 9.5 or by Theorem 11.3. Choose finite report channels as in Lemma 10.1 and rational marked trees as in Theorem 10.3, with certified errors  $\beta_E, \beta_F$  and  $\rho_E, \rho_F$ . Put*

$$a = 1 \wedge (\alpha_E + \beta_E + \rho_E), \quad c = 1 \wedge (\alpha_F + \beta_F + \rho_F).$$

*Let  $[L, U]$  be any verified local-certificate interval for  $\delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})$  at the original signature  $b$ . Then the original physical deficiency belongs to*

$$[\max\{0, L - a - c\}, \min\{1, U + a + c\}]. \quad (13.1)$$

*An upper witness is an executable simulator of the original physical experiment with exactly the declared persistent-state and selector budget. A strict lower endpoint is an impossibility certificate at that same budget; it does not refer to a convexified private feasible set.*

*For the effective assertion, the noise scales are computable reals with supplied rational positive lower bounds. Under effective graph data, including the marked cell-integral operation in Definition 11.1, there is a terminating procedure producing intervals of the form (13.1) of arbitrarily small prescribed width. Their nested intersections converge to  $\delta_b(E, F)$ . The procedure uses only finite rational certificates and certified analytic input operations. It does not assume attainment of the original nonfinite optimization.*

*Proof.* Each of the three presentation changes is implemented by a stateless, seed-free, parameter-blind and task-blind report channel. The physical trial change uses the identity report channel; the quantization change uses  $Q$  in one direction and transiently randomized  $H$  in the other; the rationalization change uses the identity finite-report channel. Their compositions give two-sided marked feedback comparisons  $E \leftrightarrow E_{\mathbb{Q}}$  with error  $a$  and  $F \leftrightarrow F_{\mathbb{Q}}$  with error  $c$ . Fresh per-report randomness in a presentation channel is integrated into a simulator row; it is not a persistent public or hidden mode. All comparisons include the joint reference of a visible selector.

For an executable finite upper witness, first pass a report of  $E$  through the forward presentation, apply the finite simulator, and pass its output through the reverse presentation for  $F$ . The stored state and persistent selector are exactly those of the finite witness. The marked feedback triangle inequality bounds its error by  $U + a + c$ . Conversely, any physical simulator with error smaller than  $L - a - c$  could be composed with the opposite stateless presentations to give a finite simulator of the same signature with error smaller than  $L$ . This contradicts the verified finite lower certificate. Equivalently, Theorem 2.9 gives

$$|\delta_b(E, F) - \delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})| \leq a + c.$$

Taking infima in this argument requires no nonfinite minimizer.

For effectiveness, first choose enriched tolerances so that the two rational upper bounds for  $\alpha$  are as small as requested. The terminating search of Theorem 11.3 supplies the trial means and their norm bounds. Choose rational  $R$  large and  $h$  small until the explicit rational upper bounds for  $\beta$  are small. There are then finitely many marked prefix masses. Request their enclosures at

a precision making the finite sum in (10.3) small, and normalize rational lower endpoints. This gives actual finite rational instruments, not floating-point transition tables.

For these fixed finite instruments, Theorem 2.7 supplies convergent dyadic local lower certificates and executable rational upper witnesses. Refine until  $U - L$  is below the requested remaining budget; convergence, or the explicit causal oscillation modulus, ensures termination. Although the finite alphabet and row dimension may grow as the physical tolerance decreases, at each stage this is a finite problem and the mesh is selected for that problem. Thus the total interval width is at most  $U - L + 2(a + c)$  and can be made arbitrarily small. Intersect each interval with its predecessors. They all contain the same true value, so the intersections are nonempty and nested, and their lengths tend to zero.  $\square$

*Remark 13.2* (What is constructive in the theorem). The quantitative implication (13.1) holds for any rigorously supplied analytic enclosures. Its effective-convergence assertion has an explicit analytic data hypothesis. A graph core known only to exist is not, by itself, a numerical integration algorithm. Neither arbitrary real preparations nor arbitrary measurable marks are silently declared computable. Section 14 gives genuine hard-sphere data for which the required calculations can be certified, including an exact continuous-report deficiency. The private certificate search remains nonconvex and may be expensive; the witness-dimension theorem reduces the number of distinct tests, not the number of cells or the cost of real quantifier elimination.

**Corollary 13.3** (Regenerative risk certificates). *Retain the actual synchronous reset and stationary-policy signature of Section 8. For losses in  $[0, 1]$ , certified marked presentation errors  $\epsilon_j$  for ages  $0 \leq j \leq N$  give a uniform stationary-risk error at most*

$$\eta \sum_{j=0}^N (1 - \eta)^j \epsilon_j + (1 - \eta)^{N+1}.$$

*The same bound holds for each lower support function of the stationary risk body. Finite-age rational local certificates may be transferred at their declared budgets by Theorem 13.1. Under its effective hypotheses the presentation part of the displayed risk bound can be made arbitrarily small, and then  $N$  can be increased.*

*Proof.* Condition the stationary regenerative law on its age. The law of age  $j$  has weight  $\eta(1 - \eta)^j$ , and the omitted ages have total weight  $(1 - \eta)^{N+1}$ . Apply the marked bounded-loss inequality to each retained age and then sum. The bound precedes optimization, so it also bounds lower support functions. Finite-age certificates concern those finite-age experiments; this corollary does not identify a finite-age optimized deficiency with an unrestricted infinite-history deficiency. The register is actually cleared at reset and any persistent selector remains charged, exactly as in the inherited theorem.  $\square$

## 14. CERTIFIED DATA FOR ACTIVE HARD SPHERES

The preceding theorems require analytic data, not only matrix regressions. We give two explicit families. The first has an exactly computable continuous-report intrinsic deficiency. The second has a nonzero graph residual obtained from a collision-compatible position observable. The hard-sphere flow and its almost-everywhere realization are the same as in the inherited active microscopic construction; no new assumption on a kicked vector's generator domain is introduced.

**14.1. A conserved mode with an exact marked deficiency.** Take  $N \geq 2$  identical hard spheres on a flat torus with a nonempty configuration region. Normalize the equilibrium law to have independent standard Gaussian velocity coordinates. Its configuration factor is uniform on

the allowed configurations. For a fixed coordinate put

$$X = N^{-1/2} \sum_{i=1}^N v_{i,1}, \quad W = \operatorname{sgn} X.$$

Then  $X$  is standard Gaussian and  $W$  is a fair actual initial mark. Elastic collisions preserve total momentum, hence  $U(t)X = X$  and  $X \in \bigcap_{j \geq 1} D(L^j)$  with  $LX = 0$ . The actions  $a \in \{-1, 1\}$  reverse all velocities or leave them fixed, so  $\bar{C}_a X = aX$ . They are physical interventions, even though the selected observable lies in a conserved sector. In the normalized space  $V = \operatorname{span}\{X\}$ , the exact Gram data are

$$G = [1], \quad J = [0], \quad B_a = [a], \quad R = [0], \quad S_a = [0]. \quad (14.1)$$

All entries are rational. These identities hold for the full colliding system, not a collision-free restriction of its equilibrium law.

Consider one acquisition after the chosen action and any fixed flight time. Source  $E$  reports a fresh standard Gaussian independent of the state, while target  $F$  reports

$$Y = aX + \xi, \quad \xi \sim N(0, 1).$$

There is no informative earlier report. Source and target carry the same actual mark  $W$ . The simulator knows the current action, not  $X$  or  $W$ ; every persistent selector is independent of the physical state.

**Theorem 14.1** (An exact continuous physical certificate). *For every nonempty finite private width profile, and every finite hidden or visible independent-selector budget,*

$$\delta_b(E, F) = \frac{1}{4}. \quad (14.2)$$

*The upper witness is the stateless simulator that emits an independent  $N(0, 2)$  report. A lower witness is the marked event  $\{W = \operatorname{sgn}(aY)\}$ . The report-sign coarsening of  $F$  has the exact rational joint table*

$$\mathbb{P}_F(W = w, Z = z \mid a) = \begin{cases} 3/8, & z = aw, \\ 1/8, & z = -aw, \end{cases} \quad Z = \operatorname{sgn} Y. \quad (14.3)$$

*It also has deficiency  $1/4$  from a source independent of  $W$ .*

*Proof.* For any source simulator its emitted report is independent of  $W$ , conditional on an exposed selector and the externally fixed action. The indicated event therefore has probability  $1/2$ . At the target,  $aY = X + a\xi$ . The plane vector  $(X, a\xi)$  is rotationally invariant; the two sign-disagreement wedges between  $X$  and  $X + a\xi$  have total angle  $\pi/2$ . Consequently agreement has probability  $3/4$  and the variation discrepancy is at least  $1/4$ . The action may be fixed in this test, so taking the feedback supremum cannot weaken the bound. This argument holds in each exposed-selector fibre and also after hiding a selector.

For the upper witness write  $\varphi_{\sqrt{2}}$  for the  $N(0, 2)$  density. The conditional Gaussian law  $X \mid Y = y$  has mean  $ay/2$  and variance  $1/2$ . Thus the target joint densities with respect to counting measure on  $w$  and Lebesgue measure on  $y$  are

$$p(w, y) = \varphi_{\sqrt{2}}(y) \Phi(way/\sqrt{2}).$$

The source witness has joint density  $q(w, y) = \frac{1}{2} \varphi_{\sqrt{2}}(y)$ . The sign of  $p - q$  is positive exactly on  $w = \operatorname{sgn}(ay)$  and negative on its complement. Hence that event attains the total variation, which is  $3/4 - 1/2 = 1/4$ . There are only the two possible actions; the same calculation holds for both, and for mixtures determined by any independent selector. The witness uses no persistent state. Integrating over the two report signs and using symmetry proves (14.3). Its variation

discrepancy from the independent fair-sign witness is again  $1/4$ , and the same event proves the lower bound.  $\square$

The sign table is an exact finite lower-witness presentation, not a two-sided zero-error reconstruction of the continuous Gaussian report. The continuous upper bound above is proved directly. For an arbitrary small two-sided bridge one must instead use the partitions and error budgets in Section 10. Confusing these two uses of a coarsening would give an incorrect certificate.

**Proposition 14.2** (Effective cell data in the conserved sector). *For finitely many repeated acquisitions of  $X$ , with the two actions above, rational noise variances bounded away from zero, and the mark  $\text{sgn } X$ , every marked Gaussian prefix mass in (10.4) admits rational enclosures of arbitrarily small prescribed width. The exact trial space has zero graph and intervention errors for every horizon. Consequently Theorem 13.1 gives a convergent same-budget finite rational certificate procedure for this active physical family.*

*Proof.* Every controlled mean is  $X$  or  $-X$ . The mass is a one-dimensional integral on one of the half-lines of the standard Gaussian density times finitely many Gaussian interval probabilities. Its tail outside  $[-M, M]$  is at most  $M^{-2}$ , by the second moment. On the remaining compact interval, each interval probability and its derivative has a computable uniform bound: its derivative is a difference of two Gaussian densities divided by its positive noise scale. Products therefore have a computable Lipschitz bound. Uniform Riemann sums have an explicit mesh error given by this bound times the interval length and mesh size. Gaussian density and distribution values at algebraic arguments have rational enclosures: integrate the power series of  $\exp(-u^2/2)$  on a bounded interval with a Taylor remainder bound, using symmetry and a Gaussian tail bound for unbounded endpoints. Square roots and the normalizing constant have elementary rational interval algorithms. Combining these enclosures with the mesh and tail errors gives the requested rational mass interval. All quantities are scalar; the configuration-volume normalizer cancels. Equation (14.1) removes the analytic trial error for every controlled word. Apply the finite-report and rational-tree constructions and then the finite local-certificate theorem. This proves an actual procedure for this family, not computability of arbitrary preparations.  $\square$

**14.2. A nonzero collision-compatible graph residual.** Consider two spheres of diameter one on the two-dimensional square torus of side ten. The allowed ordered configuration volume is  $100(100 - \pi)$ ; velocities remain independent standard Gaussians. Let  $b(u) = (1 - u^2)^2$  on  $|u| \leq 1$  and zero elsewhere, and let  $h = 1/4$ . Place squares of half-side  $h$  at  $c_1 = (2, 5)$  and  $c_2 = (8, 5)$ . Let  $\chi_1(q_1, q_2)$  be the product of the four rescaled functions  $b$  when particle one is near  $c_1$  and particle two near  $c_2$ , and put  $f = \chi_1(q_1, q_2) + \chi_1(q_2, q_1)$ . The two supports are disjoint, are separated from the collision boundary, and define a symmetric  $C^1$  position observable on the torus.

**Proposition 14.3** (Exact nonzero physical Gram data). *The vector  $f$  belongs to  $D(L)$  and  $Lf = v_1 \cdot \nabla_{q_1} f + v_2 \cdot \nabla_{q_2} f$ . With  $V = \text{span}\{\phi\}$ ,  $\phi = f/\|f\|_2$ , its exact data are*

$$J = 0, \quad R = [192], \quad B_{\pm} = 1, \quad S_{\pm} = 0, \quad \|f\|_2^2 = \frac{2h^4(256/315)^4}{100(100 - \pi)}. \quad (14.4)$$

*Here the actions still reverse all velocities or leave them fixed. For one acquisition of  $f$  with unit Gaussian noise after  $\Delta = 1/1000$ , under equilibrium preparation and with any initial finite mark channel, the static trial mean  $f$  and the true active physical acquisition have two-sided marked feedback error less than  $1/100000$ . All constants in this last assertion follow from rational Gram enclosures and  $3 < \pi < 4$ .*

*Proof.* The position support is at positive distance from contact, including periodic images; its minimal separation is greater than one. On a regular orbit  $f$  is continuously differentiable between collisions, and its transport derivative vanishes near each collision configuration. Thus the integral

transport identity holds across collisions without a jump. Its difference quotients are bounded in absolute value by a constant times the conserved total speed. This is square integrable under the Gaussian velocity law. Dominated convergence gives the strong  $L^2$  generator derivative claimed. This also proves the required collision-domain membership rather than presuming a smooth-boundary extension of a kicked function.

Direct polynomial integration gives

$$I_0 = \int_{-1}^1 b(u)^2 du = 256/315, \quad I_1 = \int_{-1}^1 b'(u)^2 du = 256/105.$$

The two disjoint product supports give  $\|f\|^2 = 2h^4 I_0^4/Z$ ,  $Z = 100(100 - \pi)$ . Independence, centering and unit variance of velocity coordinates give

$$\langle f, Lf \rangle = 0, \quad \|Lf\|^2 = \frac{4I_1}{h^2 I_0} \|f\|^2 = 192 \|f\|^2.$$

Velocity reversal fixes the position vector  $f$ , so  $B_{\pm} = 1$  and  $S_{\pm} = 0$ . The displayed Gram residual follows. Since  $f \in V$ , there is no projection error and the one-step residual is  $\Delta\sqrt{192}\|f\|$ . The two-action bound in Theorem 9.5 is therefore

$$\alpha \leq \frac{\Delta\sqrt{384}\|f\|}{\sqrt{2\pi}}.$$

Using  $I_0 < 1$ ,  $h = 1/4$  and  $Z > 9600$  gives  $\|f\|^2 < 1/1228800$ . Together with  $2\pi > 6$  this yields  $\alpha^2 < 64/(10^6 \cdot 1228800) < 10^{-10}$ . Every inequality is rational except the explicitly enclosed  $\pi$ ; no floating-point Gram estimate is used. A mark coupled to the original state stays coupled in the first-mismatch argument.  $\square$

These two examples answer different questions. A conserved sector can carry nontrivial hidden information and an exact positive deficiency; a localized transport sector supplies genuinely nonzero graph residuals and certified error bounds. The velocity interventions change the physical dynamics in both cases. Neither example is a proof of the particle-number-uniform nonlinear kinetic corrector in the historical B4 program. Their downstream edge is the explicit marked physical-to-finite certification theorem, with its own stated analytic inputs and error topology.

## 15. A NONCONSERVED COLLISION-COMPATIBLE ACQUISITION

The following example lies on the full equilibrium phase space of an interacting hard-sphere system. It is neither a collision-free restriction nor an observable constant along the flow. Its explicitly solvable collective mode permits simultaneous evaluation of a nonzero generator residual and of a marked decision deficiency. The orbit factors through total momentum, so collisions cancel from this particular observable. The residual measures projection error, not interaction strength. The collision-sensitive comparison is proved in Section 4; the solvable collective calculation is retained here as a separate exact benchmark.

Fix  $N \geq 2$  equal-mass hard spheres on the square flat torus of side  $\ell > 0$ , with positive admissible configuration volume. Velocities have independent standard Gaussian coordinates under equilibrium  $\mu$ . Let  $U(t)$  be the full hard-sphere Koopman group, and allow the two physical interventions  $v \mapsto av$ ,  $a \in \{-1, 1\}$ , before each flight of length  $\Delta$ . Define the circle-valued coordinate and the normalized total momentum

$$\vartheta = \frac{2\pi}{\ell} \sum_{i=1}^N q_{i,1} \pmod{2\pi}, \quad X = N^{-1/2} \sum_{i=1}^N v_{i,1}, \quad \omega = \frac{2\pi\sqrt{N}}{\ell}, \quad f = \cos \vartheta.$$

The exponential of the displayed coordinate is well defined on the torus; no choice of lifted particle positions enters  $f$ .

**Theorem 15.1** (Collective domain and controlled-word reduction). *The coordinate  $\vartheta$  is Haar uniform and independent of the standard normal  $X$ . On the full-measure domain of the hard-sphere flow,*

$$U(t)f = \cos(\vartheta + \omega Xt), \quad L^j f = \omega^j X^j \cos(\vartheta + j\pi/2), \quad \|L^j f\|_2^2 = \frac{\omega^{2j}(2j-1)!!}{2}. \quad (15.1)$$

Here  $f \in \bigcap_{j \geq 1} D(L^j)$  and  $(-1)!! = 1$ . In the normalized one-dimensional trial space spanned by  $\varphi = \sqrt{2}f$ , the matrices of the preceding residual theorem are

$$G = 1, \quad J = 0, \quad R = \omega^2 > 0, \quad B_{-1} = B_1 = 1, \quad S_{-1} = S_1 = 0.$$

For a word  $w = (a_1, \dots, a_k)$  put

$$s(w) = \Delta \sum_{j=1}^k \prod_{i=1}^j a_i.$$

Then  $T_{a_1} \cdots T_{a_k} f = \cos(\vartheta + \omega X s(w))$ . Exactly  $k+1$  distinct observable functions occur among the  $2^k$  words, indexed by  $s/\Delta \in \{-k, -k+2, \dots, k\}$ .

*Proof.* Simultaneous translation of every particle by  $r$  in the first torus coordinate preserves the admissible configuration measure and shifts  $\vartheta$  by  $2\pi Nr/\ell$ . These shifts exhaust the circle, so its push-forward measure is Haar. Configuration and velocity are independent, and Gaussian addition proves the assertion about  $X$ . Elastic collisions conserve total momentum and do not jump particle positions. Between collisions the derivative of  $\vartheta$  is  $\omega X$ . This proves the orbit formula, including the periodic identifications, on the classical almost-everywhere flow domain. Difference quotients of that formula, dominated successively by polynomials in  $|X|$ , converge in  $L^2(\mu)$  since all Gaussian moments are finite. Induction gives membership in every generator power domain and the derivative formula. Independence and the uniform circle integral of squared sine or cosine give the norm formula. This also establishes collision compatibility in the actual generator domain, rather than by applying a formal derivative to arbitrary boundary data.

The inner product of  $\cos \vartheta$  and  $X \sin \vartheta$  vanishes. Thus  $J = 0$  and the squared residual norm of  $L\varphi$  is  $\omega^2$ . A velocity reversal fixes  $\varphi$ , giving  $B_a = 1$  and  $S_a = 0$ . After the  $j$ th intervention the total momentum is  $(a_1 \cdots a_j)X$ , so summing flight displacements proves the word formula in the order  $T_a = C_a U(\Delta)$ . Every sequence of  $k$  signs can be the successive cumulative products, which gives all  $k+1$  clock values. They are distinct as observable functions because

$$\|\cos(\vartheta + \omega X s) - \cos(\vartheta + \omega X t)\|_2^2 = 1 - \exp\{-\omega^2(s-t)^2/2\} > 0 \quad (s \neq t).$$

□

**Theorem 15.2** (Explicit converging finite physical presentations). *Let  $m \geq 1$  and replace the mean at clock  $s$  by*

$$f_{m,s} = \sum_{j=0}^{m-1} \frac{s^j}{j!} L^j f.$$

For  $|s| \leq k\Delta$  the mean error is bounded by

$$E_{m,k} = \frac{(k\Delta\omega)^m}{m!} \sqrt{\frac{(2m-1)!!}{2}}. \quad (15.2)$$

For Gaussian report noises  $\sigma_k > 0$  and preparations satisfying  $\sup_\theta \|p_\theta\|_2 \leq D$ , the exact and polynomial acquisitions have two-sided stateless marked feedback comparison error at most

$$\alpha_{m,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{\sqrt{k+1}}{\sigma_k} E_{m,k}. \quad (15.3)$$

*The same actual initial mark and every exposed independent selector are preserved. The estimate is uniform over all consumer register widths, and  $\alpha_{m,T} \rightarrow 0$  at every fixed finite horizon. The physical trial functions and all their generator and intervention Gram entries are explicitly integrable against equilibrium.*

*Proof.* The integral Taylor remainder for the unitary orbit on  $D(L^m)$  has norm at most  $|s|^m \|L^m f\|_2 / m!$ , also for negative  $s$ . Theorem 15.1 gives (15.2). Its successive ratio is

$$\frac{E_{m+1,k}}{E_{m,k}} = k\Delta\omega \frac{\sqrt{2m+1}}{m+1} \rightarrow 0,$$

which is an effective bound for the convergence claimed here.

Couple the two acquisitions from the same initial  $x, W$  and the same independent selector. Until the first unequal report, their controls agree. Two unit-dimensional Gaussians with the same variance have variation distance at most their mean difference divided by  $\sigma_k \sqrt{2\pi}$ ; this follows by integrating the derivative of the translated Gaussian density. At fixed  $x$ , the probabilities of all matched prefixes sum to at most one. Keep those probabilities until bounding the mismatch by the largest mean error at that  $x$ . There are only  $k+1$  different errors at time  $k$ , even though there are  $2^k$  words. Cauchy–Schwarz gives

$$\int p_\theta(x) \max_s |e_{m,s}(x)| d\mu(x) \leq D \left( \sum_s \|e_{m,s}\|_2^2 \right)^{1/2} \leq D\sqrt{k+1} E_{m,k}.$$

Sum first-mismatch probabilities over  $k$ . The construction in both directions uses the identity report interface, has no added persistent state, and couples the actual mark, proving (15.3). No bound on feedback-conditioned preparation densities was used.

The trial family consists of  $X^j \cos \vartheta$  and  $X^j \sin \vartheta$  for  $j < m$ . Its inner products are products of circle integrals and Gaussian moments. Generator entries multiply by  $\omega X$  and rotate sine and cosine; the interventions replace  $X$  by  $-X$ . Every required finite entry is thus a polynomial in  $\omega$  with rational coefficients before normalization. This supplies exact symbolic expressions and arbitrarily tight rational interval enclosures. The polynomial means may be unbounded in  $X$ ; that is harmless for (15.3). For finite-report integration one first truncates  $|X|$  with its explicit Gaussian tail, then integrates bounded Gaussian report probabilities over the resulting compact cylinder. For the equilibrium and mark used below, the only mark boundaries are  $\vartheta = \pi/2, 3\pi/2$ , and can be split explicitly. On each piece trigonometric, polynomial and Gaussian distribution functions have computable uniform moduli. Rational upper and lower Riemann enclosures therefore converge. This verifies the effective marked integration hypothesis of the finite-presentation theorem for this class, without an exact hard-sphere flow integration oracle.  $\square$

*Remark 15.3.* The signed clock is an index of physical mean functions, not an unpriced extra register offered to a consumer. The width-uniform proof uses equality of mean functions to reduce a bound; it does not allow a finite-state simulator to remember its entire control word. Neither the polynomial coefficient vector nor its real dimension is a finite consumer-label count. The underlying microscopic system still has collisions; solvability here comes from the conserved total momentum that transports a *nonconserved* position observable.

**15.1. One actual mark, one nonzero residual, and a positive certificate.** Let  $W = \text{sgn} \cos \vartheta$  in the initial equilibrium state; the zero set has measure zero. Both experiments have a singleton initial report and allow  $a \in \{-1, 1\}$  before their single acquisition. The source reports an independent standard Gaussian. The target reports

$$Y = \cos(\vartheta + a\tau X) + \xi, \quad \tau = \omega\Delta, \quad \xi \sim N(0, 1)$$

with noise independent of the actual initial mark and microscopic state. Let  $\Phi$  and  $\phi$  denote the standard normal distribution and density.

**Theorem 15.4** (Exact marked physical deficiency). *Here the action precedes any informative report and its reversal symmetry leaves the relevant marked law unchanged. In the visible comparison the independent selector is retained jointly with the same selector law on the target side. For every nonempty finite private, hidden-selector, or visible-selector signature in this one-acquisition model, its intrinsic deficiency equals*

$$d(\tau) = \mathbb{E}[W\{\Phi(\cos(\vartheta + \tau X)) - 1/2\}] > 0. \quad (15.4)$$

*It is attained by the stateless simulator that draws the target's unconditional report marginal. If*

$$c_k(\tau) = 2^{-(2k+1)} \sum_{j=0}^{2k+1} \binom{2k+1}{j} \frac{2(-1)^{(|2k+1-2j|-1)/2}}{\pi|2k+1-2j|} e^{-(2k+1-2j)^2\tau^2/2},$$

*then*

$$d(\tau) = \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k c_k(\tau)}{2^k k! (2k+1)}. \quad (15.5)$$

*The error after  $k = K$  is at most  $[\sqrt{2\pi} 2^{K+1} (K+1)! (2K+3)]^{-1}$ .*

*Proof.* The action is chosen before any informative report. Reversal of the independent symmetric  $X$  does not change the joint law of  $(W, Y)$ , including conditionally on any exposed independent seed. Set  $H_\tau(\theta) = \mathbb{E} \operatorname{sgn} \cos(\theta + \tau X)$ . This is the circle heat evolution of  $\operatorname{sgn} \cos \theta$  at time  $\tau^2/2$ . It is strictly positive on  $(-\pi/2, \pi/2)$  and strictly negative on the opposite arc. To see the sign without assuming it from a Fourier expansion, reflect the heat solution oddly across both endpoints of the positive arc. Its restriction solves the heat equation on that interval with zero boundary values and initial value one. Equivalently it is the survival probability of Brownian motion killed at those endpoints. This is positive in the interior at every finite positive time. The odd reflected solution agrees with the circle solution by uniqueness (the difference has zero initial data and its squared integral has nonpositive derivative). At  $\tau = 0$  the same sign assertion is immediate.

The final angle is uniform. Reversibility of the Gaussian circle convolution gives  $\mathbb{E}[W \mid \vartheta + \tau X = \theta] = H_\tau(\theta)$ . Let  $\nu$  be the unconditional density of  $Y$ . The density of  $(W, Y)$ , minus  $\frac{1}{2}\nu(y)$  in its  $w$ th row, is

$$\frac{w}{4\pi} \int_0^{2\pi} H_\tau(\theta) \phi(y - \cos \theta) d\theta.$$

Pair points  $\theta$  and  $\theta + \pi$ . On the positive arc both  $H_\tau(\theta)$  and  $\cos \theta$  are positive, and  $\phi(y - \cos \theta) - \phi(y + \cos \theta)$  has the sign of  $y$ . Thus the density difference has the sign of  $wy$ , and is not zero. Its positive event is  $\{wy > 0\}$ . The excess probability of that event is (15.4), proving positivity and giving a stateless upper witness of that variation distance.

Every source simulator at the declared signature, conditionally on its exposed seed, produces a report independent of the fair  $W$ . For any such report law the event  $\{wy > 0\}$  has probability at most  $1/2$ , including possible mass at zero. The target probability is  $1/2 + d(\tau)$ . This supplies the matching lower bound for private and hidden randomization; the same bound in each visible fiber supplies it for visible randomization. Mixtures of the two action choices do not change this conclusion.

Finally, uniformly for  $|z| \leq 1$ ,

$$\Phi(z) - 1/2 = \frac{1}{\sqrt{2\pi}} \sum_{k \geq 0} \frac{(-1)^k z^{2k+1}}{2^k k! (2k+1)}.$$

The absolute remainder is bounded by the next term, by the alternating Taylor expansion of  $e^{-u^2/2}$  followed by integration. Expand  $\cos^{2k+1}$  into its finite complex exponential sum. For odd  $n$ ,

$$\mathbb{E}[W \cos(n\vartheta)] = \frac{2(-1)^{(|n|-1)/2}}{\pi|n|}, \quad \mathbb{E}e^{in\tau X} = e^{-n^2\tau^2/2}.$$

The sine terms vanish. These identities give  $c_k$  and the claimed series and remainder, with no unproved exchange of a nonuniform sum.  $\square$

**Corollary 15.5** (A rational, nonvacuous collision certificate). *For  $N = 2$ ,  $\ell = 10$ , any rational diameter in  $(0, 5)$  giving nonempty interior, and  $\Delta = 1/2$ , the same physical acquisition satisfies*

$$\frac{205809122887}{10^{12}} < d(\tau) < \frac{205809122889}{10^{12}}, \quad R = \frac{2\pi^2}{25} > 0.$$

*The degree-fifteen physical Taylor presentation has marked comparison error less than  $3 \cdot 10^{-11}$  for the single report.*

*Proof.* Here  $\tau^2/2 = \pi^2/100$ . Evaluate (15.5) through  $K = 10$  with outward rational arithmetic and add its stated remainder. For reproducibility the source supplies every arithmetic operation:  $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$  is bounded by alternating series; square roots are enclosed by rational bisection; exponentials are range-reduced to arguments in  $[0, 1]$ , enclosed by alternating series, and squared back with outward rounding. The resulting enclosure has lower endpoint greater than the displayed lower rational and upper endpoint smaller than the displayed upper rational. All comparisons are integer comparisons. Applying (15.2) with  $m = 16$ , and replacing  $\sqrt{2}$  in (15.3) by the larger rational 2, yields the final strict bound. The generator residual follows exactly from Theorem 15.1. The supplied checker redoes these finite rational computations; the analytic justification is the preceding domain, coupling and series proofs, not the execution of that checker.  $\square$

## 16. A JOINT PHYSICAL AND NONCONVEX CERTIFICATE THEOREM

The finite testing and microscopic approximation results now give a single lower–upper certificate for the original resource problem. The stateless transfer inequality and physical-to-rational bridge are those of Theorem 13.1; the polynomial lower module is a separate strengthening, not a new physical transfer principle. Unlike a constant mixture of tests, the polynomial lower certificate can separate a target from a private behavior image even when the target belongs to its convex hull.

**Theorem 16.1** (Constructive same-budget physical testing). *Fix a finite horizon and one of the private, hidden-selector, or visible-selector signatures. Let  $E, F$  be marked physical acquisitions with finite compatible controls and marks. Suppose that rational finite presentations  $E_j, F_j$  have executable two-sided stateless marked feedback errors  $a_j, c_j \rightarrow 0$ , with computable upper bounds. For the algorithmic assertion require the uniform enumeration and rational error enclosures of Proposition 3.6, not merely individual computability of each presentation. Let  $P_{p,L}$  be the polynomial (6.2) of the finite intrinsic deficiency problem for  $E_j, F_j$ , and let a nonnegative rational Bernstein array certify it. Let an actual rational simulator table at the same signature have finite error  $U$ . Then*

$$\max\{0, L - a_j - c_j\} \leq \delta_b(E, F) \leq \min\{1, U + a_j + c_j\}. \quad (16.1)$$

*There is a terminating dovetailed search for such an interval of any prescribed positive rational width. Every lower certificate is checked by a finite polynomial identity and sign comparisons; every upper certificate is an executable table at the original budget. The visible case retains the selector in both compared marked laws.*

*The hypotheses hold for the effective fixed-particle collision class of Theorem 12.2, and in particular for the nonconserved collective acquisitions of Theorem 15.2 with equilibrium preparation and a finite computable marked channel with the piecewise effective integration property proved there. For the sign mark of Theorem 15.4, that integration property is explicit.*

*Proof.* The finite polynomial identity gives  $L \leq \delta_b(E_j, F_j)$  by Theorem 6.1; direct evaluation of the rational table gives the reverse bound by  $U$ . Compose it with the two stateless physical

presentation channels. Their identity state costs preserve all retained widths and selector budgets. The marked triangle and presentation-stability inequalities give

$$|\delta_b(E, F) - \delta_b(E_j, F_j)| \leq a_j + c_j.$$

This proves (16.1); no attainment assertion for the original nonfinite minimum is required. The visible comparison is joint with the same selector, not its hidden marginal.

Given a target width  $\eta > 0$ , enumerate presentations until their certified sum of errors is less than  $\eta/4$ . For that presentation, Corollary 6.3 produces lower and upper endpoints with gap less than  $\eta/2$ . Retain maxima of all verified lower endpoints and minima of all verified upper endpoints. The resulting nested physical intervals have width less than  $\eta$  and converge. Zero lower bounds are always available, so this argument also covers zero deficiency and does not ask a strict-positivity certificate to prove equality at a boundary.

The cited constructive collision theorem provides effective graph approximation and marked integration; the finite-report and joint-prefix rationalization theorems then give the presentations and their error bounds. In the collective class, (15.2) replaces the unquantified graph approximation by an explicit rate. Its compact cylinder integration and Gaussian tail bounds give every finite marked-prefix probability. The retained finite-presentation theorem rationalizes these joint prefixes without dividing by a small conditional probability. Thus all links in this specialization are supplied either by an explicitly cited retained theorem or by a proof above, not by assuming a rational physical transition oracle.  $\square$

**16.1. Exponential acquisition values.** The distinction between small total variation and accuracy at an exponential scale is important for the nonlinear-observable direction of the program. The following estimate gives its precise price. It is a theorem about bounded marked acquisition functionals, not a construction of a kinetic large-deviation action.

**Theorem 16.2** (Sharp exponential transfer). *Let  $P, Q$  be two laws of the same marked transcript with  $\|P - Q\|_{TV} \leq \varepsilon$ . If  $h$  is measurable and  $A \leq h \leq B$ , then for every  $\beta > 0$ ,*

$$\left| \frac{1}{\beta} \log \mathbb{E}_P e^{\beta h} - \frac{1}{\beta} \log \mathbb{E}_Q e^{\beta h} \right| \leq \frac{1}{\beta} \log \{1 + \varepsilon(e^{\beta(B-A)} - 1)\}. \quad (16.2)$$

*The bound is sharp. If a pair of stateless marked presentation channels has error  $\varepsilon$  uniformly over a declared resource signature, the payoff is evaluated on the same compared marked output interface. Under this convention the same bound holds between optimal infimum values, and between the optimal supremum values, over that signature. The policies are transported through the channels at unchanged budgets.*

*For the effective physical class of Theorem 16.1, let  $\beta_n > 0$  and an upper bound  $M_n \geq B_n - A_n$  be computable. One can construct finite rational presentations with certified errors  $\varepsilon_n \leq e^{-\beta_n M_n}/n$ . Their optimized normalized logarithmic acquisition values then differ from the physical ones by at most  $\log(1 + 1/n)/\beta_n$ .*

*Proof.* For any bounded  $g$ , integration of its level sets gives  $|\mathbb{E}_P g - \mathbb{E}_Q g| \leq \varepsilon(\sup g - \inf g)$ . Take  $g = e^{\beta(h-A)} \in [1, e^{\beta(B-A)}]$ . Both expectations are at least one. Their larger-to-smaller ratio is at most  $1 + \varepsilon(e^{\beta(B-A)} - 1)$ , which proves the estimate. Equality holds when  $Q$  is concentrated where  $h = A$  and  $P$  moves mass  $\varepsilon$  to a point where  $h = B$ .

For any admissible physical policy transport it to the presentation and apply the uniform inequality; this gives one inequality between optimal values. Transport in the other direction for the reverse inequality. Approximate optimizing policies suffice, so attainment is unnecessary. The same reasoning works for infima and suprema, but does not replace a common policy by separate phasewise optimizers.

Finally choose a strictly positive rational tolerance below  $e^{-\beta_n M_n}/n$ , using computable exponential enclosures. The terminating presentation search of Theorem 16.1 attains that tolerance.

Substitution gives the final claim. This is an accuracy-versus-scale construction; it asserts neither an efficient complexity rate nor a large-deviation limit of the microscopic model.  $\square$

*Remark 16.3* (The downstream implication). Theorems 15.1–16.2 provide a verified fixed-particle link from a nonconserved collision-domain observable to finite-resource marked and exponential acquisition values, with rigorously computable errors. They do not assert particle-number-uniform bounds, a Boltzmann–Grad limit, a nonlinear BBGKY graph corrector on action sublevels, or a range theorem for the historical kinetic Hamiltonian. Those are distinct analytic assertions in the retained program. In particular (16.2) explains why plain unscaled total variation alone does not supply exponential-scale control. All eleven historical components remain in the dependency graph; neither component order nor the existence of an adapter is used as a proof of a missing kinetic theorem.

## APPENDIX A. THEOREM DEPENDENCIES AND INFORMATION CONVENTIONS

The comparison conventions of Section 2 supply a compact original behavior image for each fixed signature. Theorem 3.1 uses only that image and its affine tests. Lemma 3.2 and the binomial estimate then give Theorem 3.3. The strict row-cube verifier of Theorem 6.1 is a separate route for checking lower bounds. Proposition 3.6 combines finite strict certificates, executable upper tables and stateless presentation errors; it does not require an inverse constrained randomization theorem.

The collision path has a second, independent finite calculation: Theorem 4.1 solves its marked gated reference experiment, and Lemma 4.2 proves the physical separation of incoming and outgoing detector values. The stateless marked variation estimate joins these two arguments in Theorem 4.3. Proposition 4.4 verifies the effective presentation input using a separated collision root. None of these steps assumes the collective benchmark’s conserved-mode formula.

For optional projections, Bayes’ formula and maximal inequalities first prove Theorem 5.1. Conditional Gaussian change of measure then verifies its hypotheses in Theorem 5.3. The same-latent maximal coupling is used only for process convergence. Finite-register risk transport is instead a data-processing consequence of the original marked variation estimate.

The graph-domain, residual and finite-prefix sections supply the broader physical presentation machinery. Their input is an actual flow, a specified Hilbert realization, a prepared law and an observable with the declared approximation properties. The bound in Theorem 13.1 already expresses the same-budget transfer principle. Theorem 16.1 retains it with the strict polynomial lower module. The present adaptive testing and scattering results add independent-test size control and a collision-generated resource separation; they do not reclassify the transfer principle as a new result each time a certificate module changes.

Throughout the article fresh private randomization has no memory. A hidden persistent selector and a visible persistent selector are separate signatures. A visible comparison is joint with its independent target selector; marginalizing it changes the problem. The computational chart, the revealed-candidate testing game and the latent-preserving convergence coupling are proof constructions, not unpriced channels available to the physical decision maker.

## PRESERVED DEVELOPMENT: TWELFTH ARTICLE AND EARLIER MATHEMATICS

The preceding sections constitute the seventeenth canonical article. The following archival body retains the earlier mathematical development, including the twelfth canonical body, historical introductions and the annotations already made in the fourteenth development. Original files are unchanged. The seventeenth local copies only clarify the certificate terminology, collective benchmark and effective transfer hypotheses. Historical scope statements are read as historical, not as competing claims for the current article. The preservation map identifies every local change and preserves all predecessor theorem labels.

## APPENDIX 22. INTRODUCTION

A causal experiment supplies observations at specified times. A decision procedure with a finite retained register cannot in general replace this interface by an equivalent full-history experiment at no cost. Information which is supplied again by one presentation may have to be carried across time by the other. Moreover, a single encoder serving several tasks cannot be optimized separately for each task. These two constraints make the attainable risk body depend on both the experiment and its resource signature.

This paper studies that dependence for marked experiments. A mark is the actual target variable appearing in a loss, including its coupling with the observations and any latent parameter. The observation controller may know the task; the common encoder does not. Private retained labels, public seed information, and side observations supplied afresh by the physical interface are specified separately. This distinction is needed both for positive approximation results and for exact minimum-state statements.

**22.1. The structural results.** The first comparison theorem charges a simulator's persistent state. If a simulator of widths  $K_t$  and seed size  $r$  approximates the target experiment with marked feedback error  $\varepsilon$ , then an  $S_t$ -state,  $q$ -seed target consumer has a source implementation of widths  $K_t S_t$  and seed size  $r q$ . The construction retains reachable pairs and gives risk-body inclusion before optimization over consumers. Composition multiplies the state costs and adds the errors. When a consumer's state already determines the marked continuation class, the class register can instead be fused with the consumer's register. This gives the exact same-width complete-state comparison in Theorem 24.7. General consumers need not admit this cancellation.

Our main exact completion result, Theorem 25.2, identifies which width profiles preserve a declared family of decisions throughout a supported history tree. Its object is a causally closed compatible cover. Each cell must admit one optimal action for every task, and all its actual successors under a common control and report must fit in one next cell. Histories with no such successor impose no fictitious constraint. The existence of these covers is equivalent to an exact causal common encoder, whether deterministic, privately randomized, or publicly randomized. Thus neither a coarsest partition nor a positive next-report hypothesis is needed. Corollary 25.5 places this combinatorial characterization on the zero face of the minimax risk body: infeasibility has a positive decision witness  $\alpha\gamma$ , explicitly specified by the finite task table. Completing Brier prediction tests recovers the universal marked continuation quotient.

The nonfinite theory has two complementary extensions. The dominated finite-horizon compactness theorem uses weak-star policy rows and  $L^1$ -continuous loss effects. A conditional-kernel version permits a physically redisplayed nonatomic covariate, and hence singular full report laws. Theorem 28.2 then treats an unbounded time interval under coherent dominated finite-prefix effects and a summable criterion. With  $r_{j,t} \in [0, 1]$  and  $\sum a_t = 1$ , it proves compact attainable risk

bodies and the attained identity

$$\min_{\lambda} \max_j \left\{ \int \sum_{t \geq 0} a_t r_{j,t}(\pi) \lambda(d\pi) - b_j \right\} = \max_w \min_{\pi} \int \left\{ \sum_{t \geq 0} a_t r_{j,t}(\pi) - b_j \right\} w(dj).$$

No measure dominating the entire infinite path is assumed. For a fixed accuracy the finite public-seed approximation depends on row covering, not the truncation horizon. Its index is priced explicitly when stored in the private register. Prefix-coherent comparison has error  $\sum a_t \varepsilon_t$ , preserving the same product resource law.

A physical consumer connects these assertions to unbounded microscopic transport. For a contraction generator  $L$  and a finite resolved space in  $D(L) \cap D(L^*)$ , we construct the orthogonal generator on its actual domain and prove the forced-memory and Schur-complement formulas, including the contribution of the unresolved initial observable. A collision-compatible hard-sphere graph core gives strong, locally time-uniform Galerkin approximation of the full Koopman orbit. For Gaussian noisy physical acquisitions with uniformly  $L^2$ -bounded initial densities, Theorem 44.4 converts this analytic approximation into one marked comparison bound uniform over every finite register profile and feedback design. The same bound passes to the infinite-horizon dual values. Crucially, neither the Galerkin coefficients nor a resampled mark are supplied to the encoder.

Finally Theorem 45.1 joins the two sides. If an approximate experiment has resource-preserving risk and benchmark errors at most  $\varepsilon < \alpha\gamma/4$ , its minimax value separates exact causal cover feasibility from infeasibility by the threshold  $\alpha\gamma/2$ . The comparison is performed before optimization and its cost remains visible in the one-sided version. This is a precise sense in which analytic experiment approximation can certify an exact common-encoder resource profile.

**22.2. Antecedents and the comparison boundary.** Classical experiment comparison and randomization criteria are due to Blackwell, Le Cam and their successors [1, 47, 46]. Filtered statistical experiment comparison is itself an established subject [48]; the observation timing is not introduced here as a new concept. The weak-star policy topology and strategic-measure compactification used in the finite-prefix theory have close antecedents in stochastic teams [55, 56]. Common-information dynamic programming and finite-memory causal coding are also established tools [53, 54, 57]. Our encoder remains coupled across decision rows, and every memory or seed conversion has an explicit implementation cost.

The continuation-space equivalence engine is classical weighted-automaton linear algebra [58, 61]. Positive reachable state cardinality, in contrast, need not equal linear rank. The synthesis results below include polynomial count-vector families, exponential positive width at rank two, and a finite-task graph-coloring reduction. The compatible-cover device likewise belongs to partial-machine minimization. Here its support propagation is proved directly for randomized common encoders, and the resulting exact profiles are related to positive risk witnesses and marked comparison. We make no claim that the cover device or graph coloring by itself is new.

The operator section uses the classical finite-rank projection method, not a new general resolution of orthogonal dynamics. The semigroup and domain assumptions in [65, 66] are particularly relevant: a finite-rank projection does not make  $PL$  bounded unless its range lies in the adjoint domain. We impose that condition and provide the finite-block proof, the forcing and reconstruction estimates, and the width-uniform physical decision consequence. The graph-core resolvent argument, product compactness, Carathéodory's theorem and convex minimax are classical ingredients. The result proposed for assessment is their resource-sensitive connection, not a priority claim for any of these ingredients.

**22.3. Scope and organization.** All hypotheses remain attached to their statements. The infinite-horizon compactness result concerns summable criteria and dominated finite prefixes, not arbitrary average-cost control or arbitrary filtered strategic-measure closure. The exact cover theorem

concerns supported restart decision families; an arbitrary task-dependent control law is not silently replaced by an open-loop test. Public seed information is part of the compared joint interface whenever a consumer may read it. A marginal simulation bound which discards that seed is insufficient. The physical approximation proves a microscopic  $L^2$  decision limit, not a nonlinear kinetic action-sublevel or Boltzmann–Grad limit.

The theory also retains the finite common-membership separation, delayed Gaussian resource exponents, positive filtering, finite nonlinear semigroups, hidden-phase decisions, predictive trees and measure-valued consequences. The historical program is a dependency graph: a proved model consumer, an acquisition adapter and a primary theorem dependency are different relationships. In particular the primary algebraic-geometric A2 line is independent of the present comparison theory. The complete development preserves the earlier mathematical bodies with their local hypotheses. Reproducibility and source-identity records are separate from the mathematical arguments and are not used as proof premises.

### APPENDIX 23. THE MARKED DECISION SPECTRUM

We first fix the object whose finite presentations are to be compared. This requires specifying both the decision tasks and the information available to the decoder. These specifications cannot be recovered from a dimension or a cardinality alone.

**23.1. Experiments, tasks, and signatures.** An experiment is a standard Borel causal instrument, with an initial law and transition–observation kernels indexed by a parameter  $\theta$ . Actions, when present, are chosen before the corresponding transition. The same observation-based policy is used at every value of  $\theta$ . A *mark*  $d$  specifies a horizon, decision times, action spaces, a loss in  $[0, 1]$ , and a risk functional. The latter is either expectation under a declared prior or the supremum over a declared parameter set, taken after selecting one decoder. A minimax decoder does not know the unknown parameter. A mark may instead specify a fixed Bayesian prediction problem with an integrable squared loss; assertions involving total variation below apply only after a uniform loss bound is supplied. When a loss concerns a latent target, the marked experiment includes that target, and a simulation must preserve its joint law with the observations. Equality of report laws alone does not suffice for such a mark.

An encoder  $Q$  is an initialized causal transducer

$$I_t \sim q_t(\cdot \mid I_{t-1}, Y_t, A_{t-1}), \quad I_t \in \{1, \dots, S\}. \quad (23.1)$$

After an update, a decoder sees only  $(t, I_t)$  and any explicitly declared current side input. It does not see the discarded observation, an output transcript, old actions, or a data-dependent random seed. All persistent decoder state belongs to  $I_t$ . Fresh randomization is allowed. A public seed, when allowed, is independent of the experiment; a data-dependent position or modification of that seed is charged. A terminal reconstruction kernel may have a continuous output even when the physical statistic has a finite alphabet.

We use the resource signature

$$b = (S, A, L, P, p, W, \tau; R). \quad (23.2)$$

Here  $S$  counts persistent states;  $A$  counts physical terminal labels;  $L$  bounds raw symbols read per acquisition;  $P$  bounds fixed description bits;  $p$  is numerical precision;  $W$  is transient work space;  $\tau$  is update time; and  $R$  records the permitted independent randomization. A coordinate  $\infty$  means that no bound is asserted. For a precision constraint, the order is the order on permitted implementations, not the numerical order on an accuracy tolerance. The *cardinality signature* constrains only  $S$  (and  $A$  when explicitly stated). In particular, exact real constants in a fixed program are free only under that signature. All theorems below state which signature they use.

For an encoder admissible at  $b$ , let  $r_E(Q, d)$  be the infimal value of the declared risk functional over its allowed decoders. Let  $\beta_E(d)$  be the infimal risk using the full observation history under the same decision timing. The encoder is selected before the mark; its decoder may depend on the mark. For a set  $\mathcal{D}$  of marks define

$$\mathcal{A}_E(b) = \{u \in [0, 1]^{\mathcal{D}} : \text{some admissible } Q \text{ has } r_E(Q, d) \leq u_d \text{ (} d \in \mathcal{D} \text{)}\}, \quad (23.3)$$

$$\Theta_E(b; \mathcal{D}) = \inf_Q \sup_{d \in \mathcal{D}} \{r_E(Q, d) - \beta_E(d)\}. \quad (23.4)$$

Infima need not be attained. We use the displayed attainable sets, not their product-topology closure: for an infinite task family, taking that closure before a uniform task supremum can change the value. The common encoder and the order of quantifiers in (23.4) are part of the invariant. The collection of attainable sets, baselines, and resource signatures is the *marked decision spectrum*, denoted  $\Theta(E)$ . This is the meaning of the symbol  $\Theta$  in the present paper. It is a profile, not an additional physical parameter. Unbounded squared marks are treated separately without restricting the vectors to  $[0, 1]$ .

**Example 23.1** (The encoder cannot follow the task). Let  $X = (X_1, X_2)$  consist of two independent fair bits. A terminal statistic has two values. The two marks ask, respectively, for  $X_1$  and  $X_2$ , with zero-one loss. Each mark separately admits risk zero, but one common encoder has worst risk at least  $1/4$ . Indeed, for each fixed independent coding seed and each encoder label, the two optimal answers form a word in  $\{0, 1\}^2$ . Two reconstruction words have average Hamming distance at least  $1/2$  from the four equally likely input words: two words may be exact, and each remaining word is at distance at least one. Randomization does not lower the average of the two risks. Hence their maximum is at least  $1/4$ . Conversely, with a public fair coin, reveal one of the two coordinates and guess the other. Thus the value is  $1/4$  when that coin is permitted, whereas interchanging  $\inf_Q$  and the task supremum gives zero. No such interchange is made in this paper.

**23.2. Typed simulations and invariance.** A certified simulation  $S : E \Rightarrow F$  consists of initialized causal kernels, a monotone resource transformer  $\rho_S$ , and an error bound  $\epsilon_S(T)$ . The kernels do not know  $\theta$  or the latent variables. For every target-side admissible policy and every parameter, the simulated marked transcript through horizon  $T$  has total variation distance at most  $\epsilon_S(T)$  from its law in  $F$ . Moreover, composing the simulation with any target encoder of signature  $b$  produces a source encoder of signature at most  $\rho_S(b)$ . These are uniform requirements on one simulator, not permission to choose a different simulator for each task or policy. When marks have different horizons, use the corresponding error for each coordinate; a common  $\epsilon$  below denotes their supremum.

Two certified simulators represent the same morphism when they have the same transformer and error certificate and induce the same marked execution laws under every admissible continuation. Internal register names are immaterial. We use standard Borel kernels modulo this execution equivalence. The exact morphisms have error zero. A  $K$ -state cardinality simulator has the transformer  $S \mapsto KS$ ; its own output is not a second free memory register.

**Theorem 23.2** (Spectrum transport and resource equivalence). *For bounded marks and signatures as in (23.2), certified simulations have identities and associative composition up to execution equivalence. If  $S : E \Rightarrow F$  and  $T : F \Rightarrow G$ , the composite has transformer  $\rho_S \circ \rho_T$  and error at most  $\min\{1, \epsilon_S + \epsilon_T\}$ . For every target-admissible  $Q$  there is a source-admissible  $Q'$  with*

$$r_E(Q', d) \leq r_F(Q, d) + \epsilon_S, \quad \beta_E(d) \leq \beta_F(d) + \epsilon_S. \quad (23.5)$$

*If a reverse simulation  $R : F \Rightarrow E$  is also supplied, then*

$$\Theta_E(\rho_S(b); \mathcal{D}) \leq \Theta_F(b; \mathcal{D}) + \epsilon_S + \epsilon_R. \quad (23.6)$$

*In particular, mutually exact finite-state simulations give equal ideal baselines and comparable spectra under the two budget dilations. Whenever a positive power exponent of either cardinality profile exists, the other has the same exponent. These assertions also hold coordinatewise for the attainable risk sets, before taking the task supremum.*

*Proof.* Compose the kernels on the product of the two internal states. Fix one policy for  $G$ , regard its composition with  $T$  as a policy and continuation for  $F$ , and apply the certificate for  $S$ . Data processing followed by the certificate for  $T$  gives the sum of the two errors. The composition of transformers is forced by the same execution. Associativity follows from iterated kernel integration and the canonical reassociation of the triple product of registers; execution equivalence is preserved under composition because its quantifier includes all continuations. The identity kernel, identity transformer, and zero error give identities. This establishes a category for exact simulations and the corresponding error calculus for approximate ones; it is not an assertion of a new general randomization theorem.

Run  $Q$  on the simulated reports and use its decoder. Every loss in  $[0, 1]$  changes in expectation by at most the variation error. Taking infima over decoders proves the first inequality in (23.5); an arbitrarily good full-history policy proves the second. The reverse simulator implies  $\beta_F(d) - \beta_E(d) \leq \epsilon_R$ . Subtract the baselines, take the common task supremum, and only then take the infimum over  $Q$ . This proves (23.6). No compactness or attainment of an optimal encoder is used.

For exact cardinality equivalence with factors  $K_1, K_2$ , the inequalities are  $\Theta_E(K_1 M) \leq \Theta_F(M)$  and  $\Theta_F(K_2 M) \leq \Theta_E(M)$ . Monotonicity in  $M$  and  $\log(K_i M)/\log M \rightarrow 1$  sandwich any existing limit of  $-\log \Theta(M)/\log M$ . The same reasoning applies to a single bounded squared prediction mark; multiplication by its loss bound changes no positive power exponent.  $\square$

The theorem is an invariance statement for a *typed* equivalence. Ordinary statistical equivalence without a resource certificate need not preserve a causal implementation budget. Conversely, retaining every emitted label in an external transcript changes the signature and thus changes the spectrum. Products of independent observations and products of memories are different operations: the former concern the experiment, the latter the implementation.

**Proposition 23.3** (Forgetting causal and resource structure). *Fix a finite horizon and restrict to uncontrolled instruments and parameter-decision marks. Sending an instrument to its full transcript experiment, and an exact causal simulator to its transcript kernel, defines a functor to parameterized statistical experiments with parameter-independent kernels modulo equality on every parameter law. It preserves identities and composition. It forgets the resource certificate; the resulting statistical equivalence alone does not supply an equality of causal-register spectra.*

*Proof.* Iterating a simulator's initialized kernels at a fixed input transcript gives a parameter-independent Markov kernel from source transcripts to target transcripts. Iterated integration identifies the transcript kernel of a composite with the composite of these kernels. Execution equivalence implies equality on every source parameter law, so this assignment is well defined on the stated classes. The identity gives the identity transcript kernel. A general inverse transcript kernel need not be causal, and a causal inverse may store a nonfinite state; neither datum is reconstructed by the forgetful operation.  $\square$

**23.3. The terminal decision-complete slice.** In this subsection  $E = \{P_\theta : \theta \in K\}$  has standard Borel observation space  $X$ ,  $K$  is compact metric, and  $\theta \mapsto P_\theta$  is continuous in total variation. A terminal encoder is any Markov kernel  $Q : X \rightarrow [S]$ . Only its physical alphabet  $A \leq S$  is constrained; transient computation and the reconstruction kernel are unrestricted. Let  $\mathcal{D}_{\text{all}}$  consist of all finitely supported priors on  $K$  and all finite-action losses in  $[0, 1]$ . For these families we use  $\delta(F, E) = \inf_R \sup_{\theta \in K} \|RF_\theta - P_\theta\|_{\text{TV}}$ , where  $R$  is a parameter-independent reconstruction kernel and  $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$ .

**Theorem 23.4** (Decision completeness at a fixed common encoder). *For every such  $Q$ ,*

$$\sup_{d \in \mathcal{D}_{\text{all}}} \{r_E(Q, d) - \beta_E(d)\} = \delta(QE, E). \quad (23.7)$$

*Consequently*

$$\Theta_E(S; \mathcal{D}_{\text{all}}) = \inf_{Q: X \rightarrow [S]} \delta(QE, E). \quad (23.8)$$

*This is the Blackwell–Le Cam randomization criterion in a compact setting, with the common finite encoder kept outside the decision supremum. In particular, no new priority is claimed for the unconstrained criterion [1, 47, 46].*

*Proof.* A reconstruction  $R$  with variation error  $\epsilon$  transfers each bounded decision rule, proving that the left side is at most the right side. First suppose that both  $K$  and  $X$  are finite. Put  $F = QE$ . Finite-dimensional convex minimax applied to the row-stochastic matrices  $R$  and the total-variation dual gives

$$\delta(F, E) = \sup_{\pi, h} \left[ \sum_{\theta, x} \pi_{\theta} h_{\theta x} P_{\theta}(x) - \sup_R \sum_{\theta, i, x} \pi_{\theta} F_{\theta}(i) R(i, x) h_{\theta x} \right],$$

where  $\pi$  is a prior and  $0 \leq h_{\theta x} \leq 1$ . To justify convexity of the dual domain, write its variables as  $u_{\theta x} = \pi_{\theta} h_{\theta x}$  with  $0 \leq u_{\theta x} \leq \pi_{\theta}$  and  $\sum \pi_{\theta} = 1$ . Regard  $h$  as a reward with action set  $X$ . The first term is the reward of the identity decision in  $E$ , bounded above by its optimal reward; the second is the optimal reward in  $F$ . Thus every dual expression is at most one of the optimal reward differences. Changing reward to loss proves equality.

Here is the approximation needed for general  $X, K$ . A dense sequence  $(\theta_j)$  and positive weights with sum one give  $m = \sum_j w_j P_{\theta_j}$  dominating every  $P_{\theta}$ ; domination follows by total-variation continuity. The set of densities  $\{dP_{\theta}/dm : \theta \in K\}$  is compact in  $L^1(m)$ . Given  $\eta > 0$ , choose a finite  $L^1$  net and approximate its elements by simple functions on a common finite partition  $\mathcal{C}$  of  $X$ . Conditional expectation onto  $\mathcal{C}$  is an  $L^1$  contraction, so the cell reconstruction  $R_{\mathcal{C}}(\cdot | C) = m(\cdot | C)$  satisfies

$$\sup_{\theta} \|R_{\mathcal{C}} P_{\theta} - P_{\theta}\|_{\text{TV}} \leq \eta$$

after choosing the net and approximations sufficiently accurately. Zero-mass cells are irrelevant. Hence  $\delta(F, E) \leq \delta(F, \mathcal{C}E) + \eta$ . Now choose a finite parameter net on which both  $F$  and  $\mathcal{C}E$  vary by at most  $\eta$  in total variation. The supremum reconstruction error on  $K$  differs from that on the net by at most  $2\eta$ , uniformly in the reconstruction. Apply the finite result to this net. Its priors are allowed priors on  $K$ , and observing  $X$  is at least as informative as observing  $\mathcal{C}(X)$ , so the resulting decision differences for  $E$  dominate those for  $\mathcal{C}E$ . It follows that the left side of (23.7) is at least  $\delta(F, E) - 3\eta$ . Let  $\eta \downarrow 0$ . Equation (23.8) now follows by taking the infimum over  $Q$ , with no exchange of quantifiers.  $\square$

## APPENDIX 24. STATE-COSTED COMPARISON AND COMPLETE-STATE FUSION

An experiment is not identified with an implementation of its observation channel. Two equivalent experiments can expose the same information at different times. The cost of carrying that information between those times is the subject of this section. All losses in this section take values in  $[0, 1]$ ; a bound  $L$  multiplies every error term by  $L$ .

**24.1. The information interface and priced randomness.** Fix a finite horizon  $T$ , finite control and terminal decision alphabets, a parameter set, a common marked target  $W$ , and a family  $J$  of decision rows. A row specifies a parameter and a task. The task is available to its controller and terminal decision rule, but not to the common encoder. The parameter is unavailable to all three. The timing is the timing of Section 23: a report is consumed once, and only the

retained register is passed to the next checkpoint. Any current side observation supplied afresh by the instrument is part of the interface and is specified separately. It is not silently included in the register.

Write  $S = (S_0, \dots, S_T)$  for the private register bounds. A *public seed of size  $q$*  is an independent random variable with at most  $q$  values, retained and readable by the encoder and the controllers. Its law is chosen as part of the design. Fresh private randomization remains available. The notation  $q = \infty$  permits one independent uniform random variable on  $[0, 1]$ , with no bound on its information content. A seed is never allowed to depend on the source or parameter. Let

$$\mathcal{R}_E(S, q) = \{(r_j(\pi))_{j \in J} : \pi \text{ is an admissible common-encoder design on } E\}.$$

For continuous risk functions on compact  $J$ , put

$$\mathcal{U}_E(S, q) = \mathcal{R}_E(S, q) + C(J)_+, \quad h_E(S, q; w) = \inf_{r \in \mathcal{R}_E(S, q)} \int_J r_j w(dj), \quad w \in \mathcal{P}(J).$$

The lower support value  $h$  does not assert convexity of the private risk set. In particular,  $q = 1$  is not replaced by  $q = \infty$  in this notation.

**Definition 24.1** (Primitive marked simulator). A simulator  $\Gamma : E \rightarrow F$  consumes the reports of  $E$  and produces the reports of  $F$  with the same timing and control interface. At checkpoint  $t$  it retains  $J_t \in \{1, \dots, K_t\}$ , and it uses a public seed with at most  $r$  values. Its update and report kernels depend only on its old state, the current input, the currently supplied side observations, and the issued control. They do not depend on the task or parameter. The simulator has error  $\varepsilon$  when, for every parameter and every admissible randomized feedback strategy at the output interface, the total variation distance between the simulated and true joint laws of the output transcript and the *actual* mark  $W$  is at most  $\varepsilon$ . The same requirement applies after adjoining independent randomization used by the output strategy. The simulator neither observes an unavailable mark nor replaces it by an independent draw.

This definition is a constraint on a causal channel, not a claim that a channel exists whenever certain decision inequalities hold. It is often verified by a parameter-uniform coupling or by factoring the actual marked law. The abstract simulation hypothesis of Theorem 23.2 is thereby supplied with an explicit machine and a charged state space.

**Theorem 24.2** (Resource-costed comparison). *Suppose  $\Gamma : E \rightarrow F$  is a primitive simulator of cost  $(K, r)$  and error  $\varepsilon$ . Then, for every target width  $S$  and seed bound  $q$ , every target design has a source implementation retaining only reachable pairs  $(J_t, I_t)$ . In particular its width is at most  $K_t S_t$  and its public seed has at most  $r q$  values. Its risk differs from the target risk by at most  $\varepsilon$  in every row. Consequently*

$$\mathcal{U}_F(S, q) + \varepsilon \mathbf{1} \subseteq \mathcal{U}_E(KS, rq), \quad (24.1)$$

$$h_E(KS, rq; w) \leq h_F(S, q; w) + \varepsilon. \quad (24.2)$$

*Products involving  $\infty$  have their natural meaning. Two composable simulators of costs  $(K, r)$  and  $(L, s)$  and errors  $\varepsilon, \eta$  have a composite of cost  $(KL, rs)$  and error at most  $\min(1, \varepsilon + \eta)$ . Identity simulators have cost  $(1, 1)$  and error zero.*

*Proof.* Keep  $(J_{t-1}, I_{t-1})$  and the two independent seed indices. The target controller uses  $I_{t-1}$  to issue the next control. The simulator consumes the source input, draws its synthetic report and new internal state, and passes that report to the target encoder. The encoder draws  $I_t$  and then discards the report. This is one transition on the finite product state space; no earlier input is retained. The common encoder remains common because the simulator does not use the task and the only task information entering its transition is the legitimately issued control. Unreachable

pairs may be removed, which proves the stated sharper implementation bound as well as the product bound.

Definition 24.1 applies to this feedback strategy. Applying a bounded terminal loss, or a bounded measurable loss of the specified marked transcript, changes its expectation by at most total variation. The same simulated design works for all rows; it is not chosen after the row is known. This proves the risk assertion. If  $r_E$  and  $r_F$  are the resulting risks, then  $r_F + \varepsilon \mathbf{1} - r_E \geq 0$ , which gives (24.1); integration and an infimum give (24.2).

For composition retain both simulator states and both seed indices. Compare the composite first with the second simulator fed by the exact intermediate experiment. Data processing does not increase the first error. The second comparison costs at most  $\eta$ . This argument is valid under the feedback of the final consumer, since that consumer and the second simulator together are an admissible feedback strategy at the intermediate interface. The triangle inequality proves the error bound.  $\square$

*Remark 24.3* (Local error certificates). If the two conditional transition laws can be coupled at step  $t$  with failure probability at most  $e_t$ , conditional on every previously matched history, and the terminal conditional mark can be coupled with failure probability at most  $e_W$ , then the simulator error is at most  $1 - (1 - e_W) \prod_t (1 - e_t)$ . Indeed the probability that all couplings succeed is at least the product. A bound only for uncontrolled one-step marginals is not such a certificate: the conditional and feedback quantifiers are necessary.

**24.2. Deficiency frontiers and spectra.** Define  $\delta_{K,r}(E, F)$  as the infimum of the errors of the simulators in Definition 24.1 with the indicated cost bounds. Set the value to  $+\infty$  when there is no such simulator. No attainment is presupposed.

**Corollary 24.4** (Composition and decision witnesses). *For compatible interfaces,*

$$\delta_{KL,rs}(E, G) \leq \delta_{K,r}(E, F) + \delta_{L,s}(F, G).$$

*Moreover each choice of  $S, q, w$  gives the lower bound*

$$\delta_{K,r}(E, F) \geq [h_E(KS, rq; w) - h_F(S, q; w)]_+.$$

*If the two upper risk sets under consideration are closed and convex, (24.1) is equivalent to (24.2) for every  $w \in \mathcal{P}(J)$ . This last equivalence concerns risk sets, and is not an inverse theorem constructing a simulator.*

*Proof.* Use simulators within an arbitrary positive tolerance of each finite infimum, apply Theorem 24.2, and let the tolerance decrease to zero. For the risk-set assertion, separate a point outside the closed convex upper set from that set. Any separating continuous linear functional must be nonnegative on  $C(J)_+$ , since the set is upper. It is therefore integration against a finite positive measure. Normalize its mass to one. The resulting separation contradicts the asserted lower support inequalities. The other implication is immediate.  $\square$

For a fixed bounded loss family, let  $b_{E,j}$  be the full-history Bayes risk with the same observation timing and parameter-blind decision rules, and define the centered minimax spectrum

$$\Theta_E(S, q) = \inf_{r \in \mathcal{R}_E(S, q)} \sup_{j \in J} (r_j - b_{E,j}).$$

When the unrestricted baseline is defined using a prior rather than a single parameter row, the same argument applies to that prior throughout. If exact full-history simulation is possible in both directions, these baselines agree. More generally, forward and reverse simulation errors  $\varepsilon, \eta$  imply  $b_{F,j} - b_{E,j} \leq \eta$ , and hence

$$\Theta_E(KS, rq) \leq \Theta_F(S, q) + \varepsilon + \eta. \quad (24.3)$$

Here the baseline inequality follows by simulating any source full-history rule on the target and taking its infimum. Thus neither a parameter oracle nor a change of timing enters the centering. For a fixed horizon and two exact simulations with fixed finite costs, a scalar budget  $B$  controlling all  $S_t$  and the charged finite seed index is changed by at most constant factors. Whenever a positive spectrum has a power exponent, the inequalities in both directions preserve that exponent. A growing horizon with growing  $K_t$  is not covered by this constant-factor assertion.

**Proposition 24.5** (Public-to-private conversion and finite seed cost). *A design with a  $q$ -valued public seed has an exactly equivalent privately randomized implementation of width  $qS_t$ : store the sampled index with the private state. Suppose, in addition, that a family of risks satisfies  $|r_j(\pi) - r_{j'}(\pi)| \leq \omega(d(j, j'))$  for all private designs, where  $J$  is compact. For an ideal public design and any  $h, \zeta > 0$ , there is a public mixture of at most  $N(J, h) + 1$  private designs whose uniform risk error is at most  $\zeta + 2\omega(h)$ . Charging this index gives private width at most  $(N(J, h) + 1)S_t$ .*

*Proof.* The exact conversion samples the seed before observing the source and makes its index part of every register. Since the controller reads that register, all the original kernels can be executed. This is not a claim that  $qS_t$  is always necessary.

Choose an  $h$ -net of  $N = N(J, h)$  rows. At these rows the risk vector of the ideal design is an expectation of vectors in  $[0, 1]^N$ , and hence belongs to the closure of their convex hull. Approximate it within  $\zeta$  by a finite convex combination. Caratheodory's theorem reduces this combination to at most  $N + 1$  actual private designs without changing those  $N$  coordinates. Move from an arbitrary row to a net row in the target and the approximation; each move costs at most  $\omega(h)$ . This proves the claim without assuming that the private risk set itself is closed.  $\square$

**24.3. The marked quotient as a zero-overhead object.** For this subsection use the finite marked instrument of Section 29. Let  $c_t(h)$  be its marked continuation class,  $K_t$  the number of supported classes, and  $Q(E)$  the class experiment of Theorem 29.5. It has class update  $F_t(c, a, y)$  and the parameter-independent reverse report kernel  $L_t(y \mid c, a, c')$  constructed there.

**Definition 24.6** (Complete retained design). A design is complete when, at every checkpoint, there is a map  $\phi_t$  such that  $c_t(H_t) = \phi_t(U, I_t)$  almost surely for every supported preparation and open-loop control word. With a finite instrument the same maps can be chosen outside one seed-null set, simultaneously for all supported histories. On  $Q(E)$  the corresponding condition is  $C_t = \phi_t(U, I_t)$ . Let  $\mathcal{R}^c$  denote the risk body restricted to these designs.

Completeness is equivalent to preserving all the restart Brier risks of Theorem 29.3. It is not imposed on an arbitrary finite list of decision tasks.

**Theorem 24.7** (Complete-state fusion). *For the finite marked instrument, every width vector and seed signature satisfy*

$$\mathcal{R}_E^c(S, q) = \mathcal{R}_{Q(E)}^c(S, q). \quad (24.4)$$

*Both sides are empty if  $S_t < K_t$  at any required checkpoint. For general, not necessarily complete, designs the two exact simulations give instead*

$$\begin{aligned} \mathcal{U}_{Q(E)}(S, q) &\subseteq \mathcal{U}_E(KS, q), \\ \mathcal{U}_E(S, q) &\subseteq \mathcal{U}_{Q(E)}(KS, q). \end{aligned}$$

*Thus exact marked equivalence is resource-neutral on complete retained designs, while on arbitrary task designs it has an explicitly charged state distortion.*

*Proof.* Consider first a complete design on  $E$  to be run on  $Q(E)$ . At the start of a step recover the old class  $c$  from  $(U, I_{t-1})$ . When the class instrument supplies  $c'$ , sample  $y$  from  $L_t(\cdot \mid c, a, c')$  and perform the original encoder update with  $y$ . The factorization in Theorem 29.5 shows that this report has the correct conditional law, jointly with the actual future marked experiment. In

particular every positive-probability updated register symbol has class  $c'$ , by completeness of the original design. The new register therefore again recovers the new class. The sampled report and the old class need not be retained separately. Induction gives an implementation with precisely the original private labels and the original seed.

Conversely, start with a complete design on  $Q(E)$  and an actual report  $y$  from  $E$ . Recover  $c$  from the old register, compute  $c' = F_t(c, a, y)$ , and apply the class encoder with that input. Its new register recovers  $c'$ . The class update and terminal marked channel are exactly those of the class experiment, so induction again proves equality of all marked risks. At the initial checkpoint use the stipulated initial class (or the same class construction after the initial report). Neither construction inserts an additional persistent state.

The lower bound is Theorem 29.3, applied conditionally on the independent seed; the finite supported-history set permits a common null set. Finally, without completeness one retains  $c$  separately. The forward class tracker and the reverse report transducer each need at most  $K_t$  states, and Theorem 24.2 gives the last two inclusions.  $\square$

**Example 24.8** (A sharp temporal state charge). Let  $X$  be uniform on  $\{1, \dots, k\}$  and let  $W = X$ . Experiment  $E$  reports  $X$  at time one and a constant at time two; experiment  $F$  reports  $X$  at both times. These marked experiments simulate each other exactly. The forward simulator  $E \rightarrow F$  stores  $X$  between reports; the reverse one is stateless. At widths  $(S_1, S_2) = (s, k)$ , the terminal error for identifying  $W$  is zero on  $F$ , whereas on  $E$  it is at least  $1 - s/k$  when  $s \leq k$ , with equality attainable. Indeed, for a private message channel  $Q(m | x)$ , its maximal guessing success is

$$\frac{1}{k} \sum_m \max_x Q(m | x) \leq \frac{s}{k}.$$

Independent public randomization averages this bound and cannot improve it. Partitioning the  $k$  inputs into  $s$  nonempty cells attains it. The same argument applied to a simulator which must reproduce the second report shows that its exact intermediate state count is at least  $k$ . The marked continuation quotient has  $k$  states. A complete consumer already carries them, and Theorem 24.7 eliminates the additional factor for exactly that reason.

Theorem 24.2, the lower support characterization in Corollary 24.4, and Theorem 24.7 are the resource-aware comparison statement of this article. The compactness results below supply cases in which its dual risk invariant is attained; the task-completion results identify which exact state requirement a particular decision family imposes.

## APPENDIX 25. CAUSAL TASK COMPLETION ON THE ACTUAL SUPPORT

A finite task family need not induce an equivalence relation on histories. Moreover, a report which is impossible after one history must not be confused with a report having a prescribed continuation there. We give a support-sensitive characterization which addresses both issues at once. The underlying device is the compatible-cover construction for partial machines. The point here is its exact relation to a common encoder, randomized resource signatures, and the zero face of the decision spectrum.

**25.1. Supported trees and simultaneous decisions.** Let  $\mathcal{H}_t$ ,  $0 \leq t \leq T$ , be finite sets of observable histories. The initial history  $h \in \mathcal{H}_0$  is consumed by the initial encoder. Controls  $a \in \mathcal{A}_t$  and reports  $y \in \mathcal{Y}_{t+1}$  have finite alphabets. For each  $h$  and  $a$ , a nonempty set of successors  $hay$  is specified in  $\mathcal{H}_{t+1}$ . These are precisely the positive-probability successors in a fixed reference preparation of the controlled instrument. The reference preparation charges every history under consideration. Zero-probability successors are absent; their probabilities are not changed. Controls in the restart tests are fixed open-loop words. This ensures that the encoder's private randomization

remains independent of the source conditional on the supplied history. Conditional marked laws, not a parameter oracle, define all the losses below.

At time  $t$  there is a finite task set  $D_t$ . For  $d \in D_t$  let  $B_{td}$  be a finite decision alphabet and let  $\ell_{td}(h, b) \in [0, 1]$  be the conditional loss, obtained from the actual marked continuation and any continuation protocol specified by the task. Put

$$m_{td}(h) = \min_b \ell_{td}(h, b), \quad A_{td}(h) = \operatorname{argmin}_b \ell_{td}(h, b).$$

Choose a preparation  $\eta_t(h) > 0$  on every  $\mathcal{H}_t$ . The full-history restart benchmark is  $\sum_h \eta_t(h) m_{td}(h)$ . At every checkpoint the same source-consuming encoder is used for all tasks. Its initial kernel is  $q_0(m | h)$ , its updates are  $q_{t+1}(m' | m, a, y)$ , and its task readouts are  $v_{td}(b | m)$ . Time dependence is allowed and is not persistent memory. Write  $Q_t(m | h)$  for the message distribution obtained by running these kernels along the supplied history. An independent public seed, if present, is an additional named input to these kernels.

A set  $C \subseteq \mathcal{H}_t$  is *task compatible* if

$$\bigcap_{h \in C} A_{td}(h) \neq \emptyset \quad \text{for every } d \in D_t. \quad (25.1)$$

Empty sets are permitted. Define its supported successor set by

$$\operatorname{Succ}_{a,y}(C) = \{hay \in \mathcal{H}_{t+1} : h \in C\}.$$

**Definition 25.1** (Causally closed compatible cover). A width profile  $S = (S_0, \dots, S_T)$  admits a causally closed compatible cover if there are compatible sets  $C_{t,m}$ ,  $1 \leq m \leq S_t$ , which cover  $\mathcal{H}_t$  and satisfy the following condition: for every  $t < T$ ,  $m, a, y$  there is one index  $\delta_t(m, a, y)$  such that

$$\operatorname{Succ}_{a,y}(C_{t,m}) \subseteq C_{t+1, \delta_t(m, a, y)}. \quad (25.2)$$

The sets may overlap. The single index in (25.2) must work for all histories in the cell for which this successor exists.

**Theorem 25.2** (Exact causal task spectrum). *For a finite supported instrument and the finite restart tasks above, the following conditions on  $S$  are equivalent:*

- (1) *a deterministic common encoder with width profile at most  $S$  attains every full-history restart benchmark at every checkpoint;*
- (2) *a privately randomized common encoder does so, with or without an independent public seed of any prescribed positive size, including an ideal uniform seed;*
- (3)  *$S$  admits a causally closed compatible cover.*

*The equivalence is independent of the strictly positive preparations  $\eta_t$ . In particular the exact feasible width profiles are not changed by private or independent public randomization. No positivity assumption is imposed on a next-report probability which is absent from the actual support.*

*Proof.* For a fixed task the excess risk is

$$\sum_{h, m, b} \eta_t(h) Q_t(m | h) v_{td}(b | m) [\ell_{td}(h, b) - m_{td}(h)]. \quad (25.3)$$

It is a sum of nonnegative terms. If it vanishes for all tasks, then  $C_{t,m} = \{h : Q_t(m | h) > 0\}$  is compatible: every decision in the nonempty support of  $v_{td}(\cdot | m)$  belongs to every  $A_{td}(h)$  with  $h \in C_{t,m}$ . The cells cover the supported histories.

For fixed  $t, m, a, y$ , select any  $m'$  with  $q_{t+1}(m' | m, a, y) > 0$ . If  $hay$  is a supported successor of some  $h \in C_{t,m}$ , the recursion for  $Q$  includes the positive product  $Q_t(m | h) q_{t+1}(m' | m, a, y)$ . Consequently  $hay \in C_{t+1, m'}$ , proving (25.2). There is no inference about a history  $hay$  outside the support.

Conversely, choose an initial containing cell for each  $h \in \mathcal{H}_0$ . Use the maps  $\delta_t$  as deterministic updates. Induction on time shows that the retained cell always contains the actual history. For each cell and task choose an action from the intersection in (25.1). All terms in (25.3) are zero. Empty cells and empty successor sets have arbitrary decisions and updates. This constructs one machine attaining the entire width profile, rather than unrelated checkpoint statistics.

For a public seed  $U$ , condition the nonnegative expression (25.3) on  $U$ . There are finitely many checkpoints and tasks. Outside one seed-null set all their conditional excesses vanish. Because the restart preparation and the open-loop controls are independent of  $U$ , they still charge the same supported histories. Fixing one seed in this set gives a private machine to which the preceding argument applies. The converse may ignore the seed. Only positivity, not the numerical values, of  $\eta_t$  has been used.  $\square$

The theorem is a zero-excess purification statement, not purification of the entire private risk body. Positive excess can benefit from private or public randomization, as Theorem 33.1 demonstrates. Nor does a family of arbitrary checkpoint-optimal covers automatically satisfy (25.2).

**Example 25.3** (Support carries information without an extra register). Let the first report be a bit  $X$ , and let the next report again be  $X$ . The initial task has a constant optimal action and the terminal task is to predict  $X$  exactly. A one-cell initial cover and two terminal cells are causally closed: report  $y$  selects its own terminal cell. Thus  $(S_0, S_1) = (1, 2)$  is feasible. Replacing the second report by a constant leaves the separate task widths equal to 1 and 2, but the profile  $(1, 2)$  is no longer feasible. Its initial cell has both supported successors under the same report, and these require different decisions. The information timing, not merely a list of checkpoint cover numbers, determines the causal answer.

**Example 25.4** (Overlapping cells and no coarsest partition). There are three initial histories  $u, v, w$  with constant initial tasks. History  $u$  has only successor report 0, with terminal target 0;  $v$  has only report 1, with terminal target 0; and  $w$  has only report 0, with terminal target 1. The initial cover  $\{u, v\}, \{v, w\}$ , together with terminal target cells, is causally closed. Both partitions  $\{u, v\} \mid \{w\}$  and  $\{u\} \mid \{v, w\}$  are feasible, whereas their common coarsening is not. Hence no single coarsest feasible partition describes this problem. A fictitious signature for an unsupported successor would impose a condition absent from the experiment.

**25.2. A separating risk gap and prediction completion.** Choose positive numbers  $\beta_{td}$  with  $\sum_{t,d} \beta_{td} = 1$  and form the aggregate excess

$$G(\pi) = \sum_{t,d} \beta_{td} [r_{td}(\pi) - \sum_h \eta_t(h) m_{td}(h)].$$

This is a positive decision-row witness for the common risk body. Let

$$\alpha = \min_{t,d,h} \beta_{td} \eta_t(h) > 0.$$

If a strictly positive entry regret exists, let  $\gamma$  be the least such regret among  $\ell_{td}(h, b) - m_{td}(h)$ .

**Corollary 25.5** (Zero face and a finite decision witness). *If the cover condition holds, the minimum of  $G$  and the minimax restart excess are zero. If it fails, a positive entry regret exists and, for all private or independently public-randomized machines of width  $S$ ,*

$$G(\pi) \geq \alpha\gamma, \quad \max_{t,d} [r_{td}(\pi) - \sum_h \eta_t(h) m_{td}(h)] \geq \alpha\gamma. \quad (25.4)$$

*Thus causally closed covers characterize exactly the zero face of the common-encoder spectrum, with a strictly positive finite witness off that face.*

*Proof.* Fix all entries of a deterministic time-indexed encoder and all task readouts. Infeasibility means that at least one supported history at one checkpoint has a nonoptimal decision for some task. Its contribution to  $G$  is at least  $\alpha\gamma$ . If every entry regret were zero, the one-cell cover at every time would work, so  $\gamma$  exists when needed.

A behavioral private design is a mixture, for the purpose of computing its risks, of deterministic tables: sample an independent value for each initial-encoder row, each time-indexed update row, and each time/task readout row according to the design's specified kernels. Along any fixed history each such time-indexed row is used at most once, so averaging these tables reproduces every expression (25.3). A public seed can first be conditioned upon and then integrated out. Every deterministic table has the lower bound, and hence so does their mixture. This use of a table mixture is a proof of an inequality; it does not grant its table index to an admissible private encoder. The second inequality follows because  $G$  is a convex combination of task excesses.  $\square$

**Corollary 25.6** (Brier tasks and universal marked completion). *Theorem 25.2 also holds for finitely many Brier tasks whose conditional probabilities are  $p_{td}(h)$  and whose actions range over  $[0, 1]$ . A cell is then compatible precisely when each probability is constant on it. If the cover condition fails, let  $\delta > 0$  be the least positive difference between probabilities for a common time and task at two histories. Then every randomized design satisfies*

$$G(\pi) \geq \alpha\delta^2/2.$$

*For a finite marked instrument, completing the tests by a finite pair-separating family of all marked continuation-event predictions recovers the universal marked quotient and its least width profile. With strictly positive report probabilities this specializes to the signature construction in Theorem 30.4.*

*Proof.* The conditional Brier excess is  $(b - p_{td}(h))^2$ . Zero excess forces the readout distribution to be concentrated at  $p_{td}(h)$  on each message support. The support and transition proof is otherwise unchanged. For a fixed deterministic encoder, if all its cells had constant probabilities, they would form a causally closed compatible cover. In the infeasible case some cell contains two histories with probabilities  $p, p'$  separated by at least  $\delta$  for one task. For any readout  $b$ , their weighted contribution is at least  $\alpha[(b - p)^2 + (b - p')^2] \geq \alpha\delta^2/2$ . Integration over randomized readouts and the finite encoder-table mixture proves the bound. Finally, equality of all marked continuation-event probabilities is precisely equality of the continuation law. Its supported class transitions are well defined by Lemma 29.2; different classes cannot share a cell. The marked quotient tracker attains these class counts. A finite pair-separating subfamily suffices because there are finitely many classes.  $\square$

### 25.3. Coherent exact machines on an unbounded time interval.

**Corollary 25.7** (Finite-prefix criterion). *Let the supported tree have infinitely many levels but finitely many histories, controls, reports, and finite-action tasks at each level. Fix finite widths  $S_t$  in advance. One deterministic infinite machine attains all restart benchmarks if and only if every finite prefix admits a causally closed compatible cover with those widths. Independent public or private randomization does not change this criterion.*

*Proof.* The deterministic initial table and every time-indexed update and readout table take values in finite sets. Their countable product is compact. Exactness through time  $N$  defines a closed cylinder set of tables. These sets are nested and are nonempty by Theorem 25.2 and arbitrary extension of a finite table. Compactness gives a point in their intersection. The reverse implication is restriction. For a randomized infinite machine, restriction to each prefix gives the covers by the finite theorem. The argument is an existence theorem; it asserts neither a finite stopping rule for an infeasible infinite problem nor an effective bound on the prefix which witnesses infeasibility.  $\square$

## APPENDIX 26. COMPACT DUALITY FOR DOMINATED CAUSAL EXPERIMENTS

We begin with a nonfinite experiment. The observation spaces in this section need not be locally compact, and the encoder kernels need not be continuous in the observation. The hypothesis used in place of such regularity is domination of the complete marked observation law.

**26.1. The experiment and its information pattern.** Fix a horizon  $T$ , finite control sets  $A_t$ , nonempty finite registers  $I_t = [S_t]$ , and a finite task set  $D$ . Task  $d$  has a finite decision set  $B_d$ . Let  $Y_t$  and  $W$  be standard Borel spaces, with probability reference measures  $\mu_t$  and  $\nu$ , and write  $\mu = \bigotimes_{t=1}^T \mu_t$ . The mark  $W$  is the actual terminal quantity on the source probability space; it is not a reconstructed sample with the same marginal distribution.

A causal experiment is specified by a consistent family of laws of  $(Y_1, \dots, Y_T, W)$  under open-loop action words  $a = (a_0, \dots, a_{T-1})$ , indexed by a parameter  $\theta \in \Theta$ . Consistency means that the marginal law of  $Y_{1:t}$  is independent of  $a_t, \dots, a_{T-1}$ , and that feedback laws are obtained by inserting their conditional action probabilities in the open-loop laws. Suppose that these laws have densities

$$p_{\theta,a}(y, w) \quad \text{with respect to } \mu(dy)\nu(dw). \quad (26.1)$$

This hypothesis is satisfied by dominated hidden Markov instruments, including continuous observation models. It excludes, for example, an unnoised repeated real observation supported on a diagonal of product Lebesgue measure. We do not replace domination by an unproved closure claim for arbitrary strategic measures.

The encoder is a single family

$$q_0 \in \mathcal{P}(I_0), \quad q_t(i' \mid i, a, y) \in \mathcal{P}(I_{t+1}), \quad 0 \leq t < T,$$

where  $y \in Y_{t+1}$ . It is common to every task and every parameter. The task controller and terminal rule are  $g_{td}(a \mid i)$  and  $v_d(b \mid i)$ ; neither depends on  $\theta$ . The task is known to the controller, not supplied as an additional argument to  $q_t$ . The current control and observation are transient inputs. Between updates, the only private retained state is  $i \in I_t$ . Time and the prescribed tables are part of the fixed program. All private draws are fresh and conditionally independent given their arguments. A retained independent public seed, when allowed, is a separate resource. These conventions rule out both a hidden controller transcript and an uncharged retained random tape.

Let  $J = D \times \Theta$  be compact metrizable, with  $D$  discrete and  $\Theta$  compact metrizable. More generally the argument works for a compact row set with a continuous map into the finite task set. For  $j = (d, \theta)$  and  $b \in B_d$ , let  $0 \leq \ell_j(w, b) \leq 1$  be measurable. Our regularity condition is

$$f_{j,a,b}(y) := \int \ell_j(w, b) p_{\theta,a}(y, w) \nu(dw), \quad j \mapsto f_{j,a,b} \text{ continuous in } L^1(\mu) \quad (26.2)$$

on each task component, for every  $a, b$ . A continuous baseline  $b_0 \in C(J)$  is fixed. A full-history minimax baseline for task  $d$  is constant on its parameter rows, and is one permitted choice; a parameterwise oracle is a different baseline and is not substituted for it.

Write  $K_S$  for the space of all such private designs  $k = (q_0, q, g, v)$ . The risk of a design is

$$R_j(k) = \sum_{i_0, \dots, i_T} \sum_{a,b} q_0(i_0) v_d(b \mid i_T) \prod_{t=0}^{T-1} g_{td}(a_t \mid i_t) \int f_{j,a,b}(y) \prod_{t=0}^{T-1} q_t(i_{t+1} \mid i_t, a_t, y_{t+1}) \mu(dy). \quad (26.3)$$

The sums over  $b$  are in  $B_d$ . This expression follows by first fixing the control and register paths in the causal law. In particular it does not optimize the parameter rows separately.

**26.2. The topology and an executable realization.** Each encoder row is a simplex-valued element of a finite product of  $L^\infty(\mu_{t+1})$ , endowed with its weak star topology. Positivity and the constraint that its coordinates sum to one are weak star closed: test them against nonnegative  $L^1$  functions. Banach–Alaoglu therefore makes the row space compact. Since the probability space is

standard Borel,  $L^1$  is separable, and this compact space is metrizable. Give the finite controller, initialization and decoder simplices their usual topologies, and  $K_{\mathbf{S}}$  the product topology.

**Lemma 26.1** (Tensor testing). *Suppose  $u_{tn}, u_t \in L^\infty(\mu_t)$  have absolute value at most one and  $u_{tn} \rightarrow u_t$  weak star, for each  $t$ . Then*

$$\int F(y) \prod_t u_{tn}(y_t) \mu(dy) \longrightarrow \int F(y) \prod_t u_t(y_t) \mu(dy)$$

for every  $F \in L^1(\mu)$ . Convergence is uniform over every compact subset of  $L^1(\mu)$ .

*Proof.* For  $F(y) = \prod_t F_t(y_t)$  with  $F_t \in L^1(\mu_t)$  the integrals factor, and each factor converges by the definition of the topology. Finite sums of these tensors are dense in  $L^1(\mu)$ : rectangular simple functions generate the product sigma field, and a monotone class argument followed by truncation gives the density statement. An  $L^1$  error of  $\epsilon$  in the test function changes either integral by at most  $\epsilon$ . This proves convergence for one  $F$ . A finite  $\epsilon$ -net of a compact family proves the uniform assertion.  $\square$

**Lemma 26.2** (Jointly measurable representatives). *The encoder row space has a choice of representatives  $(q, y) \mapsto \hat{q}(y)$  that is jointly Borel, is simplex valued at every  $y$ , and represents each equivalence class  $q$  almost everywhere. Consequently a Borel probability on  $K_{\mathbf{S}}$  can be sampled as one independent public seed and implemented by measurable causal update rules.*

*Proof.* Choose increasing finite measurable partitions generating the report sigma field modulo completion. The coordinates of  $\mathbb{E}[q \mid \mathcal{F}_n](y)$  are finite sums of functions of the form

$$\mathbf{1}_C(y) \frac{\int_C q d\mu_t}{\mu_t(C)}, \quad \mu_t(C) > 0.$$

On each zero-measure partition cell use one fixed probability vector. Their coefficients are weak star continuous in  $q$ , so these functions are jointly Borel. For fixed  $q$ , the bounded martingale convergence theorem gives coordinatewise convergence to a representative of  $q$  almost everywhere. On the jointly Borel set where all coordinates converge use their limit, and elsewhere use one fixed probability vector. Every finite-stage vector is in the simplex, so is the limit. There are only finitely many rows. A sampled  $k$  is thus executable by inverse transforms with fresh independent uniform variables. Domination ensures that exceptional report sets have zero probability for every actual action path and parameter. Compact metrizability also ensures that every probability on  $K_{\mathbf{S}}$  is the law of a Borel function of a uniform seed. The sampled design is retained publicly; this last retention is not free private memory.  $\square$

**Theorem 26.3** (Nonfinite common-encoder duality). *Under (26.1)–(26.2), the map  $R : K_{\mathbf{S}} \rightarrow C(J)$  is continuous in the uniform norm. The attainable shared-seed risk set and its upper extension are*

$$\mathcal{A}_{\mathbf{S}} = \left\{ \int R(k) \lambda(dk) : \lambda \in \mathcal{P}(K_{\mathbf{S}}) \right\}, \quad \mathcal{U}_{\mathbf{S}} = \mathcal{A}_{\mathbf{S}} + C(J)_+.$$

The first is compact and convex and the second is closed and convex. Set

$$h_{\mathbf{S}}(w) = \min_{k \in K_{\mathbf{S}}} \int_J R_j(k) w(dj), \quad w \in \mathcal{P}(J).$$

Then  $u \in \mathcal{U}_{\mathbf{S}}$  if and only if  $\int u dw \geq h_{\mathbf{S}}(w)$  for every  $w \in \mathcal{P}(J)$ , and

$$\min_{\lambda \in \mathcal{P}(K_{\mathbf{S}})} \max_{j \in J} \left\{ \int R_j(k) \lambda(dk) - b_0(j) \right\} = \max_{w \in \mathcal{P}(J)} \left\{ h_{\mathbf{S}}(w) - \int b_0 dw \right\}. \quad (26.4)$$

Both extrema are attained. The private minimax optimum is also attained, but its attainable risk set is  $R(K_{\mathbf{S}})$ , not its convex hull.

*Proof.* Compactness of  $K_{\mathbf{S}}$  was proved above. Apply Lemma 26.1 to each term of (26.3). The effect functions form finitely many compact subsets of  $L^1$ , by (26.2). Thus convergence of designs gives uniform convergence of all row risks. This proves the first assertion without enumerating deterministic encoders.

For completeness, the barycenter map is continuous from weak convergence of  $\lambda$  to norm convergence in  $C(J)$ . Approximate the continuous map  $R$  uniformly on compact  $K_{\mathbf{S}}$  by a finite-valued continuous partition-of-unity interpolation. Integrals of the finitely many continuous coefficients converge; the uniform approximation controls the remainder. The image of compact  $\mathcal{P}(K_{\mathbf{S}})$  is therefore compact. Lemma 26.2 identifies this image with executable public mixtures, rather than only a formal relaxed set. Convexity follows by mixing the public seed.

If  $u_n = a_n + c_n \rightarrow u$ ,  $a_n \in \mathcal{A}_{\mathbf{S}}$  and  $c_n \geq 0$ , pass to a convergent subsequence of  $a_n$ . Its limit  $a$  satisfies  $u - a \geq 0$ , proving closedness of  $\mathcal{U}_{\mathbf{S}}$ . A separating continuous functional for a point outside this closed convex upper set must be nonnegative: otherwise translating an attainable risk in a nonnegative direction contradicts its lower separating bound. By the Riesz representation theorem this functional is integration against a finite positive measure, which may be normalized to a probability. Minimizing it over the upper set gives  $h_{\mathbf{S}}$ . This proves the support characterization.

The bilinear payoff

$$F(\lambda, w) = \int_J \int_{K_{\mathbf{S}}} R_j(k) \lambda(dk) w(dj) - \int_J b_0(j) w(dj)$$

is continuous separately on the two compact convex probability spaces. The classical convex minimax theorem gives equality of the two extrema. For fixed  $w$ , a linear functional on probabilities has minimum at a Dirac probability, which gives the inner minimum in (26.4). The left objective is continuous and the right objective, as an infimum of continuous functions of  $w$ , is upper semicontinuous. Compactness gives attainment. Finally the private objective is continuous on compact  $K_{\mathbf{S}}$ . No convexification was used for that assertion.  $\square$

The use of policy topologies and static reduction belongs to stochastic team theory; see Saldi [55], especially the weak star construction and continuity/existence results, and Yüksel–Saldi [56], who explain why conditional independence is not generally closed under weak convergence. Here the product domination and finite action-path expansion supply the needed continuity, while the same encoder coordinates remain coupled across all tasks and unknown-parameter rows. Convex separation and minimax are classical tools, not additional novelty claims.

**Corollary 26.4** (Effect formulation with arbitrary marks). *Theorem 26.3 still holds when there is no common reference measure for  $W$ , provided each open-loop report marginal is dominated by  $\mu$  and the signed effect measures*

$$C \mapsto \int_{C \times W} \ell_j(w, b) P_{\theta, a}(dy, dw)$$

*have  $L^1(\mu)$  densities satisfying (26.2). The actual joint marked laws, not arbitrarily selected couplings, are used to define these effects.*

*Proof.* Boundedness of the loss and report domination give absolute continuity of each effect measure. Disintegration of the actual marked law, or simply integration first in that measure, gives (26.3). All compactness and separation arguments in the theorem use only these effects. Report domination still makes the jointly measurable representatives executable. No common measure on  $W$  is needed in any of these steps. By contrast, the marked total-variation approximation theorem below retains its stronger joint-density hypothesis.  $\square$

**Corollary 26.5** (Countably many tasks). *Suppose that the task-row set is countable, every row has the dominated representation above, and every task has finitely many controller and decoder*

entries. No uniform continuity across rows is required. For a bounded baseline sequence, the shared optimum is attained and

$$\min_{\lambda} \sup_j \{\mathbb{E}_{\lambda} R_j - b_0(j)\} = \sup_{w \in \mathcal{P}_f(J)} \left\{ \min_k \sum_j w_j R_j(k) - \sum_j w_j b_0(j) \right\},$$

where  $\mathcal{P}_f(J)$  denotes the finitely supported probabilities. The right side is a supremum, not an asserted maximum.

*Proof.* The countable product of the task-controller and decoder simplices with the common encoder space is compact metrizable. Each risk is continuous by the tensor lemma; the representative construction is unchanged. On  $\mathcal{P}(K_{\mathbf{S}})$ , the objective is a supremum of continuous functions, hence lower semicontinuous and attains its minimum. If every finite subset of row constraints  $\mathbb{E} R_j - b_0(j) \leq r$  is feasible, compactness and the finite intersection property make all constraints feasible simultaneously. Finite-row convex minimax therefore gives the displayed identity, by taking  $r$  strictly above its right side and then decreasing  $r$  to the infimum. This argument makes no uniform finite-seed assertion.  $\square$

**Corollary 26.6** (The price of a finite mixture). *In Theorem 26.3, put*

$$\omega(\delta) = \sup_{k, d_J(j, j') \leq \delta} |R_j(k) - R_{j'}(k)|.$$

Then  $\omega(\delta) \rightarrow 0$ . If  $J$  has a  $\delta$ -net of size  $N$ , every attainable shared risk can be approximated within  $2\omega(\delta)$  in uniform norm by a mixture of at most  $N + 1$  private designs. Such a mixture has a private implementation with registers  $(N + 1)S_t$ , including initialization. If its weights are approximated by a probability vector with denominator  $2^p$ , an additional error at most  $(N + 1)2^{-p}$  suffices.

*Proof.* The compact subset  $R(K_{\mathbf{S}})$  of  $C(J)$  is equicontinuous. Carathéodory's theorem applied to its evaluations on the net gives  $N + 1$  designs whose mixture agrees there with the prescribed risk. Move an arbitrary row to the net; each of the two risks changes by at most  $\omega(\delta)$ . Store the chosen design index with the register label for the private implementation. Approximate weights by rounding down the first  $L - 1$  coordinates and putting the remainder in the last, where  $L \leq N + 1$ . Their total variation distance is at most  $L2^{-p}$ , which bounds the change of every  $[0, 1]$ -valued risk. This approximation can be sampled with  $p$  fair bits; exact non-dyadic weights do not have an asserted fixed worst-case fair-bit cost.  $\square$

**26.3. Finite observation approximations, uniformly in memory.** Let  $\mathcal{F}_{tn}$  be increasing finite partitions generating  $Y_t$ , and retain the actual mark  $W$ . The finite-report experiment  $E_n$  reports only the cell of each observation. Its reverse report channel samples independently from  $\mu_t$  conditioned on that cell. Cells of zero reference measure are assigned arbitrary outputs and have zero probability in the experiment. Write

$$p_{\theta, a}^{(n)} = \mathbb{E}_{\mu \otimes \nu} [p_{\theta, a} \mid \bigotimes_t \mathcal{F}_{tn} \otimes \mathcal{B}(W)], \quad \delta_n = \frac{1}{2} \sum_a \sup_{\theta} \|p_{\theta, a} - p_{\theta, a}^{(n)}\|_1.$$

**Theorem 26.7** (Width-uniform approximation). *If each family  $\{p_{\theta, a} : \theta \in \Theta\}$  is compact in  $L^1(\mu \otimes \nu)$ , then  $\delta_n \rightarrow 0$ . For every width vector, private or shared randomness, and every task family with losses in  $[0, 1]$ , the raw minimax risks of  $E$  and  $E_n$  differ by at most  $\delta_n$ . The same comparison of excess risks costs at most  $2\delta_n$  when the full-history baselines are recomputed. The simulations neither add persistent states nor reveal the mark.*

*Proof.* Conditional expectations on the generating partitions converge in  $L^1$  for each density. Their norms are at most one as operators on  $L^1$ , so a finite-net argument gives uniform convergence on each compact family. There are finitely many action words.

Reporting the cell is an exact causal channel from  $E$  to  $E_n$ . After the reverse channel, the open-loop joint density of synthetic reports and the actual  $W$  is  $p_{\theta,a}^{(n)}$ . Under an arbitrary adaptive policy and encoder, expand the joint law by its action and register paths. The conditional rule multiplying each density is between zero and one. Summing the absolute differences over action words, and integrating over the register randomization, bounds the total variation of the marked output laws by  $\delta_n$ . This bound is uniform over policies, tasks, seeds and widths. Composition of any code with either report channel needs only the original register: cell conversion or within-cell sampling is a current-report operation. Thus every code transfers with the stated risk error in the reverse direction, and with zero error in the forward direction. Taking minima and worst rows proves the raw comparison. The same argument without a width bound compares full-history baselines, and the triangle inequality gives the excess-risk bound.  $\square$

## APPENDIX 27. A SINGULAR OBSERVATION CLASS AND CONDITIONAL COMPACT DUALITY

Product domination is not invariant under repeatedly displaying a continuous observation. A physical instrument may display the same covariate at every checkpoint, without asking the private register to store it. We prove compact duality for this non-product observation class. The distinction between a repeatedly supplied observation and a privately remembered one is essential, as Example 24.8 shows.

**27.1. Conditional innovation models.** Let  $X$  take values in a standard Borel space  $\mathbf{X}$  and be supplied *currently* to the encoder, controller, and terminal decision rule at each of their operations. The distribution of  $X$  may depend on the parameter;  $X$  is not a public random seed. The successive innovation reports belong to standard Borel spaces  $\mathbf{Z}_t$ . At every fixed  $x$ , all private registers, controls, and terminal action sets are finite. Let  $\nu$  and  $\mu_t$  be reference probabilities, and set  $\mu = \bigotimes_{t=1}^T \mu_t$ . As in Section 26, for each open-loop control word  $a$ , terminal action  $b$ , and row  $j$ , define the nonnegative loss effect  $f_{j,a,b}(x, z)$ . It is the density with respect to  $\nu \otimes \mu$  of the measure obtained by integrating the bounded loss against the *actual* open-loop joint law of  $(X, Z, W)$ . We require

$$j \mapsto f_{j,a,b} \text{ continuous in } L^1(\nu \otimes \mu), \quad J \text{ compact metric.} \quad (27.1)$$

The finite collection of these effects includes the initial probabilities and all parameter-dependent weights. It need not factor in  $x, z$ , and there is no hypothesis of domination by a product measure on the full *displayed* report path  $(X, (X, Z_1), \dots, (X, Z_T))$ .

Let  $\mathcal{K}_S$  be the compact metrizable space of private innovation policies at a fixed value of  $x$ , using the weak star simplex topology of Theorem 26.3. This includes initial-state, controller, and terminal decision simplices and the common encoder kernels. A private policy with the displayed covariate is a measurable section  $x \mapsto k_x \in \mathcal{K}_S$ . A public policy is a measurable  $(x, u) \mapsto k_{x,u}$ , where  $u$  is uniform and independent of the source. The jointly measurable representatives constructed in the dominated section turn these sections into actual kernels on  $(x, u, z_t)$ .

For a fixed  $x$ , let  $R_j(x, k)$  be the finite sum over control, register, and terminal paths of their policy factors integrated against  $f_{j,a,b}(x, \cdot)$ . It is an unnormalized conditional contribution:

$$r_j(k) = \int_{\mathbf{X}} R_j(x, k_x) \nu(dx).$$

This notation avoids dividing by a possibly vanishing density of  $X$ .

**Lemma 27.1** (Conditional risk integrands). *Each  $R_j$  represents an element of  $\mathcal{B} = L^1(\nu; C(\mathcal{K}_S))$ , and the map  $j \mapsto R_j$  is continuous into  $\mathcal{B}$ . After choosing measurable versions, the integrands are measurable in  $x$  and continuous in  $k$  for almost every  $x$  (for each fixed integrand). Every measurable section of  $\mathcal{K}_S$  has an executable common-encoder representative.*

*Proof.* For almost every  $x$ , the slices  $f_{j,a,b}(x, \cdot)$  are integrable. The tensor-testing lemma in the dominated section implies continuity in  $k$  of each path integral. Its absolute value is bounded by the sum of the  $L^1(\mu)$  norms of its finitely many sliced effects, multiplied by the finite number of state paths. This is an integrable envelope. For two rows, the same estimate applied to the difference gives

$$\|R_j - R_{j'}\|_{\mathcal{B}} \leq C_S \sum_{a,b} \|f_{j,a,b} - f_{j',a,b}\|_{L^1(\nu \otimes \mu)}.$$

Measurability first follows on the countable dense family of simple effects and policies and then by the displayed bound and tensor testing. Since  $C(\mathcal{K}_S)$  is separable, this gives strong measurability in  $\mathcal{B}$ . Continuity follows from (27.1).

For executability, use the Borel representative of a weak star simplex coordinate furnished by successive conditional expectations on finite partitions, as in the dominated section. This representative is jointly Borel in the coordinate and the innovation. Composing it with the measurable map  $x \mapsto k_x$ , or  $(x, u) \mapsto k_{x,u}$ , gives measurable encoder kernels. The finite-dimensional policy coordinates require no additional choice. A section measurable for the completion of  $\nu$  has a Borel version outside a  $\nu$ -null set, since the target is standard Borel; the reference domination makes this replacement harmless for every parameter. Conversely, a measurable implemented kernel gives a measurable section by integration against a countable dense family of  $L^1(\mu_t)$  tests. Thus this parametrization neither loses policies nor adds nonexecutable ones.  $\square$

**Lemma 27.2** (Compact conditional randomization). *The set  $\mathcal{Y}$  of probability kernels  $\Lambda_x(dk)$  from  $(\mathbf{X}, \nu)$  to  $\mathcal{K}_S$ , modulo  $\nu$ -null sets, is compact metrizable for integration against  $\mathcal{B}$ . Its elements are precisely the positive continuous linear functionals  $L$  on  $\mathcal{B}$  satisfying*

$$L(g(x)\mathbf{1}(k)) = \int g \, d\nu \quad (g \in L^1(\nu)).$$

*Every such kernel has a realization  $k = F(x, U)$  with  $F$  measurable and  $U$  uniform independent of the source.*

*Proof.* Positivity and normalization imply  $|L(V)| \leq \int \|V(x, \cdot)\|_{\infty} \nu(dx)$ , so all such functionals lie in the closed unit ball of  $\mathcal{B}^*$ . Their defining equalities and inequalities are weak star closed. Banach–Alaoglu therefore gives compactness; separability of  $\mathcal{B}$  gives metrizability on this ball.

For completeness, the representation does not assume a density in the policy coordinate. For each  $\varphi \in C(\mathcal{K}_S)$ , the bounded functional  $g \mapsto L(g\varphi)$  on  $L^1(\nu)$  has a Radon–Nikodym representative  $v_{\varphi} \in L^{\infty}(\nu)$ . Choose these representatives on a countable dense rational vector lattice of continuous functions containing constants. Outside a single null set, they are linear, positive, normalized, and bounded by  $\|\varphi\|_{\infty}$ . At each remaining  $x$  extend by continuity to  $C(\mathcal{K}_S)$ . The Riesz representation theorem gives a unique probability  $\Lambda_x$ . The integrals of a countable determining family are measurable, hence  $x \mapsto \Lambda_x$  is a measurable probability kernel. Simple  $C(\mathcal{K}_S)$ -valued functions and  $L^1$  approximation now prove  $L(V) = \int V(x, k) \Lambda_x(dk) \nu(dx)$  for every  $V \in \mathcal{B}$ .

Finally a probability kernel to a standard Borel space is realized by a measurable function of its input and one independent uniform variable (the kernel randomization lemma; see [4]). One may also construct it by successively splitting  $[0, 1]$  along refining countable partitions of the compact policy space. This uses an independent seed; it does not treat  $X$  as independent of the source.  $\square$

**Theorem 27.3** (Singular conditional common-encoder duality). *Under (27.1) and the displayed-covariate interface, the ideal-public attainable risk body is the compact convex subset of  $C(J)$*

$$\mathcal{A}_S = \left\{ \left( \int R_j(x, k) \Lambda_x(dk) \nu(dx) \right)_{j \in J} : \Lambda \in \mathcal{Y} \right\}.$$

Every point is executed with the stated finite private register and one independent public seed. Its upper set is closed, and, for  $w \in \mathcal{P}(J)$ , its lower support value is

$$h_S(w) = \int_X \min_{k \in \mathcal{K}_S} \left\{ \int_J R_j(x, k) w(dj) \right\} \nu(dx). \quad (27.2)$$

In particular

$$\min_{r \in \mathcal{A}_S} \max_{j \in J} r_j = \max_{w \in \mathcal{P}(J)} h_S(w),$$

and both extrema are attained. For each fixed  $w$  the support value is attained by one measurable private section. Finite-seed approximation and its state charge are those of Proposition 24.5.

*Proof.* By Lemma 27.1, the image of  $J$  in  $\mathcal{B}$  is compact. If  $\Lambda^n \rightarrow \Lambda$  in the topology of Lemma 27.2, the risk integrals converge on each fixed integrand. Approximate the compact set of row integrands by a finite  $\mathcal{B}$ -net and use the uniform norm-one bound on the functionals. Convergence is then uniform in  $j$ . Thus the risk map is continuous into  $C(J)$ , and its image is compact and convex. The kernel realization and Lemma 27.1 prove executability. A sum of a compact set and the closed positive cone is closed.

The Bochner integral  $V_w = \int_J R_j w(dj)$  belongs to  $\mathcal{B}$ . It has a Caratheodory version  $V_w(x, k)$ . The measurable minimum theorem on a compact metric space gives a measurable minimizing section  $k_x$ ; more explicitly, its nonempty compact argmin correspondence is measurable, as can be tested by infima over a countable dense set of policies, and admits a measurable selector. Its minimum is integrable by the envelope bound. Any kernel averages values no smaller than that minimum, while the Dirac kernel at  $k_x$  attains it. This proves (27.2). The choice depends on  $x$  and  $w$ , not on the realized task or parameter; all task-dependent controls remain coupled to a single encoder inside  $k_x$ .

The payoff  $(r, w) \mapsto \int r dw$  is continuous and affine on two compact convex sets. The compact minimax theorem therefore proves the identity and attainment; equivalently use separation of the compact upper risk body. The common modulus needed for finite seeds is supplied by the compact family of integrands and its  $\mathcal{B}$ -continuity.  $\square$

**Example 27.4** (A genuinely non-product report law). Take a nonatomic  $X$  on  $[0, 1]$ , display it at times zero and one, and adjoin any finite-horizon innovation model satisfying (27.1). The full displayed law is concentrated on the diagonal  $x_0 = x_1$ . It is not dominated by any product of probability reference measures on the two display coordinates. Indeed a product measure charges the diagonal only at common atoms; there are at most countably many such atoms. Outside those atoms it assigns zero mass to the diagonal, whereas the nonatomic displayed law assigns mass one. The same conclusion holds for sigma-finite product references, by restriction to countably many finite-measure sets. Nevertheless Theorem 27.3 applies with a single reference for  $X$ .

*Remark 27.5* (What has and has not been compactified). The theorem treats arbitrary measurable dependence on the repeated covariate, not a finite quantization of it. It does not assert weak closure of all strategic measures with arbitrary information structures. Nor does it assert compactness of the entire private risk body in this conditional model. The conditional Young kernels are exactly the ideal-public relaxation, and finite retained randomness is charged by Proposition 24.5. If the instrument ceases to display  $X$ , a policy that still accesses  $X$  is no longer admissible. These are information-interface distinctions, not exceptional null-set conventions.

## APPENDIX 28. INFINITE-HORIZON RISK BODIES AND COHERENT COMPARISON

The resource profile and the comparison channel can be defined for an unbounded time interval without a single reference measure dominating all infinite trajectories. What is needed for the compactness argument below is a coherent family of dominated finite-prefix loss effects and a

uniform tail bound for the criterion. This is a different assertion from weak closedness of arbitrary filtered strategic measures.

**28.1. Finite-prefix hypotheses and executable policies.** Time is indexed by  $t \in \mathbb{N}_0$ . Fix finite private widths  $S_t$ , finite control and decision alphabets at each time, and a finite set of task labels available to controllers but not to the common encoder. Reports range over standard Borel spaces  $(Y_t, \mu_t)$ . The parameter and task rows form a compact metric space  $J$ . The controlled instrument and its actual marks are specified by coherent stochastic kernels. A mark used in the stage- $t$  loss is part of the actual prefix experiment at that checkpoint. No independently regenerated target is substituted for it.

For each  $N$ , assume the finite-prefix loss effects satisfy the hypotheses of Theorem 26.3: after integrating out the actual mark, each effect for a fixed control, register, and decision string is an  $L^1(\mu_0 \otimes \cdots \otimes \mu_N)$  function, continuous in the row  $j$ , and the losses take values in  $[0, 1]$ . These hypotheses are local in  $N$ ; no uniform density bound in  $N$  is required. Effects for shorter prefixes agree with those obtained from the specified instrument. The policy information pattern is the finite-register pattern in Section 24.

Let  $K_t$  be the compact metrizable space of stage- $t$  weak-star encoder rows and finite controller/readout simplexes. The initial stage has its corresponding initial-encoder rows. Put

$$K = \prod_{t \geq 0} K_t.$$

This notation fixes time-dependent policies, not stationary policies. For  $\pi \in K$ , denote the expected stage loss by  $r_{j,t}(\pi)$ . It depends only on the first  $t + 1$  blocks and is a continuous map into  $C(J)$  by the finite-prefix tensor-testing argument.

Choose  $a_t \geq 0$  with  $\sum_{t \geq 0} a_t = 1$ , and define

$$r_j(\pi) = \sum_{t \geq 0} a_t r_{j,t}(\pi), \quad A_N = \sum_{t > N} a_t. \quad (28.1)$$

Geometric discounting is the case  $a_t = (1 - \beta)\beta^t$ ,  $0 < \beta < 1$ . More generally all arguments apply to a row family with a common summable dominating weight sequence.

**Lemma 28.1** (Coherent policy realization). *Every point of  $K$  has executable Borel representatives, and the representatives can be chosen jointly Borel in the policy and its report arguments. One independent uniform seed executes any Borel probability on  $K$  as a publicly randomized policy. For a fixed parameter and feedback design, changing the chosen representatives on the reference null sets changes none of its finite-prefix or infinite path laws.*

*Proof.* Apply Lemma 26.2 to each of the countably many encoder rows. Conditional averages on increasing finite report partitions and the Borel convergence set give jointly measurable simplex-valued representatives; on the exceptional set choose a fixed simplex point. For each prefix there are finitely many relevant rows. Product domination makes their reference-null exceptional sets null for every actual open-loop law; the finite control-string expansion gives the same fact under feedback. A countable union over prefixes remains null for the fixed parameter and design. The resulting Borel kernels define the infinite law by successive kernel extension. Equality on every prefix implies equality of the infinite path laws.

The space  $K$  is compact metrizable, hence standard Borel. A Borel map from a uniform variable to  $K$  realizes any probability on  $K$ . Evaluating the jointly Borel representatives at the sampled policy supplies all successive encoder rows. The seed is sampled independently of the source and is available as specified at each time. This is an ideal public resource; no finite storage bound is inferred from this realization.  $\square$

**Theorem 28.2** (Summable infinite-horizon common-encoder duality). *Under the finite-prefix hypotheses above, the following statements hold. The private risk body  $\mathcal{R}(S, 1) = \{r(\pi) : \pi \in K\}$  is compact in  $C(J)$ . For each finite public seed bound  $q$ , the body  $\mathcal{R}(S, q)$  is compact. The ideal public body is the compact convex set*

$$\mathcal{R}(S, \infty) = \left\{ \int_K r(\pi) \lambda(d\pi) : \lambda \in \mathcal{P}(K) \right\}. \quad (28.2)$$

For  $b \in C(J)$  its minimax excess satisfies the attained duality

$$\min_{\lambda \in \mathcal{P}(K)} \max_{j \in J} \left[ \int r_j d\lambda - b_j \right] = \max_{w \in \mathcal{P}(J)} \min_{\pi \in K} \int_J [r_j(\pi) - b_j] w(dj). \quad (28.3)$$

There is a saddle pair of mixed policies and row probabilities. The truncated and full risk bodies, at the same width and public seed signature, have Hausdorff distance at most  $A_N$  in  $C(J)$ , after truncation is identified with arbitrary continuation of its policy. Consequently their minimax excesses differ by at most  $A_N$  for a common baseline, and by at most  $A_N + \|b - b^{(N)}\|_\infty$  for different baselines.

*Proof.* The countable product  $K$  is compact metrizable. Its finite-prefix risk maps are continuous into  $C(J)$ . Equation (28.1) gives the uniform bound

$$\left\| r(\pi) - \sum_{t=0}^N a_t r_{\cdot, t}(\pi) \right\|_\infty \leq A_N,$$

independently of  $\pi$  and of the widths. Thus  $\pi \mapsto r(\pi)$  is continuous into  $C(J)$ , proving compactness of the private body. For finite  $q$ , the map from  $\Delta_q \times K^q$  which takes weights and policies to their weighted risk is continuous and has exactly the  $q$ -seed body as its image.

For completeness, norm continuity of the barycentre map in (28.2) does not follow from scalar weak convergence alone. Cover  $K$  by finitely many sets on which  $r$  oscillates by at most  $\varepsilon$  in the  $C(J)$  norm. A continuous partition of unity subordinate to the cover approximates  $r$  uniformly by a finite sum  $\sum_i \psi_i(\pi) r(\pi_i)$ . Its barycentre depends continuously in norm on  $\lambda$  for weak convergence of probabilities. The error is at most  $\varepsilon$ . Letting  $\varepsilon \downarrow 0$  proves norm continuity, so the image of  $\mathcal{P}(K)$  is compact and convex. Executability is Lemma 28.1.

The payoff obtained by integrating  $r_j - b_j$  against  $\lambda$  and  $w$  is continuous and affine in each probability on the two compact convex spaces. The compact minimax theorem gives equality and attainment. Minimizing a linear functional over  $\mathcal{P}(K)$  is the same as minimizing it over  $K$ , which gives (28.3). A primal minimizer and a dual maximizer form a saddle pair by their two defining inequalities. Equivalently the compact upper risk body is separated by positive row functionals, as in the finite-prefix proof.

Every infinite policy restricts to a prefix policy and every prefix policy has an arbitrary admissible continuation. The uniform tail bound therefore gives both Hausdorff inclusions for the private bodies and, after integrating, for every public signature. Taking suprema, infima, and subtracting the respective baselines proves the last assertion.  $\square$

**Corollary 28.3** (Finite seed cost independent of truncation time). *The family  $\{r(\pi) : \pi \in K\}$  has a common modulus of continuity  $\omega$  on  $J$ . If  $N(J, h)$  is the size of a finite  $h$ -net of  $J$ , every ideal public risk can be approximated, in supremum norm, within  $2\omega(h)$  by a mixture of at most  $N(J, h) + 1$  private policies. Its mixture index can instead be stored privately by enlarging every register to at most  $(N(J, h) + 1)S_t$ . The number of policies does not grow with the number of time checkpoints at this fixed accuracy.*

*Proof.* A compact subset of  $C(J)$  is uniformly equicontinuous: take a finite supremum-norm approximation and use the uniform continuity of its finitely many members. This supplies  $\omega(h) \downarrow 0$ . For a net of size  $n$ , project the private risk body into  $\mathbb{R}^n$ . Its convex hull is compact and

Carathéodory's theorem expresses any projected public barycentre using at most  $n + 1$  projected private risks. The resulting mixture agrees exactly at the net points. The modulus for the original risk and for the finite mixture bounds their difference at an arbitrary row by  $2\omega(h)$ . To replace the public index by private storage, initialize the joint state  $(i, m_0)$ , keep  $i$  unchanged, and use its selected policy at all later times. The task readouts use the same index. This is the explicit product-state construction, and has no finite-horizon restriction.  $\square$

**Example 28.4** (No dominating measure for the complete path). Let the reports be independent Bernoulli variables with parameter  $p \in [1/4, 3/4]$ . With finite decision alphabets and bounded row-continuous stage losses, every finite prefix satisfies the effect hypotheses. Yet the full path laws have no common  $\sigma$ -finite dominating measure. Indeed the disjoint Borel sets  $E_p = \{y : \lim_n n^{-1} \sum_{i < n} y_i = p\}$  have probability one under their respective laws. If a  $\sigma$ -finite measure dominated all these laws, it would give positive measure to uncountably many disjoint sets. Intersecting with its countable finite-measure cover and then with positive lower measure thresholds shows this is impossible. The convergence of the empirical mean here is the ordinary strong law, not a new ingredient. Theorem 28.2 therefore has genuinely infinite-path applications without assuming a dominating measure on the complete path.

**28.2. Costed comparison at all times.** An infinite simulator is *prefix coherent* if one set of causal kernels defines all its restrictions. At time  $t$  it has state cost  $K_t$ . A public simulator seed, if used, is explicitly part of the output interface: the comparison tests its joint law with the simulated transcript and actual marks against the target experiment supplied with an independent copy of that seed. The same applies to an independent consumer seed. This joint-law convention prevents a small marginal simulation error from hiding dependence revealed by a public seed. For deterministic or privately randomized stateless observation channels there is no additional public interface to test.

**Example 28.5** (Why the simulator seed belongs to the interface). Let the source observe a fair bit  $W$ , let  $U$  be an independent public fair bit, and output  $Y = W \oplus U$ . As a law of  $(Y, W)$  this is exactly the uninformative experiment in which  $Y$  is independent of  $W$ . But a consumer reading  $U$  can use  $Y \oplus U$  to recover  $W$  from the simulated interface, whereas that decision has error  $1/2$  in the true uninformative experiment supplied with an independent  $U$ . The total variation distance of the two joint  $(Y, W, U)$  laws is  $1/2$ , not zero. Thus a zero marginal error cannot justify a zero error for this seed-aware composition. Testing the joint interface, as required here, excludes this ambiguity without forbidding public randomization.

Suppose the prefix- $t$  error, uniform in parameter and output feedback strategy in this seed-aware sense, is at most  $\varepsilon_t \leq 1$ .

**Theorem 28.6** (Coherent resource comparison). *An infinite target consumer with width  $S_t$  and seed bound  $q$ , composed with such a simulator of width  $K_t$  and seed bound  $r$ , has an executable source realization with widths  $K_t S_t$  and seed bound  $rq$ . For the criterion (28.1), every row-risk difference is at most*

$$E_a = \sum_{t \geq 0} a_t \varepsilon_t. \quad (28.4)$$

*The upper-risk inclusion and lower-support inequality of Theorem 24.2 hold with error  $E_a$ . Two prefix-coherent simulations compose with multiplied state and seed costs and prefix errors at most  $\min\{1, \varepsilon_t + \varepsilon'_t\}$ . For two-sided cost-one simulations all private and public risk bodies have Hausdorff distance at most  $E_a$ ; their support values differ by at most  $E_a$ . If the respective full-history baselines also differ by at most  $E_a$ , their centered minimax spectra differ by at most  $2E_a$ .*

*Proof.* At each checkpoint retain the reachable simulator/consumer state pair. This single infinite machine has every finite-prefix product realization in Theorem 24.2. The joint seed convention

makes its feedback rule admissible for the stated prefix comparison. A stage loss is a  $[0, 1]$ -valued measurable function of the prefix, the actual mark, and the terminal randomization. The total variation bound therefore bounds its expectation difference by  $\varepsilon_t$ . Sum through  $N$  and bound the two remaining weighted tails by their common summable majorant; letting  $N \rightarrow \infty$  gives (28.4). The risk and support statements follow by the same explicit implementation, before any optimization over consumers. Composition is prefixwise contraction of total variation and its triangle inequality. Two-sided comparison gives both inclusions. Subtracting a baseline with the stated bound adds at most one further  $E_a$ .  $\square$

**Corollary 28.7** (Geometric error accumulation). *If a conditional coupling at each step fails with probability at most  $e$ , uniformly on previously matched prefixes, then one may take  $\varepsilon_t \leq 1 - (1 - e)^{t+1}$ . Under geometric discounting,*

$$E_a \leq \frac{e}{1 - \beta + \beta e}.$$

*This statement includes any stage mark in the coupled prefix. A bound only on its marginal, with no coupling to the actual reports, does not meet the hypothesis.*

*Proof.* The conditional success bound iterates to probability at least  $(1 - e)^{t+1}$  of matching through time  $t$ . Summing the resulting bound against  $(1 - \beta)\beta^t$  is a geometric series. This is stronger than using  $\min\{1, (t + 1)e\}$  when a conditional coupling is available.  $\square$

The infinite-horizon extension uses compact policy blocks, a uniform criterion tail, and classical convex separation. It neither asserts compactness for arbitrary average-cost problems nor charges an infinite public design index as a finite register. The next section on physical operator approximation will supply a different, unconditional prefix error bound for which Theorem 28.6, but not the conditional estimate in Corollary 28.7, applies.

## APPENDIX 29. THE MINIMAL MARKED CONTINUATION QUOTIENT

We now address exact state reduction. A statistic sufficient for the parameter need not be sufficient for a latent decision target. The correct terminal datum is the joint conditional law of the parameter and the designated mark. We give both an intrinsic operational characterization and a finite construction.

Consider a finite hidden instrument with initial law  $\nu_\theta$ , transition/report probabilities  $K_{\theta,t}(z', y \mid z, a)$ , and a terminal mark channel  $L_\theta(w \mid z_T)$  on a finite set  $W$ . Here  $K$  uses the new state  $z'$  and the report jointly; no old-state observation channel is silently substituted. A finite collection of path targets can be included by adjoining their necessary finite memory to  $z$ . This augmentation, when used, is part of the input size. Fix a reference prior  $\pi(\theta) > 0$ . For every supported controlled history  $h = (a_0, y_1, \dots, a_{t-1}, y_t)$ , let  $\alpha_t(h)$  be the posterior row on  $(\Theta, Z_t)$  under the reference prior and the indicated past actions. Control probabilities cancel in Bayes' formula; histories of zero reference probability are assigned an arbitrary existing label. No extra retained label is needed for events that have probability zero under every parameter.

**Definition 29.1** (Continuation experiment). For each future open-loop action word  $v = a_t \cdots a_{T-1}$ , let  $\mathcal{C}_t(h; v)$  be the conditional law of  $(\Theta, Y_{t+1:T}, W)$  given  $h$ , with future actions  $v$ . Two supported histories are equivalent, written  $h \equiv_t h'$ , if  $\mathcal{C}_t(h; v) = \mathcal{C}_t(h'; v)$  for every such  $v$ . Denote their class by  $c_t(h)$  and the number of classes by  $K_t$ .

This definition compares experiments rather than lists of selected coordinates. At  $T$  it reduces to equality of  $\mathcal{L}(\Theta, W \mid h)$ , not only equality of  $\mathcal{L}(\Theta \mid h)$ . Since the parameter set is finite and the prior is strictly positive, changing the reference prior applies an invertible positive reweighting to every continuation law. The classes are consequently independent of this choice.

**Lemma 29.2** (Controlled continuation). *The classes satisfy*

$$h \equiv_t h' \implies hay \equiv_{t+1} h'ay$$

for every  $a, y$  of positive conditional probability. The probability of  $y$  is itself constant on a class. Equality of the open-loop continuation laws implies equality under every randomized feedback policy using the subsequent report history.

*Proof.* The marginal of the first future report is a marginal of any continuation law beginning with  $a$ . It is the same for  $h, h'$ . Condition the joint law of  $(\Theta, Y_{t+1:T}, W)$  on that report being  $y$ . Its positive common denominator is the first-report probability, and its remaining numerator is the continuation law from  $hay$  or  $h'ay$ . Varying the remaining action word gives the successor assertion. For a feedback policy, multiply each open-loop path probability by the product of its conditional control probabilities along that path and sum over action words. These factors depend only on the displayed future reports and on independent randomization. Equality is preserved. Null successors have zero probability under every parameter of positive reference prior.  $\square$

**29.1. An operational necessity theorem.** The category in which a minimality claim is made matters. At a checkpoint  $t$ , a *restart preparation* chooses a supported past history  $H$  with an arbitrary law  $\eta$ , and then runs its conditional continuation experiment. The candidate statistic is obtained by consuming that same history through a channel  $Q_t(dm | h)$ . It does not resample a new history or replace the actual target  $W$ . A restart prediction task fixes a future action word  $v$  and an event  $A \subseteq \Theta \times Y_{t+1:T} \times W$ . Immediately at the checkpoint, before the continuation is observed, the decision maker predicts the indicator of  $A$  with an action in  $[0, 1]$  and squared loss. The full-history optimal prediction is  $p_A^v(h) = \mathcal{C}_t(h; v)(A)$ .

Call a source-consuming causal statistic *restart-decision complete* when it preserves the full-history Bayes risk in every one of these tasks, for every preparation  $\eta$  and every checkpoint. The definition requires equality of operational risks. It does not require any posterior coordinate to be explicitly stored.

**Theorem 29.3** (Marked operational minimality). *A privately randomized finite-valued statistic  $M_t$  is restart-decision complete at  $t$  if and only if  $c_t(H)$  is recoverable as a function of  $M_t$  on every supported history. In particular any complete randomized causal machine has at least  $K_t$  retained states at checkpoint  $t$ . The deterministic quotient  $(c_t)$  is simultaneously complete at all checkpoints, has exactly these widths, and is minimal in this category. For a public seed  $U$ , the corresponding criterion is recovery of  $c_t(H)$  from  $(U, M_t)$ , not from  $M_t$  alone. The same lower bound on the number of private register labels holds in either signature.*

*Proof.* The optimal Brier risks before and after compression differ by

$$\mathbb{E}_\eta \left[ (p_A^v(H) - \mathbb{E}_\eta[p_A^v(H) | M_t])^2 \right]. \quad (29.1)$$

Indeed  $\mathbb{E}[\mathbf{1}_A | H, M_t] = p_A^v(H)$ , since the statistic's randomization is independent of the actual continuation given  $H$ . Orthogonal projection in  $L^2$  proves the identity. There are only finitely many histories and cylinder events in the present finite experiment. Choose a preparation with positive mass on every supported history. Vanishing of (29.1) for every  $v, A$  forces  $p_A^v(h) = p_A^v(h')$  whenever both histories give positive probability to a common value  $m$ . Thus  $h \equiv_t h'$ . Each retained value has support in one equivalence class, so the class is a function of that value and there must be at least  $K_t$  values.

Conversely, if the class is recoverable, every  $p_A^v(H)$  is measurable from  $M_t$ , and (29.1) is zero for every preparation. Lemma 29.2 implements the classes by deterministic causal updates. It also proves equality for feedback continuations. For an independent public seed  $U$ , apply the same argument to  $(U, M_t)$ ; if the declared width bounds only  $M_t$ , a free retained  $U$  is a genuinely different signature. In an exact complete public scheme with finite widths, zero nonnegative excess

also holds for almost every fixed seed, so the lower bound  $S_t \geq K_t$  remains valid. One cannot use a seed to merge distinguishable histories while preserving all these zero-loss differences.  $\square$

**Proposition 29.4** (A quantitative separation). *Suppose a continuation event has probabilities differing by  $\Delta$  at  $h, h'$ . For the equal two-history preparation, any randomized statistic has Brier excess at least*

$$\frac{\Delta^2}{4}(1 - \text{TV}(Q_t(\cdot | h), Q_t(\cdot | h'))).$$

*Proof.* Write  $x_m = Q_t(m | h)$  and  $y_m = Q_t(m | h')$ . Direct evaluation of the conditional variance in (29.1) gives  $\frac{\Delta^2}{2} \sum_m x_m y_m / (x_m + y_m)$ , with zero terms omitted. Since  $xy/(x+y) \geq \min(x, y)/2$  and  $\sum_m \min(x_m, y_m) = 1 - \text{TV}(x, y)$ , the result follows.  $\square$

This is not a minimality assertion for an arbitrary independent simulator with unpriced memory. Nor is it an assertion that a chosen small list of losses separates every latent feature. The restart family is decision complete for the stated marked continuation experiment; its loss tests force the necessary state rather than assuming a posterior coordinate representation in the definition. The continuation and linear-representation aspects have classical antecedents in automata and predictive-state theory [49, 5, 58]; the target mark and the source-consuming operational category are essential here.

**29.2. Two exact marked causal channels.** Let  $k_t^0(y | c, a)$  be the reference next-report probability and  $F_t(c, a, y)$  the successor class. Define

$$H_t^0(c' | c, a) = \sum_{y: F_t(c, a, y) = c'} k_t^0(y | c, a), \quad \lambda_\theta(c) = \mathbb{P}_\pi(\Theta = \theta | c).$$

On the support of parameter  $\theta$ , Bayes' formula gives

$$k_{\theta, t}(y | c, a) = k_t^0(y | c, a) \frac{\lambda_\theta(F_t(c, a, y))}{\lambda_\theta(c)}. \quad (29.2)$$

The marked terminal channel is  $G_\theta(w | c_T) = \mathbb{P}_\pi(W = w | \Theta = \theta, c_T)$ . It is part of the quotient; it is not discarded.

**Theorem 29.5** (Exact marked reconstruction). *The original instrument and the class instrument admit causal channels in both directions preserving the joint law of the reported transcript and the actual terminal mark, under every parameter and admissible feedback policy. The reverse report kernel is*

$$L_t(y | c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t^0(y | c, a)}{H_t^0(c' | c, a)} \quad (29.3)$$

*when the denominator is positive. It is independent of  $\theta$ . The class instrument uses transitions  $H_{\theta, t}(c' | c, a) = H_t^0(c' | c, a) \lambda_\theta(c') / \lambda_\theta(c)$  and terminal channel  $G_\theta$ . Thus the channels preserve every bounded loss on the declared marked experiment, not just parameter likelihoods.*

*Proof.* The forward channel consumes each actual report and updates  $c$  by  $F_t$ ; its current class is a finite register. For the reverse channel, first observe the new class  $c'$ , then draw  $y$  by (29.3). Multiplying this kernel by  $H_{\theta, t}$  gives exactly (29.2). The terminal conditional distribution of  $W$  given the full original history and  $\theta$  is  $G_\theta(\cdot | c_T)$  by the definition of the terminal class. Consequently the product of transition factors, report reconstruction factors and this terminal mark factor is the original joint density of  $(Y_{1:T}, W)$ , for each open-loop action word. The feedback factors can be inserted before summing action words, as in Lemma 29.2. This proves the claimed joint, parameter-uniform identity. Cases with zero  $\lambda_\theta$  or zero transition probability lie outside that parameter's support and may be assigned arbitrary kernels. The positive reference prior makes these conventions consistent across parameters.

A reverse transducer needs the previous class, the new class and the current action while forming its report. Between updates it retains only the previous class. Composing it with an  $S_t$ -state machine is therefore possible with at most  $K_t S_t$  retained labels at time  $t$ ; the new input is transient. These resource costs are separate from the bare exact-simulation identity.  $\square$

**Example 29.6** (A parameter-sufficient statistic that loses its mark). Let  $\mathbb{P}(\Theta = 0) = 9/10$ , let  $Y$  be an independent fair bit, and put  $W = Y \oplus \Theta$ . The parameter posterior is constant in  $Y$ . A terminal quotient that keeps only that posterior has one class. Yet the Bayes error in guessing the actual  $W$  is  $1/10$  with  $Y$  and  $1/2$  with that class alone. The marked terminal laws  $\mathcal{L}(\Theta, W \mid Y = 0)$  and  $\mathcal{L}(\Theta, W \mid Y = 1)$  differ, so the marked quotient has two classes. Its reverse channel preserves the relation  $W = Y \oplus \Theta$ , not merely the marginal fair-bit law of  $Y$ . This example explains why an exact reconstruction theorem for reports alone cannot be used for arbitrary latent-target losses.

### APPENDIX 30. TASK-INDEXED EXACT COMPRESSION AND ITS COMPLETION

The least state preserving a finite set of optimal decisions need not be the least state preserving their entire joint experiment. Even at one checkpoint, the former object is generally a compatible cover rather than a quotient. We first give its exact operational characterization, then identify the condition under which it becomes a causal quotient.

**30.1. Finite decision families.** Let  $\mathcal{H}$  be a finite supported-history set at a restart checkpoint, and let  $\eta(h) > 0$  be a preparation. A task  $d \in D$  has a finite action set  $B_d$  and conditional loss  $\ell_d(h, b)$ . The latter is obtained by conditioning the actual marked continuation law and integrating the specified bounded loss; any prescribed continuation protocol is part of the task. Define

$$m_d(h) = \min_{b \in B_d} \ell_d(h, b), \quad A_d(h) = \{b : \ell_d(h, b) = m_d(h)\}.$$

The full-history benchmark for this restart problem is  $\sum_h \eta(h) m_d(h)$ . A subset  $C \subseteq \mathcal{H}$  is *compatible* when

$$\bigcap_{h \in C} A_d(h) \neq \emptyset \quad \text{for every } d \in D. \quad (30.1)$$

Let  $\tau_D$  be the smallest number of compatible sets covering  $\mathcal{H}$ . Empty sets are irrelevant.

**Theorem 30.1** (Exact common-task width). *The least number of labels in a source-consuming randomized statistic which attains the full-history benchmark for every task is exactly  $\tau_D$ . Independent public randomization does not reduce this number. The answer is independent of the choice of strictly positive preparation. If all the conditional losses factor through a marked continuation quotient with  $K$  classes, then*

$$\max_{d \in D} \tau_{\{d\}} \leq \tau_D \leq \min \left\{ K, \prod_{d \in D} |B_d| \right\}.$$

*These are checkpoint widths. Separate checkpoint optima are not asserted to be implementable by one causal machine.*

*Proof.* Let  $Q(m \mid h)$  be the common message channel and  $v_d(b \mid m)$  the task decoder. Its excess risk is

$$\sum_{h, m, b} \eta(h) Q(m \mid h) v_d(b \mid m) [\ell_d(h, b) - m_d(h)]. \quad (30.2)$$

Every summand is nonnegative. Equality to zero implies that the support of  $v_d(\cdot \mid m)$  is contained in  $A_d(h)$  for every  $h$  with  $Q(m \mid h) > 0$ . Thus each message support  $C_m = \{h : Q(m \mid h) > 0\}$  is compatible. These supports cover  $\mathcal{H}$ , and so there are at least  $\tau_D$  labels.

Conversely, from a compatible cover choose one containing cell for each history, and encode that cell deterministically. For each task and cell choose one action in the intersection in (30.1). Every

excess term is then zero. The argument uses only positivity of  $\eta$ , proving preparation independence. With a public seed condition (30.2) on that seed. Its conditional value is nonnegative, and its integral is zero for each of the finitely many tasks. For almost every seed all these equalities therefore hold, so the same cover lower bound applies.

The lower bound by individual tasks follows by removing constraints. Each marked class is compatible whenever the losses factor through the class. For the other upper bound, choose at each history an optimal action for each task and group histories by the resulting action vector. There are at most  $\prod_d |B_d|$  vectors.  $\square$

**Example 30.2** (Compatibility is not pairwise equality). For one task with three actions, suppose three histories have optimal sets  $\{1, 2\}, \{2, 3\}, \{1, 3\}$ . Every pair is compatible, but the three histories together are not. The exact width is two, not one. Consequently pairwise indistinguishability alone does not describe general finite-task compression. This is the obstruction to replacing Theorem 30.1 by an arbitrary task-indexed equivalence relation.

**30.2. Prediction tasks and a canonical completion.** For an event  $A$  in a marked continuation experiment with a fixed open-loop word  $v$ , put  $p_{A,v}(h) = \mathbb{P}(A \mid h, v)$ . The Brier task asks for  $a \in [0, 1]$  with loss  $(a - \mathbf{1}_A)^2$ . Its conditional excess is exactly  $(a - p_{A,v}(h))^2$ . Although its action space is not finite, the same nonnegative-summand argument applies because the minimizer is unique.

**Corollary 30.3** (Finite features and universal completion). *For finitely many Brier tasks with probability vector  $s(h) = (p_1(h), \dots, p_m(h))$ , the least checkpoint width, with arbitrary private or independent public randomization, equals  $|s(\mathcal{H})|$ . Taking all marked continuation event tasks gives exactly the marked continuation quotient of Theorem 29.3. For finite instruments, a finite subfamily already distinguishes all its classes.*

*Proof.* If one retained label is possible at two histories, zero excess forces all their probabilities  $p_i$  to agree. Conversely storing the probability vector attains all the benchmarks. Equality for all events and words is equality of the finite marked continuation laws, which is the definition of the universal quotient. For each pair of distinct classes choose one word and event whose probabilities differ. There are finitely many pairs, so these choices give a finite determining family. The public-seed argument is the same conditional zero-excess argument as before.  $\square$

The completion in this corollary does not enlarge a finite task list by fiat. It identifies precisely the extra tests that turn task-optimal compression into marked-experiment sufficiency. The possible gap is arbitrarily large: if all declared losses have the same optimal constant action,  $\tau_D = 1$ , even when  $K$  is arbitrarily large.

**30.3. A task-indexed causal quotient.** We now impose strict positivity of each next-report probability for every supported history and issued control. This makes all common continuations available and avoids identifying a null continuation with a supported one. At checkpoint  $t$  let  $s_t(h)$  be a finite vector of conditional Brier probabilities for the specified restart tasks. Define signatures backward:

$$\begin{aligned} \sigma_T(h) &= s_T(h), \\ \sigma_t(h) &= (s_t(h), (\sigma_{t+1}(hay))_{a \in \mathcal{A}_t, y \in \mathcal{Y}_{t+1}}), \quad t < T. \end{aligned} \tag{30.3}$$

Histories with equal signatures form a right congruence, so a register containing  $\sigma_t$  can be updated from its old value and  $(a, y)$ . Denote the number of distinct signatures at time  $t$  by  $K_t^D$ .

**Theorem 30.4** (Exact causal prediction width). *Among causal source-consuming machines that preserve the specified restart Brier risks at every checkpoint and for all supported open-loop control histories, the coordinatewise least width profile is  $(K_t^D)_{t=0}^T$ . Private or independent public*

randomization does not improve it. These signatures form the coarsest right congruence refining the probability features  $s_t$ . For the decision-complete family of all marked continuation event tasks, the signature quotient is the universal marked continuation quotient.

*Proof.* If one register symbol has positive probability at two histories, zero Brier excess forces equality of their current feature vectors. At time  $T$  this proves equality of signatures. Suppose the corresponding claim is proved at  $t + 1$ , and a symbol  $m$  is possible at histories  $h, h'$  at time  $t$ . Issue the same control  $a$  and consider the same report  $y$ . Strict positivity ensures that  $hay$  and  $h'ay$  are both supported. The machine uses the same next-register kernel at  $(m, a, y)$ ; at least one of its outputs  $m'$  has positive probability. Thus the next-register supports overlap at these two successor histories. By backward induction their signatures at time  $t + 1$  are equal. Doing this for all  $(a, y)$  and using equality of  $s_t$  proves equality of  $\sigma_t$ . Hence disjoint signatures require disjoint nonempty register supports, which gives the lower bound  $K_t^D$  simultaneously at each time.

The deterministic signature tracker attains the entire profile, because (30.3) makes its update well defined. Any right congruence refining the current features must also refine all successor features, and backward induction shows it refines  $\sigma_t$ ; this proves the coarsest assertion. For a public seed there are only finitely many histories and specified tests, so all zero-excess and overlap statements hold outside one null set, where the preceding proof applies.

For the complete event family, equality of present features is already equality of marked continuation law, and the latter is a right congruence by Lemma 29.2. The backward refinement therefore makes no further identifications or distinctions.  $\square$

*Remark 30.5.* With zero report probabilities, the backward construction can be performed on a suitably specified partial transition system, but the preceding randomized lower-bound proof cannot simply compare unsupported successor histories. The theorem deliberately states its positivity hypothesis. The unrestricted universal result, including its treatment of supports, remains Theorem 29.3.

## APPENDIX 31. SYNTHESIS FROM CONTINUATION SPACES

The continuation quotient can be computed without first expanding all controlled histories. Its linear dimension and its number of positive states are different quantities; the algorithm uses the former to test equivalence and the latter to bound its output.

Assume the finite marked instrument of Section 29 has rational data. At time  $t$  write  $n_t = |\Theta| |Z_t|$  and  $n = \max_t n_t$ . Its block-diagonal transition matrices are

$$M_{t,a,y}((\theta, z), (\theta', z')) = \mathbf{1}_{\{\theta=\theta'\}} K_{\theta,t}(z', y \mid z, a).$$

The terminal columns are  $e_{\theta,w}(\theta', z) = \mathbf{1}_{\{\theta=\theta'\}} L_{\theta}(w \mid z)$ . Define rational continuation spaces backwards:

$$V_T = \text{span}\{e_{\theta,w} : \theta, w\}, \quad V_t = \text{span}\{M_{t,a,y}v : a, y, v \in V_{t+1}\}. \quad (31.1)$$

They have dimensions at most  $n_t$ , and contain the all-ones column. For the latter assertion sum the terminal columns, and then use  $\sum_y M_{t,a,y} \mathbf{1} = \mathbf{1}$  for any fixed  $a$ .

**Lemma 31.1** (Finite linear test for a marked class). *Two normalized posterior rows  $\alpha_t(h), \alpha_t(h')$  determine the same marked continuation class if and only if they agree on  $V_t$ . It suffices to compare their evaluations on any rational basis  $B_t$  of that space.*

*Proof.* The probability of a specified future report word and terminal pair  $(\theta, w)$  under an action word is the row  $\alpha_t(h)$  applied to the column

$$M_{t,a_t,y_{t+1}} \cdots M_{T-1,a_{T-1},y_T} e_{\theta,w}.$$

These columns span exactly  $V_t$ , by induction in (31.1). Equality of all these probabilities is precisely Definition 29.1. Equality on their span is equivalent to equality on a basis.  $\square$

**Theorem 31.2** (Output-sensitive exact synthesis). *Let  $B$  be the total binary length of the rational source description, including any finite target augmentation and the positive reference prior. The marked quotient, one representative posterior per class, and its deterministic transition tables can be constructed in time polynomial in*

$$B + T + n + \sum_t |A_t| |Y_{t+1}| + \sum_{t=0}^T K_t.$$

*No factor equal to the number of expanded histories is required. The stored transition table has  $\sum_{t < T} K_t |A_t| |Y_{t+1}|$  entries, including null entries. A bit-serial implementation retains  $\lceil \log_2 K_t \rceil$  state bits, with finite transient workspace and a finite table-lookup running-time bound determined by this table. This is an exact-state synthesis theorem, not a polynomial-time algorithm for optimal lossy common-encoder design.*

*Proof.* Compute the spaces (31.1) backwards by rational elimination. At each stage there are at most  $|A_t| |Y_{t+1}| n$  candidate columns. Retain independent columns, rather than repeatedly replacing the basis by arbitrary normalized linear combinations. Each retained column is then a product of at most  $T$  input matrices with a terminal column. A common denominator for all input probabilities bounds the binary length of its entries by a polynomial in  $B, T, n$ . Fraction-free elimination, or determinant bounds for rational elimination, therefore takes polynomial bit time.

Start at the initial posterior  $\alpha_0 = (\pi(\theta)\nu_\theta(z))$ . At stage  $t$  maintain one posterior representative  $\alpha$  for each class found. For every  $a, y$ , form

$$r = \alpha M_{t,a,y} \mathbf{1}.$$

If  $r = 0$ , record the null transition. Otherwise form  $\alpha' = \alpha M_{t,a,y}/r$ , evaluate  $\alpha' B_{t+1}$ , and compare this vector with the stored keys at  $t + 1$ . Reuse its label if equal; otherwise store a new label and the representative  $\alpha'$ . By Lemma 31.1, the key is the marked continuation class. The successor result ensures that all histories represented by  $\alpha$  have this same transition, even though their full hidden posteriors can differ. Thus retaining one representative loses no reachable class. Induction over  $t$  proves that precisely the  $K_t$  classes are generated.

There are only the displayed number of transitions to examine. Every representative is the normalized posterior of some history of length at most  $T$ . The numerator and denominator are finite sums of products of rational input probabilities. Their binary lengths are polynomial in  $B, T, n$ , even if their magnitudes are very small. Exact comparison of keys, either by pairwise search or rational sorting, and all matrix operations consequently take polynomial bit time in the stated input and output parameters.

The executor needs only the class label and a fixed time-indexed table. A sequential scan of that finite table, with binary counters and the current action/report held transiently, is one explicit implementation. It has program length bounded by the table length times the binary length of an entry, and polynomial transient workspace and scan time in the table description. A more efficient random-access convention may improve this bound, but is not needed for finite effectivity. Report acquisition remains a separate interface cost.  $\square$

The invariant-space computation is the classical weighted-automaton equivalence method; see Kiefer [58], Proposition 2.1 and Theorem 2.3, and Tzeng [61]. What is being synthesized here is a *positive, normalized, marked causal quotient*, not a minimal signed linear representation. Linear dimension alone would not give its persistent-state cost.

**Example 31.3** (Small linear rank and growing positive width). Let  $\Theta \in \{0, 1\}$  have positive prior and, conditional on it, let the reports be independent Bernoulli variables with parameters  $p \neq q$  in  $(0, 1)$ . Retain  $\Theta$  as a terminal mark. The hidden linear dimension is two at every time. After  $t$

reports, the posterior odds for a history with  $k$  successes are the initial odds multiplied by

$$(q/p)^k ((1-q)/(1-p))^{t-k}.$$

These  $t+1$  values are distinct since  $p \neq q$ . The parameter posterior is a marginal of the marked continuation law, and conditional future laws depend only on this posterior. Hence  $K_t = t+1$ , whereas the expanded report tree has  $2^t$  leaves. The algorithm constructs the quotient in polynomial time in  $T$ , without mistaking rank two for width two.

## APPENDIX 32. POSITIVE-STATE GROWTH AND TASK-SYNTHESIS COMPLEXITY

The continuation-space algorithm of Theorem 31.2 separates two quantities: the linear dimension used to test equality, and the number of positive normalized classes that must actually be retained. We give polynomial and exponential families at fixed linear dimension, and distinguish both from task-optimal synthesis.

**Proposition 32.1** (Count-controlled positive-state growth). *Let  $\theta$  be a static hidden variable with finite support. Conditional on  $\theta$ , reports produced under a control  $a$  have time-homogeneous probabilities  $p_\theta(y | a)$  and are independent. Suppose the terminal mark depends only on  $\theta$ , through a fixed channel. For  $r = |\mathcal{A}||\mathcal{Y}|$ , the marked continuation quotient satisfies*

$$K_t \leq \binom{t+r-1}{r-1}.$$

*For fixed  $r$ , rational input, and unary horizon  $T$ , exact positive-state synthesis is polynomial in the horizon and input bit length.*

*Proof.* The posterior weight of  $\theta$  after a supported history is proportional to  $\pi_\theta \prod_{a,y} p_\theta(y | a)^{n_{a,y}}$ , where  $n_{a,y}$  is the number of occurrences of that control-report pair. Its future marked law depends only on this posterior, so histories with the same count vector are equivalent. There are  $\binom{t+r-1}{r-1}$  nonnegative count vectors with total  $t$ . This is an upper bound; different counts can give the same posterior. Products of  $t$  rational input entries have bit length at most linear in  $t$  times the input bit bound, and normalization and equality testing preserve a polynomial bit bound. Enumerating the count vectors and their successors therefore gives the claimed synthesis bound. Alternatively apply the output-sensitive compiler to the same polynomial output bound.  $\square$

**Proposition 32.2** (Exponential positive width at linear rank two). *There are strictly positive rational binary-report instruments with two hidden states, polynomial description length in  $T$ , and continuation rank two for which  $K_t = 2^t$  at every checkpoint  $t \leq T$ .*

*Proof.* Let  $\theta \in \{0, 1\}$  have a strictly positive rational prior and let  $W = \theta$ . There are no controls. Choose distinct primes  $p_1, \dots, p_T$ , and conditionally independently set

$$\mathbb{P}(Y_i = 1 | \theta = 0) = \frac{1}{2}, \quad \mathbb{P}(Y_i = 1 | \theta = 1) = \frac{p_i}{1 + p_i}.$$

The posterior odds after  $y_1, \dots, y_t$  equal the prior odds times

$$\prod_{i=1}^t \frac{2}{1 + p_i} \prod_{i=1}^t p_i^{y_i}.$$

Unique factorization makes the last products distinct for all binary words. Since the continuation contains the mark  $\theta$ , different posterior odds imply different marked continuation laws. All  $2^t$  words have positive probability, proving the exact count.

The hidden linear state space has dimension two. The terminal marked vectors are the two coordinate vectors, and their positive diagonal backward images remain independent, so the continuation rank is exactly two, not merely bounded by two. The first  $T$  primes have bit length

$O(T)$ , for example by the elementary bound obtained by successive use of Bertrand's postulate. Thus the rational description has polynomial length. The exponential output is not concealed in exponentially long input probabilities.  $\square$

**Theorem 32.3** (Zero-excess task synthesis contains graph coloring). *For finite rational one-checkpoint marked decision problems, deciding whether three retained labels suffice to attain every task's full-history Bayes risk is NP-complete. This remains true when all task action sets are binary and private and independent public randomization are allowed.*

*Proof.* Given a finite graph  $G = (V, E)$ , let the observed history  $X$  be uniform on  $V$  and let  $W = X$ . For each edge  $e = \{u, v\}$  make one binary task. At  $u$  its loss is zero only for action zero; at  $v$  it is zero only for action one; at every other vertex both actions have zero loss. The full-history Bayes risk is zero for each task. A set of histories is compatible in the sense of (30.1) precisely when it contains no edge of  $G$ . Compatible covers are therefore covers by independent sets, equivalently colorings after assigning each vertex to one containing set. Theorem 30.1 gives  $\tau_D = \chi(G)$  even for randomized encoders. This is a polynomial reduction from three-colorability.

For general explicitly tabulated rational losses, a certificate consists of a label for each history and one optimal action for each task and label. Check all nonempty cells against the conditional minima by exact rational comparisons. The certificate has polynomial size and can be checked in polynomial time. Randomization does not enlarge the class of yes instances by Theorem 30.1. This proves membership in NP and completes the proof.  $\square$

Graph-coloring descriptions of zero-error information problems have classical antecedents, notably [62]; the reduction above is given to specify the precise common-task loss formulation, not to claim that the connection with coloring is new. Likewise linear continuation equivalence is classical [61, 58]. The distinction established here is between polynomial linear tests, possibly exponential positive-state output, and a separate task-optimal combinatorial problem. None of these three should be described as a polynomial solution to the other two.

### APPENDIX 33. A COMMON-ENCODER OBSTRUCTION WITH AN INTERIOR DUAL WITNESS

The following example puts the task-coupling and randomization issues in one experiment. It is not a difference between distinct source families or between incompatible loss normalizations.

The source reveals  $X = \theta \in \{1, 2, 3\}$  without noise. Task  $d \in \{1, 2, 3\}$  asks whether  $\theta = d$ , with binary output and zero-one loss. The encoder has two labels, is not told  $d$ , and is common to the nine rows  $(d, \theta)$ . The task is supplied to the decoder. All full-history task baselines are zero. A task-specific two-label encoder would answer its task exactly.

**Theorem 33.1** (Three tasks, two labels). *For the common encoder, the best deterministic worst-row risk is one, the best private-randomized risk is  $1/2$ , and the best shared-seed risk is  $2/9$ . The uniform probability on all nine rows is a dual optimizer for the shared problem. Every row receives positive dual weight, and all nine risks of the displayed optimal shared scheme are equal.*

*Proof.* A deterministic two-label encoder merges at least two source symbols. For the membership task of either symbol, the decoder makes an error on at least one of them. Thus its worst-row risk is one. Moreover, under the uniform distribution on the nine rows every deterministic design makes at least two errors: each of the two membership tasks for a merged pair has an error. The uniform risk is at least  $2/9$ . Freezing all encoder and decoder randomization gives this same lower bound for every shared mixture of complete designs.

For the matching scheme choose a singleton  $k$  uniformly from the three symbols using the public seed. Encode whether  $X = k$ . If  $d = k$ , answer the task exactly. If  $d \neq k$ , answer zero on the singleton label; on the pair label answer one with probability  $2/3$ . For a positive row  $d = \theta$ , error occurs when  $k \neq d$  and the decoder answers zero, with probability  $(2/3)(1/3) = 2/9$ . For a

negative row  $d \neq \theta$ , error occurs only when  $k$  is the third symbol and the decoder answers one, with probability  $(1/3)(2/3) = 2/9$ . This establishes optimality and the asserted strictly positive dual witness. The additional decoder coin may be fresh private randomness; it may also be frozen into the public mixture of deterministic designs.

For a private encoder write  $p_x = \mathbb{P}(M = 1 \mid X = x)$ . For each task  $d$ , the probability of answering one is an affine function  $F_d(p_x)$ , since it is obtained by applying two decoder probabilities to the two encoder labels. Choose an index  $d$  for which  $p_d$  is a median of the three values. For this task,  $p_d$  lies between the other two values, including the possibility of equality. An affine function cannot be greater than  $1/2$  at this middle value and less than  $1/2$  at both end values. Hence some row of task  $d$  has error at least  $1/2$ . Ignoring the message and making a fair binary guess attains  $1/2$ . A common retained coin shared with the encoder would invalidate this affine argument, which is exactly why it is a different resource.  $\square$

For a single task, independent randomization cannot improve a scalar expected-loss minimum, as in the strategic-measure results of [56]. The worst-row criterion here is a maximum of linear risks, not one linear risk. Convexification can therefore strictly improve it. There is no contradiction with that classical irrelevant-randomization principle.

#### APPENDIX 34. CONTROLLED COMMON ENCODERS AND THE MINIMAX RISK BODY

The causal constraint is a constraint on the encoder, not on each task separately. We now give a variational characterization with this constraint inside the feasible set. The distinction between a shared random seed and private fresh randomization is essential to the statement.

**34.1. Finite instruments and the information pattern.** Fix a horizon  $T$ , finite parameter set  $K$ , finite hidden spaces  $Z_t$ , report alphabets  $Y_t$ , and a finite control alphabet  $A$ . The instrument is specified by probability vectors  $\nu_\theta$  and stochastic matrices

$$K_{\theta,t}(z', y \mid z, a), \quad 0 \leq t < T.$$

The hidden variable is not available to either the encoder or the controller. A finite collection  $D$  of tasks has finite terminal action sets  $B_d$  and losses  $\ell_{d\theta}(z, b) \in [0, 1]$ . Any bounded loss on a finite trajectory, including losses at several decision times, is of this form after adjoining that trajectory to the hidden state. This enlargement is a description of the experiment, not memory supplied to its controller.

The encoder has widths  $\mathbf{S} = (S_0, \dots, S_T)$ , with  $S_0 \geq 1$ . A parameter-independent initialization law chooses the initial label before the first control; this label is visible to the controller and is charged in  $S_0$ . Deterministic initialization can be labelled 1 by relabelling states. Before transition  $t$ , the task- $d$  controller chooses  $a$  from its current label  $i$ . The encoder then observes  $(a, y)$  and updates the label. At the last time, a task- $d$  terminal rule chooses  $b$ . The encoder's update table is the *same* for all tasks and parameters; the controller's tables may depend on  $d$  but not on  $\theta$ . The current control is an input to the ensuing update, not a retained action history. Task-dependent restrictions on legal controls may be imposed on the controller tables. The controller has no additional persistent register.

For a minimax task retain one row  $j = (d, \theta)$  per parameter. For a Bayesian task retain one row  $j$  with its specified prior  $\kappa_j$  on  $K$ . More generally a row has a task  $d(j)$  and a probability vector  $\kappa_j$ ; point masses give the minimax rows. Let  $J$  be this finite row set,  $M = |J|$ , and let  $\beta_d$  be the full-history optimal risk for the *whole* task  $d$ . In particular, the baseline of a minimax task is not a separate parameterwise optimum. Write  $b_j = \beta_{d(j)}$ .

A *deterministic design*  $\omega$  consists of one family of updates  $f_t(i, a, y) \in [S_{t+1}]$ , together with all controller tables  $a_t^d(i) \in A$  and terminal tables  $b^d(i) \in B_d$ . Let  $\Omega_{\mathbf{S}}$  be the finite set of these designs.

Its raw risk vector is denoted  $R(\omega) = (R_j(\omega))_{j \in J}$ , where

$$R_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} \ell_{d(j)\theta}(Z_T, b^{d(j)}(I_T)). \quad (34.1)$$

The expectation uses the actual closed-loop law. In particular, controls are not optimized separately inside hidden-parameter components.

There are two randomization conventions. In the *shared-seed* convention an independent variable is available to the encoder and the task controller throughout the run. It is chosen independently of the task and parameter; the risk averages its realization before taking the supremum over tasks or parameters. The adversary does not select a new parameter after seeing that realization. In the *private* convention there is no such retained shared variable. Each declared kernel uses fresh randomness, with conditional independence given its allowed inputs. Any seed remembered across updates is charged in the label. We write  $\Theta_{\text{sh}}$  and  $\Theta_{\text{pr}}$  for these two specified slices of (23.4). The term shared seed here has the precise meaning of the public independent seed permitted in Section 23; it never contains data from the experiment.

**34.2. The variational theorem.** Define the upper risk body and its lower support function by

$$\mathcal{U}_E(\mathbf{S}; J) = \text{co}\{R(\omega) : \omega \in \Omega_{\mathbf{S}}\} + \mathbb{R}_+^J, \quad (34.2)$$

$$h_E(\mathbf{S}; w) = \min_{\omega \in \Omega_{\mathbf{S}}} w \cdot R(\omega), \quad w \in \Delta_J. \quad (34.3)$$

Here  $\Delta_J$  is the probability simplex. The body is defined from controlled execution risks, not from a presumed predictive quotient.

**Theorem 34.1** (Controlled common-encoder minimax theorem). *For the finite instrument and information pattern just specified:*

- (1) *The shared-seed attainable upper risk set is exactly  $\mathcal{U}_E(\mathbf{S}; J)$ . The common-encoder excess satisfies*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (34.4)$$

$$= \max_{w \in \Delta_J} \{h_E(\mathbf{S}; w) - w \cdot b\}. \quad (34.5)$$

*An optimum uses at most  $M + 1$  deterministic designs. Thus the encoder is chosen once, before the task, in both formulae.*

- (2) *The functions  $h_E(\mathbf{S}; \cdot)$  determine the whole upper risk body. For two matched task/row systems,*

$$\mathcal{U}_F(\mathbf{S}'; J) \subseteq \mathcal{U}_E(\mathbf{S}; J) \iff h_E(\mathbf{S}; w) \leq h_F(\mathbf{S}'; w) \quad (w \in \Delta_J).$$

*In particular equality of all these support functions is equivalent to equality of the attainable upper bodies. Relabelling histories, hidden states or machine labels changes neither assertion.*

- (3) *For constant width  $S$ , cardinality alone being charged,*

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D). \quad (34.6)$$

*The first inequality is implemented by retaining the design index in the actual register, including the permitted randomized initialization. If a different signature forbids randomized initialization, this inequality refers to its explicitly initialization-enlarged signature. For a fixed finite row family, any existing positive cardinality exponent is consequently the same under these two randomization conventions.*

*Proof.* First include the decoder and controller tables in the design. For one fixed common encoder their choices for distinct tasks are independent optimization variables, whereas the same choice for one task must work for all its parameter rows. Compactness of the finite products of probability simplexes implies attainment. Hence minimizing jointly over the common encoder and all its task rules gives exactly the original order  $\inf_Q \max_d (r_E(Q, d) - \beta_d)$ ; it does not give a decoder indexed by the unknown parameter.

Realize every update and controller kernel by a uniform variable at its specified time, independent of the instrument. For a shared-seed scheme, condition on the entire shared seed and these variables. One obtains a deterministic common update family and one deterministic controller for each task. Although different tasks then generate different observation laws, the frozen maps are defined on *all* their possible inputs. They therefore form one element of  $\Omega_{\mathbf{S}}$ . Its conditional risk is computed under the corresponding controlled instrument, not under an incorrectly frozen uncontrolled history law. Averaging gives a convex combination of the finite vectors  $R(\omega)$ .

Conversely sample a design index with probabilities  $\lambda$ , expose that independent index to encoder and controller, and execute its tables. The encoder law is independent of which task will be requested. Its risk vector is exactly  $\sum \lambda_\omega R(\omega)$ . Taking coordinatewise upper bounds proves (34.2) and (34.4). Finite-dimensional convex minimax gives (34.5). Equivalently, the primal is the linear programme minimizing  $v$  subject to  $\lambda \geq 0$ ,  $\sum \lambda = 1$ , and  $\sum \lambda_\omega R_j(\omega) \leq v + b_j$  for every row. Carathéodory's theorem in  $\mathbb{R}^M$  gives support at most  $M + 1$  without changing the optimum risk vector. This argument freezes a tape to form its *distribution of designs*; it never moves a worst-task supremum through that distribution.

The body in (34.2) is closed and convex. If a point lies outside it, finite-dimensional separation supplies a linear functional whose infimum on the body exceeds its value at the point. The functional must be nonnegative on every coordinate direction, because the body contains every positive ray starting at one of its points. Normalizing its coefficients places it in  $\Delta_J$ . Its infimum is (34.3). This proves the comparison assertion, including both directions of equality. A relabelling gives a bijection of the same execution laws and of their deterministic designs.

Every private scheme is also a shared-seed scheme that ignores its extra input. Conversely an optimal shared scheme can use  $k \leq M + 1$  designs. Sample its index privately before the first observation and retain  $(r, I_t)$ , which has at most  $kS$  values. Each subsequent update, control and readout uses this actual pair. This has the same law as the shared scheme; no external transcript or hidden seed is used. This proves (34.6). Monotonicity, together with  $\log((M + 1)S)/\log S \rightarrow 1$ , proves the exponent assertion. The constant  $M + 1$  is not uniform over growing or infinite task families.  $\square$

The theorem identifies a finite controlled spectrum by one variational functional. It is more than the transport of a supplied simulation: its feasible set is constructed from the instrument and the prescribed memory, and the converse recovers every admissible execution. Convex minimax, separation and finite mixed strategies are classical tools. Their use here does not reassign their priority. The substantive information restriction is that all tasks share  $f$ , while all parameters within a task share its controller, and every retained design index is accounted for when the shared-seed convention is removed.

**34.3. Controlled continuation partitions.** Let  $\mathcal{H}_t$  be the full finite tree of action–report histories, including histories null under a particular parameter or task. Histories null under every possible execution can be assigned arbitrary updates. A family of partitions  $\mathcal{P}_t$  is controlled successor compatible when

$$h \sim_t h' \implies hay \sim_{t+1} h'ay \quad (a \in A, y \in Y_{t+1}). \quad (34.7)$$

**Proposition 34.2** (Intrinsic feasible partitions). *Deterministic common encoders with widths  $\mathbf{S}$  are, up to label names and unused states, exactly the partition families satisfying (34.7) and*

$|\mathcal{P}_t| \leq S_t$ . A deterministic design adds, for each task, a control constant on every current cell and a terminal decision constant on every terminal cell. Thus the risk body and both sides of (34.5) can equivalently be defined on controlled continuation partitions with these cellwise rules.

*Proof.* Equal machine states have equal successors on the same action and report, proving necessity. Conversely label the cells and send  $(C, a, y)$  to the cell containing  $hay$  for any  $h \in C$ . Condition (34.7) makes this independent of  $h$ . The cellwise controller rule supplies the legal control, and induction identifies the actual label with the actual history's cell for every parameter and task. Defining the maps on the whole tree ensures that an action absent under one task has not made its update task dependent.  $\square$

**34.4. Private randomization is a different finite problem.** For completeness there is also an exact, nonconvex description without a shared seed. Write  $g_t^d(a \mid i)$  for the private controller,  $q_t(i' \mid i, a, y)$  for the common private update, and  $v^d(b \mid i)$  for its terminal rule. Let  $q_0$  be a parameter-independent initialization law when randomized initialization is permitted. Then

$$\begin{aligned} p_0^{d\theta}(z, i) &= \nu_\theta(z) q_0(i), \\ p_{t+1}^{d\theta}(z', i') &= \sum_{z, i, a, y} p_t^{d\theta}(z, i) g_t^d(a \mid i) K_{\theta, t}(z', y \mid z, a) q_t(i' \mid i, a, y). \end{aligned} \quad (34.8)$$

The risk of row  $j$  is

$$\sum_{\theta, z, i, b} \kappa_j(\theta) p_T^{d(j)\theta}(z, i) v^{d(j)}(b \mid i) \ell_{d(j)\theta}(z, b).$$

**Proposition 34.3** (Private flow characterization). *Minimizing the largest baseline-subtracted row risk over the probability simplexes in (34.8) gives exactly  $\Theta_{\text{pr}}(\mathbf{S}; D)$ . The minimum is attained. With rational input this is a finite polynomial optimization problem, not in general the linear programme (34.4).*

*Proof.* Induction on the actual execution gives the displayed occupancies and risk. Conversely their kernels implement precisely that law with no extra retained variable. The variables form a compact product of finite simplexes, and every row risk is a polynomial in them. Continuity gives attainment. The shared occurrences of  $q_t$  cannot be split by task or parameter; doing so would change the information pattern.  $\square$

**Example 34.4** (A strict cost for hiding the seed). Observe  $X = \theta \in \{1, 2, 3\}$  without error, retain two labels, and finally guess  $\theta$  with zero-one loss under worst-parameter risk. The full-information baseline is zero. With a shared seed, choose uniformly one of the three two-element subsets and encode the two selected symbols without error. The omitted symbol is necessarily wrong. Each parameter is omitted with probability  $1/3$ , so the risk is  $1/3$ . For any fixed seed, the sum of the three correct-reconstruction probabilities of a two-label experiment is at most two; averaging proves the matching lower bound.

Without a shared seed let  $r_1, r_2$  be the two decoder distributions on three symbols. Correctness for symbol  $x$  is at most  $\max\{r_1(x), r_2(x)\}$ . If all three correctness probabilities exceeded  $1/2$ , one of two probability vectors would exceed  $1/2$  at two coordinates, an impossibility. Thus the private risk is at least  $1/2$ . It is attained by assigning symbol 1 to the first label and symbols 2, 3 to the second, whose decoder tosses a fair coin. Therefore

$$\Theta_{\text{sh}}(2) = \frac{1}{3}, \quad \Theta_{\text{pr}}(2) = \frac{1}{2}.$$

The gap is not removed by allowing randomized private encoding. A single unknown parameter is selected before the seed, as required above.

### APPENDIX 35. ONE VARIATIONAL PRINCIPLE: OCCUPANCIES, BAYES ENVELOPES AND PRESSURE

The support function in Theorem 34.1 can be computed without first constructing a quotient or listing full decision trees. The relevant state for this calculation is an *offline* joint occupancy. It is not an additional observation available during execution.

**35.1. An exact support-function recursion.** For  $w \in \Delta_J$  put

$$z_0^{d\theta}(z, i) = \mathbf{1}_{\{i=1\}} \nu_\theta(z) \sum_{j:d(j)=d} w_j \kappa_j(\theta).$$

For arbitrary nonnegative arrays  $z = (z^{d\theta}(x, i))$ , a prescription consists of actions  $a^d(i)$  and a *single* table  $f(i, a, y)$ . Define its linear occupancy update by

$$(\mathcal{T}_{a,f} z)^{d\theta}(x', i') = \sum_{x,i,y} z^{d\theta}(x, i) K_{\theta,t}(x', y \mid x, a^d(i)) \mathbf{1}_{\{f(i, a^d(i), y)=i'\}}. \quad (35.1)$$

The time  $t$  is understood from the two spaces in this formula. Define

$$V_T(z) = \min_{b^d(i) \in B_d} \sum_{d, \theta, x, i} z^{d\theta}(x, i) \ell_{d\theta}(x, b^d(i)), \quad (35.2)$$

$$V_t(z) = \min_{a,f} V_{t+1}(\mathcal{T}_{a,f} z), \quad t = T-1, \dots, 0. \quad (35.3)$$

**Theorem 35.1** (Occupancy representation of the spectrum). *For the finite controlled instrument,*

$$h_E(\mathbf{S}; w) = V_0(z_0(w)), \quad \Theta_{\text{sh}}(\mathbf{S}; D) = \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (35.4)$$

*Every  $V_t$  is a positively homogeneous, concave, piecewise-linear function on its nonnegative occupancy cone. A minimizing sequence of prescriptions compiles into an actual common-encoder design with the specified widths. No realized task index or hidden parameter is supplied to  $f$ .*

*Proof.* Fix a deterministic sequence of prescriptions. Induction in (35.1) shows that  $z_t^{d\theta}$  is the joint law of  $(Z_t, I_t)$  under that task and parameter, multiplied by its initial weight. Its terminal linear cost is therefore  $w \cdot R(w)$ . At the last time the terminal tables in (35.2) give precisely all legal terminal rules:  $b^d(i)$  is common to every  $\theta$  in task  $d$ .

At an earlier time, choosing  $a^d(i)$  and one common  $f$  determines the entire next occupancy. Conversely every possible first prescription of a deterministic continuation has this form. Backward induction therefore proves that (35.3) minimizes over exactly all legal future designs. This proves the first equality in (35.4); Theorem 34.1 proves the second. For any fixed continuation the cost is linear in the initial occupancy. There are finitely many continuations, so  $V_t$  is the minimum of finitely many linear forms. The stated geometric properties follow.

The occupancy is propagated during design, starting at the known array  $z_0(w)$ . Once a sequence of minimizing tables is chosen, its future occupancies are deterministic functions of that sequence. Only the finite tables and the actual label are used online. In particular, one must not store the occupancy array without charging its size, nor allow a different prescription for the unobserved realized parameter.  $\square$

This recursion is a designer's dynamic programme with coupled information constraints. Common-information prescriptions and belief-state dynamic programming have substantial antecedents [53, 54]. In the cited common-information preprint, Corollary 5 already treats absent common memory and finite local memories; its unconditional information state is an offline occupancy. The additional assertion here is the exact equality with the *common-encoder minimax spectrum*, including its dual adversarial weights, controlled continuation constraint and explicit seed cost. The recursion does not assert polynomial complexity: the number of linear pieces and the number of finite designs can be very large.

**35.2. The common quadratic task family.** There is a particularly transparent intrinsic form when reports are uncontrolled. Let the finite history law  $p_t(h)$  be fixed, let  $U_{dt}$  be square-integrable Hilbert-valued targets, and let  $\alpha_{dt} \geq 0$ . Task  $d$  has loss  $\sum_t \alpha_{dt} \|U_{dt} - \hat{U}_{dt}\|^2$ . The decoder's answers do not affect later observations or updates. Set  $m_{dt}(h) = \mathbb{E}[U_{dt} \mid H_t = h]$ . For a successor-compatible partition family  $\mathcal{P}$ , define

$$D_d(\mathcal{P}) = \sum_t \alpha_{dt} \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_{dt}(h) - \bar{m}_{dt,C}\|^2, \quad \bar{m}_{dt,C} = \frac{\sum_{h \in C} p_t(h) m_{dt}(h)}{p_t(C)}. \quad (35.5)$$

Zero-mass cells are omitted. Define the single weighted geometry

$$M_{w,t}(h) = (\sqrt{w_d \alpha_{dt}} m_{dt}(h))_{d \in D}$$

to lie in the Hilbert direct sum of the target spaces.

**Corollary 35.2** (Common-encoder variance duality). *For a shared independent seed and the declared widths,*

$$\begin{aligned} \Theta_{\text{sh}}(\mathbf{S}; D) &= \min_{\lambda \in \Delta(\mathcal{P})} \max_d \sum_{\mathcal{P}} \lambda_{\mathcal{P}} D_d(\mathcal{P}) \\ &= \max_{w \in \Delta_D} \min_{\mathcal{P}} \sum_t \mathbb{E} \|M_{w,t}(H_t) - \mathbb{E}[M_{w,t}(H_t) \mid \mathcal{P}_t]\|^2. \end{aligned} \quad (35.6)$$

*The minima range over one successor-compatible family of width at most  $S_t$  at each time. This is an exact characterization of the maximum over tasks, not a statement only about one scalarized loss. Its private version lies between the values given by this formula at  $S$  and by its implementation with  $(|D| + 1)S$  states.*

*Proof.* Conditional expectation gives the orthogonal decomposition of each squared risk into its full-history Bayes risk and prediction of  $m_{dt}(H_t)$ . On a deterministic state cell its unique optimal readout is the displayed cell mean. All tasks can use their respective means simultaneously without changing the common encoder. Thus the coordinatewise best risk vector of a deterministic partition is exactly  $D(\mathcal{P})$ . There are finitely many history partitions, although the target Hilbert spaces need not be finite dimensional.

Conditioning on an independent seed gives a distribution of these partitions; conversely their mixtures are executable under the shared convention. The proof of Theorem 34.1 now applies to this finite vector set. The identity  $\sum_d w_d D_d(\mathcal{P}) = \sum_t \mathbb{E} \|M_{w,t} - \mathbb{E}(M_{w,t} \mid \mathcal{P}_t)\|^2$  is the direct-sum norm formula. Storing the mixture index proves the last assertion exactly as in (34.6).  $\square$

For  $|D| = 1$ , (35.6) reduces to the earlier conditional-variance characterization, Theorem 47.1. For several tasks it supplies its missing minimax convexification. The outer maximizing weight is common to every time; selecting a separate favourable weight or encoder at each checkpoint is not permitted. Compatible predictive trees are one class of feasible partitions of this same functional. Their energy estimates in the supporting theory supply quantitative bounds for that class; they are not hypotheses of Theorems 34.1 or 35.1.

**Corollary 35.3** (A1: maximum over finite query and checkpoint tasks). *Consider the actual finite-command accepted-cell/failure protocol of Corollary 47.2, with its fixed prior, finite horizon and parameter-independent command laws. Fix any finite family of weighted quadratic query/checkpoint tasks. Its optimal shared-seed common causal encoder is given by (35.6), using the accepted and summed failure likelihoods of that protocol. At most  $|D| + 1$  partition designs are needed. Retaining their index gives the same optimum with at most  $(|D| + 1)S$  private states.*

*Proof.* The observation likelihoods and posterior means are derived in Corollary 47.2 from  $g_j k_j(\theta)$  on acceptance and  $\sum_j (1 - g_j) k_j(\theta)$  on the one failure report. These give one finite history law for all tasks. Insert its posterior query means in (35.5) and apply Corollary 35.2. The unseen failed

raw-cell label is never used in a partition update. Unlike the earlier scalarization, the present formula retains the maximum over the whole prescribed task family.  $\square$

**Proposition 35.4** (A1 with private randomness at unchanged width). *In the same finite-command protocol, let  $p_t(h)$  be its actual history law, and let  $m_{dt}(h)$  be the finite vector of posterior query means for task  $d$ , with query weights absorbed into that vector. Write  $\eta_t^q(i | h)$  for the conditional label law generated recursively by one common private encoder  $q$ . The private maximum-over-tasks excess at unchanged width is exactly*

$$\min_{q,v} \max_{d \in D} \sum_{t,h,i} \alpha_{dt} p_t(h) \eta_t^q(i | h) \|m_{dt}(h) - v_{dt}(i)\|^2. \quad (35.7)$$

*The readouts can be restricted to the compact boxes containing the weighted query means. This is a finite polynomial optimization problem in the stochastic update rows and readout coordinates, with coefficients given by the actual likelihood integrals. It has a minimum. No finite arithmetic claim is made when those coefficients are arbitrary reals.*

*Proof.* Given a supported history, the label law is a finite sum of products of the encoder's transition probabilities. The same row is used whenever the preceding label and observed command/report agree, in particular on the common failure report. Conditional orthogonality subtracts the full-history Bayes risk, leaving precisely the displayed expression. Conversely every listed stochastic row and readout is an admissible private machine, so both infima are equal. Projecting readouts into the coordinatewise range of the query means cannot increase squared error. The feasible set is compact, and its objective is the maximum of finitely many polynomials. Continuity proves attainment.  $\square$

**35.3. Exponential criteria and hidden-phase control.** The same support calculation has a multiplicative form. Let  $\gamma > 0$  and let  $C_{d\theta}$  be a bounded cumulative cost on the finite trajectory, including a terminal decision. Put

$$M_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} e^{\gamma C_{d(j)\theta}}.$$

For task  $d$  the risk is  $\gamma^{-1} \log \max_{j:d(j)=d} M_j$ ; a Bayesian task has its single prior row. Let  $\beta_d^{\exp}$  denote its corresponding full-history baseline.

**Corollary 35.5** (Risk-sensitive common-encoder duality). *The shared-seed excess under these exponential risks equals*

$$\Theta_{\text{sh}}^{\exp}(\mathbf{S}; D) = \frac{1}{\gamma} \log \left[ \max_{w \in \Delta_J} \min_{\omega \in \Omega_{\mathbf{S}}} \sum_j w_j e^{-\gamma \beta_{d(j)}^{\exp}} M_j(\omega) \right]. \quad (35.8)$$

*For additive costs, the inner minimum is computed by (35.1)–(35.3) after replacing the transition by  $K_{\theta,t} e^{\gamma C_{d\theta,t}}$  and the terminal loss by  $e^{\gamma C_{d\theta,T}}$ . The initial row weights are multiplied by  $e^{-\gamma \beta_d^{\exp}}$ . These are nonnegative occupancies, not normalized posterior probabilities.*

*Proof.* For a mixture of designs, the exponential moment is  $\sum_{\omega} \lambda_{\omega} M_j(\omega)$ . It is *not* the exponential of an average of logarithmic risks. Since logarithm is increasing, subtracting the baseline and maximizing rows gives

$$\frac{1}{\gamma} \log \max_j \left[ e^{-\gamma \beta_{d(j)}^{\exp}} \sum_{\omega} \lambda_{\omega} M_j(\omega) \right].$$

Minimize over  $\lambda$  and apply finite minimax to the bracketed positive moment table. This proves (35.8). Expanding the product of transition weights proves the Feynman–Kac occupancy assertion by the same induction as before. Positivity, not normalization, is what that induction requires.  $\square$

For example, let an unobserved phase  $\theta \in \{0, 1\}$  have probabilities  $1/2, 1/2$ , provide no report, and charge  $C = \mathbf{1}_{\{a \neq \theta\}}$ . Every observation-admissible action, randomized or not, has moment  $(1 + e^\gamma)/2$  and risk  $\gamma^{-1} \log((1 + e^\gamma)/2) > 0$ . Optimizing separately within each phase would instead give zero. The single  $a^d(i)$  in the occupancy recursion forbids precisely that exchange. This supplies a finite-model resolution of the common-policy issue arising in phase mixtures; it does not presume that a latent phase decomposition of a microscopic system has already been constructed.

### APPENDIX 36. MINIMAL POSTERIOR-AND-REPORT-PREDICTIVE COORDINATES

The variational theorem does not require a sufficient statistic. When such a statistic is desired, the finite instrument admits a direct construction. It distinguishes the cost of prediction from the cost of preserving the whole parameter experiment.

Fix the finite instrument of Section 34. For an action–report history  $h = (a_0, y_1, \dots, a_{t-1}, y_t)$  define the unnormalized forward vector by

$$\begin{aligned} u_{\theta,0}(z) &= \nu_\theta(z), \\ u_{\theta,t+1}(hay, z') &= \sum_z u_{\theta,t}(h, z) K_{\theta,t}(z', y \mid z, a), \quad L_\theta(h) = \sum_z u_{\theta,t}(h, z). \end{aligned} \quad (36.1)$$

Choose a reference prior  $\pi_\theta > 0$ . A history is supported if  $\bar{L}(h) := \sum_\theta \pi_\theta L_\theta(h) > 0$ . On supported histories put

$$\begin{aligned} \lambda_\theta(h) &= \pi_\theta L_\theta(h) / \bar{L}(h), \\ k_t(y \mid h, a) &= \bar{L}(hay) / \bar{L}(h). \end{aligned} \quad (36.2)$$

The likelihoods in these formulae do not contain action-selection probabilities. Under any parameter-independent observation-based policy, those probabilities cancel from the posterior and from likelihood ratios. Thus the formulae apply to every policy on its supported histories, not only to a policy used to explore the history tree.

A history coordinate  $s_t(h)$  is called *saturated and predictive* if it determines  $\lambda(h)$ , determines every vector  $k_t(\cdot \mid h, a)$ , and has a deterministic update from  $(s_t, a, y)$  on supported continuations. At time  $T$  only the first condition is imposed. This is the finite, terminal-parameter version of the three conditions in Section 46. Additional latent-target marks require the corresponding conditional kernels as further decorations; parameter sufficiency alone does not imply their sufficiency. Sections 29–31 supply the marked reconstruction theorem and an output-sensitive construction.

Construct partitions backwards. At level  $T$  group histories with the same vector  $\lambda(h)$ . Having constructed the class map  $c_{t+1}$ , let  $c_t(h)$  be the class determined by the finite tuple

$$\left( \lambda(h), (k_t(y \mid h, a), c_{t+1}(hay))_{a,y} \right). \quad (36.3)$$

For an unsupported continuation its class entry is one fixed cemetery symbol. Let  $K_t$  be the number of classes at level  $t$ .

**Theorem 36.1** (Minimal posterior-and-report-predictive coordinate). *For every finite controlled instrument the partitions (36.3) form a saturated predictive coordinate. Every other such deterministic history coordinate refines them at each level. Consequently  $K_t$  is its least possible width, simultaneously at every level. The partitions, as equivalence relations on supported histories, are independent of the choice of the strictly positive reference prior.*

*There is a parameter-independent finite reconstruction kernel  $L_t(y \mid c, a, c')$  which, from successive quotient classes and the chosen controls, reconstructs the report stream under every admissible policy. The original and quotient parameterized report experiments have mutually exact causal*

*simulations. This assertion concerns reports and parameter likelihoods only; latent-target preservation is the separate marked statement of Theorem 29.5. Their additional persistent state is at most  $\max_t K_t$  labels, so composing either simulation with an  $S$ -state encoder costs at most  $(\max_t K_t)S$  states.*

*If the instrument and prior have rational entries, the classes and reconstruction kernels can be computed by exact finite arithmetic. The calculation is polynomial in the size of the expanded finite history tree and its rational input length. That tree, and the least width  $K_t$ , may grow exponentially with the horizon.*

*Proof.* Equality of the keys at level  $t$  implies equality of the posterior vectors, of all reference predictive probabilities, and of the class of every common positive-probability successor. Sending  $(c, a, y)$  to that successor class therefore defines an update. Unsupported continuations may be assigned arbitrarily. This proves the three required properties.

For minimality, at time  $T$  any saturated coordinate separates unequal posterior vectors, so its fibres refine  $c_T$ . Suppose the assertion has been proved at  $t + 1$ . Equal coordinates at time  $t$  have equal posterior and predictive vectors. Their supported successors under any fixed  $(a, y)$  have equal next coordinates by recursion, and hence equal  $c_{t+1}$  by the induction hypothesis. Their complete keys (36.3) coincide. Backward induction proves the claimed simultaneous refinement, rather than just a bound on a terminal statistic.

To check prior independence, first observe that equality of posterior vectors under a positive prior is equivalent to proportionality of the nonnegative likelihood vectors  $(L_\theta(h))_\theta$ . This relation is independent of the prior. For a fixed parameter with  $\lambda_\theta(h) > 0$ , the actual next-report probabilities are

$$P_\theta(y \mid h, a) = k_t(y \mid h, a) \frac{\lambda_\theta(hay)}{\lambda_\theta(h)}. \quad (36.4)$$

In an existing class the right side is constant on histories for every supported successor, because the successor class determines its posterior. The set of parameters with positive likelihood is also constant on the class. Under a different positive prior, its new posterior weights are functions of its old posterior weights; averaging these common parameter-specific predictive kernels therefore gives a common new reference predictive kernel. The existing partition is saturated, predictive and recursive for the new prior. Minimality in both directions proves equality of the two partitions.

For reconstruction, write  $F_t(c, a, y)$  for the class update and define

$$H_t^0(c' \mid c, a) = \sum_{y: F_t(c, a, y) = c'} k_t(y \mid c, a),$$

$$L_t(y \mid c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t(y \mid c, a)}{H_t^0(c' \mid c, a)} \quad \text{if } H_t^0(c' \mid c, a) > 0.$$

Choose an arbitrary probability vector when the denominator is zero. The posterior at the next class is constant over the summed reports. Equation (36.4) thus factors the actual joint law of report and next class into this same  $L_t$  and the parameter-specific class-transition probability. Conditioning further on the next class cancels the likelihood factor. This gives a single parameter-independent report reconstruction, including when parameters have different supports. Histories having zero likelihood under a parameter are not visited under that parameter.

Starting from the initial class, the reconstruction stores the previous class, receives the next class after the actual chosen action, samples its report using  $L_t$ , and replaces its stored class. A target policy then chooses its next action from the reconstructed reports. Induction in time proves equality of the resulting action-report laws for every parameter and policy. In the other direction, apply  $F_t$  to the original reports. Product registers give the displayed storage bound; neither simulation presumes a cost-free transcript.

Finally, (36.1) enumerates the rational forward vectors on a finite tree. Formulae (36.2) and (36.3) use only finite sums, products, divisions by positive numbers, and equality tests of rational tuples. Class labels can be assigned by sorting those tuples backwards. To see the stated bit-complexity bound, take a common denominator for all rational input entries. Its bit length is at most their total input length. At level  $t$ , a forward entry can be represented with the  $t$ th power of this denominator times the initial denominator; sums add at most the bit length of the number of summands at each step. Thus every intermediate integer has bit length polynomial in horizon and input length. Posterior divisions and tuple comparisons preserve a polynomial bound. There are only as many forward and backward steps as nodes and edges of the expanded tree, times its finite hidden and parameter dimensions. This proves effectivity without any oracle for exact real comparisons.  $\square$

The theorem constructs the object assumed by the causal factorization in this finite class. Its minimality is among the *specified* likelihood-saturated, raw-report-predictive coordinates. It is not a claim that every exact simulator, or every task-specific lossy encoder, must use these coordinates. The variational principle optimizes the latter directly and may use substantially less memory.

**Example 36.2** (Saturation can have an exponential price). Let the parameter range over  $\{0, 1\}^d$ , and reveal it noiselessly one bit at a time during the first  $d$  observations. Thereafter all reports are independent fair bits. At time  $d$  all future-report predictions coincide, so the future-report predictive quotient has one class. Nevertheless the posterior on the parameter is a different point mass for each word. Every saturated coordinate at that time has at least  $2^d$  classes, and the construction above has exactly  $2^d$ . Predictive closure alone therefore supplies no controlled-cost parameter saturation. This is a cost calculation in a concrete model, not an assumption that the missing saturation has bounded size.

**Example 36.3** (Likelihoods alone need not close prediction). There is one parameter. Let  $Y_1$  be a fair bit and put  $Y_2 = Y_1$ . The constant coordinate is recursive and contains every parameter likelihood, but its one-step predictive law at time 1 is not a function of that coordinate. The least saturated predictive widths are  $(K_0, K_1, K_2) = (1, 2, 1)$ . The last quotient class can discard the bit, because reconstruction of  $Y_2$  still uses the previous class. This example separates the hypotheses of the factorization theorem; it does not assert impossibility of other simulators with other charged resources.

### APPENDIX 37. FINITE COMPILATION AND SEVERAL CHARGED RESOURCES

The finite variational principle also has an effective implementation. This is a separate source of multi-resource control from the energy oracle used in the predictive-tree construction. The price of synthesis is explicit and can be large; it is not moved into a free online oracle.

Use the finite instrument and finite decision alphabets of Section 34, with rational probabilities and losses. Let  $S = \max_t S_t$ ,  $Y = \max_t |Y_t|$ ,  $B = \max_d |B_d|$ , and  $m = |D|$ . A table implementation of one deterministic design has description length bounded, up to an absolute coding constant, by

$$\begin{aligned}
 P_0 = & \sum_{t=0}^{T-1} S_t |A| |Y_{t+1}| \lceil \log_2(S_{t+1} + 1) \rceil \\
 & + m \sum_{t=0}^{T-1} S_t \lceil \log_2(|A| + 1) \rceil + m S_T \lceil \log_2(B + 1) \rceil \\
 & + O((T + m) \log_2(T + S + |A| + Y + B + 2)).
 \end{aligned} \tag{37.1}$$

Unused states and forbidden controller actions can simply be retained as redundant table entries. The externally specified time index is the same clock convention as in (23.1). If no external clock is permitted, its  $T + 1$  values must also be included in the register.

**Theorem 37.1** (Effective finite-source realization). *An optimal shared-seed common-encoder scheme for the rational finite instrument can be synthesized by a finite exact calculation. It can use  $k \leq M + 1$  deterministic tables with rational mixing weights. For every integer  $p \geq 1$  there is a dyadic implementation whose excess is at most*

$$\Theta_{\text{sh}}(\mathbf{S}; D) + k 2^{-p}. \quad (37.2)$$

*Its fixed description has  $O(kP_0 + kp)$  bits. Initialization uses exactly  $p$  independent fair bits; no random bits are required subsequently. With an explicitly permitted read-only shared seed its persistent register has width at most  $S$ . Without that convention the same scheme has width at most  $kS$ , including the stored design index.*

*Each update reads one supplied finite-alphabet report. On a bit-serial table implementation, transient space can be bounded by  $O(p + \log(P_0k + 2) + \log(S + |A| + Y + B + T + m + 2))$  bits and the running time of an update or initialization by  $O(kP_0 + kp + p + 1)$  bit operations, after increasing the universal coding constant. No unit-cost access to an arbitrary real constant is required. Reading the physical report itself remains an input cost, rather than a claim about how an underlying continuous measurement was acquired.*

*Proof.* Enumerate the finite deterministic designs. Their risks are rational numbers obtained from finite products and sums of rational transition entries and losses. Full-history baselines can likewise be computed by enumerating full-history policies and solving their finite minimax linear programmes; for a Bayesian task their minimum is already a finite linear optimization. Thus the primal programme (34.4) has rational data and has a rational optimal basic solution. Rational linear algebra and the usual elimination of affine dependence among its risk vectors give a rational optimal mixture of at most  $M + 1$  designs. This gives a finite synthesis algorithm, not a bound polynomial in the horizon or the number of reports. Integer determinants of the finite rational constraint matrix give finite bit bounds for all weights, so synthesis does not require a probability oracle.

Let its weights be  $\lambda_1, \dots, \lambda_k$ . For  $r < k$  set  $\tilde{\lambda}_r = 2^{-p} \lfloor 2^p \lambda_r \rfloor$  and let the last weight be the nonnegative remainder. Then

$$\|\lambda - \tilde{\lambda}\|_{\text{TV}} = \sum_{r < k} (\lambda_r - \tilde{\lambda}_r) \leq (k - 1) 2^{-p}.$$

For every row, its risk lies in  $[0, 1]$ , so the difference of its mixed risks is at most this total variation. Taking the maximum and subtracting the unchanged baseline proves (37.2).

A uniform  $p$ -bit integer, compared to the cumulative integer weights, selects the dyadic design exactly. Its deterministic tables then implement the scheme. There are  $k$  copies of the tables counted in (37.1) and  $k$  integer weights of at most  $p + 1$  bits, which proves the description bound. In the private implementation the index is part of  $(r, I_t)$  and hence multiplies the width by  $k$ . In the shared implementation the seed is explicitly allowed throughout the run; its index can instead be reread from that permitted input. A modified or data-dependent seed would not qualify for this exception.

For a conservative machine model, store the immutable description as a sequential binary table with self-delimited addresses. Locate an update, a controller action, or a terminal answer by scanning it. Counters need only logarithmic space in its length and in the finite alphabet sizes. Comparing the  $p$ -bit initialization integer with its cumulative weights requires  $O(p)$  extra space and  $O(kp)$  time by a sequential scan. Copying the selected finite answer and updating the label has the same bounds. Thus the displayed time and space bounds are valid without a random-access assumption. They are deliberately not optimized.  $\square$

**Proposition 37.2** (Stability of finite synthesis). *Suppose two finite instruments have matched alphabets, losses and timing. Their initial hidden laws have total variation distance at most  $\eta_0$  for*

every parameter, and the corresponding transition–report rows have distance at most  $\eta_t$  uniformly in hidden state and control. Then, for every admissible common–encoder/controller design, the difference of each bounded–loss risk is at most  $\varepsilon = \min\{1, \eta_0 + \sum_{t < T} \eta_t\}$ . Their full–history baselines differ by at most  $\varepsilon$ , and their shared or private excess spectra at the same signature differ by at most  $2\varepsilon$ . An additional uniform error  $\zeta$  in the loss adds  $2\zeta$  to this last bound.

*Proof.* Couple the initial hidden states maximally. Until the two trajectories first disagree, choose the same controls, report–driven updates and random bits in both executions. Couple the corresponding hidden/report transition rows maximally. The probability of a disagreement by the last time is at most  $\eta_0 + \sum \eta_t$ . A bounded loss changes in expectation by at most that probability. This argument is uniform over every parameter and every legal controller, including full–history controllers. Infima over controllers and common encoders, and maxima over tasks or parameters, are nonexpansive in uniform error. Apply that observation once to the constrained risk and once to its baseline. A uniformly perturbed loss changes each of these by at most  $\zeta$ .  $\square$

This provides a multi–resource finite theorem with computable synthesis, finite randomization, finite description and finite execution cost. Together with Theorem 36.1, it also makes finite sufficient reconstruction effective whenever the expanded finite source is given rationally. It does not compute multiplicative energies for an arbitrary infinite predictive tree, nor does it turn a merely measurable sufficient statistic into a finite algorithm. Those are different input classes with different resource signatures.

Exact rational reconstruction tables need not have dyadic probabilities. Their computability does not imply exact sampling with a fixed number of fair bits. A fixed–bit implementation rounds their rows and uses Proposition 37.2 to charge the resulting path error; exact non–dyadic sampling requires a different randomization–time convention.

#### APPENDIX 38. A GAUSSIAN EXPERIMENT WITH A GENUINE TEMPORAL PENALTY

The append–register Gaussian theorem in the supporting development concerns a terminal vector experiment. The following experiment instead forces information to survive until the question that uses it arrives. Its final target is scalar, but its causal exponent depends on the number of coordinates acquired before that question.

Fix  $d \geq 2$  and  $\sigma > 0$ . Let  $\theta_1, \dots, \theta_d$  be independent and uniform on  $[-1, 1]$ , and observe sequentially

$$X_j = \theta_j + \sigma Z_j, \quad 1 \leq j \leq d,$$

where the  $Z_j$  are independent standard Gaussians. After these observations have been discarded, an independent uniform query  $J \in \{1, \dots, d\}$  is revealed. The machine then updates its register and estimates  $\theta_J$ . There is no external observation transcript. The query is an input to that final update, not an additional persistent decoder register. The cost is squared loss, bounded by four after projecting the answer onto  $[-1, 1]$ . Write

$$m(x) = \mathbb{E}[\theta_1 \mid X_1 = x], \quad \beta = \mathbb{E} \text{Var}(\theta_1 \mid X_1).$$

Let  $\mathcal{E}_S$  be the infimal excess above  $\beta$  when the total register has at most  $S$  states at every stage. Independent shared randomness is permitted; the same conclusions hold without it.

**Definition 38.1** (Delayed–query information signature). There are  $d$  sequential scalar–acquisition updates  $I_j = F_j(I_{j-1}, X_j, U_j)$ , followed by one query update  $I_{d+1} = F_{d+1}(I_d, J, U_{d+1})$  and a terminal output  $\hat{\theta} = D(I_{d+1}, U_{d+2})$ . Each register has at most  $S$  values. The  $U_j$  are independent private draws; an optional independent retained public seed may be an argument of every rule. The query is independent of the pre–query history, and neither  $J$  nor an old report is available to the terminal decoder except through  $I_{d+1}$ . The cardinality signature permits arbitrary measurable real arithmetic. Proposition 39.2 gives a distinct finite–acquisition and finite–table implementation.

**Theorem 38.2** (Delayed Gaussian query law). *There are constants  $0 < c_{d,\sigma} \leq C_{d,\sigma} < \infty$  such that for every  $S \geq 2^d$ ,*

$$c_{d,\sigma} S^{-2/d} \leq \mathcal{E}_S \leq C_{d,\sigma} S^{-2/d}. \quad (38.1)$$

*The same order holds for the maximum over the  $d$  individual coordinate query risks, with a common encoder before the query.*

*In contrast, an encoder which receives the query together with the full observation vector at its sole checkpoint has excess of order  $S^{-2}$ . Thus the scalar terminal target has a strictly worse cardinality exponent under the delayed-query causal constraint. The lower bound uses the register before the query and is not a final-output dimension bound.*

*Proof.* The posterior density on  $[-1, 1]$  is proportional to  $\exp(-(x - u)^2/(2\sigma^2))$ . Differentiation under the integral on this compact interval gives

$$m'(x) = \frac{\text{Var}(\theta_1 \mid X_1 = x)}{\sigma^2} > 0. \quad (38.2)$$

The posterior variance is positive because its density is positive on an interval. The function  $m$  is smooth, odd, and strictly increasing. Choose  $b > 0$  and put  $a = m(b) > 0$ . The density of  $X_1$  is positive and continuous, and  $m'$  is positive and continuous on  $[-b, b]$ . The change-of-variables formula therefore gives a density for  $m(X_1)$  which is bounded below by a positive constant on  $[-a, a]$ . Independence implies that  $M = (m(X_1), \dots, m(X_d))$  has density bounded below by a positive constant on the cube  $Q = [-a, a]^d$ .

We record an elementary volume bound. If  $c_1, \dots, c_S$  are arbitrary points in  $\mathbb{R}^d$ , and  $v_d$  is the volume of the unit ball, choose  $r = (|Q|/(2Sv_d))^{1/d}$ . The union of their radius- $r$  balls has volume at most  $|Q|/2$ . Consequently

$$\int_Q \min_{i \leq S} \|x - c_i\|^2 dx \geq \frac{|Q|}{2} \left( \frac{|Q|}{2Sv_d} \right)^{2/d}. \quad (38.3)$$

This bound does not require cells to be convex or centres to lie in  $Q$ .

Consider any admissible machine and condition on its independent shared random seed, when present. Its pre-query register  $I_d$  has at most  $S$  values. Subsequent private updates and readouts can do no better than being given  $(I_d, J)$  and using its conditional posterior mean. Let  $c_i = (\mathbb{E}[\theta_j \mid I_d = i])_{j=1}^d$ , with the seed included in all these conditional expectations. The Markov relation from the full observation to the register, and independence of the random seed, imply  $c_i = \mathbb{E}[M \mid I_d = i]$ . Orthogonality of conditional expectation shows that the resulting excess for the uniform query is

$$\frac{1}{d} \mathbb{E} \|M - c_{I_d}\|^2 \geq \frac{1}{d} \mathbb{E} \min_{i \leq S} \|M - c_i\|^2.$$

For each fixed seed the law of  $M$  is unchanged. Its density lower bound on  $Q$  and (38.3) give  $c_{d,\sigma} S^{-2/d}$ . Averaging the seed preserves this lower bound. The argument permits randomized private encoding: replacing its selected centre by the nearest centre is only a lower bound, not an assumption that the encoding is deterministic. The maximum coordinate risk is at least the uniform average, proving its lower bound as well.

For the upper bound put  $L = \lfloor S^{1/d} \rfloor \geq 2$  and partition  $[-1, 1]$  into  $L$  equal intervals. At observation  $j$  compute the interval label of  $m(X_j)$  and append it in base  $L$  to the stored labels. At that time there are at most  $L^j \leq S$  states. After the query, replace the whole label by the interval label of the requested coordinate; the terminal decoder outputs its midpoint. This last update leaves only  $L$  possible labels and retains no separate query. Its squared excess in every coordinate is at most  $L^{-2} \leq 4S^{-2/d}$ . Conditional orthogonality again subtracts the same full-observation baseline  $\beta$ . The construction is deterministic and works with or without a shared seed.

For the noncausal checkpoint comparison, the encoder sees  $(X, J)$  and quantizes only  $m(X_J)$  into  $S$  equal intervals. This has excess at most  $S^{-2}$ . The one-dimensional version of the same

density lower bound and (38.3) gives the matching lower bound, even with randomized encoding and a shared seed. Its  $S$  terminal centres are arbitrary real answers in  $[-1, 1]$ , so no additional discreteness assumption has entered. This proves all assertions.  $\square$

The theorem is a Bayesian slice of the parameterized Gaussian experiment with the specified product prior, not a claim about uniform deficiency over the whole Gaussian family. Its constants depend on  $d$  and  $\sigma$ . It controls persistent cardinality and finite acquisition count, not the arithmetic cost of evaluating the real posterior mean. Under a finite numerical signature, an approximation of  $m$  requires a separate accuracy and computation bound. The obstruction nevertheless holds for every such implementation, since it was proved for the larger unrestricted-arithmetic class.

The lower bound is the same conditional-variance geometry as Corollary 35.2, now evaluated directly for a continuous history law. In the finite theorem, successor compatibility determines which cells can be executed. Here the late query forces all coordinate readouts to factor through the same pre-query cell, and the volume bound computes its cost. This explains both the lossless append-only regime and a regime where forgetting before a future question changes the exponent, without identifying either one with static terminal dimension.

### APPENDIX 39. MARKED PROJECTIONS AND FINITE NUMERICAL ACQUISITION

The common object behind the decision problems is the attainable family of joint laws under one admissible encoder, with its task controllers and its actual target marks. Evaluating a bounded loss is a linear projection of this family. A risk body for one selected task family is therefore a projection, not by itself a classification of the whole experiment. We record the exact links to the quantitative regimes used later in the article.

**Proposition 39.1** (Quadratic and metric projections). *For an uncontrolled standard Borel history  $H$  and a square-integrable Hilbert-valued target  $U$ , write  $m(H) = \mathbb{E}[U \mid H]$ . Any statistic  $I$  formed from  $H$  with independent randomization has optimal quadratic excess*

$$\mathbb{E} \|m(H) - \mathbb{E}[m(H) \mid I]\|^2. \quad (39.1)$$

*For finitely many such tasks the common-encoder weighted support problem is the conditional variance of the direct sum of their posterior means multiplied by the square roots of their task weights. On finite histories this is precisely the compatible-partition functional in Corollary 35.2. On the predictive trees of Theorem 53.1, its within-cell variances are comparable to the retained tree-energy sums under the hypotheses of that theorem.*

*For a revealed law  $\pi \in \mathcal{P}([0, 1])$ , take as actual target  $\pi$  and as output a realized probability law  $\hat{\pi}$ . The loss  $W_1(\pi, \hat{\pi})$  yields exactly the law-presentation profile of Theorem 49.1. Replacing the output by a sample, and testing only its marginal, is a different task and does not preserve this profile.*

*Proof.* Conditional on  $(H, I)$  the target mean is  $m(H)$ . Orthogonal projection gives

$$\mathbb{E} \|U - a(I)\|^2 = \mathbb{E} \|U - m(H)\|^2 + \mathbb{E} \|m(H) - a(I)\|^2.$$

The second term is minimized by  $a(I) = \mathbb{E}[m(H) \mid I]$ , proving (39.1). Weighted sums are squared norms in the Hilbert direct sum. For a deterministic finite partition, conditioning on  $I$  is averaging over its cell, giving the stated finite functional. The lower and upper comparisons with tree energy are exactly the separation and diameter estimates of Theorem 53.1; no new equality between energy and variance is asserted. The continuous Gaussian volume argument is another evaluation of (39.1), not an invocation of the finite-history partition theorem on a continuous history space.

For the second assertion, the joint output loss of an encoder and its random decoder is, by definition,  $\mathbb{E} W_1(\pi, D(Q(\pi)))$ . The decoder must output a law on each run. This is the profile in (49.1), whose minimax order is  $1/\log(S+1)$  under its stated common-code convention. The

single-marginal probe in Proposition 49.3 has a different output and a different rate. Integrating out the realized law before applying the distance changes the loss, so does not commute with this projection.  $\square$

The revealed-law family is generally nondominated. It belongs to the marked-spectrum definition, but is not being smuggled into the product-dominated compactness theorem. Conversely the latter has a proved closed attainable body, whereas a general infinite family of tasks need not have one without hypotheses. Terminal statistical deficiency uses the decision-complete task family and compactness specified in Theorem 23.4; a finite selection of task supports is not sufficient to infer Blackwell equivalence. These distinctions allow the tree, Wasserstein, finite Gaussian and count bounds to be used together without equating their output spaces.

**39.1. A finite table for the delayed Gaussian problem.** Retain the notation and information pattern of Definition 38.1. The following bound separates acquisition, numerical evaluation and persistent storage.

**Proposition 39.2** (Finite-precision Gaussian implementation). *Let  $L = \lfloor S^{1/d} \rfloor \geq 2$ . An acquisition interface returns one of the interior bins of a grid of width at most  $h$  on  $[-R, R]$  and two tail symbols, where  $R > 1$ . Let the stored table approximate  $m(x)$  at each interior bin midpoint to error at most  $\varepsilon$ , with values clipped to  $[-1, 1]$ . Quantize those approximations into  $L$  equal intervals and store the resulting labels in the base- $L$  register. Then the squared excess in every coordinate is at most*

$$\left(L^{-1} + \frac{h}{2\sigma^2} + \varepsilon\right)^2 + 8 \exp\left(-\frac{(R-1)^2}{2\sigma^2}\right). \quad (39.2)$$

*For fixed computable  $\sigma > 0$ , there are rational uniform-grid choices with  $1/(2L) \leq h \leq 1/L$ ,  $\varepsilon \leq 1/L$ , and  $1 + \sigma\sqrt{6\log L} \leq R \leq 2 + 2\sigma\sqrt{6\log L}$  that give excess  $O_\sigma(L^{-2})$ . The number of acquisition symbols is  $O_\sigma(L\sqrt{\log L})$ , the table has  $O_\sigma(L(\log L)^{3/2})$  bits, and each scalar acquisition uses  $O_\sigma(\log L)$  report bits. For rational  $\sigma^2$  the table is computable by a terminating certified rational procedure. Its offline evaluation cost is separate from the bit-serial lookup and register-update costs.*

*Proof.* Equation (38.2) and  $|\theta| \leq 1$  imply  $0 < m'(x) \leq \sigma^{-2}$ . An interior-bin midpoint differs from  $X$  by at most  $h/2$ , so its posterior mean differs by at most  $h/(2\sigma^2)$ . Table evaluation and the final equal-interval quantization add at most  $\varepsilon$  and  $1/L$ . On a tail event, any clipped output differs from  $m(X)$  by at most two. Since  $X = \theta + \sigma Z$  and  $|\theta| \leq 1$ ,

$$\mathbb{P}(|X| > R) \leq 2 \exp(-(R-1)^2/(2\sigma^2)).$$

The conditional-variance identity subtracts the same full-observation baseline and gives (39.2). This estimate holds coordinatewise; no union bound over all coordinates is needed for a single queried coordinate. The pre-query and post-query register updates are exactly the base- $L$  construction of Theorem 38.2.

There are  $O(R/h)$  interior bins. Only their final labels in  $[L]$  need be stored after certified preprocessing; thus the table uses  $O((R/h) \log L)$  bits. One may include the rational grid endpoints and the two prescribed tail labels in the fixed program. Choosing rational upper and lower approximations with constant-factor slack gives the asserted bounds without requiring an exact logarithm or square root in the online machine. A sequential lookup scans the finite table; binary arithmetic then appends or extracts a base- $L$  digit. It uses  $O(d \log L + \log(R/h))$  transient bits besides the read-only program and polynomial time in the table length and the register bit length. The physical analog-to-bin acquisition is an explicit interface operation, not claimed to be implemented for free by the register machine.

For completeness, certified values of  $m$  are available without an exact real oracle when  $\sigma^2$  and the grid are rational. They are ratios of the integrals

$$\int_{-1}^1 u e^{-(x-u)^2/(2\sigma^2)} du, \quad \int_{-1}^1 e^{-(x-u)^2/(2\sigma^2)} du.$$

The denominator is at least  $2e^{-(R+1)^2/(2\sigma^2)} > 0$  for  $|x| \leq R$ . Rational Taylor bounds for the exponential, combined with Riemann-sum error bounds from uniform derivative bounds on the compact rectangle, give rational intervals for both integrals of arbitrarily small width. Refine until the quotient interval has width at most  $\varepsilon$ , and choose its rational midpoint. If it lies near an equal-interval threshold, either adjacent label still has error at most  $1/L + \varepsilon$  after enlarging the certified tolerance by a fixed factor, or the rational midpoint may simply be compared exactly to the rational thresholds. Termination follows from the denominator lower bound. The same construction for fixed computable  $\sigma$  uses certified intervals for it. No polynomial bound in the unspecified description of a general computable real is asserted.  $\square$

The cardinality converse remains valid for this finite numerical signature because its implementations form a subclass of the unrestricted measurable machines used in that converse. Finite acquisition therefore preserves the delayed-query exponent when its error parameters are scaled as above, while displaying the additional resources that the cardinality theorem alone does not price.

#### APPENDIX 40. POSITIVE OBSERVATION MODELS AND HISTORY REDUCTION

We give explicit model consequences in which the marked law, continuity and contraction hypotheses can be checked from primitive data. These results concern the models specified here. They do not identify a finite-state model with an unconstructed microscopic billiard or kinetic model.

Let  $Z_t \in [n]$ ,  $a \in A$  with  $A$  finite, and suppose

$$\epsilon \leq P_a(z, z') \leq M, \quad Y_{t+1} = m_{a, Z_{t+1}} + \sigma_{a, Z_{t+1}} \xi_{t+1}, \quad \xi_t \sim N(0, 1), \quad (40.1)$$

where  $0 < \epsilon < M$ ,  $|m_{a,z}| \leq m_*$  and  $0 < \sigma_- \leq \sigma_{a,z} \leq \sigma_+$ . All primitive coefficients may depend continuously on a compact unknown parameter, with the same bounds. The initial hidden probabilities and any finite marking kernel also depend continuously on this parameter. The observation is of the *new* state. Its joint transition/report kernel is

$$K_a(z', dy \mid z) = P_a(z, z') g_{a, z'}(y) dy.$$

Finite terminal or path marks are permitted.

**Theorem 40.1** (A continuous-report consumer). *For every fixed horizon, the instruments (40.1) satisfy the compact dominated hypotheses of Theorem 26.3 for bounded losses continuous in the primitive parameter. Their common-encoder minimax problems have attaining private designs and attaining shared mixtures.*

*Partition  $[-R, R]$  into intervals of length at most  $h$ , where  $R > m_*$ , and use two additional tail cells. Reconstruction is uniform on interior intervals and uses any fixed probability density on each tail. For every memory width and every admissible controller, the resulting marked finite-report approximation has risk error at most  $Te(h, R)$ , where*

$$e(h, R) = \frac{h}{2\sigma_-} \sqrt{\frac{2}{\pi}} + 2 \exp\left(-\frac{(R - m_*)^2}{2\sigma_+^2}\right). \quad (40.2)$$

*The corresponding excess-risk error is at most  $2Te(h, R)$  when baselines are recomputed.*

*Proof.* Use a positive Gaussian probability as the reference measure on each report space and counting probability on finite marks. With an open-loop action word fixed, the joint marked density is a finite sum over hidden paths of products of transition probabilities, Gaussian densities and terminal marking probabilities. Gaussian location-scale densities depend continuously in  $L^1$  on their parameters when the variance is bounded away from zero. This follows, for example, by pointwise convergence and equality of their integrals, or directly by truncating uniformly Gaussian tails and using uniform continuity on a compact interval. Products of probability densities are continuous in  $L^1$  by telescoping, and so are finite mixtures. Compactness of the primitive parameter set gives compactness of the path-density family in  $L^1$ . Multiplication by bounded continuously parameterized losses preserves the required effect continuity. Theorem 26.3 applies.

If  $I$  is an interval of length at most  $h$  and  $\bar{g}_I$  is its average density, the fundamental theorem of calculus gives

$$\int_I |g - \bar{g}_I| \leq h \int_I |g'|.$$

Indeed average  $|g(x) - g(y)|$  over  $y \in I$  and bound the difference by the integral of  $|g'|$  between  $x, y$ . For a Gaussian density,  $\int |g'| = \sqrt{2/\pi}/\sigma$ . On the two tails, replacing a conditional density changes total variation by at most their total probability, bounded by the exponential term in (40.2). Thus each conditional observation channel has total variation error at most  $e(h, R)$  uniformly in hidden state, control and parameter. Couple the hidden transition identically and maximally couple the original and reconstructed observations. As long as observations and random tapes have matched, both controllers make the same decisions. The probability of a first mismatch during  $T$  steps is at most  $Te(h, R)$ . On its complement the entire hidden path, all reports and the actual mark agree. Bounded losses change by at most this probability. The report conversions are transient operations, so the memory width is unchanged. Comparing the unrestricted baselines in the same way finishes the proof.  $\square$

**40.1. A proved forgetting estimate.** For positive probability rows  $u, v$ , put

$$d_H(u, v) = \max_i \log(u_i/v_i) - \min_i \log(u_i/v_i).$$

Multiplying both rows by the same positive diagonal matrix and then normalizing leaves  $d_H$  unchanged. We use the elementary positive matrix contraction in its explicit form.

**Lemma 40.2** (Positive-matrix contraction). *If the entries of a matrix  $P$  belong to  $[\epsilon, M]$ , then*

$$d_H(uP, vP) \leq \rho d_H(u, v), \quad \rho = \frac{M - \epsilon}{M + \epsilon} < 1.$$

*For arbitrary nonnegative probability rows, the first images satisfy  $d_H(uP, vP) \leq 2 \log(M/\epsilon)$ .*

*Proof.* Set  $x_i = \log(u_i/v_i)$  and interpolate  $v_i \exp(sx_i)$ ,  $0 \leq s \leq 1$ . The derivative of the logarithm of output coordinate  $j$  is the expectation of  $x_i$  under weights proportional to  $v_i \exp(sx_i) P_{ij}$ . For two output coordinates  $j, k$ , the likelihood ratio of their normalized weight distributions has maximum divided by minimum at most  $(M/\epsilon)^2$ . If two probability distributions have likelihood ratios in an interval  $[a, b]$  with  $b/a \leq C$ , their total variation is at most  $(\sqrt{C} - 1)/(\sqrt{C} + 1)$ . To verify this bound, put all ratios below one at the lower endpoint and all ratios above one at the upper endpoint; the mass constraint makes the resulting variation at most  $(1 - a)(b - 1)/(b - a)$ . Maximizing over  $b/a \leq C$ ,  $a \leq 1 \leq b$ , gives  $a = C^{-1/2}$ ,  $b = C^{1/2}$  and the displayed value. Differences of the two derivative expectations are therefore at most  $\rho \text{osc}(x)$ . Integrate in  $s$  and maximize over  $j, k$ . At  $s = 0$  all output log ratios vanish, proving the first claim. For the second, every coordinate of either probability row times  $P$  lies in  $[\epsilon, M]$ ; its coordinatewise ratio lies in  $[\epsilon/M, M/\epsilon]$ .  $\square$

**Theorem 40.3** (Uniform finite-history recovery). *Fix the primitive parameter. Two filters for (40.1), started with arbitrary initial probabilities and fed the same actions and observations for  $L \geq 1$  steps, satisfy*

$$\mathrm{TV}(\pi_L, \tilde{\pi}_L) \leq \frac{1}{2} \log(M/\epsilon) \rho^{L-1}.$$

*In particular a last- $L$ -history filter, initialized with a fixed probability at its left endpoint, approximates the full filter with this error uniformly in the earlier history. With  $N$  report cells, a literal suffix implementation uses at most  $(|A|N)^L$  labels, besides a finite beginning-of-stream convention. For any  $[0, 1]$ -valued terminal loss on the current hidden state, an optimal suffix-based action has risk at most twice this filtering error above the full-filter optimum.*

*Proof.* Prediction multiplies a posterior row by  $P_a$ . Conditioning on the current report multiplies it by the positive diagonal matrix with entries  $g_{a,z'}(y)$  and normalizes. The latter operation preserves  $d_H$ , and Lemma 40.2 contracts the former. The first prediction gives diameter at most  $2\log(M/\epsilon)$  and the next  $L - 1$  predictions give the factor  $\rho^{L-1}$ . For probability rows,  $\mathrm{TV}(u, v) \leq \tanh(d_H(u, v)/4) \leq d_H(u, v)/4$ ; the first inequality follows from the same two-endpoint likelihood-ratio bound in the lemma. Restart the truncated filter with the fixed prior at time  $t - L$  and apply this result to obtain the suffix estimate. Its raw suffix is updated by shifting and appending the current control/report cell. Finally every  $[0, 1]$  loss expectation changes by at most the total variation of the two posteriors. Compare the optimal action under each posterior through the other posterior, which costs at most twice that distance.  $\square$

This contraction is not asserted for arbitrary past-path target marks: augmenting the hidden state to retain such a mark can destroy strict positivity. The general marked duality retains these marks without using the forgetting assertion.

The parameter must be fixed in this forgetting assertion. A static unknown phase need not be forgotten, and applying the assertion to its posterior would be false. The compact robust theorem applies to the unknown-parameter experiment without that extra assertion. Also, summable variations alone have not been used to infer exponential decay: here it follows from the explicit positive-matrix contraction. The distinction is important for history-kernel applications.

#### APPENDIX 41. NONLINEAR FINITE-STATE SEMIGROUPS

Let  $Q^a$  be conservative generators on  $[n]$ ,  $a \in A$  finite, and let  $v_i^a$  be bounded potentials. Define on  $\mathbb{R}^n$

$$H_i(u) = \max_{a \in A} \left\{ v_i^a + \sum_{j \neq i} q_{ij}^a (e^{u_j - u_i} - 1) \right\}. \quad (41.1)$$

This is a nonlinear Hamiltonian. The following proof does not apply a linear resolvent identity to it.

**Theorem 41.1** (Global semigroup and nonlinear resolvent). *The equation  $\dot{u} = H(u)$  has a global solution for every initial condition and defines a semigroup  $S_t$  that is order preserving, additively homogeneous and nonexpansive in the sup norm. If  $V = \max_{i,a} |v_i^a|$ , then  $\|S_t u\|_\infty \leq \|u\|_\infty + tV$ . For every  $\lambda > 0$  the equation*

$$u - \lambda H(u) = f$$

*has a unique solution  $J_\lambda f$ . The resolvent is order preserving, additively homogeneous and nonexpansive. In particular  $H$  is maximal dissipative with this sign convention.*

*Proof.* The map  $H$  is locally Lipschitz as the maximum of finitely many smooth functions. If  $u_i - v_i = \max_j (u_j - v_j)$ , then  $u_j - u_i \leq v_j - v_i$  for every  $j$ , and hence  $H_i(u) \leq H_i(v)$ . The upper right derivative of the maximum coordinate of the difference of two solutions is therefore nonpositive. Applying the same argument to the reverse difference proves sup-norm contraction

and comparison. Since  $H(u + c\mathbf{1}) = H(u)$ , uniqueness gives additive homogeneity. Comparing with the time-dependent constant barriers  $(\max_i u_i + tV)\mathbf{1}$  and  $(\min_i u_i - tV)\mathbf{1}$  gives the norm bound. It prevents finite-time escape, so the locally defined solution is global; uniqueness gives the semigroup law.

For the resolvent, consider the damped equation  $\dot{u} = f - u + \lambda H(u)$ . The interval of constant barriers

$$(\min_i f_i - \lambda V)\mathbf{1} \leq u \leq (\max_i f_i + \lambda V)\mathbf{1}$$

is invariant. At a maximal difference between two solutions the upper derivative is at most the negative of that difference. Thus its flow contracts distances by  $e^{-t}$ . At time one this is a strict contraction of a complete invariant rectangle into itself, so it has a unique fixed point. Commutation with the flow shows that the fixed point is stationary: every later image is another fixed point of the time-one map. It therefore solves the resolvent equation. The same maximal-coordinate comparison applied to two resolvent equations gives  $\max_i (u_i - \tilde{u}_i) \leq \max_i (f_i - \tilde{f}_i)$ , which proves uniqueness, comparison and nonexpansiveness. Translation by constants proves homogeneity.  $\square$

**Theorem 41.2** (Marked mesh approximation and stability). *Define the log-moment Bellman step*

$$(F_\Delta u)_i = \max_a \log \left( e^{\Delta(Q^a + \text{diag } v^a)} e^u \right)_i.$$

*For fixed  $t$  and bounded initial  $u$ , with  $\Delta = t/m$ ,  $F_\Delta^m u \rightarrow S_t u$  with error  $O(\Delta)$ , uniformly on bounded sets. Each mesh problem, after acquisition of the initial state, has finite report and control spaces and a bounded actual path-reward mark. Its bounded moment effects therefore have the common-encoder risk-body duality of Corollary 26.4, for compact continuously parameterized primitive data. The seed average is taken on these moments before taking a logarithm.*

*If  $\tilde{Q}^a, \tilde{v}^a$  are another such family and*

$$\delta_Q = \max_{i,a} \sum_{j \neq i} |q_{ij}^a - \tilde{q}_{ij}^a|, \quad \delta_v = \max_{i,a} |v_i^a - \tilde{v}_i^a|,$$

*then on a common oscillation bound  $B$  for the trajectories,*

$$\|S_t u - \tilde{S}_t u\|_\infty \leq t\{\delta_v + (e^B + 1)\delta_Q\}.$$

*For the resolvents the same bound holds with  $t$  replaced by  $\lambda$  and  $B$  bounding their solutions.*

*Proof.* The matrix exponential is a positive Feynman–Kac operator. On a bounded set of  $u$ , its Taylor expansion and the logarithm give  $F_\Delta u = u + \Delta H(u) + O(\Delta^2)$ , uniformly for the finite family of generators. Positivity makes  $F_\Delta$  order preserving, and  $F_\Delta(u + c\mathbf{1}) = F_\Delta u + c\mathbf{1}$ . These two properties imply sup-norm nonexpansiveness. Its iterates and the exact flow remain in the same bounded region up to time  $t$ , by the constant barriers. The local flow expansion has the same first-order term and a uniform  $O(\Delta^2)$  remainder. Telescope the errors over  $m$  steps using nonexpansiveness; the bound is  $mO(\Delta^2) = O(t\Delta)$ .

For a fixed control during one interval, the positive kernel  $e^{\Delta(Q^a + \text{diag } v^a)}$  is the expectation of the exponential integrated reward with the terminal state indicator. Keeping the actual bounded reward integral as a mark, rather than interpreting this nonstochastic matrix as a transition probability, gives its proper probabilistic representation. Concatenation gives the mesh instrument. Finite sampled states and controls have a finite report representation; continuous reward marks can be integrated into its bounded effect functions. For this reward-maximizing problem, use a decreasing affine normalization of the bounded moment as a loss in  $[0, 1]$  when applying the risk theorem. This reverses the optimization in the required way. For finite state/action paths this integration needs no assumption that reward marks have a common density: Corollary 26.4 applies directly. For finite parameter and task sets, one may also augment the finite hidden state by the sampled state/control path and replace the moment loss by its conditional expectation

given that path. This gives a finite instrument with exactly the same moment risks, to which Theorem 34.1 applies. It is an equivalence for these moment effects, not a finite-state realization of every continuous reward mark. Continuity of the effects on a compact primitive family follows from continuity of the finite matrix exponentials and products.

On  $\text{osc}(u) \leq B$ , direct subtraction of (41.1) gives  $\|H(u) - \tilde{H}(u)\|_\infty \leq \delta_v + (e^B + 1)\delta_Q$ . The maximal-coordinate comparison used above now bounds the upper derivative of the solution difference by this constant. Integration proves the flow estimate. Subtracting the two resolvent equations at a maximizing coordinate proves its resolvent counterpart. The logarithm of an average exponential moment is not an average of logarithms; none of these steps changes that ordering.  $\square$

The semigroup construction is a finite-state instance of the logarithmic control-semigroup method of Nisio [59] and risk-sensitive control. Its role here is an explicit nonlinear consumer with a proved range condition and a marked finite-mesh interface, not a claim that finite-dimensional comparison settles the domain and coercivity questions of an infinite-dimensional kinetic generator.

#### APPENDIX 42. HIDDEN PHASES AND RISK-SENSITIVE CONTINUATION

Let  $C$  be a finite latent phase chosen once with positive initial weights, and let  $Z_t$  follow a controlled hidden model conditional on  $C$ . Its joint transition/report densities are  $K_{c,a}(z', y | z)$  with respect to a probability measure  $\mu(dy)$ . Let  $r_{c,a}(z, z')$  and the terminal reward  $h_c(z)$  be bounded, and fix  $\gamma > 0$ . One controller must choose each action without observing  $C$ . The objective in this section is to minimize the positive exponential moment. A logarithm may be applied afterwards.

The exponential information-state mechanism is classical; see Bäuerle and Rieder [60]. We use the following explicit unnormalized version so that no phasewise policy is hidden in the notation. Define a nonnegative row  $\eta$  on  $(c, z)$  and the tilted update

$$(\eta K_a^\gamma(y))(c, z') = \sum_z \eta(c, z) K_{c,a}(z', y | z) e^{\gamma r_{c,a}(z, z')}. \quad (42.1)$$

Unlike an ordinary phase posterior,  $\eta$  includes the accumulated reward weight. Its total mass must not be discarded without a corresponding multiplicative value factor.

**Theorem 42.1** (One-policy phase recursion). *For full-history controls, the minimum remaining exponential moment has the recursion*

$$V_T(\eta) = \sum_{c,z} \eta(c, z) e^{\gamma h_c(z)}, \quad V_t(\eta) = \min_{a \in A} \int V_{t+1}(\eta K_a^\gamma(y)) \mu(dy).$$

*The value is positively homogeneous and Borel, and an optimal measurable selector exists. The minimum is outside the phase sum: replacing it by separate phasewise minima is in general incorrect. For finite-register common encoders, the exact moment risk body is instead obtained from the coupled design of Theorem 26.3 (or its finite specialization), not by providing  $\eta$  as free online memory. The model satisfies that theorem whenever its marked path densities and bounded effects satisfy the stated domination and continuity hypotheses; in particular finite phases with positive Gaussian observations and bounded finite-path rewards do so for fixed primitives, or for compact continuous primitive families as in Theorem 40.1, using Corollary 26.4 when only effect domination is needed.*

*Proof.* For a fixed report/action history,  $\eta_t(c, z)$  is the joint density of phase, current state and that report history, multiplied by the exponential accumulated reward and summed over past hidden states. This interpretation is true initially and propagates exactly by (42.1). At the terminal time summing the remaining exponential terminal factor gives  $V_T$ . At time  $t$  the action is one measurable choice from the current history; it cannot depend on an unobserved phase. For a fixed

choice  $a$ , integrate the remaining value of the updated density against the reference observation law. Backward induction gives the recursion. The minimum of finitely many Borel integrals is Borel and the least-index minimizer is a measurable selector. Induction also gives positive homogeneity. This is the unnormalized information-state form of risk-sensitive dynamic programming; it accounts for the probability and reward weights together.

For a finite-register design, fix its action and register paths and expand the true marked path law. Its exponential reward is a bounded positive loss after affine rescaling. The same encoder kernels appear in every task and phase row, so the compact risk theorem applies without giving the controller the unbounded-dimensional history or an uncharged exact real vector. The Gaussian model's domination and compactness follow from the finite mixture proof in Theorem 40.1. This proves the last assertion.  $\square$

**Example 42.2** (Strict failure of phasewise optimization). There is no observation. The equally likely phase is  $C \in \{0, 1\}$ , the action is  $a \in \{0, 1\}$ , and the cost is  $\mathbf{1}_{\{a \neq C\}}$ . The best actual exponential moment is  $(1 + e^\gamma)/2$ , for any mixture of the two actions. Its logarithmic value is  $\gamma^{-1} \log((1 + e^\gamma)/2)$ . Choosing  $a = C$  separately in each hidden phase would give moment one, but is not an admissible policy. The tilted recursion and the common-encoder risk body both retain the actual value.

#### APPENDIX 43. A MICROSCOPIC GRAPH CORE AND WIDTH-UNIFORM ACQUISITION TRANSFER

The comparison theorem can be used before a kinetic limit is taken. Here the underlying trajectories are actual hard-sphere trajectories, not a finite hidden-state surrogate. We construct a collision-compatible core for their unbounded transport generator and show that replacing observables by this core preserves every bounded-register decision problem under noisy acquisition. The topology of the core is specified explicitly: it is a weighted  $L^p$  graph topology, not local uniform convergence on kinetic action sublevels.

**43.1. The microscopic flow and invariant spaces.** Let  $\Omega_{N,\epsilon}$  be the nonoverlap phase space of  $N$  identical hard spheres of diameter  $0 < \epsilon < 1/4$  in the unit flat torus, with unbounded velocities. At a binary nongrazing collision the incoming and outgoing velocities are related by elastic specular reflection. We use the classical almost-everywhere hard-sphere flow  $\Phi_t^{N,\epsilon}$ , defined for all real times outside the invariant singular null set. The requisite existence statement excludes multiple collisions, grazing pathologies, and finite-time accumulation for almost every initial state; see the hard-sphere flow construction in [63]. This classical flow existence theorem, not a kinetic-limit theorem, is an explicit input here. The cited construction is stated in Euclidean space. Its local exceptional-set argument applies to the periodic system as follows. On a bounded-energy set and a bounded time interval there are only finitely many periodic images that can enter a collision chart. Near a contact,  $\epsilon < 1/4$  gives an unambiguous local lift to a Euclidean binary-collision surface. The volume estimate for states with two collisions in an interval of length  $\delta$  is therefore the same  $O(\delta^2)$  estimate, with a finite sum of chart constants; this is the estimate in Lemma 4.1.1 of the cited construction. A union over  $O(T/\delta)$  time intervals has volume  $O(T\delta)$ . Letting  $\delta$  decrease to zero excludes collision accumulation, together with the multiple and grazing exceptional sets. Exhaust bounded energies and time intervals countably. This supplies the periodic version used here; no uniform-in- $N$  collision estimate is inferred from it.

The normalized equilibrium measure

$$d\mu_{N,\epsilon,\beta} = Z_{N,\epsilon,\beta}^{-1} \mathbf{1}_{\Omega_{N,\epsilon}} \exp(-\beta|V_N|^2/2) dX_N dV_N$$

is invariant. Free flight preserves phase volume, elastic reflection has unit absolute Jacobian and preserves kinetic energy, and their concatenation gives invariance on the regular flow. Null sets are completed in the usual way. We restrict to permutation-symmetric observables.

One may use a fixed nonempty system, or a countable disjoint family of nonempty systems  $(N, \epsilon_N)$  with positive summable weights  $a_N$ . In the latter case put  $\bar{\mu} = \sum_N a_N \mu_{N, \epsilon_N, \beta}$ , normalized to mass one. This is a family of particle sectors, not a claim that infinitely many positive-radius spheres fit into a fixed torus. Additional particle-number or energy weights may be absorbed into the invariant measure whenever their total mass is finite. Fix  $1 \leq p < \infty$  and let  $\mathcal{B}_p = L^p_{\text{sym}}(\bar{\mu})$ .

**Lemma 43.1** (Strong transport and a dense observable class). *The operators  $U_t f = f \circ \Phi_t$  form a strongly continuous isometric group on  $\mathcal{B}_p$ . Let  $A$  be its generator. The symmetric smooth observables with finite sector support and compact support in the interior phase space are dense in  $\mathcal{B}_p$ . The generator is the closed transport operator associated with the physical collision flow; in the interior it acts as  $\sum_i v_i \cdot \nabla_{x_i}$ .*

*Proof.* Invariance gives  $\|U_t f\|_p = \|f\|_p$  and the flow law gives the group property. First truncate the sector sum, then truncate velocities and remove a neighborhood of the collision boundary. The discarded mass tends to zero. Ordinary smooth approximation on the remaining open phase space, followed by permutation averaging, proves density. For a bounded continuous interior test,  $f(\Phi_t z) \rightarrow f(z)$  for almost every initial state as  $t \rightarrow 0$ : before its first collision an interior trajectory is a free flight. Dominated convergence gives strong continuity on the dense test class; isometry extends it to every  $f$ . The generator of a strongly continuous group is closed. Testing against interior smooth functions and differentiating free flight identifies its interior distributional action. The collision identification is inherited from the flow, rather than prescribed by an arbitrary boundary extension.  $\square$

Choose a nonnegative  $\varphi \in C_c^\infty(\mathbb{R})$  with integral one and write  $\varphi_\delta(s) = \delta^{-1} \varphi(s/\delta)$ . Define the orbit regularization

$$B_\delta f = \int_{\mathbb{R}} \varphi_\delta(s) U_s f \, ds. \quad (43.1)$$

The integral is a Bochner integral in  $\mathcal{B}_p$ .

**Theorem 43.2** (Collision-compatible graph core). *For every  $f \in \mathcal{B}_p$ ,  $B_\delta f \in \bigcap_{m \geq 1} D(A^m)$ , and*

$$A^m B_\delta f = (-1)^m \int \varphi_\delta^{(m)}(s) U_s f \, ds, \quad (43.2)$$

$$\|A^m B_\delta f\|_p \leq \delta^{-m} \|\varphi^{(m)}\|_1 \|f\|_p.$$

*The linear span of  $B_\delta f$ , with  $f$  in the dense class of Lemma 43.1 and  $\delta > 0$ , is a graph core for  $A$ . There is one diagonal sequence of such observables converging to any  $g \in D(A)$  in the graph norm. These observables have equal incoming and outgoing traces at regular specular collisions, in the local collision-flux almost-everywhere sense.*

*Proof.* A change of variables gives

$$U_h B_\delta f = \int \varphi_\delta(s - h) U_s f \, ds.$$

Difference quotients of the scalar compactly supported smooth function converge in  $L^1(ds)$ . Isometry therefore permits passage to the limit in  $\mathcal{B}_p$ , proving (43.2) for  $m = 1$  and its norm estimate. Iterate to get every  $m$ . Strong continuity and the approximate-identity property give  $B_\delta f \rightarrow f$  in  $\mathcal{B}_p$ . If  $g \in D(A)$ , differentiation under the Bochner integral gives  $AB_\delta g = B_\delta Ag$ , so  $B_\delta g \rightarrow g$  also in graph norm.

To obtain a core with the stated dense tests, choose  $\delta_n \downarrow 0$  and a smooth finite-sector  $f_n$  with  $\|f_n - g\|_p \leq \delta_n^2$ . Then

$$\|B_{\delta_n}(f_n - g)\|_p \leq \delta_n^2, \quad \|AB_{\delta_n}(f_n - g)\|_p \leq \|\varphi'\|_1 \delta_n.$$

Together with graph convergence of  $B_{\delta_n}g$  this proves the single sequence assertion. There is no passage from an iterated limit to a uniform kinetic estimate in this argument.

It remains to verify the boundary assertion, which is why orbit regularization is used instead of convolution in particle coordinates. For almost every regular orbit,  $s \mapsto f(\Phi_s z)$  is locally integrable. Equation (43.1), evaluated on that orbit, is convolution in its time coordinate; it is a smooth function of time. At a nongrazing binary collision, the incoming point and its reflected outgoing point are the same orbit at the same collision time. The two one-sided values of this smooth time function therefore agree. To pass from interior almost-everywhere statements to collision traces, use a local free-flight tube transverse to a regular collision surface. Its phase-volume element is the collision surface element times the absolute normal relative velocity times  $ds$ . Fubini's theorem transfers the orbit statement to collision-flux almost every boundary point. Exhaust bounded-velocity, nongrazing, single-collision tubes by countably many such charts. Their complement consists of the excluded singular or grazing sets. This gives the asserted trace convention and no stronger claim of global smoothness across grazing configurations.  $\square$

The theorem solves an exact microscopic domain problem in a stated weighted norm. It does not assume a bounded finite-state generator. It also does not identify the transport generator with a nonlinear collision Hamiltonian: the deterministic chain rule remains the transport chain rule.

**43.2. Noisy acquisitions of physical observables.** Let the initial microscopic state  $Z$  have density  $\rho_\theta$  with respect to  $\bar{\mu}$ . Assume

$$\sup_{\theta} \|\rho_\theta\|_{p'} \leq M, \quad 1/p + 1/p' = 1. \quad (43.3)$$

A finite acquisition protocol chooses, by feedback, an index  $a \in \mathcal{A}_t$  before its  $t$ th report. Its scalar report is

$$Y_t = f_{t,a}(\Phi_{t_t} Z) + \sigma_t \xi_t, \quad \xi_t \sim N(0, 1), \quad \sigma_t > 0,$$

where the  $\xi_t$  are mutually independent and independent of  $Z$ . Each  $f_{t,a}$  is a bounded symmetric observable. An actual mark  $W$  is sampled from a fixed kernel given  $Z$ , independently of acquisition noise; in particular it may be any measurable functional of the entire microscopic trajectory. The acquisition choice does not alter the hard-sphere dynamics. Denote this marked experiment by  $E_f$ .

**Theorem 43.3** (Graph-core acquisition comparison). *For another finite observable family  $g_{t,a}$ , the identity report transducers in both directions between  $E_f$  and  $E_g$  have one internal state, no public seed, and marked feedback error at most*

$$e(f, g) = \min \left\{ 1, \frac{M}{\sqrt{2\pi}} \sum_t \frac{1}{\sigma_t} \sum_{a \in \mathcal{A}_t} \|f_{t,a} - g_{t,a}\|_p \right\}. \quad (43.4)$$

Consequently, for every width profile, every seed bound and every bounded common-task family,

$$\begin{aligned} |h_{E_f}(S, q; w) - h_{E_g}(S, q; w)| &\leq e(f, g), \\ |\Theta_{E_f}(S, q) - \Theta_{E_g}(S, q)| &\leq 2e(f, g). \end{aligned}$$

Taking  $g_{t,a} = B_\delta f_{t,a}$  makes these bounds tend to zero uniformly over all register widths and feedback designs, while all approximating observables belong to the microscopic graph domain and satisfy its collision compatibility.

*Proof.* Couple the two experiments with the same  $Z$ , the same mark conditional on  $Z$ , and the same private and public randomization of the feedback design. Conditional on a matched report history and on  $Z$ , the issued control is the same. The total variation distance between the two next Gaussian report laws is

$$2\Phi\left(\frac{|f_{t,a}(\Phi_{t_t} Z) - g_{t,a}(\Phi_{t_t} Z)|}{2\sigma_t}\right) - 1 \leq \frac{|f_{t,a}(\Phi_{t_t} Z) - g_{t,a}(\Phi_{t_t} Z)|}{\sigma_t \sqrt{2\pi}}.$$

Use a maximal coupling at each step. The probability of a first mismatch is bounded by the expectation of the sum of these bounds on previously matched histories. Bound the selected absolute difference by the sum over all possible  $a$ ; this removes the selection's dependence on  $Z$  and on earlier noises. By Hölder's inequality and invariance,

$$\int \rho_\theta(z) |f - g|(\Phi_{tt} z) d\bar{\mu}(z) \leq M \|U_{tt}(f - g)\|_p = M \|f - g\|_p.$$

Summing proves (43.4). The mark has been coupled identically throughout. This proves the required uniform marked feedback bound, not just convergence of open-loop marginal observations. The identity channel retains no state. Apply Theorem 24.2 in both directions and (24.3) to get the two decision bounds. Finally strong convergence of  $B_\delta$  applies to the finitely many observables; Theorem 43.2 supplies the domain and boundary assertions.  $\square$

**Corollary 43.4** (A physical consumer of compact duality). *Suppose the parameter space is compact,  $\theta \mapsto \rho_\theta$  is  $L^1(\bar{\mu})$  continuous, and the mark kernel is parameter independent. Let the bounded loss family be uniformly continuous in its compact row index in supremum norm. Then  $E_f$  satisfies the dominated loss-effect hypothesis of Theorem 26.3. Its ideal-public risk body is compact, its common-encoder minimax duality is attained, and its graph-core approximations obey Theorem 43.3 under (43.3).*

*Proof.* For a fixed open-loop control word, condition on  $Z$  and multiply the Gaussian report densities. The mixture is absolutely continuous with respect to Lebesgue measure, and hence to a product of standard Gaussian reference probabilities, whose density is everywhere positive. Integrate the actual conditional marked loss before taking this density. A change of preparation by  $\|\rho_\theta - \rho_{\theta'}\|_1$  changes the  $L^1$  loss effect by at most that amount times the loss bound. A change of loss costs at most its supremum norm. The row-effect map is therefore  $L^1$  continuous. This verifies the hypothesis rather than presuming it from smoothness of the microscopic dynamics. The stated conclusions now follow from the two cited theorems.  $\square$

**43.3. The precise historical implication.** The B4 microscopic program requires both exact generator-domain objects and a model-specific identification of their kinetic limit. The preceding results supply a boundary-compatible weighted  $L^p$  graph core and a width-uniform decision transfer for noisy physical acquisitions. They neither prove the BBGKY recollision estimates nor turn  $L^p$  graph convergence into convergence locally uniformly on action sublevels. The latter requires additional uniform integrability and model-specific corrector estimates. In particular the full historical B4 target is not an automatic consequence of the present consumer.

There is a second distinction worth recording. A nonlinear resolvent  $J_\lambda = (I - \lambda H)^{-1}$ , when it exists uniquely, satisfies

$$J_\lambda f = J_\mu \left( \frac{\mu}{\lambda} f + \left( 1 - \frac{\mu}{\lambda} \right) J_\lambda f \right). \quad (43.5)$$

Indeed, put  $u = J_\lambda f$  and substitute  $Hu = (u - f)/\lambda$  in  $u - \mu Hu$ . A linear difference/product resolvent identity cannot simply be carried over to  $H$ . Exponential resolvents and the required domain extension have an established theory; see [64], Theorem 3.4 and Proposition 3.6. Equation (43.5) is the identity to use in a nonlinear continuation of the microscopic graph argument, not a proof of full range from the existence of the linear Koopman group.

#### APPENDIX 44. UNBOUNDED GENERATORS, FORCED MEMORY, AND PHYSICAL DECISION LIMITS

We now connect the causal comparison results to the microscopic transport and compressed-memory objects. The analytic input is a generator-domain construction, not a claim that a finite positive model has replaced the physical dynamics. The finite-rank Mori–Zwanzig mechanism is classical; in particular the domain and bounded-perturbation issues are treated in [65, 66]. We give

the block proof in the form needed for a quantitative marked-acquisition comparison, including the unresolved initial component and the full observable.

**44.1. A common domain for resolved and orthogonal dynamics.** Let  $H$  be a complex Hilbert space, with inner product linear in the first argument. Let  $L$  generate a strongly continuous contraction semigroup  $U(t)$ , and let  $P$  be an orthogonal finite-rank projection such that

$$R = PH \subseteq D(L) \cap D(L^*), \quad Q = I - P. \quad (44.1)$$

The restriction  $LP$  is bounded. The restriction  $PL$  on  $D(L)$  extends to a bounded map on  $H$ : if  $e_1, \dots, e_k$  is an orthonormal basis of  $R$ , its extension is  $f \mapsto \sum_i \langle f, L^* e_i \rangle e_i$ . Define

$$A = PLP : R \rightarrow R, \quad B = \overline{PL}Q : QH \rightarrow R, \quad C = QLP : R \rightarrow QH, \quad D = QLQ|_{QD(L)}. \quad (44.2)$$

Here  $D$  acts in  $QH$ . The bar denotes the bounded extension, not a new choice of a generator domain.

**Theorem 44.1** (Forced memory with an explicit orthogonal generator). *Under (44.1),  $D$  generates a strongly continuous contraction semigroup  $T_Q(t)$  on  $QH$ . For every  $f \in H$ , the resolved trajectory  $x(t) = PU(t)f$  is continuously differentiable and satisfies*

$$x'(t) = Ax(t) + \int_0^t K(t-s)x(s) ds + \eta_f(t), \quad x(0) = Pf, \quad (44.3)$$

$$K(t) = BT_Q(t)C, \quad \eta_f(t) = BT_Q(t)Qf. \quad (44.4)$$

For  $\sigma > 0$ , with  $k = \|B\|\|C\|$ ,

$$\|K(t)\| \leq k, \quad \int_0^\infty e^{-\sigma t} \|K(t)\| dt \leq k/\sigma. \quad (44.5)$$

The compressed resolvent  $\mathbf{C}(z) = P(z - L)^{-1}P|_R$  is invertible for  $\Re z > 0$ , and

$$\mathbf{C}(z)^{-1} = zI_R - A - B(z - D)^{-1}C, \quad (44.6)$$

$$\|\mathbf{C}(z)^{-1}\| \leq |z| + \|A\| + k/(\Re z). \quad (44.7)$$

The forcing transform is both  $B(z - D)^{-1}Qf$  and  $\mathbf{C}(z)^{-1}P(z - L)^{-1}Qf$ . If  $L$  is skew-adjoint,  $D$  is skew-adjoint on  $QH$ ,  $B = -C^*$ , and

$$K(t) = -C^*T_Q(t)C, \quad K(0) = -C^*C. \quad (44.8)$$

*Proof.* Since  $PH \subseteq D(L)$ ,  $D(L) = R \oplus QD(L)$ . On this domain  $L$  has blocks  $A, B, C, D$ . The off-diagonal operator  $F = PBQ + QCP$ , with the evident inclusions of its ranges in  $H$ , is bounded. Equivalently it acts by  $F(u + v) = Bv + Cu$ . The bounded-perturbation construction shows that  $L_0 = L - F$  on  $D(L)$  is a generator. One can obtain its semigroup directly by the norm-convergent Dyson series on bounded time intervals; the  $n$ th perturbation term is bounded by  $\|F\|^n t^n / n!$ .

For  $u \in R$  and  $v \in QD(L)$ ,

$$\Re \langle L_0(u + v), u + v \rangle = \Re \langle Lu, u \rangle + \Re \langle Lv, v \rangle \leq 0.$$

For domain initial data the derivative of the squared norm of the  $L_0$  orbit is nonpositive. Density extends the contraction bound to all initial data. The projection  $P$  preserves the domain and commutes with  $L_0$ , and hence with its resolvent and semigroup. Its restrictions are  $e^{tA}$  and a contraction semigroup on  $QH$  with generator precisely  $D$ . This proves the asserted existence and domain, without writing an undefined exponential of  $QLQ$ .

Bounded off-diagonal perturbation and variation of constants now give, for  $x(t) = PU(t)f$  and  $y(t) = QU(t)f$ ,

$$\begin{aligned} x(t) &= e^{tA}Pf + \int_0^t e^{(t-s)A}By(s) ds, \\ y(t) &= T_Q(t)Qf + \int_0^t T_Q(t-s)Cx(s) ds. \end{aligned}$$

The first expression is continuously differentiable because  $A$  is bounded and  $By(s)$  is continuous. In particular this differentiation uses the bounded block integral formula, not a derivative of  $U(t)f$  for an arbitrary  $f \notin D(L)$ . Substituting the second proves (44.3)–(44.4) for every  $f \in H$ . Contractivity gives (44.5). The operator-valued kernel is norm continuous, since  $C$  has finite-dimensional domain and  $T_Q$  is strongly continuous.

For  $\Re z > 0$ ,  $z - D$  has a bounded inverse of norm at most  $(\Re z)^{-1}$ . Solving the lower block of  $(z - L)(u + v) = g + h$  yields  $v = (z - D)^{-1}(h + Cu)$ . The upper block is

$$[zI_R - A - B(z - D)^{-1}C]u = g + B(z - D)^{-1}h.$$

The bracket is injective: a null vector would give a null vector of  $z - L$  in its domain. It is therefore invertible in the finite-dimensional space  $R$ . Taking  $h = 0$  identifies its inverse with  $C(z)$ ; taking  $g = 0$  proves the forcing identity. The norm estimate follows from the resolvent bound for  $D$ .

If  $L^* = -L$ , the off-diagonal operator  $F$  is bounded and skew-adjoint, so  $L_0 = L - F$  is skew-adjoint on  $D(L)$ . Its reducing restrictions are skew-adjoint, and the adjoint relation of the off-diagonal blocks gives  $B = -C^*$ . This proves (44.8).  $\square$

Two distinctions in this statement matter. First,  $Qf = 0$  suffices to remove the forcing, but this cannot be assumed for an arbitrary physical observable. Second, (44.5) is a discounted bound on the right half-plane. It does not assert decay of  $K(t)$  or meromorphic continuation across the imaginary axis. Those require additional spectral information. In particular no finiteness of left-half-plane transmission zeros is inferred from finite rank alone.

**44.2. A stability estimate which retains the full observable.** Let  $\hat{x}$  solve a finite-dimensional Volterra equation with data  $\hat{A}, \hat{K}, \hat{\eta}, \hat{x}_0$ , where the kernel and forcing are locally integrable. Suppose  $\hat{x}$  is continuous on  $[0, T]$  and put  $M_T = \sup_{t \leq T} \|\hat{x}(t)\|$ . Define

$$\begin{aligned} B_T &= \|Pf - \hat{x}_0\| + T\|A - \hat{A}\|M_T + \int_0^T \|\eta_f(s) - \hat{\eta}(s)\| ds \\ &\quad + M_T \int_0^T (T-s)\|K(s) - \hat{K}(s)\| ds. \end{aligned} \tag{44.9}$$

**Proposition 44.2** (Resolved and reconstructed error). *For  $k = \|B\|\|C\|$ ,*

$$\sup_{t \leq T} \|x(t) - \hat{x}(t)\| \leq B_T \cosh(\sqrt{k} T). \tag{44.10}$$

*If  $\hat{y}(t) \in QH$  satisfies*

$$\chi_T = \sup_{t \leq T} \left\| \hat{y}(t) - T_Q(t)Qf - \int_0^t T_Q(t-s)C\hat{x}(s) ds \right\| < \infty,$$

*then the reconstructed full observable obeys*

$$\sup_{t \leq T} \|U(t)f - \hat{x}(t) - \hat{y}(t)\| \leq (1 + T\|C\|)B_T \cosh(\sqrt{k} T) + \chi_T. \tag{44.11}$$

*Proof.* Write the difference equation with the unperturbed  $A$  as its local part and use the contractivity of  $e^{tA}$ . Fubini and  $\|K(s)\| \leq k$  show that  $e(t) = \|x(t) - \hat{x}(t)\|$  satisfies

$$e(t) \leq B_T + k \int_0^t (t-s)e(s) ds \quad (0 \leq t \leq T).$$

Iteration yields the series  $B_T \sum_{n \geq 0} k^n t^{2n} / (2n)!$ ; the remainder tends to zero because  $e$  is bounded on the compact interval. This is (44.10). Subtract the expression for  $y(t)$  in the preceding proof from the specified reconstruction. Contractivity of  $T_Q$  gives  $\|y(t) - \hat{y}(t)\| \leq T\|C\| \sup e + \chi_T$ . Adding the resolved error proves (44.11).  $\square$

The last term cannot be dropped merely because a resolved trajectory is accurate. A projected observable  $PU(t)f$  is not the actual observable  $U(t)f$  unless its orthogonal component is accounted for. This is the operator counterpart of preserving the actual marked target in an experiment comparison.

**44.3. Galerkin convergence from the microscopic graph core.** We specialize to the weighted symmetric  $L^2$  space of the almost-everywhere hard-sphere flow in Lemma 43.1. The measure  $\mu$  is an invariant probability; it may be the normalized invariant measure of a finite particle sector or a summable mixture of such sectors with the stated weights. The Koopman group is unitary and its generator  $L$  is skew-adjoint. The graph core in Theorem 43.2 supplies a countable real core: take countable dense smooth interior inputs, rational orbit-smoothing scales, and their finite linear combinations. Separability of the graph norm allows a countable selection. Gram–Schmidt on an enumeration gives nested finite-dimensional spaces  $V_n$  such that

$$V_n \subset D(L), \quad \bigcup_n V_n \text{ is a graph core for } L.$$

Let  $P_n$  be their orthogonal projections and set  $G_n = P_n L P_n$  on  $H$ . It is bounded and skew-adjoint; on  $V_n^\perp$  it is zero.

The stable graph-core resolvent approximation below is a specialization of Trotter’s classical approximation theorem [67, Theorem 5.2]; the proof spells out its core consistency, uniform contraction bound, and resolvent-to-orbit implication.

**Theorem 44.3** (Microscopic graph-core Galerkin limit). *For every  $f \in H$  and every finite  $T$ ,*

$$\sup_{0 \leq t \leq T} \|e^{tG_n} P_n f - U(t)f\|_2 \longrightarrow 0. \quad (44.12)$$

*Moreover  $(z - G_n)^{-1}$  converges strongly to  $(z - L)^{-1}$  for  $\Re z > 0$ . Every finite  $V_n$  satisfies the common-domain hypothesis (44.1), and hence has the forced-memory representation of Theorem 44.1.*

*Proof.* Let  $f$  belong to the union of the  $V_n$ . For sufficiently large  $n$ ,  $G_n f = P_n L f \rightarrow L f$ . The resolvent bounds for skew-adjoint operators and the identity

$$(z - G_n)^{-1}(z - L)f - f = (z - G_n)^{-1}(G_n - L)f$$

give convergence to zero. The set of vectors  $(z - L)f$  for  $f$  in the core is dense in  $H$ : approximate  $(z - L)^{-1}g$  in the graph norm. The common resolvent bound extends convergence to every  $g \in H$ .

We also give an exponential-formula argument for the time-uniform conclusion. For any contraction generator  $A$  and  $f \in D(A)$ ,

$$(I - tA/m)^{-m} f = \mathbb{E}[e^{sA} f] \quad \text{with } s \sim \text{Gamma}(m, t/m).$$

The formula follows by iterating the Laplace resolvent representation. Since  $\mathbb{E}s = t$ ,  $\text{Var } s = t^2/m$ , and  $\|e^{sA} f - e^{tA} f\| \leq |s - t| \|A f\|$ , it gives

$$\|(I - tA/m)^{-m} f - e^{tA} f\| \leq t \|A f\| / \sqrt{m}. \quad (44.13)$$

For a fixed core vector,  $\|G_n f\| \leq \|L f\|$  eventually. Strong resolvent convergence gives convergence of the  $m$ th powers in (44.13) for fixed  $m$  and  $t > 0$ ; their factors are uniformly bounded. First let  $n \rightarrow \infty$  and then  $m \rightarrow \infty$  to obtain pointwise time convergence. The orbit Lipschitz bounds for this core vector are uniform in  $n$ , so a finite time net makes the convergence uniform on  $[0, T]$ . Density and contractivity extend it to all  $f \in H$ . Finally  $P_n f \rightarrow f$ , and  $e^{tG_n}$  is unitary, giving (44.12). Since  $D(L^*) = D(L)$ , each  $V_n$  satisfies (44.1).  $\square$

This proof is a graph-core version of the classical resolvent-to-semigroup approximation argument. It is included to verify, rather than assume, the approximation used in the next theorem. It is not a uniform estimate in particle number, a Boltzmann–Grad limit, or a nonlinear kinetic Trotter–Kato theorem.

**44.4. The common-encoder limit for actual physical observations.** *Scope of this preserved physical theorem: fixed microscopic flow, passive observation control, fixed particle number and positive noise; no kinetic or Boltzmann–Grad limit. The active intervention theorem of Section 9 is a separate result.*

Fix a family of initial densities  $f_\theta$  with respect to  $\mu$  such that

$$f_\theta \geq 0, \quad \int f_\theta d\mu = 1, \quad \sup_\theta \|f_\theta\|_2 \leq D_0 < \infty. \quad (44.14)$$

At checkpoint  $t$ , control  $a$  selects a real observable  $F_{t,a} \in H$  and noise level  $\sigma_{t,a} > 0$ . It does not change the microscopic flow. The exact and approximating reports are, respectively,

$$\begin{aligned} Y_t &= (U(\tau_t)F_{t,a})(Z_0) + \sigma_{t,a}\xi_t, \\ Y_t^{(n)} &= (e^{\tau_t G_n} P_n F_{t,a})(Z_0) + \sigma_{t,a}\xi_t, \end{aligned} \quad (44.15)$$

where the displayed normal variables are independent conditional observation noises, not a requirement that the two laws use a common additive-noise coupling. The control is selected before its report. There are finitely many controls at each time. Actual stage marks are sampled by one fixed Borel channel from  $Z_0$  and any specified latent randomness and are shared in a comparison coupling. These latent marks are not observed by an encoder unless the interface expressly supplies them. A mark may be an observable of the microscopic future; the latent sample and the flow determine it before the comparison is made.

Put

$$\begin{aligned} d_{t,a}^{(n)} &= \|U(\tau_t)F_{t,a} - e^{\tau_t G_n} P_n F_{t,a}\|_2, \\ \zeta_t^{(n)} &= \frac{D_0}{\sqrt{2\pi}} \sum_{a \in \mathcal{A}_t} \frac{d_{t,a}^{(n)}}{\sigma_{t,a}}, \quad \varepsilon_t^{(n)} = \min\{1, \sum_{s \leq t} \zeta_s^{(n)}\}. \end{aligned} \quad (44.16)$$

**Theorem 44.4** (Width-uniform physical decision limit). *For the instruments in (44.15), identity report channels give two-sided, state-cost-one, marked prefix comparison with errors  $\varepsilon_t^{(n)}$ . The comparison is uniform in the parameter, the common-encoder width profile, all admissible feedback designs, and each private or independent public seed signature. For the weights in (28.1), define*

$$E_n = \sum_{t \geq 0} a_t \varepsilon_t^{(n)}. \quad (44.17)$$

*Then  $E_n \rightarrow 0$ . For every finite width profile, and indeed uniformly over all such profiles, the full and approximating risk bodies have Hausdorff distance at most  $E_n$ , their lower support values differ by at most  $E_n$ , and their centered minimax spectra differ by at most  $2E_n$  when the full-history baselines use the same information pattern. These statements also hold for a finite acquisition horizon by setting later weights to zero.*

*Proof.* Couple the two experiments with the same  $Z_0$ , the same actual marks, and the same independent policy randomization. Until the first report mismatch, their controls and retained states agree. Two real normal laws of equal variance  $\sigma^2$  and means  $u, v$  have total variation distance at most  $|u - v|/(\sigma\sqrt{2\pi})$ . This follows by integrating the absolute derivative of their translated density, with the factor one half in the total variation norm. A measurable maximal coupling of the successive report kernels therefore has this conditional mismatch bound.

The conditional law of  $Z_0$  given a past report can have a large density; we do not apply (44.14) to that posterior. Instead, integrate the first-mismatch contribution under the original marginal  $f_\theta\mu$ , bound the indicator of previous matching by one, and sum over the finitely many possible controls at the step. Cauchy–Schwarz gives the unconditional contribution at time  $s$  at most  $\zeta_s^{(n)}$ . The union bound proves (44.16). The actual marks remain matched whenever the transcript does, because they were coupled through the same latent sample. The bound includes the joint public seed, which is identical and independent in the two experiments. The identity comparison channels themselves require no persistent state or additional seed and do not reveal this proof coupling to a consumer.

For each fixed  $t, a$ , Theorem 44.3 gives  $d_{t,a}^{(n)} \rightarrow 0$ , and hence  $\varepsilon_t^{(n)} \rightarrow 0$ . Because  $0 \leq \varepsilon_t^{(n)} \leq 1$  and the  $a_t$  sum to one, finite-prefix truncation proves  $E_n \rightarrow 0$ . Apply Theorem 28.6 in both directions. The same argument applies to unrestricted full-history consumers, so the optimal full-history baselines differ by at most  $E_n$  without changing their control or information pattern. The final centered bound follows.  $\square$

**Corollary 44.5** (Combining compact duality with microscopic approximation). *In this corollary only the preparation density and the bounded losses vary with the row. The flow, observables, noise laws, and actual mark channel are fixed. Assume in addition that the density family is continuous in  $L^1(\mu)$  on a compact parameter space, and that the bounded row losses are uniformly continuous in their row index in supremum norm. With the finite task alphabets of Theorem 28.2, both physical instruments satisfy its finite-prefix effect hypotheses. They therefore have attained infinite-horizon public duality, and their attained values obey the uniform error bound in Theorem 44.4. A full observable approximation certified by (44.11) can replace the Galerkin approximation in (44.16) with the corresponding  $L^2$  error.*

*Proof.* For a fixed finite control word, the joint report law is a mixture over  $Z_0$  of products of positive normal densities. Choose at each report time any fixed strictly positive probability density as reference. The resulting loss effects are in product  $L^1$ , since the bounded loss is integrated against a probability density. Changing  $f_\theta$  changes any such effect in  $L^1$  by at most  $\|f_\theta - f_{\theta'}\|_1$  times the loss bound: integrate out the conditional normal and marked kernels. A supremum-norm change in the row loss gives the analogous bound. Thus the required effect map is continuous. The arguments do not require bounded observables or a density on the infinite report path. The last assertion follows because the coupling proof uses only the full observable error in  $L^2$ , not how it was obtained.  $\square$

The two dimensions in this construction must not be identified.  $\dim V_n$  is an analytic approximation parameter of the physical observation presentation. The retained register remains  $S_t$ ; no access to a Galerkin coefficient vector is supplied to the consumer. The theorem proves a uniform comparison before optimizing those registers. The historical kinetic program additionally requires nonlinear action-sublevel bounds, hierarchy correctors, and a joint microscopic limit. Neither a finite-rank memory identity nor strong  $L^2$  approximation proves those additional assertions.

#### APPENDIX 45. STABLE CERTIFICATES FOR EXACT RESOURCE PROFILES

The task-cover theorem and the analytic comparison theorem concern the same invariant: the least common-encoder excess at a prescribed width profile. The following consequence states

precisely when an approximate experiment certifies an exact resource answer. This is also the point at which the microscopic decision limit can be used without exchanging a model limit with optimization over encoders.

For the finite restart system of Section 25, write

$$V_E(S, q) = \inf_{\pi} \max_{t,d} \left[ r_{td}^E(\pi) - \sum_h \eta_t(h) m_{td}^E(h) \right].$$

If a positive entry regret exists, define  $g_E = \alpha\gamma$  as in Corollary 25.5; for a Brier task family use  $g_E = \alpha\delta^2/2$ . This gap is defined by the target task table, not estimated from a lossy experiment. In the trivial case of no positive regret every resource profile is feasible and no gap test is necessary.

**Theorem 45.1** (Comparison, completion, and certification). *Let  $E$  be a finite restart experiment with  $g_E > 0$ , and let  $F$  be an experiment with the same decision rows and loss scale. Suppose two-sided cost-one comparison at the declared seed signature gives risk-body Hausdorff error at most  $\varepsilon$ , and the corresponding full-history benchmarks differ by at most  $\varepsilon$ . Then*

$$|V_F(S, q) - V_E(S, q)| \leq 2\varepsilon. \quad (45.1)$$

If  $\varepsilon < g_E/4$ , the single test

$$V_F(S, q) < g_E/2 \quad (45.2)$$

is equivalent to existence of a causally closed compatible cover for  $E$  with width profile  $S$ . Equality of the exact feasible profiles holds for all private and independent public seed signatures, even though the positive-excess values for those signatures may differ.

More generally, suppose only a simulator from  $E$  to  $F$  is given, with cost  $(K, r)$ , and suppose its risk and benchmark errors are at most  $\varepsilon$ . If

$$V_F(S, q) + 2\varepsilon < g_E,$$

then  $E$  has a causally closed compatible cover of width at most  $KS$ . The public seed bound of the implemented design is  $rq$ , but it is unnecessary for exactness of that cover.

*Proof.* The two risk-body inclusions bound the infimum of the maximum centered row by the opposite value plus the risk error and the benchmark error. This proves (45.1). By Corollary 25.5,  $V_E(S, q)$  is either zero or at least  $g_E$ , and it is zero precisely for a causally closed compatible cover. In the first case  $V_F(S, q) \leq 2\varepsilon < g_E/2$ ; in the second case  $V_F(S, q) \geq g_E - 2\varepsilon > g_E/2$ . This proves the test without requiring an optimizer for  $F$ . The one-sided implementation gives  $V_E(KS, rq) \leq V_F(S, q) + 2\varepsilon < g_E$ , forcing its value to be zero. Theorem 25.2 then supplies a deterministic cover machine at width  $KS$ .  $\square$

**Corollary 45.2** (Certified approximation sequence). *For any family  $F_n$  satisfying the hypotheses of the preceding theorem with a known error bound  $\varepsilon_n \rightarrow 0$ , every fixed resource profile of  $E$  is classified correctly by (45.2) once  $\varepsilon_n < g_E/4$ . If the computed value has an additional certified numerical error  $\kappa_n$ , a sufficient condition is  $2\varepsilon_n + \kappa_n < g_E/2$ , with the corresponding interval test. In applications of Theorem 44.4, the error bound is  $E_n$  whenever the entire declared restart family, including its preparations, is among the compared physical experiments.*

*Proof.* Apply (45.1) and add the numerical error. The last qualification ensures that the physical comparison controls these restart rows, rather than a different initial preparation or only a single uncontrolled experiment.  $\square$

The gap may be small, and computing the finite common-encoder optimum can be difficult. The statement is a certification theorem, not a polynomial-time algorithm. In particular it does not make a constrained inverse randomization assertion: a risk witness need not manufacture a simulator of a prescribed cost. Its content is the stability of exact causal task completion under an independently verified, resource-priced comparison.

## APPENDIX 46. A PRIOR-INDEPENDENT EXPERIMENT BEHIND PREDICTIVE COORDINATES

A future-test quotient relative to one Bayesian preparation need not preserve a parameterized experiment. The additional requirement is *likelihood saturation*. This section gives both a terminal construction and the extra recursion needed for a causal reconstruction. All assertions about measurable coordinates are modulo the stated common null sets.

**Theorem 46.1** (A saturated posterior quotient). *Let  $K$  be compact metric, let  $X$  be standard Borel, and let  $E = \{P_\theta : \theta \in K\}$  be continuous in total variation. Choose a dense sequence  $(\theta_j)$  and weights  $w_j > 0$  with  $\sum_j w_j = 1$ . Put  $m = \sum_j w_j P_{\theta_j}$  and define*

$$T(x) = \left( w_j \frac{dP_{\theta_j}}{dm}(x) \right)_{j \geq 1} \in [0, 1]^{\mathbb{N}}. \quad (46.1)$$

*There is one parameter-independent kernel  $R$  such that*

$$R(T_{\#}P_\theta) = P_\theta \quad (\theta \in K). \quad (46.2)$$

*Thus  $E$  and  $TE$  are statistically equivalent. The completed  $\sigma$ -field generated by  $T$  is minimal among sufficient statistics. A different dense sequence and strictly positive weights give the same minimal sufficient  $\sigma$ -field modulo  $m$  and the same experiment equivalence class, not necessarily the same coordinates. For the terminal cardinality signature, their marked decision spectra are identical for every budget and every family of bounded terminal parameter-decision marks, whose losses depend on the parameter and the action. An additional latent-target mark requires its own joint-law sufficiency condition. No computability or finite recursive representation of  $T$  is asserted.*

*Proof.* If  $m(B) = 0$ , then  $P_{\theta_j}(B) = 0$  for all  $j$ ; continuity and density imply  $P_\theta(B) = 0$  for every  $\theta$ . Choose Borel versions of the countably many densities. They give  $\sum_j T_j = 1$   $m$ -almost surely; modify them on one common null set if necessary. Disintegrate  $m$  given  $T$  to obtain the standard Borel conditional kernel  $R(t, dx)$ . For every  $j$ , the density of  $P_{\theta_j}$  is  $T_j/w_j$ , so

$$P_{\theta_j}(B) = \int \frac{t_j}{w_j} R(t, B) T_{\#}m(dt) = R(T_{\#}P_{\theta_j})(B).$$

For arbitrary  $\theta$ , choose  $\theta_{j_k} \rightarrow \theta$ . Both applying  $T$  and applying  $R$  contract total variation. Taking limits proves (46.2).

For minimality, let  $S$  be sufficient, in the conditional-law sense that one kernel  $R_S$  is a conditional law of  $X$  given  $S$  under every  $P_\theta$ . Mixing shows that it is also the conditional law under  $m$ . The factorization

$$\frac{dP_{\theta_j}}{dm}(x) = \frac{d(S_{\#}P_{\theta_j})}{d(S_{\#}m)}(S(x))$$

follows by integrating both sides on arbitrary measurable sets using  $R_S$ . Thus every coordinate of  $T$  is  $\sigma(S)$ -measurable. Conversely the preceding disintegration proves sufficiency of  $T$ . Mixtures obtained from different dense sequences are mutually absolutely continuous: each dominates every member of  $E$ , hence also the other mixture. Minimality then proves equality of the completed sufficient  $\sigma$ -fields.

Finally, any terminal encoder from  $TE$  is an encoder from  $E$  by composition with  $T$ . Conversely, an  $S$ -label encoder  $Q$  from  $E$  is implemented from  $TE$  by  $QR$ , with exactly the same label laws for all parameters. This uses only transient reconstruction randomization and at most  $S$  terminal labels. It is a kernel assertion, not an algorithm that stores a sampled infinite object in a finite register. The two sets of attainable terminal risks, and hence their spectra, coincide.  $\square$

The theorem makes explicit a classical dominated-sufficiency mechanism [47, 46]. Its role here is to identify the precise terminal slice on which the prior-relative and parameter-uniform notions have the same resource profile. A predictive-test quotient has this property when its generated

$\sigma$ -field contains the likelihood coordinates (46.1). Without that inclusion the conclusion is false, even for a two-point parameter set: observe  $Y = \theta \in \{0, 1\}$  once, and allow only constant future tests. Their quotient is a point although the observed experiment distinguishes both parameters exactly. Predictive equivalence always refers to a declared family of future questions.

**46.1. The causal factorization.** We give a finite-horizon formulation to avoid any assertion that a terminal sufficient statistic is automatically a causal sufficient state. Let  $H_t = (A_0, Y_1, \dots, A_{t-1}, Y_t)$  be the observed history. All policies have the same parameter-independent action kernels. Under a reference prior on  $\theta$ , write  $\bar{P}$  for the mixture law. Suppose Borel coordinates  $S_t = S_t(H_t)$  satisfy the following *saturation conditions*, with versions fixed for every admissible history and action where they are used:

$$S_t = F_t(S_{t-1}, A_{t-1}, Y_t), \quad (46.3)$$

$$\bar{P}(dy \mid H_{t-1}, a) = k_t(dy \mid S_{t-1}, a), \quad (46.4)$$

$$\frac{dP_\theta|_{H_t}}{d\bar{P}|_{H_t}} = \ell_{\theta,t}(S_t). \quad (46.5)$$

The likelihood identity is required for each parameter, initially on a dense reference family and then for all parameters by finite-horizon variation continuity. These conditions hold uniformly over the policies under consideration. Alternatively one may assume domination and the identity directly for every parameter. The reference prior is not allowed to change with the policy during a reconstruction.

**Theorem 46.2** (Causal reconstruction of a saturated quotient). *Under (46.3)–(46.5), there are parameter-independent kernels  $L_t(dy \mid s, a, s')$  which reconstruct the observation stream from its successive quotient states, with the correct joint law under every admissible policy. Hence the original and quotient instruments have mutually exact causal simulations. If  $S_t$  has at most  $K_0$  values at each time, both simulations may be implemented with  $K_0$  internal states and the cardinality spectra obey the budget dilations of Theorem 23.2. If the quotient is not finite, the theorem asserts statistical and causal reconstruction only, with the storage of  $S_{t-1}$  explicitly charged; it does not assert a finite-register resource equivalence.*

*Proof.* Let  $H_t^0(ds' \mid s, a)$  be the pushforward of  $k_t(\cdot \mid s, a)$  by  $F_t(s, a, \cdot)$ . Disintegration, including the Borel variables  $(s, a)$ , gives

$$k_t(dy \mid s, a) = \int L_t(dy \mid s, a, s') H_t^0(ds' \mid s, a), \quad (46.6)$$

with  $F_t(s, a, y) = s'$  under the resulting joint kernel. On histories with  $\ell_{\theta,t-1}(s) > 0$ , change of measure gives the actual joint conditional law of  $(Y_t, S_t)$  as

$$L_t(dy \mid s, a, s') \frac{\ell_{\theta,t}(s')}{\ell_{\theta,t-1}(s)} H_t^0(ds' \mid s, a).$$

In particular, conditioning further on  $s'$  removes the likelihood factor. On zero-likelihood histories the kernel may be chosen arbitrarily; such histories are not visited under that parameter. For parameters outside a countable reference support, equality of the constructed finite-horizon laws extends by total variation, or follows directly from the assumed domination identity.

Start at the declared initial state. Given a target policy on the reconstructed reports, feed its chosen action into the quotient instrument; on receiving  $s'$ , draw  $y$  with  $L_t(\cdot \mid s, a, s')$ , then replace  $s$  by  $s'$ . Induction proves equality of every finite parameter-indexed report/action law. This induction is essential: it reconstructs responses after the same actions, not after actions selected with hidden parameter information. In the forward direction, apply  $F_t$  recursively to the real reports. Both executions retain only the previous quotient state in addition to the target

encoder's own register. This costs  $K_0 S$  states if that state has  $K_0$  possible values. The general standard Borel version has no such finite bound, exactly as stated.  $\square$

A likelihood statistic alone supplies (46.5), not (46.4). Conversely, a Bayesian predictive state supplies future report predictions, not necessarily all parameter likelihoods. The conjunction is the bridge. Sequential comparison of filtered experiments already has a substantial decision-theoretic history [48]; the present statement fixes the additional register cost of the reconstruction rather than claiming that sequential comparison itself is new.

For marks involving an additional latent target rather than the parameter, its conditional kernel must also be included in the predictive saturation condition. The causal reconstruction theorem does not infer equality of latent-target laws from equality of report laws alone.

#### APPENDIX 47. A NECESSITY THEOREM FOR CAUSAL REALIZATION

We now characterize the causal constraint without imposing a tree geometry. The time dependence of a machine is retained, so the result does not replace a free clock by an uncharged stationary-state assumption. Fix a horizon  $T$ , finite report alphabets, one Bayesian law, and square-integrable Hilbert targets  $U_t$ . There is no control in this section. Let  $\mathcal{H}_t$  be the finite set of positive-probability histories,  $p_t(h) = \mathbb{P}(H_t = h)$ , and  $m_t(h) = \mathbb{E}[U_t \mid H_t = h]$ . Let  $\beta_t \geq 0$  be fixed weights.

A family of partitions  $(\mathcal{P}_t)_{t=0}^T$ , with  $\mathcal{P}_0 = \{\emptyset\}$ , is *successor compatible* if

$$h, h' \in C \in \mathcal{P}_t, \quad hy, h'y \in \mathcal{H}_{t+1} \implies hy, h'y \text{ belong to the same cell of } \mathcal{P}_{t+1}. \quad (47.1)$$

Only common positive-probability continuations enter this condition. For  $C \in \mathcal{P}_t$  put  $\bar{m}_C = p_t(C)^{-1} \sum_{h \in C} p_t(h) m_t(h)$  and define

$$\mathcal{V}(\mathcal{P}) = \sum_{t=1}^T \beta_t \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_t(h) - \bar{m}_C\|^2. \quad (47.2)$$

**Theorem 47.1** (Recursive partition characterization). *For the cardinality signature, allowing randomized time-dependent updates and decoders and independent public randomness, the optimal weighted excess over the full-history Bayes risk equals*

$$\inf_{\#I_t \leq S} \sum_{t=1}^T \beta_t \mathbb{E} \|m_t(H_t) - \hat{U}_t\|^2 = \min_{\substack{\mathcal{P} \text{ successor compatible} \\ |\mathcal{P}_t| \leq S}} \mathcal{V}(\mathcal{P}). \quad (47.3)$$

*Thus same-order realization of a sequence of static checkpoint optima is equivalent to the existence of successor-compatible partitions with the corresponding widths and variance bounds. This is a necessary as well as a sufficient condition. It is invariant under relabelling of histories; suffix closure is one sufficient construction of these partitions, not a necessary choice of coordinates.*

*Proof.* A deterministic initialized machine partitions  $\mathcal{H}_t$  according to  $I_t$ . Its update depends only on  $(t, I_t, y)$ , so its partitions satisfy (47.1). Given a state cell, the mean  $\bar{m}_C$  minimizes its squared error by Hilbert orthogonality. The resulting excess is precisely (47.2).

Conversely, label each partition's cells by at most  $S$  labels. For each cell  $C$  and symbol  $y$ , send its label to the cell containing  $hy$  for any supported  $h \in C$  with supported extension  $hy$ . Compatibility makes the answer independent of the choice of  $h$ . If there is no such extension, define the update arbitrarily. At time  $t$  decode the label as  $\bar{m}_C$ . Induction identifies the actual label with the cell of the actual history. There are finitely many partition families, so the minimum exists.

To include randomization, realize the finitely many update and decoder kernels by independent uniforms, and condition on their entire tape. For each tape, the rule is a deterministic time-dependent  $S$ -state machine. Independence of the tape and the experiment preserves the history law

and conditional means. The average of its weighted costs cannot be smaller than the deterministic minimum. Public independent randomness is handled in the same way. This freezing argument concerns one fixed Bayesian sum; it does not interchange a minimax task supremum with the average over tapes.

Unrestricted checkpoint encoders correspond to arbitrary partitions at each time, without (47.1). Hence they have cost  $\sum_t \beta_t e_S^2(m_t(H_t))$ . Formula (47.3) gives exactly the asserted equivalence of order comparisons, including the case when a static cost is zero.  $\square$

**Corollary 47.2** (Finite-command cell instruments). *Let  $k_1, \dots, k_J$  be Borel functions on a parameter space with  $k_j \geq 0$  and  $\sum_j k_j = 1$ , and fix a prior  $\mu$ . At each time the raw cell is drawn independently conditional on the parameter. A command  $g$  from a prescribed finite menu in  $[\eta, 1 - \eta]^J$ ,  $0 < \eta < 1/2$ , has a prescribed parameter-independent law  $\nu_t$ . Cell  $j$  is reported as  $j$  with probability  $g_j$  and otherwise as the single failure symbol  $\dagger$ . Only the command and the reported symbol are observed. Fix a finite horizon, finite future-query menus with known likelihoods  $H_{tq} \in [0, 1]$ , independent query weights  $\omega_{tq}$ , and nonnegative checkpoint weights  $\beta_t$ .*

*For the persistent-cardinality signature, the optimal common causal encoder for the weighted sum of quadratic prediction excesses is exactly (47.3), with the finite history law and query means computed from the actual cell likelihoods. This applies to the finite-command specialization of the A1 finite-cell protocol [52]. It is an exact scalarized criterion, not an asserted solution of the different maximum-over-checkpoints criterion with private randomization.*

*Proof.* For the observed pair  $(g, x)$  define

$$\lambda_{g,x}(\theta) = \begin{cases} g_x k_x(\theta), & x \in \{1, \dots, J\}, \\ \sum_j (1 - g_j) k_j(\theta), & x = \dagger. \end{cases}$$

For a history  $h = ((g_s, x_s))_{s \leq t}$  its probability is  $\prod_{s \leq t} \nu_s(g_s)$  times  $\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)$ . On positive-probability histories the mean of query  $q$  is

$$m_{tq}(h) = \frac{\int H_{tq}(\theta) \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}{\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}.$$

Take the Hilbert prediction vector with coordinates  $\sqrt{\omega_{tq}} m_{tq}(h)$  and apply Theorem 47.1. All histories are finite, and the law used there is exactly the actual acquisition law above. A failure history contains  $\dagger$ , not the unobserved cell  $j$ ; the successor compatibility condition therefore requires one common update for that report. Integration of the given Borel likelihoods may require arbitrary real constants or offline work. Neither is claimed finite under the cardinality signature. Zero-evidence histories are omitted rather than given artificial posteriors.  $\square$

**Corollary 47.3** (Continuation obstruction). *Let  $g$  be a bounded readout on all finite words over a finite alphabet. For stationary deterministic machines required to reproduce  $g$  on every word, an exact  $S$ -state realization exists if and only if the relation*

$$h \equiv_g h' \iff g(hw) = g(h'w) \text{ for every finite } w \quad (47.4)$$

*has at most  $S$  classes. More generally put  $d_*(h, h') = \sup_w \|g(hw) - g(h'w)\|$ . Every stationary deterministic realization with uniform error at most  $\epsilon$  has state fibres of  $d_*$ -diameter at most  $2\epsilon$ , and thus at least as many states as any  $2\epsilon$ -separated set of histories.*

*Proof.* Histories in one machine state have the same state after every common continuation. Exact output equality implies (47.4); for approximate outputs the triangle inequality gives the diameter bound. The relation (47.4) is a right congruence. Its classes, updated by appending the new symbol, form an exact machine, whose output is  $g(h)$ , well defined by the empty continuation. This is the output version of the classical finite-automaton congruence argument [49]; the finite-horizon, weighted-distortion characterization in Theorem 47.1 is the form needed here.  $\square$

**Example 47.4** (A two-valued checkpoint requiring exponentially many persistent states). At the first time observe a uniformly distributed word  $X \in \{0, 1\}^d$ . At the second time observe an independent uniform index  $J \in \{1, \dots, d\}$  and predict  $U = X_J$ . The raw first acquisition costs  $d$  bits; it is not called a one-bit observation. The full-history conditional mean at the second checkpoint has only two values, so its static two-label error is zero. Nevertheless a causal exact predictor requires  $S \geq 2^d$ . Two different first-stage words in one state differ in a coordinate which can be queried with positive probability. The second-stage update cannot separate them. Retaining the whole word is sufficient.

There is also a quantitative obstruction. If  $e$  is the smallest error probability of an  $S$ -state predictor,  $2 \leq S \leq 2^d$ , then

$$e \geq h_2^{-1} \left( 1 - \frac{\log_2 S}{d} \right), \quad (47.5)$$

where  $h_2$  is the binary entropy and its inverse takes values in  $[0, 1/2]$ . Indeed, condition also on every independent public coding variable  $R$ . Since the first register has at most  $S$  values,  $H(X \mid I_1, R) \geq d - \log_2 S$ . Predicting each coordinate optimally from  $(I_1, R)$  gives errors  $e_j \leq 1/2$ , and the binary entropy inequality and concavity give

$$H(X \mid I_1, R) \leq \sum_{j=1}^d H(X_j \mid I_1, R) \leq \sum_{j=1}^d h_2(e_j) \leq d h_2 \left( d^{-1} \sum_j e_j \right).$$

The average on the right is a lower bound for any actual queried-bit error. This proves (47.5). If squared loss is used instead, its optimal value is at least  $e/2$ , since  $p(1-p) \geq \min(p, 1-p)/2$ . Thus the discrepancy persists for the Hilbert-loss slice of the spectrum. It is a temporal obstruction, not poor scalar quantization. The query must be answered after the update;  $J$  is not an additional persistent decoder input.

#### APPENDIX 48. FINITE-PREFIX, FINITE-DESCRIPTION REALIZATION

The recursive partition theorem isolates a necessity condition. Predictive-energy trees provide a verifiable sufficient construction. The full tree theorem and its geometric hypotheses are proved in Section 53 below. We now strengthen its implementation: exact ordering of nearly tied real energies is unnecessary, and the raw acquisition need not expose an infinite word.

Use the factorial legal language, stationary cylinder masses  $p_v$ , energies  $a_v$ , scales  $D_v = (a_v/p_v)^{1/2}$ , and constants  $a_-, \rho, U, C_0, c_0$  of that theorem. In particular  $a_\emptyset = 1$ ,  $a_{vi} \leq \rho a_v$  with  $\rho < 1$ ,  $a_{vi} \geq a_- a_v$ ,  $D_{vi} \leq D_v$ , and  $a_{uv} < a_v$  for nonempty  $u$ . Legality and the child list are decidable. Suppose an oracle returns a positive rational upper estimate  $u_v$  satisfying, for a fixed  $\kappa \geq 1$ ,

$$a_v \leq u_v \leq \kappa a_v. \quad (48.1)$$

No uniform cost for this oracle is hidden in a cardinality assertion. Set

$$\bar{a}_v = \min\{u_w : w \text{ is a contiguous factor of } v, \text{ including } \emptyset\}. \quad (48.2)$$

For  $0 < \eta < 1$ , split precisely the legal vertices for which  $\bar{a}_v > \eta$ . Retain all their legal children. Denote the resulting full finite tree by  $\mathcal{T}_\eta$ , its leaf cut by  $\mathcal{C}_\eta$ , and its number of leaves by  $l_\eta$ .

**Theorem 48.1** (Robust tree realization with charged acquisition). *The rational-threshold construction is finite, suffix closed, and supports both deletion and prepend-and-parse insertion. Its depth and state count satisfy*

$$D_\eta \leq 1 + \left\lceil \frac{\log(\kappa/\eta)}{|\log \rho|} \right\rceil, \quad (48.3)$$

$$V_\eta = 1 + \sum_{v \text{ internal}} \deg^+(v) \leq (U+1)(2l_\eta - 1). \quad (48.4)$$

For constants depending only on the displayed structural bounds and  $\kappa$ ,

$$\sum_{v \in \mathcal{C}_\eta} a_v \asymp e_{l_\eta}^2(Z) \asymp G_{l_\eta}. \quad (48.5)$$

In the deletion experiment, at most  $D_\eta$  symbols of a snapshot acquisition suffice; no later access to that snapshot is used. The same average-risk order as the exact greedy construction follows. For every sufficiently large state budget  $S$ , a dyadic threshold can be selected with  $V_\eta \leq S$ ,  $l_\eta \geq cS$ , and  $D_\eta = O(\log(S+1))$ .

If the online readouts are in a fixed bounded subset of  $\mathbb{R}^d$  and representatives can be computed to prescribed accuracy, the selected machine has a finite rational presentation. Readout precision  $p = O(\log(S+1))$  suffices to retain the risk order. A word list, transition table and readout table have total description

$$P = O(S \log(S+1) + Sdp) \quad (48.6)$$

for a fixed alphabet. Its online persistent storage is one of at most  $S$  states; a transition-table implementation uses transient storage  $O(\log(S+1) + dp)$  bits. An acquisition uses  $O(\log(S+1))$  symbol queries. Online table accesses can be implemented in word-RAM time proportional to these query counts; a sequential bit-table implementation gives the conservative bound  $O(S(\log(S+1) + dp) \log(S+1))$  bit operations per acquisition. Offline oracle time is separate. The Hilbert-space cardinality assertion does not, by itself, encode an infinite-dimensional readout in finitely many bits.

*Proof.* If  $w$  is a factor of  $v$ , removing a prefix and then a suffix cannot decrease energy. Consequently

$$a_v \leq \bar{a}_v \leq \kappa a_v, \quad \bar{a}_{iv}, \bar{a}_{vi} \leq \bar{a}_v. \quad (48.7)$$

The second assertion follows by inclusion of factor sets and requires no accurate comparison between two distinct true energies. Thus the internal vertices form a prefix-closed and suffix-closed set. Each internal vertex is split into all its legal children. A proper suffix of a leaf is a legal child of a suffix of its parent, hence is present; repeating this argument proves full suffix closure. If  $iv$  had no leaf prefix when  $v$  is a leaf, it would be internal, forcing its suffix  $v$  to be internal, a contradiction. These are exactly the two update operations required.

At depth  $n$ ,  $\bar{a}_v \leq \kappa \rho^n$ , which proves finiteness and (48.3). Every nonroot leaf has an internal parent, so

$$a_{-}\eta/\kappa < a_v \leq \eta. \quad (48.8)$$

The exact vertex count follows by counting edges; contracting unary chains gives (48.4) as in the full-tree lemma below.

Here is a lower bound not depending on exact greedy ordering. Refine all leaves by  $h = 3(U+1)$  further legal levels. Each leaf has at least  $2^3$  descendants at that level, because there are at most  $U$  consecutive unary vertices. The resulting incomparable cylinders number at least  $8l_\eta$ . Every one has energy at least  $a_-^{h+1}\eta/\kappa$ . The separation hypothesis and monotonicity of  $D$  make the open tubes of radii  $\delta D_w$  about their images disjoint for  $2\delta < c_0$ . At most  $l_\eta$  tubes can contain one of  $l_\eta$  arbitrary proposed Hilbert centres. Each unoccupied cylinder contributes at least  $\delta^2 a_w$  to the squared error. Hence  $e_{l_\eta}^2(Z) \geq cl_\eta \eta$ . One representative per original leaf gives  $e_{l_\eta}^2(Z) \leq C_0^2 \sum a_v$ , and (48.8) proves the first comparison in (48.5). The second is the static conclusion of Theorem 53.1, whose proof does not use the present theorem. Its oscillation estimate in the deletion orientation, or its cylinder diameter estimate in the insertion orientation, also proves the stated risk bounds for this cut.

A prefix parser stops on reaching a leaf, after at most  $D_\eta$  queries. It never reads or stores the rest of the snapshot. This is a finite-latency acquisition convention: the snapshot is available during that update and the prediction deadline permits these microsteps. No claim is made for a

moving physical state observed after an unmodelled read delay. Insertion and deletion then use the already proved closure.

For budget selection, use  $\eta_j = 2^{-j}$  and take the last threshold with  $V_{\eta_j} \leq S$ . There is a constant bound on  $V_{\eta_{j+1}}/V_{\eta_j}$ . In fact, beneath a leaf at threshold  $\eta$ , all descendants at depth  $k$  have estimated energy at most  $\kappa\rho^k\eta$ , and so the next threshold adds at most  $k$  levels once  $\kappa\rho^k \leq 1/2$ . The ratio is bounded by a fixed geometric sum of powers of the alphabet size. It follows that the chosen  $V_\eta$  is at least a fixed multiple of  $S$ , and then (48.4) gives  $l_\eta \geq cS$ . For completeness, a leaf at depth  $n$  has  $a_v \geq a_-^n$ , so  $n \geq \log(1/\eta)/|\log a_-|$ . Every legal path branches at least once in  $U + 1$  levels. The complete cut therefore has at least  $2^{\lfloor \log(1/\eta)/((U+1)|\log a_-|) \rfloor}$  leaves. Since  $l_\eta \leq S$ , this bounds  $\log(1/\eta)$  by  $O(\log(S + 1))$ . Combine with (48.3). Fixed-dilation comparison for  $G$  then converts (48.5) into the same- $S$  order.

Finally replace each finite-dimensional readout by a rational vector within Euclidean distance  $\delta$ . Pointwise squared loss is at most twice its previous value plus  $2\delta^2$ . Take  $\delta^2 \leq \eta$  and use  $\sum a_v \geq a_- l_\eta \eta / \kappa$ . Bounded readouts require  $p = O(\log(1/\eta) + \log(d + 1))$  binary digits per coordinate. Store all vertex words, their legal child/deletion/insertion transitions, and these rational vectors. There are  $O(S)$  transitions for a fixed alphabet, each state index costs  $O(\log(S + 1))$  bits, and the word-list depth is  $O(\log(S + 1))$ . This proves (48.6). A streaming parser uses only its current vertex index during the acquisition; no prefix string survives the update. The stated time bounds follow by random access, or by a full sequential scan for each table access. The number and precision of the offline queries can be listed, but (48.1) alone gives no uniform running-time bound for answering them. This is why offline time is not absorbed into (48.6).  $\square$

The use of all factors in (48.2) is essential. Independent relative approximations  $u_v$  need not respect suffix monotonicity, even when the true energies do. Thresholding the raw approximations can therefore produce a tree that cannot execute its own suffix update. The envelope repairs that defect while changing energies by at most the fixed factor  $\kappa$ .

#### APPENDIX 49. THE MEASURE-VALUED SLICE: SHARP PRESENTATION ENTROPY

The decoder's output type is part of a mark. In particular, outputting a probability law close in expected Wasserstein distance is stronger than simulating one randomly chosen observation from that law. This separation is important for a genuinely measure-valued predictive state.

Let  $\mathcal{M} = \mathcal{P}([0, 1])$  with  $W_1$ . A *uniform law presentation* receives  $\pi \in \mathcal{M}$  as a transient input and returns one of  $S$  persistent labels. Its decoder outputs a law  $\hat{\pi}$ . The fixed code, including its representative laws, is chosen once for all  $\pi$ . In particular the unknown input is not inserted into the decoder's free program constants. Let

$$H_S = \inf_{Q, D} \sup_{\pi \in \mathcal{M}} \mathbb{E} W_1(\pi, D(Q(\pi))). \quad (49.1)$$

Both maps may use independent private or public randomness. The criterion is the expected distance of the *realized* decoded law. Only persistent cardinality is charged here; the input-law interface and general real-law computation are unrestricted. Regard  $\pi$  as the unknown parameter and the uncompressed observation as  $\pi$  itself. Then  $H_S$  is the spectrum of the single bounded minimax law-presentation mark, with full-observation baseline zero. It is not a supremum of point-prior Bayes risks: such a prior would reveal the unknown law to the decoder and change the problem.

**Theorem 49.1** (Uniform presentation of a probability measure). *There are absolute  $c, C > 0$  such that*

$$\frac{c}{\log(S + 1)} \leq H_S \leq \frac{C}{\log(S + 1)}, \quad S \geq 1. \quad (49.2)$$

Moreover no continuous map from  $\mathcal{M}$  into a fixed  $\mathbb{R}^k$  can be an exact coordinate sufficient to evaluate every bounded Lipschitz integral on all of  $\mathcal{M}$ . This statement does not exclude noncontinuous Borel encodings of a standard Borel space into one real coordinate.

*Proof.* For the upper bound use the explicit net in Lemma 55.1 below: equally weighted  $n$ -point empirical laws with points on the grid  $\{0, 1/n, \dots, 1\}$ , in increasing order. There are  $\binom{2n}{n} \leq 4^n$  such laws. Quantile discretization followed by grid rounding gives covering radius  $3/(2n)$ . Taking  $n = \lfloor \log S / \log 4 \rfloor$  proves the upper bound for  $S \geq 4$ ; increase  $C$  for smaller  $S$ .

For the lower bound, partition  $[0, 1]$  into  $n$  equal intervals. In each interval put mass  $1/n$  at either its quarter point or its three-quarter point. Denote the resulting law by  $\pi_\omega$ ,  $\omega \in \{0, 1\}^n$ . Their quantile functions give the exact identity

$$W_1(\pi_\omega, \pi_{\omega'}) = \frac{d_H(\omega, \omega')}{2n^2}. \quad (49.3)$$

Greedily choose binary words and delete Hamming balls of radius  $\lfloor n/4 \rfloor$ . The selected words have mutual distance greater than  $n/4$ , and their number is at least  $2^{c_0 n}$ , where  $c_0 = 1 - h_2(1/4) > 0$ . Indeed  $\sum_{j \leq n/4} \binom{n}{j} \leq 2^{nh_2(1/4)}$ , as follows by bounding these binomial terms using the binomial expansion at parameter  $1/4$ . Thus the selected measures have mutual  $W_1$  distance greater than  $1/(8n)$ .

Choose  $n \leq C_0 \log(S + 1)$  sufficiently large that the packing has at least  $2S$  members. Give it the uniform prior. For any  $S$  proposed law centres, their open balls of radius  $1/(16n)$  contain at most  $S$  packing members in total. At least half the prior therefore incurs distance at least  $1/(16n)$ , no matter how it is encoded. Fixing all independent coding randomness gives this same bound for each tape; integration preserves it. The worst input risk is at least the uniform prior risk, which proves the lower bound in (49.2).

For the last assertion, bounded Lipschitz integrals separate probability measures, so a sufficient coordinate map must be injective. The measures supported on any  $k + 2$  distinct points form a  $(k + 1)$ -simplex. A small sphere about its barycentre lies in its relative interior and is homeomorphic to  $S^k$ . By the Borsuk–Ulam theorem every continuous map of this sphere to  $\mathbb{R}^k$  identifies two antipodal points [51]. Hence no such injective coordinate exists. For  $k = 0$  two distinct measures already give the contradiction.  $\square$

The entropy mechanism is consistent with the power-exponential geometry of Wasserstein spaces studied in [50]; the packing argument is supplied here because the exact presentation loss, including randomization, matters for the result.

**Corollary 49.2** (A nonlinear posterior class with a matching bound). *Consider the nonlinear binary instrument of Corollary 55.3 below, with*

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + h_0(x), \quad \Phi_y \pi = \frac{3}{4}(h_y)_\# \pi + \frac{1}{4}(h_{1-y})_\# \pi.$$

For each fixed report  $y$ ,

$$\frac{1}{5} W_1(\pi, \nu) \leq W_1(\Phi_y \pi, \Phi_y \nu) \leq \frac{6}{25} W_1(\pi, \nu). \quad (49.4)$$

Consequently, for every fixed  $t$ , a single finite-register scheme that must present the posterior for every input prior has worst expected  $W_1$  error at least  $c5^{-t}/\log(S + 1)$  at time  $t$ , even if the decoder is additionally told the reports. A uniform recursive presentation has error  $O(1/\log(S + 1))$  at all times, when the initial law is encoded at the same accuracy and the update error is of that order. No fixed finite-dimensional continuous exact predictive coordinate exists on this all-priors posterior class. Constants in the lower bound are not asserted uniform as  $t \rightarrow \infty$  for one fixed preparation.

*Proof.* The images of  $h_0$  and  $h_1$  lie in two disjoint ordered intervals. The mixture weights for a fixed  $y$  agree for  $\pi$  and  $\nu$ . Quantile coupling therefore gives

$$W_1(\Phi_y \pi, \Phi_y \nu) = \frac{3}{4} W_1((h_y)_\# \pi, (h_y)_\# \nu) + \frac{1}{4} W_1((h_{1-y})_\# \pi, (h_{1-y})_\# \nu).$$

The derivatives of both increasing maps lie between  $1/5$  and  $6/25$ , proving (49.4). Iterate it along any length- $t$  report word and apply the same packing as in the preceding proof. The reports of this instrument are independent fair bits, with a law independent of the initial prior. Conditioning on the report word thus does not change the uniform packing prior. The centre-counting lower bound survives even with this word supplied freely to the decoder; integrating it over words and coding tapes proves the assertion for the actual, less informed decoder.

The recursive upper bound follows from Theorem 55.2 with  $\rho = 6/25$ , initializing by a net point rather than by an arbitrary fixed law. The initial error is at most  $3/(2n)$  and the accumulated error is bounded by  $[3/(2n) + \eta]/(1 - 6/25)$ . Every fixed-word map on laws is continuous and injective by the lower bound in (49.4). Its restriction to the compact simplex used above is a homeomorphism onto its image. Applying the same sphere argument proves the coordinate obstruction.  $\square$

**Proposition 49.3** (Marginal-probe simulation is a different presentation problem). *Suppose only one Bernoulli probe is required: after compression a function  $g : [0, 1] \rightarrow [0, 1]$  with Lipschitz constant at most one is specified, and the desired output law is  $\text{Bern}(\pi g)$ . The probe is fixed independently of the realized encoder randomness. For  $S \geq 2$  there is a common  $S$ -label encoder whose output-law error is at most  $1/[2(S-1)]$ , uniformly in  $\pi, g$ . Thus this marginal simulation mark cannot be substituted for the realized-law mark in (49.1).*

*Proof.* Sample  $X \sim \pi$  transiently, store the nearest point  $Z$  of the  $S$ -point uniform grid, and discard  $X$ . On receiving  $g$ , output a Bernoulli variable with parameter  $g(Z)$ . Its unconditional law is Bernoulli with parameter  $\mathbb{E}g(Z)$ , and  $|\mathbb{E}g(Z) - \pi g| \leq \mathbb{E}|Z - X| \leq 1/[2(S-1)]$ . The encoder is chosen before  $g$ . It does not produce a realized law close to  $\pi$  in  $W_1$ : its natural decoded law is  $\delta_Z$ . Nor does the argument cover probes chosen after observing  $Z$ , or an unlimited transcript of independent samples. Those are different tasks and interfaces. This output-law simulation criterion is not identified with a realized-law decision loss.  $\square$

## APPENDIX 50. A DYNAMIC FINITE-EXPERIMENT CONSEQUENCE

The decision-complete slice of the same spectrum has a different geometry. We first make its temporal register convention explicit, and then connect it to the physical count experiment.

Fix  $n, r \geq 1$ , a compact  $K \subset \mathbb{R}^r$  with nonempty interior, and a positive definite covariance  $\Gamma$ . At successive times observe independent

$$Y_t \sim N(\theta_t, \Gamma), \quad \theta = (\theta_1, \dots, \theta_n) \in K^n. \quad (50.1)$$

Each  $Y_t$  is discarded after the update. No earlier output labels are retained outside the register. At the terminal time only its final label is supplied to a parameter-independent reconstruction kernel. This is not an  $S$ -symbol channel whose entire transcript is available.

**Theorem 50.1** (Sequential accumulation under a total register budget). *For the cardinality signature with persistent  $S$  states and at most  $S$  terminal outcomes, the decision-complete terminal spectrum of (50.1) satisfies*

$$c_n S^{-2/(nr)} \leq \Theta_{G^{(n)}}(S; \mathcal{D}_{\text{all}}) \leq C_n S^{-2/(nr)}. \quad (50.2)$$

*The infimum is over causal encoders obeying the total state budget at every time. Lower bounds include arbitrary randomized, time-dependent encoders. A deterministic causal encoder attains the upper order. Constants are for fixed  $n, r, K, \Gamma$ , not uniform in a growing horizon. The readout/reconstruction randomness is external and fresh and private, while the physical alphabet*

and the persistent register remain finite. No public tape or other side information is supplied alongside the final label in this theorem.

*Proof.* The full observation vector is a Gaussian location experiment of dimension  $nr$ , with compact mean set  $K^n$  and covariance  $I_n \otimes \Gamma$ . Every final register gives an experiment with at most  $S$  outcomes, regardless of its causal implementation. The lower bound of Theorem 56.1, proved below, applies to all such experiments. Because a register is a garbling of the full observation, its forward deficiency is zero. Theorem 23.4 therefore identifies its reverse deficiency with its worst normalized Bayes regret, proving the lower bound in (50.2).

For the upper bound put  $J = \lfloor S^{1/(nr)} \rfloor$ . At time  $t$  compactify each coordinate of  $Y_t$  by  $\tau(x) = 1/2 + \arctan(x)/\pi$ , quantize into  $J$  cells per coordinate, and append the resulting  $r$ -tuple of cell indices to the current register word. At that time there are  $J^{rt} \leq J^{rn} \leq S$  possible words. The known absolute time determines their length; it carries no data-dependent information. Thus a single  $S$ -value register, not  $n$  separately uncharged registers, implements the rule. At the final time apply the positive tensor-hat reconstruction of Lemma 56.3 in dimension  $nr$ . For  $J \geq 2$  its error is  $O(J^{-2}) = O(S^{-2/(nr)})$ . Enlarge the constant for the remaining finite budget range. This proves the upper bound using a genuine causal encoder. Its real-valued input computation has no finite-time claim under this signature.  $\square$

If instead the sequence of emitted cell labels were supplied to the terminal decoder, its alphabet would be their product. In that different model a small current register could leave a large data-bearing transcript. The lower bound above is about the total retained experiment, not the size of the most recent communication symbol. This is one reason zero-delay channel coding and total-register coding require separate resource signatures [33, 34].

**50.1. A theorem-level downstream arrow.** We now give the adapter in the coordinates of the retained known-mark exposure protocol, rather than identifying all of the algebraic paper with this statistical subexperiment.

**Corollary 50.2** (Known-mark exposure adapter). *Fix positive exposure centre  $\lambda^0$ , known marks, a derivative matrix  $D$  of full row rank, and  $\Lambda = \text{diag}(\lambda^0)$ . Put  $\Sigma = D\Lambda D^\top$  and  $B_* = \Lambda D^\top \Sigma^{-1}$ . Let  $L$  be a fixed normal-compression map of rank  $r \geq 1$ , let  $F$  be an orthonormal frame for its range, and set*

$$C = F^\top L D, \quad A = F^\top L, \quad \Gamma = C A C^\top.$$

*On a fixed compact index set  $H$ , observe independent counts of means  $N\lambda_N(h)$ , where  $\lambda_N(h) = \lambda^0 + N^{-1/2}B_*h$ . If  $AH$  contains an  $r$ -dimensional ball, the deterministic unjittered statistic  $T_{N,J}$  of Section 57, applied to the score  $C(N - N\lambda^0)/\sqrt{N}$ , has at most  $J^r \leq S$  outputs. For  $\mathcal{G} = \{N(Ah, \Gamma) : h \in H\}$ , it satisfies*

$$cS^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_K S^{-2/r}, \quad J = \lfloor S^{1/r} \rfloor, \quad 2^r \leq S \leq KN^{r/6}. \quad (50.3)$$

*Only the physical output cardinality and the stated joint sample/register window are bounded here. Reconstruction randomization is external, parameter independent and private; no finite-bit complexity of the raw real score computation is asserted.*

*Proof.*  $DB_* = I$ , so  $CB_* = A$ . Since  $F^\top L$  has row rank  $r$  and  $D$  is surjective,  $C$  has row rank  $r$  and  $\Gamma > 0$ . The compact local alternatives have uniformly positive exposures for all sufficiently large  $N$ . Thus every hypothesis of Theorem 57.1 is verified with these actual score weights, mean set and covariance. That theorem supplies both inequalities in (50.3); its lower bound allows every experiment with at most  $S$  outcomes, not only this statistic. Theorem 23.4 gives the all-decision interpretation of the limiting obstruction, and Theorem 23.2 transports each bounded task through the comparison kernels. No algebraic primary-decomposition theorem is a premise of this argument.  $\square$

The left inequality concerns every experiment with at most  $S$  terminal outputs approximating this specified Gaussian family, and therefore also the physical statistic. The upper inequality concerns that explicit statistic and its positive reconstruction kernel. In particular, achieving Gaussian-family distance at most  $CN^{-1/5}$  forces  $S \geq cN^{r/10}$ , and budgets of that order suffice for this statistic. These are label-budget statements in the displayed joint window; they are not a sample-optimality theorem over all finite- $N$  physical objectives.

Theorem 23.4 supplies a decision interpretation of the limiting experiment's finite-output obstruction, while Theorem 23.2 transports each bounded task through the Poisson comparison kernels. These are genuine arrows between specified experiments. They neither require nor assert a Sinai local limit theorem, a particle path large-deviation principle, or a physical phase construction.

## SUPPORTING THEORY AND MODEL VERIFICATION

The following sections give the complete proofs used above. Unless a more detailed signature is specified, they use the cardinality signature of Section 23: only persistent states or physical terminal outcomes are bounded, with arbitrary transient computation and independent reconstruction randomness. In the tree application, Theorem 48.1 additionally replaces the full-word acquisition by a charged finite-prefix interface. In the measure application the new matching lower bound is for the uniform realized-law presentation task, not for a single marginal probe. The Gaussian and Poisson inequalities concern the indicated terminal experiment family. These local conventions are part of their use here.

### APPENDIX 51. POSITIVE EXPERIMENTS AND PREDICTIVE QUOTIENTS

**51.1. Actual histories.** All spaces of states, observations and actions are standard Borel. For an unknown parameter  $\lambda$ , a controlled experiment has a probability kernel

$$K_t^\lambda(dx', dy, dc \mid x, a), \quad (51.1)$$

where  $c$  is the cost of the actual step. Failure, rejection and stopping are outcomes, not missing probability mass. A policy chooses  $a_t$  from the observed past and independent randomization. Preparation has a specified law and cost. With an absorbing symbol after stopping, repeated kernel integration constructs one probability law on the infinite history space. Its finite marginals are consistent because every kernel in (51.1) has mass one. This is the usual kernel extension theorem [4]; conditioning on success would define a different experiment unless its preparation and waiting costs were separately charged.

For a fixed Bayesian preparation, let  $\pi$  be a reachable belief state on the hidden state, including the parameter when it is random. The positive instrument associated with an action and observation event is

$$\mathsf{T}_{a,B}\pi(E) = \int K(E, B, dc \mid x, a) \pi(dx).$$

After an observed event of positive probability, normalization gives the updated belief. For nonatomic reports we use a specified disintegration, with all equalities interpreted outside the appropriate null sets. A formula for an event of probability zero is not an additional observation channel.

**51.2. The future-test quotient.** A future test consists of a finite observation-based policy, a stopping rule, and a bounded measurable terminal payoff. Its conditional expectation at a belief  $\pi$  is written  $\pi(v)$ , where  $v$  is the corresponding backward test function. Fix a family  $\mathcal{V}$  containing

constants and closed under the one-step pullbacks of the allowed tests. It may, for example, contain all bounded finite-horizon executable tests. Define

$$\pi \sim_{\mathcal{V}} \pi' \iff \pi(v) = \pi'(v) \quad (v \in \mathcal{V}). \quad (51.2)$$

The set of classes is the predictive quotient. The choice of tests is part of the experiment specification: identifying a hidden state is not required if none of the designated tests distinguishes it.

**Proposition 51.1** (Minimality and descent). *The quotient (51.2) is the coarsest set-valued statistic that determines all designated predictions. If the test family is one-step closed, normalized Bayesian updates descend to the quotient where their normalization is positive. If, in addition, the reachable belief domain  $B$  is compact metrizable, a countable family of continuous tests determines (51.2), and each fixed-report update is continuous on its positive-probability domain, the quotient has a canonical metrizable realization as a subset of  $\mathbb{R}^{\mathbb{N}}$ , and each such update descends continuously on compact subdomains on which its denominator is bounded away from zero.*

*Proof.* Map  $\pi$  to the vector of all its test expectations. Its fibres are exactly (51.2). Any statistic determining the expectations has fibres contained in these fibres, so the vector factors through that statistic as a set map. For the update, the numerator of a future prediction is a one-step pulled-back test and the denominator is the pulled-back constant. Equivalent states have equal numerators and denominators, hence equal normalized predictions.

In the countable continuous case use  $q(\pi) = (\pi(v_j))_{j \geq 1}$ . After scaling the tests to be bounded by one, its range is a compact subset of  $[-1, 1]^{\mathbb{N}}$ , with its product metric. A continuous map from a compact space into a Hausdorff space is a quotient map. The fibre-respecting continuous update therefore factors continuously on the stated compact subdomain. No assertion of a Borel cross-section, or of a smooth quotient for arbitrary Borel tests, is used. The minimality statement in the first paragraph alone would not imply either assertion.  $\square$

The proposition is relative to the prepared Bayesian experiment. An identity for one prior is not a parameter-uniform comparison of the family  $\{K^\lambda\}_\lambda$ . All experiment distances below explicitly take the supremum over the parameter when that stronger meaning is intended.

**51.3. Attainable Bayes geometry.** Let  $U$  be square integrable in a separable Hilbert space  $H$ , let  $\mathcal{H}$  be the actual information, and put  $m = \mathbb{E}[U \mid \mathcal{H}]$ . Define

$$e_M^2(m) = \inf_{z_1, \dots, z_M \in H} \mathbb{E} \min_i \|m - z_i\|^2.$$

Independent encoder/decoder randomization is permitted, but a retained observation is not a free input after encoding.

**Proposition 51.2** (Checkpoint geometry and information loss). *The optimum squared loss of a single  $M$ -label checkpoint statistic is*

$$\mathbb{E}\|U - m\|^2 + e_M^2(m). \quad (51.3)$$

*For any sub- $\sigma$ -field  $\mathcal{C} \subset \mathcal{H}$ , posterior laws of  $\lambda$  satisfy the barycentre identity  $\pi_{\mathcal{C}} = \mathbb{E}[\pi_{\mathcal{H}} \mid \mathcal{C}]$ . Consequently convex posterior functionals decrease under coarsening. Whenever the quantities are finite,*

$$I(\lambda; \mathcal{H}) - I(\lambda; \mathcal{C}) = \mathbb{E} D_{\text{KL}}(\pi_{\mathcal{H}} \parallel \pi_{\mathcal{C}}). \quad (51.4)$$

*In a differentiable dominated family, the score of a statistic  $S$  is  $\mathbb{E}[s_\lambda \mid S]$ , and the lost Fisher information matrix is  $\mathbb{E} \text{Cov}(s_\lambda \mid S)$ , provided differentiation and conditioning are justified in  $L^2$ .*

*Proof.* Conditional orthogonality gives  $\mathbb{E}\|U - z\|^2 = \mathbb{E}\|U - m\|^2 + \mathbb{E}\|m - z\|^2$  for every  $\mathcal{H}$ -measurable output, and also after adding independent randomness. Freeze that randomness. There are at most  $M$  decoder outputs, so nearest-centre assignment is no worse than the given encoding. Conversely every finite codebook has a measurable nearest assignment, with deterministic tie-breaking. Taking infima proves (51.3), without assuming attainment of an optimal codebook. Conditional expectation against bounded parameter tests gives the barycentre identity. Conditional Jensen proves the convex assertion. The chain rule for the logarithm of posterior densities gives (51.4); finite partitions followed by monotone approximation give the same assertion under the stated finiteness condition. Differentiating the marginal density of  $S$  gives the score identity. The matrix law of total variance gives the Fisher identity. These are the classical decision and information identities in [1, 45]; their role here is to specify the geometry being implemented, not to certify recursive closure.  $\square$

**51.4. Normalized response and observation modules.** Let  $b_\theta$  be a differentiable finite positive measure with  $b_\theta(1) > 0$ , and let  $\pi_\theta = b_\theta/b_\theta(1)$ . Differentiating this actual normalization gives

$$\partial_\theta \pi_\theta(f) = \frac{\partial_\theta b_\theta(f) - \pi_\theta(f) \partial_\theta b_\theta(1)}{b_\theta(1)}. \quad (51.5)$$

If the derivative has score density  $s$  relative to  $b_\theta$ , the right side is  $\text{Cov}_{\pi_\theta}(f, s)$ . Thus a response direction is invisible exactly when its centered score annihilates the linear span of the designated tests. If the observation protocol consists of products from two declared linear test spaces  $V, W$ , its null space is  $(VW)^\perp$ , with  $VW = \text{span}\{vw : v \in V, w \in W\}$ . This last statement follows immediately from linearity, but it requires that those products really are the observed probes. It does not identify an algebraic multiplicity with a physical noise variance. Formula (51.5) and this annihilator are the normalization and observation-module inputs used to type a contact experiment.

## APPENDIX 52. CAUSAL MORPHISMS AND CHARGED APPROXIMATION

A resource signature specifies ordered coordinates, for example persistent cardinality, numerical precision, transient workspace, program description, preparation count and physical time. A budget transformer is a monotone map between two such signatures, accompanied by an actual implementation. A coordinate may explicitly be unlimited. The persistent-cardinality results below leave fixed programs, fixed real constants, time and transient computation unlimited unless a more specific implementation is given. There is no claim that these resources are equivalent.

A causal simulation from an experiment  $\mathcal{E}$  to  $\mathcal{F}$  consists of a source controller, a causal transducer and its initialized state. For every target policy the joint construction must produce the target observation/cost history within a declared parameter-uniform total-variation error. The transducer never sees the unknown parameter or the unobserved physical state. Resource bounds are imposed on this joint construction, not just on the terminal observation kernel. We use  $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$ .

**Theorem 52.1** (Typed simulation and recursive approximation). *The following statements hold.*

- (i) *Serial causal simulations of errors  $\epsilon_1, \epsilon_2$  and budget transformers  $\rho_1, \rho_2$  compose with error at most  $\epsilon_1 + \epsilon_2$  and transformer  $\rho_1 \circ \rho_2$ . A simulator with  $K$  persistent states and a target observer with  $M$  states can be run with  $KM$  persistent states. For losses in  $[0, L]$ , the risk changes by at most  $L(\epsilon_1 + \epsilon_2)$ . Other resource coordinates are transformed according to the supplied implementations. The identity simulation has zero error and the identity transformer. Independent parallel simulations have additive TV error and the product register, with preparations and other costs charged by their joint implementation.*

(ii) Suppose the true predictive state lies in a metric space  $(S, d)$  and updates by  $s_t = F_t(s_{t-1}, y_t, a_t)$ . For all reports and actions suppose  $F_t$  is  $\rho$ -Lipschitz in its state,  $\rho < 1$ . Let an implementable approximate update  $\tilde{F}_t$  have local error at most  $\eta_t$ , and let  $Q_t : S \rightarrow C_t$  be a measurable finite quantizer with radius  $\epsilon_t$ . The register

$$\hat{s}_t = Q_t \tilde{F}_t(\hat{s}_{t-1}, y_t, a_t) \quad (52.1)$$

has at most  $|C_t|$  states at time  $t$  and, on every common observation path,

$$d(s_t, \hat{s}_t) \leq \rho^t d(s_0, \hat{s}_0) + \sum_{j=1}^t \rho^{t-j} (\epsilon_j + \eta_j) =: b_t. \quad (52.2)$$

A data-dependent persistent numerical residue must be part of  $C_t$ ; it cannot be placed in an uncharged side register. If the report kernels are  $L_0$ -Lipschitz in total variation as functions of the predictive state, the length- $n$  simulated history differs by at most  $L_0 \sum_{t < n} \mathbb{E} b_t$ . The same statement holds up to a bounded stopping time, with the stopped sum. If the true conditional mean is  $g(s_t)$  with  $g$   $L_1$ -Lipschitz into a Hilbert space, its implemented prediction has Bayes excess at most  $L_1^2 \mathbb{E} b_t^2$ .

*Proof.* Compose the two transducers on their joint initialized state. Applying a Markov kernel contracts total variation, so the triangle inequality gives the sum of errors, uniformly over the policy and parameter for which the two certificates were supplied. The joint register is the Cartesian product. This proves the cardinality bound, and integration of a loss with range  $[0, L]$  proves the risk bound. The identity is the unchanged history. For independent parallel laws, insert one product factor at a time and use TV contraction to obtain the same sum bound. The composition order of the budget maps follows from first realizing the target in the intermediate experiment and then realizing that implementation in the source.

For (ii), insert  $F_t(\hat{s}_{t-1}, y_t, a_t)$  between the two updates. Contraction, the local defect and the quantization radius give  $d(s_t, \hat{s}_t) \leq \rho d(s_{t-1}, \hat{s}_{t-1}) + \eta_t + \epsilon_t$ . Induction proves (52.2). For the law comparison, telescope the products of the history kernels, replacing one kernel at a time. The prefix before the replaced kernel is distributed according to the true experiment; its integrated local TV error is at most  $L_0 \mathbb{E} b_t$ . All suffix kernels are contractions in TV. Summing proves the bound. An absorbing stopped state proves the stopped version by the identical calculation. The policy is evaluated on the same history in each local comparison: no continuity of a possibly discontinuous policy is presumed. Finally apply conditional orthogonality as in Proposition 51.2 and the Lipschitz bound for  $g$ .  $\square$

**Corollary 52.2** (Resource transport of an attainable profile). *If an  $M$ -state target observer has risk  $R_M$  and a  $K$ -state causal simulation has error  $\epsilon$  for its bounded decision problem, then the source has a  $KM$ -state observer of risk at most  $R_M + L\epsilon$ . Two simulations give the corresponding two directed inequalities; they need not preserve the same cardinality. A  $K$ -valued decoder-side event can always be absorbed into a register at a factor  $K$ , but a continuous side observation admits no such finite-factor assertion in general.*

*Proof.* Use the product construction. For the finite event store its current value as the second register coordinate. For the last assertion, an exact uniform real observation supplied directly to a decoder has zero squared estimation error with one otherwise empty state. If it is available only before an  $M$ -label encoding, scalar quantization has error  $1/(12M^2) > 0$ . Thus no finite factor simulates that side interface for all accuracies.  $\square$

The theorem joins quotient approximation to experiment comparison. Contractivity is a property to be verified on the true quotient, and  $Q_t$  is an executable representation of it. A finite-dimensional formula for the conditional mean is sufficient for one application, but is not a hypothesis of the

theorem. In Section 55 the true state is an entire probability measure. In the next section a different structure replaces metric contraction by an exact finite-tree update.

### APPENDIX 53. A STRUCTURAL THEOREM FOR PREDICTIVE TREES

**53.1. Local hypotheses and the finite profile.** Let  $\mathcal{L}$  be a pruned factorial language on a finite alphabet of size  $b \geq 2$ . Thus prefixes and suffixes of legal words are legal, every word has an infinite continuation, and only legal children occur in its rooted tree. Write  $[v]$  for a cylinder in the boundary  $\Omega$  and  $\emptyset$  for the empty word. Assume that along any branch there are at most  $U$  consecutive vertices with only one legal successor. This condition allows unary vertices but excludes an infinite deterministic branch. Let  $\mu$  be a stationary probability on  $\Omega$ , positive on every legal cylinder, and put  $p_v = \mu[v]$ .

A predictive energy is a positive family  $a_v$ , with  $a_\emptyset = 1$ . Put  $D_v = (a_v/p_v)^{1/2}$ . The following are the hypotheses of the structural theorem. There are constants  $a_- > 0$ ,  $\rho < 1$ , and  $c_0, C_0 > 0$  such that

$$a_- a_v \leq a_{vi} \leq \rho a_v, \quad \sum_{i: vi \in \mathcal{L}} a_{vi} \leq \rho a_v, \quad (53.1)$$

$$a_{uv} < a_v \quad (u \neq \emptyset), \quad D_{vi} \leq D_v. \quad (53.2)$$

These statements concern legal words only. A bounded Borel map  $Z$  from  $\Omega$  into a separable Hilbert space satisfies

$$\text{diam } Z([v]) \leq C_0 D_v, \quad (53.3)$$

$$\|Z(\omega) - Z(\omega')\| \geq c_0 D_u \quad \text{if } u \text{ is their longest common prefix.} \quad (53.4)$$

The image may be infinite dimensional. Separation is an explicit geometric hypothesis, not a consequence of the word notation.

Starting with the root, repeatedly split a leaf of maximal energy into *all* its legal children. Fix a tie rule. Let  $\mathcal{T}_N$  be the last tree in this sequence with at most  $N$  leaves,  $\mathcal{P}_N$  its leaf cut, and

$$G_N = \sum_{v \in \mathcal{P}_N} a_v. \quad (53.5)$$

A split at a unary vertex is an actual step. Finiteness of  $\mathcal{T}_N$  will follow below. The construction is a mathematical finite algorithm; computable implementation of arbitrary real energies is an additional input, not part of their Borel specification.

There are two causal interpretations of these data.

**Deletion model.** Independent acquisitions  $\omega \sim \mu$  occur at renewal times of independent positive integer length  $L$ , with  $\mathbb{E}L < \infty$ . The acquired sequence is observed exactly. At age  $k$  the desired output is  $g(\sigma^k \omega)$ , where  $g$  is bounded and Hilbert valued. Write

$$w_k = \mathbb{P}(L > k)/\mathbb{E}L, \quad W(n) = \sum_{k \geq n} w_k, \quad Z(\omega) = (\sqrt{w_k} g(\sigma^k \omega))_{k \geq 0}.$$

In addition to (53.1)–(53.4), assume the local oscillation bound

$$\sum_{k < |v|} w_k (\text{osc}_{[v_k:]} g)^2 + 4 \|g\|_\infty^2 W(|v|) \leq C_1 D_v^2. \quad (53.6)$$

A register sees the renewal event only during its update; the decoder receives neither that event nor a stored raw observation for free.

**Insertion model.** Let  $(Y_t)_{t \in \mathbb{Z}}$  be a stationary finite-alphabet observed process whose reversed past  $\omega_t = (Y_t, Y_{t-1}, \dots)$  has law  $\mu$  and language  $\mathcal{L}$ . For a square-integrable stationary Hilbert target  $U_t$ , suppose

$$m_t = \mathbb{E}[U_t \mid Y_s : s \leq t] = Z(\omega_t), \quad B = \mathbb{E}\|U_t - m_t\|^2. \quad (53.7)$$

The finite register may start at zero with a fixed state; it is not initialized with the infinite past. Infinite stationary pasts here specify the reference prediction and its law. A finite warm-up is included in the limiting average. No renewal or independent-observation assumption is imposed in this interpretation.

For either model let  $R_M$  be the infimum, over all  $M$ -value registers with causal standard Borel update and decoding kernels, of the upper limiting expected average squared loss. Absolute time and independent coding randomization are permitted. In the insertion model put  $E_M = R_M - B$ .

**Theorem 53.1** (Predictive-tree realization in two orientations). *Under (53.1)–(53.4),*

$$cG_M \leq e_M^2(Z) \leq CG_M. \quad (53.8)$$

*Every complete greedy tree is suffix closed. A tree with  $L$  leaves has exactly  $1 + \sum_{v \text{ internal}} d^+(v)$  vertices, and at most  $(U + 1)(2L - 1)$  vertices; all of them may be used as designated states. For every fixed  $B_0 \geq 1$ ,  $G_{\lceil B_0 M \rceil} \asymp_{B_0} G_M$ .*

*In the deletion model, with (53.6),*

$$cG_M \leq e_M^2(Z) \leq R_M \leq CG_M. \quad (53.9)$$

*In the insertion model,*

$$cG_M \leq e_M^2(Z) \leq E_M \leq CG_M. \quad (53.10)$$

*The converses apply to the lower limiting average of each randomized, time-dependent observer. The upper bounds are achieved in order by stationary deterministic updates. In particular a logarithmic age or history multiplier is not required. Constants depend on the displayed structural hypotheses, not on the register budget.*

### 53.2. Balance, dilation and exact closure.

**Lemma 53.2** (Greedy balance). *For all sufficiently large  $N$ ,  $N/b \leq |\mathcal{P}_N| \leq N$ . If  $\theta_N$  is the largest energy of a current leaf, then*

$$a_- \theta_N \leq a_v \leq \theta_N \quad (v \in \mathcal{P}_N). \quad (53.11)$$

*The trees are finite and their profiles have the fixed-dilation property in Theorem 53.1.*

*Proof.* Suppress unary vertices in any finite complete tree. A rooted tree with  $L$  leaves and no unary internal vertex has at most  $2L - 1$  vertices, since its number of edges is the sum of its internal outdegrees. Each vertex of the suppressed tree accounts for itself and at most  $U$  unary vertices on its incoming path, including the initial root path. Thus the original tree has at most  $(U + 1)(2L - 1)$  vertices. An infinite sequence of splits with bounded leaf number is impossible, proving finiteness. The split following  $\mathcal{T}_N$  crosses the budget and adds at most  $b - 1$  leaves. Hence  $|\mathcal{P}_N| > N - b + 1$ .

The maximal leaf energy never increases. Every present leaf was born with energy at least  $a_-$  times an earlier maximal energy. This proves (53.11), including any unary splits. Set  $\ell = U + 1$ . Every vertex has at least two distinct descendants at depth  $\ell$ , and at least  $2^j$  at depth  $\ell j$ . Choose  $j$  with  $2^j > b(B_0 + 2)$ . If the later maximal energy at budget  $\lceil B_0 N \rceil$  were less than  $a_-^{\ell j + 1} \theta_N$ , every original leaf would have been split through all these  $\ell j$  levels: all intervening vertices have energy at least that threshold. The resulting number of leaves would exceed  $\lceil B_0 N \rceil$ , a contradiction. Therefore the later maximum is at least a fixed multiple of  $\theta_N$ . Applying (53.11) at the later budget gives  $G_{\lceil B_0 N \rceil} \geq c_{B_0} N \theta_N \geq c_{B_0} G_N$ . Splitting decreases the energy sum by (53.1), giving the reverse inequality. Adjusting constants covers the finite initial range.  $\square$

**Lemma 53.3** (Two exact operations on one tree). *Every proper suffix of an internal greedy-tree vertex is internal, and every suffix of any vertex is present. Consequently deletion of the first letter stays in the tree. If  $v$  is a leaf and  $iv$  is legal, then  $iv$  has a leaf prefix in the same tree. It can therefore be updated by prepending  $i$  and reading only the letters of  $v$  until that leaf is reached.*

*Proof.* Suppose an internal word  $uv$ ,  $u \neq \emptyset$ , is about to be split. If its suffix  $v$  has not been split, some current leaf  $t$  is a prefix of  $v$ . Child decrease and strict prefix-attachment decrease give  $a_t \geq a_v > a_{uv}$ , contrary to maximality. Thus the suffix is already internal. For a leaf  $i_1 \cdots i_n$  with  $n \geq 2$ , its parent is internal; the suffix  $i_2 \cdots i_{n-1}$  of that parent is internal, so its legal child  $i_2 \cdots i_n$  is present. The one-letter case ends at the root. Iteration gives closure of every vertex.

If there were no leaf prefix of  $iv$ , the complete tree would contain  $iv$  as an internal vertex. Its suffix  $v$  would then be internal by the first assertion, contradicting that it is a leaf. This proves the insertion assertion. Only legal children were used in either proof.  $\square$

### 53.3. Static lower bound and the causal converse.

**Lemma 53.4** (Arbitrary centres). *The bounds (53.8) hold for arbitrary Hilbert-space centres, not only centres on  $Z(\Omega)$ .*

*Proof.* Choose one image point in each leaf cylinder to obtain  $e_N^2(Z) \leq C_0^2 G_N$ . For the lower bound use  $\mathcal{P}_{4bM}$ , which has at least  $4M$  leaves for large  $M$ . The open tubes about  $Z([v])$  of radii  $\delta D_v$  are disjoint for a sufficiently small fixed  $\delta > 0$ . Indeed for incomparable  $v, v'$  their common prefix  $u$  has  $D_u \geq \max(D_v, D_{v'})$ , and (53.4) separates the two images by at least  $c_0 D_u$ . Take  $2\delta < c_0$ .

At most  $M$  of the tubes contain one of  $M$  proposed centres. On every unoccupied cylinder the squared error is at least  $\delta^2 D_v^2$ . Its integrated contribution is  $\delta^2 a_v$ . By balance, the at least three-quarters unoccupied leaves carry at least  $3a_-/4$  of the total energy sum. Thus  $e_M^2(Z) \geq cG_{4bM} \geq c'G_M$  by Lemma 53.2. The finite initial range is harmless: (53.4) and positive cylinder masses at unbounded branching depths give infinitely supported image law, so every fixed finite-centre error is positive. Alternatively the same tube proof at a larger fixed cut proves that positivity directly.  $\square$

**Lemma 53.5** (Renewal converse). *In the deletion experiment every randomized, nonstationary  $M$ -state observer has lower limiting average loss at least  $e_M^2(Z)$ .*

*Proof.* A standard Borel probability kernel is realized as a measurable function of its input and an independent uniform random variable [4]. Applying this at the countably many update and decode times gives an independent coding sequence. A data-dependent persistent position in such a sequence, if used, belongs to the charged state; the representation itself supplies no history channel.

Condition on an acquisition at a fixed time, its pre-acquisition history, and the independent coding variables. The acquired  $\omega$  still has law  $\mu$ . Along a hypothetical continuation with no new acquisition, each of the at most  $M$  post-acquisition labels determines one output string. For a truncation at age  $n$ , the survival-weighted loss is therefore at least  $\mathbb{E}L e_M^2(Z^{<n})$ , where  $Z^{<n}$  is the first  $n$  coordinates of the weighted orbit. At a fixed physical time, events specifying the last acquisition are disjoint. Summing this bound over acquisition times at least  $n$  before the end of a horizon and using the Cesàro renewal count gives lower limiting average at least  $e_M^2(Z^{<n})$ . For completeness, the elementary renewal theorem gives expected renewals per unit time tending to  $1/\mathbb{E}L$ ; a fixed truncation removes at most  $n$  terminal time indices, each with bounded contribution. No stationary coding rule is presumed.

Finally  $e_M^2(Z^{<n}) \uparrow e_M^2(Z)$ . Monotonicity follows by projection. The reverse limiting inequality follows by appending zero tails to an arbitrarily good truncated code: the extra mean squared norm is at most  $\|g\|_\infty^2 W(n) \rightarrow 0$ . This completes the proof.  $\square$

*Proof of Theorem 53.1.* The static assertion, finite count, dilation and suffix closure are proved above. In the deletion experiment retain every tree vertex as a designated state. At acquisition store the unique leaf prefix of  $\omega$ . Between acquisitions delete its first letter until the empty word is reached, and then keep that word. Choose a representative in each nonempty cylinder and output

its  $g$  value; use a fixed bounded output at the empty word. For an acquired leaf  $v$  the weighted error is bounded by the left side of (53.6). Averaging and applying renewal reward gives at most  $C_1 G_N$ . Lemma 53.5 gives the unrestricted converse.

For insertion, after reaching a leaf use the prepend-and-parse rule of Lemma 53.3. During initialization store the observed reversed word while it is internal, or its first leaf prefix when one appears. This uses only vertices of the same tree and after at most its maximum depth gives the correct leaf prefix of the full observed past. Decode a representative  $Z$  value in that leaf cylinder. Its stationary excess loss is at most  $C_0^2 G_N$  by (53.3) and conditional orthogonality. The finite warm-up has zero limiting average contribution. The converse holds at each time: freeze independent coding randomness, apply orthogonality against  $m_t$ , and use the at most  $M$  decoder values to lower bound excess by  $e_M^2(Z)$ . Stationarity makes this the same number at every time, even for time-dependent decoders.

In either orientation choose  $N$  a fixed sufficiently small multiple of  $M$ , so that  $(U+1)(2|\mathcal{P}_N|-1) \leq M$ . Dilation converts  $G_N$  into  $O(G_M)$  without a time-depth multiplier. One fixed state handles the remaining bounded range of budgets. This proves the two risk assertions.  $\square$

The static part of the proof uses familiar separated-cylinder quantization ideas [43, 36]. Maximal-energy prefix splitting is likewise a classical coding device [37]. The additional conclusion is that a strict *predictive* energy inequality makes the adaptive tree executable in both orientations, with its whole persistent state counted.

## APPENDIX 54. TWO MODEL VERIFICATIONS

**54.1. Markov-renewal orbit geometry.** Let  $A$  be a primitive zero-one matrix on  $b$  symbols, with Perron root greater than one. Its legal one-sided sequences have a stationary fully supported law  $\mu$  such that every legal conditional child probability is at least  $\zeta > 0$ . Let  $h_i$  map a compact set of diameter at most one into pairwise  $g$ -separated subsets of itself. Assume word scales  $r_v$ ,  $r_\emptyset = 1$ , satisfy

$$\begin{aligned} D_0^{-1} r_v |x - y| &\leq |h_v x - h_v y| \leq r_v |x - y|, \\ D_0^{-1} r_u r_v &\leq r_{uv} \leq r_u r_v, \quad 0 < r_- \leq r_i \leq r_+ < 1. \end{aligned} \quad (54.1)$$

The injective coding map identifies the legal sequences with a compact set  $K$ , on which  $T$  deletes the first symbol. Let  $f : K \rightarrow \mathbb{R}^m$  obey

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad \sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta}, \quad 0 < \beta \leq 1. \quad (54.2)$$

The renewal age weights satisfy  $0 < q_- \leq w_{k+1}/w_k \leq q_+ < 1$ . Set  $\tau = 2\beta$  and

$$V(v) = \sum_{k < |v|} w_k r_{v_k}^\tau + W(|v|), \quad a_v = \mu[v]V(v), \quad Z(x) = (\sqrt{w_k} f(T^k x))_{k \geq 0}. \quad (54.3)$$

**Corollary 54.1** (Markov renewal as a structural instance). *The data (54.1)–(54.3) satisfy Theorem 53.1 in its deletion orientation. In particular  $e_M^2(Z) \asymp R_M \asymp G_M$ , with a stationary finite-tree observer and an unrestricted randomized, nonstationary converse. Forbidden edges are not completed, and unary vertices are charged.*

*Proof.* Every suffix scale is at most one and appending a letter decreases its scale. If  $|u| = m$  and  $|v| = n$ , monotonicity of  $w$  gives

$$\begin{aligned} V(v) - \left( \sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \sum_{k < m} w_k, \\ V(uv) - \left( \sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\leq r_+^\tau r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

Therefore  $V(uv) < V(v)$ . Stationarity gives only  $\mu[uv] \leq \mu[v]$ , which is enough to give the strict energy inequality. Equality of cylinder masses at a unique predecessor causes no problem.

Appending a legal letter multiplies old spatial terms by a factor between  $D_0^{-\tau} r_-^\tau$  and  $r_+^\tau$ . The old tail  $W(n)$  becomes  $w_n r_i^\tau + W(n+1)$ . Since  $w_n \geq (1 - q_+)W(n)$  and  $W(n+1) \geq q_-W(n)$ , we have

$$\ell V(v) \leq V(vi) \leq \rho V(v), \quad \ell = \min(D_0^{-\tau} r_-^\tau, q_-), \quad \rho = 1 - (1 - r_+^\tau)(1 - q_+) < 1.$$

Use  $a_- = \zeta \ell$  and sum the conditional masses to verify (53.1). These inequalities also give monotonicity of  $D_v = \sqrt{V(v)}$ . In a primitive graph with Perron root greater than one, a deterministic path cannot run for  $b$  repeated unary vertices: a repeated vertex would generate a closed deterministic cycle. Thus one can take  $U = b$ .

Introduce  $d_\beta(x, y)^2 = \sum_k w_k |T^k x - T^k y|^{2\beta}$ . Hölder continuity bounds the orbit distance from above by  $Hd_\beta$ . Summing the delayed lower bound (54.2), and using  $w_{k-j} \leq q_-^j w_k$ , bounds  $d_\beta$  from above by a constant times the orbit distance. Points in one cylinder have squared  $d_\beta$  distance at most  $V(v)$ . Points whose longest common prefix is  $u$  have, before its end, distances at least  $D_0^{-1} g r_{u_k}$ , and at its end distance at least  $g$ . The last term controls  $W(|u|)$  since  $W(n) \leq w_n/(1 - q_+)$ . This proves both image comparisons. Finally the Hölder bound on each suffix and the bounded tail prove (53.6). The structural theorem now supplies all combinatorial, quantization, and causal conclusions.  $\square$

This corollary recovers the earlier Markov-renewal profile without repeating its tree argument as a new model-specific proof. Gibbs pressure identifies the profile exponent under the additional thermodynamic assumptions of the previous Markov theorem, whose full proof is retained in the companion. Pressure is a computation of this instance, not an assumption of the foundational realization theorem. For clarity, the passage from power bounds to an exponent uses only the following elementary fact.

**Proposition 54.2** (Logarithmic squeeze). *If for every  $\delta > 0$  a positive sequence  $R_M$  satisfies  $c_\delta M^{-\alpha-\delta} \leq R_M \leq C_\delta M^{-\alpha+\delta}$  eventually, then  $-\log R_M / \log M \rightarrow \alpha$ .*

*Proof.* Taking logarithms gives lower and upper limits between  $\alpha - \delta$  and  $\alpha + \delta$ , because fixed logarithmic constants divided by  $\log M$  vanish. Let  $\delta$  decrease to zero. In a pressure-root proof one first chooses the auxiliary root parameter on either side of its zero and only then lets  $M$  tend to infinity; root continuity then supplies the arbitrary  $\delta$  used here.  $\square$

**54.2. An infinite hidden shift with noisy continual reports.** Let  $(B_t)_{t \in \mathbb{Z}}$  be independent identically distributed Bernoulli variables. The hidden state at time  $t$  is the whole sequence  $X_t = (B_t, B_{t-1}, \dots)$ . A binary memoryless channel reports  $Y_t$  about  $B_t$ . Write

$$\gamma = \mathbb{P}(Y_t = 1) \in (0, 1), \quad a_i = \mathbb{P}(B_t = 1 \mid Y_t = i), \quad a_0 \neq a_1.$$

Fix  $0 < r < 1$ . The Hilbert-valued prediction target and its mean are

$$U_t = (\sqrt{1 - r^2} r^j B_{t-j})_{j \geq 0}, \quad m_t = (\sqrt{1 - r^2} r^j a_{Y_{t-j}})_{j \geq 0}. \quad (54.4)$$

This squared loss can equally be implemented by drawing a fresh terminal query  $j$  with probability  $(1 - r^2)r^{2j}$  and scoring the prediction of  $B_{t-j}$ . The independent query is a task index, not a new observation of the hidden state. Its distribution and interface are fixed in advance.

**Theorem 54.3** (Sharp register law for an infinite posterior). *Put  $p_0 = 1 - \gamma$ ,  $p_1 = \gamma$  and let  $s \in (0, 1)$  be the unique solution of*

$$r^{2s}(p_0^s + p_1^s) = 1, \quad \alpha = (1 - s)/s. \quad (54.5)$$

*For the actual noisy shift experiment,*

$$R_M = B + \Theta((a_1 - a_0)^2 M^{-\alpha}), \quad B = \sum_{i=0}^1 p_i a_i (1 - a_i). \quad (54.6)$$

*The constants in the excess comparison depend on  $r, p_0, p_1$ , not on  $a_0, a_1$ . No posterior truncation is assumed in deriving this law. For a fair hidden bit and a binary symmetric channel of error  $\varepsilon < 1/2$ , it becomes*

$$R_M = \varepsilon(1 - \varepsilon) + \Theta((1 - 2\varepsilon)^2 M^{2 \log r / \log 2}).$$

*At  $\varepsilon = 1/2$  the predictive mean is constant and the excess is zero with one state.*

*Proof.* The full posterior is the product of Bernoulli laws with parameters  $a_{Y_{t-j}}$ . Thus (54.4) is the actual conditional mean, and orthogonality gives the displayed  $B$ . Normalize its nonconstant part by  $a_1 - a_0$ . On the full binary past language take  $p_v = \prod p_{v_j}$ ,  $D_v = r^{|v|}$  and  $a_v = p_v r^{2|v|}$ . Children multiply energy by  $p_i r^2$ , their sum multiplies it by  $r^2$ , and every nonempty prefix attachment strictly decreases it. If two pasts first differ at position  $n$ , their normalized squared image distance lies between  $(1 - r^2)r^{2n}$  and  $r^{2n}$ . All hypotheses of Theorem 53.1 hold in the insertion orientation.

For the exact order of  $G_M$ , set  $q_i = (p_i r^2)^s$ . Equation (54.5) says  $q_0 + q_1 = 1$ . For every finite complete cut, consistency of this Bernoulli measure gives  $\sum_v a_v^s = 1$ . By greedy balance, for a cut with  $L$  leaves and maximum  $\theta$ ,  $L(a_- \theta)^s \leq 1 \leq L\theta^s$ . Hence  $G_M \asymp L\theta \asymp L^{-(1-s)/s} \asymp M^{-\alpha}$ . The root exists uniquely because the left side of (54.5) decreases continuously from two to  $r^2 < 1$ . The fair case solves it explicitly, and the constant-mean case is immediate. If only reports from time zero are supplied, the unobserved past contributes at most  $r^{2(t+1)}$  to the squared target norm at time  $t$ . Its average vanishes; the finite-tree initialization in the structural theorem still gives (54.6).  $\square$

This is not an exactly closed finite list of conditional means. Every past report changes its own posterior coordinate, and arbitrarily many coordinates can be queried. It also has no fixed-step common minorizer: after  $k$  hidden updates the old tail remains intact, and two different initial tails give disjoint supports. A measure dominated by all those kernels must be zero. As always, a bare Borel encoding of an infinite sequence into one real number is possible; it is not a finite-cardinality register and does not change the theorem.

## APPENDIX 55. FINITE PRESENTATIONS OF MEASURE-VALUED FILTERS

The preceding sharp example has a separated tree geometry. We next give a constructive approximation theorem requiring no such separation and no finite-dimensional exact posterior coordinate.

**Lemma 55.1** (A finite Wasserstein presentation). *For  $n \geq 1$  let  $\mathcal{C}_n$  consist of empirical probability measures with  $n$  equally weighted atoms on  $\{0, 1/n, \dots, 1\}$ . Then*

$$|\mathcal{C}_n| = \binom{2n}{n} \leq 4^n, \quad \sup_{\pi \in \mathcal{P}([0,1])} W_1(\pi, Q_n \pi) \leq \frac{3}{2n} \quad (55.1)$$

*for a specified Borel quantizer  $Q_n$ .*

*Proof.* Take the generalized quantiles of  $\pi$  at  $(j - 1/2)/n$ ,  $1 \leq j \leq n$ , and round them to nearest grid points, breaking ties toward zero. Quantiles are Borel functions of a probability law. If  $q$  is its monotone quantile function, the integral of the error from replacing  $q$  by its midpoint value on each of the  $n$  equal subintervals is at most  $1/n$ : bound each integral by its interval length times the oscillation and sum the monotone variations, at most one. Rounding adds at most  $1/(2n)$ . In one dimension  $W_1$  is the  $L^1$  distance of quantile functions, giving the bound. The possible sorted lists are multisets of size  $n$  from  $n + 1$  points, counted by  $\binom{2n}{n}$ . The binomial theorem bounds this by  $4^n$ .  $\square$

**Theorem 55.2** (A charged measure-valued recursive approximation). *Let the actual filter obey  $\pi_t = \Phi_{y_t, a_t}(\pi_{t-1})$  on  $\mathcal{P}([0, 1])$ , and suppose every allowed update is  $\rho$ -contractive in  $W_1$ , with  $\rho < 1$ . Suppose an implementable update has  $W_1$  defect at most  $\eta$ . The initialized register*

$$\hat{\pi}_t = Q_n \tilde{\Phi}_{y_t, a_t}(\hat{\pi}_{t-1}), \quad \hat{\pi}_0 \in \mathcal{C}_n$$

*has at most  $\binom{2n}{n}$  persistent values and satisfies, for every actual observation path,*

$$W_1(\pi_t, \hat{\pi}_t) \leq \rho^t + \frac{3/(2n) + \eta}{1 - \rho}. \quad (55.2)$$

*This bounds the error of every 1-Lipschitz test expectation, uniformly over the choice of test. Thus for  $M \geq 4$ ,  $n = \lfloor \log M / \log 4 \rfloor$  gives error  $O((\log M)^{-1} + \eta)$  and squared error  $O((\log M)^{-2} + \eta^2)$  after the initialization term. If these test expectations are conditional means, the latter is a Bayes excess bound. Report-law comparison additionally requires the report Lipschitz hypothesis of Theorem 52.1; it is not automatic from  $W_1$  contraction.*

*Proof.* Apply Theorem 52.1 with the true metric state  $S = \mathcal{P}([0, 1])$ , radius  $3/(2n)$ , and diameter one. The dual formula for  $W_1$  gives the assertion for every test; independent selection of a terminal test does not enlarge its bound. The codebook count and the logarithmic substitution are exact. Only an index of a codebook law persists between updates, not its exact unquantized posterior or a numerical residual.  $\square$

**55.1. A nonlinear physical realization.** Let  $B_t$  be independent fair bits and define

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + x/5 + x^2/50, \quad X_t = h_{B_t}(X_{t-1}). \quad (55.3)$$

The observation  $Y_t$  reports  $B_t$  correctly with probability  $3/4$ , independently at each time. A permitted terminal probe  $g : [0, 1] \rightarrow [0, 1]$  returns a Bernoulli variable of probability  $g(X_t)$ ; the tested family is 1-Lipschitz probes and their bounded finite executable continuations. The actual update-report instrument on measures is

$$\mathsf{T}_y \pi = \frac{1}{2} \left\{ \frac{3}{4} (h_y)_{\#} \pi + \frac{1}{4} (h_{1-y})_{\#} \pi \right\}.$$

Its mass is  $1/2$ , independent of the old state. The normalized filter is therefore

$$\Phi_y \pi = \frac{3}{4} (h_y)_{\#} \pi + \frac{1}{4} (h_{1-y})_{\#} \pi. \quad (55.4)$$

**Corollary 55.3** (Nonlinear posterior, explicit register). *The actual measure-valued filter (55.4) is  $6/25$ -contractive in  $W_1$ . Theorem 55.2 applies with  $\eta = 0$  to an exact rational implementation on  $\mathcal{C}_n$ . The model has no fixed-step nonzero common minorizer. For every integer  $D \geq 1$ , the first  $D$  polynomial moments do not determine their own exact next-step update on the space of all prior laws.*

*Proof.* The two images lie respectively in  $[0, 11/50]$  and  $[3/5, 41/50]$ . Each derivative is between  $1/5$  and  $6/25$ . Couple two old laws optimally and use the same branch draw in their mixtures; this gives  $W_1(\Phi_y \pi, \Phi_y \pi') \leq (6/25) W_1(\pi, \pi')$ . For a codebook input, (55.4) has at most  $2n$  atoms, at rational points with denominator dividing  $50n^2$ , with weights multiples of  $1/(4n)$ . Sorted cumulative weights give the exact quantiles, which are rounded back to the fixed grid. Thus no

numerical limit is being hidden in  $\eta = 0$ . The transient list of at most  $2n$  atoms is discarded; its entries and weights have  $O(\log n)$ -bit rational descriptions. The designated persistent object is the single codebook index. Approximate arithmetic instead obeys the separately charged  $\eta$  bound of the theorem.

After  $k$  hidden transitions the law from a fixed  $x$  has finite support  $\{h_v(x) : |v| = k\}$ . For a second initial point  $x'$ , each equality  $h_v(x') = h_u(x)$  excludes at most one  $x'$ , since  $h_v$  is strictly increasing. Choose  $x'$  outside this finite set. The two supports are disjoint, so a measure dominated by all  $k$ -step laws must be zero.

For the moment assertion, the  $D$ th updated moment integrates  $\frac{3}{4}h_y(x)^D + \frac{1}{4}h_{1-y}(x)^D$ , a polynomial of degree  $2D$  with nonzero leading coefficient  $50^{-D}$ . It is not in the span of  $1, x, \dots, x^D$ . Linear algebra on finitely many distinct points therefore supplies a signed measure annihilating this span but not the new polynomial. Its positive and negative parts have equal mass; normalizing them yields two prior probabilities with identical first  $D$  moments and different updated  $D$ th moment. The argument is about ordinary finite moment closure, not the impossibility of a Borel encoding of measures by real numbers.  $\square$

The quotient here is genuinely the posterior law for the specified probe family: bounded Lipschitz functions separate probability measures on a compact interval, and constants do not affect that separation. Updates respect this quotient by (55.4). The advancing reports have the exact state-independent fair law, while a terminal Bernoulli probe has total-variation defect at most  $W_1(\pi_t, \hat{\pi}_t)$ . Thus the same finite presentation defines a causal experiment simulation with a controlled terminal error and an exactly charged register. Multiple probes would update the posterior again; no claim for an unverified enlarged probe protocol is implicit. This example connects positive instruments, predictive quotient, recursive approximation and resource transformation within one model.

## APPENDIX 56. SHARP FINITE-EXPERIMENT BUDGETS

For experiments  $\mathcal{E} = \{P_\theta\}$  and  $\mathcal{F} = \{Q_\theta\}$  with the same parameter set, write

$$\delta(\mathcal{E}, \mathcal{F}) = \inf_K \sup_\theta \|KP_\theta - Q_\theta\|_{\text{TV}}, \quad \Delta(\mathcal{E}, \mathcal{F}) = \max\{\delta(\mathcal{E}, \mathcal{F}), \delta(\mathcal{F}, \mathcal{E})\}.$$

All parameterized laws in this section are Borel families. The infimum is over parameter-independent Markov kernels. These are terminal experiment comparisons in the sense of [45, 46]; only the encoding statistic, not the comparison randomization, is part of the physical data reduction.

Let  $\mathcal{G} = \{N(z, \Gamma) : z \in K\}$ , where  $K \subset \mathbb{R}^r$  is compact and  $\Gamma$  is positive definite. In the lower bound assume that  $K$  contains a nonempty Euclidean ball.

**Theorem 56.1** (Gaussian finite-experiment cardinality). *There are constants  $c, C > 0$ , depending on  $K, \Gamma, r$ , such that*

$$cM^{-2/r} \leq \inf_{\mathcal{F}: |\mathcal{F}| \leq M} \Delta(\mathcal{F}, \mathcal{G}) \leq CM^{-2/r}, \quad M \geq 1. \quad (56.1)$$

*The lower bound ranges over all experiments on at most  $M$  outcomes, not just deterministic Gaussian quantizers. The upper bound is attained in order by a deterministic statistic with  $J^r$  labels, where  $J = \lfloor M^{1/r} \rfloor$ , followed by a specified positive reconstruction kernel. No overflow label is needed.*

### 56.1. The cardinality obstruction.

**Lemma 56.2** (A decision-theoretic lower bound). *Under the preceding assumptions,  $\Delta(\mathcal{F}, \mathcal{G}) \geq cM^{-2/r}$  whenever  $\mathcal{F}$  has at most  $M$  outcomes.*

*Proof.* Restrict the mean parameter to a closed cube  $K_0$  of positive side length contained in  $K$  and put the uniform prior on  $K_0$ . Restriction can only decrease an experiment distance. Use squared-error estimation of  $z$ , with actions restricted to  $K_0$ ; its loss range is bounded by  $L = \text{diam}(K_0)^2$ . Let  $m(y) = \mathbb{E}[z \mid y]$  in this Gaussian experiment. Differentiation under the compact prior integral gives

$$Dm(y) = \text{Cov}(z \mid y)\Gamma^{-1}. \quad (56.2)$$

The posterior density is positive on the cube. Hence its covariance is positive definite, and  $Dm(y)$  is nonsingular for every  $y$ . The inverse function theorem on a small neighborhood of any fixed  $y_0$ , together with the positive continuous marginal density of  $y$ , shows that the law of  $m(y)$  has a density bounded below by a positive constant on some smaller cube in  $\mathbb{R}^r$ .

This implies  $e_M^2(m) \geq c_0 M^{-2/r}$ . To see it directly, put balls of radius  $t = (|Q|/(2Mv_r))^{1/r}$  about the  $M$  proposed centres, where  $Q$  is the smaller cube and  $v_r$  the volume of the unit ball. At least half of  $Q$  remains outside them, and contributes at least a fixed multiple of  $t^2$ . This argument permits arbitrary centres.

Every  $M$ -outcome garbling  $S$  of the Gaussian observation therefore has Bayes risk at least  $B_{\mathcal{G}} + c_0 M^{-2/r}$ , by conditional orthogonality. If  $\Delta(\mathcal{F}, \mathcal{G}) < \epsilon$ , choose a kernel  $K$  from  $\mathcal{G}$  to  $\mathcal{F}$  with error below  $\epsilon$ . The experiment  $K\mathcal{G}$  has at most  $M$  outputs, and bounded-loss comparison gives

$$B_{K\mathcal{G}} \leq B_{\mathcal{F}} + L\epsilon \leq B_{\mathcal{G}} + 2L\epsilon.$$

The second inequality uses a kernel in the reverse direction. Comparing these bounds and taking the infimum over  $\epsilon$  proves the result. Randomized actions do not improve a Bayes squared-error minimum. Notice that  $\mathcal{F}$  itself was not assumed to be a Gaussian garbling.  $\square$

## 56.2. Positive second-order reconstruction. Put

$$\tau(x) = \frac{1}{2} + \pi^{-1} \arctan x, \quad u = \tau^{\otimes r}(y) \in (0, 1)^r. \quad (56.3)$$

Let  $p_z$  be the transformed Gaussian density. If  $x_j = \tan \pi(u_j - 1/2)$ , then

$$p_z(u) = \phi_{\Gamma}(x - z) \prod_{j=1}^r \pi(1 + x_j^2). \quad (56.4)$$

It extends by zero across every face to a smooth periodic density. Indeed each fixed-order derivative is a polynomial in  $x$  times the Gaussian density, which tends to zero faster than any polynomial as a face is approached. Uniformity holds for  $z$  in the fixed compact set. For use below define the finite constants

$$H_2 = \sup_{z, u} \|D^2 p_z(u)\|_{\text{op}}, \quad H_0 = \max_j \sup_{z, u_j} p_{z, j}(u_j), \quad (56.5)$$

where  $p_{z, j}$  is a transformed marginal density. These suprema, or any upper bounds obtained from (56.4), fix all the reconstruction and boundary constants; they do not depend on  $M$  or sample size. For instance, differentiating twice introduces only polynomials and the eigenvalue bounds of  $\Gamma$ , so bounded mean norm and fixed covariance conditioning suffice.

Partition the flat unit torus into  $J^r$  cubes of side  $h = 1/J$ , with centres  $c_i$ , and write  $Q_J(u)$  for the cube label. For  $J \geq 2$  let  $\Phi_i$  be the periodic tensor product of the piecewise-linear hat functions with value one at  $c_i$ , zero at its neighboring mesh centres, and support within one mesh step in each coordinate. Then

$$\Phi_i \geq 0, \quad \int \Phi_i = h^r, \quad \sum_i \Phi_i(u) = 1.$$

On each mesh interval the neighbouring centres reproduce the affine coordinate, using local periodic lifts across the seam. Define the reconstruction from label  $i$  to have density  $h^{-r} \Phi_i(u)$ , then apply  $\tau^{-1}$  coordinatewise. Values on the faces have probability zero.

**Lemma 56.3** (A quadratic reconstruction estimate). *Uniformly in  $z \in K$ , this kernel  $H_J$  satisfies*

$$\|H_J Q_J \tau^{\otimes r} N(z, \Gamma) - N(z, \Gamma)\|_{\text{TV}} \leq \frac{r H_2}{12 J^2}. \quad (56.6)$$

*Proof.* Let  $p_i$  be the probability of cube  $i$ . Symmetric Taylor expansion about its centre gives  $|p_i/h^r - p_z(c_i)| \leq r H_2 h^2/24$ . The linear interpolant  $I_h p_z = \sum_i p_z(c_i) \Phi_i$  has error at most  $r H_2 h^2/8$ : its weights reproduce constants and affine functions and the sum of their coordinate variances is at most  $r h^2/4$ . The second Taylor remainder is half this variance times  $H_2$ . Both calculations hold across the seam because  $p_z$  has a smooth periodic extension. The reconstructed density is  $\sum_i (p_i/h^r) \Phi_i$ . Nonnegativity and the partition of unity give supremum error at most  $r H_2 h^2/6$ . Integrating over volume one and using the factor  $1/2$  for total variation proves the claim. Inverting (56.3) preserves this distance.  $\square$

*Proof of Theorem 56.1.* Lemma 56.2 gives the lower bound. The deterministic statistic  $Q_J \tau^{\otimes r}$  has  $J^r$  outputs, so the forward deficiency is zero; Lemma 56.3 gives the reverse deficiency. For  $M \geq 2^r$ ,  $J = \lfloor M^{1/r} \rfloor \geq M^{1/r}/2$ . For the remaining finite range use any one-outcome experiment and the trivial bound  $\Delta \leq 1$ , increasing  $C$ . The reconstruction uses positive probabilities; it is not a signed interpolation operator.  $\square$

The use of a compactified grid avoids a growing Gaussian overflow radius. The use of hats rather than constant cells cancels the first Taylor term. Both choices are necessary for the stated proof of a logarithm-free quadratic budget rate.

#### APPENDIX 57. A SHARP BUDGET FOR THE PHYSICAL COUNT-TO-CONTACT EXPERIMENT

We now specify the actual local protocol, following the known-mark experiment of A2 [42]. Fix  $\lambda_0 \in (0, \infty)^m$ , a matrix  $B \in \mathbb{R}^{m \times d}$ , a compact set  $H \subset \mathbb{R}^d$ , and a full-row-rank matrix  $C \in \mathbb{R}^{r \times m}$ . Observe independent counts

$$K_{N,j} \sim \text{Pois}(N \lambda_{0,j} + \sqrt{N}(Bh)_j), \quad h \in H. \quad (57.1)$$

Write  $\Lambda = \text{diag } \lambda_0$ ,  $A = CB$ , and  $\Gamma = C \Lambda C^T$ . The contact score and limiting experiment are

$$S_N = C(K_N - N \lambda_0)/\sqrt{N}, \quad \mathcal{G} = \{N(Ah, \Gamma) : h \in H\}.$$

The physical finite statistic is

$$T_{N,J} = Q_J(\tau^{\otimes r}(S_N)), \quad |T_{N,J}| \leq J^r. \quad (57.2)$$

There is no physical jitter, no retained real score, and no omitted overflow event. Every real score belongs to one of the  $J^r$  cells of the open compactified cube, with deterministic boundary tie-breaking.

**Theorem 57.1** (Joint sample and register law). *For the fixed protocol (57.1), there are  $N_0, C_* < \infty$  such that uniformly for  $N \geq N_0$  and  $J \geq 2$ ,*

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_* \left( N^{-1/2} + \frac{J}{\sqrt{N}} + J^{-2} \right), \quad (57.3)$$

where  $\mathcal{E}_{N,J} = \{\text{Law}_h(T_{N,J}) : h \in H\}$ . If  $AH$  contains an  $r$ -dimensional ball, every experiment with at most  $M$  outputs has distance from  $\mathcal{G}$  at least  $c_* M^{-2/r}$ . Consequently the statistic with  $J = \lfloor M^{1/r} \rfloor$  satisfies

$$c_* M^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_{*,K} M^{-2/r}, \quad 2^r \leq M \leq K N^{r/6}, \quad (57.4)$$

for every fixed  $K < \infty$ . In particular  $M \asymp N^{r/6}$  gives distance of order  $N^{-1/3}$ ; achieving distance of order  $N^{-1/5}$  requires and is achieved by a budget of order  $N^{r/10}$ . All constants may depend on the fixed matrices, compact parameter set and covariance conditioning. No uniformity at rank degeneration or unknown marks is asserted.

We prove the uniform estimates in five steps, keeping comparison randomization separate from (57.2).

**Lemma 57.2** (Jittered Poisson local normal comparison). *There is an absolute  $C_P < \infty$  such that for  $\mu \geq 1$ , independent  $U \sim \text{Unif}(-1/2, 1/2)$ , and  $K \sim \text{Pois}(\mu)$ ,*

$$\|\text{Law}(K + U) - N(\mu, \mu)\|_{\text{TV}} \leq C_P \mu^{-1/2}. \quad (57.5)$$

*Proof.* It suffices to consider  $\mu \geq 64$ , increasing the constant for the remaining range. The jittered density is  $p_\mu(k) = e^{-\mu} \mu^k / k!$  on  $[k - 1/2, k + 1/2)$ ,  $k \geq 0$ . On the central cells  $|k - \mu| \leq \mu^{2/3}/4$ , put  $t = k - \mu$ ,  $x = k + u$ ,  $|u| \leq 1/2$ , and  $z = (x - \mu)/\sqrt{\mu}$ . Stirling's formula with remainder  $0 < R_k < 1/(12k)$  gives

$$\log \frac{p_\mu(k)}{\phi_\mu(x)} = -\frac{1}{2} \log(k/\mu) - R_k - \{k \log(k/\mu) - k + \mu\} + \frac{(x - \mu)^2}{2\mu},$$

where  $\phi_\mu$  is the  $N(\mu, \mu)$  density. For  $|t| \leq \mu/2$ , Taylor's theorem applied to  $(1 + v) \log(1 + v) - (1 + v) + 1$  gives remainder at most  $|t|^3/\mu^2$  after its quadratic term. Also  $|\log(1 + t/\mu)| \leq 2|t|/\mu$  and  $|(x - \mu)^2 - t^2| \leq |t| + 1/4$ . Substitution therefore gives the convenient bound

$$\left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 32\mu^{-1/2}(1 + |z|^3), \quad \left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 64. \quad (57.6)$$

The constants are deliberately loose. For example  $|z| \leq \mu^{1/6}/4 + 1/(2\sqrt{\mu})$  on these cells verifies the second bound directly. The inequality  $|e^v - 1| \leq e^{|v|}|v|$  now bounds the integrated central density error by  $32e^{64}\mu^{-1/2}(1 + \mathbb{E}|G|^3)$ ,  $G \sim N(0, 1)$ .

For completeness the omitted tails are exponentially smaller. For  $|s| \leq 1$ , the centered Poisson log moment generating function is  $\mu(e^s - 1 - s) \leq \mu s^2$ . Exponential Markov with  $s = \mu^{-1/3}/16$ , and its negative, bounds the probability outside the central cells, allowing the half-unit jitter margin, by  $2 \exp(-\mu^{1/3}/128)$  after adjusting an absolute factor for  $64 \leq \mu$  in the bounded initial interval. The Gaussian bound follows by its moment generating function. Thus an absolute choice such as  $C_P = 10^8 e^{64}$  dominates the central error, the tails and the finite initial range. This proves a uniform  $\mu^{-1/2}$  bound rather than an unquantified local-limit assertion.  $\square$

**Lemma 57.3** (Variance replacement and projected comparison). *Let  $U_j$  be independent uniform half-cell jitters, used only in this proof, and set  $S_N^o = S_N + CU/\sqrt{N}$ . Uniformly for  $h \in H$ ,*

$$\|\text{Law}_h(S_N^o) - N(Ah, \Gamma)\|_{\text{TV}} \leq C_0 N^{-1/2}. \quad (57.7)$$

*One may choose  $C_0$  as a function only of  $m$ , the minimum of  $\lambda_0$ ,  $\max_{h \in H} \|Bh\|$ , and fixed matrix norms.*

*Proof.* Put  $b_* = \max_{h \in H} \|Bh\|_\infty$  and  $\lambda_* = \min_j \lambda_{0,j}$ . Choose  $N_0 \geq \max(128/\lambda_*, 4b_*^2/\lambda_*^2)$ . Then each count mean is at least  $N\lambda_*/2$ . Product comparison using Lemma 57.2, followed by centering and scaling, gives distance at most  $mC_P\sqrt{2/\lambda_*} N^{-1/2}$  from the normal vector with mean  $Bh$  and covariance  $\Lambda + N^{-1/2} \text{diag}(Bh)$ .

For the replacement of this covariance by  $\Lambda$ , the relative variance eigenvalues are  $1 + t_j$ , with  $|t_j| \leq b_*/(\lambda_*\sqrt{N}) \leq 1/2$ . The exact Gaussian relative entropy is  $\frac{1}{2} \sum_j \{t_j - \log(1 + t_j)\} \leq \sum_j t_j^2$ . The elementary entropy-variation inequality  $\|P - Q\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P\|Q)/2}$  gives a further bound  $\sqrt{m/2} b_*/(\lambda_*\sqrt{N})$ . Taking  $C_0 = mC_P\sqrt{2/\lambda_*} + \sqrt{m/2} b_*/\lambda_*$  is sufficient. Projection by  $C$  contracts total variation and sends the fixed Gaussian covariance to  $\Gamma$ , proving the assertion.  $\square$

**Lemma 57.4** (Deterministic cell crossing). *Let  $H_0$  be the transformed marginal bound in (56.5), now for the means  $AH$ , and put  $d_N = \|C\|_{\infty,1}/(2\pi\sqrt{N})$ . Then*

$$\sup_{h \in H} \|\text{Law}_h(T_{N,J}) - Q_J \tau^{\otimes r} N(Ah, \Gamma)\|_{\text{TV}} \leq 2C_0 N^{-1/2} + 2rH_0 J d_N. \quad (57.8)$$

Here  $\|C\|_{\infty,1}$  is the maximum absolute row sum.

*Proof.* The coupled scores satisfy  $\|S_N - S_N^o\|_{\infty} \leq \|C\|_{\infty,1}/(2\sqrt{N})$ . Since  $|\tau'| \leq 1/\pi$ , their compactified coordinates differ by at most  $d_N$ . Distinct grid labels require a coordinate of the jittered score to be within  $d_N$  of a grid boundary. On the torus there are  $J$  boundaries per coordinate; under the transformed Gaussian their total probability is at most  $2rH_0 J d_N$ , or its truncation by one. Replacing the jittered law by that Gaussian adds at most  $C_0 N^{-1/2}$ , by (57.7). The coupling inequality then compares theunjittered and jittered labels. One more data-processing comparison of the jittered labels to the Gaussian labels adds the second  $C_0 N^{-1/2}$ . Boundary tie events are already in the strips. This argument never asserts that the unjittered real score has a Gaussian law in total variation.  $\square$

*Proof of Theorem 57.1.* Use the deterministic Gaussian-to-cell kernel for the direction  $\mathcal{G} \rightarrow \mathcal{E}_{N,J}$ . Lemma 57.4 bounds its error by  $b_{N,J} = 2C_0 N^{-1/2} + 2rH_0 J d_N$ . For the reverse direction, use the positive kernel  $H_J$  from Lemma 56.3. Kernel contraction and the triangle inequality give

$$\sup_h \|H_J \text{Law}_h(T_{N,J}) - N(Ah, \Gamma)\|_{\text{TV}} \leq b_{N,J} + rH_2/(12J^2).$$

This proves (57.3), with, for example,  $C_* = 2C_0 + rH_0\|C\|_{\infty,1}/\pi + rH_2/12$  increased to one. Every kernel is parameter independent. Jitter was introduced only to bound a difference of laws, not to modify the physical statistic.

If  $AH$  contains a ball, restrict to an affine parameter section over a smaller mean cube and apply Lemma 56.2. More generally one may simply index the Gaussian family by its mean and choose a Borel representative preimage for each mean in the cube (the compact fibres of the continuous map on  $H$  admit a measurable section); the lower comparison uses only that subfamily. For  $J = \lfloor M^{1/r} \rfloor$  and  $M \geq 2^r$ ,  $J^{-2} \leq 4M^{-2/r}$ . If  $M \leq KN^{r/6}$ , then  $M^{3/r}/\sqrt{N} \leq K^{3/r}$ ; hence  $J/\sqrt{N}$  and  $N^{-1/2}$  are also at most fixed multiples of  $M^{-2/r}$ . This proves the window. Taking  $M$  respectively of order  $N^{r/6}$  and  $N^{r/10}$  gives the two rates. Conversely, an error at most  $CN^{-1/5}$  forces  $M \geq cN^{r/10}$  by the universal cardinality lower bound.  $\square$

**57.1. The A2 matrices and the scope of the consequence.** In A2's known-mark local protocol, the moment derivative  $D$  has  $\Sigma = D\Lambda D^T > 0$ , and its calibrated exposure direction is  $B = \Lambda D^T \Sigma^{-1}$ , so  $DB = I$ . If  $L$  maps to the contact coordinates and  $F$  is an orthonormal frame of its range, the matrices above are

$$C = F^T L D, \quad A = CB = F^T L, \quad \Gamma = C\Lambda C^T > 0.$$

Here  $A$  has full row rank  $r$ . A compact local parameter neighborhood therefore has an image containing a mean ball. Both sides of Theorem 57.1 apply to that actual protocol. This invokes a structural experiment-cardinality theorem verbatim, not just an analogy between a root singularity and a contact observation.

The result is a matched budget theorem in the displayed joint window. It is not a lower bound saying that  $N^{-1/3}$  is the best sample-size comparison achievable with arbitrarily many labels. It also does not assign a stochastic noise law to a conductor scheme or establish uniformity at a rank-changing physical stratum. The geometric source, the exposure protocol, the finite statistic and the comparison kernels are distinct typed objects throughout.

## PRESERVED HISTORICAL INTRODUCTIONS AND SUPPORTING BODIES

## APPENDIX 58. INTRODUCTION

Statistical comparison asks which decisions can be transferred from one experiment to another. When observations arrive sequentially and the decision procedure has a fixed retained register, comparison has an additional constraint: the randomization used to transfer an experiment must itself obey the information and memory interface. A hidden persistent selector, an exposed selector, and transient private randomness define different feasible sets. Equality of unpriced experiments does not imply equality of their fixed-register risk bodies.

We develop comparison at the original resource signature. The central finite object is the image of a product of stochastic-row simplexes under the map taking an executable causal simulator to its marked feedback laws. This image is generally nonconvex. Convex separation of its convex hull can therefore lose precisely the private-resource obstruction one wishes to measure. Our main finite result replaces a fixed mixture of tests by polynomial proof functions on the simulator rows. The resulting strictly positive inequalities admit finite exact certificates without adding a persistent selector to the simulator.

The same lower certificates can be attached to genuine physical acquisitions. This requires more than approximating an observable in norm: the physical approximation must preserve the actual mark, respect feedback, produce finite reports, and rationalize their joint laws. We retain each of those steps and prove an explicit realization in which a nonconserved collision-compatible observable has both nonzero generator residual and a positive exactly characterized marked deficiency. The statistical and microscopic conclusions are joined at the same register and selector budgets.

**58.1. The main theorem and its components.** For a fixed finite marked experiment and resource signature  $b$ , write

$$\delta_b = \min_{q \in \mathcal{P}_b} \max_i f_i(q).$$

Here  $q$  is an executable stochastic-row table, the  $f_i$  are the finite marked event discrepancies under deterministic feedback, and  $\mathcal{P}_b$  is a product of row simplexes. Visible selectors remain part of the target reference as well as the simulated joint law. An onto polynomial stick-breaking parameterization  $q = q(x)$  places the domain on  $[0, 1]^D$ . With  $h_i = f_i \circ q$  and  $g_i = (1 + h_i)/2$ , including the empty test, we prove

$$\delta_b = \sup_{p \geq 1} \min_{x \in [0, 1]^D} \frac{\sum_i g_i(x)^p h_i(x)}{\sum_i g_i(x)^p}. \quad (58.1)$$

The minima increase to  $\delta_b$  with error at most  $2 \log m / (p + 1)$ , where  $m$  is the number of tests. For rational  $L$ , the strict inequality  $L < \delta_b$  is equivalent to a strictly positive finite tensor Bernstein expansion of  $\sum_i g_i^p(h_i - L)$  for some finite  $p$  and degree. The proof includes an explicit degree bound in terms of the monomial coefficients and the positivity margin. This is Theorem 6.1.

The weights in (58.1) depend on a candidate table; they are not independent test probabilities in an exchanged minimax. A two-time example has private width-one deficiency  $\sqrt{2} - 1$ , hidden-two-selector deficiency zero, and visible-selector deficiency  $\sqrt{2} - 1$  at every finite selector size. Every fixed test mixture has minimum at most zero there. Thus the distinction is visible already in a concrete exact problem, not just in a formal choice of dual notation. The source supplies a strictly positive rational Bernstein certificate for the private lower bound  $2/5$ .

For physical acquisitions, Theorem 16.1 combines the polynomial certificate with stateless marked presentations. A finite lower-upper interval  $[L, U]$  gives the original physical interval

$$[\max\{0, L - a - c\}, \min\{1, U + a + c\}],$$

where  $a, c$  include analytic approximation, report quantization and joint-prefix rationalization. There is no change in the resource signature. Under the stated effective data hypotheses these intervals converge. The finite search and the analytic presentation search are distinguished; neither an abstract graph core nor a decimal transition table is called a rigorous physical input.

Our explicit microscopic realization uses  $N \geq 2$  interacting hard spheres in equilibrium on a torus. For  $\vartheta = (2\pi/\ell) \sum_i q_{i,1}$  and normalized total momentum  $X$ , the nonconserved observable  $f = \cos \vartheta$  obeys  $U(t)f = \cos(\vartheta + \omega X t)$  with  $\omega = 2\pi\sqrt{N}/\ell$ . All generator powers and their Gram entries are computed on the actual collision-compatible domain. Reversal interventions give exactly  $k + 1$  distinct mean functions among  $2^k$  words. A Taylor presentation then has a computable error tending to zero, uniformly over consumer width. The actual initial sign mark yields an exact physical deficiency series; for the parameters of Corollary 15.5 its value lies strictly between 0.205809122887 and 0.205809122889, while the normalized one-dimensional residual is  $2\pi^2/25 > 0$ . The same example supplies both quantities.

Finally, for a bounded marked payoff of oscillation  $M$ , a variation error  $\varepsilon$  changes a normalized logarithmic value at scale  $\beta$  by at most  $\beta^{-1} \log(1 + \varepsilon(e^{\beta M} - 1))$ . Theorem 16.2 proves sharpness and transports optimized common-policy values. This identifies the additional accuracy needed at an exponential scale rather than asserting an unproved large-deviation conclusion from unscaled comparison.

**58.2. Classical ingredients and the comparison of scope.** Blackwell–Le Cam comparison and its decision-theoretic formulations provide the unpriced background [1, 47, 46]. Filtered comparison is a direct antecedent, not a remote analogy. Weisshaupt’s generalized filtered randomization theorem uses a compact convex class of filtration-preserving stochastic operators and a separation/minimax argument [68]. Our private feasible image is not that convex class. The two-time example above states exactly what ordinary constant convex testing loses, and the positive polynomial theorem states exactly what replaces it. We do not claim that polynomial positivity, power-mean approximation, or Bernstein basis elevation is new; their elementary proofs are included to expose the hypotheses on which the new causal application depends.

The retained existence theory credits weak-star policy topology and strategic-measure approaches [55, 56]. Common-information and finite-memory formulations are also prior structural tools [54, 53]. State equivalence, incomplete-machine minimization, and weighted-automaton methods are relevant to the retained exact task and synthesis results [49, 70, 58]. Those results are not relabelled as new polynomial duality. The accompanying comparison records which original sources were inspected and which proof-level priority questions remain unverified, including the Norberg original. The present novelty assertion is the stated resource-sensitive theorem and its physical realization, not an assertion of an exhaustive priority search across every form of filtered comparison.

The microscopic input is the classical almost-everywhere hard-sphere flow [63]. The new collective calculation uses a solvable mode of that interacting system; it is not a solution of its kinetic limit. This distinction is mathematical: fixed-particle  $L^2$  control of noisy observations does not give a particle-uniform BBGKY corrector on action sublevels. The program is represented as a typed dependency graph, with theorem implications, operational adapters and independent primary branches kept distinct.

**58.3. Organization.** We first develop intrinsic finite causal deficiency, its local and behavior-dimension certificates, and the polynomial testing theorem. We retain the endogenous task and stationary regenerative results, the active graph-domain comparison, finite marked presentations, enriched residual convergence and constructive collision data. The last sections prove the nonconserved collective realization and the joint physical certificate theorem. Every mathematical section of the preceding canonical article is retained. The complete-development view additionally contains

the earlier mathematical bodies and historical introductions; no earlier theorem is removed to create the new comparison structure.

## APPENDIX 59. INTRODUCTION

Comparison of statistical experiments becomes a different problem when the comparison channel must itself run with a prescribed finite memory. A causal channel may preserve the filtration and nevertheless require more persistent state than the experiment's consumer possesses. Moreover, randomizing among complete designs can enlarge the achievable set even when each design uses the same number of labels. The information exposed by the random selector, and the cost of retaining its value, are therefore part of the mathematical object being compared.

We study marked causal experiments with this resource structure. A mark is an actual random variable coupled to the source, not an independent copy with the same marginal law. The intrinsic deficiency is the smallest worst-feedback variation error among simulators at the declared state and randomization budget. Private, hidden-selector and visible-selector budgets lead to distinct feasible sets. The private image is generally nonconvex. Its exact finite certificates must consequently test that image itself rather than replace it by the convex hull of deterministic implementations.

The main result is a same-budget physical certification theorem. It starts with an active microscopic acquisition, passes through a finite trial presentation, quantizes its reports, constructs a coherent rational marked experiment, and transfers finite private certificates back to the original physical experiment. Each presentation map is stateless. The register cost paid by the final physical simulator is therefore the original finite budget, without an unpriced model-selection index. The analytic error and the finite optimization error are both explicit.

**59.1. The principal theorem.** Let  $b$  be a fixed private or finite-selector signature. For a pair of finite-horizon active Gaussian experiments  $E, F$ , write  $\delta_b(E, F)$  for their intrinsic marked feedback deficiency. Let  $E_{\mathbb{Q}}, F_{\mathbb{Q}}$  be finite rational marked presentations constructed by the procedures of this paper. Suppose their two-sided presentation errors are at most  $a, c$ , respectively. If finite local testing and an executable same-budget row table give bounds  $L, U$  for  $\delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})$ , then

$$\max\{0, L - a - c\} \leq \delta_b(E, F) \leq \min\{1, U + a + c\}. \quad (59.1)$$

Theorem 13.1 proves this statement and a convergent construction under effective graph and marked-integration data. The lower bound remains valid without assuming attainment for the original nonfinite experiment. The upper bound comes with an actual physical simulator. The resulting intervals can be nested and made arbitrarily short. This is a finite-horizon effective theorem with specified analytic inputs, not a computability assertion for arbitrary measurable models.

There are three substantive steps. First, report truncation and histogram reconstruction are compared before controller optimization, using the original preparation and a coupled actual mark. This avoids unjustified bounds on feedback-conditioned densities. Second, rationalization is performed on joint marked prefix masses. A weighted normalization inequality cancels the small-prefix denominator. It gives a coherent rational causal tree and a two-sided feedback error without deciding which original prefix probabilities vanish. Third, piecewise affine graph-core paths give enriched residual bounds whose values themselves converge to zero. The intervention is applied after this domain-controlled path calculation; no invariance of the generator domain under kicks is assumed.

Two further results connect the finite geometry to state complexity. Theorem 2.10 bounds the number of distinct testing witnesses by the affine dimension of the attainable behavior image and the strict error gap. It keeps the original nonconvex private image; its bound is on tests, not the number of policy cells or computational running time. Theorem 7.7 gives packing lower bounds and covering upper bounds for the positive-error width of a finite family of marked prediction tasks at a

source-consuming restart. Completing that family recovers the universal marked continuation state at zero error. Causal update constraints remain necessary for simultaneous checkpoint realizations.

The effective data are not left as an integration oracle. Theorem 12.2 constructs an enumerable microscopic graph core and verifies its Gram and marked probability operations by certified full-measure collision-chart exhaustion. It applies to fixed-particle hard spheres with algebraic velocity kicks and bounded computable input data, without requiring conserved observables. No particle-number-uniform rate is inferred.

The physical data also admit explicit closed-form instances. Theorem 14.1 gives an exact deficiency  $1/4$  for a continuous Gaussian acquisition of a conserved momentum coordinate in an actively intervened colliding hard-sphere system. Its trial Gram data are rational and exact. Proposition 14.3 then gives a collision-compatible localized position observable with a genuinely nonzero graph residual  $R = 192$  and a rigorously bounded physical error. The conserved-sector family admits the complete finite-rational procedure at every fixed horizon, including rigorous scalar integration.

**59.2. Classical antecedents and the distinction of resources.** The general comparison of experiments and its risk interpretation are classical [1, 47]. Filtered randomization criteria predate the present finite-state formulation [48, 68]. In particular, Weisshaupt’s Lemma 1 and Theorem 5 use a compact convex set of filtered stochastic operators with values in finitely additive measures; their proof reduces the filtered statement to convex separation. They do not price a fixed persistent register. That comparison cannot be applied to our generally nonconvex private image by silently convexifying it. We retain the classical result as an antecedent, not a theorem displaced by this paper.

Common-information control and real-time coding provide essential information-state precedents [53, 54]. Compatible-cover language has an antecedent in incomplete-machine minimization [70]. The distinction here is the stated marked, resource-constrained comparison problem and its error transfer, not the invention of dynamic programming or compatible covers. A complete proof-level priority comparison with the original Norberg and Paull–Unger texts remains to be established; we make no assertion of exhaustive priority over those works.

Separate-affine vertex interpolation, subdivision, normed-space packing, real quantifier elimination [69], variation of constants, and graph-core approximation [67] are also classical. Their roles in the proofs are identified where used. The contribution claimed here is the original-budget physical certificate chain, including its actual marked feedback bridges, zero-prefix-stable rationalization and convergent enriched certificates, together with the stated resource-image and task-resolution consequences. None of these claims relies on counting source files or passing a diagnostic suite as a mathematical argument.

**59.3. Organization and the program of applications.** Section 2 develops intrinsic resource comparison, its finite local certificates and the new behavior-dimension witnesses. Section 7 treats endogenous common decisions and positive-error restart complexity. Section 8 retains the stationary regenerative theorem with its actual physical and register reset. The active microscopic construction and the original Galerkin residual inequalities follow. Sections 10 and 11 supply the two missing approximation steps; Section 12 constructs effective microscopic data; Section 13 proves the joint theorem and Section 14 instantiates certified physical data.

The associated eleven-component program is a dependency graph. Its primary geometric A2 branch is independent of the present comparison theory; an acquisition adapter is a different relation from theorem dependence. The present microscopic B4 edge proves fixed-horizon marked certification. It does not turn  $L^2$  transport estimates into the particle-number-uniform nonlinear action-sublevel corrector of the historical kinetic program. Those larger targets and their derivations

are preserved in the source history and accompanying complete development. This article states precisely the new implications, without using repository order as a substitute for proof.

## APPENDIX 60. INTRODUCTION

A causal simulation has two outputs: the report delivered to its consumer and the state that must survive until the next report. Classical comparison records the first and may leave the second unpriced. We keep both. The quantity of interest is the least error attainable by one task- and parameter-independent simulator at a prescribed persistent-state budget. The error compares the entire feedback transcript together with its actual latent mark. In particular, equality of report marginals does not replace equality of a marked experiment.

This distinction leads to three optimization problems. A private simulator retains only its declared state and uses fresh randomness. A hidden-selector simulator first selects a policy but does not reveal that choice to the consumer. A visible-selector simulator reveals it and includes it in the compared joint interface. Hidden convexification has a finite testing dual, but its realization may require storing a selector index. At a fixed private width this is not a harmless convention: a delayed three-symbol channel has zero hidden-selector deficiency and strictly positive private deficiency at width two. If the selector is visible, its optimized zero set is again the private zero set.

**60.1. The main statement.** For finite marked causal experiments  $E, F$ , prescribed widths  $K = (K_0, \dots, K_T)$  and a fixed finite seed signature, let  $\delta_K(E, F)$  be the infimum of the marked feedback error over all admissible simulators. The definitions and exact signature-dependent variational formulas are given in Section 2. The central result can be stated as follows.

At each fixed private or finite-selector budget, intrinsic causal deficiency has a complete family of finite local testing certificates and original-budget upper witnesses. These bounds converge with an explicit causal modulus. They compose across stateless marked presentation changes without increasing the budget, and active microscopic approximation supplies such changes with finite graph-residual bounds.

More precisely, Theorem 2.7 constructs rational lower and upper bounds  $L_M \leq \delta_K \leq U_M$  from clipped-simplex policy cells. In the private case, with at most  $d_t$  joint output–state labels in a row at time  $t$ , their gap is at most

$$1 - \prod_{t=0}^T (1 - \min\{1, d_t/(2M)\}),$$

with a zero contribution from one-point rows. The modulus depends on depth and row alphabets, not on the number of histories. Enumeration of the cells can nevertheless be large; no polynomial-time synthesis conclusion is drawn. The lower bound assigns a marked feedback test to each cell. The upper bound is a single admissible row table at the original budget. This is a nonlinear testing converse, not an unpriced application of convex separation. For dyadic  $M$  the bounds are monotone and converge. Strict feasible and infeasible thresholds have finite witnesses; exact boundary decisions use classical real algebraic elimination.

The optimized spectra have the composition law

$$\delta_{KL}(E, G) \leq \delta_K(E, F) + \delta_L(F, G),$$

with selector costs multiplied when present. Theorem 2.9 then transfers an entire intrinsic interval across two-sided stateless presentation changes. Its lower statement excludes all original-budget simulators, even when the original carriers are nonfinite. Its upper statement constructs a simulator

by composing the rational row table with the bridge channels. Thus the scope of an assertion is visible in both its error and its persistent resources.

The other parts of the article give operational and analytic realizations of this comparison. Theorem 7.3 identifies the exact common-encoder face for endogenously controlled finite tasks by reachable compatible covers, and relates the lossy problem to Bellman regrets. Theorem 8.1 treats nonsummable average criteria for stationary finite-register controllers with actual simultaneous physical and register resets; deterministic or singular physical transitions are permitted between resets. Theorem 9.2 supplies active hard-sphere interventions and convergent graph-core approximations. Theorem 9.5 strengthens their finite-horizon comparison by an a posteriori bound involving only graph and intervention Gram data. Its root-sum-square word estimate retains the actual mark and precedes optimization over every finite consumer width. A finite age cutoff adds an explicit geometric remainder for average risks.

**60.2. Comparison with classical results.** Blackwell–Le Cam comparison, convex minimax, and finite-dimensional convex representation are classical ingredients [1, 47]. Filtered comparison is not new as a subject. Weisshaupt’s filtered stochastic-operator compactness lemma and randomization theorem [68, Lemma 1 and Theorem 5] give an explicit nearby convex comparison result with a finitely additive target space. They do not impose the persistent-state budget considered here. We distinguish that retrieved statement from Norberg’s closely related filtered comparison paper [48]; the full original-source comparison with the latter is not claimed complete.

The common-information dynamic program of Nayyar–Mahajan–Teneketzis [53, Theorem 3 and Corollary 5] provides a classical coordinator formulation, including a finite-memory specialization. The compatible-cover device also has classical incomplete-machine antecedents [70]. Our exact task theorem adds the common encoder, policy-dependent reachable support and Bellman regret conditions; the cover terminology by itself is not a novelty claim. Real quantifier elimination is used only for an exact decision corollary [69]. Similarly, graph-core semigroup approximation and the well-defined microscopic hard-sphere flow are credited to their classical sources [67, 63]. The residual proof below uses a finite-dimensional variation-of-constants formula and does not propose a new abstract semigroup approximation theorem.

The mathematical distinction retained after those ingredients are removed is the simultaneous original-budget testing and realization statement: local feedback tests certify the private feasible set that a convex filtered comparison can enlarge, and the same optimized quantity accepts active microscopic error bounds without a hidden consumer-memory charge. The finite delayed-channel separation makes that distinction explicit. It is not a claim that all ingredients, or their entire conjunction, have an exhaustively certified priority. The source-level literature audit records which original theorems were inspected and which remain to be compared.

**60.3. Organization and scope.** Section 2 develops the signature-indexed deficiency, its local certificates and its composition-stability theorem. The next sections treat endogenous tasks, regenerative average criteria, and active microscopic realization. All five sections include the proofs they use. The complete development retains the preceding mathematical bodies, including dominated finite-prefix and summable infinite-time duality, marked continuation quotients, exact synthesis, operator memory, Gaussian resource laws, and the historical model interfaces.

The physical results fix particle number, action alphabet, observation noise, and the initial-mark channel. They do not imply the historical nonlinear kinetic action-sublevel corrector or a Boltzmann–Grad limit. Likewise an intrinsic lower certificate is a mathematical inequality over a specified finite model, not an assertion that floating-point Gram entries certify a physical system. The eleven-component program is a mathematical dependency graph with adapters and independent lines; repository chronology and its separately frozen audit are not hypotheses of any theorem in this article.

## APPENDIX 61. INTRODUCTION

An exact simulation of a statistical experiment need not preserve the memory budget of a decision maker. Conversely, convexifying a collection of finite-state simulations may introduce a persistent choice whose storage was absent from each member of the collection. The distinction is particularly sharp when the choice is revealed to the decision maker: a cancellation in a mixture of report laws then need not be a cancellation in the joint law of the choice and the reports. These observations suggest that the error, the retained state, and the visibility of randomization should be coordinates of a single comparison object.

We make this object explicit. Given marked controlled experiments  $E, F$ , a width profile  $K = (K_0, \dots, K_T)$ , and a seed signature, the *intrinsic causal deficiency* is the infimum of the marked feedback simulation error over all simulators with that signature. The mark is an actual hidden variable of the experiment, not a newly drawn variable having a convenient marginal. A simulator is independent of the unknown parameter and of the eventual decision task. Controls are passed through at the declared time, and the simulator can retain only its specified state. These restrictions are part of the object being compared.

**61.1. The main theorem.** For finite experiments, write  $\mathcal{K}_K$  for the product of the simulator's stochastic row simplexes and  $b(q)$  for the vector of all marked transcript probabilities against deterministic feedback trees. The map  $q \mapsto b(q)$  is polynomial, but its image need not be convex. Put

$$a_K = \dim \operatorname{aff}\{b(q) : q \in \mathcal{K}_K\}.$$

Theorem 2.2 proves the following statement. *The hidden-mixture deficiency has a complete dual in weighted causal tests and an exact realization using at most  $a_K + 1$  selector values; this realization costs at most  $(a_K + 1)K$  private states. In contrast, arbitrary visible selection has zero deficiency exactly when a private  $K$ -state simulator has zero deficiency. All three signatures have multiplicative state-cost composition and uniform decision transport.* The private optimization and every fixed finite selector problem are attained. Thus the theorem compares optimal costs, rather than merely composing a simulator supplied in advance.

The state price in this statement is substantive. In a two-checkpoint experiment, with the input consumed before the output is produced, zero private deficiency at width  $K$  is equivalent to nonnegative rank at most  $K$  of the required stochastic channel (Proposition 2.4). A three-row channel has positive private two-state deficiency but zero hidden-selector deficiency at the same within-mode width. The selector can be stored, but it cannot be discarded. Consequently a convex decision dual cannot silently be interpreted as a same-width private inverse randomization theorem. Our converse gives the implementation and its cost together.

**61.2. Endogenous decisions and nonsummable risks.** A common encoder must also be shared by tasks that choose different controls. The set of histories relevant to a task then depends on the controller being synthesized. Theorem 7.3 characterizes zero excess relative to the full-history Bellman value using task-indexed reachable cells, task-dependent actions, and one task-independent update map. Closure is required only under the chosen actions and actual supported successors. Randomization leaves this exact feasibility set unchanged. An occupation identity gives quantitative upper bounds for approximate covers and a positive finite-table lower bound for infeasible profiles. The latter is a certificate for a fixed experiment, not a uniform margin over perturbed models.

The average-risk theorem uses an explicit physical regeneration. A reset refreshes the physical state and clears the within-policy register. Between resets the controlled physical kernel may be singular or deterministic. Uniform regeneration yields a compact risk image for stationary common finite-register policies, an attained minimax theorem after priced selection, and a vanishing-discount limit (Theorem 8.1). It is the reset of the joint physical and retained state, not an unproved

assertion about the mixing of a hidden plant, that gives the required uniformity. This result treats a nonsummable criterion; it is different from the dominated finite-prefix, summable infinite-horizon theory retained in the complete development.

**61.3. Active microscopic realization.** Our physical example is a finite hard-sphere system. At observation times an action applies an orthogonal velocity intervention, followed by the hard-sphere flow. The action therefore changes the subsequent trajectory. The interventions are bounded composition operators and need not preserve the generator domain. A graph-core Galerkin approximation of the unbounded Koopman generator, followed by projected interventions, approximates each finite controlled word. Gaussian acquisition converts the resulting  $L^2$  errors into two-sided marked comparison bounds before optimizing any finite-register decision rule (Theorem 9.2). A sum over action words is essential: feedback conditioning does not preserve a uniform posterior density bound. After physical regeneration, the same comparison extends to average risks by a geometrically weighted cycle argument, even though no rate uniform in the word length is assumed.

The physical system, noise levels, and finite action alphabet are fixed. The approximation dimension is a continuous physical state dimension, not a bound on the cardinality of a decision maker’s register. We do not identify this active, regenerative microscopic theorem with a Boltzmann–Grad limit or with a nonlinear kinetic action theorem.

**61.4. Relation to earlier theories.** Blackwell comparison and Le Cam deficiency supply the classical statistical background; filtered experiments add the timing of information and decisions [1, 47, 48]. The finite minimax step, separation, and Carathéodory’s theorem are classical. What is retained here through these operations is the simulator’s *persistent-state and visibility signature*. The nonnegative-factorization example makes the difference from unconstrained randomization concrete. A full proof-level comparison with Norberg’s original filtered-experiment theorem is not asserted: the original full text was not available in the present source audit. In particular, we do not claim novelty for filtered deficiency as such.

Finite-memory control and the common-information method organize controllers through information states and prescriptions [54, 53]. Our cover theorem addresses a different finite question: exact Bellman optimality for several endogenous tasks with one retained encoder. Compatible covers in incomplete machine minimization are classical, and the cover device itself is not claimed as new. We give its decision-regret interpretation, its shared-update constraint, and its quantitative connection to the cost-indexed comparison theorem. The corresponding source-audit limitations are recorded outside the article.

The operator ingredients are likewise not presented as a new Mori–Zwanzig identity or a new abstract semigroup approximation theorem. Trotter’s approximation theorem [67, Theorem 5.2] is the classical antecedent of our graph-core argument. Recent domain-conscious projection results include [65, 66]. The new physical conclusion concerns feedback intervention and marked finite-register comparison, including its regenerative average-risk limit. The microscopic flow input is the standard almost-everywhere hard-sphere flow theorem [63].

**61.5. Organization.** Section 2 defines and proves the intrinsic comparison theorem, including its nonconvex and exposed-seed boundaries. Section 7 connects it to endogenous exact and lossy common decisions. Section 8 treats average risks, and Section 9 proves the active microscopic realization. The canonical article is self-contained relative to the expressly cited classical inputs. The accompanying complete development retains every predecessor mathematical body; those additional results are not silently made hypotheses of the present main theorem. The research program is a typed dependency graph, not a chain of implications inferred from paper names or repository chronology.

## APPENDIX 62. INTRODUCTION

Statistical equivalence does not specify the memory needed to use an experiment. A report supplied at two different times and the same report supplied only once contain the same full-history information, but they need not support the same finite-register decisions. Nor does the state needed for a finite list of optimal decisions necessarily determine the state needed to preserve a whole marked continuation experiment. We study these distinctions through causal comparison with an explicit persistent state cost.

The object is a marked causal experiment together with an information interface. A single parameter-blind encoder must serve all declared tasks. A task controller may use its own task index but cannot reveal it as an additional uncharged encoder input. A consumed report is discarded; only the specified register survives. Independent public randomness is a separate resource. In particular a public variable taking an arbitrary point of a compact policy space is not identified with a finite index stored in the register.

**62.1. The structural result.** Let  $\mathcal{R}_E(S, q)$  be the risk functions attainable on  $E$  with private widths  $S = (S_t)$  and a retained independent public seed of at most  $q$  values. The value  $q = \infty$  denotes an ideal uniform seed. The central comparison statement has three parts.

First, a causal marked simulator with internal widths  $K = (K_t)$ , public seed bound  $r$ , and feedback-uniform error  $\varepsilon$  transfers every target design to a source design with widths at most  $KS$ , seed bound  $rq$ , and row-risk error at most  $\varepsilon$  (Theorem 24.2). The actual implementation uses the reachable pairs of the simulator and consumer states. Composed simulators multiply these costs and add their errors. Lower support values of the common-encoder risk bodies give witnesses against an insufficient simulation budget (Corollary 24.4). These are concrete implementations of the abstract spectrum transport statement retained in Theorem 23.2, not a new name for that statement.

Second, for a finite marked instrument let  $Q(E)$  be the quotient by conditional laws of the parameter, future reports, and actual target mark. A design is complete if its retained register, together with its independent seed, determines the current continuation class. Then

$$\mathcal{R}_E^c(S, q) = \mathcal{R}_{Q(E)}^c(S, q).$$

Theorem 24.7 proves this equality by fusing the simulator's class register into the consumer's already complete register. Without completeness the same simulation can require  $K_t S_t$  labels. The repeated report example gives a sharp temporal cost: exact experiment equivalence can coexist with error  $1 - s/k$  at an intermediate width  $s$ , while the other presentation has zero error. Thus the cost distortion and its zero-overhead subcategory are both part of the comparison invariant.

Third, completeness has an operational decision meaning. For a finite family of restart decisions, the least exact width is a compatible-cover number (Theorem 30.1). It is generally not the number of equivalence classes of a pairwise relation. For Brier prediction tasks the unique optimal predictions turn that cover into equality of conditional probability features. Under positive reports their causal closure is the least right congruence refining those features (Theorem 30.4). Completion under all marked event predictions recovers the universal quotient, and hence the complete-state fusion theorem.

**62.2. Analytic realization and computational distinctions.** The lower support invariant is attained in two nonfinite observation classes. The product-dominated finite-horizon theorem retained from the preceding development proves compactness of the private risk set and compact convexity of its ideal-public body for standard Borel reports with an  $L^1$ -continuous family of complete loss effects (Theorem 26.3). Its tensor-testing and measurable representative arguments remain essential: merely naming a standard Borel carrier would not supply compactness.

Theorem 27.3 adds a singular displayed-covariate class. A continuous physical observation is supplied afresh at every checkpoint, while the innovations satisfy a conditional product-effect hypothesis. The full observation path can be supported on a nonatomic diagonal and have no product dominating measure. A compact space of conditional probability measures on private innovation policies gives an exact executable ideal-public risk body. This is a conditional Young measure argument with the private state still bounded, not an assumption that arbitrary information constraints are weakly closed. Finite stored randomness is recovered at the explicit row-covering cost of Proposition 24.5.

The exact continuation compiler uses classical weighted-automaton linear algebra to test equality, but outputs positive normalized states. We prove a polynomial count-vector bound for homogeneous conditionally independent controlled reports and an exponential  $K_t = 2^t$  example at continuation rank two. We also realize graph coloring as the exact zero-excess common-task width. Thus linear equivalence testing, positive-state realization, and task-optimal compression are distinct computational questions, not three descriptions of one polynomial algorithm.

**62.3. A physical consequence and the place of the other results.** The microscopic hard-sphere application makes the comparison theorem operative on an unbounded generator. Orbit regularization constructs a collision-compatible weighted  $L^p$  graph core, with an explicit single approximation sequence (Theorem 43.2). For noisy finite acquisitions of actual physical observables, its approximation error gives stateless marked simulators in both directions. Their decision bounds are uniform over all memory widths and feedback rules (Theorem 43.3). This is a microscopic domain and acquisition result; it does not assert the separate action-sublevel BBGKY corrector or the nonlinear kinetic limit. The physical model and its operator topology are not replaced by a finite positive hidden-state model to obtain this conclusion.

The common-membership separation, delayed-query Gaussian exponent, positive filtering, finite nonlinear semigroups, hidden-phase moments, predictive trees, probability-measure geometry, and finite statistical comparison are retained with their full proofs and local hypotheses. They give distinct ways to evaluate or test the same risk construction. The canonical article contains this development; the complete-development edition additionally retains the earlier mathematical bodies and historical introductions. Earlier results are not deleted or declared consequences of stronger hypotheses they do not satisfy.

**62.4. Antecedents and the claim of distinction.** Blackwell–Le Cam comparison and positive convex separation underlie the risk-set viewpoint [1, 47, 46]. Filtered-experiment comparison is already present in Norberg [48]. Weak star team-policy topology, existence, and strategic-measure convexification are established methods [55, 56]. Theorems 6, 7, and 9 of [55] provide the relevant continuity, existence, and static-reduction antecedents. We do not claim any of these mechanisms as new, or infer unrestricted filtered compactness from the conditional extension proved here.

The common-information dynamic program of [53], Theorem 3 and Corollary 5, and causal coding reductions such as [57] identify useful designer states and policy structures. An offline belief can be an excellent optimization variable without being free online memory. Our costed simulators count the state needed to execute the comparison and identify exactly when it is already carried by a complete consumer.

Automaton continuation and predictive equivalence have their own long history [49, 5, 61, 58]. Proposition 2.1 and Theorem 2.3 of [58] describe the linear span engine used here. Graph coloring in zero-error information is also classical [62]. The new statements are therefore submitted as resource-sensitive refinements and their operational connection, not as the invention of minimax, causal comparison, state minimization, or graph coloring. The distinction sought is the interaction between executable state cost, the task family, and exact marked reconstruction. No claim of priority for unrestricted filtered comparison is made.

A model becomes a consumer of the comparison theory only after its information interface and analytic hypotheses have been verified. An acquisition adapter does not, by itself, derive that model's independent geometric or operator-theoretic assertions. This distinction is especially important for microscopic limits, where the topology and the scale of approximation form part of the theorem.

### APPENDIX 63. INTRODUCTION

A statistic may preserve every parameter likelihood and yet fail to preserve the target of a decision. A compressor may be optimal for each task separately and yet fail when its update rule must be chosen before the task. Both distinctions become structural when information arrives sequentially and retained memory is bounded. We study them through the attainable risks of an executable causal experiment, with its target marks and information pattern specified together.

The principal object is the set of risk functions obtained by one common encoder and a family of task controllers. The encoder may use the current report, the current control and its retained label, but neither the task index nor the unknown parameter is an additional input. A task controller may depend on its task and retained label; it must work across all parameter rows of that task. Randomness is also part of the specification. Fresh private draws do not provide a free retained random variable, whereas an independent shared seed may correlate all stages of an execution. Expectations over that seed are taken before the worst decision row is selected.

**63.1. The main results.** Fix finite controls, decisions and a horizon  $T$ , but allow standard Borel report spaces and a compact parameter set. For a width vector  $\mathbf{S} = (S_0, \dots, S_T)$  let  $K_{\mathbf{S}}$  be the set of all private common-encoder designs and let  $R(k) \in C(J)$  be their risk functions on a compact set  $J$  of task-parameter rows. Under the dominated loss-effect condition of Section 26, Theorem 26.3 gives

$$\min_{\lambda \in \mathcal{P}(K_{\mathbf{S}})} \max_{j \in J} \{\mathbb{E}_{\lambda} R_j - b_0(j)\} = \max_{w \in \mathcal{P}(J)} \left\{ \min_{k \in K_{\mathbf{S}}} \int R_j(k) w(dj) - \int b_0(j) w(dj) \right\}. \quad (63.1)$$

The private attainable risk set is compact, and its shared-seed barycenter set is compact and convex. Positive measures on  $J$  characterize the closed upper risk body. These conclusions concern all measurable finite-state updates, not just a compact family postulated in advance or encoders enumerated on a finite report tree.

The proof uses a product of weak star simplex spaces. Its main analytic step is a tensor-testing lemma: convergence of one bounded encoder factor in each report coordinate can be tested against an arbitrary  $L^1$  path effect. Compact families of effects give uniform convergence of the risk functions. A separate representative lemma makes public mixtures of weak star classes jointly measurable and executable. Thus compactness is proved within the common-encoder information pattern. Corollary 26.5 treats countably many tasks by compactness and finite-subproblem duality; in that setting the dual is a supremum and no uniform finite seed is asserted.

The finite observation theorem, Theorem 26.7, compares the actual joint report-mark laws by causal cell conversion and within-cell reconstruction. Its error tends to zero uniformly in the memory width, task and controller. Finite experiments are therefore an approximating family for the dominated theory, not its sole domain. Corollary 26.6 gives the price of replacing a public mixture by a finite privately retained index.

Our second main result concerns the state carried by a marked experiment. For a finite instrument, two histories have the same continuation state when their conditional laws of the parameter, future reports and terminal mark agree under every future control word. Theorem 29.3 derives this state from decision risks: among all source-consuming randomized causal statistics, it has the least number of labels preserving every restart prediction risk. No prescribed posterior coordinates appear in that operational definition. The proof uses a strict conditional-variance

identity for binary-event prediction. Theorem 29.5 then constructs parameter-independent causal report channels preserving the actual terminal mark. An explicit binary example shows why parameter sufficiency alone does not give this conclusion.

For rational finite instruments, Theorem 31.2 computes these classes by backward continuation spaces and forward exploration of distinct positive states. Its cost is polynomial in the input bit length and the number of output states, rather than in the entire expanded history tree. The dimension of a linear representation is used to test equivalence; it is not confused with the number of positive retained states. A binary source with linear dimension two and  $t + 1$  distinct states at time  $t$  illustrates the difference.

Two quantitative examples make the information pattern visible. Theorem 33.1 has three individually lossless two-label tasks, but common-encoder worst risk  $1/2$  with private randomness and  $2/9$  with shared randomness. All nine decision rows receive positive mass in an optimal dual witness. The delayed-query Gaussian theorem, Theorem 38.2, has pre-query error of order  $S^{-2/d}$  while its query-aware checkpoint has order  $S^{-2}$ . Proposition 39.2 gives a finite acquisition and table implementation retaining the causal exponent.

**63.2. Related work and the distinction of scope.** Weak star policies and existence in stochastic teams are developed by Saldi [55]; strategic measures, public mixtures and failures of weak closure are treated by Yüksel and Saldi [56]. In particular, convexification by common randomness and compactness methods are not new principles here. Our result specifies a bounded retained register shared by several tasks, proves continuity for its coupled encoder factors, and obtains the risk-function body and its width-uniform marked approximations. Nonconvexity of the private set is consistent with the classical fact that randomization cannot improve a single fixed linear team objective: our strict example concerns a worst-row objective.

Causal source coding and common-information control provide the closest information-structure antecedents. See Witsenhausen [54], Yüksel [57], and Nayyar, Mahajan and Teneketzis [53]. An offline occupancy or belief recursion does not by itself constitute an online finite-memory implementation; we keep that distinction in the finite support recursion and the compiler. Comparison of filtered experiments, as in Norberg [48], is likewise part of the background of causal comparison, rather than a novelty claimed for the word “causal”.

Continuation equivalence is related to automata and predictive-state constructions [49, 5]. The linear span computation uses the weighted-automaton equivalence method presented by Kiefer [58]; neither rational elimination nor linear automaton minimization is asserted as new. The operational minimality here is for the declared marked continuation and for source-consuming statistics under all restart preparations. It is not minimality among independent simulators with unpriced retained memory. The target mark and this comparison category cannot be omitted from the statement.

The positive-model applications include established mechanisms: positive-matrix filter contraction, the logarithmic control-semigroup construction of Nisio [59], and the risk-sensitive information-state augmentation of Bäuerle and Rieder [60]. We prove the needed estimates here and specify how their actual marked effects enter the common-encoder theorem. No priority claim is made for these individual mechanisms.

**63.3. Consequences and organization.** The marked decision spectrum of Section 23 fixes the experiment, the task family, the information pattern and the resource budget before optimization. The compact theorem and the finite common-encoder theorem give exact variational descriptions in their stated classes. Quadratic targets give conditional variance; complete terminal loss families give statistical comparison; tree geometry gives weighted distortion estimates; and presenting an actual probability measure gives a Wasserstein law task. These are typed consequences of one risk construction, not an assertion that one scalar rate or one domination assumption describes

every model. In particular, presenting a realized measure and sampling a single marginal probe are different tasks.

The following sections develop the compact theorem, marked quotient, compiler and common-task example, followed by the finite variational and saturation theory. The Gaussian temporal example and numerical acquisition result precede three model applications: positive hidden Markov observations and history reduction, nonlinear finite-state semigroups, and hidden-phase exponential moments. The remaining sections supply the predictive-tree, measure, terminal Gaussian and count developments with full proofs. The separate complete-development edition also includes the earlier mathematical bodies and their historical introductions. None of their local hypotheses is removed when the results are used here.

#### APPENDIX 64. INTRODUCTION

An observation can be compressed only relative to questions and an information pattern. When the question is known before compression, one may retain only its answer. When it arrives later, the same machine must preserve the possibility of answering several incompatible questions. With feedback, the retained information also determines which experiment is performed next. The resulting problem is neither a static covering problem nor a collection of separate control problems.

This paper studies the marked decision spectrum of a causal experiment. Its encoder is fixed before the task, its controller and decoder may know the task but not the unknown parameter, and its persistent state is charged throughout the execution. The spectrum records the attainable risk vectors and their dependence on a resource signature. Our principal result is a controlled common-encoder variational theorem. It constructs the risk body from the finite instrument itself, characterizes its feasible continuation partitions, and identifies its lower support function by an exact occupancy recursion. The maximum over tasks is retained; it is not replaced by one favourable scalarization.

To describe the result, let  $E$  be a finite controlled hidden-state instrument on a finite horizon, and let  $D$  be a finite family of bounded decision tasks. A row  $j \in J$  records a parameter within a minimax task, or the declared prior of a Bayesian task. Put  $M = |J|$ . A deterministic design  $\omega$  consists of one common causal encoder together with one observation-admissible controller and decoder for each task. Write  $R_j(\omega)$  for its row risk, and  $b_j$  for the corresponding task's full-history optimal baseline. These definitions are made precise in Section 34; in particular, a minimax task has one baseline, not a parameterwise oracle.

**Theorem 64.1** (The common-encoder variational principle). *For prescribed finite register widths  $\mathbf{S}$ , let  $\Omega_{\mathbf{S}}$  be the deterministic designs. Under an independent shared seed, averaged before the worst task and parameter are selected,*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (64.1)$$

$$= \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (64.2)$$

Here  $V_0$  is the finite prescription recursion (35.2)–(35.3), whose update table is common to all tasks and parameters. Its support values determine the whole attainable upper risk body. Deterministic feasible encoders are exactly the controlled successor-compatible history partitions.

An optimal mixture has at most  $M + 1$  designs. Removing the shared seed while retaining its design index in the actual register gives

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D).$$

Private randomization at unchanged width instead has the nonconvex flow characterization (34.8); it need not give the convex body in (64.1).

The complete proof is given in Theorems 34.1 and 35.1 and Propositions 34.2 and 34.3. The underlying convex minimax and separation arguments are classical. The theorem uses them with a coupled feasible set: the encoder is common across tasks, each controller is common across its parameter rows, and a remembered randomization index is an actual resource when it is not an explicitly permitted shared input. These constraints are indispensable. In Example 34.4, the two-label minimax error of a three-symbol experiment is  $1/3$  with a shared seed and  $1/2$  without one. Thus replacing private kernels by an unpriced convexification would compute a different spectrum.

The same variational principle has two useful specializations. For a family of Bayesian quadratic tasks it is exactly a maximization over one task weight vector of the least conditional variance of a weighted posterior-mean vector, subject to causal compatibility. This strengthens the single-objective partition formula to the common-encoder maximum criterion. It applies directly to the accepted-cell/failure protocol of the A1 manuscript, with a single update on the observed failure symbol. For exponential criteria it is a duality for moments before taking logarithms, with a multiplicative occupancy recursion. In a hidden-phase example it excludes optimizing separately with knowledge of the phase. These are specializations of one feasible execution body, not independent assumptions that their controllers can be chosen jointly.

We also construct a least likelihood-saturated predictive coordinate for every finite instrument. A backward refinement of posterior vectors, next-report probabilities and successor classes gives its exact minimal widths. For rational input, this construction and its reconstruction kernel are computable by finite arithmetic. It makes the finite sufficiency bridge constructive, while identifying its possible cost: a one-class future-report quotient can require  $2^d$  saturated classes after  $d$  parameter-revealing observations. Statistical saturation is therefore not a cost-free completion of a predictive quotient.

Finite synthesis yields control of several resources at once. Theorem 37.1 compiles an optimal finite design into a finite program and a dyadic initial lottery. It charges program length, persistent labels, initialization bits, transient work and bit-serial running time. Its excess differs from the optimum by at most  $k2^{-p}$  for  $k \leq M + 1$  designs and  $p$  lottery bits. Synthesis can be exponential in the horizon; no multiplicative-energy oracle is used in this finite source theorem. The predictive-tree compiler retained later addresses a different, infinite symbolic input class under its explicit geometric and energy-certification assumptions.

A continuous Gaussian example shows that the same causal constraint has quantitative force beyond finite enumeration. Acquire  $d$  noisy scalar coordinates before an independent coordinate query is disclosed. The final target is one scalar parameter, but its optimal causal excess is  $S^{-2/d}$ , whereas a query-aware static checkpoint has excess  $S^{-2}$ . The lower bound uses the common register *before* the query, the conditional-variance identity, and a density-volume bound on the vector of posterior means. This is a temporal penalty, not the terminal dimension argument for an append-only Gaussian vector experiment.

**64.1. Relation to existing theory.** Comparison of experiments by all bounded decisions is the theory of Blackwell and Le Cam [1, 47, 46]. The terminal deficiency representation included below is stated with its compactness hypotheses and the common encoder outside the task supremum. It does not claim a new unconstrained randomization criterion. Filtered experiment comparison is itself an established subject [48]. Likewise, predictive equivalence and continuation congruences have the classical antecedents [5, 49].

Real-time source coding and decentralized control are particularly close neighbours [54, 33, 34]. Nayyar, Mahajan and Teneketzis [53] derive common-information prescription recursions. In their preprint, Theorem 3 gives the dynamic programme, Corollary 5 treats absent common memory with finite local memory, and Theorem 4 simplifies its information state. Our offline occupancy is in this established designer/common-information tradition. The assertion made here is the coupled

minimax risk-body identity, its controlled partition converse, and the priced passage between shared and private randomization. An ordinary designer's dynamic programme, by itself, is not asserted to be new.

The retained quantitative sections use functional and separated-cylinder quantization, symbolic trees, nonlinear filtering and experiment approximation [35, 36, 43, 37, 44]. Their geometric hypotheses remain local to those theorems. The new conditional-variance variational principle includes the finite partition problem and explains the delayed-query Gaussian obstruction. It does not identify Wasserstein law presentation with decision-complete statistical deficiency, which retain different output marks.

**64.2. Organization and scope.** Section 23 fixes the marked spectrum and its transport and terminal comparison properties. Sections 34 and 35 prove the central variational principle. Sections 36–38 construct saturation, compile finite sources and exhibit the Gaussian temporal penalty. The subsequent sections retain the complete sufficient-statistic, partition, predictive-tree, measure-presentation, Gaussian and count arguments. The companion development additionally retains the earlier mathematical bodies and historical introductions with their original hypotheses; none is silently replaced by a stronger assertion.

The general marked spectrum includes standard Borel experiments, whereas the exact controlled minimax theorem proved here has finite horizon, finite hidden/report/control spaces and finitely many decision rows. The Gaussian example is established separately by its displayed continuous argument. In particular, the factor  $M + 1$  is not a uniform statement about a growing or infinite task family. A finite-horizon exponential moment formula is not an LDP, an unbounded-operator theorem, or a proof of a microscopic phase decomposition. Those further structures require their own model theorems. The present foundation supplies a controlled common-encoder converse without importing such theorems as unnamed premises.

## APPENDIX 65. INTRODUCTION

Finite descriptions of an experiment can be compared only after the questions to be answered and the permitted execution have been fixed. A two-valued checkpoint target may require exponentially many causal states when the relevant question arrives after the observation has been discarded. A finite statistic may preserve every bounded decision problem approximately without producing an accurate realization of a whole posterior measure. These are not competing definitions of the same error. They are different slices of one constrained decision problem.

We attach to a causal experiment its *marked decision spectrum*: the attainable risk vectors, indexed by the retained resources and by families of decision tasks. One common encoder is chosen before the task, whereas its decoder may depend on the task. The spectrum retains this order of quantifiers. It separates persistent state, physical output, acquisition length, program description, numerical precision and transient work. Under a resource-controlled causal equivalence it is invariant, up to the certified budget transformations. For a compact dominated terminal experiment, its decision-complete slice is exactly the optimal reverse deficiency of a common finite-output encoding. Its quadratic Bayesian slice is a compatible-partition distortion problem.

**Theorem 65.1** (Representation and realization of the decision spectrum). *The spectrum defined in Section 23 has the following three properties.*

- (1) *Initialized, parameter-independent causal simulations transport its attainable risk sets with their full resource transformers. Mutually exact finite-state simulations preserve every existing positive cardinality exponent.*
- (2) *For a compact total-variation-continuous terminal family,*

$$\Theta_E(S; \mathcal{D}_{\text{all}}) = \inf_{Q: X \rightarrow [S]} \delta(QE, E).$$

*A posterior coordinate formed with any positive dense reference prior is sufficient for the entire parameter family and leaves this terminal spectrum unchanged. Its causal analogue requires likelihood saturation and recursive predictive closure; a finite quotient then supplies a finite-state equivalence with an explicitly charged budget dilation.*

- (3) *For finite report alphabets and a fixed finite-horizon Bayesian quadratic problem, the optimal causal excess is exactly the minimum conditional-variance functional over successor-compatible partitions of the history levels, with width at most  $S$ .*

*The precise hypotheses and proofs are Theorems 23.2, 23.4, 46.1, 46.2 and 47.1.*

The third assertion supplies a necessity theorem rather than another sufficient coding scheme. In Example 47.4, a random  $d$ -bit word is followed by a random coordinate query. The checkpoint mean has two values, but zero-error causal realization needs  $2^d$  states. Its positive error is bounded below by the binary entropy inverse. Thus static predictive geometry alone cannot determine a causal budget. Compatibility of the partitions is the missing datum.

We then prove three quantitative consequences. First, Theorem 48.1 replaces exact real-energy ordering in a predictive tree by a rational threshold construction. Taking the minimum of certified energy bounds over every contiguous factor gives exact suffix closure despite ties and numerical uncertainty. Its finite tree has optimal distortion order,  $O(\log S)$  acquisition depth and an  $O(S \log S + Sdp)$ -bit description for a  $d$ -dimensional readout. The acquisition is a finite prefix of a snapshot, not a free infinite word.

Second, Theorem 49.1 and Corollary 49.2 give a matching  $1/\log S$  presentation law for the entire space of probability measures on an interval and for a genuine nonlinear posterior class with arbitrary input prior at each fixed time. A continuous finite-dimensional exact coordinate is impossible on this class. Yet a single marginal Bernoulli probe admits an  $O(1/S)$  simulation. The separation proves that forgetting the output mark can change the rate, even when the latent measure is unchanged.

Third, Theorem 50.1 gives the decision-complete terminal law  $S^{-2/(nr)}$  for  $n$  sequential  $r$ -dimensional Gaussian innovations when only the current register is retained. The register must hold the accumulated finite experiment; previous labels cannot be stored in a free transcript. The positive Gaussian reconstruction and the physical Poisson count-to-contact comparison are proved in full in the supporting sections. The latter gives a direct downstream instance with matched label budgets in a joint sample-size window.

**65.1. Relation to previous theory.** Predictive equivalence and minimal predictive states are classical ideas; Shalizi and Crutchfield [5] establish optimality and uniqueness properties of causal-state representations. The continuation congruence used here has the classical finite-automaton antecedent [49]. We use neither of these observations as a claim of a new exact minimal-state principle. The extra constraint is a charged recursive presentation, with a fixed common encoder and a quantitative loss. The finite-history variance formula identifies its feasible partitions exactly.

The comparison of experiments through all bounded decision problems is likewise the theory of Blackwell and Le Cam [1, 47, 46]. Norberg [48] treats comparison with filtered probability spaces. Our terminal duality is proved to fix its compactness assumptions and its encoder quantifiers, not to reassign priority to that criterion. The new organization retains resource transformers as part of a causal morphism and distinguishes saturated statistical quotients from prior-relative predictive quotients.

Zero-delay coding of Markov sources has a substantial structural literature. Linder–Yüksel [33] and Wood–Linder–Yüksel [34] study causal coding policies, including stationary optimal policies and finite-memory approximations. Their communication alphabet is not automatically the cardinality of all retained information. In the latter paper, Theorems 3–4 of the author preprint

give stationary average-cost optimality and periodic finite-memory approximations. The sequential Gaussian result here concerns a different, explicitly total-register terminal problem.

Functional quantization and separated-cylinder quantization provide the static geometry [35, 36, 43]; greedy variable-to-fixed trees have the classical antecedent [37]. We retain those attributions. The robust factor envelope proves that a near-optimal finite tree can be compiled without an oracle for exact comparisons, while its suffix closure makes both causal updates executable. Nonlinear-filter contraction and finite-model approximation have extensive precedents, including [44]. Our additional lower bound concerns the full posterior presentation task, not a new proof of average-cost control optimality. The topological and entropy arguments also distinguish continuous finite-dimensional coordinates from arbitrary Borel codings.

**65.2. Organization and quantitative conventions.** Sections 23–47 establish the common spectrum, its sufficient-statistic representation and its causal compatibility obstruction. Sections 48–50 prove the quantitative consequences. The remaining sections supply the complete predictive-tree, filtering, Gaussian and count-experiment proofs used by these consequences. They are mathematical dependencies, not substitutes for a proof of the new representation statements.

The notation  $a_S \asymp b_S$  means comparison by strictly positive constants independent of  $S$  in the stated parameter regime. Constants for the tree class are uniform only when its separation, child-energy, unary-chain and distortion constants have common bounds. Horizon, covariance conditioning and rank dependencies are retained. Explicit Poisson normal-approximation constants in the supporting proof are nonoptimized. The physical A2 consequence is order-optimal in the label budget inside its displayed joint window, not sample-optimal over all observation protocols. The paper’s title names the continuing programme; the invariant proved here is the typed marked spectrum, not a claim that unrelated spectral, kinetic or phase theorems are automatic corollaries.

## APPENDIX 66. INTRODUCTION

A statistic that preserves present predictions need not admit a finite recursive implementation. Conversely, a small implementable register need not preserve the decision problems of the underlying experiment. The purpose of this paper is to relate these three objects: a predictive quotient, its attainable geometry, and a charged causal realization. The order of the argument is

$$\begin{aligned} &\text{positive causal experiment} \longrightarrow \text{future-test quotient} \\ &\longrightarrow \text{causal simulation} \longrightarrow \text{finite-resource realization.} \end{aligned}$$

The quotient is always relative to a declared family of executable future tests. Resource preservation is always relative to a declared resource signature. Neither a measurable re-encoding by real numbers nor a terminal randomization is, by itself, a finite-memory implementation.

The principal structural result is Theorem 53.1. It starts with a factorial language, a stationary law, and a positive energy on its finite words. The hypotheses are local child comparisons, strict energy decrease under prefix attachment, bounded unary chains, and geometric separation of the predictive images. No quantization dimension is an input. If  $G_M$  is the energy of the greedy cut with at most  $M$  leaves, then its static squared quantization error is comparable to  $G_M$ . Every greedy tree is suffix closed. With  $L$  leaves it has at most  $(U + 1)(2L - 1)$  vertices, where  $U$  bounds unary chains. These vertices support two different causal operations. Deleting the oldest available future symbol realizes a renewal orbit code; prepending the newest observed symbol realizes a code of the entire observed past. Both operations have the same-order risk as the unrestricted static problem. The lower bounds allow randomized, time-dependent observers.

The two orientations are not merely two descriptions of one reset model. The Markov-renewal verification in Section 54 includes forbidden transitions and noninjective, delay-identifying observations. The second verification is a noisy hidden shift register whose tail persists at every time. Its complete posterior has infinitely many observation-dependent coordinates. Theorem 54.3

gives a matched whole-register excess-risk law in that experiment, including its noise dependence. Thus one structural realization theorem has a regenerative and a genuinely continually observed application.

A complementary approximation theorem treats predictive quotients that are measure valued rather than tree separated. Theorem 55.2 constructs a finite recursive register for contracting filters on  $\mathcal{P}([0, 1])$ . Its codebook has at most  $4^n$  states and error at most  $3/(2n(1 - \rho))$ , apart from initialization and explicitly charged numerical defects. The nonlinear example has no fixed-step common minorizer and no closed recursion of any prescribed finite list of polynomial moments. This is an upper approximation theorem, not a claim that every nonlinear filter has this contraction or that its logarithmic budget rate is universally optimal.

The experiment-comparison application is two-sided. Theorem 56.1 proves that an  $r$ -dimensional nondegenerate Gaussian location experiment, over a compact parameter set containing a ball, has optimal  $M$ -outcome approximation error  $\Theta(M^{-2/r})$  in Le Cam distance. The upper bound uses deterministic cells and positive second-order reconstruction, not first-order constant histograms. The lower bound applies to every experiment on at most  $M$  outcomes. For the actual local count-to-contact protocol of A2, Theorem 57.1 proves

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C\{N^{-1/2} + JN^{-1/2} + J^{-2}\}, \quad |\mathcal{E}_{N,J}| \leq J^r.$$

It follows that  $\Delta \asymp M^{-2/r}$  throughout the growing-budget window  $1 \ll M \lesssim N^{r/6}$ . The proof includes the local normal, variance replacement, boundary crossing, reconstruction, and decision comparison arguments. The physical statistic is deterministic throughout.

**66.1. Relation to previous work.** The existence of path laws, conditional-probability quotients, Bayes orthogonality, information loss under coarsening, and the composition of statistical randomizations are classical; see [4, 1, 45, 46]. Sections 51–52 give the forms needed here, with the simulator and its persistent state included. We do not claim novelty for these identities or for their categorical notation.

Static graph-directed quantization is a particularly close antecedent. Kesseböhmer–Zhu [43], Theorem 1.1, determine Markov-type quantization dimension by the spectral radius of a matrix of  $(p_{ij}c_{ij}^r)^{s/(s+r)}$  and characterize finiteness of the upper coefficient through maximal strongly connected components. Their spatial theorem is not a theorem about a causal register. Our tree argument separates that familiar static mechanism from the additional suffix-closure condition and proves realization in both time orientations. The Markov application retains the renewal-weighted orbit and survival contribution; its spatial pressure is not a new discovery of graph-directed quantization. The earlier comparisons with functional quantization, dynamical rate distortion and zero-delay coding remain relevant [35, 33, 34]. A finite channel alphabet does not generally bound the entire persistent state of its coding policy.

Demirci–Kara–Yüksel [44], Theorems 3.5–3.6 of the cited preprint, obtain quantized-belief and finite-window policy approximations in their contraction-based control analysis. Our measure-valued result uses the classical contraction-plus-quantization mechanism, but provides a concrete entire-register cardinality and a same-observation recursive error bound for a specified nonlinear experiment. Its objective is prediction and terminal experiment simulation, not average-cost control optimality. The sharp hidden-shift theorem is instead a two-sided predictive-tree result. Neither theorem purports to subsume general partially observed control.

The new contribution is therefore not “finite memory”, “pressure”, or “Bayes reduction” in isolation. It is the combination of a checkable energy criterion, exact two-orientation realization, unrestricted register converses, and resource-typed experiment comparison. The Gaussian/A2 application supplies a separate decision-theoretic cardinality obstruction and an attainable joint sample/budget law. No priority claim for every component of these arguments is required.

**66.2. Reading convention.** This article is the canonical seventh revision of Volume I. Its proofs are self-contained at the level of the stated models. A separate *Complete development* contains this article followed by the unchanged earlier theorem bodies, including pressure, critical renewal, calibration, collision, and control results. The companion is a preservation and extended-reference object, not an alternative definition of the main paper. Source identities, historical consultation, and responses to the referee are recorded outside the mathematical text.

## APPENDIX 67. INTRODUCTION

The information retained by a causal observer must serve predictions whose time of use is not known when that information is acquired. A description of the current state can therefore be inadequate even when it is an optimal quantization at the acquisition time. There is also a converse difficulty: a good finite description of an entire future orbit need not admit a closed update on the same finite state set. These two questions link the geometry of prediction to the implementation of a finite memory.

This paper develops two realization principles for that link. The first is an exact suffix-closure principle for renewal-weighted cylinder codes on a mixing Markov repeller. The second is a contracting posterior principle for continual partial observation on an unbounded state space. Both are compared with the full class of randomized, time-dependent observers having a bounded persistent register.

**67.1. The intrinsic law.** Let  $\Sigma_A$  be a topologically mixing subshift of finite type, with forbidden transitions allowed, and let  $K$  be its strongly separated geometric realization by contractions of bounded conformal distortion. The class includes similarities in any Euclidean dimension and nonconstant smooth inverse derivatives in dimension one. The measure on  $K$  is a stationary Hölder Gibbs law. A Hölder observable may be noninjective provided a finite delay vector recovers the geometric state with a uniform snowflake lower bound.

Exact acquisitions occur at independent renewal times. The survival law need not be geometric. A finite cylinder energy combines the mass of the acquired cylinder, the successive suffix contraction scales, and the tail of the stationary renewal age. Splitting maximal-energy cylinders defines a finite profile  $G_M$ . Our first conclusion is

$$\mathcal{R}_M \asymp e_M^2(Z_w^f) \asymp G_M,$$

where  $Z_w^f$  is the weighted observable orbit and  $\mathcal{R}_M$  is the optimal long-time average prediction risk. The constants hold for every budget, not merely along a sequence of cylinder depths.

The realization argument must accommodate vertices with a unique successor and suffix cylinders with a unique predecessor. Strict inequality of cylinder masses is then unavailable. We instead prove strict decrease of the renewal distortion under prefix attachment. Together with stationarity, it forces the complete greedy tree to be suffix closed. Unary vertices are counted, and the whole state set has size at most a graph-dependent constant times the number of leaves. The lower bound uses separated tubes in Hilbert space and is valid for arbitrary off-image centres. Thus neither admissibility nor causal execution is hidden in the definition of a static codebook.

When the renewal weights have exponential rate  $q$ , the squared-error exponent is determined by the larger of two pressure roots,

$$P_A(s_g\varphi + 2\beta s_g\psi) = 0, \quad P_A(s_t\varphi) + s_t \log q = 0.$$

The first combines mass with geometric contraction; the second combines mass with survival through unobserved dynamics. We prove the existence of the quantization exponent from these equations. It is not an additional hypothesis on an infinite-dimensional orbit image. For a Markov source, zeros of the transition matrix remain zeros in the transfer matrices.

The first-order exponent does not determine the critical correction. For the Parry law on any primitive graph and equal-ratio similarities, renewal survival  $q^k(k+1)^{-\kappa}$  gives, at the common critical value, corrections  $(\log M)^{1-\kappa}$  for  $\kappa < 1$ ,  $\log \log M$  for  $\kappa = 1$ , and a bounded factor for  $\kappa > 1$ . The underlying geometry and pressure threshold can be kept fixed. A uniform finite-size profile limit describes the window  $q_n = q_c e^{x/n}$ . These statements describe actual renewal laws and matching risk orders; they do not assert an optimal leading constant from a pair of inequalities.

**67.2. A nonregenerative realization principle.** For continual observations, let  $m_t$  denote the true complete-history conditional mean. Suppose it obeys a finite-dimensional Borel recursion which contracts conditionally in fourth mean in its previous mean coordinate. Individual updates may expand. An inward rational quantizer preserves fourth-moment bounds on an unbounded state space. The resulting recursive observer attains excess squared risk  $O(M^{-2/d})$  using at most  $M$  persistent values. If the stationary posterior mean has positive density on a cube, a matching lower bound holds against all observers. The Bayes floor here is the *full-history* Bayes error, not the error of the latest observation.

For a stationary Gaussian autoregression with Gaussian observation noise, all hypotheses are verified and the risk is  $P + \Theta(M^{-2})$ , where  $P$  is the steady conditional variance. Every fixed-step hidden transition kernel in this example lacks a nonzero global common minorizing measure. The realization proof is therefore genuinely different from inserting an independent reference experiment at each acquisition. The general theorem states its posterior contraction hypothesis explicitly; it is not a contraction theorem for all nonlinear filters.

**67.3. The resource and its invariances.** Only the number of values of the persistent state is constrained. Fixed programs, fixed real constants, the absolute clock, and transient computation are separate resources. A renewal indicator is an input to the update, not a free post-update observation. Theorem 73.1 proves exact operational simulations: adding a  $K$ -valued event to the decoder costs at most a factor  $K$  in state count; splitting registers costs their product; binary checkpoint encodings are related by  $M = 2^B$ . These transformations preserve the exponent and, under fixed-budget dilation, the risk order. A real-valued free observation at decoding is a different experiment, as the explicit uniform example shows.

**67.4. Relation to previous work.** Thermodynamic formulas for *spatial* quantization predate the present work. Lindsay and Mauldin [38] relate quantization dimension for finite conformal systems to the temperature function. Atnip, Roychowdhury and Urbański [39, Theorem 3.1] extend that relation to an infinite conformal family under their summability and strong open set hypotheses. Roychowdhury [36] treats Gibbs-like measures on cookie-cutter sets. Those spatial results include systems beyond the finite, strongly separated class used here. We do not claim a generalization of their geometric hypotheses. The additional object here is the survival-weighted observable *future orbit*, with an unrestricted causal converse and an admissible, suffix-closed finite-state realization. Classical pressure and greedy variable-to-fixed trees [32, 37] are inputs, not inventions of this paper.

Finite-memory and zero-delay approximation likewise have a substantial literature. Linder and Yüksel [33] and Wood, Linder and Yüksel [34] establish structural and stationary coding results. Ghomi, Linder and Yüksel [40, Theorem 4] prove stationary optimality for linear vector Markov sources and control finite-horizon approximation to the infinite-horizon distortion. Their channel alphabet and their history-dependent coding policy are not a single finite persistent register. Our filtering result is not a new proof of stationary zero-delay coding optimality.

Kara and Yüksel [19, Theorem 12] prove exponentially accurate finite-memory feedback under their POMDP stability hypotheses. Cregg, Alajaji and Yüksel [41, Theorem 2 and Corollary 2] bound the discounted suboptimality of sliding-window source-channel policies by predictor stability, with an exponential window bound under sufficient Dobrushin conditions. Thus stability-based

finite-memory approximation is already established. In contrast, Theorem 72.1 concerns the number of retained values, a matched squared-error exponent against the full-history Bayes floor, and an unbounded continuous observation model. It is a quantitative realization result in that resource, not a Q-learning or optimal-control theorem. Recursive filter quantization itself is also classical; see Pagès and Pham [8].

Static functional quantization [35], dynamical rate distortion [29], and data-rate control [27] constrain different objects. The distinction is operational: a block code or information-rate bound does not by itself specify a closed update on a bounded persistent state set. Conversely, the state-count theorems here give no runtime, program-description, or channel-capacity bound.

**67.5. Organization and dependencies.** Sections 68–70 prove the Markov profile and pressure theorems. Section 71 gives the critical classification and uniform window. Sections 72 and 73 prove the nonregenerative and interface results. The general posterior-orbit converse is included with its complete proof in Section 74. The complete edition additionally retains the earlier Gibbs, dependent-acquisition, noisy expansion, calibration, root-collision, filter, control and foundational developments. They have their own hypotheses; the current Markov and filtering proofs do not depend on the singular-statistics or unbounded-operator conclusions.

The programme's T07–T08 question is the passage from acquired predictive geometry to actual finite-state evolution. The two realization mechanisms proved here resolve that question in their stated classes, including a noncompact nonregenerative model. The historical physical models require further, model-specific results. The active A2 conductor and nonreduced-generator edition [42] is an algebraic input to a different observation interface: its scheme multiplicities do not specify a noisy observation kernel. The accompanying snapshot audit records that interface at its current source edition rather than identifying algebraic multiplicity with a statistical exponent. The complete edition also proves a joint sample–label deficiency bound for its actual known-mark count-to-contact protocol. The overflow alphabet and the comparison randomization are explicitly charged and distinguished from the physical statistic.

## APPENDIX 68. MARKOV GEOMETRY AND THE FINITE-STATE LAW

We first specify the geometry independently of the observation apparatus. Let  $A$  be a primitive zero–one matrix on an alphabet  $\mathcal{A}$  of cardinality  $b \geq 2$ , with Perron root  $\lambda_A > 1$ . Write  $\Sigma_A$  for its one-sided shift and  $\mathcal{L}_n$  for its words of length  $n$ . The empty word is denoted by  $\emptyset$ . Forbidden transitions are permitted; in particular, a vertex may have only one successor.

Let  $D \subset \mathbb{R}^d$  be compact, of diameter at most one, and let  $h_i : D \rightarrow D$  be injective contractions. Assume that the compact sets  $h_i(D)$  have pairwise distance at least  $g > 0$ . For  $v = i_0 \cdots i_{n-1}$  write  $h_v = h_{i_0} \circ \cdots \circ h_{i_{n-1}}$ . We require scales  $r_\emptyset = 1$ ,  $0 < r_- \leq r_i \leq r_+ < 1$ , and  $D_0 \geq 1$  such that

$$D_0^{-1} r_v |x - y| \leq |h_v x - h_v y| \leq r_v |x - y|, \quad (68.1)$$

$$D_0^{-1} r_u r_v \leq r_{uv} \leq r_u r_v. \quad (68.2)$$

These conditions are imposed for words for which the compositions are used. They hold for similarities in every dimension, and for increasing one-dimensional  $C^{1+\gamma}$  contractions with uniformly positive derivative and bounded logarithmic distortion, with  $r_v = \sup |h'_v|$ . For higher-dimensional conformal maps, (68.1) is an explicit metric bounded-distortion hypothesis; no global co-Lipschitz conclusion is inferred merely from a pointwise derivative estimate.

The coding map  $\pi : \Sigma_A \rightarrow K \subset D$  is injective. Set  $T\pi(\omega) = \pi(\sigma\omega)$  and  $K_v = \pi([v])$ . Let  $\mu$  be the image of a stationary, fully supported Hölder Gibbs measure on  $\Sigma_A$ . Normalize its potential  $\varphi$  by  $P_A(\varphi) = 0$ . In particular, for a fixed  $C_G$ ,

$$C_G^{-1} e^{S_n \varphi(\omega)} \leq \mu[v] \leq C_G e^{S_n \varphi(\omega)} \quad (\omega \in [v]). \quad (68.3)$$

We use the classical Gibbs and variational pressure theorems for a primitive finite shift in the form recorded in [32]. The profile theorem below needs only stationarity and a uniform positive lower bound on each admissible one-letter conditional probability.

**68.1. Observables and acquisition times.** Fix  $0 < \beta \leq 1$ , put  $\tau = 2\beta$ , and let  $f : K \rightarrow \mathbb{R}^m$  satisfy

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad (68.4)$$

$$\sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta} \quad (x, y \in K) \quad (68.5)$$

for some finite  $J$  and  $c_o > 0$ . Thus the present output need not identify the state; finitely many successive outputs must do so quantitatively. Condition (68.5) concerns the actual observable on  $K$ , not an assigned distance on symbolic words.

At a renewal an independent state  $U$  of law  $\mu$  is observed exactly. Between renewals the state evolves by  $T$ . Cycle lengths are independent copies of a positive integer  $L$ , independent of the acquired states, with finite mean. The age weights and their tails are

$$w_k = \frac{\mathbb{P}(L > k)}{\mathbb{E}L}, \quad W(n) = \sum_{k \geq n} w_k, \quad 0 < \underline{q} \leq \frac{w_{k+1}}{w_k} \leq \bar{q} < 1. \quad (68.6)$$

An initial acquisition occurs at zero; the next renewal time is not announced. Nongeometric lengths satisfying (68.6) are allowed. Write  $X_t$  for this process.

An observer keeps  $I_t$  in a set of at most  $M$  elements. Its update may use  $I_{t-1}$ , absolute time, the renewal indicator, and the acquired state at a renewal. The post-update decoder uses  $I_t$ , absolute time, and independent coding randomness, but does not separately receive the renewal indicator or a retained real-valued observation. All observationally dependent persistent variables are counted in  $I_t$ . Stochastic kernels, private randomness, arbitrary fixed real constants, and transient computation are allowed. Section 73 compares this convention with other finite-memory interfaces.

Set

$$\begin{aligned} Z_w^f(x) &= (\sqrt{w_k} f(T^k x))_{k \geq 0} \in \ell^2(\mathbb{N}_0; \mathbb{R}^m), \\ e_M^2(Z_w^f) &= \inf_{c_1, \dots, c_M} \int \min_{i \leq M} \|Z_w^f(x) - c_i\|^2 d\mu(x), \\ \mathcal{R}_M &= \inf_{\mathcal{C}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t < N} \mathbb{E} |f(X_t) - \hat{f}_t|^2. \end{aligned} \quad (68.7)$$

Centres in the quantization problem may lie anywhere in the Hilbert space. The infimum in the last line ranges over the entire preceding observer class, not just stationary or cylinder observers.

**68.2. A computable geometric profile.** For  $v \in \mathcal{L}_n$  and its suffix  $v_k := i_k \cdots i_{n-1}$  define

$$V_w(v) = \sum_{k < n} w_k r_{v_k}^\tau + W(n), \quad a_w(v) = \mu[v] V_w(v), \quad a_w(\emptyset) = 1. \quad (68.8)$$

Starting from the root, split a leaf of maximal  $a_w$  into all its admissible one-symbol successors, with a fixed rule for ties. Let  $\mathcal{T}_N$  be the last tree in this sequence having at most  $N$  leaves, let  $\mathcal{P}_N$  be its leaves, and put

$$G_N(w) = \sum_{v \in \mathcal{P}_N} a_w(v). \quad (68.9)$$

For the finitely many budgets preceding the first root split, the tree is the root. Unary splits are genuine operations. They do not increase the number of leaves, but they cannot occur in an infinite uninterrupted chain. Consequently the last tree in this definition exists.

The construction is a finite mathematical algorithm on cylinder masses, scales and renewal tails. Effective numerical evaluation additionally requires certified access to those real inputs; no computability assertion for arbitrary Borel or Hölder data is implicit.

**Theorem 68.1** (Markov renewal profile). *Under (68.1)–(68.6), there are positive constants  $c, C$ , independent of  $M$ , such that*

$$cG_M(w) \leq e_M^2(Z_w^f) \leq \mathcal{R}_M \leq CG_M(w) \quad (M \geq 1). \quad (68.10)$$

*Every randomized time-dependent observer satisfies the middle lower bound with its lower limiting average loss. A stationary deterministic observer attains the upper order.*

*More precisely, the complete admissible tree  $\mathcal{T}_N$  is suffix closed. If it has  $L$  leaves, all its vertices can be retained as states, and their number is at most*

$$(b+1)(2L-1). \quad (68.11)$$

*Its exact number is  $1 + \sum_{v \text{ internal}} d^+(v)$ , where  $d^+(v)$  is the number of children actually present. No time-depth factor is required. The constants in (68.10) may depend on the graph, distortion, separation, conditional masses, observable, and the two ratios in (68.6).*

For the pressure statement assume additionally that a Hölder potential  $\psi$  satisfies  $\log r_v = S_n\psi(\omega) + O(1)$  uniformly on  $[v]$ , and that

$$\lim_{n \rightarrow \infty} n^{-1} \log w_n = \log q < 0. \quad (68.12)$$

The inverse-derivative potential gives  $\psi$  in the one-dimensional smooth case. For similarities,  $\psi(\omega) = \log r_{\omega_0}$ .

**Theorem 68.2** (Intrinsic pressure exponent). *There are unique roots  $s_g, s_t \in (0, 1)$  of*

$$P_A(s_g\varphi + \tau s_g\psi) = 0, \quad P_A(s_t\varphi) + s_t \log q = 0. \quad (68.13)$$

*Writing  $s_* = \max(s_g, s_t)$ , one has*

$$\lim_{M \rightarrow \infty} \frac{-\log \mathcal{R}_M}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z_w^f)}{\log M} = \frac{1 - s_*}{s_*}. \quad (68.14)$$

*The spatial and survival roots exchange dominance at  $q_c = \exp\{-P_A(s_g\varphi)/s_g\}$ . This formula uses the exponential rate of the renewal tail. Polynomial factors in that tail can change the critical correction without changing either root.*

## APPENDIX 69. RENEWAL ENERGIES AND ADMISSIBLE SUFFIX TREES

The main combinatorial fact is a strict energy inequality, not strict inequality of cylinder masses. The distinction matters at vertices having a unique predecessor.

**Lemma 69.1** (Strict suffix domination). *If  $u$  is nonempty and  $uv$  is admissible, then*

$$V_w(v) - V_w(uv) \geq (1 - r_+^\tau) r_v^\tau \sum_{k < |u|} w_k > 0, \quad a_w(uv) < a_w(v). \quad (69.1)$$

*There are constants  $a_0 > 0$  and  $\rho < 1$  such that each admissible child satisfies*

$$a_0 a_w(v) \leq a_w(vi) \leq \rho a_w(v), \quad \sum_{i: vi \text{ admissible}} a_w(vi) \leq \rho a_w(v). \quad (69.2)$$

*The same inequalities hold at the root after decreasing  $a_0$ .*

*Proof.* Write  $m = |u|$ ,  $n = |v|$  and suppress the subscript  $w$ . Every suffix of  $v$  has scale at least  $r_v$ . Since the weights decrease,

$$\begin{aligned} V(v) - \left( \sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \left( \sum_{j < n} (w_j - w_{m+j}) + W(n) - W(m+n) \right) \\ &= r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

The remaining prefix terms in  $V(uv)$  are at most  $r_v^\tau \sum_{k < m} w_k r_{u_k}^\tau$ , which is at most  $r_+^\tau r_v^\tau \sum_{k < m} w_k$ . This proves the first inequality, also for  $v = \emptyset$ . Stationarity gives  $\mu[uv] \leq \mu\{\omega : \omega_{m:m+n} = v\} = \mu[v]$ . Both masses are positive, so the second inequality follows. Neither argument completes an inadmissible word by an artificial transition.

For  $|v| = n$ , appending  $i$  multiplies each old spatial scale by a factor between  $D_0^{-1}r_-$  and  $r_+$ . The old tail  $W(n)$  is replaced by  $w_n r_i^\tau + W(n+1)$ . The weight assumptions imply

$$(1 - \bar{q})W(n) \leq w_n, \quad W(n+1) \geq \underline{q}W(n).$$

It follows that

$$\ell_0 V(v) \leq V(vi) \leq \rho V(v), \quad \ell_0 = \min(D_0^{-\tau} r_-^\tau, \underline{q}), \quad \rho = 1 - (1 - r_+^\tau)(1 - \bar{q}) < 1.$$

A uniform lower bound  $\zeta > 0$  on admissible conditional cylinder masses now gives  $a_0 = \zeta \ell_0$  in (69.2). The last assertion follows by summing the masses of the children. Such a  $\zeta$  follows directly from (68.3) and the bounded potential, for admissible extensions only.  $\square$

**Lemma 69.2** (Balanced trees and controlled dilation). *Let  $L_N = |\mathcal{P}_N|$ . For all sufficiently large  $N$ ,  $N/b \leq L_N \leq N$ . If  $\theta_N$  is the energy of the last split vertex, then every leaf satisfies*

$$a_0 \theta_N \leq a_w(v) \leq \theta_N. \quad (69.3)$$

For every fixed  $B \geq 1$ , with constants allowed to depend on  $B$ ,

$$G_{\lceil BN \rceil}(w) \asymp_B G_N(w). \quad (69.4)$$

The complete tree with  $L$  leaves has at most  $(b+1)(2L-1)$  vertices.

*Proof.* Maximal split energies never increase, by (69.2). A present leaf was born with energy at least  $a_0$  times an earlier split energy, which proves the lower bound in (69.3); the upper bound follows from the final maximal split. Each split adds between zero and  $b-1$  leaves. The split following  $\mathcal{T}_N$  increases the number beyond  $N$ ; hence  $L_N > N - b + 1$ , which implies the stated coarser bound.

Here are the finiteness details needed in the presence of unary vertices. A path of  $b$  successive vertices all having outdegree one would repeat a vertex and enter a closed deterministic cycle. This contradicts primitivity and  $\lambda_A > 1$ . Thus a unary chain has length at most  $b$ . After suppressing unary vertices, any finite tree with  $L$  leaves and no internal vertex of degree one has at most  $2L-1$  vertices. Reinserting the unary chains, including an initial chain, gives the asserted bound. In particular an infinite sequence of splits with bounded leaf number is impossible.

Choose  $\ell$  such that  $A^\ell$  has all entries positive. From every last symbol there are at least two distinct continuations of length  $\ell$ . Thus every vertex has at least  $2^j$  descendants at depth  $\ell j$ . In continuing a greedy tree with threshold  $\theta_N$ , if all later leaves have energy less than  $a_0^{\ell j+1} \theta_N$ , then every original leaf must have been split through those  $\ell j$  levels. This would give at least  $2^j L_N$  later leaves. Choosing  $j$  so that  $2^j > b(B+2)$  shows that the final maximum leaf energy at budget  $\lceil BN \rceil$  is at least a fixed multiple of  $\theta_N$ . The balance estimate at that budget bounds its minimum leaf energy by another fixed multiple of  $\theta_N$ . Since  $L_{\lceil BN \rceil}$  is at least a fixed multiple of  $N$ , its energy sum is at least  $c_B N \theta_N \geq c_B G_N$ . Conversely splitting decreases the sum by (69.2), so  $G_{\lceil BN \rceil} \leq G_N$ . Increasing constants handles the finitely many initial trees.  $\square$

**Lemma 69.3** (Observable orbit cylinders). *The image  $Z_w^f(K_v)$  has squared diameter at most  $CV_w(v)$ . If two different sequences have longest common prefix  $u$ , then*

$$\|Z_w^f(x) - Z_w^f(y)\|^2 \geq cV_w(u). \quad (69.5)$$

*All constants depend only on the displayed geometric and observable hypotheses, not on the word.*

*Proof.* Introduce the auxiliary snowflake orbit distance  $d_\beta(x, y)^2 = \sum_{k \geq 0} w_k |T^k x - T^k y|^{2\beta}$ . The Hölder estimate bounds  $\|Z_w^f(x) - Z_w^f(y)\|^2$  from above by  $H^2 d_\beta(x, y)^2$ . Multiplying (68.5) at  $T^k x, T^k y$  by  $w_k$  and summing gives

$$c_o d_\beta(x, y)^2 \leq \sum_{j=0}^J \sum_{k \geq 0} w_k |f(T^{k+j}x) - f(T^{k+j}y)|^2 \leq \left( \sum_{j=0}^J q^{-j} \right) \|Z_w^f(x) - Z_w^f(y)\|^2.$$

For points in  $K_v$ , the first  $n$  terms of  $d_\beta^2$  are bounded by  $w_k r_{v_k}^{2\beta}$  and its remaining terms by  $W(n)$ . If  $u$  is their longest common prefix, then at time  $|u|$  they lie in different first-level images, and at time  $k < |u|$  their distance is at least  $D_0^{-1} g r_{u_k}$ . The first post-prefix term is  $w_{|u|} g^{2\beta}$ , and  $W(|u|) \leq w_{|u|}/(1 - \bar{q})$ . These estimates prove both assertions, including the empty prefix.  $\square$

**Lemma 69.4** (Unrestricted quantization and suffix realization). *The quantization errors satisfy  $e_M^2(Z_w^f) \asymp G_M(w)$ . Every complete greedy tree is suffix closed, including its leaves. A tree with  $L$  leaves gives a deterministic observer with at most  $(b+1)(2L-1)$  states and average loss at most  $C \sum_{v \in \mathcal{P}} a_w(v)$ .*

*Proof.* Taking one orbit representative from each leaf cylinder proves  $e_N^2 \leq CG_N$ . For the reverse direction use the leaf partition  $\mathcal{P}_{4bM}$ , which has at least  $4M$  members, apart from finitely many budgets absorbable in the constants. Around  $Z_w^f(K_v)$  take its open Hilbert tube of radius  $\delta \sqrt{V_w(v)}$ . These tubes are disjoint for a sufficiently small fixed  $\delta$ : for incomparable leaves the separation in (69.5) is at their common prefix, and appending symbols only decreases  $V_w$ . Any  $M$  Hilbert centres, whether or not on the orbit image, occupy at most  $M$  tubes. On every other cylinder the nearest-centre squared error is at least  $\delta^2 V_w(v)$ . The balance estimate shows that these unoccupied cylinders carry at least a fixed fraction of the energy sum. Hence  $e_M^2 \geq cG_{4bM} \geq c'G_M$  by (69.4). For each of the finitely many omitted budgets this error is positive: the delay condition and infinitely supported Gibbs law exclude support on finitely many orbit points.

Suppose an internal word  $uv$ , with  $u$  nonempty, is about to be split. If its suffix  $v$  has not yet been split, some current leaf  $t$  is a prefix of  $v$ . By (69.2) and (69.1),  $a_w(t) \geq a_w(v) > a_w(uv)$ , contradicting maximality. Thus every proper suffix of an internal word is already internal. If  $v = i_0 \cdots i_{n-1}$  is a leaf with  $n \geq 2$ , the word  $i_1 \cdots i_{n-2}$  is a suffix of its internal parent, so it has been split. Its admissible child  $i_1 \cdots i_{n-1}$  is therefore present. For  $n = 1$  the suffix is the root. Iteration proves closure of the entire tree; it does not require that all symbols follow all others.

Choose  $x_v \in K_v$  for every nonempty tree vertex, and decode that vertex as  $f(x_v)$ . At an acquisition store the unique leaf prefix of the observed sequence. At each nonacquisition delete the first symbol. Decode the empty word as any fixed  $f(x_*)$ , and keep it until the next acquisition. Each update stays within the same tree. If the acquired word is  $v$  of length  $n$ , the squared loss at age  $k < n$  is at most  $H^2 r_{v_k}^{2\beta}$ ; after exhaustion it is bounded by a constant. The renewal-reward formula therefore bounds the average by  $C \sum_v \mu[v] V_w(v)$ . All vertices are charged, with the count established in Lemma 69.2.  $\square$

*Proof of Theorem 68.1.* The retained posterior-orbit theorem, Theorem 74.2, applies with exact observation and Bayes floor zero. It gives  $e_M^2 \leq \mathcal{R}_M$  and the stated lower-limit assertion for arbitrary stochastic kernels. Its full proof is included in both reading editions. Lemma 69.4 gives the other lower bound. Choose a leaf budget  $N$  a fixed sufficiently small fraction of  $M$  so that (68.11) is at most  $M$ . The same lemma and fixed-budget dilation give an upper bound

$CG_N \leq C'G_M$ . A one-state observer handles the remaining bounded range of  $M$ . This proves (68.10).  $\square$

## APPENDIX 70. TWO PRESSURES ON AN ADMISSIBLE LANGUAGE

All pressures in this section are taken on  $\Sigma_A$ . A Hölder potential  $F$  has uniformly bounded oscillation of  $S_n F$  on cylinders, and its partition sums are comparable to  $e^{nP_A(F)}$ , uniformly in  $n$ . These are classical finite-state thermodynamic facts [32]. The new issue is the renewal-weighted sum of suffix scales, which is not an additive potential.

**Lemma 70.1** (Mixed partition pressure). *For  $0 < s < 1$ , under (68.12),*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{v \in \mathcal{L}_n} a_w(v)^s = \max\{P_A(s\varphi + \tau s\psi), P_A(s\varphi) + s \log q\}. \quad (70.1)$$

*Proof.* Put  $P_g(s) = P_A(s\varphi + \tau s\psi)$  and  $P_0(s) = P_A(s\varphi)$ . The first and last terms in (68.8), separately, imply

$$\sum_{|v|=n} a_w(v)^s \geq c_s \max\{w_0^s e^{nP_g(s)}, W(n)^s e^{nP_0(s)}\}.$$

For the upper bound use  $(\sum x_k)^s \leq \sum x_k^s$  and split an admissible word after  $k$  letters. Its mass and suffix scale are, within fixed factors, the exponential of the corresponding Birkhoff sums. Dropping the constraint joining the two pieces can only increase the sum. Thus

$$\sum_{|v|=n} a_w(v)^s \leq C_s \left( \sum_{k < n} w_k^s e^{kP_0(s)} e^{(n-k)P_g(s)} + W(n)^s e^{nP_0(s)} \right). \quad (70.2)$$

No factorization of arbitrary words across a forbidden transition is asserted. Given  $\delta > 0$ , the weight assumption gives  $w_k \leq C_\delta e^{k(\log q + \delta)}$ , and  $W(n) \asymp w_n$ . The upper sum has at most  $n + 1$  terms with exponential rate at most  $\max(P_g(s), P_0(s) + s \log q + s\delta)$ . The lower bound has rate at least the same maximum with  $\delta = 0$ . Let  $\delta \downarrow 0$ .  $\square$

*Proof of Theorem 68.2.* We first verify uniqueness of the roots. Every vertex has at least two continuations of a common length  $\ell$ . By the positive conditional mass bound, no particular continuation of that length has conditional probability larger than  $1 - \zeta^\ell$ . Hence  $\mu[v] \leq C\eta^{|v|}$  for an  $\eta < 1$ . The Gibbs estimate gives  $S_n \varphi \leq n \log \eta + O(1)$  on every cylinder. Similarly  $S_n \psi \leq n \log r_+ + O(1)$ . The pressure functions in (68.13) are continuous and strictly decreasing, with positive value  $\log \lambda_A$  at zero. At one their values are strictly negative, since  $P_A(\varphi) = 0$ . Their roots lie in  $(0, 1)$ .

For  $s > s_*$ , both exponents in (70.2) are negative. Choose  $q'' > q$  close enough that they remain negative with  $q''$  in place of  $q$ . Bounding  $w_k$  and  $W(k)$  by a constant times  $(q'')^k$  and summing (70.2) over all  $n$  yields

$$\sum_{v \in \mathcal{L}} a_w(v)^s < \infty. \quad (70.3)$$

For a balanced partition with  $L \asymp N$  leaves, this implies  $L(a_0 \theta_N)^s \leq C_s$ , and therefore  $G_N \leq L \theta_N \leq C'_s N^{-(1-s)/s}$ .

For the lower bound at  $s_g$ , use the Gibbs probability  $\nu_g$  for the zero-pressure potential  $s_g \varphi + \tau s_g \psi$ . On each cylinder,

$$\nu_g[v] \leq C \mu[v]^{s_g} r_v^{\tau s_g} \leq C' a_w(v)^{s_g}.$$

Summing over  $\mathcal{P}_N$  gives  $1 \leq C' L \theta_N^{s_g}$ . Balance then gives  $G_N \geq c N^{-(1-s_g)/s_g}$ .

For the survival root, a polynomial correction to  $w_n$  prevents using  $q^n$  as a uniform lower bound. Instead fix  $0 < q' < q$ . There is  $c_{q'} > 0$  with  $W(n) \geq c_{q'} (q')^n$  for all  $n$ . Let  $s_t(q')$  be the zero of  $P_A(s\varphi) + s \log q'$ , and use the Gibbs measure of this zero-pressure potential. Its cylinder masses are

bounded by  $C\mu[v]^{s_t(q')}(q')^{|v|s_t(q')}$ , hence by  $C'a_w(v)^{s_t(q')}$ . The same balanced-partition argument proves the lower bound with exponent  $(1 - s_t(q'))/s_t(q')$ . The roots vary continuously with  $q'$ , by strict monotonicity. Letting  $q' \uparrow q$  and  $s \downarrow s_*$  proves the common exponent for  $G_N$ . Theorem 68.1 transfers it to both errors. Finally  $s_t = s_g$  precisely when  $P_A(s_g\varphi) + s_g \log q = 0$ , proving the threshold formula.  $\square$

**Corollary 70.2** (Markov matrices with zero entries). *Suppose the  $h_i$  are similarities and  $\mu$  is a stationary Markov measure with transition probabilities  $p_{ij} > 0$  exactly when  $A_{ij} = 1$ . The roots in (68.13) are characterized by*

$$\text{spr}((A_{ij}p_{ij}^{s_g}r_i^{\tau s_g})_{ij}) = 1, \quad q^{s_t} \text{spr}((A_{ij}p_{ij}^{s_t})_{ij}) = 1. \quad (70.4)$$

*Only admissible entries are powered; other entries are zero. At  $s = 0$  the limiting matrix is  $A$ , not a matrix of ones.*

*Proof.* Take  $\varphi(\omega) = \log p_{\omega_0\omega_1}$  on admissible edges and  $\psi(\omega) = \log r_{\omega_0}$ . The nonuniform stationary initial factor and the terminal factor are bounded above and below, so do not change pressure. The finite transfer matrices of these edge potentials are exactly those in (70.4). Perron–Frobenius theory and Theorem 68.2 give the assertion.  $\square$

**Example 70.3** (A noninjective planar readout with a forbidden transition). On  $[0, 1]^2$  take

$$h_0(x, y) = (ry, rx), \quad h_1(x, y) = (ry + 2/3, rx), \quad r = 1/3, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Rescaling the ambient domain makes its diameter one without changing the argument. Both maps are similarities and their images have a positive horizontal gap. Give the golden-mean shift any stationary Hölder Gibbs law. For  $f(x, y) = x$ , the relation  $f(T(x, y)) = y/r$  on  $K$  gives

$$|f(x, y) - f(x', y')|^2 + |f(T(x, y)) - f(T(x', y'))|^2 = (x - x')^2 + r^{-2}(y - y')^2.$$

Thus (68.5) holds with  $J = 1$  and  $\beta = 1$ . The sequences  $000\cdots$  and  $01000\cdots$  are admissible, have the same first coordinate, and have different second coordinates. Hence  $f$  is not injective. This is a higher-dimensional geometric system with a genuinely forbidden transition and a delayed, rather than instantaneously identifying, observable.

## APPENDIX 71. CRITICAL CORRECTIONS AND A UNIFORM RENEWAL WINDOW

The exponential rate of the acquisition tail does not determine the critical second order. We give a classification in a family allowing arbitrary primitive transition graphs, every ambient dimension in the similarity construction, and delay-observable outputs.

Assume that all branch ratios equal  $r$ , and that  $\mu$  is the Parry measure of  $\Sigma_A$ . Write  $\lambda = \lambda_A$  and  $a = r^{2\beta}$ . For  $q \in (0, 1)$  and  $\kappa \geq 0$  take the actual survival law

$$\mathbb{P}(L > k) = q^k(k+1)^{-\kappa}, \quad c_{q,\kappa} = \left( \sum_{k \geq 0} q^k(k+1)^{-\kappa} \right)^{-1}. \quad (71.1)$$

Its weights satisfy (68.6), with lower ratio  $q2^{-\kappa}$  and upper ratio  $q$ . The common depth- $n$  energy scale is

$$V_{q,\kappa}(n) = c_{q,\kappa} \left[ a^n \sum_{k < n} (q/a)^k(k+1)^{-\kappa} + \sum_{k \geq n} q^k(k+1)^{-\kappa} \right]. \quad (71.2)$$

**Theorem 71.1** (Critical power–log classification). *Let  $n_M = \lfloor \log M / \log \lambda \rfloor$ . In this family,*

$$\mathcal{R}_M \asymp V_{q,\kappa}(n_M). \quad (71.3)$$

For fixed  $\kappa$  the constants can be chosen uniformly for  $q$  in a compact subinterval of  $(0, 1)$ . The critical value is  $q_c = a$ . With  $\alpha = \log(1/a)/\log \lambda$ , at this value, as  $M \rightarrow \infty$ ,

$$\mathcal{R}_M \asymp M^{-\alpha} \begin{cases} (\log M)^{1-\kappa}, & 0 \leq \kappa < 1, \\ \log \log M, & \kappa = 1, \\ 1, & \kappa > 1. \end{cases} \quad (71.4)$$

In particular the same graph, geometry, observable, pressure roots, and critical exponential survival rate admit different critical corrections. The specification of the renewal law is essential.

*Proof.* Perron eigenvectors give, uniformly in  $v \in \mathcal{L}_n$ ,  $\mu[v] \asymp \lambda^{-n}$  and  $|\mathcal{L}_n| \asymp \lambda^n$ . All these cylinders have scale (71.2). A fixed-depth complete tree has  $O(\lambda^n)$  vertices (sum the geometric level counts), is suffix closed, and gives average loss at most  $CV_{q,\kappa}(n)$  by the pointwise construction in Lemma 69.4. Choose  $n = n_M - C_0$  so that its state count is at most  $M$ .

For a lower bound choose  $n = n_M + C_1$  with at least  $4M$  cylinders. The disjoint tubes used in Lemma 69.4 then leave at least three quarters of the cylinders unoccupied by any  $M$  centres. Their masses are comparable and their scales equal, giving  $e_M^2 \geq cV_{q,\kappa}(n)$ . A fixed change in depth changes  $V$  by factors bounded above and below by (69.2)'s scale proof, uniformly for  $q$  in the specified compact set. The converse gives (71.3). This argument neither treats the Parry measure as independent nor fills forbidden transitions.

At  $q = a$ , the first term in (71.2) is  $c_{a,\kappa} a^n \sum_{j=1}^n j^{-\kappa}$ ; the tail is  $O(a^n(n+1)^{-\kappa})$ . Integral comparison gives the three possibilities in (71.4). Also  $P_A(s\varphi) = (1-s)\log \lambda$  and  $P_A(s\varphi + 2\beta s \log r) = (1-s)\log \lambda + s \log a$ . Thus the two roots coincide at  $q = a$ , as asserted.  $\square$

**Theorem 71.2** (Uniform finite-size profile window). *Fix  $\kappa \geq 0$  and put  $q_n = ae^{x/n}$ . Uniformly for  $x$  in a bounded real interval, the following limits hold:*

$$\frac{V_{q_n,\kappa}(n)}{a^n n^{1-\kappa}} \longrightarrow c_{a,\kappa} \int_0^1 t^{-\kappa} e^{xt} dt, \quad 0 \leq \kappa < 1, \quad (71.5)$$

$$\frac{V_{q_n,1}(n)}{a^n \log n} \longrightarrow c_{a,1}, \quad \kappa = 1, \quad (71.6)$$

$$\frac{V_{q_n,\kappa}(n)}{a^n} \longrightarrow c_{a,\kappa} \sum_{j \geq 1} j^{-\kappa}, \quad \kappa > 1. \quad (71.7)$$

The optimal risk is uniformly comparable to these profiles at budgets of order  $\lambda^n$ . The displayed leading coefficients are profile coefficients, not asserted exact optimal-risk coefficients.

*Proof.* The normalizing constants tend uniformly to  $c_{a,\kappa}$  by a common geometric majorant. For  $0 \leq \kappa < 1$ , split the normalized sum at  $k = \delta n$ . Away from zero, its summands form a uniformly convergent Riemann sum for  $t^{-\kappa} e^{xt}$ . The part below  $\delta n$  is bounded uniformly by  $C\delta^{1-\kappa}$ ; then let  $\delta \downarrow 0$ . The normalized tail is  $O(n^{-1})$  uniformly, using  $q_n \leq q_+ < 1$  and  $q_n^n/a^n = e^x$ . For  $\kappa = 1$ ,

$$\left| \sum_{k < n} \frac{e^{xk/n} - 1}{k+1} \right| \leq C$$

uniformly on bounded  $x$ , while the unperturbed harmonic sum is  $\log n + O(1)$ . The tail divided by  $a^n \log n$  vanishes. For  $\kappa > 1$ , dominated convergence applies to the summable sequence  $(k+1)^{-\kappa}$ , and the tail divided by  $a^n$  is  $O(n^{-\kappa})$ . Finally use the uniform comparison in Theorem 71.1.  $\square$

The classification concerns the stated Markov–Parry similarity family and the renewal tails (71.1). For unequal Hölder derivative cocycles the finite profile theorem and pressure exponent still apply; (71.4) is not imposed on that larger class. In the geometric case  $\kappa = 0$  the first window profile is  $(1-a)(e^x - 1)/x$ , continuously extended at  $x = 0$ .

## APPENDIX 72. FINITE REGISTERS WITHOUT A COMMON REGENERATION LAW

A different realization mechanism applies when observations arrive continually and the posterior itself contracts. It does not use an independent reference component of a transition kernel.

Let  $(U_t, Y_t)_{t \in \mathbb{Z}}$  be stationary, with  $U_t \in \mathbb{R}^d$  square integrable and  $Y_t$  standard Borel. Write

$$\mathcal{Y}_t^- = \sigma(Y_s : s \leq t), \quad m_t = \mathbb{E}[U_t \mid \mathcal{Y}_t^-], \quad B = \mathbb{E}\|U_0 - m_0\|^2.$$

An observer starting at zero receives  $Y_t$  at every update and retains at most  $M$  states. Its decoder receives only its post-update state, absolute time, and independent coding randomness. Let  $\mathcal{R}_M^{\text{fil}}$  be the infimum of its upper limiting average squared loss. The reference  $B$  uses the full two-sided past; initial lack of that past will be handled by contraction, not by a fictitious initial sufficient statistic.

**Theorem 72.1** (Contracting posterior realization). *Suppose  $m_0 \in L^4$  and a Borel function  $F$  has a measurable Lipschitz coefficient  $\ell(y) \geq 0$  such that*

$$\begin{aligned} m_t &= F(m_{t-1}, Y_t), & \|F(x, y) - F(x', y)\| &\leq \ell(y)\|x - x'\|, \\ \mathbb{E}[\ell(Y_t)^4 \mid \mathcal{Y}_{t-1}^-] &\leq \rho^4 < 1, & K_4 &:= \|F(0, Y_0)\|_{L^4} < \infty. \end{aligned} \quad (72.1)$$

*A uniform bound  $\ell(y) \leq \rho$  is sufficient but not necessary; individual observation updates may expand. If the law of  $m_0$  has a Lebesgue density bounded below by a positive constant on a cube of positive side length, then*

$$B + cM^{-2/d} \leq \mathcal{R}_M^{\text{fil}} \leq B + CM^{-2/d} \quad (M \geq 1). \quad (72.2)$$

*The lower bound holds for the lower limiting average loss of every randomized time-dependent observer. The upper observer is stationary and deterministic. It uses no compact support or minorization hypothesis on either the hidden dynamics or the observation process.*

*For a product register of  $(2n - 1)^d$  states,  $n \geq 2$ , its excess error has the explicit upper bound*

$$\limsup_{t \rightarrow \infty} \mathbb{E}\|m_t - \hat{U}_t\|^2 \leq \frac{d(1 + K_4/(1 - \rho))^4}{n^2(1 - \rho)^2}. \quad (72.3)$$

**Lemma 72.2** (A finite inward quantizer). *For  $n \geq 2$  define*

$$j_n(x) = \min\left(n - 1, \left\lfloor \frac{n|x|}{1 + |x|} \right\rfloor\right), \quad Q_n(x) = \text{sgn}(x) \frac{j_n(x)}{n - j_n(x)}.$$

*It has  $2n - 1$  values and satisfies*

$$|Q_n(x)| \leq |x|, \quad |x - Q_n(x)| \leq n^{-1}(1 + |x|)^2. \quad (72.4)$$

*Its coordinatewise extension to  $\mathbb{R}^d$  satisfies  $\|Q_n x\| \leq \|x\|$  and  $\|x - Q_n x\| \leq \sqrt{d} n^{-1}(1 + \|x\|)^2$ .*

*Proof.* For  $x \geq 0$  put  $u = x/(1 + x)$ , so the inverse is  $g(u) = u/(1 - u)$ . The quantized argument  $j_n(x)/n$  is at most  $u$  and differs from it by less than  $1/n$ , including the saturated last bin. The increasing derivative  $g'(u) = (1 - u)^{-2}$  bounds the difference by  $(1 + x)^2/n$ . Odd reflection proves the scalar assertion. Sum the squared coordinate errors and use  $|x_i| \leq \|x\|$  for the vector bound.  $\square$

*Proof of Theorem 72.1.* Conditional orthogonality, also after adjoining randomness independent of the experiment, gives

$$\mathbb{E}\|U_t - \hat{U}_t\|^2 = B + \mathbb{E}\|m_t - \hat{U}_t\|^2. \quad (72.5)$$

At any fixed  $t$ , freeze all independent randomization. The decoder has at most  $M$  possible values, even though the encoder has seen a long observation history. On a cube where the density of  $m_t$  is at least  $c_0$ , take balls of radius  $\delta M^{-1/d}$  around those values. A sufficiently small  $\delta$ , independent of their locations, makes their total intersection volume at most half the cube's volume. The

remaining volume incurs squared distance at least  $\delta^2 M^{-2/d}$ . Integrating the density lower bound and then the randomization gives the lower bound at each time, and hence for lower limiting averages.

For the upper bound start from  $z_{-1} = 0$  and iterate

$$x_t = F(z_{t-1}, Y_t), \quad z_t = Q_n x_t, \quad \hat{U}_t = z_t. \quad (72.6)$$

Only the index of  $z_t$  is stored;  $x_t$  is transient. The values of  $Q_n$  are fixed rational vectors, part of the fixed decoder table. Conditional fourth-moment contraction gives  $\|\ell(Y_t)z_{t-1}\|_{L^4} \leq \rho\|z_{t-1}\|_{L^4}$ . The inward property and Minkowski's inequality therefore give, without independence of successive observations,

$$\|z_t\|_{L^4} \leq \rho\|z_{t-1}\|_{L^4} + K_4 \leq A_4 := K_4/(1 - \rho), \quad \|x_t\|_{L^4} \leq A_4.$$

Lemma 72.2 therefore implies  $\|x_t - Q_n x_t\|_{L^2} \leq \sqrt{d}(1 + A_4)^2/n$ . Couple the exact stationary recursion to the same observations. The previous error is measurable with respect to  $\mathcal{Y}_{t-1}^-$ ; conditional Cauchy–Schwarz implies  $\mathbb{E}[\ell(Y_t)^2 \mid \mathcal{Y}_{t-1}^-] \leq \rho^2$ . Thus

$$\|m_t - z_t\|_{L^2} \leq \rho\|m_{t-1} - z_{t-1}\|_{L^2} + \sqrt{d}(1 + A_4)^2/n.$$

The finite initial error is multiplied by  $\rho^{t+1}$ . Summing this recursion proves (72.3). Equation (72.5) gives the upper average bound. For large  $M$  choose  $n = \lfloor (M^{1/d} + 1)/2 \rfloor$ , so  $(2n - 1)^d \leq M$  and  $n \asymp M^{1/d}$ . For the remaining finite range, the one-state output zero has finite loss, and an enlarged constant completes (72.2).  $\square$

**Corollary 72.3** (An unbounded hidden Gaussian chain). *Let*

$$X_t = aX_{t-1} + \xi_t, \quad Y_t = X_t + \eta_t, \quad 0 < |a| < 1, \quad \xi_t \sim N(0, v), \quad \eta_t \sim N(0, \omega), \quad (72.7)$$

where  $v, \omega > 0$ , the noises are independent, and  $X$  has its stationary law. Let  $P$  be the unique positive solution of

$$P = \frac{\omega(a^2 P + v)}{a^2 P + v + \omega}. \quad (72.8)$$

Then  $\mathcal{R}_M^{\text{fil}} = P + \Theta(M^{-2})$ . For  $d$  independent such coordinates the law is  $\sum_i P_i + \Theta(M^{-2/d})$ . For every fixed  $k \geq 1$ , the hidden  $k$ -step kernel admits no nonzero finite measure dominated by all its transition probabilities.

*Proof.* Put  $S = v/(1 - a^2)$ . Conditional Gaussian projection gives the one-step prediction variance  $a^2 P + v$  and the update (72.8). The map on its right sends  $[0, S]$  into itself and has derivative at most  $a^2 < 1$ ; its unique fixed point lies in  $(0, S)$ . Iterating from a finite past and letting that past recede gives the stationary conditional variance  $P$  and the mean recursion

$$m_t = (1 - K)am_{t-1} + KY_t, \quad K = \frac{a^2 P + v}{a^2 P + v + \omega}.$$

This passage is also the  $L^2$  martingale convergence of the increasing observation sigma fields. Gaussian conditioning supplies the displayed finite-past recursion at each step. Thus no distributional filter approximation is being substituted for the true conditional mean. The mean is Gaussian with variance  $S - P > 0$ , so has all required moments and a strictly positive density on every bounded interval. The contraction constant is  $|(1 - K)a| < 1$ . Theorem 72.1 applies; independent coordinates give the vector statement with the Euclidean contraction constant equal to the largest coordinate constant.

The hidden  $k$ -step transition from  $x$  is a Gaussian with mean  $a^k x$  and fixed finite positive variance. If a nonzero finite measure  $\nu$  were dominated by every such probability, some bounded Borel set  $E$  would have  $\nu(E) > 0$ . Sending  $x$  to infinity gives  $Q^k(x, E) \rightarrow 0$ , a contradiction. The

proof of the risk law therefore cannot be an application of a global one-step or fixed-multistep common-reference minorization.  $\square$

The contraction and Gaussian conditioning in this section are classical mechanisms. The quantitative assertion is the matched error in the number of values of the *whole persistent register*, with a uniform-in-time implementation on an unbounded state space. Constants in (72.3) display their deterioration as  $\rho \uparrow 1$ . No assertion at  $\rho = 1$ , nor a reinterpretation of this theorem as the  $\varepsilon \downarrow 0$  limit of a minorization bound, is needed.

### APPENDIX 73. OPERATIONAL CHANGES OF THE FINITE-MEMORY INTERFACE

The following comparison is distribution-free. In particular it applies to both realization mechanisms above.

**Theorem 73.1** (Finite interface equivalence). *Consider any sequential experiment with a standard Borel update input. Suppose an event  $E_t$  in an alphabet of size  $K < \infty$  is available to the update. Let  $R_M^{\text{side}}$  be the optimum when the post-update decoder also receives  $E_t$ , and let  $R_M^{\text{reg}}$  be the optimum when it does not. With the same choices of time and independent randomness, for every nonnegative path loss and every horizon, and likewise for upper or lower limiting average loss,*

$$R_{KM}^{\text{reg}} \leq R_M^{\text{side}} \leq R_M^{\text{reg}}. \quad (73.1)$$

*These inequalities arise from exact pathwise simulations. An  $M$ -value register has an equivalent binary checkpoint encoding in  $\lceil \log_2 M \rceil$  bits with unused strings allowed. Several finite registers of sizes  $M_1, \dots, M_s$  are equivalent to one register of size  $\prod_i M_i$ .*

*Proof.* An observer can ignore a free event, giving the second inequality. For the first, store the pair  $(I_t, E_t)$ . Its transition discards the previous event, performs the original update using the current input, and stores the new event. The original decoder is then evaluated on this pair. The outputs agree for each input and each realization of all coding random variables. Thus all path losses and their expectations agree, not just a selected one-time distortion. Applying a fixed bijection to the finite state set transports the transition kernels and decoder to binary strings or to a product state set. The inverse bijections give the remaining assertions.  $\square$

Consequently a positive squared-error exponent is invariant under any fixed finite event interface, fixed regrouping of finite registers, and finite-state renaming. When the risk or its Bayes excess satisfies a fixed-budget-dilation comparison, its full order is invariant as well. For a binary checkpoint budget  $B$  the corresponding register scale is  $M = 2^B$ ; this changes a polynomial in  $M$  to an exponential in  $B$ , rather than changing the theorem's physical resource silently.

**Example 73.2** (An unbounded side interface is a different experiment). Let  $X_t$  be independent uniform random variables on  $[0, 1]$  and let the current update input be  $Y_t = X_t$ . A decoder with the real-valued  $Y_t$  as a free post-update side input has zero loss with one persistent state. Without that side input the optimal  $M$ -state loss is  $1/(12M^2)$ : midpoint quantization attains it, and for cells of masses  $p_i$  the uniform interval moment inequality gives  $\sum_i p_i^3/12 \geq 1/(12M^2)$ . Randomization does not improve the pointwise nearest-centre minimum. A finite  $K$  in (73.1) cannot be replaced by an arbitrary standard Borel alphabet. The distinction is the presence of a new measurement at the decoding boundary, not a hidden age-state convention.

### APPENDIX 74. POSTERIOR ORBITS AND FINITE-STATE REALIZATION

Let  $E$  and  $\mathcal{Y}$  be standard Borel spaces, let  $\mu$  be a probability on  $E$ , and let  $K$  be a probability kernel from  $E$  to  $\mathcal{Y}$ . Let  $T : E \rightarrow E$  and  $f : E \rightarrow \mathbb{R}^r$  be Borel, with  $\|f\| \leq F < \infty$ . At each renewal a fresh pair  $(U, Y)$  has law  $\mu(du)K(u, dy)$ . The observer acquires  $Y$ , not necessarily  $U$ . Between

renewals the hidden state follows  $T$ . The successive cycle lengths  $L_0, L_1, \dots$  are independent copies of a positive integer-valued random variable  $L$ , independent of all acquired pairs, and

$$\ell = \mathbb{E}L < \infty, \quad s_k = \mathbb{P}(L > k), \quad w_k = \frac{s_k}{\ell}, \quad W(n) = \sum_{k \geq n} w_k. \quad (74.1)$$

Thus  $\sum_{k \geq 0} w_k = 1$ . There is an acquisition at time 0. Subsequent acquisition times are  $\tau_j = L_0 + \dots + L_{j-1}$ . At time  $t = \tau_j + k < \tau_{j+1}$ , the target is  $f(T^k U_j)$ . The renewal indicator is observed, but the length of the current cycle is not announced at its beginning.

An  $M$ -state observer retains  $I_t \in \{1, \dots, M\}$ . Its update is an arbitrary Borel stochastic kernel of the preceding state, absolute time, renewal indicator, and, at a renewal only, the new observation. After the update, the decoder has the state, absolute time, and independent public randomness. It does not separately retain the last observation, the renewal indicator, or the renewal age. Private randomness is allowed, but any data-dependent persistent register, including an age or a random-seed position, is part of  $I_t$ . Program descriptions, real constants, and transient computation are not charged to  $M$ . This is a persistent-cardinality constraint, not a bound on total computational storage.

Choose simultaneous Borel versions

$$m_k(y) = \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0} \in \mathcal{H} := \ell^2(\mathbb{N}_0; \mathbb{R}^r). \quad (74.2)$$

A countable union of null sets makes these versions simultaneous; values there may be set to zero. Write

$$e_M^2(Z) = \inf_{c_1, \dots, c_M \in \mathcal{H}} \mathbb{E} \min_{i \leq M} \|Z(Y) - c_i\|_{\mathcal{H}}^2, \quad B = \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \quad (74.3)$$

The number  $B$  is the unavoidable error of the complete acquisition history. Let  $A_t = t - \max\{\tau_j : \tau_j \leq t\}$ , and let  $Y_t^*$  be the observation at that last renewal. Define

$$E_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2, \quad (74.4)$$

$$R_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|f(X_t) - \hat{f}_t\|^2. \quad (74.5)$$

Here and below an infimum is over the specified machine class; it need not be attained. All statements about  $e_M^2$  use squared distortion.

**Lemma 74.1** (Kernel realization). *The preceding machine class can be realized using a countable family of independent uniform random variables, independent of the experiment. After fixing this family and the history before a renewal, the first  $n$  outputs along a continuation with no further renewal belong to a set of at most  $M$  vectors in  $(\mathbb{R}^r)^n$ .*

*Proof.* For a finite-state transition kernel the inverse cumulative probabilities give a Borel function of the input and one uniform variable. For a general standard Borel output, embed its range as a Borel subset of  $[0, 1]$ , use the measurable inverse distribution function, and map back; a fixed default value handles the null exceptional set. Applying this construction to each time-indexed kernel uses only countably many uniforms. The product probability space supplies them jointly. Public uniforms are included in the same construction, with their prescribed sharing between update and decoder. Once they are fixed, the update and decoder are deterministic. Given the state immediately after a renewal, the no-renewal input sequence and absolute times determine the subsequent states and outputs. There are only  $M$  possible starting states. The argument uses a fixed time-indexed family, not an uncharged data-dependent position in a random tape.  $\square$

**Theorem 74.2** (Posterior-orbit converse and realization). *For the experiment just defined,*

$$R_M^{\text{av}} = B + E_M^{\text{av}}, \quad e_M^2(Z) \leq E_M^{\text{av}}. \quad (74.6)$$

*In fact every randomized, time-dependent  $M$ -state machine satisfies the latter lower bound with its  $\liminf$  time-average excess loss. Conversely, for all integers  $N, n \geq 1$ ,*

$$E_{Nn+1}^{\text{av}} \leq e_N^2(Z^{<n}) + F^2W(n) \leq e_N^2(Z) + F^2W(n), \quad (74.7)$$

*where  $Z^{<n}$  is the first  $n$  coordinates of  $Z$ . The upper bound is realized, up to an arbitrary positive error, by a stationary deterministic machine with states  $\{1, \dots, N\} \times \{0, \dots, n-1\} \cup \{\partial\}$ . In particular, for  $M \geq 2$ ,*

$$e_M^2(Z) \leq E_M^{\text{av}} \leq \inf_{1 \leq n \leq M-1} \left\{ e_{\lfloor (M-1)/n \rfloor}^2(Z) + F^2W(n) \right\}. \quad (74.8)$$

*Proof.* Conditional on the full acquired history, the renewal indicators, and independent coding randomness, the conditional mean of  $f(X_t)$  is  $m_{A_t}(Y_t^*)$ . Indeed, the current acquired pair is independent of previous cycles, and survival of its cycle is independent of that pair. Conditional orthogonality gives, for every machine,

$$\mathbb{E}\|f(X_t) - \hat{f}_t\|^2 = \mathbb{E}\|f(X_t) - m_{A_t}(Y_t^*)\|^2 + \mathbb{E}\|m_{A_t}(Y_t^*) - \hat{f}_t\|^2.$$

The Cesàro renewal-reward identity makes the average of the first term tend to  $B$ . Bounded rewards and  $\mathbb{E}L < \infty$  suffice, including arithmetic or deterministic cycle lengths. This proves the equality in (74.6).

Fix  $n$ . At a renewal at time  $s$ , condition on the past and then use Lemma 74.1. Along the first  $n$  no-renewal steps the outputs are chosen from at most  $M$  decoder strings. The new  $Y$  has its original marginal law under this conditioning. Since survival to age  $k$  has probability  $s_k$ , the expected excess reward in those steps is at least

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_i \sum_{k < n} s_k \|m_k(Y) - c_{i,k}\|^2 = \ell e_M^2(Z^{<n}).$$

Randomizing the selection of a string cannot reduce its pointwise minimum. Integrating the conditioned variables preserves the bound. Let  $u_s = \mathbb{P}(s \text{ is a renewal})$ . Rewards arising from distinct last renewals are disjoint. Summing over  $0 \leq s \leq N-n$  therefore gives a lower bound  $\ell e_M^2(Z^{<n}) \sum_{s=0}^{N-n} u_s$  for the total excess loss up to time  $N-1$ . The elementary renewal theorem, in its Cesàro form, gives  $N^{-1} \sum_{s < N} u_s \rightarrow 1/\ell$ . Consequently the lower limit of average loss is at least  $e_M^2(Z^{<n})$ .

Truncation cannot increase quantization error. Conversely, extending a codebook for the truncation by zero gives

$$e_M^2(Z^{<n}) \leq e_M^2(Z) \leq e_M^2(Z^{<n}) + F^2W(n).$$

Letting  $n \rightarrow \infty$  proves the converse, without a compactness assumption on the orbit image or a restriction to on-image centres.

For the upper bound choose an  $\varepsilon$ -optimal  $N$ -centre codebook for  $Z^{<n}$ . Coordinatewise projection permits its  $k$ th coordinate to be written  $\sqrt{w_k}c_{i,k}$  with  $\|c_{i,k}\| \leq F$ ; a zero-weight coordinate is immaterial. At a renewal, choose the first nearest centre, a Borel choice, and set the state to  $(i, 0)$ . At a nonrenewal change  $(i, k)$  to  $(i, k+1)$  until  $k = n-1$ , after which the state is  $\partial$ . The latter state is absorbing between renewals. Decode  $(i, k)$  as  $c_{i,k}$  and  $\partial$  as zero. The average excess reward is at most  $e_N^2(Z^{<n}) + \varepsilon + F^2W(n)$  by the same renewal-reward identity. This is an actual finite-state update; its age information is precisely the charged second coordinate. Let  $\varepsilon \downarrow 0$ . Unused states are permitted, which proves the last assertion.  $\square$

**Corollary 74.3** (Invariance of the quantization exponent). *Suppose  $W(n) \leq A \exp(-cn^\beta)$  for some  $A, c, \beta > 0$ , and*

$$\alpha = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z)}{\log M} \quad \text{exists in } (0, \infty). \quad (74.9)$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log E_M^{\text{av}}}{\log M} = \alpha. \quad (74.10)$$

If  $e_M^2(Z) \asymp M^{-\alpha}$ , more explicitly

$$c_1 M^{-\alpha} \leq E_M^{\text{av}} \leq C_1 M^{-\alpha} (\log M)^{\alpha/\beta} \quad (M \geq 2),$$

with constants depending on the displayed hypotheses. For geometric renewals with parameter  $p > 0$ , the exponent conclusion also holds for the infimum of the supremum-in-time excess risk.

*Proof.* Choose any  $s > \alpha$  and  $n = \lceil ((s/c) \log M)^{1/\beta} \rceil$ . Then  $\log[(M-1)/n]/\log M \rightarrow 1$ , whereas  $W(n) \leq AM^{-s}$ . Apply (74.8); the two logarithmic bounds converge to  $\alpha$ . Under the two-sided power bound the same substitution gives the explicit inequalities, enlarging constants for finitely many small  $M$ .

For geometric renewals put  $q = 1 - p$ , so  $w_k = pq^k$ . A stationary reset-based machine has a nonnegative age-loss sequence  $d_k$ . With an initial acquisition at zero its loss at time  $t$  is

$$p \sum_{k < t} q^k d_k + q^t d_t \leq p^{-1} \sum_{k \geq 0} w_k d_k.$$

Use the upper machines in Theorem 74.2; the converse follows because a supremum is at least the lower time-average.  $\square$

The corollary is an exponent theorem, not a uniform equivalence of errors for arbitrary orbits. It applies to arbitrary Borel dynamics and arbitrary noisy or partial reset observations. Its geometric invariant is the quantization exponent of the *posterior* future orbit; the tail assumption controls the cost of making a finite codebook causal. Computing that invariant in a particular dynamical system remains a separate mathematical task.

**Proposition 74.4** (Exact autonomous realization). *Let a ray mean a sequence  $c = (c_k)_{k \geq 0}$  in the ball  $\{z \in \mathbb{R}^r : \|z\| \leq F\}$ . Among stationary deterministic machines, the optimal average excess risk is*

$$\inf_{\substack{|C| \leq M \\ SC \subseteq C}} \mathbb{E} \min_{c \in C} \sum_{k \geq 0} w_k \|m_k(Y) - c_k\|^2, \quad (Sc)_k = c_{k+1}. \quad (74.11)$$

The reset update may depend on the previous state; allowing that dependence does not alter this infimum.

*Proof.* For a fixed autonomous no-renewal update  $a$  and decoder  $d$ , a state  $i$  determines the ray  $(d(a^k i))_{k \geq 0}$ . These at most  $M$  rays are closed under  $S$ . Any reset rule selects one of them, possibly using the previous state; its reward is no smaller than the pointwise minimum over all the rays. Conversely, for a finite shift-closed set, retain the chosen ray as the state, apply  $S$  on a nonrenewal, and decode its first coordinate. The first nearest ray is a Borel reset rule independent of the previous state. Renewal reward proves the formula. This is a characterization of the stated autonomous class, not an assertion that all optimal randomized nonstationary machines are autonomous.  $\square$

**Example 74.5** (Orbit cardinality does not include an age register). Let  $E = \{0, 1\}$ ,  $U = 1$  deterministically,  $T(0) = T(1) = 0$ , and  $f(x) = x$ . Use geometric renewals with  $0 < p < 1$  and a constant reset observation. The posterior orbit is the single ray  $(1, 0, 0, \dots)$ , so  $e_1^2(Z) = 0$  and  $B = 0$ . Nevertheless a one-state decoder, having only absolute time and independent public coins,

does not know whether a renewal occurred at that time. For every  $t \geq 1$  the minimum squared error is  $p(1-p)$ , since the target is an independent Bernoulli( $p$ ) renewal indicator. Two states suffice for zero error: write 1 at each renewal and 0 otherwise. Thus the orbit converse cannot be reversed with the same cardinality without a realizability condition. The finite-chain overhead in (74.7) and the closure in (74.11) address a genuine obstruction.

#### APPENDIX 75. A FINITE-REGISTER REFINEMENT OF THE CURRENT CONTACT EXPERIMENT

This application uses the actual known-mark count-to-contact protocol of A2 revision 118 [42, Theorems on contact experiments and quantized scores]. It is independent of the preceding dynamical hypotheses. We give the probability estimates explicitly so that no historical local-limit assertion is needed.

Fix  $\lambda^0 \in (0, \infty)^m$ , a matrix  $B \in \mathbb{R}^{m \times p}$ , a compact parameter set  $H \subset \mathbb{R}^p$ , and a rank- $r$  matrix  $C \in \mathbb{R}^{r \times m}$ , with  $r \geq 1$ . Observe independent counts

$$N_j \sim \text{Pois}(N\lambda_j^0 + \sqrt{N}(Bh)_j), \quad h \in H, \quad S_N = C(N - N\lambda^0)/\sqrt{N}.$$

All means are positive for sufficiently large  $N$ . Put  $\Lambda = \text{diag}(\lambda^0)$ ,  $A_* = CB$ , and  $\Gamma = C\Lambda C^\top > 0$ . In A2's notation,  $B = B_* = \Lambda D^\top (D\Lambda D^\top)^{-1}$  and  $C = F^\top L D$ , so  $A_* = F^\top L$ . The exposure centre, mark law, compression and range frame are fixed before observing the counts. They are not estimated by an uncharged pilot.

For integers  $J \geq 1$  and  $R \geq 1$ , partition  $[-R, R]^r$  into  $J^r$  cubes of side  $\eta = 2R/J$ . Retain the cell containing  $S_N$ , or one overflow symbol. This is a deterministic physical statistic with a designated alphabet of  $J^r + 1$  labels (some may have zero probability). For experiments indexed by  $h \in H$ , write  $\Delta$  for the maximum of the two deficiencies in total variation, with parameter-independent randomization kernels.

**Theorem 75.1** (Joint sample and label bound). *There are positive constants  $C_0, c_0, R_0$ , depending on the fixed matrices, centre and compact parameter set, such that, for  $R \geq R_0$  and  $0 < \eta \leq 1$ , the finite-label experiment  $\mathcal{Q}_{N,J,R}$  satisfies*

$$\Delta(\mathcal{Q}_{N,J,R}, \{N(A_*h, \Gamma) : h \in H\}) \leq C_0 \left\{ N^{-1/5} + \frac{J}{R\sqrt{N}} + \frac{R}{J} + e^{-c_0 R^2} \right\}. \quad (75.1)$$

Thus, for  $M \geq 2$ , one may impose a prescribed register budget by choosing  $J = \lfloor (M-1)^{1/r} \rfloor$ , whenever  $2R/J \leq 1$ . In particular, for sufficiently large fixed  $K$ , the choices  $R = K\sqrt{\log N}$  and  $J = \lceil 2RN^{1/4} \rceil$  give deficiency  $O(N^{-1/5})$  with

$$M = O(N^{r/4}(\log N)^{r/2}). \quad (75.2)$$

The rank  $r$  is the rank of the actual contact score. This is a sufficient finite-label budget, not a claim of its minimax necessity.

*Proof.* Adjoin independent  $U_j$  uniform on  $(-1/2, 1/2)$  for comparison only. The law of  $X_N^\circ = (N - N\lambda^0 + U)/\sqrt{N}$  is within  $CN^{-1/5}$  in total variation of  $N(Bh, \Lambda)$ , uniformly on  $H$ . Here is a direct bound. In one coordinate let  $\mu_N = N\lambda_j^0 + \sqrt{N}(Bh)_j$ . On cells with  $|k - \mu_N| \leq \sqrt{\mu_N}N^{1/10}$ , Stirling's formula with its  $O(k^{-1})$  remainder gives the ratio of the piecewise constant jittered Poisson density to the normal density with matching mean and variance as  $1 + O(N^{-1/5})$ . Indeed the logarithmic error is  $O((1 + |z|^3)/\sqrt{N})$  for  $|z| \leq CN^{1/10}$ ; variation of the normal within a cell contributes only a smaller term. The discarded mass is  $O(N^{-1/5})$  by Chebyshev's inequality, uniformly in  $h$ . Replacing variance  $\mu_N/N$  by  $\lambda_j^0$  costs  $O(N^{-1/2})$  by the normal density derivative bound. Product total variation is at most the sum of the coordinate errors. This proves the assertion. Applying  $C$  contracts total variation, so  $S_N^\circ = CX_N^\circ$  is within  $CN^{-1/5}$  of  $N(A_*h, \Gamma)$ .

Under this coupling,  $\|S_N - S_N^\circ\|_\infty \leq d_N = C_1/\sqrt{N}$ . The finite quantizers differ only when a coordinate crosses a grid boundary, including a boundary of the retained cube. For a normal marginal density  $f$  and any translated grid,

$$\eta \sum_{j \in \mathbb{Z}} \sup_{[j\eta, (j+1)\eta]} f \leq 1 + \eta \int |f'|.$$

The total probability within  $d_N$  of that grid is therefore at most  $Cd_N/\eta$  if  $d_N \leq \eta/2$ . For larger  $d_N$  the same bound follows from the trivial bound one. A union bound and the preceding total variation comparison show that the physical labels and the labels of the limiting Gaussian have distance at most  $C(N^{-1/5} + d_N/\eta)$ .

In the reverse direction, from a finite label generate a uniform point in its cell, and send the overflow label to a fixed point. On the retained cells the resulting histogram differs from the normal density in  $L^1$  by at most  $C\eta \int \|\nabla f\| \leq C'\eta$ ; this follows by integrating the mean-value estimate over pairs of points in each cube. Since  $H$  is compact and  $\Gamma$  fixed positive definite, the Gaussian mass outside  $[-R, R]^r$  is at most  $Ce^{-c_0 R^2}$  for  $R \geq R_0$ . Its replacement by the overflow point costs at most this mass in total variation. Quantization in the other direction is a fixed kernel. Composing these two comparisons proves (75.1). Choosing the stated  $R, J$  makes  $d_N/\eta$  and  $\eta$  of order  $N^{-1/4}$  and the tail at most  $N^{-1/5}$ , proving (75.2).  $\square$

For every bounded decision problem, these parameter-independent kernels also transfer the corresponding optimal risks with additive error bounded by (75.1) times the loss range. The physical statistic remains deterministic; the random kernels occur only in the comparison of experiments. Across contact strata the rank and conditioning of  $CAC^T$  must be reevaluated. The constants are not uniform through a collapsing positive eigenvalue. The conductor flags and nonreduced determinant powers of the current A2 edition specify algebraic failure schemes. They do not, without this specified observation protocol, supply the Gaussian covariance or a noise exponent. This application consequently uses the current statistical experiment, not an analogy with its scheme multiplicity.

## APPENDIX 76. INTRODUCTION

A finite representation of a present observation need not be a finite representation of the predictions that observation will support. Expansion separates initially close states, and an interval without new data forces an observer to retain information about several future readouts. There are consequently two questions. One asks for the geometry of the weighted future orbit. The other asks whether a code for that geometry admits a closed update on a finite persistent state space. This paper gives matching answers on separated Gibbs repellers and extends the resulting prediction laws to acquisition kernels that retain hidden memory.

**76.1. The principal results.** Let  $T$  be the expanding map defined by finitely many separated  $C^{1+\gamma}$  inverse branches, let  $\mu$  be a Gibbs probability, and suppose an acquisition is followed by  $k$  unobserved iterates with probability proportional to  $q^k$ . Write  $\varphi$  for the normalized Gibbs potential and  $\psi$  for the logarithmic derivative of an inverse branch. The principal results identify the optimal  $M$ -state squared prediction risk  $R_M$  with a finite cylinder profile  $G_M(q)$ :

$$R_M \asymp e_M^2(Z_q) \asymp G_M(q), \quad Z_q(x) = (\sqrt{1-q} q^{k/2} T^k x)_{k \geq 0}.$$

Here  $G_M$  is obtained by successively splitting a cylinder of largest mass-weighted orbit diameter. It is a finite construction from cylinder probabilities and branch derivatives, not a restatement of an optimization over orbit codebooks.

Theorem 77.1 supplies this equivalence with the same state budget, up to constants. Its essential realization lemma is that the entire greedy tree is suffix closed: a full  $b$ -ary tree with  $N$  leaves

needs exactly  $(bN - 1)/(b - 1)$  designated states. The proof uses stationarity of the Gibbs law to compare the energy of a word with the energy of any suffix. This is different from simply attaching an age register to each codeword, which in general costs the number of codewords times the truncation depth. A disjoint-tube estimate on the orbit image supplies the matching lower bound for arbitrary Hilbert-space centres and hence for all randomized, time-dependent causal observers.

Theorem 77.2 computes the exponent of this profile. If  $P$  denotes full-shift pressure, let

$$P(s_g\varphi + 2s_g\psi) = 0, \quad P(s_t\varphi) + s_t \log q = 0.$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log R_M}{\log M} = \frac{1 - \max\{s_g, s_t\}}{\max\{s_g, s_t\}}.$$

The two roots distinguish spatial resolution from survival through unobserved expansion. Their pressure balance determines the transition, including nonconstant derivative cocycles and nonuniform Markov branch laws. A variational formula expresses them through entropy, information and expansion rates. Positive-matrix formulas and finite pressure enclosures make the invariant explicit in specialized and general Hölder data, respectively.

Theorem 80.1 addresses dependence across acquisitions. The state transition at an acquisition may retain an arbitrary component of the preceding hidden state, subject to a common invariant law and a uniform minorization. A lower bound is obtained by conditional domination, not by declaring the acquisition to be a regeneration. A resetting observer has exactly the same expected loss as in the independent reference experiment. These two facts transfer the intrinsic pressure law and the noisy expanding-torus law to genuinely dependent experiments. In the latter case old observations can affect the current posterior; no identification of that posterior with a latest-observation posterior is needed.

**76.2. Resource convention.** The constrained resource is the cardinality of one persistent register. The program, fixed real constants, absolute clock and transient workspace are separate resources. Every data-dependent quantity used after a checkpoint must be encoded in the register. In particular, an acquisition indicator is available to the update kernel but not separately to the post-update decoder. Changing this interface can change exact zero-error cardinalities, as Example 74.5 demonstrates. All principal lower bounds allow stochastic kernels and arbitrary time dependence. Our upper constructions are stationary finite-state machines. We prove their order optimality, not the exact optimality of stationary policies for every member of the unrestricted class.

**76.3. Relation to the literature.** Classical quantization, causal coding and dynamical information theory provide different pieces of the problem. They should not be identified by a shared use of the word “memory.”

Linder and Yüksel [33] formulate zero-delay coding of vector Markov sources by controlled quantizers; their encoder and decoder can use transmitted histories or posterior states. Wood, Linder and Yüksel [34, Theorem 3] prove deterministic stationary policy optimality for finite irreducible aperiodic Markov sources. A stationary policy on a belief space is not, by itself, a machine with a fixed finite number of persistent values. Here acquisitions are intermittent and all retained observation-dependent information is charged to that finite number. This explains why a separate suffix-cardinality argument is necessary.

Greedy prefix-tree construction has classical antecedents in Tunstall’s variable-to-fixed coding [37]. The present priority is the mass-weighted *orbit distortion*, not merely word probability. Neither greedy tree construction nor the elementary vertex count is claimed as a new coding principle; the issue is to prove that this particular distortion tree is shift closed and matches the unrestricted causal risk.

Hilbert-space functional quantization is also classical. For example, Luschgy and Pagès [35] derive sharp Gaussian functional-quantization asymptotics from covariance eigenvalue decay. We use finite-centre Hilbert quantization as a converse, but must additionally compute a non-Gaussian dynamical orbit geometry and realize its codes causally. Pressure formulas for spatial quantization on cookie-cutter sets already occur in Roychowdhury [36]. The new calculation concerns the *weighted future orbit*: its partition exponent is the maximum of a spatial pressure and a survival pressure, and its adaptive codes have linear-sized suffix closure. The spatial pressure alone is not presented as a new principle.

Finite-rate control [26, 27, 28] constrains communication over time, whereas dynamical rate distortion [29] constrains information in orbit representations. Those resource models are different from the persistent-cardinality model here. The following comparison records the particular distinctions used in the proofs.

| Reference or result           | Constrained object                                | Structural conclusion used for comparison                                                                |
|-------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Linder–Yüksel                 | Per-time transmitted symbol; vector Markov source | Existence and structure of causal quantizer policies on posterior state spaces.                          |
| Wood–Linder–Yüksel, Theorem 3 | Zero-delay symbols; finite Markov source          | Deterministic stationary policy optimality and finite-horizon comparison.                                |
| Luschgy–Pagès                 | Static finite-centre Hilbert code                 | Gaussian quantization asymptotics from covariance eigenvalues; no finite-state update constraint.        |
| Roychowdhury                  | Spatial code for a cookie-cutter measure          | Quantization dimension related to thermodynamic temperature; no future survival cost.                    |
| Theorems 77.1–77.2            | All persistent observer states                    | Matched cylinder profile, arbitrary-coder converse, linear suffix realization and two-pressure exponent. |
| Theorem 80.1                  | Same finite register, dependent acquisitions      | Conditional minorization lower bound and exact expected-loss transfer for resetting observers.           |

**76.4. Organization and dependencies.** Sections 77–79 state and prove the intrinsic profile and pressure theorem. Section 80 proves the dependent-acquisition comparison. The posterior-orbit and noisy expanding results needed in these arguments are proved in full in Sections 74–81. Those sections also retain the exact uniform critical crossover for the integer expanding experiment.

The subsequent quantitative development retains the calibrated collision-moment theorem, the exact Gaussian nuisance quotient, and the checkpoint, filtering and control results, followed by the full measurable foundations. These have independent assumptions; the pressure theorem does not rely on the collision theorem or on a historical spectral, particle, or operator limit. The separate principal-theorem edition is compiled from the same core files and includes every proof on the causal-memory dependency chain. The complete edition remains the preservation and cross-reference record for the full Foundations I manuscript. The foundational connection is precise: acquired-law quantization provides the converse, the suffix theorem provides the resource-aware causal realization, and thermodynamic pressure computes the resulting invariant. The assumptions of a separated finite-branch repeller and of an invariant minorized acquisition kernel are proved or stated where they are used; they are not replacements for the model-specific hypotheses of a Sinai, hard-particle, or unbounded- operator problem.

## APPENDIX 77. AN INTRINSIC LAW ON GIBBS REPELLERS

Let  $b \geq 2$ , let  $0 < \gamma \leq 1$ , and let  $h_1, \dots, h_b$  be increasing  $C^{1+\gamma}$  maps of  $[0, 1]$  into itself. We assume

$$0 < r_- \leq h'_i(x) \leq r_+ < 1, \quad \text{dist}(h_i([0, 1]), h_j([0, 1])) \geq g > 0 \quad (i \neq j). \quad (77.1)$$

Write  $\Sigma = \{1, \dots, b\}^{\mathbb{N}_0}$ , let  $S$  denote the left shift, and put

$$\pi(\omega) = \lim_{n \rightarrow \infty} h_{\omega_0} \cdots h_{\omega_{n-1}}(0), \quad K = \pi(\Sigma), \quad T = \pi S \pi^{-1}.$$

Strong separation makes  $\pi$  a homeomorphism onto  $K$ . In particular,  $T$  is a well-defined continuous expanding map on the repeller. Our readout is the ambient coordinate  $f(x) = x$ .

For a word  $w = w_0 \cdots w_{n-1}$ , set

$$h_w = h_{w_0} \circ \cdots \circ h_{w_{n-1}}, \quad r_w = \sup_{[0,1]} h'_w, \quad K_w = h_w(K), \quad r_\emptyset = 1.$$

The notation  $w_{k\cdot}$  means the suffix  $w_k \cdots w_{n-1}$ . Choose a Hölder potential  $\varphi$  on  $\Sigma$ , normalized by  $P(\varphi) = 0$ , and let  $\mu$  be its shift-invariant Gibbs probability. We use  $\mu$  also for its image on  $K$ , and write  $\mu[w] = \mu(K_w)$ . Here  $P$  is topological pressure on the full shift. Thus, for a constant  $C_G \geq 1$ ,

$$C_G^{-1} e^{S_n \varphi(\omega)} \leq \mu[w] \leq C_G e^{S_n \varphi(\omega)} \quad (\omega \in [w]), \quad S_n u = \sum_{j=0}^{n-1} u \circ S^j. \quad (77.2)$$

The derivative potential is

$$\psi(\omega) = \log h'_{\omega_0}(\pi(S\omega)). \quad (77.3)$$

It is Hölder and is bounded between  $\log r_-$  and  $\log r_+$ .

Fix  $0 < q < 1$ . At time zero and at each subsequent acquisition a fresh  $\mu$ -distributed state is observed exactly. At every other time the hidden state is advanced by  $T$ . The acquisition indicators after time zero are independent Bernoulli variables with probability  $1 - q$  and are independent of all fresh states. After the update, the decoder receives only the persistent state, its absolute time, and independent coding randomness. In particular, it does not receive the current acquisition indicator as a separate uncharged input. An  $M$ -state observer has one persistent register with at most  $M$  values. Its Borel update and decoding kernels may be randomized and time dependent; no other data-dependent quantity survives a checkpoint. Define

$$R_M = \inf_{\mathcal{A}} \limsup_{H \rightarrow \infty} \frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E} |X_t - \hat{X}_t^{\mathcal{A}}|^2. \quad (77.4)$$

The infimum is over this entire observer class. Its deterministic stationary subclass will suffice for the upper bounds. Program constants and transient computation are not charged to  $M$ .

The weighted orbit and its squared quantization error are

$$Z_q(x) = (\sqrt{1-q} q^{k/2} T^k x)_{k \geq 0}, \quad e_M^2(Z_q) = \inf_{c_1, \dots, c_M \in \ell^2} \int \min_j \|Z_q(x) - c_j\|^2 d\mu(x). \quad (77.5)$$

The centres in this definition need not lie on the orbit image.

**77.1. A finite cylinder algorithm.** For every finite word define

$$V_q(w) = (1-q) \sum_{k=0}^{|w|-1} q^k r_{w_{k\cdot}}^2 + q^{|w|}, \quad a_q(w) = \mu[w] V_q(w). \quad (77.6)$$

Both quantities at the empty word are one. They require only finite cylinder masses and derivatives of finite compositions. Set

$$\rho_q = q + (1-q)r_+^2 < 1. \quad (77.7)$$

View all finite words as the vertices of a full  $b$ -ary tree. Starting from the empty word, repeatedly replace a leaf of maximal  $a_q$  by its  $b$  children. Resolve ties in lexicographic order. For  $N \geq 1$ , let  $\mathcal{P}_N$  be the last leaf partition in this sequence having at most  $N$  leaves, and put

$$G_N(q) = \sum_{w \in \mathcal{P}_N} a_q(w). \quad (77.8)$$

This convention also defines  $G_N = 1$  for  $N < b$ . No optimization over Hilbert-space centres or over observer policies enters the algorithm.

**Theorem 77.1** (Intrinsic profile and finite-state realization). *Under (77.1)–(77.2), for every  $0 < q < 1$  there are constants  $0 < c \leq C < \infty$  such that, for every integer  $M \geq 1$ ,*

$$cG_M(q) \leq e_M^2(Z_q) \leq R_M \leq CG_M(q). \quad (77.9)$$

*The lower bound holds also with the liminf time-average loss of any individual randomized, time-dependent observer. The upper bound is attained to the displayed factor by a deterministic stationary observer which, at each acquisition, selects a leaf of a greedy partition and thereafter deletes the first symbol of its stored word. For every  $N$ , the full greedy tree is suffix closed and the constructed observer has exactly  $(b|\mathcal{P}_N| - 1)/(b - 1)$  designated states and average loss at most  $G_N(q)$ . In particular, it uses at most  $(bN - 1)/(b - 1) \leq 2N - 1$  states.*

*The same order  $G_M(q)$  holds when the objective in (77.4) is replaced by  $\inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E}|X_t - \hat{X}_t^{\mathcal{A}}|^2$ . Constants may depend on the branch, distortion, Gibbs and separation data and on  $q$ . They can be chosen uniformly on compact  $q$ -intervals in  $(0, 1)$  when those data are uniformly controlled.*

There are two separate assertions here. The cylinder algorithm computes the relevant distortion profile; its suffix closure computes an implementable observer. The number of suffix states is proportional to the number of leaves, not to the number of leaves times their maximum depth. This is the point at which a geometric code becomes a causal code without a growing multiplicative overhead.

**77.2. The pressure law.** Let  $s_g$  and  $s_t(q)$  be the unique numbers in  $(0, 1)$  satisfying

$$P(s_g\varphi + 2s_g\psi) = 0, \quad P(s_t(q)\varphi) + s_t(q) \log q = 0, \quad s_*(q) = \max\{s_g, s_t(q)\}. \quad (77.10)$$

Existence and uniqueness are part of the next theorem.

**Theorem 77.2** (Pressure determines the causal exponent). *In the setting of Theorem 77.1,*

$$\lim_{M \rightarrow \infty} \frac{-\log R_M}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z_q)}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log G_M(q)}{\log M} = \frac{1 - s_*(q)}{s_*(q)}. \quad (77.11)$$

*More precisely, for  $0 < s < 1$ ,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{|w|=n} a_q(w)^s = \max\{P(s\varphi + 2s\psi), P(s\varphi) + s \log q\}. \quad (77.12)$$

*Consequently the transition occurs at*

$$q_c = \exp\{-P(s_g\varphi)/s_g\} \in (0, 1). \quad (77.13)$$

*For  $q \leq q_c$  the exponent is  $(1 - s_g)/s_g$ ; for  $q > q_c$  it is  $(1 - s_t(q))/s_t(q)$ . At the transition, the finite-budget profile is still given, up to constants, by (77.8) and (77.9).*

Formula (77.12) is a maximum of two pressures, not pressure evaluated at the pointwise maximum of two potentials. The derivative cocycle enters the first term; survival across unobserved iterates enters the second. The theorem computes the quantization exponent from these data rather than postulating its existence. It does not identify a universal logarithmic correction or an optimal leading constant at  $q_c$  for all Gibbs repellers. The finite cylinder profile retains such corrections without discarding them in the exponent limit.

## APPENDIX 78. CYLINDER GEOMETRY AND CAUSAL REALIZATION

Throughout this section  $q$  and the data in (77.1)–(77.2) are fixed. We suppress the subscript  $q$  in  $V$  and  $a$ . A comparison constant never depends on a word or a state budget.

**Lemma 78.1** (Distortion and orbit separation). *There is  $D \geq 1$  such that*

$$\sup h'_w \leq D \inf h'_w, \quad D^{-1}r_u r_v \leq r_{uv} \leq r_u r_v. \quad (78.1)$$

*If  $x, y \in K$  have longest common symbolic prefix  $w$ , then*

$$c_1 V(w) \leq \|Z_q(x) - Z_q(y)\|^2 \leq V(w), \quad c_1 = (1 - q)g^2 D^{-2}. \quad (78.2)$$

*For arbitrary words  $u, v$ ,*

$$V(uv) \leq r_v^2(V(u) - q^{|u|}) + q^{|u|}V(v), \quad (78.3)$$

$$V(uv) \leq V(u)V(v), \quad a(uv) \leq a(v). \quad (78.4)$$

*There exists  $0 < a_0 < \rho_q$  such that each one-symbol child  $wi$ ,  $1 \leq i \leq b$ , satisfies*

$$a_0 a(w) \leq a(wi) \leq \rho_q a(w), \quad \sum_{i=1}^b a(wi) \leq \rho_q a(w). \quad (78.5)$$

*Proof.* Let  $H$  bound the  $\gamma$ -Hölder seminorms of  $\log h'_i$ . The chain rule, read from the innermost factor, gives

$$|\log h'_w(x) - \log h'_w(y)| \leq H \sum_{j \geq 0} r_+^{j\gamma} \leq H/(1 - r_+^\gamma).$$

This proves the first assertion with  $D = \exp\{H/(1 - r_+^\gamma)\}$ . The upper multiplicative bound follows from the chain rule; for the lower bound evaluate the inner derivative at a maximizer and bound the outer derivative below by  $r_u/D$ .

Put  $n = |w|$ . For  $k < n$ , the points  $T^k x, T^k y$  belong to the same suffix cylinder. Their distance is at most  $r_{w_k}$ . Immediately after that suffix they belong to two different first-level cylinders, so the mean value theorem also bounds this distance below by  $gr_{w_k}/D$ . At  $k = n$  their distance is at least  $g$ . The remaining squared distances are at most one. Multiplying by  $(1 - q)q^k$  and summing proves (78.2), including the term  $q^n$  by means of its first summand  $(1 - q)q^n g^2$ . The same upper bound  $V(w)$  holds for any two points of  $K_w$ .

Split the defining sum for  $V(uv)$  at  $|u|$  and apply (78.1) to its first part to obtain (78.3). Each suffix derivative  $r_{v_k}$  is at least  $r_v$ , since the omitted outer maps are contractions. Consequently  $V(v) \geq r_v^2$ . Equation (78.3) therefore gives  $V(uv) \leq V(u)V(v)$ . Since  $V(u) \leq 1$  and shift invariance gives

$$\mu[uv] \leq \mu(S^{-|u|}[v]) = \mu[v],$$

we obtain the exact suffix domination  $a(uv) \leq a(v)$ . No independence of the symbols is used.

For a one-symbol child,  $V(i) = q + (1 - q)r_i^2 \leq \rho_q$ ; submultiplicativity gives  $V(wi) \leq \rho_q V(w)$ . Conversely, split at  $|w|$  and apply the lower distortion estimate to the old summands. Since  $V(i) \geq r_i^2 \geq r_-^2$ ,

$$V(wi) \geq D^{-2}r_-^2(V(w) - q^{|w|}) + r_-^2 q^{|w|} \geq D^{-2}r_-^2 V(w).$$

The Gibbs bounds imply a positive lower bound  $\beta$  on  $\mu[wi]/\mu[w]$ , uniformly in  $w, i$ : divide the lower bound for  $[wi]$  by the upper bound for  $[w]$  at the same point, and use  $\varphi \geq \inf \varphi$ . Take  $a_0 = \beta D^{-2}r_-^2/2$ , decreased if necessary so that  $a_0 < \rho_q$ . Finally the children partition  $[w]$ ; summing  $\mu[wi]V(wi) \leq \rho_q \mu[wi]V(w)$  proves the final assertion.  $\square$

**Lemma 78.2** (Balanced greedy partitions). *Let  $L_N = |\mathcal{P}_N|$ . Then  $N/b \leq L_N \leq N$ . For  $N \geq b$ , let  $\theta_N$  be the energy of the last leaf split in constructing  $\mathcal{P}_N$ . Every  $w \in \mathcal{P}_N$  satisfies*

$$a_0 \theta_N \leq a(w) \leq \theta_N, \quad a_0 L_N \theta_N \leq G_N \leq L_N \theta_N. \quad (78.6)$$

For every fixed  $A \geq 1$  there is  $c_A > 0$  such that

$$c_A G_N \leq G_{\lceil AN \rceil} \leq G_N \quad (N \geq 1). \quad (78.7)$$

Moreover, for each  $c > 0$ , the number of all finite words  $v$  with  $a(v) \geq c\theta_N$  is at most  $C_c N$ .

*Proof.* The leaf counts are  $1 + j(b-1)$ . A maximal energy selected for splitting never increases, since each new child has at most  $\rho_q$  times its parent's energy. A leaf still present at the last split was born from a parent of energy at least  $\theta_N$ , and every current leaf has energy at most  $\theta_N$ . This proves (78.6); the root-only cases are kept separate. The sum  $G_N$  is nonincreasing by (78.5).

For completeness, the reverse comparison in (78.7) does not require a preassigned power law. Choose an integer  $j$  with  $b^{j+1} > b(A+1)$ . If  $\theta_{\lceil AN \rceil} < a_0^{j+1} \theta_N$ , every descendant through depth  $j$  of each old leaf would have been split: all these nodes have energy at least  $a_0^{j+1} \theta_N$ . The new partition would then have at least  $b^{j+1} L_N$  leaves, which exceeds  $\lceil AN \rceil$ . Thus  $\theta_{\lceil AN \rceil} \geq a_0^{j+1} \theta_N$ . Combining this with (78.6) proves the reverse comparison for  $N \geq b$ . Finitely many smaller budgets can be absorbed into  $c_A$ ; their energies are positive.

The finite full tree ending at  $\mathcal{P}_N$  has at most  $bL_N/(b-1)$  vertices. Every other vertex descends from a leaf. At depth  $j$  below a leaf its energy is at most  $\rho_q^j \theta_N$ . Only a bounded number of additional *generations* can therefore have energy at least  $c\theta_N$ ; each has at most  $b^j$  descendants per leaf. This bounds the total number of such words by  $C_c N$ .  $\square$

**Lemma 78.3** (Quantization of the orbit image). *There are  $0 < c_2 \leq C_2 < \infty$  such that*

$$c_2 G_M \leq e_M^2(Z_q) \leq C_2 G_M \quad (M \geq 1).$$

*The lower bound allows arbitrary off-image centres in  $\ell^2$ .*

*Proof.* Choosing one orbit point from each cylinder of  $\mathcal{P}_M$  uses at most  $M$  centres and has squared error at most  $G_M$ , by the cylinder diameter estimate.

For the reverse bound take  $\mathcal{P}_{4bM}$ . It has at least  $4M$  leaves. For each leaf  $w$  let

$$U_w = \{z \in \ell^2 : \text{dist}(z, Z_q(K_w)) < \delta \sqrt{V(w)}\}, \quad 0 < \delta < \sqrt{c_1}/(2\sqrt{2}).$$

These open sets are pairwise disjoint. Indeed, two different leaves have a common prefix  $u$  strictly shorter than either leaf. Any points in their two orbit cylinders have distance at least  $\sqrt{c_1} V(u)$ , whereas  $V(w), V(v) \leq 2V(u)$ . The sum of the two tube radii is smaller than this separation. Thus  $M$  centres meet at most  $M$  of the tubes. Every unoccupied cylinder contributes at least  $\delta^2 a(w)$  to the distortion. By (78.6), the sum over the unoccupied leaves is at least a fixed multiple of  $G_{4bM}$ . Lemma 78.2 compares this with  $G_M$ . The argument uses distances to entire cylinders, so it does not require any centre to belong to the orbit image. The finitely many root-only cases can again be absorbed: the measure is nonatomic and the orbit's first coordinate separates distinct states, so each fixed finite quantization error is positive.  $\square$

**Lemma 78.4** (Exact suffix closure of the greedy tree). *Let  $\mathcal{T}_N$  be the set of all vertices, including the root, in the greedy tree with leaf set  $\mathcal{P}_N$ , and write  $L_N = |\mathcal{P}_N|$ . Then  $\mathcal{T}_N$  is closed under deletion of the first letter and*

$$|\mathcal{T}_N| = \frac{bL_N - 1}{b - 1} \leq \frac{bN - 1}{b - 1}. \quad (78.8)$$

*In particular, every suffix of every leaf is already a vertex of this same tree. No additional age states are required.*

*Proof.* Full support and stationarity give, for every nonempty  $u$ ,

$$\mu[uv] < \sum_{|z|=|u|} \mu[zv] = \mu[v].$$

Together with  $V(uv) \leq V(v)$ , this yields the strict inequality  $a(uv) < a(v)$ . Suppose a vertex  $w$  is selected for splitting. Every proper suffix  $v$  of  $w$  was already an internal vertex at that time. Otherwise some prefix  $z$  of  $v$  would be a leaf of the current tree, and the decreasing child energies would give  $a(z) \geq a(v) > a(w)$ , contradicting the maximality of  $w$ . Thus the internal vertices are suffix closed.

If  $w$  is a leaf, the parent of each nonempty proper suffix of  $w$  is a suffix of the parent of  $w$ . This parent is internal by the preceding argument (with the empty word interpreted as the root). Its children are vertices of the tree, so the suffix itself is also a vertex. The root and root-only tree cause no exception. This proves closure for all vertices. Finally, a finite full  $b$ -ary tree with  $I$  internal vertices has  $bI$  edges and  $I + L_N$  vertices. Hence  $bI = I + L_N - 1$ , giving (78.8).  $\square$

*Proof of Theorem 77.1.* The posterior-orbit converse, Theorem 74.2, applied to exact observations and geometric cycles, gives  $e_M^2(Z_q) \leq R_M$ , with a liminf bound for each observer. Its randomization representation is Lemma 74.1. Equivalently, at each acquisition one freezes independent coding randomness and the past; the  $M$  possible post-update states supply at most  $M$  no-acquisition decoder strings. Quantization bounds their weighted squared loss below, and the disjoint last-acquisition events give the time-average bound. This does not restrict the competing observers to symbolic codes.

For achievability, use  $\mathcal{T}_N$  as the state set. At an acquisition  $x$ , write its unique leaf word. On a nonacquisition update delete its first letter; the empty word is absorbing until the next acquisition. Decode a nonempty word  $v$  by  $\pi(v111\cdots)$  and the empty word by  $\pi(111\cdots)$ . At age  $k < |w|$  after acquisition in  $K_w$ , the squared error is at most  $r_{w_k}^2$ ; after exhaustion it is at most one. The average squared loss is at most  $G_N$ . Every piece of information used by this rule is in its suffix state. Neither reset age nor a precision index is supplied from outside the register.

By Lemma 78.4, this machine uses exactly  $(bL_N - 1)/(b - 1)$  designated states and at most  $C_0N$  states, with  $C_0 = b/(b - 1)$ . Take  $N = \lfloor M/C_0 \rfloor$  for sufficiently large  $M$ . The fixed-dilation comparison gives  $G_N \leq CG_M$ . For the remaining finitely many budgets use the one-state constant observer and enlarge  $C$ . Lemma 78.3 proves the rest of (77.9).

Let  $d_k$  be the expected age- $k$  loss of the constructed machine under one acquisition. At time  $t$  its expected loss is

$$\sum_{k < t} (1 - q)q^k d_k + q^t d_t \leq \frac{1}{1 - q} \sum_{k \geq 0} (1 - q)q^k d_k.$$

The right side is at most  $G_N/(1 - q)$ . The supremum objective is bounded below by the average objective, proving its stated order. When  $q$  stays in a compact subinterval of  $(0, 1)$  and the branch and Gibbs data are controlled,  $\rho_q$  stays below one and the separation and child-energy constants have common bounds. All constants in the proof can then be chosen uniformly.  $\square$

## APPENDIX 79. PRESSURE, DISTORTION AND SURVIVAL

We use the Gibbs and variational principles on a finite full shift in their usual Hölder form [32, Chapters 1–2]. For a Hölder function  $u$ , these principles give a shift-invariant probability  $\nu_u$  and  $C_u \geq 1$  with

$$C_u^{-1} e^{S_n u(\omega) - nP(u)} \leq \nu_u[w] \leq C_u e^{S_n u(\omega) - nP(u)}, \quad P(u) = \sup_{\nu \in \mathcal{M}_S} \left( h_\nu(S) + \int u d\nu \right). \quad (79.1)$$

Here  $\mathcal{M}_S$  is the set of shift-invariant probabilities. Only these finite-alphabet statements, not a countable-state spectral or suspension theorem, are used below.

**Lemma 79.1** (The mixed partition sum). *Fix  $0 < s < 1$ , and set*

$$A(s) = P(s\varphi + 2s\psi), \quad B_q(s) = P(s\varphi) + s \log q.$$

There are constants  $c_s, C'_s > 0$ , independent of  $n$ , such that

$$c_s \max\{e^{nA(s)}, e^{nB_q(s)}\} \leq \sum_{|w|=n} a_q(w)^s \leq C'_s \sum_{k=0}^n e^{kB_q(s) + (n-k)A(s)}. \quad (79.2)$$

*Proof.* For a word  $w$  of length  $n$  and  $\omega \in [w]$ , the chain rule and bounded distortion give

$$r_{w_k} \asymp e^{S_{n-k}\psi(S^k\omega)}, \quad \mu[w] \asymp e^{S_n\varphi(\omega)},$$

with constants independent of  $n, k, w$ . The term  $k = 0$  and the terminal term in (77.6), together with (79.1), give the two lower bounds in (79.2).

For the upper bound use  $(\sum x_j)^s \leq \sum x_j^s$  for nonnegative  $x_j$ . At a fixed splitting time  $k < n$ , the resulting sum, up to a constant and the factor  $(1 - q)^s$ , is

$$\sum_{|w|=n} \exp \left\{ s S_k \varphi(\omega) + k s \log q + S_{n-k}(s\varphi + 2s\psi)(S^k\omega) \right\}.$$

The variations of  $S_k u$  inside any  $k$ -cylinder are uniformly bounded for a Hölder  $u$ . Since every prefix can be concatenated with every suffix on the full shift, this sum is comparable to the product of the length- $k$  partition sum for  $s\varphi + s \log q$  and the length- $(n - k)$  partition sum for  $s\varphi + 2s\psi$ . Each is comparable to its pressure exponential by (79.1). The terminal term is the same calculation with  $k = n$ . Summing over  $k$  proves the assertion.  $\square$

*Proof of Theorem 77.2.* Both pressures at  $s = 0$  equal  $\log b$ . At  $s = 1$  they are negative:  $A(1) \leq 2 \log r_+ < 0$  and  $B_q(1) = \log q < 0$ . There is a uniform positive lower bound on every one-symbol conditional cylinder mass. Since there are at least two symbols, there is consequently  $\eta < 1$  such that  $\mu[w] \leq \eta^{|w|}$ . Equation (77.2) implies  $S_n \varphi \leq n \log \eta + \log C_G$ . Integration against any invariant probability gives  $\int \varphi d\nu \leq \log \eta < 0$ . The variational principle now shows that both  $A$  and  $B_q$  are strictly decreasing, with their slopes bounded above by a negative constant. Their unique zeros therefore lie in  $(0, 1)$ . Equation (79.2) proves (77.12): the sum on its right has at most  $n + 1$  terms, each at most the exponential of  $n \max\{A(s), B_q(s)\}$ .

We prove the exponent without assuming any quantization dimension. Let  $s > s_*(q)$ ,  $s < 1$ . Both pressures are negative, so (79.2) is summable in  $n$ . Hence

$$H_s := \sum_{w \text{ finite}} a_q(w)^s < \infty.$$

Balance in the greedy tree yields  $L_N(a_0\theta_N)^s \leq H_s$ , and thus

$$G_N \leq C_s'' N^{-(1-s)/s}. \quad (79.3)$$

For the reverse estimate choose a Gibbs measure  $\nu_*$  for a zero-pressure potential as follows. When  $s_* = s_g$ , use  $s_g\varphi + 2s_g\psi$ ; otherwise use  $s_t(q)\varphi + s_t(q)\log q$ . The spatial summand or the terminal summand of  $V_q$ , respectively, shows that

$$\nu_*[w] \leq C_* a_q(w)^{s_*} \quad \text{for every word } w.$$

Since  $\mathcal{P}_N$  is a cylinder partition,  $1 \leq C_* L_N \theta_N^{s_*}$ . Therefore

$$G_N \geq c_* N^{-(1-s_*)/s_*}. \quad (79.4)$$

Let  $s$  decrease to  $s_*$  in (79.3) and use (79.4). This proves the exponent limit for  $G_N$ , and Theorem 77.1 proves the other limits. Finally  $P(s_g\varphi) > 0$  because  $s_g < 1$  and  $P(\varphi) = 0$ . Thus the number  $q_c$  in (77.13) lies in  $(0, 1)$ . Strict monotonicity shows that  $s_t(q) \leq s_g$  precisely when  $B_q(s_g) \leq 0$ , which is equivalent to  $q \leq q_c$ .  $\square$

**Corollary 79.2** (Entropy–expansion ratios). *Put  $I_\nu = -\int \varphi d\nu$  and  $\chi_\nu = -\int \psi d\nu$ . Then*

$$s_g = \sup_{\nu \in \mathcal{M}_S} \frac{h_\nu(S)}{I_\nu + 2\chi_\nu}, \quad s_t(q) = \sup_{\nu \in \mathcal{M}_S} \frac{h_\nu(S)}{I_\nu + \log(1/q)}. \quad (79.5)$$

*All denominators are strictly positive. The quantities  $\chi_\nu$  are the mean expansion rates of  $T$  in the coding coordinates.*

*Proof.* The preceding proof gives  $I_\nu \geq -\log \eta > 0$ , and  $\chi_\nu \geq -\log r_+ > 0$ . By the variational principle,  $A(s) = \sup_\nu \{h_\nu - s(I_\nu + 2\chi_\nu)\}$ . Its zero is the supremum of the displayed ratios. For example, at that supremum all the affine expressions are nonpositive, while ratios approaching the supremum make their values approach zero, since the denominators are uniformly bounded above as well as below. The second formula follows in the same way from  $B_q$ . The chain rule identifies  $-S_n\psi$  with the logarithmic expansion of  $T^n$  on its cylinder.  $\square$

**Proposition 79.3** (Finite pressure enclosures). *For a Hölder potential  $u$ , define*

$$Z_n^-(u) = \sum_{|w|=n} \exp\{\inf_{[w]} S_n u\}, \quad Z_n^+(u) = \sum_{|w|=n} \exp\{\sup_{[w]} S_n u\}.$$

*If  $\text{var}_j(u) = \sup\{|u(x) - u(y)| : x_{0:j} = y_{0:j}\}$ , then*

$$\frac{1}{n} \log Z_n^-(u) \leq P(u) \leq \frac{1}{n} \log Z_n^+(u), \quad \frac{1}{n} \log \frac{Z_n^+(u)}{Z_n^-(u)} \leq \frac{1}{n} \sum_{j \geq 1} \text{var}_j(u). \quad (79.6)$$

*Here  $x_{0:j} = y_{0:j}$  means equality of the first  $j$  symbols. Thus certified cylinder bounds on the potentials give convergent finite enclosures for the two pressure roots and for  $q_c$ .*

*Proof.* Concatenation gives  $Z_{n+k}^- \geq Z_n^- Z_k^-$  and  $Z_{n+k}^+ \leq Z_n^+ Z_k^+$ . Their normalized logarithms have the same limit  $P(u)$ , since the oscillation of  $S_n u$  in a cylinder is at most  $\sum_{j=1}^n \text{var}_j(u)$ . Superadditivity and subadditivity prove the enclosure, and the same oscillation bound gives its width. Strict monotonicity of the pressures then converts pressure enclosures into root enclosures.  $\square$

These are finite enclosures in the intrinsic potentials. As usual with arbitrary real-valued smooth input data, an effective numerical algorithm requires certified access to those data. Neither the resource model nor this proposition asserts a Turing complexity bound for an unspecified real function.

### 79.1. Markov branch weights and unequal multipliers.

**Corollary 79.4** (A Perron–Frobenius formula). *Suppose  $h_i(x) = c_i + r_i x$  and the symbols have a stationary Markov law with a strictly positive stochastic matrix  $P = (p_{ij})$ . Then  $s_g$  and  $s_t(q)$  are the unique roots*

$$\rho((p_{ij}^{s_g} r_i^{2s_g})_{ij}) = 1, \quad q^{s_t(q)} \rho((p_{ij}^{s_t(q)})_{ij}) = 1. \quad (79.7)$$

*In particular,*

$$q_c = \rho((p_{ij}^{s_g})_{ij})^{-1/s_g}.$$

*For independent symbols with probabilities  $p_i > 0$ , these equations reduce to  $\sum_i (p_i r_i^2)^{s_g} = 1$  and  $q^{s_t} \sum_i p_i^{s_t} = 1$ .*

*Proof.* Take  $\varphi(\omega) = \log p_{\omega_0 \omega_1}$  and  $\psi(\omega) = \log r_{\omega_0}$ . The Markov cylinder probability has a stationary initial factor and one fewer transition than  $e^{S_n \varphi}$ , so the Gibbs ratio is bounded above and below. The pressure of an edge potential  $u(i, j)$  is  $\log \rho((e^{u(i,j)})_{ij})$ , by summing products along words and applying Perron–Frobenius to the positive matrix. This yields (79.7). When all rows equal  $(p_i)_i$ , the matrix has rank one and its nonzero eigenvalue is the displayed sum.  $\square$

The powers of  $p_{ij}$  in (79.7) are *entrywise powers*. Replacing them by a matrix power of  $P$  would erase precisely the information pressure at issue.

**Example 79.5** (A nonlinear variable-expansion family). On  $[0, 1]$  take

$$h_1(x) = \frac{x}{5} + \varepsilon x(1 - x), \quad h_2(x) = \frac{3}{4} + \frac{x}{6} + \varepsilon x(1 - x), \quad 0 \leq \varepsilon \leq \frac{1}{100}.$$

Their image intervals are  $[0, 1/5]$  and  $[3/4, 11/12]$ . The derivatives lie, respectively, in  $[19/100, 21/100]$  and  $[1/6 - 1/100, 1/6 + 1/100]$ . Thus all hypotheses, including a common positive separation and distortion bound, hold uniformly. Equip the symbols, for example, with the stationary Markov law

$$P = \begin{pmatrix} 4/5 & 1/5 \\ 7/20 & 13/20 \end{pmatrix}.$$

At  $\varepsilon = 0$ , Corollary 79.4 gives two explicit positive-matrix equations. At  $\varepsilon > 0$ , Theorem 77.2 and Proposition 79.3 apply to the genuinely nonconstant derivative potential.

This family is not obtained by a smooth change of coordinate from a map with a common multiplier at every fixed point. The two inverse branches have fixed points with derivatives in the two disjoint intervals just displayed. A  $C^1$  conjugacy with nonzero derivative at those points preserves the fixed-point multipliers, which are their reciprocals. Their inequality is therefore an obstruction to such a conjugacy. There remains a topological full-shift coding, as there must for this repeller class; the assertion concerns the metric expansion, not the absence of symbolic coordinates.

#### APPENDIX 80. ACQUISITIONS WITH PERSISTENT HIDDEN STATE

The preceding realization can be used when acquisition does not regenerate the physical state. The appropriate comparison concerns raw prediction risk: the full-history posterior need not equal the posterior based on the latest observation alone.

Let  $E$  and  $\mathbf{Y}$  be standard Borel spaces, let  $T : E \rightarrow E$  be Borel, let  $f : E \rightarrow \mathbb{R}^r$  be Borel with  $\|f\| \leq F$ , and let  $\mu$  be a probability with  $T_{\#}\mu = \mu$ . Acquisitions occur at  $\tau_0 = 0$  and  $\tau_{j+1} = \tau_j + L_j$ , where the  $L_j$  are independent positive integer random variables of common finite mean  $\ell$ . The acquisition clock is independent of the hidden transition and observation randomizations. Between acquisitions  $X_t = TX_{t-1}$ . At an acquisition  $t > 0$  the transition and observation are specified, in this order, by

$$X_t \mid X_{t-1} = x \sim Q(x, du), \quad Y_t \mid X_t = u \sim K(u, dy). \quad (80.1)$$

At time zero  $X_0 \sim \mu$  and  $Y_0 \sim K(X_0, \cdot)$ . We assume

$$\mu Q = \mu, \quad Q(x, A) \geq \epsilon \mu(A) \quad (x \in E, A \text{ Borel}), \quad 0 < \epsilon \leq 1. \quad (80.2)$$

The whole state  $X_{t-1}$  can influence  $X_t$  through the remaining part of  $Q$ . No regeneration indicator is supplied to the observer. The observer sees the acquisition indicators and the  $Y_t$  only when acquisitions occur; the persistent-state convention is that of Section 77.

Use a single pair  $(U, Y) \sim \mu(du)K(u, dy)$  to define

$$\begin{aligned} s_k &= \mathbb{P}(L_0 > k), \quad w_k = s_k/\ell, \quad W(n) = \sum_{k \geq n} w_k, \\ m_k(y) &= \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0}, \\ B_0 &= \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \end{aligned} \quad (80.3)$$

Let  $R_M^{\text{dep}}$  denote the infimum of limsup average *raw* squared loss in this dependent experiment.

**Theorem 80.1** (Minorized acquisition comparison). *Under (80.1)–(80.2),*

$$R_M^{\text{dep}} \geq \epsilon(B_0 + e_M^2(Z)) \quad (M \geq 1). \quad (80.4)$$

*The same lower bound holds for the liminf average loss of each randomized, time-dependent  $M$ -state observer. For every  $N, n \geq 1$  with  $Nn + 1 \leq M$ ,*

$$R_M^{\text{dep}} \leq B_0 + e_N^2(Z^{<n}) + F^2W(n). \quad (80.5)$$

*A one-state zero decoder always has risk at most  $F^2$ . More generally, every observer that overwrites its entire state using only the latest observation at each acquisition has exactly the same sequence of expected one-time losses as in the independent-acquisition experiment with law  $\mu K$  and the same clock, provided it is initialized in the same way.*

*Proof.* Condition on the entire acquisition-indicator sequence. At time zero the hidden marginal is  $\mu$ . Each intervening transition is either  $T$  or  $Q$ , both of which preserve  $\mu$ . Induction therefore shows that the hidden state at each acquisition has marginal  $\mu$ , even after this conditioning. Its observation pair has law  $\mu K$ . Future survival for  $k$  steps has probability  $s_k$ , independent of that pair. Thus an observer which restarts from the current observation has the same age-dependent expected loss as in the independent experiment, and hence the same expected one-time loss after averaging over the common clock. This proves the last assertion. Applying the  $Nn + 1$ -state construction of Theorem 74.2 proves (80.5); the constant decoder proves the remaining upper bound.

For the converse fix  $n$ . At an acquisition condition on the entire hidden and observed past immediately before the transition, and use the independent uniform representation of the observer's kernels from Lemma 74.1. Freeze its coding uniforms, including those used on the hypothetical next  $n$  nonacquisition steps. The  $M$  possible post-update states determine at most  $M$  length- $n$  decoder strings. Conditional on this past, the new observation pair has law  $Q(x, du)K(u, dy)$ , which dominates  $\epsilon\mu(du)K(u, dy)$ . Squared loss is nonnegative, so its expected contribution up to the next acquisition, truncated at  $n$ , is at least

$$\epsilon \inf_{i(\cdot), c_{i,k}} \sum_{k < n} s_k \mathbb{E}_{\mu K} \|f(T^k U) - c_{i(Y),k}\|^2 = \epsilon \ell(B_{0,<n} + e_M^2(Z^{<n})).$$

For the equality, condition on  $Y$  and use orthogonality around  $m_k(Y)$ ; the remaining infimum is precisely finite-dimensional quantization. Enlarging the possible codebooks to all vectors only weakens this lower bound. The encoder is permitted to depend on the conditioned past, but the resulting bound does not.

For a horizon  $H$ , sum these truncated contributions only over acquisitions at times at most  $H - n$ . Each contribution stops at the next acquisition, so distinct terms charge disjoint time indices, path by path. The expected number of these acquisitions, divided by  $H$ , converges to  $1/\ell$  by the elementary renewal theorem. Taking liminf in  $H$  proves the lower bound with  $B_{0,<n}$  and  $Z^{<n}$ . Finally let  $n \rightarrow \infty$ . The noise terms converge by boundedness. The quantization terms increase to  $e_M^2(Z)$ : projection gives one inequality, and extending a truncated codebook by zero gives

$$e_M^2(Z) \leq e_M^2(Z^{<n}) + \mathbb{E}\|Z - Z^{<n}\|^2 \leq e_M^2(Z^{<n}) + F^2W(n).$$

This proves (80.4). □

The reference number  $B_0$  in this theorem is *not* asserted to be the Bayes risk conditional on the entire dependent history. Older observations can improve that posterior. The theorem uses minorization to obtain a lower bound and invariant marginals to obtain an upper bound; it does not replace a dependent conditional law by an independent one.

**Corollary 80.2** (The pressure law survives partial refreshment). *Take the repeller, Gibbs law and geometric acquisition clock of Section 77, but use any kernel  $Q$  satisfying (80.2) at acquisitions. Observe the current state exactly at those times. Then, for each fixed  $\epsilon > 0$ ,*

$$R_M^{\text{dep}} \asymp G_M(q), \quad \lim_{M \rightarrow \infty} \frac{-\log R_M^{\text{dep}}}{\log M} = \frac{1 - s_*(q)}{s_*(q)}.$$

*The same order holds for supremum one-time loss. An explicit example is*

$$Q(x, du) = (1 - \epsilon)\delta_{Tx}(du) + \epsilon\mu(du). \quad (80.6)$$

*For  $\epsilon < 1$  the physical state is not regenerated at every acquisition. The phase boundary is governed by the acquisition survival probability  $q$ , not by the unobserved physical-refresh survival probability  $1 - \epsilon(1 - q)$ .*

*Proof.* Exact observations give  $B_0 = 0$  and the orbit  $Z_q$ . Theorem 80.1 supplies the lower bound  $\epsilon e_M^2(Z_q)$ . The resetting suffix observer in Theorem 77.1 supplies the upper bound  $CG_M(q)$  with the same expected losses in the dependent experiment. The pressure formula and the supremum statement follow from the corresponding results there. Kernel (80.6) preserves  $\mu$  because  $T$  does, and has the displayed minorization. Its transition coin is not an observation.  $\square$

**Corollary 80.3** (Noisy acquisition with retained hidden memory). *On  $\mathbb{T}^d$  let  $T_b x = bx \pmod{1}$ , with integers  $b \geq 2, d \geq 1$ , let  $\mu$  be Haar measure, and use any acquisition kernel  $Q$  satisfying (80.2). At each acquisition observe  $Y = X + \sigma G \pmod{1}$ , where the observation noises are independent standard Gaussian vectors. Predict  $\Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d$ . Assume  $W(n) \leq Aw_n$  for every  $n$  and some  $A < \infty$ . With*

$$a_k = e^{-2\pi^2 \sigma^2 b^{2k}}, \quad v_k = w_k a_k^2, \quad B_0 = d \sum_{k \geq 0} w_k (1 - a_k^2),$$

*and  $\Psi_v, n_M$  as in (81.4) and (81.6), for all  $M \geq 1$  and  $\sigma \geq 0$ ,*

$$\epsilon \left( B_0 + \frac{\Psi_v(n_M)}{8Ab^6} \right) \leq R_M^{\text{dep}} \leq B_0 + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (80.7)$$

*For geometric acquisitions this also bounds the optimal supremum raw loss. In particular  $R_M^{\text{dep}} \asymp B_0 + \Psi_w(n_M)$ , with constants independent of  $M$  and  $\sigma$  when  $\epsilon$  is bounded below. The critical-window comparison of Corollary 81.4 remains valid for this dependent experiment, and the explicit restarting suffix observer has the same exact crossover coefficient as there.*

*Proof.* The single-acquisition posterior under  $\mu K$  is  $m_k(y) = a_k \Phi(b^k y)$ , by (81.3). Lemma 81.2 bounds its orbit quantization error below by  $\Psi_v(n_M)/(8Ab^6)$ ; use (80.4). The suffix observer proving Theorem 81.1 overwrites its state at acquisitions, so its exact expected losses, including the formula (81.9), transfer unchanged. This proves the upper bound and the supremum assertion. Finally  $0 \leq \Psi_w(n) - \Psi_v(n) \leq B_0/d$  proves the raw-profile comparison. Its constants are uniform in the same compact  $q$ -windows as in Corollary 81.4; the explicit observer has exactly the same loss, rather than merely comparable loss, so its coefficient limit transfers as well.  $\square$

## APPENDIX 81. NOISY EXPANDING EXPERIMENTS AND A UNIFORM CRITICAL WINDOW

The realization overhead in Theorem 74.2 is not always necessary. We now identify a class in which symbolic closure yields matching errors, including the acquisition noise and the renewal tail. Let  $b \geq 2$  and  $d \geq 1$  be integers, let  $\mathbb{T}^d = (\mathbb{R}/\mathbb{Z})^d$ , and put

$$T_b x = bx \pmod{1}, \quad \Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d. \quad (81.1)$$

At a renewal  $U$  is Haar uniform and the observation is

$$Y = U + \sigma G \pmod{1}, \quad G \sim N(0, I_d), \quad \sigma \geq 0. \quad (81.2)$$

The noise, the fresh states, and the renewal lengths are mutually independent. The target is  $\Phi(X_t) \in \mathbb{R}^{2d}$ , not a discontinuous choice of coordinates on the torus. Both average raw risk and average excess risk have the meanings of Section 74.

Haar invariance implies that  $Y$  and  $G$  are independent: conditional on  $G$ , the translate of  $U$  is Haar. Consequently the posterior law of  $U$  given  $Y = y$  is that of  $y - \sigma G$  modulo one, and

$$m_k(y) = a_k \Phi(b^k y), \quad a_k = \exp(-2\pi^2 \sigma^2 b^{2k}), \quad B = d \sum_{k \geq 0} w_k (1 - a_k^2). \quad (81.3)$$

This calculation uses the Gaussian characteristic function, so it also specifies exactly which noise information is used by the decoder. Write

$$v_k = w_k a_k^2, \quad \Psi_v(n) = \sum_{k < n} v_k b^{-2(n-k)} + \sum_{k \geq n} v_k, \quad n \geq 0. \quad (81.4)$$

For  $v = w$ , use the notation  $\Psi_w$ . We impose the quantitative renewal condition

$$W(n) \leq A w_n \quad (n \geq 0) \quad (81.5)$$

for some finite  $A \geq 1$ . In terms of the cycle length this is a uniform bound on the mean residual lifetime, conditional on survival. It includes geometric renewals, mixtures with geometric parameters bounded away from zero, and bounded renewal lengths. It does not include arbitrary polynomial renewal tails.

Put  $a = b^d$  and

$$S_n = 1 + a + \cdots + a^n, \quad n_M = \max\{n \geq 0 : S_n \leq M\}. \quad (81.6)$$

The distinction between  $a^n$  and  $S_n$  counts, rather than suppresses, the persistent information about remaining precision.

**Theorem 81.1** (Sharp posterior-orbit law). *Under (81.5), for every  $M \geq 1$  and  $\sigma \geq 0$ ,*

$$\frac{\Psi_v(n_M)}{8Ab^6} \leq E_M^{\text{av}} \leq \frac{d\pi^2}{3} \Psi_v(n_M), \quad (81.7)$$

$$B + \frac{\Psi_v(n_M)}{8Ab^6} \leq R_M^{\text{av}} \leq B + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (81.8)$$

*The lower bounds hold for all randomized, nonstationary machines. The upper bounds are attained by the finite suffix machine described below. Its exact average raw risk with  $S_n$  states is*

$$U_{n,\sigma} = B + d \sum_{k < n} v_k \left[ 1 - \text{sinc}^2(\pi b^{-(n-k)}) \right] + d \sum_{k \geq n} v_k, \quad (81.9)$$

*where  $\text{sinc}(z) = \sin(z)/z$  and  $\text{sinc}(0) = 1$ . For geometric renewals, (81.8) also holds for  $R_M^{\text{sup}} := \inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E} \|\Phi(X_t) - \hat{f}_t\|^2$ , with an initial acquisition at zero. The supremum raw risk of the displayed suffix machine is exactly  $U_{n,\sigma}$ .*

**Lemma 81.2** (Orbit separation and small balls). *Suppose  $v_k = w_k a_k^2$  with  $a_k$  nonincreasing and (81.5) holds. For Haar  $Y$  the quantization error of  $(\sqrt{v_k} \Phi(b^k Y))_{k \geq 0}$  satisfies*

$$e_M^2 \geq \frac{\Psi_v(n_M)}{8Ab^6}. \quad (81.10)$$

*Proof.* First,  $\sum_{k \geq n} v_k \leq A v_n$ . Also  $\Psi_v$  is nonincreasing, and

$$\Psi_v(n+1) \geq b^{-2} \Psi_v(n). \quad (81.11)$$

These follow directly from (81.4) and monotonicity of  $a_k$ . For  $x, y \in \mathbb{T}^d$ , write  $\delta = \max_j \text{dist}_{\mathbb{T}}(x_j, y_j) \in [0, 1/2]$ . If  $\delta > 0$ , set  $K = \lfloor \log_b(1/(2\delta)) \rfloor$ . In a coordinate attaining  $\delta$ , for  $0 \leq k \leq K$  the unreduced difference has modulus  $b^k \delta \leq 1/2$ . Hence  $|\sin(\pi b^k \delta)| \geq 2b^k \delta$ , and the squared orbit distance  $D^2(x, y)$  obeys

$$\begin{aligned} D^2(x, y) &= 4 \sum_{k \geq 0} v_k \sum_{j=1}^d \sin^2(\pi b^k (x_j - y_j)) \\ &\geq 16\delta^2 \sum_{k=0}^K v_k b^{2k} \geq \frac{4}{b^2} \sum_{k=0}^K v_k b^{-2(K-k)} \geq \frac{4}{Ab^2} \Psi_v(K). \end{aligned} \quad (81.12)$$

The middle inequality uses  $\delta > b^{-(K+1)}/2$ ; the last uses the residual-tail bound and  $A \geq 1$ .

Let  $h = [2(2M)^{1/d}]^{-1}$  and  $K_h = \lfloor \log_b(1/(2h)) \rfloor = \lfloor \log_a(2M) \rfloor$ . Since  $M < S_{n_M+1}$  and  $a \geq 2$ , one has  $K_h \leq n_M + 2$ . Whenever  $\delta \geq h$ , the associated  $K \leq K_h$ ; thus (81.11) and (81.12) give

$$D^2(x, y) \geq \gamma \Psi_v(n_M), \quad \gamma = \frac{4}{Ab^6}.$$

If  $\Psi_v(n_M) = 0$ , the conclusion is immediate. Otherwise take a Hilbert ball of radius  $r$ , with  $r^2 = \gamma \Psi_v(n_M)/16$ . Two preimages in the same ball have orbit distance at most  $2r$ , so their torus sup-distance is less than  $h$ . Fixing one preimage, all preimages lie in a torus cube of side  $2h$ , whose Haar measure is  $1/(2M)$ . This argument applies even when the centre of the Hilbert ball is not on the orbit image. The union of  $M$  such balls has preimage measure at most  $1/2$ . Every  $M$ -centre quantizer therefore has mean squared error at least  $r^2/2 = \Psi_v(n_M)/(8Ab^6)$ .  $\square$

*Proof of Theorem 81.1.* The lower bound follows from Theorem 74.2, (81.3), and Lemma 81.2. The orthogonality identity then adds  $B$  to the excess risk.

For the upper bound use the alphabet  $\{0, \dots, b-1\}^d$ . The states are all words of lengths  $0, \dots, n$ , exactly  $S_n$  states. At a renewal retain the first  $n$  base- $b$  digits of each coordinate of  $Y$ , grouped into one word. Use the terminating expansion at ambiguous endpoints; those endpoints have Haar measure zero. At a nonrenewal delete the first letter, leaving the empty word fixed. For a nonempty word  $w$  of length  $l$ , let  $c(w)$  be the midpoint of its  $b$ -adic cube, and decode it as

$$a_{n-l} \text{sinc}(\pi b^{-l}) \Phi(c(w)).$$

Decode the empty word as zero. The age before exhaustion is  $n-l$ , so there is no separate age register. The update does not use the renewal law.

Conditional on this suffix at age  $k < n$ , the vector  $b^k Y$  is uniform on its cube. Averaging each trigonometric coordinate over an interval gives its midpoint value times  $\text{sinc}(\pi b^{-(n-k)})$ . The displayed output is therefore the conditional mean of  $\Phi(b^k U)$  under the suffix. Its raw squared error is

$$d \left[ 1 - a_k^2 \text{sinc}^2(\pi b^{-(n-k)}) \right].$$

After exhaustion the raw error is  $d$ . Renewal reward proves (81.9). For  $0 \leq t \leq 1/2$ ,

$$\frac{4}{3}t^2 \leq 1 - \text{sinc}^2(\pi t) \leq \frac{\pi^2}{3}t^2. \quad (81.13)$$

Indeed, if  $V, V'$  are independent uniform variables on an interval of length  $t$ , the middle expression is  $2\mathbb{E} \sin^2(\pi(V - V'))$ . Use  $2|u| \leq |\sin(\pi u)| \leq \pi|u|$  for  $|u| \leq 1/2$  and  $\mathbb{E}(V - V')^2 = t^2/6$ . This proves the average upper bounds, including  $n = 0$ .

For geometric renewals put  $q = 1 - p$ . The raw age-loss sequence just computed is nondecreasing:  $a_k$  decreases and, as  $k$  increases before exhaustion, the sinc factor decreases as well. The age at time  $t$  has the law of  $\min(G_0, t)$ , where  $\mathbb{P}(G_0 = k) = pq^k$ . Therefore its expected raw loss increases

to  $\sum_k pq^k d_k = U_{n,\sigma}$  and never exceeds this number. The lower bound for supremum loss follows from the average converse.  $\square$

**Corollary 81.3** (Joint statistical and persistent-state law). *Under the hypotheses of Theorem 81.1,*

$$R_M^{\text{av}} \asymp_{b,d,A} B + \Psi_w(n_M). \quad (81.14)$$

*The same holds for raw supremum risk under geometric renewals. Moreover a suffix decoder which does not know  $\sigma$  or the renewal law attains this order of raw risk.*

*Proof.* Every coefficient in  $\Psi_v$  is between zero and one. Thus

$$0 \leq \Psi_w(n) - \Psi_v(n) \leq \sum_k w_k(1 - a_k^2) = B/d.$$

This and (81.8) prove the equivalence. For the noise-independent decoder use  $\text{sinc}(\pi b^{-l})\Phi(c(w))$ , omitting  $a_{n-l}$ . Compare with the hypothetical target  $\Phi(b^k Y)$ . At age  $k$ ,

$$\mathbb{E}\|\Phi(b^k U) - \Phi(b^k Y)\|^2 = 2d(1 - a_k) \leq 2d(1 - a_k^2).$$

The squared triangle inequality and the noiseless suffix bound give average raw risk at most  $4B + (2d\pi^2/3)\Psi_w(n)$ . For geometric renewals the same bound holds for supremum loss: both the noise-discrepancy sequence  $1 - a_k^2$  and the noiseless suffix raw-loss sequence are nondecreasing, so each time expectation is bounded by its stationary-age average. The machine uses only  $b, d, n$  and the observed digits.  $\square$

**81.1. Geometric renewals and the critical window.** For  $w_k = pq^k$ ,  $p = 1 - q$ , the noiseless profile is

$$\Psi_q(n) = pb^{-2n} \sum_{k=0}^{n-1} (b^2 q)^k + q^n. \quad (81.15)$$

Theorem 81.1 is uniform for  $q$  in any fixed compact subset of  $(0, 1)$ , since  $A = 1/p$  is then bounded. In particular it remains valid when  $q$  and  $\sigma$  depend on  $M$ .

**Corollary 81.4** (Phase regimes and crossover). *For each fixed  $0 < q < 1$ , at  $\sigma = 0$  the optimal average and supremum raw risks have orders*

$$\begin{cases} M^{-2/d}, & q < b^{-2}, \\ (1 + \log M)M^{-2/d}, & q = b^{-2}, \\ M^{-\log(1/q)/(d \log b)}, & q > b^{-2}. \end{cases} \quad (81.16)$$

*Let  $q_n = b^{-2} \exp(x/n)$  and  $H(x) = (e^x - 1)/x$ , with  $H(0) = 1$ . Uniformly for  $x$  in bounded intervals,*

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) \longrightarrow (1 - b^{-2})H(x). \quad (81.17)$$

*For the explicit  $S_n$ -state suffix machine, its exact noiseless raw risk satisfies the stronger coefficient limit*

$$\frac{b^{2n}}{n} U_{n,0} \longrightarrow \frac{d\pi^2}{3} (1 - b^{-2})H(x). \quad (81.18)$$

*The optimal risks are bounded above and below by constant multiples of the profile in (81.17), uniformly in this window; no equality of their leading constants is asserted.*

*Proof.* Since  $S_n \asymp_b b^{dn}$ , evaluating the geometric sum in (81.15) proves (81.16). In the critical window,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) = (1 - q_n) \frac{1}{n} \sum_{k=0}^{n-1} e^{xk/n} + \frac{e^x}{n}.$$

The Riemann sum tends uniformly on bounded  $x$ -intervals to  $\int_0^1 e^{xt} dt = H(x)$ . For the exact suffix risk use  $1 - \text{sinc}^2(\pi t) = (\pi^2/3)t^2 + O(t^4)$ , uniformly for  $0 \leq t \leq 1/2$ . After multiplication by  $b^{2n}$ , the error sum is bounded by a constant times

$$\sum_{k < n} e^{xk/n} b^{-2(n-k)},$$

which is uniformly bounded on bounded  $x$ -intervals. The exhaustion term  $db^{2n}q_n^n = de^x$  is bounded as well. Dividing by  $n$  proves (81.18). The final assertion is exactly the uniform two-sided bound of Theorem 81.1.  $\square$

For completeness, the noise floor has the same transition. For  $0 < \sigma \leq 1$ , set  $n_\sigma = \lfloor \log_b(1/\sigma) \rfloor$ ; for  $\sigma > 1$  set  $n_\sigma = 0$ . From  $1 - e^{-4\pi^2 z^2} \asymp \min(1, z^2)$  and  $b^{-(n_\sigma+1)} < \sigma \leq b^{-n_\sigma}$  one obtains

$$B \asymp_{b,d} \Psi_q(n_\sigma), \quad R_M \asymp_{b,d,q} \Psi_q(n_\sigma) + \Psi_q(n_M). \quad (81.19)$$

At  $\sigma = 0$  the first term is interpreted as zero. The comparison in the second formula is uniform on compact  $q$ -intervals. Thus the small-noise orders of  $B$  are respectively  $\sigma^2$ ,  $\sigma^2(1 + \log(1/\sigma))$ , and  $\sigma^{\log(1/q)/\log b}$  in the three regimes. This describes filtering uncertainty and expanding transport in the same experiment, including joint limits, rather than adding an unrelated noisy model.

**81.2. Nonconstant slopes and other readouts.** The preceding law is not confined to a constant-slope coordinate presentation. The next statement records both a genuine robustness class and its precise boundary.

**Proposition 81.5** (Transport under conjugacy and metric readouts). *Let  $h : \mathbb{T}^d \rightarrow E'$  be a Borel isomorphism. Transport the reset law, observation (81.2), dynamics, and readout by  $h$ ; thus the new observation is  $h(Y)$ , the dynamics are  $hT_b h^{-1}$ , and the readout is  $\Phi h^{-1}$ . All the preceding finite-state risks are unchanged. Suppose instead that  $E'$  is compact and the new continuous readout  $g$  satisfies, for some  $0 < c \leq C < \infty$ ,*

$$c \|\Phi(x) - \Phi(y)\| \leq \|g(hx) - g(hy)\| \leq C \|\Phi(x) - \Phi(y)\|. \quad (81.20)$$

*Then its optimal raw squared risks, for the same observations and state budgets, lie between  $c^2/4$  and  $4C^2$  times those for  $\Phi$ . This assertion concerns raw risk, not equality of posterior excess risk for nonlinear readouts.*

*Proof.* Apply  $h^{-1}$  to each acquired observation within the transient update computation. This and its inverse transform machines without changing a persistent state. For the readout assertion, replace any output  $z$  for  $\Phi$  by a nearest point  $\Phi(x_z)$  on its compact image, choosing measurably. For a true state  $x$ ,  $\|\Phi(x_z) - \Phi(x)\| \leq 2\|z - \Phi(x)\|$ . Output  $g(hx_z)$  and use the upper inequality in (81.20). The inverse construction projects onto the compact image of  $g$  and uses the lower inequality. Both transformations act only on the decoder, use no extra states, and are valid conditionally on any coding randomness.  $\square$

For example, in one dimension take

$$h_\varepsilon(x) = x + \frac{\varepsilon}{2\pi} \sin(2\pi x) \pmod{1}, \quad |\varepsilon| < \frac{b-1}{b+1}.$$

This is a smooth circle diffeomorphism and the conjugate map has nonconstant derivative

$$(h_\varepsilon T_b h_\varepsilon^{-1})'(h_\varepsilon x) = b \frac{1 + \varepsilon \cos(2\pi b x)}{1 + \varepsilon \cos(2\pi x)} > 1.$$

Coordinatewise products give higher-dimensional examples, with nonuniform physical reset density and transported nonadditive observation noise. The ordinary circle embedding of  $h_\varepsilon x$  satisfies (81.20). These are nonconstant-slope expanding experiments, but their Lyapunov exponent remains  $\log b$  by conjugacy. We do not identify their law with a pressure theorem for arbitrary nonconjugate

expanding maps. The general posterior-orbit theorem still applies to such maps; its quantization invariant would have to be evaluated separately.

## APPENDIX 82. CALIBRATED MOMENT OBSERVATIONS AT ROOT COLLISIONS

We next derive, rather than postulate, a singular observation modulus. Let

$$\Theta_d = \{\theta \in [0, 1]^d : \theta_1 \leq \cdots \leq \theta_d\}, \quad P_j(\theta) = \sum_{i=1}^d \theta_i^j, \quad 1 \leq j \leq d,$$

and write  $P = (P_1, \dots, P_d)$ . The data are the actually observed pair

$$Y = P(\theta) + \eta + \varepsilon_Y, \quad W = \eta + \varepsilon_W, \quad \begin{pmatrix} \varepsilon_Y \\ \varepsilon_W \end{pmatrix} \sim N(0, \sigma^2 \Sigma_0), \quad (82.1)$$

where  $\eta \in \mathbb{R}^d$  is unrestricted,  $\sigma \geq 0$  is known, and  $\Sigma_0$  is a fixed positive-definite  $2d$  by  $2d$  matrix. Let

$$V_0 = (\Sigma_0)_{YY} + (\Sigma_0)_{WW} - (\Sigma_0)_{YW} - (\Sigma_0)_{WY} > 0.$$

An encoder of the pair retains one of  $M$  labels. Its decoder estimates  $\theta$ , with squared Euclidean loss. Public and private randomness are permitted under the same hard label constraint as above. Denote the minimax risk, taking a supremum over both  $\theta$  and  $\eta$ , by  $\mathcal{R}_M(\Theta, \sigma)$  when the parameter set is  $\Theta$ . Projection permits decoder values to lie in  $\Theta_d$ , so the loss is bounded by  $d$ .

Theorem 85.1, proved in the retained development, reduces this problem *exactly for every*  $M$  to

$$D = Y - W \sim N(P(\theta), \sigma^2 V_0). \quad (82.2)$$

For  $\sigma > 0$  this is its correlated Gaussian nuisance quotient; for  $\sigma = 0$  the difference is noiseless and the reverse comparison follows by fixing the nuisance to any constant. The reduction is an equality of minimax risks for decisions about  $\theta$ . It is not a Blackwell equivalence for the full parameter  $(\theta, \eta)$ .

**Theorem 82.1** (Collision strata and label resolution). *For every fixed  $d \geq 1$  and positive-definite  $\Sigma_0$ , there are  $0 < c \leq C < \infty$ , independent of  $M$  and  $\sigma$ , such that*

$$c \left\{ \min(1, \sigma^{2/d}) + M^{-2/d} \right\} \leq \mathcal{R}_M(\Theta_d, \sigma) \leq C \left\{ \min(1, \sigma^{2/d}) + M^{-2/d} \right\}. \quad (82.3)$$

*More generally, let  $I_1, \dots, I_s \subset [0, 1]$  be ordered closed intervals of positive lengths, with positive gaps between consecutive intervals. Prescribe  $r_j \geq 1$  roots in  $I_j$ , with  $\sum_j r_j = d$ , and let  $\Theta(I, r)$  be the resulting ordered parameter set. If  $r_* = \max_j r_j$ , then*

$$\mathcal{R}_M(\Theta(I, r), \sigma) \asymp_{d, V_0, I, r} \min(1, \sigma^{2/r_*}) + M^{-2/d}. \quad (82.4)$$

*The constants in the stratified assertion may deteriorate when an interval width or a gap tends to zero. Neither assertion assumes a two-sided global contact equivalence.*

The model is genuinely coupled. The determinant of the derivative of  $P$  at distinct ordered roots is

$$\det DP(\theta) = d! \prod_{i < j} (\theta_j - \theta_i). \quad (82.5)$$

Thus the rank changes on the collision strata. There is no invertible smooth global coordinate change that makes this a uniformly regular  $\theta$ -experiment. The next lemma supplies the inverse modulus needed for the statistical theorem. Its root-perturbation argument is classical in method; the point here is to carry it through the actual calibrated observation and hard-label decision problem.

**Lemma 82.2** (Derived inverse moment moduli). *There is a finite  $C_d$  such that for all  $\theta, \theta' \in \Theta_d$ ,*

$$\|\theta - \theta'\| \leq C_d \|P(\theta) - P(\theta')\|^{1/d}. \quad (82.6)$$

*For  $\theta, \theta' \in \Theta(I, r)$  and  $\delta = \|P(\theta) - P(\theta')\|$  sufficiently small, the coordinates in cluster  $j$  satisfy*

$$\|\theta^{(j)} - \theta'^{(j)}\| \leq C_{d,I,r} \delta^{1/r_j}. \quad (82.7)$$

*Both statements include multiple roots and preserve their multiplicities.*

*Proof.* Let  $e_j(\theta)$  be the elementary symmetric polynomials, with  $e_0 = 1$ . Newton's identities give

$$je_j = \sum_{i=1}^j (-1)^{i-1} e_{j-i} P_i.$$

On  $[0, 1]^d$ ,  $|P_i| \leq d$  and  $|e_j| \leq \binom{d}{j}$ . For an explicit finite Lipschitz bound define  $L_0 = 0$  and

$$L_j = \frac{1}{j} \sum_{i=1}^j \left( dL_{j-i} + \binom{d}{j-i} \right), \quad 1 \leq j \leq d.$$

Induction gives  $|e_j(\theta) - e_j(\theta')| \leq L_j \delta$ . For the monic polynomials  $Q_\theta(z) = \prod_i (z - \theta_i)$  and  $Q_{\theta'}$ , it follows that

$$|Q_\theta(z) - Q_{\theta'}(z)| \leq A_d \delta \quad (|z| \leq 2), \quad A_d = 2^d \max_{1 \leq j \leq d} L_j. \quad (82.8)$$

If  $\delta = 0$ , Newton's identities identify the two multisets of roots, and their ordered vectors coincide. Suppose  $\delta > 0$  is small and put  $\rho = (2A_d \delta)^{1/d} < 1$ . Take the union of the complex disks of radius  $\rho$  about the roots of  $Q_\theta$ . At its boundary every factor has modulus at least  $\rho$ , so  $|Q_\theta| \geq \rho^d = 2A_d \delta > |Q_\theta - Q_{\theta'}|$ . Rouché's theorem on each connected component gives the same number of zeros, counted with multiplicity, for the two polynomials. One may increase  $\rho$  arbitrarily slightly to avoid tangent boundaries and then take a limit; the piecewise smooth boundary version follows equivalently from the argument principle. A component containing at most  $d$  disks has diameter at most  $2d\rho$ . Match the roots within each component. For real ordered vectors, sorting a matching cannot increase its maximum displacement: an inverted pair can be uncrossed with no larger maximum distance. Thus  $\|\theta - \theta'\| \leq 2d\sqrt{d}\rho$ . For larger  $\delta$ , compactness and the bound  $\|\theta - \theta'\| \leq \sqrt{d}$  extend the inequality with a larger constant.

For the stratified assertion let  $\Delta > 0$  be the smallest gap; when  $s = 1$  use the preceding argument with  $r_1 = d$ . Around roots in  $I_j$  use disks of radius

$$\rho_j = \left[ 2A_d (2/\Delta)^{d-r_j} \delta \right]^{1/r_j}.$$

For small  $\delta$  these radii are less than  $\min(1, \Delta/3)$ . At the boundary of their union the  $r_j$  factors from this cluster have modulus at least  $\rho_j$ , and all other factors have modulus at least  $\Delta/2$ . Hence the product again exceeds  $A_d \delta$ . Rouché gives equal root counts in each component, and its diameter is at most  $2r_j \rho_j$ . Matching and sorting within the cluster proves (82.7).  $\square$

**Lemma 82.3** (Indistinguishable colliding configurations). *For every  $r \geq 1$  there exist two distinct real vectors  $a_1 < \dots < a_r$  and  $b_1 < \dots < b_r$  in  $(-1/4, 1/4)$  such that*

$$\sum_i a_i^j = \sum_i b_i^j \quad (0 \leq j < r), \quad \sum_i a_i^r \neq \sum_i b_i^r. \quad (82.9)$$

*If these vectors are scaled by  $h$  and translated to an interior point  $x_0$  of a prescribed cluster, their root separation is a positive constant times  $h$ , whereas the difference of all moments of orders  $1, \dots, d$  is  $O(h^r)$ , for fixed  $d \geq r$ .*

*Proof.* Choose any distinct  $a_i$  in the indicated interval and let  $Q(z) = \prod_i (z - a_i)$ . Around each  $a_i$  choose a small closed interval, with disjoint interiors and endpoints still in  $(-1/4, 1/4)$ , on which  $Q$  has opposite nonzero signs at the two endpoints. For any sufficiently small nonzero constant  $c$ , the polynomial  $Q + c$  retains these sign changes. It therefore has one real root in each interval, and these are all its  $r$  roots. Call them  $b_i$ . The monic coefficients through degree one of  $Q$  and  $Q + c$  agree; only the constant coefficient changes. Newton's identities imply (82.9), with the difference at order  $r$  equal up to sign to  $rc \neq 0$ . For every  $j \leq d$ , expand  $\sum_i (x_0 + ha_i)^j - \sum_i (x_0 + hb_i)^j$  by the binomial theorem. All terms of degree less than  $r$  in  $h$  vanish. The remaining terms are  $O(h^r)$ . The Euclidean separation of the two root vectors is  $h\|a - b\| > 0$ .  $\square$

*Proof of Theorem 82.1.* Work first in the difference experiment (82.2). On either compact parameter set take a measurable least-squares minimizer

$$\tilde{\theta} \in \operatorname{argmin}_{\vartheta \in \Theta} \|D - P(\vartheta)\|.$$

Existence follows from compactness and continuity; the measurable minimum theorem gives a Borel selection. With  $D = P(\theta) + \sigma V_0^{1/2} G$ ,  $G \sim N(0, I_d)$ , minimality gives

$$\|P(\tilde{\theta}) - P(\theta)\| \leq 2\sigma \|V_0^{1/2} G\|. \quad (82.10)$$

The global modulus and Gaussian moments yield mean squared error at most  $C\sigma^{2/d}$ , also bounded by  $d$ . On  $\Theta(I, r)$ , (82.7), extended by a bounded constant outside its small- $\delta$  range, yields an upper bound  $C \sum_j \sigma^{2/r_j} \leq C'\sigma^{2/r_*}$  for  $0 < \sigma \leq 1$ . For clarity, that extension is valid since  $\|\theta^{(j)} - \theta'^{(j)}\|^2 \leq r_j$  and  $\delta \geq \delta_0$  implies  $r_j \leq r_j \delta_0^{-2/r_j} \delta^{2/r_j}$ . The cap by the diameter handles  $\sigma > 1$ .

Let  $N = \lfloor M^{1/d} \rfloor \geq M^{1/d}/2$ . The  $N^d$  midpoint product grid in  $[0, 1]^d$  has squared covering radius at most  $d/(4N^2)$ . Project its centres onto  $\Theta$ , which in both cases is a closed convex subset. This does not increase their distances to points of  $\Theta$ . Quantize  $\tilde{\theta}$  to the first nearest projected centre and retain its index. The squared triangle inequality adds at most  $C_d M^{-2/d}$  and multiplies the statistical bound by at most two. This proves the upper bounds with at most  $M$  labels, including the noiseless case.

For a statistical lower bound take a cluster of size  $r_*$ , or the single cluster of size  $d$  for the global assertion. Place the configurations of Lemma 82.3 at an interior point of that cluster; fix all other roots in their intervals. For sufficiently small  $\sigma > 0$  take  $h = c_0 \sigma^{1/r_*}$ . Their squared parameter distance is at least  $c_1 h^2$  and the Kullback–Leibler divergence of the two difference laws is

$$\frac{1}{2\sigma^2} \|V_0^{-1/2} (P(\theta) - P(\theta'))\|^2 \leq C h^{2r_*} / \sigma^2.$$

Choose  $c_0 > 0$  so that this is at most  $1/8$ . Pinsker's inequality and the two-point testing bound give risk at least  $c_2 \sigma^{2/r_*}$  even without a label constraint. For example, classifying an estimator by the closer of the two parameters shows that its maximum squared risk is at least one eighth of their squared distance times one minus their total variation distance. For larger  $\sigma$ , the Gaussian experiment is a garbling of the one at any smaller positive fixed noise level: add an independent normal with covariance  $(\sigma^2 - \sigma_0^2)V_0$  before encoding. Minimax risk is therefore nondecreasing in  $\sigma$ , giving the cap  $c \min(1, \sigma^{2/r_*})$ .

It remains to prove a representation lower bound independent of noise and calibration. The parameter set contains a  $d$ -dimensional box of positive volume, obtained by placing all roots in disjoint small subintervals inside their prescribed intervals. Put the uniform prior on that box. Conditional on independent public randomness, a decoder has at most  $M$  possible deterministic centres; private output randomness can be replaced by its mean for squared loss. A union of  $M$  Euclidean balls of radius  $c_3 M^{-1/d}$  covers at most half the box, with  $c_3 > 0$  depending only on its volume and  $d$ . Pointwise, every distribution of an encoded label has loss at least the

squared distance to the nearest centre. Integrate first over the box and then over the public randomness. The Bayes, hence minimax, risk is at least  $c_4 M^{-2/d}$ . Combining the two lower bounds by  $\max(u, v) \geq (u + v)/2$  proves the result. Finally apply the exact nuisance quotient to return to the observed pair.  $\square$

*Remark 82.4* (Statistical meaning of the collision exponent). The exponent  $1/r_*$  is derived from moment cancellation and multiplicity-sensitive root stability; it is not an assumed contact order. The representation exponent remains  $2/d$  because each allowed stratum-domain contains an open set of  $d$  distinct roots. The paired experiment has equal known weights and noisy observations of its first  $d$  power sums. It is not the experiment of independent samples from a finite mixture with unknown weights. Rates in the latter experiment, such as those of Heinrich and Kahn [31], concern a different likelihood and different local parameter sets. Nor is a polynomial root argument a substitute for an unsmoothed billiard local limit theorem. The present theorem verifies an actual changing-rank observation geometry within the calibrated reduction, without importing either of those assertions.

*Remark 82.5* (Uniformity in the retained contact theorem). The constants of Theorem 85.2 depend on its displayed metric-equivalence constants  $\lambda, \Lambda$ . Their independence of contact orders, widths, or noise scales is conditional on uniform control of  $\lambda, \Lambda$  in those regimes. It is not a claim of uniform physical conditioning. In the moment model the dependence on dimension, covariance conditioning, cluster gaps, and interval widths is made explicit at the level of the hypotheses in Theorem 82.1; no uniform limit through a collapsing cluster gap is inferred from the stratified statement.

## COMPLEMENTARY QUANTITATIVE DEVELOPMENT

The following sections retain the preceding quantitative theory and its proofs. The scalar doubling result provides an exact checkpoint comparison; the correlated nuisance quotient is also used in Section 82. The later filter and control results have separate assumptions and are not premises of the posterior-orbit realization theorem.

### APPENDIX 83. REGENERATION AND THE GEOMETRY OF FUTURE PREDICTIONS

The obstruction in this section concerns information already acquired and subsequently discarded. It does not arise from estimating an unknown transition probability. We first give a converse for a general regenerative experiment and then compute its consequence for an expanding map.

**83.1. A retained-label experiment.** Let  $E$  be a standard Borel space,  $T : E \rightarrow E$  a Borel map,  $\mu$  a probability measure, and  $f : E \rightarrow [0, 1]$  a Borel readout. Fix  $0 < p < 1$  and write  $q = 1 - p$ . At time zero an observer acquires  $U_0 \sim \mu$  and  $X_0 = U_0$ . At each integer time  $t \geq 1$ , an independent mode  $R_t$  is observed, with  $\Pr(R_t = 1) = p$ . If  $R_t = 1$ , a fresh independent  $U_t \sim \mu$  is acquired and  $X_t = U_t$ . Otherwise  $X_t = T(X_{t-1})$ ; there is no further observation of  $X_t$ . All acquisitions and modes are independent of the coding randomness. The full-history prediction is  $f(X_t)$ .

A causal  $M$ -label machine has a state  $L_t \in \{1, \dots, M\}$ . At time zero it encodes  $U_0$ . Subsequently its update can use the previous label, the clock, the current mode, and, at a reset, the current  $U_t$ . After the update only its label, the clock, and independent public coding randomness are available for decoding. In particular the current acquisition is no longer available at the checkpoint. Private coding randomness is allowed. Past modes, reset times, previous acquisitions, and previous decoder outputs are not additional persistent registers. Time-dependent programs are allowed in the lower bound. Any random-generator state or data-dependent tape position that persists also counts as part of the label. Private coins may equivalently be supplied as independent, time-indexed uniforms.

One may realize the loss physically by selecting, after this compression, a Bernoulli terminal probe of mean  $f(X_t)$ . The probe is not an additional observation in the continuing experiment. Conditional expectation then identifies squared prediction excess with

$$L_t^{\text{pred}} = |f(X_t) - \hat{f}_t|^2.$$

Projection of the decoder onto  $[0, 1]$  cannot increase this loss. Define

$$\mathcal{R}_M^{\text{ca}}(p; T, \mu, f) = \inf_{\text{causal } M\text{-label machines}} \sup_{t \geq 0} \mathbb{E} L_t^{\text{pred}}.$$

The clock is free; the elapsed time since the last reset is not free.

**83.2. The future-orbit converse.** For  $u \in E$  set

$$Z_q(u) = (\sqrt{p} q^{k/2} f(T^k u))_{k \geq 0} \in \ell^2, \quad e_M^2(Z_q) = \inf_{c_1, \dots, c_M \in \ell^2} \int \min_i \|Z_q(u) - c_i\|_{\ell^2}^2 \mu(du). \quad (83.1)$$

The map is Borel and has norm at most one. Its geometry records every future readout that must be recovered from one label if no new acquisition occurs. In contrast, the checkpoint geometry concerns only  $f(X_t)$ .

**Theorem 83.1** (Regenerative future-orbit lower bound). *Every causal  $M$ -label machine in the preceding experiment satisfies*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbb{E} L_t^{\text{pred}} \geq e_M^2(Z_q). \quad (83.2)$$

Consequently  $\mathcal{R}_M^{\text{ca}}(p; T, \mu, f) \geq e_M^2(Z_q)$ . The assertion holds for arbitrary independent public and private coding randomization and does not require a stationary machine or an invariant law for  $T$ .

*Proof.* Let  $e_{M,K}^2$  be the quantization error of the first  $K+1$  coordinates of  $Z_q$ . Appending the coordinates  $\sqrt{p} q^{k/2}/2$ ,  $k > K$ , to any truncated codebook shows that

$$e_M^2(Z_q) - \frac{1}{4} q^{K+1} \leq e_{M,K}^2 \leq e_M^2(Z_q). \quad (83.3)$$

Approximate minimizers suffice for this argument; existence is not needed.

Fix a possible reset time  $j \geq 1$ . Condition on its entire past, on the occurrence of that reset, and on the independent coding random tapes. The new  $U_j$  still has law  $\mu$ . The update assigns it one of at most  $M$  labels. Along the subsequent word of  $K$  nonreset modes, every such label determines a sequence of  $K+1$  decoder values. Future random choices are part of the fixed coding tapes; the nonreset word does not contain new data. The weighted squared error of these values, integrated over  $U_j$ , is therefore at least  $e_{M,K}^2$ . This remains true when the assignment itself is randomized, since fixing its independent random tape produces an ordinary assignment to the same number of centres. Integrating back over the past and the tapes gives

$$p \sum_{k=0}^K q^k \mathbb{E}[L_{j+k}^{\text{pred}} \mid R_j = 1, R_{j+1} = \dots = R_{j+k} = 0] \geq e_{M,K}^2.$$

The displayed conditional expectations also integrate over the unconditioned past at  $j$ ; future modes beyond  $j+k$  play no role.

For each checkpoint  $t$ , the events that its last reset occurred at  $j = t - k$ ,  $0 \leq k \leq K$ , are disjoint. Sum their nonnegative contributions over  $1 \leq t \leq N$  and retain only  $1 \leq j \leq N - K$ . Independence of the modes gives

$$\sum_{t=1}^N \mathbb{E} L_t^{\text{pred}} \geq (N - K) e_{M,K}^2.$$

First let  $N \rightarrow \infty$  at fixed  $K$ , and then let  $K \rightarrow \infty$  in (83.3). The supremum of expected checkpoint losses dominates their lower limiting average, proving the consequence.  $\square$

The theorem is a converse for arbitrary causal machines, not only for machines obtained by quantizing  $E$ . The centres in (83.1) may lie off the orbit image. In particular, replacing their quantization problem by a covering problem for the current state would lose information needed for this lower bound.

#### APPENDIX 84. A SHARP CAUSAL-MEMORY TRANSITION FOR AN EXPANDING EXPERIMENT

Take

$$E = [0, 1), \quad \mu = \text{Leb}|_{[0,1)}, \quad T(x) = 2x - \lfloor 2x \rfloor, \quad f(x) = x. \quad (84.1)$$

The fresh acquisition is an analog uniform variable. A machine in the upper bound below reads only a finite binary prefix of it. The map is piecewise expanding; it is not asserted to be globally Lipschitz for the Euclidean metric on  $[0, 1)$ . The proof will account explicitly for the jumps at dyadic boundaries.

Let  $\mathcal{R}_M^{\text{cp}}(t)$  be the optimal expected squared excess when an encoder can use the entire acquisition history at checkpoint  $t$  and then send one of  $M$  labels. Put

$$\Phi_q(M) = \begin{cases} M^{-2}, & 0 < q < 1/4, \\ (1 + \log M)M^{-2}, & q = 1/4, \\ M^{-\kappa(q)}, & 1/4 < q < 1, \end{cases} \quad \kappa(q) = \frac{\log(1/q)}{\log 2}. \quad (84.2)$$

All logarithms except those explicitly marked otherwise are natural.

**Theorem 84.1** (Identical checkpoint geometry, three causal rates). *For the regenerative experiment (84.1), every integer  $M \geq 1$ , and every  $t \geq 0$ ,*

$$\mathcal{R}_M^{\text{cp}}(t) = \frac{1}{12M^2}. \quad (84.3)$$

For each  $0 < q < 1$ , constants  $0 < c_q \leq C_q < \infty$  give

$$c_q \Phi_q(M) \leq \mathcal{R}_M^{\text{ca}}(1 - q; T, \mu, f) \leq C_q \Phi_q(M) \quad (M \geq 1). \quad (84.4)$$

The lower bound already holds for the lower limiting temporal mean in (83.2). The upper bound is attained in order by a single deterministic finite program which does not use  $p$  or  $q$ . The program has at most  $M$  persistent labels, including every state used to record the number of retained binary digits.

The threshold is  $p = 3/4$ . Below that reset frequency the causal label exponent is smaller than the checkpoint exponent; at the threshold a logarithmic factor is necessary. The current acquired prediction is uniform in all these regimes. Thus its dimension and even its entire probability law do not determine the optimal causal memory law.

**84.1. An explicit finite machine.** For an integer  $B \geq 0$  use the label set

$$\mathcal{W}_B = \bigcup_{b=0}^B \{0, 1\}^b, \quad |\mathcal{W}_B| = 2^{B+1} - 1,$$

including the empty word. At each reset retain the first  $B$  binary digits of  $U_t$ , using the terminating expansion at a dyadic rational. At a nonreset delete the leading digit, or leave the empty word unchanged. Decode a word of length  $b$  as the midpoint of its binary cylinder; the empty word decodes as  $1/2$ .

**Lemma 84.2** (Exact risk of the suffix machine). *For this machine,  $\sup_{t \geq 0} \mathbb{E} L_t^{\text{pred}}$  equals*

$$U_B(q) = \frac{1}{12} \left[ p 4^{-B} \sum_{k=0}^{B-1} (4q)^k + q^B \right], \quad (84.5)$$

with the sum empty when  $B = 0$ . Equality holds at every  $t \geq B$ .

*Proof.* A uniform variable has independent fair binary digits outside the null set of dyadic rationals. If the most recent reset was  $k$  steps ago, then the label retains its next  $(B - k)_+$  digits. Conditional on this label and the age, the current state is uniform on the corresponding cylinder, so its midpoint has squared error  $4^{-(B-k)_+}/12$ . At time  $t$  the age probabilities are  $pq^k$  for  $0 \leq k < t$ , and  $q^t$  for  $k = t$ , because time zero was an acquisition. Equivalently this age has the law of  $\min(G, t)$ , where  $\Pr(G = k) = pq^k$ . The function  $k \mapsto 4^{-(B-k)_+}/12$  is nondecreasing and constant for  $k \geq B$ . Its expectation increases to (84.5) and is already equal to it for  $t \geq B$ .  $\square$

Choose the largest  $B$  such that  $2^{B+1} - 1 \leq M$ . Then  $(M + 1)/4 < 2^B \leq (M + 1)/2$ . If  $4q < 1$ , the sum in (84.5) is bounded, and  $q^B \leq 4^{-B}$ . If  $4q = 1$ , the expression is  $4^{-B}(pB + 1)/12$ . If  $4q > 1$ , it is at most  $[1 + p/(4q - 1)]q^B/12$ . These estimates prove the upper bound in (84.4) for every integer budget, not just powers of two.

The label can be stored as an integer whose leading sentinel records its word length. Only  $\lceil \log_2(2^{B+1} - 1) \rceil$  persistent bits are needed. Binary prefix acquisition and output formatting use  $O(B + 1)$  transient bits; deletion can be performed in  $O(B + 1)$  bit operations. The only read-only integer parameter is  $B$ . Neither a real-valued codebook nor the last reset time is hidden in the program. The cost of observing the reset mode and reading the new prefix is an acquisition cost, distinct from these memory bounds.

**84.2. The lower bound without a restriction on encoders.** Define, for  $0 < \delta \leq 1$ ,

$$F_q(\delta) = \begin{cases} \delta^2, & q < 1/4, \\ \delta^2(1 + \log(1/\delta)), & q = 1/4, \\ \delta^{\kappa(q)}, & q > 1/4. \end{cases}$$

Each function is increasing on  $(0, 1]$ .

**Lemma 84.3** (Separation of future orbits). *For  $u, v \in [0, 1]$ ,*

$$\|Z_q(u) - Z_q(v)\|_{\ell^2}^2 \geq \frac{p}{32} F_q(|u - v|). \quad (84.6)$$

*Proof.* The case  $u = v$  is immediate. Write  $\delta = |u - v|$ . If  $\delta \leq 1/2$ , let  $n = \lfloor \log_2(1/(2\delta)) \rfloor$ . For  $0 \leq k \leq n$ , the difference of the fractional parts of  $2^k u$  and  $2^k v$  has absolute value either  $2^k \delta$  or  $1 - 2^k \delta$ . Both are at least  $2^k \delta$ , since  $2^k \delta \leq 1/2$ . Consequently

$$\|Z_q(u) - Z_q(v)\|^2 \geq p\delta^2 \sum_{k=0}^n (4q)^k. \quad (84.7)$$

For  $q < 1/4$  the  $k = 0$  term alone suffices, for every  $\delta$ . For  $q = 1/4$  and  $\delta \leq 1/4$ , the integer  $n + 1$  is at least  $\log(1/\delta)/(2 \log 2)$  and hence at least  $(1 + \log(1/\delta))/4$ . For  $\delta \geq 1/4$ , the initial term  $p\delta^2$  is at least  $p\delta^2(1 + \log(1/\delta))/(1 + \log 4)$ . This proves a stronger constant than the stated one in the critical case.

For  $q > 1/4$  and  $\delta \leq 1/2$ , one has  $2^n \delta > 1/4$  and  $q^n \geq (2\delta)^{\kappa(q)}$ . The last term in (84.7) is therefore at least  $p\delta^{\kappa(q)}/16$ . For  $\delta > 1/2$ , the initial term gives  $p\delta^2 \geq p\delta^{\kappa(q)}/4$ , since  $0 < \kappa(q) < 2$ . All cases imply (84.6).  $\square$

*Proof of Theorem 84.1.* Lebesgue measure is preserved by the doubling map, and a reset also has Lebesgue law. Thus  $X_t$  is uniform for every  $t$ . Given the complete history it is known exactly. Scalar quantization of the uniform law by  $M$  midpoints has error  $1/(12M^2)$ . Conversely, after ordering arbitrary centres their Voronoi cells are intervals of lengths  $a_i$  with  $\sum_i a_i = 1$ ; the integral error on an interval is at least  $a_i^3/12$ . Convexity gives  $\sum_i a_i^3 \geq M^{-2}$ . Random assignment cannot improve on the nearest centre, and conditioning on an independent public seed proves the same bound for randomized encoders and decoders. This establishes (84.3).

For the causal lower bound put  $h = 1/(2M)$  and  $r = \frac{1}{4}\sqrt{pF_q(h)}/32$ . Consider any radius- $r$  ball in  $\ell^2$ , centred anywhere. Two points of the orbit image lying in this ball have distance at most  $2r$ . By Lemma 84.3, their preimages cannot be separated by  $h$  or more. The preimage of the ball therefore has Lebesgue measure at most  $h$ . A union of  $M$  such balls covers at most half the acquired mass, and hence

$$e_M^2(Z_q) \geq r^2/2 = \frac{p}{1024}F_q(1/(2M)).$$

For  $q \leq 1/4$  this last profile is at least  $\Phi_q(M)/4$ ; for  $q > 1/4$  it is  $2^{-\kappa(q)}\Phi_q(M) \geq \Phi_q(M)/4$ . Thus  $c_q = p/4096$  is one possible lower constant. Theorem 83.1 applies to every causal machine and proves the lower assertion including the temporal-mean version. The suffix machine and Lemma 84.2 give the upper bound as already computed.  $\square$

The converse uses Euclidean differences only up to the first relevant dyadic separation; it does not replace the doubling map by an invalid global Lipschitz estimate. Nor does it assume that a general machine stores an interval or a binary word. These restrictions are used only for the explicit upper construction. The same algorithm works for all  $p \in (0, 1)$ , so an unknown reset frequency does not change its order of performance at any fixed true  $p$ .

At  $p = 1$  every step is a fresh checkpoint and the exact risk is  $1/(12M^2)$ . The assertion above concerns  $p > 0$ ; its constants need not remain uniform as  $p \downarrow 0$ . This dependence is displayed in the finite geometric sum (84.5), rather than suppressed in an unspecified stability constant.

## APPENDIX 85. AN EXACT CALIBRATION QUOTIENT AND SINGULAR RESOLUTION

The regenerative experiment concerns excess prediction loss relative to a full-history observer. We now study squared error relative to an unknown parameter. The distinction is essential: calibration uncertainty persists even before any label constraint is imposed.

**85.1. Correlated paired measurements.** Let  $\Theta$  be a compact metric space, let  $f : \Theta \rightarrow \mathbb{R}^m$  be continuous, and observe

$$Y = f(\theta) + \eta + \varepsilon_Y, \quad W = \eta + \varepsilon_W, \quad \theta \in \Theta, \quad \eta \in \mathbb{R}^m. \quad (85.1)$$

The joint centred Gaussian noise has a known positive definite covariance

$$\Sigma = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YW} \\ \Sigma_{WY} & \Sigma_{WW} \end{pmatrix}.$$

No independence between the two channels or their coordinates is imposed. The nuisance  $\eta$  is unrestricted and is not available to the observer. Both measurements are part of the acquisition. The procedure retains one of  $M$  labels and decodes an action in a standard Borel action space; independent public and private randomization is allowed.

Put

$$D = Y - W, \quad V = \Sigma_{YY} + \Sigma_{WW} - \Sigma_{YW} - \Sigma_{WY}, \quad (85.2)$$

$$C = \Sigma_{WY} - \Sigma_{WW}, \quad A = CV^{-1}, \quad R = \Sigma_{WW} - CV^{-1}C^\top. \quad (85.3)$$

The matrices  $V$  and  $R$  are positive definite. A direct Gaussian regression calculation gives the invertible transformation

$$(Y, W) \mapsto (D, Z), \quad Z = W - AD, \quad D \sim N(f(\theta), V), \quad Z \sim N(\eta - Af(\theta), R), \quad (85.4)$$

where  $D$  and  $Z$  are independent for fixed  $(\theta, \eta)$ .

For a bounded nonnegative Borel loss  $\ell(\theta, a) \leq L$ , write  $\mathcal{V}_M^{\text{pair}}$  for the minimax risk in (85.1), with supremum over  $(\theta, \eta)$ , and  $\mathcal{V}_M^D$  for the minimax risk from the sole observation  $D \sim N(f(\theta), V)$ , with the same action space, loss, label budget, and randomization convention.

**Theorem 85.1** (Exact minimax elimination of additive calibration). *For every integer  $M \geq 1$ ,*

$$\mathcal{V}_M^{\text{pair}} = \mathcal{V}_M^D. \quad (85.5)$$

*Thus correlated measurement noise enters the decision problem through the covariance of the observed difference, not through the two marginal variances considered separately.*

*Proof.* Taking differences implements every  $D$ -procedure on the paired experiment without changing its label count, so  $\mathcal{V}_M^{\text{pair}} \leq \mathcal{V}_M^D$ . For the reverse inequality fix any paired procedure and any  $T > 0$ . Average its risk over an independent nuisance  $\eta \sim N(0, T^2 I_m)$ . For each  $\theta$ , this average is bounded above by its worst-nuisance risk. Under this averaging, (85.4) has product law

$$N(f(\theta), V) \otimes N(-Af(\theta), T^2 I_m + R).$$

Replace the second factor by the parameter-independent law  $N(0, T^2 I_m + R)$ . The relative entropy between the two second factors is at most  $\|Af(\theta)\|_2^2 / (2T^2)$ , so their total variation distance is at most  $\|Af(\theta)\|_2 / (2T)$ . This follows also when  $A$  is singular. Since  $f(\Theta)$  is compact, the bound is uniform in  $\theta$ .

A  $D$ -encoder can sample this parameter-independent replacement  $Z$  using private randomness, reconstruct  $W = Z + AD$  and  $Y = D + W$ , and apply the original paired encoder. Its decoder is unchanged and still sees at most  $M$  labels. Data processing and boundedness of the loss imply that its maximum risk is at most the maximum paired risk plus

$$\frac{L}{2T} \sup_{\theta \in \Theta} \|Af(\theta)\|_2.$$

Let  $T \rightarrow \infty$  and then take the infimum over paired procedures. This proves the opposite inequality in (85.5).  $\square$

This is an equality of minimax decision problems for  $\theta$  under an unrestricted additive nuisance. It is not a sufficiency assertion for  $(\theta, \eta)$ , nor an equivalence of the complete experiments for estimating  $\eta$ . The distinction avoids identifying a nuisance-dependent conditional law with a parameter-independent postprocessing kernel.

**85.2. Contact geometry in the noise metric.** Let now

$$\Theta = \prod_{j=1}^d [0, a_j], \quad a_j > 0, \quad k_j \in \mathbb{N}, \quad k_j \geq 1, \quad v_j > 0.$$

Assume that, for fixed constants  $0 < \lambda \leq \Lambda < \infty$ ,

$$\lambda \|b(\theta) - b(\theta')\|_2 \leq \|V^{-1/2}(f(\theta) - f(\theta'))\|_2 \leq \Lambda \|b(\theta) - b(\theta')\|_2, \quad b_j(\theta) = \theta_j^{k_j} / v_j. \quad (85.6)$$

This is a two-sided condition on the actual observation map in the noise metric. It is stronger than a rank statement at one point, but it allows nonlinear, nonseparable embeddings of the contact coordinates. No coordinate independence of the statistical experiment is required.

With  $a_{(1)} \geq \dots \geq a_{(d)}$  the ordered widths, put

$$S = \sum_{j=1}^d \min\{a_j^2, v_j^{2/k_j}\}, \quad Q_M = \max_{1 \leq l \leq d} \left( \frac{\prod_{i=1}^l a_{(i)}}{M} \right)^{2/l}. \quad (85.7)$$

Write  $\mathcal{R}_M$  for the minimax squared Euclidean parameter risk in (85.1), with estimates projected onto  $\Theta$ . Let  $\omega_l$  denote the volume of the Euclidean unit ball in  $\mathbb{R}^l$ .

**Theorem 85.2** (Correlated, nonseparable contact and label law). *Under (85.6), for every integer  $M \geq 1$ ,*

$$c_{d,\Lambda}(S + Q_M) \leq \mathcal{R}_M \leq 2K_{m,\lambda}S + 8dQ_M, \quad K_{m,\lambda} = \max\{1, 4m/\lambda^2\}. \quad (85.8)$$

One may take

$$c_{d,\Lambda} = \frac{1}{2} \min \left\{ \frac{3c^2}{32}, \min_{1 \leq l \leq d} \frac{1}{2} (2\omega_l)^{-2/l} \right\}, \quad c = \min\{1/2, 1/(2\Lambda)\}.$$

The comparison constants are independent of the contact orders, widths, noise scales, label budget, and unknown calibration. A deterministic minimum-distance estimate followed by one fixed product codebook attains the upper bound.

*Proof.* The squared loss is bounded on  $\Theta$ , so Theorem 85.1 reduces the minimax problem exactly to  $D \sim N(f(\theta), V)$ .

*Estimation.* Choose a measurable minimizer

$$\tilde{\theta} \in \operatorname{argmin}_{u \in \Theta} \|V^{-1/2}(D - f(u))\|_2.$$

The minimum exists by compactness. For an explicit Borel selection, subdivide the parameter box dyadically. At each level choose the first closed child of the previously chosen box on which the minimum equals the global minimum. Such a child exists. The minimum over any fixed compact box is continuous in  $D$ , so all these finitely many equality tests are Borel. The selected nested boxes have diameters tending to zero; their unique limiting point is a Borel minimizer. For  $G = V^{-1/2}(D - f(\theta)) \sim N(0, I_m)$ , minimality and the triangle inequality imply

$$\|V^{-1/2}(f(\tilde{\theta}) - f(\theta))\|_2 \leq 2\|G\|_2.$$

It follows from (85.6) and  $|x - y|^{k_j} \leq |x^{k_j} - y^{k_j}|$  for nonnegative  $x, y$  that

$$|\tilde{\theta}_j - \theta_j|^2 \leq \min\{a_j^2, (2v_j \|G\|_2 / \lambda)^{2/k_j}\}.$$

Since  $\mathbb{E} \|G\|_2^{2/k_j} \leq m^{1/k_j}$ , summing gives  $\sup_{\theta} \mathbb{E} \|\tilde{\theta} - \theta\|_2^2 \leq K_{m,\lambda}S$ .

*Compression.* Set  $\delta = 2\sqrt{Q_M}$  and divide coordinate  $j$  into  $n_j = \max\{1, \lfloor a_j/\delta \rfloor\}$  equal intervals. Their product has at most  $M$  cells: if  $l$  widths exceed  $\delta$ , then  $\prod_j n_j \leq \prod_{i \leq l} a_{(i)}/\delta^l \leq M/2^l$ , and the case  $l = 0$  uses one cell. Each side length is at most  $2\delta$ . The midpoint codebook therefore has squared radius at most  $4dQ_M$ . Quantize  $\tilde{\theta}$  to it. The squared triangle inequality proves the upper bound in (85.8).

*Statistical lower bound.* Set  $r_j = \min\{a_j, v_j^{1/k_j}\}$  and use the  $2^d$  parameter vertices  $\theta_j(\xi) = cr_j\xi_j$ ,  $\xi \in \{0, 1\}^d$ . For adjacent vertices differing only in coordinate  $j$ , (85.6) bounds their Gaussian relative entropy by

$$\frac{1}{2}\Lambda^2(cr_j)^{2k_j}/v_j^2 \leq \frac{1}{2}\Lambda^2c^{2k_j} \leq \frac{1}{8}.$$

Their total variation distance is at most  $1/4$ . Thresholding an estimate at  $cr_j/2$  gives a test of bit  $j$ ; an incorrect bit entails squared error at least  $c^2r_j^2/4$ . Pairing adjacent vertices gives average error probability at least  $3/8$ , even after compression. Summing over  $j$  yields a minimax lower bound  $3c^2S/32$ .

*Representation lower bound.* Put uniform probability on  $\Theta$ . For each independent public seed an averaged decoder has at most  $M$  centres, whatever noisy observation produced its label. Projection onto the  $l$  widest coordinates shows that  $M$  balls of radius  $r$  cover at most  $M\omega_l r^l / \prod_{i \leq l} a_{(i)}$  of the parameter mass. With  $r = (\prod_{i \leq l} a_{(i)} / (2M\omega_l))^{1/l}$ , at least half the mass has squared error at least  $r^2$ . Maximizing over  $l$  and integrating over the seed gives  $\mathcal{R}_M \geq [\min_l \frac{1}{2}(2\omega_l)^{-2/l}]Q_M$ . Combining this bound with the statistical bound proves the stated lower constant.  $\square$

**Example 85.3** (Nonseparable contact observations). In two dimensions let

$$b(\theta) = (\theta_1^2/v_1, \theta_2^3/v_2), \quad H(x_1, x_2) = (x_1, x_2 + \alpha \sin x_1), \quad f(\theta) = V^{1/2}H(b(\theta)).$$

Both  $H$  and its inverse have Lipschitz constant at most  $1+|\alpha|$ . Thus (85.6) holds with  $\lambda = (1+|\alpha|)^{-1}$  and  $\Lambda = 1+|\alpha|$ . Theorem 85.2 gives a matched law with statistical term  $\min(a_1^2, v_1) + \min(a_2^2, v_2^{2/3})$  and the two-dimensional anisotropic label floor, despite the coupling of the two mean coordinates. For an actual correlated pair with any chosen positive definite  $V$ , take independent Gaussian variables  $\xi_D \sim N(0, V)$  and  $\xi_R \sim N(0, R)$ , with  $R > 0$ , and any matrix  $A$ . Set  $\varepsilon_W = A\xi_D + \xi_R$  and  $\varepsilon_Y = (I + A)\xi_D + \xi_R$ . The joint noise covariance is positive definite, and its observed difference has exactly covariance  $V$ .

For a scalar channel, correlation coefficient  $\rho$  gives  $V = \sigma^2 + \tau^2 - 2\rho\sigma\tau$ . Ignoring the cross covariance can therefore change the resolution scale. The independent-coordinate experiment, including its zero-noise endpoints, remains Theorem 88.1. The positive definite hypothesis here is used for the exact Gaussian regression and does not replace those previously proved endpoints.

The observation map and covariance in this section are specified parts of the experiment. The unknown quantity is the additive calibration, not an arbitrary unknown map  $f$  or an unknown covariance. The theorem derives the statistical and label rates from the two-sided contact geometry; it does not infer that geometry from a finite derivative rank.

## APPENDIX 86. ACQUIRED GEOMETRY AND NONUNIFORM CAUSAL RESOLUTION

**86.1. The experiment, the metric, and the resource.** Let  $D_t$  be separable metric spaces with metrics  $d_t$  and their Borel sigma fields. A prepared experiment has a known initial predictive state  $s_0$  and jointly Borel updates

$$s_{t+1} = F_t(s_t, a_t, y_{t+1}), \quad F_t(D_t, a, y) \subset D_{t+1}.$$

All actions admitted at a time have a common continuation interface on  $D_t$ . The maps must be defined at every representative used below, not only almost surely under the exact experiment. The current input may include a publicly observed mode, but its past realizations are not available to a decoder unless charged to the retained label.

The physical prediction is  $z_t = \Pi_t(s_t) \in \mathbb{R}^{q_t}$ , with

$$\|\Pi_t(s) - \Pi_t(s')\|_2 \leq K_t d_t(s, s').$$

For a finite query distribution, coordinates are multiplied by the square roots of the query probabilities. The excess squared prediction loss is then exactly  $\|z_t - \hat{z}_t\|_2^2$ . This follows by conditioning on the full acquisition history; the cross term with the future outcome has conditional mean zero. The query is drawn only after compression.

Fix an exploration law independent of coding randomness and denote the resulting acquired law of  $z_t$  by  $\mu_t$ . A lower bound always refers to this law. An upper bound may hold for a larger family of feedback policies; in that case its stability hypotheses must hold for that family. A parameter-uniform statement requires the same implemented update for all parameters, as in Section F. Alternatively, a Bayesian unknown parameter can be included in the predictive state.

For each  $t$ , choose a fixed Borel quantizer  $q_t : D_t \rightarrow C_t \subset D_t$ , where  $|C_t| \leq M_t$  and

$$d_t(s, q_t(s)) \leq r_t \quad (s \in D_t).$$

Finite covers with a first-minimum tie rule give such maps. Let the implemented update  $\tilde{F}_t$  take values in  $D_{t+1}$  and have uniform error at most  $\eta_{t+1}$  at representatives. The machine is

$$\hat{s}_0 = s_0, \quad \hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (86.1)$$

Only the index of  $\hat{s}_t$  persists. A prescribed sequence of arbitrary codebooks is not automatically a finite uniform program. An infinite-time finite-program assertion below uses a fixed codebook, a periodic finite list, or a separately specified finite algorithm generating and decoding the codebooks.

For a fixed visible input word, let  $\Phi_{j,t} : D_j \rightarrow D_t$  be the exact composition of updates from time  $j$  to time  $t$ , and  $\Phi_{t,t} = \text{id}$ . Assume specified measurable bounds  $\Lambda_{j,t} \geq 0$  satisfy

$$d_t(\Phi_{j,t}(x), \Phi_{j,t}(x')) \leq \Lambda_{j,t} d_j(x, x'), \quad \Lambda_{t,t} = 1. \quad (86.2)$$

The bounds can depend on that word. Taking an uncountable supremum and silently declaring it measurable is unnecessary. The hypothesis is on specified measurable bounds valid for every pair of states in the domain.

**86.2. A resolution theorem.** Write  $[x]_+ = \max(x, 0)$ . If  $\nu \leq \mu_t$  is a subprobability measure of mass  $b$ , put

$$A_\nu(r) = \sup_{c \in \mathbb{R}^{qt}} \nu(B(c, r)), \quad \mathcal{L}_t(M) = \sup_{\nu \leq \mu_t, r > 0} r^2 [b - M A_\nu(r)]_+.$$

One may use any collection of specified submeasures in place of the supremum over all of them; no existence of an optimizing submeasure is asserted or required.

**Theorem 86.1** (Acquired-law resolution and exact-suffix stability). *Under the preceding hypotheses the following statements hold.*

(i) *Every checkpoint encoder with at most  $M$  labels, including independent public and private coding randomization, has expected excess at least  $\mathcal{L}_t(M)$ . Consequently every one-machine procedure with stage budgets  $M_t$  has maximum expected excess at least  $\sup_t \mathcal{L}_t(M_t)$ .*

(ii) *Along every admitted common input word the machine (86.1) satisfies*

$$d_t(s_t, \hat{s}_t) \leq \sum_{j=1}^t \Lambda_{j,t} (r_j + \eta_j). \quad (86.3)$$

*In particular its expected checkpoint excess is at most*

$$K_t^2 \left\| \sum_{j=1}^t \Lambda_{j,t} (r_j + \eta_j) \right\|_{L^2}^2 \leq K_t^2 \left( \sum_{j=1}^t \|\Lambda_{j,t}\|_{L^2} (r_j + \eta_j) \right)^2. \quad (86.4)$$

*The probability and all the norms in this assertion are those of the actual run of the implemented controller.*

(iii) *Suppose the same integer budget  $M$  is used at every time. Let  $\psi_t(M) > 0$ . If fixed constants  $a, A, b > 0$  give covers with  $r_t + \eta_t \leq A\psi_t(M)$  and acquired submeasures  $\nu_t$  of mass at least  $b$  such that*

$$M A_{\nu_t}(a\psi_t(M)) \leq b/2, \quad \sup_{t,M} \sum_{j=1}^t K_t \|\Lambda_{j,t}\|_{L^2} \frac{\psi_j(M)}{\psi_t(M)} \leq G < \infty,$$

*then checkpoint expected risks are bounded below by  $ba^2\psi_t(M)^2/2$ , and one causal machine has expected excess at most  $A^2G^2\psi_t(M)^2$  at each time. Thus the optimal maximum expected checkpoint risk is comparable to  $\sup_t \psi_t(M)^2$ . Replacing the  $L^2$  bounds by bounds valid for all histories gives the analogous worst-history upper assertion.*

*Proof.* Condition on an independent public seed. Averaging a randomized decoder conditional on its label and that seed cannot increase squared loss. There are at most  $M$  prediction centres  $c_1, \dots, c_M$ . Randomized assignment cannot improve on the distance to the nearest centre. The  $\nu$ -mass of their  $r$ -balls is at most  $MA_\nu(r)$ ; all remaining mass has squared error at least  $r^2$ . This proves (i), first for a fixed seed and then after integration. The seed is independent of the acquired law. Every causal label is a checkpoint encoding of the entire prefix, so the same lower bounds hold before taking the infimum over causal machines.

For (ii), freeze the actual input word and set  $w_j = \Phi_{j,t}(\hat{s}_j)$  for  $0 \leq j \leq t$ . Then  $w_0 = s_t$ ,  $w_t = \hat{s}_t$ , and

$$w_{j-1} = \Phi_{j,t}(F_{j-1}(\hat{s}_{j-1}, a_{j-1}, y_j)).$$

The two arguments compared here belong to  $D_j$ . Their distance is at most  $r_j + \eta_j$ . Applying (86.2) and summing the  $t$  triangle inequalities proves (86.3). No hypothetical controller is reevaluated at a different state: all suffixes use the same actual word. The Lipschitz readout and the triangle inequality in  $L^2$  give (86.4). The finite-sum argument also applies at each time of an infinite run. Substituting the hypotheses of (iii) into (i) and (ii) proves (iii). In particular the conclusion is  $\sup_t \mathbb{E}L_t$ , not  $\mathbb{E}\sup_t L_t$ .  $\square$

The theorem deliberately separates the upper and lower geometries. A small-dimensional chart can supply a converse without covering the whole domain; conversely, a small cover does not imply that the experiment acquires mass at that scale. Certified perturbations of the model enter  $\eta_j$  only when measured on the stated domain in its stated metric.

**Corollary 86.2** (Finite-horizon anisotropic transfer). *Assume the global geometry, acquired rectangular marginals, and causal compatibility hypotheses (G), (A), (C) of the attainable-geometry theorem in [15]. Its all-integer-budget checkpoint profile and its finite-horizon causal conclusion follow from Theorem 86.1. Explicitly, with its notation,*

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left( M^{-1} \prod_{i=1}^{\ell} s_{t,i} \right)^{2/\ell}, \quad d_t(s_t, \hat{s}_t) \leq C \sum_{j=1}^t \left( \prod_{i=j}^{t-1} L_i \right) Q_j(M_j)^{1/2}.$$

*The resulting comparison constants are those of the fixed horizon.*

*Proof.* The bounded-format thin-rectangle covering lemma of [15], whose statement and proof are reproduced in the source supplement, gives  $r_t(M) \leq C\sqrt{Q_t(M)}$  with representatives in the reachable set. To recall the integer-budget step, its entropy bound is  $C \sum_{j=0}^{p_t} r^{-j} \prod_{i \leq j} s_{t,i}$ . At  $r = L\sqrt{Q_t(M)}$ , all terms of positive degree sum to at most  $M/2$  for a sufficiently large fixed  $L$ . For  $M \geq 2C$  the constant term fits the other half; the remaining finitely many budgets are covered by one point and the bound  $\sqrt{Q_t(M)} \geq s_{t,1}/M$ .

For each acquired rectangular marginal in (A), the mass in a union of  $M$  balls of radius  $r$  is at most  $C_\ell M r^\ell / \prod_{i \leq \ell} s_{t,i}$  of its normalized mass. Choose  $r$  a sufficiently small multiple of  $(\prod_{i \leq \ell} s_{t,i}/M)^{1/\ell}$ . A fixed fraction of the actual subprobability remains. The bounded left inverse of the physical query map and the bounded coordinate transformation transfer this lower bound to prediction error. Maximizing over  $\ell$  gives  $cQ_t$ . Zero products need no converse; if every scale vanishes the domain is a singleton. Causal compatibility gives  $\Lambda_{j,t} \leq \prod_{i=j}^{t-1} L_i$ . Apply (ii), and use the finite number of times for the maximum-risk bound. This reproduces the transfer conclusion; the experiment-specific proofs of (G) and (A) remain those of [15].  $\square$

**86.3. Intermittent contraction.** Let  $\mathcal{H}_t$  contain the visible past through time  $t$ . The next criterion is useful when some steps are not contracting at all.

**Theorem 86.3** (A conditional block criterion). *Assume each same-input map is nonexpansive. Fix an integer  $b \geq 1$  and  $0 < p \leq 1$ ,  $0 \leq \rho < 1$ . Starting at any deterministic time  $j$ , partition the*

subsequent inputs into consecutive blocks of length  $b$ . For each block starting at  $k$ , suppose there is an event  $E_k \in \mathcal{H}_{k+b}$  with

$$\Pr(E_k \mid \mathcal{H}_k) \geq p, \quad \text{Lip}(\Phi_{k,k+b}) \leq \rho \text{ on } E_k,$$

and the block is nonexpansive otherwise. These hypotheses refer to the actual policy or hold uniformly over the specified policy family. Set  $q = 1 - p(1 - \rho^2)$ . Then valid suffix bounds satisfy

$$\|\Lambda_{j,t}\|_{L^2} \leq q^{\lfloor (t-j)/b \rfloor / 2}, \quad \sum_{j=1}^t \|\Lambda_{j,t}\|_{L^2} \leq \frac{b}{1 - \sqrt{q}}.$$

Thus a fixed radius- $r$  codebook, with numerical defect at most  $\eta$ , gives

$$\sup_{t \geq 1} \mathbb{E} d_t(s_t, \hat{s}_t)^2 \leq \frac{b^2(r + \eta)^2}{(1 - \sqrt{q})^2}.$$

No independence between blocks is required. If every block contracts by  $\rho$  on every history, the corresponding pathwise bound is  $b(r + \eta)/(1 - \rho)$ .

*Proof.* For a complete block use the factor  $Z_k = \rho \mathbf{1}_{E_k} + \mathbf{1}_{E_k^c}$ . Its conditional second moment is at most  $q$ . For the first  $m$  complete blocks after  $j$ ,

$$\mathbb{E} \prod_{\ell=0}^{m-1} Z_{j+\ell b}^2 \leq q^m.$$

Indeed the product through the penultimate block is measurable at the start of the last, so conditional expectation removes the last factor with multiplier at most  $q$ ; induction completes the argument. The unfinished final block is nonexpansive. Products of these factors therefore give the required measurable suffix bounds. Each possible value of the integer part occurs at most  $b$  times in the sum over  $j$ . Sum the geometric series and apply (86.4) with the identity readout. The deterministic assertion is the same calculation without expectations.  $\square$

Arbitrarily long strings of unsuccessful blocks have positive probability in many examples satisfying this theorem. A uniform expected checkpoint bound therefore cannot be reinterpreted as a pathwise bound by suppressing the norm in the hypothesis. Conversely, expansion at individual steps is allowed by Theorem 86.1 whenever the exact suffix bounds are summable; the nonexpansive block criterion is a sufficient condition, not the definition of the general class.

## APPENDIX 87. POSITIVE AND INTERMITTENT HIDDEN MARKOV EXPERIMENTS

**87.1. Projective stability.** For positive probability vectors define

$$h(p, q) = \max_i \log(p_i/q_i) - \min_i \log(p_i/q_i).$$

It is a metric on the interior of the simplex. For a positive likelihood vector  $r$ , normalization of  $r_i p_i$  does not change these log-ratios apart from a common additive constant. Bayes multiplication is consequently an isometry in  $h$ .

**Lemma 87.1** (Positive-matrix contraction). *For a positive matrix  $A$ , put*

$$R(A) = \max_{i,j,k,l} \frac{A_{ik}A_{jl}}{A_{il}A_{jk}}, \quad \vartheta(A) = \frac{\sqrt{R(A)} - 1}{\sqrt{R(A)} + 1}.$$

*The normalized map  $p \mapsto Ap/(\mathbf{1}^\top Ap)$  is  $\vartheta(A)$ -Lipschitz in  $h$ . A nonnegative matrix mapping positive vectors to positive vectors is nonexpansive. Also  $\|p - q\|_1 \leq h(p, q)$ .*

*Proof.* We include the finite-dimensional argument for the classical projective estimate [18]. Write  $p_i = e^{x_i} / \sum e^{x_l}$ . The derivative of  $\log \sum_k A_{ik} e^{x_k}$  is the probability row  $w_{ik} = A_{ik} e^{x_k} / \sum_l A_{il} e^{x_l}$ . For two rows  $i, j$ , the ratio  $w_{ik}/w_{jk}$  has maximum divided by minimum at most  $R(A)$ . If probability vectors  $u, v$  have ratios in  $[m, M]$ , with  $m \leq 1 \leq M$ , convexity of  $|x - 1|$  gives

$$\|u - v\|_{\text{TV}} \leq \frac{(M - 1)(1 - m)}{M - m}.$$

Subject to  $M/m \leq R$ , the right side is at most  $(\sqrt{R} - 1)/(\sqrt{R} + 1)$ : increasing the endpoint ratio and then putting  $M = Rm$  reduces the maximization to  $m = R^{-1/2}$ . Consequently the difference of two derivative rows applied to a vector  $v$  is at most  $\vartheta(A) \text{osc}(v)$  in absolute value. Integrate along the segment joining two log-coordinate vectors and take the maximum difference between rows. Normalization subtracts a common constant and does not change oscillation. This proves the contraction. For nonnegative matrices the same derivative argument uses only  $\|w_i - w_j\|_{\text{TV}} \leq 1$ .

For the last assertion join  $q$  to  $p$  by the normalized exponential path with logarithmic increment  $v_i = \log(p_i/q_i)$ . Its derivative has coordinates  $p_i(s)(v_i - \sum_l p_l(s)v_l)$ , whose  $\ell^1$  norm is at most  $\text{osc}(v)$ . Integrating gives the assertion.  $\square$

Use the experiment and physical terminal probe in Section G, now with

$$d \geq 1, \quad 0 < \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (87.1)$$

All known constants and both binary report signs retain their original meaning. Write  $n = d + 1$ ,  $\tau = 1/4$ , and define

$$\rho_{d,\gamma} = \frac{1 - \gamma}{1 - \gamma + 2\gamma/n}, \quad B_\epsilon = \log \frac{1 + \epsilon}{1 - \epsilon}, \quad R_\epsilon = \frac{B_\epsilon}{1 - \rho_{d,\gamma}}, \quad (87.2)$$

$$D_{d,\gamma} = 2^{-d}(5/4)^{d+2}(n/\gamma)^{d+1} \cosh^{2d}(1). \quad (87.3)$$

Let  $v_d$  denote the volume of the Euclidean unit ball in  $\mathbb{R}^d$ .

**Theorem 87.2** (The memory law for every positive refresh rate). *For (87.1), every integer  $M \geq 1$ , and every  $t \geq 1$ ,*

$$\begin{aligned} c_{d,\gamma} \epsilon^2 M^{-2/d} &\leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_{d,\gamma} \epsilon^2 M^{-2/d}, \\ c_{d,\gamma} \epsilon^2 M^{-2/d} &\leq \mathfrak{R}_{M,\infty}^{\text{av}} \leq \mathfrak{R}_{M,\infty}^{\text{max}} \leq C_{d,\gamma} \epsilon^2 M^{-2/d}. \end{aligned}$$

One possible choice is

$$c_{d,\gamma} = \frac{\tau^2}{2d} (2v_d D_{d,\gamma})^{-2/d}, \quad C_{d,\gamma} = \frac{\tau^2}{d} \left(\frac{64}{3}\right)^2 (1 - \rho_{d,\gamma})^{-4}. \quad (87.4)$$

*One deterministic fixed-codebook machine gives the upper bounds for all histories and all times. The lower bounds use independent uniform commands and allow independent coding randomization. At  $\epsilon = 0$ , one label is exact. If the terminal amplitude is multiplied by  $\epsilon$ , all bounds acquire the additional factor  $\epsilon^2$ .*

*Proof.* Put  $a = 1 - \gamma$ ,  $b = \gamma/n$ . The transition matrix has diagonal entries  $a + b$  and off-diagonal entries  $b$ . Its maximum cross-ratio is  $((a + b)/b)^2$ , attained by two distinct diagonal positions. Lemma 87.1 gives coefficient  $a/(a + 2b) = \rho_{d,\gamma}$ . Multiplication by the likelihood does not alter  $h$ . Since  $P_\gamma \nu = \nu$ ,

$$h(F(p, u, y), \nu) \leq \rho_{d,\gamma} h(p, \nu) + B_\epsilon.$$

Thus  $D = \{p : h(p, \nu) \leq R_\epsilon\}$  is an invariant compact subset of the positive simplex. The initialization  $\nu$  belongs to  $D$ .

In log-odds coordinates  $x_i = \log(p_i/p_0)$ , the domain is exactly  $\text{osc}(0, x_1, \dots, x_d) \leq R_\epsilon$  and lies in  $[-R_\epsilon, R_\epsilon]^d$ . Divide this cube into  $k^d$  equal cells,  $k = \lfloor M^{1/d} \rfloor$ . Select one point of  $D$  in each

occupied cell. Two points in a cell have projective distance at most  $4R_\epsilon/k$ , so the fixed codebook has radius  $r_M \leq 8R_\epsilon M^{-1/d}$ . Its points are admissible normalized beliefs, although they need not be reachable at a particular time. Quantize after each exact update. The suffix bound is  $\rho_{d,\gamma}^{t-j}$ , whence

$$\sup_t h(p_t, \hat{p}_t) \leq \frac{8B_\epsilon}{(1 - \rho_{d,\gamma})^2} M^{-1/d}.$$

For  $0 \leq \epsilon \leq 1/4$ , integration of  $2/(1 - \epsilon^2)$  gives  $B_\epsilon \leq (8/3)\epsilon$ . The physical loss is  $(\tau^2/d) \|p_{t,+} - \hat{p}_{t,+}\|_2^2$  and is at most  $(\tau^2/d) h(p_t, \hat{p}_t)^2$ . This proves the upper constant.

For the converse the pre-report coordinates satisfy  $q_i \geq \gamma/n$ . The command-to-posterior Jacobian in Lemma G.4, with this floor in place of  $m_*$ , gives a density bounded by  $D_{d,\gamma} \epsilon^{-d}$  for  $p_{t,+}$ . For clarity, the actual joint density of command and sign is  $2^{-(d+1)} D_y$  and

$$|\det D_u F(p, u, y)_+| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \operatorname{sech}^2 u_i.$$

The odds  $p'_i/p'_0 = q_i(1 + y\epsilon \tanh u_i)/q_0$  make each sign's map injective. Changing variables, summing the two signs, and integrating over every previous history gives exactly (87.3); no conditioning cost is removed. The mass of  $M$  Euclidean balls of radius  $r$  is at most  $M v_d D_{d,\gamma} \epsilon^{-d} r^d$ . Take  $r = \epsilon(2M v_d D_{d,\gamma})^{-1/d}$ . At least half the mass remains, and the physical factor is  $\tau^2/d$ . This proves the lower constant and, by checkpoint relaxation, its causal version. At zero signal the posterior stays  $\nu$ . Scaling the terminal probe scales its squared loss, proving the last assertion.  $\square$

The dependence on mixing and dimension in (87.4) is part of the statement. Since  $1 - \rho_{d,\gamma} = 2\gamma/[n(1 - \gamma) + 2\gamma]$ , the upper constant is explicit even for weak refresh. The theorem is uniform in  $M, t$  and  $\epsilon$  for each positive  $\gamma$ ; it does not assert matching powers of  $\gamma$  in a simultaneous  $\gamma \downarrow 0$  limit. This is a strict enlargement of the refresh interval, not a relabeling of the old  $\ell^1$  estimate. At zero refresh the finite-horizon theorem remains available whenever its separate geometry and stability hypotheses hold.

**87.2. An experiment with unbounded noncontracting intervals.** At each time first draw an observed mode  $I_t$ , independently of the hidden state and of all past variables, with  $\Pr(I_t = 1) = p > 0$ . For  $I_t = 1$  execute the preceding refresh and commanded observation. For  $I_t = 0$  apply any prescribed permutation of the hidden states and return an uninformative fair sign. Commands are independent uniform for the converse. The upper construction accepts every admitted command word. The mode is read in the current update, then discarded except for its effect on the retained label. The terminal probe is unchanged.

**Corollary 87.3** (Intermittent-refresh expected resolution). *In this experiment, for every integer  $M \geq 1$ ,*

$$p c_{d,\gamma} \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,\infty}^{\text{av}} \leq \frac{\tau^2}{d} \frac{(64/3)^2 \epsilon^2 M^{-2/d}}{(1 - \rho_{d,\gamma})^2 (1 - \sqrt{1 - p(1 - \rho_{d,\gamma}^2)})^2}.$$

*The same lower bound holds at every positive checkpoint. The upper bound uses one fixed codebook. For  $0 < p < 1$ , arbitrarily long noncontracting intervals are admitted.*

*Proof.* Permutations fix  $\nu$  and preserve  $h$ ; hence the invariant domain  $D$  and its cover from the preceding proof work in both modes. A good step contracts by  $\rho_{d,\gamma}$ , a bad step is an isometry. Theorem 86.3 with  $b = 1$  gives the upper bound. Conditional on the current good mode, the density calculation in the preceding proof is uniform over the prior history. The good-mode submeasure has mass  $p$  and density at most  $p D_{d,\gamma} \epsilon^{-d}$ . The same balls leave at least  $p/2$  of its mass, proving the lower bound. Independent Bernoulli modes permit every finite string of bad steps.  $\square$

**87.3. Discrete and singular acquisition.** For a finite command alphabet of size  $K$ , a fixed time  $t$  has at most  $(2K)^t$  command/sign histories in the ordinary refreshing experiment. Its checkpoint prediction is therefore exactly encodable at that budget. This elementary fact precludes a positive  $M^{-2/d}$  lower bound for all  $M$  at fixed  $t$  in that different experiment.

A singular command law has a different outcome. Let  $d = 1$ ,  $\gamma = 1$ , and let  $u = 2V - 1$ , where  $V$  is the standard middle-thirds Cantor random variable. Set  $s = \log 2 / \log 3$ . The hidden state is freshly prepared at each step, but the binary command observation and the terminal probe are still physical observations as above.

**Proposition 87.4** (A singular acquired dimension). *For this experiment there are constants  $0 < c < C < \infty$  such that for every  $t \geq 1$ ,  $M \geq 1$ , and  $0 \leq \epsilon \leq 1/4$ ,*

$$c\epsilon^2 M^{-2/s} \leq R_{M,t}^{\text{av}} \leq \mathfrak{R}_{M,\infty}^{\text{max}} \leq C\epsilon^2 M^{-2/s}.$$

*Here worst histories use commands in the support of the prescribed Cantor law. One fixed codebook attains the upper bound.*

*Proof.* A level- $m$  Cantor cylinder has length  $3^{-m}$  and mass  $2^{-m}$ . At that scale an interval of length comparable to  $3^{-m}$  meets only a bounded number of cylinders. Thus the mass of an interval of radius  $r$  is at most  $Cr^s$ , while the support has a cover by  $2^m$  intervals of length  $3^{-m}$ . Affine rescaling to  $[-1, 1]$  changes only constants.

For either sign the posterior coordinate is  $(1 + y\epsilon \tanh u)/(2 + y\epsilon \tanh u)$ . Its derivative in absolute value is  $\epsilon \text{sech}^2 u / (2 + y\epsilon \tanh u)^2$ , between two fixed positive multiples of  $\epsilon$  when  $\epsilon > 0$ . Each sign therefore gives a bi-Lipschitz image at scale  $\epsilon$ . The actual sign weight is between  $3/8$  and  $5/8$ . The sum of the two image measures has concentration at most  $C(r/\epsilon)^s$ ; each image has the corresponding covers. Allocate half the available centres to each image, adjusting the fixed constant for the finitely many budgets below four. The acquired-law lower bound and this cover give the asserted rate. Since  $\gamma = 1$ , the new posterior does not depend on the old label. The same quantizer is an exact one-step causal construction at every time. At  $\epsilon = 0$  it is constant.  $\square$

Neither the domain dimension of the command nor a Jacobian rank alone determines these three rates. For a mixture with finitely many additional atoms, assign one centre to each atom and use the remaining centres for the singular component; the submeasure converse retains its actual mixture weight. More generally Theorem 86.1 uses the full concentration profile without assigning a single dimension.

## APPENDIX 88. SINGULAR STATISTICAL RESOLUTION WITH MEASURED CALIBRATION

We now change the target and the loss explicitly. Let  $\theta \in \Theta = \prod_{j=1}^d [0, a_j]$ ,  $a_j > 0$ , and let  $\eta \in \mathbb{R}^d$  be an unrestricted unknown additive calibration. Fix integers  $k_j \geq 1$ . The experiment consists of the  $2d$  observations

$$Y_j = \theta_j^{k_j} + \eta_j + \sigma_j Z_j, \quad W_j = \eta_j + \tau_j Z'_j, \quad 1 \leq j \leq d, \quad (88.1)$$

with independent standard normal variables and known  $\sigma_j, \tau_j \geq 0$ . A procedure observes the pair, retains one of  $M$  labels, and decodes an estimate of  $\theta$ . Only the label and independent public coding randomness are available to the decoder. Both  $Y$  and  $W$  count as acquisitions. In particular  $W$  is not a free read-only value of  $\eta$ .

Put  $v_j = (\sigma_j^2 + \tau_j^2)^{1/2}$ , and let  $a_{(1)} \geq \dots \geq a_{(d)}$  be the ordered widths. Define

$$S = \sum_{j=1}^d \min\{a_j^2, v_j^{2/k_j}\}, \quad Q_M = \max_{1 \leq \ell \leq d} \left( M^{-1} \prod_{i=1}^{\ell} a_{(i)} \right)^{2/\ell}. \quad (88.2)$$

The risk here is squared parameter error, not the prediction excess used in the preceding sections.

**Theorem 88.1** (A matched contact, calibration, and label law). *For the actual experiment (88.1), write*

$$\mathcal{R}_M = \inf_{M\text{-label procedures}} \sup_{\theta \in \Theta, \eta \in \mathbb{R}^d} \mathbb{E}_{\theta, \eta} \|\hat{\theta} - \theta\|_2^2.$$

For every integer  $M \geq 1$ ,

$$c_d(S + Q_M) \leq \mathcal{R}_M \leq 2S + 8dQ_M, \quad (88.3)$$

where one may take

$$c_d = \frac{1}{2} \min \left\{ \frac{3}{128}, \min_{1 \leq \ell \leq d} \frac{1}{2} (2\omega_\ell)^{-2/\ell} \right\},$$

where  $\omega_\ell$  denotes the volume of the Euclidean unit ball in  $\mathbb{R}^\ell$ . The constants are independent of  $a_j, k_j, \sigma_j, \tau_j, M$  and the unknown calibration. The upper construction is deterministic. The lower bound allows independent public and private randomization.

*Proof. Estimation before compression.* The actually observed differences  $D_j = Y_j - W_j$  have laws  $N(\theta_j^{k_j}, v_j^2)$  independent of  $\eta$ . Define

$$\tilde{\theta}_j = (\text{proj}_{[0, a_j^{k_j}]} D_j)^{1/k_j}.$$

For nonnegative  $x, y$ ,  $|x^{1/k} - y^{1/k}| \leq |x - y|^{1/k}$ . Projection cannot increase the distance to  $\theta_j^{k_j}$ . Concavity of  $x^{1/k_j}$  and  $\mathbb{E}Z^2 = 1$  consequently give

$$\mathbb{E}|\tilde{\theta}_j - \theta_j|^2 \leq \min\{a_j^2, v_j^{2/k_j} \mathbb{E}|Z|^{2/k_j}\} \leq \min\{a_j^2, v_j^{2/k_j}\}.$$

This is a finite-sample bound, including degenerate zero-noise coordinates.

*An integer-budget cover.* Set  $\delta = 2\sqrt{Q_M}$  and partition coordinate  $j$  into  $n_j = \max\{1, \lfloor a_j/\delta \rfloor\}$  equal intervals. Let  $\ell$  be the number of widths greater than  $\delta$ . If  $\ell = 0$  the product of the  $n_j$  is one. Otherwise

$$\prod_j n_j \leq \prod_{i \leq \ell} a_{(i)}/\delta^\ell \leq M/2^\ell \leq M.$$

Every interval has length at most  $2\delta$ , so its midpoint product codebook has squared covering radius at most  $d\delta^2 = 4dQ_M$ . Quantize  $\tilde{\theta}$  to this codebook. The squared triangle inequality proves the upper bound in (88.3).

*The statistical obstruction with unknown calibration.* Write  $r_j = \min\{a_j, v_j^{1/k_j}\}$ . Coordinates with  $r_j = 0$  contribute zero and can be omitted from this step. For  $\omega \in \{0, 1\}^d$ , set

$$\theta_j(\omega) = \frac{1}{2}r_j\omega_j, \quad m_j = (r_j/2)^{k_j}, \quad \eta_j(\omega) = -\frac{\tau_j^2}{v_j^2}m_j\omega_j.$$

These are admissible alternatives in the same unknown-calibration experiment. Adjacent vertices differ in the mean pair at coordinate  $j$  by  $(m_j\sigma_j^2/v_j^2, -m_j\tau_j^2/v_j^2)$ . Their Gaussian divergence is

$$\frac{1}{2} \left( \frac{m_j^2\sigma_j^2}{v_j^4} + \frac{m_j^2\tau_j^2}{v_j^4} \right) = \frac{m_j^2}{2v_j^2} \leq \frac{1}{8}.$$

A zero-variance channel in this expression has identical means and is simply omitted. Pinsker's inequality bounds the variation distance by  $1/4$ . Any estimate determines a test of bit  $j$  by thresholding at  $r_j/4$ . An incorrect test has squared coordinate error at least  $r_j^2/16$ . Pair adjacent vertices and average: their average error probability is at least  $(1 - 1/4)/2 = 3/8$ . Summing over coordinates gives average squared risk at least  $3 \sum_j r_j^2/128$ . Compression and independent randomization cannot improve the testing distance. Since maximum risk dominates this finite average,  $\mathcal{R}_M \geq 3S/128$ .

*The representation obstruction.* Fix  $\eta = 0$  and put uniform probability on the parameter box. At each fixed public seed an averaged decoder has at most  $M$  centres. Regardless of what noisy observations led to the label, its error at  $\theta$  is at least the distance to the nearest centre. Project onto the  $\ell$  widest coordinates. The fraction of this rectangular uniform law covered by  $M$  balls of radius  $r$  is at most  $M\omega_\ell r^\ell / \prod_{i \leq \ell} a_{(i)}$ . Choose  $r = (\prod_{i \leq \ell} a_{(i)} / (2M\omega_\ell))^{1/\ell}$ . At least half the mass has squared error at least  $r^2$ . Maximize over  $\ell$  and integrate over the seed. This proves  $\mathcal{R}_M \geq [\min_\ell \frac{1}{2}(2\omega_\ell)^{-2/\ell}]Q_M$ . Combining the two lower bounds by  $\max(x, y) \geq (x + y)/2$  proves the result.  $\square$

The testing alternatives explain why the calibration variance adds to the signal variance. Holding the nuisance artificially fixed in that step would prove only the weaker obstruction from  $\sigma_j$ , and would miss the cost of measuring calibration. The encoder never reads the unknown nuisance. Its read-only constants are the specified noise levels, contact orders, and parameter box.

For  $d = 1$ ,  $a_1 = 1$ , and  $v_1 \asymp N^{-1/2}$ , the theorem gives  $\mathcal{R}_M \asymp N^{-1/k_1} + M^{-2}$ . Hence the statistical amplitude scale is  $N^{-1/(2k_1)}$ , and order  $N^{1/(2k_1)}$  labels suffice to reach it. When  $k_1 = 2$ , this recovers the quadratic-contact amplitude scale in the physical-ray result of [16], now with an explicit finite-label constraint and an actually measured unrestricted additive calibration in (88.1). It is a comparison of rates in these specified experiments, not an assertion of their Le Cam equivalence. The nonnegative branch is part of the parameter space: an even-power observation on a signed branch would have an additional identifiability obstruction.

#### APPENDIX 89. OPTIMIZATION UNDER A RETAINED-LABEL CONSTRAINT

The following result supplies a policy comparison in addition to the common-controller statements of Section F. Let  $D$  be a compact metric predictive-state domain and  $\mathcal{A}$  a finite action set. The report kernel is  $g(dy | s, a)$  and the state update is  $F(s, a, y) \in D$ . Assume these are Borel, and, uniformly in  $a, y$ ,

$$0 \leq c(s, a) \leq C_0, \quad |c(s, a) - c(s', a)| \leq L_c d(s, s'), \quad d(F(s, a, y), F(s', a, y)) \leq l d(s, s'),$$

$$\|g(\cdot | s, a) - g(\cdot | s', a)\|_{\text{TV}} \leq \kappa d(s, s').$$

The current conditional expected physical cost is  $c(s, a)$ . The predictive state is sufficient for future costs and experiments under all admitted policies. Thus the belief-state dynamic program represents the optimum among all history-dependent policies. Fix discount  $0 < \beta < 1$  with  $\beta l < 1$ , and define

$$L_* = \frac{L_c + \beta \kappa C_0 / (1 - \beta)}{1 - \beta l}.$$

Let  $\mathcal{C} \subset D$  have at most  $M$  points and a radius- $r$  Borel quantizer  $q$ . Assume the initial state is in  $\mathcal{C}$ ; it can be reserved as a centre. The finite reference decision process uses cost  $c(c, a)$  and next state  $qF(c, a, y)$  with report distribution  $g(dy | c, a)$ . It has an optimal stationary policy  $\hat{a}(c)$  because its state and action spaces are finite. Its transition probabilities are the specified integrals of these kernels; numerical approximation is addressed below.

**Theorem 89.1** (A finite reference policy versus unrestricted control). *Implement  $a_t = \hat{a}(\hat{s}_t)$  in the actual experiment and update  $\hat{s}_{t+1} = qF(\hat{s}_t, a_t, y_{t+1})$ . Let  $V^*$  be the unrestricted optimal discounted value and  $J(\hat{a})$  the actual cost of this retained-label controller. Then*

$$0 \leq J(\hat{a}) - V^*(s_0) \leq 2L_* \sum_{t \geq 0} \beta^t \mathbb{E} d(s_t, \hat{s}_t) + \frac{2\beta L_* r}{(1 - \beta)^2}. \quad (89.1)$$

*In particular, if the fixed-codebook construction has  $\sup_t \mathbb{E} d(s_t, \hat{s}_t) \leq Gr$ , the upper bound is  $2L_* Gr / (1 - \beta) + 2\beta L_* r / (1 - \beta)^2$ . Only one of  $M$  labels persists; no exact value function or full history is an input to the implemented policy.*

*Proof.* The exact Bellman operator on bounded Borel functions is

$$TV(s) = \min_a \{c(s, a) + \beta \int V(F(s, a, y))g(dy \mid s, a)\}.$$

It is a contraction in the uniform norm. Starting at zero, its iterates have values in  $[0, C_0/(1 - \beta)]$ . If an iterate has Lipschitz constant  $L$ , the next has constant at most  $L_c + \beta lL + \beta \kappa C_0/(1 - \beta)$ : compare the same report argument first and then the two report laws, using the range bound on  $V$ . The minimum over the same finite action set preserves the bound. It follows that  $V^*$  is  $L_*$ -Lipschitz, and each optimal action-value function  $Q^*(\cdot, a)$  is also  $L_*$ -Lipschitz.

Let  $\widehat{V}$  be the finite reference value. At a centre, replacing  $F(c, a, y)$  by  $qF(c, a, y)$  in  $V^*$  changes a Bellman expression by at most  $\beta L_* r$ . The finite Bellman contraction therefore implies

$$\sup_{c \in \mathcal{C}} |\widehat{V}(c) - V^*(c)| \leq \frac{\beta L_* r}{1 - \beta} =: D_0.$$

For every centre and action, the exact action value and the finite reference action value differ by at most  $\beta(L_* r + D_0) = \beta L_* r/(1 - \beta) = D_0$ . Minimizing the latter consequently yields

$$Q^*(c, \widehat{a}(c)) - V^*(c) \leq 2D_0.$$

For an actual state  $s$  and retained centre  $c$ , Lipschitz continuity of both  $Q^*$  and  $V^*$  gives  $Q^*(s, \widehat{a}(c)) - V^*(s) \leq 2L_* d(s, c) + 2D_0$ .

Along the actual trajectory, the conditional definition of  $Q^*$  makes the discounted sum of these residuals equal to  $J(\widehat{a}) - V^*(s_0)$ . More explicitly the finite sum telescopes, and its final term has absolute value at most  $\beta^{T+1} C_0/(1 - \beta)$ , tending to zero. Sum the bound and let  $T$  increase. This proves (89.1). The policy is a finite table indexed by the retained representative. The exact value entered only this comparison proof.  $\square$

A stationary centre may be reserved at a cost of one label: for  $M \geq 2$  use the preceding covers with  $M - 1$  centres and add  $s_0$ . A power-law covering rate is unchanged up to its stated fixed factor; for  $M = 1$  the bounded-cost estimate is available. If an implemented finite table is within  $\xi$  in each action value of the exact reference table, choosing its minimizing action adds at most  $2\xi/(1 - \beta)$  to the bound. This numerical certificate is separate from computing that table. Theorems 86.1 and 86.3 provide  $G$  when their assumptions hold under the lifted policy. A prediction lower bound is not a matching lower bound for every control cost; no such identification is used here.

## APPENDIX 90. FINITE DESCRIPTIONS AND NUMERICAL IMPLEMENTATION

We distinguish five resource coordinates: persistent labels, program description, transient workspace, numerical error, and physical acquisitions. The first coordinate is determined by the resolution laws; the others are not silently identified with it. We give a finite-description construction for the positive-refresh upper bound, so its codebook does not require an independently stored list of  $M$  arbitrary real vectors.

Suppose  $\gamma, \epsilon$  are rational known inputs and  $\epsilon > 0$ . Choose a rational  $R'$  with  $R_\epsilon \leq R' \leq 2R_\epsilon$ . Such a number is computable from the displayed elementary formula. For an integer  $k \geq 1$ , set  $\delta = R'/k$  and use the log-odds grid

$$\mathcal{J}_k = \{j \in \{-k, \dots, k\}^d : \text{osc}(0, j_1, \dots, j_d) \leq k\}, \quad p_i(j) = \frac{e^{\delta j_i}}{1 + \sum_{l=1}^d e^{\delta j_l}}, \quad (90.1)$$

with  $p_0$  the reciprocal denominator. Its domain is the invariant ball of radius  $R'$ .

**Proposition 90.1** (An implicit codebook). *There is a finite-description codebook in this ball with at most  $M$  representatives and projective radius at most  $12R'M^{-1/d}$ . Its persistent state uses at most  $\lceil \log_2 M \rceil$  bits. Its read-only description consists of  $d$ , the rational calibration,  $R'$ ,*

*an integer  $k$ , and fixed algorithms for enumeration and elementary functions; it contains no list of  $M$  arbitrary real numbers. Certified update error  $\eta$  and quantization error at most  $r + \eta$  give projective filtering error at most  $(r + 2\eta)/(1 - \rho_{d,\gamma})$ .*

*Proof.* For  $M \geq 3^d$  choose  $k = \lfloor (M^{1/d} - 1)/2 \rfloor$ . Then  $(2k+1)^d \leq M$  and  $k \geq M^{1/d}/6$ . Round each log-odds coordinate toward zero in multiples of  $\delta$ . The result remains in the oscillation ball: when both signs occur, both extreme absolute values decrease; when all coordinates have one sign, their extreme absolute value decreases. Each coordinate changes by at most  $\delta$ , so projective distance changes by at most  $2\delta \leq 12R'M^{-1/d}$ . For  $M < 3^d$  use only  $\nu$ ; its radius  $R'$  is at most  $3R'M^{-1/d}$ . Lexicographic enumeration of the finite integer set labels the grid. The model and this enumeration are finite descriptions of all its representatives. The stated error is the geometric-series estimate, with total one-step defect  $r + 2\eta$ .  $\square$

All log-odds and softmax functions in this construction are computable on their compact domains. With digitally represented commands, interval arithmetic can evaluate the update and finitely many distances until a representative within  $r + \eta$  is certified, for any prescribed positive rational  $\eta$ . The fixed positive slack ensures termination even at a tie; exact comparison of two computable reals is not required. This is an existence and accuracy construction, not a uniform polynomial-time claim. The bound charges the counter needed to enumerate the grid and the precision of the computed numbers as transient workspace. The required precision can depend on  $d, \gamma, \epsilon, M, \eta$ .

A command rounded with coordinate error at most  $h$  changes each log likelihood by at most  $\epsilon h/(1 - \epsilon)$  and hence changes the posterior projective distance by at most  $2\epsilon h/(1 - \epsilon)$ . This supplies an explicit additional one-step defect. It does not replace the analog acquisition law in the lower bound by a finite alphabet while keeping the same converse. The sampling law and the numerical interface must be declared separately. For a noncomputable real calibration, an exact read-only real convention would be a different resource model. The finite-description assertion above uses rational calibration and a numerical certificate instead.

The paired singular experiment uses  $2d$  scalar measurements and one terminal label. Its box codebook is generated by  $d$  integer subdivision counts rather than an arbitrary centre list. Computing the differences and roots uses transient numerical workspace. A controller in Theorem 89.1 additionally has a read-only policy table; its construction and precision are not included in the label count.

## APPENDIX 91. COMPARISON OF RESULTS AND MATHEMATICAL DEPENDENCIES

The relationship to earlier work is most precise at the level of the conclusion and the retained information.

| Antecedent                               | Existing conclusion used for comparison                                                 | Conclusion proved here                                                                                                          |
|------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| [15], attainable-geometry transfer       | Anisotropic acquired-law profile and a compatible finite-horizon label machine          | Exact-suffix bounds and intermittent infinite-horizon resolution; the old transfer is Corollary 86.2                            |
| [8]                                      | Quantization methods for nonlinear filtering                                            | A matching converse for acquired prediction labels, not merely a finite signal grid                                             |
| [19, Theorems 12, 16, 17]                | Finite-memory policy approximation from filter stability                                | A policy computed on a finite representative model with an explicit retained-label budget and actual-state error term           |
| [20, Theorems 9, 27]                     | Approximate information states and approximate dynamic programs with performance bounds | Theorem 89.1 is a specialized quantitative policy-transfer result, used together with the acquired-law converse                 |
| [16], physical-ray and score experiments | Quadratic-contact scale, Gaussian comparison, information retained by contact scores    | Finite-sample contact orders, unrestricted measured additive calibration, and a matched label floor in one specified experiment |

Recent finite-window developments include continuous-space approximations [21] and refined window-policy bounds [22]. The fixed-policy belief-cost comparison in [23] is another nearby, but distinct, target. None is replaced by a claim that the stability recurrence, quantization principle, or Bellman comparison was invented here. The specific mathematical assertions to compare are the joint hypotheses and conclusions of Theorems 86.1, 86.3, 87.2, and 88.1.

There is no backward dependence on a later theta manuscript. The only model-specific imported geometric estimate used in a corollary is the bounded-format covering result of the fixed A1 edition; the new principal theorems have proofs here. The A2 comparisons keep its physical cone and noise experiment distinct from (88.1). The earlier Sinai/vector-roof, particle large-deviation, kinetic-semigroup, unbounded operator, and labelled-phase problems keep their original dependency order. A proof of finite-label filtering does not assert any of those model-specific conclusions. The contribution to those chains is a proved resolution and decision interface, usable after their own experiment and domain hypotheses have been established.

## FOUNDATIONAL DEVELOPMENT AND BOUNDARY EXAMPLES

The following appendices give the measurable, statistical, and asymptotic foundations used above. Their statements and proofs are retained from the first edition. The strong-refresh result in Appendix G is extended by Theorem 87.2; the additional statistical results are Theorems 85.1 and 85.2. The final interface appendix distinguishes these results from separate model-specific spectral, large-deviation, and operator problems.

### APPENDIX A. POSITIVE CAUSAL EXPERIMENTS

**A.1. Types and primitive data.** A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. All primitive spaces below are standard Borel. Time is initially  $t = 0, \dots, N$ , with the countable-time extension stated separately. The unknown parameter is  $\lambda \in \Lambda$ ; it is not a random variable until a prior is supplied. The hidden state, applied action, and next report have spaces  $X_t, A_t, Y_{t+1}$ . A visible history is

$$h_t = (a_0, y_1, \dots, a_{t-1}, y_t) \in H_t.$$

A public initial report, clock increments, and observed cost records can be included in this notation. An action-availability rule, when present, is part of the interface. We usually write a common action space to avoid encoding that rule repeatedly.

**Definition A.1** (Raw experiment). A raw experiment consists of a parameter space  $\Lambda$ , preparations  $\mu_\lambda \in \mathcal{P}(X_0)$ , and jointly measurable probability kernels

$$K_t^\lambda(dx', dy, dc \mid x, a) \quad (\text{A.1})$$

on  $X_{t+1} \times Y_{t+1} \times \mathcal{C}$ , together with a class of executable future tests and a resource interface. The nonnegative vector  $c$  records charged increments, such as raw preparations or physical time. When it is observed, it is included in  $y$ ; when it is not observed, it is integrated in the visible law but is still counted in the resource account. Every kernel has total mass one. A failed measurement and a terminating measurement are reports, not missing mass.

The following six requirements will be referred to as the foundational axioms. They specify the admissible language rather than analytic conclusions.

- A0 (measurable interfaces).** Primitive spaces are standard Borel; histories and continuation availability are specified.
- A1 (positive experiment).** Preparation and transition–observation kernels are nonnegative and normalized as in (A.1).
- A2 (causality).** The next action uses only the visible past and permitted independent randomness. Stopping is nonanticipative.
- A3 (executable tests).** The designated future tests have actual protocols and are closed under the one-step extensions needed for prediction.
- A4 (preparation and accounting).** Average risks and acquisition laws refer to a declared preparation and exploration strategy. Failures, pilots, and accepted-event probabilities are not removed from their costs.
- A5 (charged implementation).** Persistent labels, numerical precision, workspace, read-only calibration, and simulator state have separate accounts.

In particular these axioms do not assert strict positivity of each report, finite-dimensionality, stationarity, differentiability, a spectral gap, or a large-deviation principle. If the physical state is not Markov in the coordinates initially proposed, its entire necessary past can be included in  $X_t$ . This does not prove closure for a smaller observable state.

**A.2. Strategies, paths, and stopping.** A strategy is a sequence of measurable kernels  $\alpha_t(da \mid h_t)$  supported on available actions. It cannot depend on the unknown true parameter. A finite-state strategy may have an additional explicit internal state; an unrestricted history strategy need not satisfy a finite-memory budget.

**Proposition A.2** (Path construction). *For every parameter and causal strategy there is a unique consistent finite-dimensional path law with iterated-kernel expression*

$$\mu_\lambda(dx_0) \prod_{t=0}^{N-1} \alpha_t(da_t \mid h_t) K_t^\lambda(dx_{t+1}, dy_{t+1}, dc_{t+1} \mid x_t, a_t). \quad (\text{A.2})$$

*For countably many discrete times these laws extend uniquely to the product path sigma field. Deterministic evolution and deterministic observations are included by taking Dirac kernels.*

*Proof.* For finite  $N$ , integrate bounded measurable cylinder functions from the last variable backwards. At each step measurability follows from the kernel property, positivity is preserved, and the integral of one is one. The resulting probability measures have consistent marginals because the last kernel integrates to one. The standard kernel-extension theorem gives the countable product law; this is the Ionescu–Tulcea construction, as treated in [4]. Uniqueness follows first on

cylinder events and then on their generated sigma field. A Dirac kernel makes the corresponding integral substitution along the deterministic trajectory, without changing the remaining preparation or measurement randomness.  $\square$

A bounded stopping time is implemented by an absorbing stopped symbol and zero subsequent increments. Thus the same construction handles optional termination. For an unbounded stopping time, an almost-surely finite stopped path can be considered by truncation; quantitative infinite-horizon error bounds still require a summable bound or a separate limiting argument. A random number of collisions is not silently substituted for a deterministic large-deviation speed.

**A.3. Prepared and parameter-uniform questions.** For statistical decision theory we retain the whole family  $(P_\lambda^\alpha)_\lambda$ . A decision rule is one rule, independent of the true parameter. For Bayesian prediction we instead specify a prior  $\eta(d\lambda)$  and use the joint hidden state  $(\lambda, x_t)$  with preparation  $\eta(d\lambda)\mu_\lambda(dx_0)$ . Fixed known calibration is permitted, but it is not a substitute for an unknown parameter.

This distinction also applies to infima over encoders. The infimum for a fixed preparation need not be achieved by the same encoder for all priors. Likewise, a family of codebooks indexed by a known experimental calibration is not an estimator of an unknown calibration. Every quantitative theorem below states which situation is intended.

**A.4. Instruments and evidence.** After a preparation has been fixed, write the hidden state simply as  $x$ . For a report event  $B \subset Y_{t+1}$ , define the positive instrument

$$(\mathbb{T}_{t,a,B}f)(x) = \int \mathbf{1}_B(y)f(x')K_t(dx', dy \mid x, a).$$

It is countably additive in  $B$  for nonnegative  $f$ , positive in  $f$ , and  $\mathbb{T}_{t,a,Y}1 = 1$ . With finite reports write  $\mathbb{T}_{t,a,y} = \mathbb{T}_{t,a,\{y\}}$ . Starting with  $b_0 = \mu$ , define the unnormalized history functional by

$$b_{t+1}(f) = b_t(\mathbb{T}_{t,a_t,y_{t+1}}f), \quad Z_t = b_t(1). \quad (\text{A.3})$$

These functionals do not include the probability of the chosen command. They are likelihood/evidence functionals conditional on the realized command sequence; the actual joint history law also includes the strategy kernels in (A.2).

**Proposition A.3** (Normalized update). *For finite reports and  $Z_t > 0$ , let  $\pi_t = b_t/Z_t$ . If  $q_t(y \mid \pi_t, a) = \pi_t(\mathbb{T}_{t,a,y}1) > 0$ , then*

$$\pi_{t+1}(f) = \frac{\pi_t(\mathbb{T}_{t,a,y}f)}{\pi_t(\mathbb{T}_{t,a,y}1)}, \quad Z_{t+1} = Z_t q_t(y \mid \pi_t, a). \quad (\text{A.4})$$

*The formula remains valid for adaptive actions. When the evidence is zero, the unnormalized functional is zero and no normalized posterior is specified by this formula.*

*Proof.* Divide (A.3) by its value at one. Positivity shows that a positive functional of total mass zero vanishes on every bounded test. For adaptive actions, condition on the visible history: the strategy's probability of the current action is fixed by that history and is independent of the hidden state conditional on it. This factor cancels in the posterior. An induction in (A.2) verifies the conditional expectation identity for every bounded hidden-state test.  $\square$

For a non-atomic report space, conditioning is by disintegration of the joint next-state/report law, not by dividing the mass of a singleton. Under a fixed preparation a standard Borel conditional distribution exists almost everywhere. A single Borel update jointly in all proposed states, inputs, and reports is additional implementation data, or must be justified by an applicable parametrized disintegration theorem. We will not use a simultaneous version across a nondominated parameter family without that justification. All explicit quantitative models below have finite reports and Borel formulas.

**A.5. Preparing a conditional experiment.** If an event has acceptance probability  $p > 0$ , independently repeating a preparation until acceptance implements the conditional law. Its preparation count  $T$  satisfies  $\mathbb{P}(T > n) = (1-p)^n$  and  $\mathbb{E}T = 1/p$ . A cap  $B$  instead creates an abort symbol with probability  $(1-p)^B$ . This is a different experiment from exact conditioning. The cap is necessary for a uniform pathwise preparation budget. No lower bound or implementation claim below replaces an actual report word by a probability-one conditioning event.

## APPENDIX B. FUTURE TESTS AND PREDICTIVE QUOTIENTS

**B.1. Operator words and adaptive protocols.** A finite future protocol starts from the present time, uses available actions causally, and returns a bounded terminal readout. Its conditional expected readout, as a function of the current hidden state, is a future test. A terminal hidden-state probe is allowed only if an actual measurement kernel producing it is part of the protocol. For finite report alphabets, a deterministic depth- $m$  protocol is a tree of actions and terminal readouts. Its test has the recursion

$$f = \sum_y \mathsf{T}_{t,a,y} f_y. \quad (\text{B.1})$$

Randomized protocols integrate such recursions. Let  $\mathcal{V}_t$  be the real linear space generated by the designated future tests, including constants. We require the one-step closure

$$\mathsf{T}_{t,a,y} \mathcal{V}_{t+1} \subset \mathcal{V}_t \quad (\text{B.2})$$

for the continuations being retained. Where full future laws are intended, the tests include indicator readouts from a determining event class and its needed bounded-monotone closure.

A fixed finite menu used to assess prediction loss can be much smaller than  $\mathcal{V}_t$ . Such a menu does not automatically define a recursively sufficient state. One must either prove its closure/completeness or keep a larger predictive state for the online construction.

**Definition B.1** (Predictive equivalence). Two prepared beliefs at time  $t$  are predictively equivalent when they have the same continuation interface and

$$\pi \equiv_t \tilde{\pi} \iff \pi(f) = \tilde{\pi}(f) \quad (f \in \mathcal{V}_t).$$

The continuation interface includes the legal future action tree, remaining budget, and any public clock that affects future tests. The predictive quotient is the set of these equivalence classes, or equivalently the image of  $\pi \mapsto (\pi(f))_{f \in \mathcal{V}_t}$  together with that interface.

**Theorem B.2** (Minimal future prediction). *Under (B.2), predictively equivalent beliefs give the same laws for all designated future protocols. After an action and a report of positive probability, their updated beliefs remain predictively equivalent. Conversely, equality of those protocol laws implies equality of their future tests. If a deterministic history statistic  $r_t$  determines all these predictions and the continuation interface, the predictive state factors uniquely through  $r_t$  on its attained image.*

*Proof.* For a one-step report, its probability is the value of  $\mathsf{T}_{t,a,y} 1 \in \mathcal{V}_t$ , and is therefore common. For a next-state test  $f$ , both the numerator and denominator of (A.4) have the same values under equivalent beliefs. The conditional next predictions are equal. Induct over the depth of a protocol tree using (B.1). Randomized action choices use the same conditional randomization at corresponding visible histories and can be integrated. Bounded monotone convergence extends equality from finite determining tests to the designated generated events. Conversely, expectations of each designated bounded readout are determined by its law, hence agree; equality extends linearly to  $\mathcal{V}_t$ .

If  $r_t(h) = r_t(\tilde{h})$ , sufficiency of the statistic for these tasks implies equality of all future predictions and interfaces. Thus each fibre of  $r_t$  is contained in a predictive-equivalence class. Define the

factor on the image of  $r_t$  by the predictive class of any history in that fibre. It is well defined and is the unique such map. This last assertion is a set-theoretic universal property. Its measurable version requires an actual measurable quotient realization, as specified below.  $\square$

The theorem does not give a normalized continuation at a zero-probability report. One can extend an update arbitrarily there for a chosen measurable implementation, but such an extension is not canonical information. For worst-history statements, admissible histories and these extensions must be explicitly specified.

**B.2. Safe measurable realizations.** The complete posterior on a standard Borel hidden space is a safe measurable state under a fixed experiment and appropriate conditional versions. Its predictive quotient may be simpler, but not every abstract measurable quotient is standard Borel. We use the following concrete sufficient condition.

**Theorem B.3** (Compact continuous realization). *Let  $B_t$  be compact metric admissible belief domains and let  $(f_{t,j})_{j \geq 1}$  be continuous bounded evaluation functions on  $B_t$  that determine predictive equivalence. Put*

$$e_t(b) = (f_{t,j}(b))_{j \geq 1}, \quad S_t = e_t(B_t),$$

*after rescaling the functions into  $[-1, 1]$ . Suppose action spaces are compact metric, reports finite, and  $F_t : B_t \times A_t \times Y_{t+1} \rightarrow B_{t+1}$  is continuous for each report and respects predictive equivalence. Then  $S_t$  is compact metrizable and the induced update  $\bar{F}_t : S_t \times A_t \times Y_{t+1} \rightarrow S_{t+1}$  is continuous for each report. Every continuous evaluation constant on predictive fibres descends to a continuous function on  $S_t$ .*

*Proof.* The countable product  $[-1, 1]^{\mathbb{N}}$  is compact metrizable, and  $e_t$  is continuous. Its image is compact, hence closed and metrizable. A continuous surjection from a compact space to a Hausdorff space is a closed quotient map. The same holds for  $e_t \times \text{id}_{A_t}$  because its domain is compact and its target Hausdorff. For each report,  $e_{t+1} \circ F_t$  is continuous and constant on fibres of this quotient map. The definition of quotient topology therefore gives a unique continuous descended map. The final assertion is the same argument without the action factor.  $\square$

If the hidden state space is compact and the belief domain is compact in weak convergence, continuous hidden-state tests give the required continuous evaluations. With continuous positive likelihoods bounded away from zero, Bayes updating is continuous. In concrete finite models this is verified directly. A countable dense set of actions can determine all finite words when the corresponding evaluations are continuous; this must be checked, not inferred from standard Borel structure alone.

For finite histories the quotient is finite and the measurable issue is trivial. For noncompact or nondominated settings we state results on an already supplied Borel predictive realization rather than claiming a universal smooth quotient.

**B.3. Finite remaining-test coordinates.** The following elementary result is useful when a finite representation exists, including representations with redundant coordinates.

**Proposition B.4** (Positive finite-coordinate update). *Let  $\psi_t = (1, \psi_{t,1}, \dots, \psi_{t,q_t})$  be bounded tests. Suppose*

$$\mathbb{T}_{t,a,y} 1 = b_{t,a,y} \psi_t, \quad \mathbb{T}_{t,a,y} \psi_{t+1,i} = c_{t,a,y,i} \psi_t,$$

*with uniformly bounded coefficient rows and jointly Borel dependence on the input. On a domain of beliefs where every relevant report probability is at least  $\kappa > 0$ , the coordinate vector  $v = (\pi \psi_{t,i})_{i \geq 1}$  updates by*

$$v'_i = \frac{c_{t,a,y,i}(1, v)^{\top}}{b_{t,a,y}(1, v)^{\top}}. \quad (\text{B.3})$$

*It is uniformly Lipschitz on segments joining such moment vectors when the same report floor holds on their belief mixtures. Linear independence of the tests is not required.*

*Proof.* Equation (B.3) follows from (A.4). A segment between two belief moment vectors represents their convex mixtures. The denominator is a report probability under that mixture and is at least  $\kappa$ . The numerator, coefficient rows, and coordinate vectors are bounded. Differentiating the quotient bounds its derivative by a constant depending only on these bounds and  $\kappa$ . Integrate that derivative along the segment. Redundant coefficient representations have the same evaluations on actual moment vectors and their mixtures, so they give the same update there.  $\square$

In the static latent-variable case the instruments are multiplication by report likelihoods. In a dynamic model they are positive transfer operators. In either case the update uses current coordinates, command, and report, not a recovered history tape. An observer who stores the entire vector  $v$  at arbitrary real precision has not yet complied with a finite-label budget; that is the problem of Sections E and F.

**B.4. Three different fibres.** It is useful to record the types before adding geometry. Predictive fibres are fibres of  $h_t \mapsto s_t(h_t)$ . Statistical fibres are equivalence classes of parameters with equal laws under the specified strategy class. Feature fibres are fibres of a chosen map  $\mathbf{O} : \Lambda \rightarrow \mathcal{Z}$ . Equality of features can be strictly weaker than equality of laws, and equality of a fixed-prior posterior prediction need not be parameter-uniform sufficiency. Their derivatives, when defined, have different domains.

## APPENDIX C. EXECUTABLE REDUCTIONS AND CHARGED RESOURCES

**C.1. Finite-state implementations.** A synchronized causal machine between experiments has an internal state  $r_t$ . At a target step it receives the target action  $a'_t$ , chooses a source action according to a parameter-independent kernel depending on  $(r_t, a'_t)$ , receives the source report, and outputs a target report while updating  $r_t$ . Random seeds are independent of the parameter and physical preparation. A stored seed that encodes previous reports is part of  $r_t$ . Every raw source step and every retained state is charged.

This synchronized model suffices for the results below. A macro-step that uses a random number of source steps requires an explicit causal stopping protocol and its preparation certificate. Such a protocol can be included, but a bound for one synchronized source step is not silently reused for it.

A target controller with state  $z_t$  receives the target reports and updates  $z_t$ . Its joint implementation with the machine stores  $(r_t, z_t)$ . Thus  $|r_t| \leq R_t$  and  $|z_t| \leq M_t$  give the direct product bound  $R_t M_t$ . This is an upper bound for one realization, not a claim that no smaller implementation is possible.

**Definition C.1** (Causal reduction certificate). For experiments over a common parameter space, a certificate  $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$  consists of a parameter-independent causal machine, a lift  $L_\Gamma$  of every admitted target controller to its source implementation, and a monotone resource map  $\rho_\Gamma$ . Under target controller  $\alpha$ , write  $\hat{P}_\lambda^{\Gamma, \alpha}$  for the simulated target-history law obtained from the joint machine/source execution. An  $\varepsilon$ -certificate satisfies

$$\sup_{\lambda, \alpha} \left\| \hat{P}_\lambda^{\Gamma, \alpha} - P_{\mathfrak{F}, \lambda}^\alpha \right\|_{\text{TV}} \leq \varepsilon, \quad (\text{C.1})$$

and implementing a target budget  $B$  costs at most  $\rho_\Gamma(B)$ . The default certificate is pathwise. Expected-cost and tail-budget certificates are different types and must be stated separately.

For randomized strategy lifts, the right object is the joint causal execution. It need not be representable by a terminal kernel on the raw source history alone that is independent of the

controller. Once the lift and controller are fixed, its visible pushforward is unambiguous. This avoids a misleading notation that would hide feedback inside a supposedly static garbling.

An exact certificate has  $\varepsilon = 0$ . The identity machine gives the identity reduction. A reparametrization  $r : \Lambda' \rightarrow \Lambda$  pulls back an experiment family and is kept separate from data reduction. If the reparametrization is not injective it introduces parameter redundancy; it is not loss of observed data.

**C.2. The path-kernel estimate.** We use the convention  $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$ . For  $0 \leq f \leq L$  this implies  $|Pf - Qf| \leq L \|P - Q\|_{\text{TV}}$ .

**Lemma C.2** (Predictable telescoping). *Let  $P$  and  $Q$  be laws on the same finite visible history tree, with the same initial law and the same feedback action kernels. Suppose their conditional report kernels at history  $h_t$  and action  $a_t$  are  $p_t(\cdot \mid h_t, a_t)$  and  $q_t(\cdot \mid h_t, a_t)$ . If*

$$\|p_t(\cdot \mid h_t, a_t) - q_t(\cdot \mid h_t, a_t)\|_{\text{TV}} \leq d_t(h_t, a_t),$$

then

$$\|P - Q\|_{\text{TV}} \leq \min \left\{ 1, \mathbb{E}_P \sum_{t=0}^{N-1} d_t(H_t, A_t) \right\}. \quad (\text{C.2})$$

If both systems become absorbing under the same visible stopping rule  $\sigma$ , the summand can be multiplied by  $\mathbf{1}_{\{t < \sigma\}}$ .

*Proof.* Fix a terminal measurable test  $0 \leq f \leq 1$ . Replace the report kernels one at a time, using a  $P$ -prefix and a  $Q$ -suffix. The suffix conditional expectation of  $f$  is a measurable function between zero and one. The change at step  $t$ , after integrating the common feedback action kernel, is therefore bounded in absolute value by  $\mathbb{E}_P d_t(H_t, A_t)$ . Summing the telescoping differences proves the bound for  $f$ . Take the supremum over event indicators. After the visible rule has stopped, the kernels are the same absorbing kernel and their local error is zero. Histories of zero probability under  $P$  contribute nothing; the kernels are nevertheless specified on all admitted histories so that all intermediate laws exist.  $\square$

The expectation is under the actual reference experiment. In particular a rare report carrying a large conditional error is not automatically charged as if it occurred at every time. No maximal-coupling measurability theorem is needed for this proof. The action policy may be discontinuous: the same kernel is applied on the same history in each telescoping term.

**Theorem C.3** (Composition and risk transfer). *Suppose  $\Gamma_1 : \mathfrak{E} \rightarrow \mathfrak{F}$  and  $\Gamma_2 : \mathfrak{F} \rightarrow \mathfrak{G}$  have certificates  $(\varepsilon_1, \rho_1)$  and  $(\varepsilon_2, \rho_2)$  uniformly over the controller classes needed by the lifts. Their serial implementation has certificate*

$$(\min\{1, \varepsilon_1 + \varepsilon_2\}, \rho_1 \circ \rho_2).$$

*Exact reductions compose associatively as executions, up to relabeling of product states. For a common target  $\tau(\lambda)$  and bounded loss  $0 \leq \ell \leq L$ ,*

$$\mathfrak{R}_{\mathfrak{E}}(\rho_{\Gamma}(B)) \leq \mathfrak{R}_{\mathfrak{F}}(B) + L\varepsilon \quad (\text{C.3})$$

*for either minimax risk or risk under a common prior.*

*Proof.* Execute the second machine as a controller of the first. Their stored states form a product and their source actions remain nonanticipative and parameter independent. The first certificate applies to this enlarged controller class by hypothesis. The subsequent machine/controller execution is a Markov postprocessing of its visible history together with its independent randomization, so it contracts total variation. Compare first with an exact  $\mathfrak{F}$  run and then with the intended  $\mathfrak{G}$  run. The triangle inequality gives the displayed defect. Budget maps compose in the same order

because they map target requirements to source requirements. Regrouping three nested machine states changes only their names, proving associativity.

For a target procedure within budget  $B$ , run its source implementation and use the same final decision. Its risk at each parameter differs by at most  $L\varepsilon$ . The resource certificate makes it admissible under  $\rho_\Gamma(B)$ . Take the supremum over parameters, or integrate against the common prior, and then take the infimum over target procedures. An approximating sequence of procedures suffices if an optimum is not attained.  $\square$

A risk certificate concerns the same decision target and loss on both sides. It does not compare excess errors relative to different full-history predictors. An expected preparation constraint cannot be transported by a pathwise certificate without checking its type. Conversely, total-variation closeness alone does not transfer an unbounded cost expectation.

### C.3. Checking an exact state reduction locally.

**Proposition C.4** (One-step realization). *Let  $r_t : X_t \rightarrow X'_t$ ,  $g_t : Y_t \rightarrow Y'_t$ , and an action lift  $\iota_t : A'_t \rightarrow A_t$  be measurable. Suppose the preparations push forward correctly and, for every admitted source state and target action,*

$$(r_{t+1}, g_{t+1})_\# K_t^\lambda(\cdot \mid x, \iota_t a') = K_t'^\lambda(\cdot \mid r_t x, a'). \quad (\text{C.4})$$

*Suppose also that continuation interfaces, recorded costs, and designated readouts match. Then applying the lifted actions and report maps gives an exact causal reduction for every target feedback strategy.*

*Proof.* At time zero the reduced state laws agree. Conditional on a source state and on a lifted action, (C.4) gives the joint reduced next-state/report law. Integrate against the common current reduced law. Since the target controller reads the reduced history, its next action law is the same in both systems. Induction proves equality on all finite cylinders, and hence on the path sigma field. Readout and cost matching give the asserted experiment and resource identity.  $\square$

The state map need not be observable to implement the reduction: the machine only needs the action lift and report map. The hidden-state map is a proof device for the law identity. If the proposed lift instead requires the unobserved state, it is not an executable reduction of this type.

**C.4. Terminal and causal equivalence.** For a fixed terminal family, a garbling  $G$  is reversible if there exists a parameter-independent kernel  $R$  such that  $P_\lambda G R = P_\lambda$ . This is statistical equivalence. Requiring  $R$  to operate causally and charging its memory gives a stronger notion. Requiring suitable bounds in both directions gives resource-aware causal equivalence. These notions are not identified by definition.

Independent parallel experiments may be implemented by product machines when their preparations and physical randomness are genuinely independent conditional on the parameter. Persistent cardinalities multiply and costs add. Shared nuisance variables or coupled preparations require the joint experiment; replacing them by an independent product changes the object.

## APPENDIX D. POSTERIOR GEOMETRY AND INFORMATION LOSS

**D.1. Posterior barycentres.** The following statements concern a common preparation. Let  $Z$  be a hidden target,  $Y$  an original observation, and  $W$  a parameter-independent postprocessing of  $Y$ . Conditional independence is denoted by the Markov chain  $Z \rightarrow Y \rightarrow W$ .

**Theorem D.1** (Posterior barycentre and convex order). *For bounded measurable  $f_1, \dots, f_q$  of  $Z$ , put*

$$p_Y = (\mathbb{E}[f_i(Z) \mid Y])_i, \quad p_W = (\mathbb{E}[f_i(Z) \mid W])_i.$$

Then

$$p_W = \mathbb{E}[p_Y | W]. \quad (\text{D.1})$$

Consequently  $\mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$  for every convex function for which the expectations are defined. Moreover,

$$\text{Cov}(p_Y) - \text{Cov}(p_W) = \mathbb{E} \text{Cov}(p_Y | W) \succeq 0. \quad (\text{D.2})$$

*Proof.* The Markov property gives  $\mathbb{E}[f_i(Z) | Y, W] = \mathbb{E}[f_i(Z) | Y]$ . Taking conditional expectation with respect to  $W$  proves (D.1). Conditional Jensen yields the convex-order inequality. Subtract the common mean and expand the orthogonal decomposition  $p_Y - \mathbb{E}p_Y = (p_Y - p_W) + (p_W - \mathbb{E}p_W)$ ; its cross covariance is zero by conditioning on  $W$ . This gives (D.2).  $\square$

For a measure-valued posterior, the same identity holds against every bounded test. It is a conditional barycentre, not a function sending each original posterior to a single coarser posterior. In particular the image of a finite set of original posteriors may generate a continuous family of conditional barycentres. This is why convex-order monotonicity is not a statement about the cardinality of a predictive representation.

**D.2. Relative entropy and recovery.** For  $P \ll Q$ , set  $D(P\|Q) = \int \log(dP/dQ) dP$ , allowing  $+\infty$ ; set it to infinity otherwise. Let  $G : X \rightsquigarrow W$  be a Markov kernel and form the joint laws  $\bar{P}(dx, dw) = P(dx)G(dw | x)$  and  $\bar{Q}(dx, dw) = Q(dx)G(dw | x)$ .

**Theorem D.2** (Entropy loss and a common recovery kernel). *If  $D(P\|Q) < \infty$ , then*

$$D(P\|Q) = D(PG\|QG) + \int D(\bar{P}(dx | w)\|\bar{Q}(dx | w)) PG(dw). \quad (\text{D.3})$$

*In particular, entropy decreases under  $G$ . Suppose a family  $P_\lambda$  is dominated by one  $P_0$ , every  $D(P_\lambda\|P_0)$  is finite, and*

$$D(P_\lambda G\|P_0 G) = D(P_\lambda\|P_0) \quad \text{for every } \lambda.$$

*Then  $R(w, dx) = \bar{P}_0(dx | w)$  is a parameter-independent recovery kernel satisfying  $P_\lambda GR = P_\lambda$  for every  $\lambda$ .*

*Proof.* The joint likelihood ratio is  $d\bar{P}/d\bar{Q}(x, w) = dP/dQ(x)$ . Disintegrating both joint laws gives the factorization into the marginal likelihood ratio on  $W$  and the conditional likelihood ratio on  $X$ . Take logarithms and integrate. Standard truncation of the densities justifies the formula when logarithms are not bounded; finite total entropy makes the nonnegative conditional entropy integrable. This proves (D.3). Alternatively the marginal ratio is the conditional expectation under  $\bar{Q}$  of  $dP/dQ$ , and conditional Jensen proves the inequality directly, also for infinite entropy.

For the family, equality makes the conditional entropy vanish  $P_\lambda G$ -almost everywhere. Hence the two conditional distributions coincide there. The conditional law under the one reference  $P_0$  is the same measurable kernel for all parameters;  $P_\lambda G \ll P_0 G$  handles its null-set convention. Integrating that conditional law against  $P_\lambda G$  recovers  $P_\lambda$ .  $\square$

This is a sufficient recovery criterion under a common reference. An isolated equality for one pair does not certify an unrelated parameter family. Nor does it make the recovery kernel causal or memory free.

**D.3. Scores and Fisher information.** To avoid hiding regularity in notation, first take a finite observation space with  $p_\lambda(y) > 0$  and continuously differentiable probabilities on an open parameter domain in  $\mathbb{R}^k$ . Let

$$s_\lambda(y) = \nabla_\lambda \log p_\lambda(y), \quad \mathcal{I}_\lambda = \mathbb{E}_\lambda[s_\lambda s_\lambda^\top].$$

If  $G(w | y)$  is a parameter-independent finite channel, restrict to output symbols having positive probability; identically impossible symbols can be removed.

**Theorem D.3** (Score projection). *The processed score and information satisfy*

$$s_\lambda^G(W) = \mathbb{E}_\lambda[s_\lambda(Y) \mid W], \quad (\text{D.4})$$

$$\mathcal{I}_\lambda - \mathcal{I}_\lambda^G = \mathbb{E}_\lambda \text{Cov}_\lambda(s_\lambda(Y) \mid W) \succeq 0. \quad (\text{D.5})$$

For a direction  $v$ , equality of information in that direction is equivalent to  $v^\top s_\lambda(Y)$  being determined by  $W$  almost surely.

*Proof.* Differentiate  $q_\lambda(w) = \sum_y p_\lambda(y)G(w \mid y)$  and divide by  $q_\lambda(w)$ . The coefficients are precisely the conditional probabilities of  $Y$  given  $W$ , proving (D.4). Differentiating  $\sum_y p_\lambda(y) = 1$  gives  $\mathbb{E}_\lambda s_\lambda = 0$ . The conditional variance identity now gives (D.5). Its quadratic form in  $v$  is the mean conditional variance of  $v^\top s_\lambda$ , which vanishes exactly under the stated measurability condition.  $\square$

The same proof works in a dominated differentiable model if differentiation under the integral is justified and the score lies in  $L^2$ . More generally, the QMD version is the classical score-projection theorem described by Pollard [11]; when it is invoked, quadratic-mean differentiability is an additional statistical hypothesis. The raw experiment axioms by themselves impose no differentiability.

Information equality in (D.5) is a statement at a parameter point. It is not the recovery conclusion of Theorem D.2. Even information preservation throughout a model need not make a statistic sufficient without stronger assumptions; see [11].

Under a smooth parameter map  $\lambda = r(\zeta)$ , the information pulls back as  $Dr^\top \mathcal{I} Dr$ . Under a deterministic linear map of a random vector, its covariance pushes forward as  $A \Sigma A^\top$ . These are different operations on different objects. Neither formula says that an arbitrary covariance equals inverse Fisher information.

**D.4. Sequential information in finite positive models.** A useful controlled specialization follows without differentiating the controller. Suppose the conditional report probabilities are finite, positive, and differentiable, and the feedback policy has no explicit unknown-parameter dependence. Along a history define the local score

$$\xi_{t+1} = \nabla_\lambda \log p_\lambda(Y_{t+1} \mid H_t, A_t).$$

The policy factors contribute no score. Conditional normalization gives  $\mathbb{E}_\lambda[\xi_{t+1} \mid H_t, A_t] = 0$ .

**Proposition D.4** (Sequential score decomposition). *If the local scores are square integrable, the path score is  $S_N = \sum_{t < N} \xi_{t+1}$ , and*

$$\mathbb{E}_\lambda[S_N S_N^\top] = \sum_{t < N} \mathbb{E}_\lambda[\xi_{t+1} \xi_{t+1}^\top]. \quad (\text{D.6})$$

For bounded stopping, each term is restricted to active steps. No independence of successive reports is required.

*Proof.* The likelihood factors into conditional report probabilities and parameter-independent policy factors. Its logarithmic derivative is the displayed sum. For  $s < t$ ,  $\xi_{s+1}$  is measurable before  $\xi_{t+1}$ , so their cross moment is zero by the conditional mean-zero property. Expanding the square proves (D.6). A bounded stopping indicator is predictable for the next increment; the same argument applies.  $\square$

This is an information identity for the actual controlled law. It does not make an information-optimal strategy automatically prediction-memory optimal, and it does not supply an asymptotic normal approximation.

## APPENDIX E. ATTAINED PREDICTION AND CHECKPOINT QUANTIZATION

**E.1. The physical loss.** Fix a preparation and an exploration law, independently of coding randomness. At time  $t$  choose a future query  $J$  independently after the label is formed, with probabilities  $w_j > 0$ ,  $\sum_j w_j = 1$ . Conditional on history and query, a bounded response  $Z_j$  has mean  $m_j(h)$ . Squared loss is used, and only the selected query is executed. Set

$$p(h) = (\sqrt{w_j} m_j(h))_{j=1}^q, \quad \nu = \text{Law}(p(H_t)).$$

The weighted decoder vector is denoted by  $c$ . Subtracting the complete history predictor's risk leaves  $\|p(h) - c\|^2$ .

An  $M$ -label checkpoint encoder reads the complete prefix once and retains one of  $M$  symbols. Its decoder receives the label and the subsequent query, not the discarded history or the acquisition commands. A public seed may specify a codebook but is independent of the preparation. Private coins are allowed and are either discarded or included in the retained label. We assume a bounded closed convex prediction domain containing all  $p(h)$ ; projection into it can only improve squared error.

**Theorem E.1** (Exact checkpoint identity). *The optimal mean checkpoint excess risk equals*

$$R_M^{\text{chk}} = e_M^2(\nu) := \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp). \quad (\text{E.1})$$

*Independent public or private coding randomization does not decrease this infimum. An arbitrarily near-optimal deterministic nearest-centre encoder is measurable.*

*Proof.* Condition on history, query, and the code randomness. The elementary identity

$$\mathbb{E}[(Z_j - a)^2 \mid h, J = j] = \text{Var}(Z_j \mid h, J = j) + (m_j(h) - a)^2$$

reduces the loss to the displayed Euclidean distortion. For each label, replace a randomized decoder by its conditional mean over decoder coins; convexity of squared norm does not increase the loss. Given fixed decoder centres, random assignment by the encoder is no better than the nearest centre. Hence every fixed public-seed codebook has risk at least the right-hand side of (E.1); averaging over the independent seed preserves this inequality.

Conversely choose centres within any prescribed positive tolerance of the infimum. Encode by the first index attaining the least distance. Finite comparisons of continuous distance functions make this rule Borel. The resulting distortion is the integrand in (E.1). Let the tolerance decrease to zero. A codebook indexed by an unknown true parameter is not produced by this argument:  $\nu$  is the one declared preparation law.  $\square$

For worst-history loss the analogous deterministic checkpoint quantity is a covering radius, not the integral in (E.1). The mean and worst-history laws have the same order only under additional geometric and mass conditions.

**E.2. A small-ball lower bound.**

**Lemma E.2** (Mass obstruction). *Suppose a probability law  $\nu$  in  $\mathbb{R}^d$  satisfies  $\nu(B(c, r)) \leq Ar^d$  for every centre  $c$  and radius  $r > 0$ . Then for every integer  $M \geq 1$ ,*

$$e_M^2(\nu) \geq \frac{1}{2}(2AM)^{-2/d}. \quad (\text{E.2})$$

*In particular, a Lebesgue density bounded by  $B\epsilon^{-d}$  gives  $e_M^2(\nu) \geq c_{d,B}\epsilon^2 M^{-2/d}$  for  $\epsilon > 0$ .*

*Proof.* The union of  $M$  balls of radius  $r = (2AM)^{-1/d}$  has probability at most one half. On its complement every centre is at least distance  $r$  away. Integrating the nearest-centre squared distance gives at least  $r^2/2$ . For the density assertion use  $A = Bv_d\epsilon^{-d}$ , where  $v_d$  is the volume of the Euclidean unit ball.  $\square$

The lemma permits centres outside the support and therefore already covers arbitrary decoders. It does not condition away a small probability of reaching the difficult region. If the useful density lower geometry instead occurs only on an event of probability  $\beta$ , the eventual risk lower bound must carry that mass.

**Lemma E.3** (A positive attained chart). *Suppose a coordinate projection of a prediction law dominates  $\beta$  times uniform probability on a  $d$ -dimensional cube of side  $a$ , with  $\beta, a > 0$ . If that projection has operator norm at most  $L$ , then*

$$e_M^2(\nu) \geq c_d \beta a^2 L^{-2} M^{-2/d}.$$

*Proof.* For any centres in the original prediction space, project the centres onto the cube's coordinate space. Distances there are at most  $L$  times the original distances. The uniform cube law has density  $a^{-d}$  and satisfies Lemma E.2. Domination by  $\beta$  and rescaling prove the assertion.  $\square$

**E.3. Whole-image covers and all integer budgets.** Let  $D \subset \mathbb{R}^d$  be nonempty, compact, and contained in a cube of side  $2a$ . Divide each coordinate interval into  $k = \lfloor M^{1/d} \rfloor$  pieces. There are at most  $k^d \leq M$  cells. Choose one point of  $D$  in each nonempty cell. Any point is within Euclidean distance  $2a\sqrt{d}/k$  of its representative. Since  $k \geq M^{1/d}/2$ , we obtain a radius  $4a\sqrt{d}M^{-1/d}$  for every integer  $M \geq 1$ . The corresponding  $\ell^1$  bound is  $4adM^{-1/d}$ . Neither a large-budget asymptotic nor an integer-budget relaxation is needed.

More generally, finitely many Lipschitz images of compact cubes have such covers with exponent equal to the largest domain dimension. Allocate centres between the finitely many charts, use the preceding grid, and enlarge the constant for budgets smaller than the number of charts. This is a sufficient global description, not a consequence of local rank alone. A nonlinear or disconnected image cannot be replaced by a tangent plane without checking its entire cover.

**E.4. Causal lower bounds use one exploration law.** Let  $\mathcal{C}_M$  be causal encoders with at most  $M_t$  labels at time  $t$ . Under the fixed exploration strategy, their label is also a measurable, possibly randomized, encoding of the complete time- $t$  prefix. Therefore

$$\inf_{C \in \mathcal{C}_M} \max_{1 \leq t \leq N} \mathbb{E} \ell_t^C \geq \max_{1 \leq t \leq N} e_{M_t}^2(\nu_t). \quad (\text{E.3})$$

Indeed the checkpoint lower inequality holds for every single  $C$  and all  $t$  before taking the infimum. The same law of exploration supplies all marginals  $\nu_t$ . Changing the exploration strategy with the encoder, or with the checkpoint after the infimum, would define another problem.

**E.5. Calibrated anisotropy.** A useful model interface is provided by [15]. There, under whole-image covering, positive attained mass for each initial scale flag, and uniformly stable raw updates, the scale profile is

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left( \frac{s_{t,1} \cdots s_{t,\ell}}{M} \right)^{2/\ell}.$$

The physical prediction metric is fixed; small singular values are not removed by an unbounded coordinate whitening. The finite-horizon causal bound follows by propagating the covering radii, as in the next section. We cite this refined profile as a conditional model interface, not as an unproved universal consequence of a list of singular values. The dynamic model of Section G will establish its own isotropic scale, density bound, and invariant cover directly from the experiment.

## APPENDIX F. CAUSAL QUANTIZATION WITH FEEDBACK AND STOPPING

**F.1. Admissible state domains.** Consider a predictive realization  $s_t(h_t)$  with a known initial state  $s_0$ . Let  $(D_t, d_t)$  be admissible metric state domains containing every attainable state. They may also contain normalized beliefs that are not attained at that precise time. This causes no

problem if all updates and readouts are genuinely defined on  $D_t$ . What is forbidden is evaluating an update at an artificial point where it has no experimental meaning or no stated extension.

Assume common action availability on each domain, or work within a fixed continuation-interface component. For each input  $(a, y)$ , let  $F_t(\cdot, a, y) : D_t \rightarrow D_{t+1}$  be Borel and satisfy

$$d_{t+1}(F_t(s, a, y), F_t(\tilde{s}, a, y)) \leq L_t d_t(s, \tilde{s}). \quad (\text{F.1})$$

The known-calibration formulas implementing  $F_t$  are independent of the unknown parameter. If there is a parameterized law, write the next-report kernel as  $g_t^\lambda(dy \mid s, a)$  and assume

$$\left\| g_t^\lambda(\cdot \mid s, a) - g_t^\lambda(\cdot \mid \tilde{s}, a) \right\|_{\text{TV}} \leq \kappa_t d_t(s, \tilde{s}) \quad (\text{F.2})$$

uniformly over the relevant parameter domain. The same Borel state update must represent every law for a parameter-uniform statement. For a single prepared Bayesian experiment there is no additional parameter supremum.

Choose finite representative sets  $C_t \subset D_t$  with  $|C_t| \leq M_t$  and Borel quantizers  $q_t : D_t \rightarrow C_t$  satisfying  $d_t(s, q_t s) \leq r_t$ . A numerical update  $\tilde{F}_t(c, a, y) \in D_{t+1}$  may replace the exact update at a representative, with

$$d_{t+1}(\tilde{F}_t(c, a, y), F_t(c, a, y)) \leq \eta_{t+1}. \quad (\text{F.3})$$

Set  $\hat{s}_0 = s_0$  and implement

$$\hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (\text{F.4})$$

The persistent state is only the index of  $\hat{s}_t$  in  $C_t$ .

**Theorem F.1** (Finite-state emulation). *Under (F.1)–(F.3), define*

$$b_0 = 0, \quad b_{t+1} = L_t b_t + r_{t+1} + \eta_{t+1}, \quad b_t = \sum_{j=1}^t (r_j + \eta_j) \prod_{i=j}^{t-1} L_i. \quad (\text{F.5})$$

*Then the following assertions hold.*

(i) *On every admitted common command/report sequence,  $d_t(s_t, \hat{s}_t) \leq b_t$ . If the physical prediction readout is  $K_t$ -Lipschitz, its excess squared error is at most  $K_t^2 b_t^2$ .*

(ii) *Define the finite-state reference experiment to use the same representative update (F.4), but to draw its next report from  $g_t^\lambda(\cdot \mid \hat{s}_t, a_t)$ . For any common causal feedback controller, the visible path laws of the actual experiment with the representative machine and of this finite-state reference experiment satisfy*

$$\left\| P_{N,\lambda}^\alpha - \hat{P}_{N,\lambda}^\alpha \right\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=0}^{N-1} \kappa_t b_t \right\}. \quad (\text{F.6})$$

*The histories here include the applied actions, reports, and retained labels. A shared visible stopping rule replaces the deterministic sum by  $\mathbb{E}_{P_\lambda^\alpha} \sum_{t < \sigma \wedge N} \kappa_t b_t$ .*

(iii) *Any additional causal simulator is charged as in Theorem C.3. A feedback controller with  $D_t$  own states is implemented with at most  $D_t M_t$  persistent pairs. For a common bounded path-dependent loss, the risk-transfer error is at most its range times the right side of (F.6).*

*Proof.* For (i), insert successively the exact update from the retained representative and its numerical approximation. The triangle inequality, (F.1), the numerical error, and the covering radius give

$$d_{t+1}(s_{t+1}, \hat{s}_{t+1}) \leq L_t d_t(s_t, \hat{s}_t) + \eta_{t+1} + r_{t+1}.$$

Induction proves (F.5). The physical readout bound follows by its Lipschitz property and the conditional squared-loss identity. This argument remains valid when the actions depend on the retained label: it compares the exact posterior and the representative along the actual same sequence of applied commands and reports.

For (ii), at any common visible prefix the applied controller has the same action kernel. The representative  $\hat{s}_t$  is the same measurable function of that prefix in both experiments. In the actual law, the exact predictive state is  $s_t$ ; in the finite-state reference law, the next-report kernel is evaluated at  $\hat{s}_t$ . Their conditional variation distance is at most  $\kappa_t b_t$  by (i) and (F.2). The next label is a common deterministic function of the action, report, and current label, so appending it does not increase the conditional variation distance. Apply Lemma C.2. Stopping makes the conditional kernels identical after the stopping time, which gives the predictable active-step bound.

For (iii), pair the retained representative with the controller's internal state. Each next action and update then reads only this pair and current inputs. The state cardinality is bounded by the product. The bounded-loss and extra-simulator statements are precisely the conclusions of Theorem C.3.  $\square$

The theorem is a directional emulation statement. It does not construct a reverse kernel from the finite-state reference experiment to the raw physical data. It therefore does not claim two-sided Le Cam equivalence. Nor does it estimate an unknown predictive update. The implemented update and representatives are known; uniformity over a statistical family is available only under the explicitly common update conditions above.

**F.2. What feedback does and does not require.** There are two different uses of a finite filter. One can predict on an arbitrarily selected actual input sequence; this uses only Theorem F.1(i). One can also use a compressed controller to choose inputs. That still gives accurate filtering along its own actual run, but comparison with the original full-information optimal controller is another problem. Part (ii) compares a common controller across two experiments and so supplies a valid route through experiment simulation. It does not require the controller to be Lipschitz, because identical prefixes produce identical action laws. A discontinuous policy evaluated at two different approximate states is not being assumed to choose the same action.

For example, minimizing a terminal common decision loss over all  $D_t$ -state controllers in the finite-state reference experiment gives a source controller with  $D_t M_t$  states and the corresponding (F.6) risk allowance. This need not approximate the unrestricted optimal source policy, whose comparison would require a separate richness theorem for the controller class.

### F.3. Uniform prediction and finite-horizon path comparison.

**Corollary F.2** (Contractive implementation). *If  $L_t \leq \rho < 1$ ,  $r_t \leq r$ , and  $\eta_t \leq \eta$  uniformly, then*

$$\sup_t d_t(s_t, \hat{s}_t) \leq \frac{r + \eta}{1 - \rho}.$$

*For a bounded physical readout with uniform Lipschitz constant  $K$ , the prediction excess is at most  $K^2(r + \eta)^2/(1 - \rho)^2$  at every time. If also  $\kappa_t \leq \kappa$ , the path-law error on  $N$  steps is at most  $\min\{1, N\kappa(r + \eta)/(1 - \rho)\}$ .*

*Proof.* Sum the geometric series in (F.5), then apply the two parts of Theorem F.1.  $\square$

A uniform per-checkpoint prediction bound is not uniform total-variation closeness of two infinite path laws. The last display may grow with  $N$ . Two processes that are arbitrarily close at each step can remain statistically distinguishable from a sufficiently long record. If a general construction uses a different arbitrary codebook at each time, it is a time-indexed machine description, not automatically a finite uniform program. The dynamic construction in the next section uses one fixed invariant domain and one fixed codebook, so that distinction does not obstruct its infinite-time statement.

**F.4. Model or calibration perturbations.** Suppose an implementable reference update  $F_t^0$  has  $d(F_t(s, a, y), F_t^0(s, a, y)) \leq \delta_{t+1}$  on the relevant domain, and its report law differs by at most  $\zeta_t$  at the same state. Repeating the proof replaces  $r_j + \eta_j$  by  $r_j + \eta_j + \delta_j$  in (F.5) and  $\kappa_t b_t$  by  $\kappa_t b_t + \zeta_t$  in (F.6). This is a deterministic certificate, not an unpaid pilot. If a preliminary estimate yields such bounds only on an event of probability  $1 - p$ , bounded risks receive an additional allowance of at most their range times  $p$ , and the pilot's data and persistent state must be charged.

The perturbation must remain in the same physical metric and admissible domain. A reparametrization with an unbounded inverse near a degeneracy cannot be inserted while keeping the old constants.

## APPENDIX G. A DYNAMIC MODEL WITH A UNIFORM MEMORY LAW

**G.1. The experiment and its resource convention.** Fix  $d \geq 1$ , let  $E_d = \{0, 1, \dots, d\}$ , and write  $\nu_i = 1/(d+1)$ . The known parameters satisfy

$$4/5 \leq \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (\text{G.1})$$

Prepare  $X_0 \sim \nu$ . At each acquisition step the hidden state is updated by the transition matrix

$$P_\gamma(i, j) = (1 - \gamma)\mathbf{1}_{\{i=j\}} + \gamma\nu_j. \quad (\text{G.2})$$

A command  $u_t \in [-1, 1]^d$ , chosen before the report, gives a binary report  $Y_t \in \{-1, 1\}$  with likelihoods

$$\ell_y(u, i) = \begin{cases} 1/2, & i = 0, \\ (1 + y\epsilon \tanh u_i)/2, & 1 \leq i \leq d. \end{cases} \quad (\text{G.3})$$

Conditional on the hidden path and commands, the report randomizations are independent. The resulting joint transition/report kernel is  $P_\gamma(i, j)\ell_y(u, j)$ . It is an instance of Definition A.1; all report probabilities lie between  $3/8$  and  $5/8$ .

For the acquisition lower bound the commands are independent uniform on  $[-1, 1]^d$ , independent of the hidden process and coding randomness. The upper-bound filter will work on every admitted command/report sequence, including adaptively chosen commands. It does not change the fixed exploration law used for the lower bound.

At any selected time  $t \geq 1$  the acquisition can stop and one terminal probe is executed without an intervening refresh. After the retained label is formed, draw  $J$  uniformly from  $\{1, \dots, d\}$ . Probe  $j$  produces  $Z \in \{0, 1\}$  with

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau\mathbf{1}_{\{i=j\}}, \quad \tau = 1/4. \quad (\text{G.4})$$

This is one noisy binary readout, not observation of  $X_t$  or of the full posterior. Its choice is independent of the acquisition. Each acquisition and the terminal probe costs one recorded experimental step. Evaluating maximum-checkpoint risk means that a run executes only its selected probe; it does not execute counterfactual probes at all times.

Let  $p_t = \text{Law}(X_t \mid H_t)$ , and write  $p_{t,+} = (p_{t,1}, \dots, p_{t,d})$ . The probe's full-history success probability is  $1/2 + \tau p_{t,j}$ . If a decoder uses candidate coordinates  $c_j$ , its excess loss is

$$\frac{\tau^2}{d} \|p_{t,+} - c\|_2^2. \quad (\text{G.5})$$

Arbitrary probe predictions are allowed; projecting them onto the interval  $[1/2, 1/2 + \tau]$  cannot worsen loss. Thus the decoder coordinate formulation loses no candidates for the infimum.

Let  $R_{M,t}^{\text{av}}$  and  $R_{M,t}^{\text{max}}$  be optimal checkpoint risks under the stated exploration and over all admitted histories, respectively. Let  $\mathfrak{R}_{M,N}^{\text{av}}$  and  $\mathfrak{R}_{M,N}^{\text{max}}$  be the corresponding best maximum risks for one causal  $M$ -label encoder at times  $1, \dots, N$ . The clock,  $d, \gamma, \epsilon$ , and the finite read-only program are known; no history or unknown calibration is stored in those constants. At time zero

one fixed initial state is allowed. For  $N = \infty$  replace maxima by suprema and use one machine for all times.

**Theorem G.1** (Refreshing experiment memory law). *For every fixed  $d \geq 1$ , constants  $0 < c_d \leq C_d < \infty$  exist such that, for every (G.1), every integer  $M \geq 1$ , every  $N \in \mathbb{N} \cup \{\infty\}$ , and every  $1 \leq t \leq N$ ,*

$$c_d \epsilon^2 M^{-2/d} \leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}, \quad (\text{G.6})$$

$$c_d \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,N}^{\text{av}} \leq \mathfrak{R}_{M,N}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}. \quad (\text{G.7})$$

*The upper bounds are realized by one deterministic finite-label filter using a fixed codebook for all positive times. The lower bounds allow independent public and private coding randomization. At  $\epsilon = 0$  all four risks are zero with one label. Constants are uniform in time, horizon, calibration within (G.1), and the integer budget.*

The proof will derive the cover, the actual attained density, and the stable update separately. This also supplies a concrete dynamic realization of the general emulation theorem.

**G.2. Explicit posterior recursion.** Set  $p_0 = \nu$ . Given the previous posterior  $p$ , the distribution just before the next report is

$$q = (1 - \gamma)p + \gamma\nu, \quad q_i \geq m_* := \frac{4}{5(d+1)}. \quad (\text{G.8})$$

For a command and report put

$$r_0 = 1, \quad r_i = 1 + y\epsilon \tanh u_i \quad (i \geq 1), \quad D_y = q_0 + \sum_{i=1}^d q_i r_i.$$

Then the probability of that report is  $D_y/2$ , and

$$F(p, u, y)_0 = \frac{q_0}{D_y}, \quad F(p, u, y)_i = \frac{q_i r_i}{D_y} \quad (i \geq 1). \quad (\text{G.9})$$

Every denominator belongs to  $[1 - \epsilon, 1 + \epsilon]$ . These formulas hold for every belief in the full simplex and every admitted input. Predictable actions reveal no further hidden-state information once their past is conditioned upon, as in Proposition A.3.

**Lemma G.2** (Uniform contraction). *For any two simplex beliefs and the same  $(u, y)$ ,*

$$\|F(p, u, y) - F(\tilde{p}, u, y)\|_1 \leq \rho \|p - \tilde{p}\|_1, \quad \rho = 2 \frac{1 + \epsilon}{1 - \epsilon} (1 - \gamma) \leq \frac{2}{3}. \quad (\text{G.10})$$

*Proof.* For fixed positive likelihood vector  $r$ , write  $B_r(q) = \text{diag}(r)q/(q^T r)$ . Subtract the two normalized vectors by adding and subtracting  $\text{diag}(r)\tilde{q}/(q^T r)$ . The first term has norm at most  $(r_{\max}/r_{\min}) \|q - \tilde{q}\|_1$ . The second has norm  $|q^T r - \tilde{q}^T r|/(q^T r)$ , at most the same bound. Thus  $\text{Lip}_1(B_r) \leq 2r_{\max}/r_{\min}$ . Equation (G.8) multiplies differences by  $1 - \gamma$ . Use  $r_{\max} \leq 1 + \epsilon$ ,  $r_{\min} \geq 1 - \epsilon$ , and (G.1) to obtain the result.  $\square$

**Lemma G.3** (An invariant shrinking domain). *Let*

$$a_\epsilon = \frac{2\epsilon}{\gamma(1 - \epsilon)}, \quad D_\epsilon = \{p \in \Delta_d : \|p - \nu\|_1 \leq a_\epsilon\}. \quad (\text{G.11})$$

*Then  $\nu \in D_\epsilon$ ,  $F(D_\epsilon, u, y) \subset D_\epsilon$ , and  $a_\epsilon \leq (10/3)\epsilon$ . Every attainable posterior belongs to  $D_\epsilon$ , for every time and every admitted history.*

*Proof.* For any pre-report belief  $q$ ,

$$\|B_r(q) - q\|_1 = \sum_i q_i \frac{|r_i - q^T r|}{q^T r} \leq \frac{2\epsilon}{1 - \epsilon}.$$

Consequently

$$\|F(p, u, y) - \nu\|_1 \leq (1 - \gamma) \|p - \nu\|_1 + \frac{2\epsilon}{1 - \epsilon}.$$

The right side is at most  $a_\epsilon$  when  $p \in D_\epsilon$ , by its definition. The numerical bound follows from  $\gamma \geq 4/5$  and  $1 - \epsilon \geq 3/4$ . Start the induction at  $\nu$ .  $\square$

The set  $D_\epsilon$  is a compact convex polytope for each fixed calibration. Its representatives need not all be attainable at a particular time; they are normalized beliefs with valid updates and belong to a proved invariant domain. No extrapolation outside the probability simplex is used. At  $\epsilon = 0$  the domain is the singleton  $\{\nu\}$ .

**G.3. The actual acquisition density.** Fix a preceding history, so its pre-report belief  $q$  is fixed with (G.8). For  $\epsilon > 0$  and a fixed sign  $y$ , the command map

$$\Phi_{q,y} : u \mapsto F(p, u, y)_+ \in \mathbb{R}^d$$

is one-to-one. Indeed its odds satisfy

$$\frac{p'_i}{p'_0} = \frac{q_i}{q_0} (1 + y\epsilon \tanh u_i), \quad (\text{G.12})$$

so  $u_i$  is recovered by applying  $\text{artanh}$  to  $\{p'_i q_0 / (p'_0 q_i) - 1\} / (y\epsilon)$ . The image of the command interior has a smooth inverse. Boundary commands have zero probability under the uniform acquisition law.

**Lemma G.4** (Posterior Jacobian and unconditional density). *For  $\epsilon > 0$ ,*

$$|\det D\Phi_{q,y}(u)| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \text{sech}^2 u_i. \quad (\text{G.13})$$

*Under independent uniform commands and actual reports, the law of  $p_{t,+}$  has a Lebesgue density bounded by  $B_d \epsilon^{-d}$  at every  $t \geq 1$ , where one possible uniform constant is*

$$B_d = 2^{-d} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1). \quad (\text{G.14})$$

*The bound is conditional on every preceding history as well as unconditional.*

*Proof.* Write  $r'_i = y\epsilon \text{sech}^2 u_i$ . In the  $d$  nonzero-state coordinates,

$$D\Phi_{q,y} = \text{diag} \left( \frac{q_i r'_i}{D_y} \right) - \left( \frac{q_i r'_i}{D_y} \right)_i \left( \frac{q_j r'_j}{D_y} \right)_j^\top.$$

The determinant lemma gives the determinant of the diagonal matrix times

$$1 - \sum_{i=1}^d \frac{q_i r'_i}{D_y} = \frac{q_0}{D_y}.$$

Taking absolute values gives (G.13).

Before any conditioning on the sign, the joint density of the uniform command and that sign is  $2^{-d} D_y / 2$ . Change variables through the injective command map. Its contribution to the posterior density is

$$\frac{2^{-(d+1)} D_y^{d+2}}{q_0 \prod_{i=1}^d q_i \epsilon^d \prod_{i=1}^d \text{sech}^2 u_i} \leq 2^{-(d+1)} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1) \epsilon^{-d}.$$

Sum the two signs, allowing their images to overlap, to obtain (G.14). Finally integrate the conditional bounds against the actual preceding-history law. The same bound survives because it is independent of that history. Thus no event probability or complementary command volume is discarded.  $\square$

This is a full-density upper bound for the actual law, which is sufficient for a quantization lower bound. It is stronger than merely exhibiting a nonzero differential at one command, but it does not assert that every point of the invariant polytope is attained. Those are different statements.

**G.4. A fixed codebook and the proof of the theorem.** Project  $D_\epsilon$  to its  $d$  coordinates  $p_+$ . It lies in the cube  $\nu_+ + [-a_\epsilon, a_\epsilon]^d$ . Partition this cube into  $k^d$  cells,  $k = \lfloor M^{1/d} \rfloor$ , and select one belief in  $D_\epsilon$  from each occupied cell. Fix those representatives once and for all, along with a first-minimum tie convention. Empty cells need no label. The discarded coordinate satisfies  $p_0 = 1 - \sum_{i=1}^d p_i$ . Hence within a cell the full-simplex  $\ell^1$  distance is at most twice the  $\ell^1$  distance of its  $d$  coordinates. The resulting codebook has at most  $M$  points and covering radius

$$r_M \leq \frac{4da_\epsilon}{k} \leq \frac{80d}{3}\epsilon M^{-1/d}. \quad (\text{G.15})$$

Choose the nearest representative in full  $\ell^1$  distance. This preserves the covering bound and is Borel. The initial state is exactly  $\nu$ ; thereafter the same codebook and the same update apply at every step.

*Proof of Theorem G.1.* For  $\epsilon > 0$ , use (G.9) at the retained belief and then quantize with (G.15). Lemmas G.2 and G.3 justify the whole procedure on every history. By Corollary F.2,

$$\sup_{t \geq 1} \|p_t - \hat{p}_t\|_1 \leq 3r_M \leq 80d\epsilon M^{-1/d}. \quad (\text{G.16})$$

Equation (G.5) bounds the excess by  $(\tau^2/d)(80d)^2\epsilon^2 M^{-2/d}$ . This proves a pathwise upper bound for one machine, hence all the required mean and worst-history upper bounds. A checkpoint encoder can use the same final nearest-centre procedure.

For the lower bound, Lemma G.4 and Lemma E.2 give

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_j \|p_{t,+} - c_j\|_2^2 \geq \frac{1}{2}(2B_d v_d)^{-2/d} \epsilon^2 M^{-2/d}.$$

Multiply by  $\tau^2/d$  and use Theorem E.1. The bound holds for every  $t \geq 1$  under the one fixed exploration law, with coding randomness allowed. Every causal encoder is a checkpoint encoder at each time, proving the causal lower bound for finite or infinite horizons. The inequality between mean and worst-history risks is immediate. When  $\epsilon = 0$ , both reports have probability  $1/2$  independent of the state and command. Refreshing preserves the uniform preparation, so  $p_t = \nu$  on every history. A single label predicts the terminal probe exactly relative to the full-history oracle. This proves the zero case and completes the uniform statement.  $\square$

**Corollary G.5** (Persistent bits and double attenuation). *For  $b$  persistent bits, take  $M = 2^b$  in (G.7). With the fixed terminal probe the squared excess has order  $\epsilon^2 2^{-2b/d}$ . If the actual terminal probe is instead*

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau\epsilon \mathbf{1}_{\{i=j\}},$$

*the order is  $\epsilon^4 2^{-2b/d}$ , with the same uniformity and the same filter. At  $\epsilon = 0$ , zero bits suffice in both models.*

*Proof.* The first assertion substitutes the number of labels. In the second experiment the physical readout difference is multiplied by  $\epsilon$ , while the acquisition process, posterior law, and filter are unchanged. Its squared excess is therefore multiplied by  $\epsilon^2$ . For the lower bound the same scaling applies to arbitrary decoder centres after clipping to the physical mean interval. The zero case was established above.  $\square$

**G.5. Feedback-uniform emulation in this model.** For fixed command  $u$ , the probability of the positive report under belief  $p$  is

$$g_+(p, u) = \frac{1}{2} \left( 1 + \epsilon \sum_{i=1}^d ((1 - \gamma)p_i + \gamma\nu_i) \tanh u_i \right).$$

For binary laws their total variation is the difference of their positive probabilities. Thus

$$\|g(\cdot \mid p, u) - g(\cdot \mid \tilde{p}, u)\|_{\text{TV}} \leq \frac{1}{2}\epsilon(1 - \gamma) \|p - \tilde{p}\|_1. \quad (\text{G.17})$$

By Theorem F.1, the actual report-path law and the finite-state reference report-path law obey

$$\|P_N^\alpha - \hat{P}_N^\alpha\|_{\text{TV}} \leq \min\{1, 40dN\epsilon^2(1 - \gamma)M^{-1/d}\} \quad (\text{G.18})$$

for every common feedback controller. This statement is uniform over controllers but is finite-horizon; it does not identify infinite path laws. If a terminal probe is included in the compared observed path, its additional conditional error is at most  $\tau \|p_t - \hat{p}_t\|_1$ , averaged over its random index. This term is to be added to (G.18). Prediction regret for that probe instead uses its squared error, as in (G.5); the two claims are different.

The bound vanishes for the acquisition report law when  $\gamma = 1$ : each new state is independently uniform, so previous belief errors have no effect on the next acquisition report. The current posterior can still have nontrivial geometry and require finite prediction memory for the immediate terminal probe. This limiting case is a useful check on the meaning of each estimate.

**G.6. Calibration and finite precision.** The theorem is uniform over (G.1), but the codebook and update may use the known values of  $\gamma$  and  $\epsilon$ . It is not a calibration-blind theorem. Numerically approximating each belief update by an admissible belief within  $\eta$  in  $\ell^1$  changes (G.16) to  $3(r_M + \eta)$ . The physical excess is at most  $9\tau^2(r_M + \eta)^2/d$ . Thus choosing  $\eta$  proportional to  $\epsilon M^{-1/d}$  preserves the order. This is an accuracy requirement; it does not assert zero workspace or a universal bit-complexity bound for computing the update.

For rational calibration one can specify the invariant polytope by rational linear inequalities. Representatives can be selected in its occupied grid cells by a finite feasibility procedure; centres and a membership/tie rule form a finite read-only program. With nonrational calibration the theorem uses the declared calibrated real-number model; a numerical implementation must supply its own certified approximation. The only persistent acquired information after a report is its representative label.

Finally, the refresh lower bound and the fixed positive state floor are substantive. If  $\gamma$  tends to zero, the constant  $2/3$  need not remain valid and the argument does not give a horizon-uniform theorem. That case is not hidden in the calibration interval.

## APPENDIX H. NORMALIZED RESPONSE AND OBSERVATION MODULES

**H.1. The covariance pairing.** Let  $\pi$  be a probability measure and  $f, g$  bounded measurable functions. For  $u$  near zero define

$$\pi_u(dx) = \frac{e^{ug(x)}\pi(dx)}{\pi(e^{ug})}.$$

**Proposition H.1** (Normalized response). *The derivative of a normalized prediction is*

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \pi(fg) - \pi(f)\pi(g) = \text{Cov}_\pi(f, g). \quad (\text{H.1})$$

More generally, if  $L_u > 0$  is differentiable with the dominated derivatives needed for the quotient rule, then the derivative at  $u_0$  of  $\pi(fL_u)/\pi(L_u)$  is the covariance of  $f$  with  $\partial_u \log L_u|_{u_0}$  under the normalized law at  $u_0$ .

*Proof.* Differentiate numerator and denominator. At zero their values are  $\pi(f)$  and one, and their derivatives are  $\pi(fg)$  and  $\pi(g)$ . This proves (H.1). Apply the same quotient rule at  $u_0$  and divide the derivative of  $L_u$  by  $L_{u_0}$  to obtain the general statement. Boundedness, or the stated domination in the general case, justifies integration of the derivatives.  $\square$

For vector tests and several commands this gives a cross-covariance matrix. One index may represent an acquisition direction and the other a future test. That matrix is not automatically the Fisher information for an unknown physical parameter. In (G.9), command changes are selected by the observer; in Theorem D.3, parameter changes vary the family of physical laws. The similar algebra does not equate the two statistical problems.

**H.2. Finite positive functionals and exact product fibres.** Suppose a real algebra of functions contains finite-dimensional spaces  $U_1, \dots, U_L$ , and let  $V$  contain all their products. A parameter is a linear functional  $\ell \in \Omega \subset V^*$ , where  $\Omega$  is open in  $V^*$ . Positivity of measured blocks may define this open set. The observed bilinear forms are

$$B_\nu(f, g) = \ell(fg), \quad f, g \in U_\nu.$$

There are no cross-contact products unless the experiment actually measures them. Set

$$W = \sum_{\nu=1}^L \text{span}\{fg : f, g \in U_\nu\} \subset V.$$

**Theorem H.2** (Product-observation fibre). *The observation fibre through  $\ell$  is*

$$(\ell + W^\perp) \cap \Omega. \tag{H.2}$$

*Its local dimension is  $\dim V - \dim W$ . Recovery of every functional in  $\Omega$  is equivalent to  $W = V$ . If the parameter domain is relatively open in an affine subspace with direction  $L \subset V^*$ , the local invisible space is instead  $L \cap W^\perp$ .*

*Proof.* Two observations agree precisely when the difference functional vanishes on every generating product, equivalently on  $W$ . This gives (H.2). Openness allows all sufficiently small perturbations in the annihilator, so the local dimension is its dimension. The affine-domain version restricts those perturbations to  $L$ .  $\square$

This is linear duality. The difficult model-specific work can lie in proving that these forms are actually observed, determining  $W$ , describing the locus where its dimension drops, and controlling real positive realizability. For the calibrated Hankel family of [16],  $V$  is a polynomial space and the observation is a normal compression of an inverse information matrix. The theorem supplies its abstract fibre mechanism, not the paper's complete degeneracy classification.

**H.3. The contact calculation.** For clarity, the local geometric interface can also be derived directly. Let  $M \subset \mathbb{R}^k$  have an embedded smooth germ at  $x_0$ , and let  $R > 0$ . Assume the local minimizing projection onto that germ computes the global distance in a neighborhood of  $x_0$ . Define

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b).$$

Let  $Z$  have orthonormal columns spanning the Euclidean normal space at  $x_0$ , and put  $B = Z^\top R Z$ .

**Proposition H.3** (Normal second jet). *Under these local projection assumptions,*

$$C_R(x_0 + h) = h^\top Z B^{-1} Z^\top h + o(\|h\|^2), \quad D^2 C_R(x_0) = 2 Z B^{-1} Z^\top.$$

*Consequently the full second jet determines exactly the positive normal block  $B$  and has the tangent space as its kernel.*

*Proof.* Replace the germ by its tangent space  $T$  to compute the quadratic term. The first-order minimizing condition for  $\min_{v \in T} (h - v)^\top R^{-1} (h - v)$  says  $R^{-1} (h - v) = Zn$  for some  $n$ . Hence  $h - v = RZn$ , and normal projection gives  $Z^\top h = Bn$ . The minimum is  $(Z^\top h)^\top B^{-1} (Z^\top h)$ . A smooth local parametrization of the germ has derivative spanning  $T$ ; its tangent Gram matrix in the positive  $R^{-1}$  metric is invertible. The implicit function theorem gives a differentiable minimizing parameter with the stated linear term. The quadratic expansion and the projection assumption identify it with  $C_R$  near  $x_0$ . Positive block inversion is one-to-one, proving the observation assertion.  $\square$

If  $R = JH(\theta)J^\top$  with Hankel entries  $H(\theta)_{ab} = \theta_{a+b}$ , then  $v^\top R w$  evaluates the moment functional on the product of the two corresponding coefficient polynomials. Theorem H.2 applies after this explicit identification. A higher-order distance jet has additional curvature and projection terms; replacing  $\text{Sym}^2$  by  $\text{Sym}^m$  is not a derivation of a higher-order experiment.

#### H.4. Why general dynamic tests need not form a finite algebra.

**Proposition H.4** (Finite continuous algebras have finite quotients). *Let  $X$  be compact Hausdorff and let  $W \subset C(X, \mathbb{R})$  be a finite-dimensional unital subalgebra. Then  $W$  is the algebra of all functions constant on a finite clopen partition of  $X$ . The number of nonempty cells is  $\dim W$ .*

*Proof.* For  $f \in W$ , sufficiently many powers of  $f$  are linearly dependent. A nonzero polynomial therefore vanishes on  $f(X)$ , so that image is finite. A finite basis of  $W$  has only finitely many joint values; its fibres form a finite clopen partition. Every member of  $W$  is constant on these fibres. Conversely, for a joint value  $a$ , choose for every distinct joint value  $b$  a basis function  $f_{i(a,b)}$  separating them. The product

$$\prod_{b \neq a} \frac{f_{i(a,b)} - f_{i(a,b)}(b)}{f_{i(a,b)}(a) - f_{i(a,b)}(b)}$$

is in  $W$  and is the indicator of the fibre at  $a$ . These indicators are linearly independent and span all functions on the partition, which proves the result.  $\square$

Thus a finite multiplication-closed continuous algebra is a useful model, but it cannot by itself represent arbitrary continuous predictive information. Growing test spaces and positive operator words are the more general primitive. Neither the algebra dimension nor one tangent rank replaces the actual predictive law or its quantization geometry.

### APPENDIX I. EXPONENTIAL TRANSFORMS AND CONDITIONAL LIMIT INTERFACES

**I.1. Random channels have nonlinear exponential pullbacks.** For a Markov kernel  $G : X \rightsquigarrow Y$ , bounded measurable  $f : Y \rightarrow \mathbb{R}$ , and a fixed  $\beta > 0$ , define

$$\mathcal{T}_{G,\beta} f(x) = \beta^{-1} \log \int e^{\beta f(y)} G(dy \mid x). \quad (\text{I.1})$$

For a probability  $P$  on  $X$  put  $\mathcal{P}_{P,\beta}(h) = \beta^{-1} \log P(e^{\beta h})$ . These expressions concern finite experiments and bounded sources; no long-time limit is presumed.

**Theorem I.1** (Exponential composition and entropy duality). *For  $H : Y \rightsquigarrow Z$  and bounded  $f$ ,*

$$\mathcal{T}_{GH,\beta} f = \mathcal{T}_{G,\beta}(\mathcal{T}_{H,\beta} f), \quad (\text{I.2})$$

$$\mathcal{P}_{PG,\beta}(f) = \mathcal{P}_{P,\beta}(\mathcal{T}_{G,\beta} f), \quad (\text{I.3})$$

$$\mathcal{P}_{P,\beta}(h) = \sup_{Q \ll P} \{Qh - \beta^{-1} D(Q \| P)\}. \quad (\text{I.4})$$

*The supremum is attained at  $dP^h/dP = e^{\beta h}/P(e^{\beta h})$ . The source derivatives are*

$$D\mathcal{P}_{P,\beta}(h)[u] = P^h u, \quad D^2\mathcal{P}_{P,\beta}(h)[u, v] = \beta \text{Cov}_{P^h}(u, v). \quad (\text{I.5})$$

*Proof.* Exponentiate (I.2). Its right side becomes  $G(H(e^{\beta f}))$ , which is the definition of the composed kernel's exponential integral. Integrating the same identity against  $P$  proves (I.3). For any  $Q \ll P$  of finite entropy,

$$D(Q \| P^h) = D(Q \| P) - \beta Qh + \log P(e^{\beta h}) \geq 0.$$

Rearrangement gives the upper inequality in (I.4), and  $Q = P^h$  attains equality. Measures of infinite entropy do not improve the supremum for bounded  $h$ . Finally boundedness permits differentiation of the exponential integral in the directions  $u, v$ ; one quotient rule gives the mean and the second gives the covariance with the factor  $\beta$ .  $\square$

If  $G$  is the deterministic map  $g$ , the exponential pullback is exactly  $f \circ g$ . If  $G$  is random, it is generally not  $Gf$ . Jensen gives  $\mathcal{T}_{G,\beta}f \geq Gf$ , with equality precisely when  $f$  is constant  $G(\cdot | x)$ -almost surely at each relevant  $x$ . The entropy variational principle is classical [13]; the proof fixes its conventions for this experiment category rather than asserting a new variational principle.

**Proposition I.2** (Second-order chain rule for a random channel). *Put  $h = \mathcal{T}_{G,\beta}f$  and*

$$G_f(dy | x) = \frac{e^{\beta f(y)} G(dy | x)}{G(e^{\beta f})(x)}.$$

*Then the tilted joint law is  $P^h(dx)G_f(dy | x)$  and*

$$\begin{aligned} D\mathcal{T}_{G,\beta}(f)[u] &= G_f u, \\ D^2\mathcal{T}_{G,\beta}(f)[u, v] &= \beta \text{Cov}_{G_f}(u, v), \\ D^2\mathcal{P}_{PG,\beta}(f)[u, v] &= \beta \text{Cov}_{P^h}(G_f u, G_f v) + \beta \mathbb{E}_{P^h} \text{Cov}_{G_f}(u, v). \end{aligned}$$

*The last expression is  $\beta \text{Cov}_{P^h G_f}(u, v)$ .*

*Proof.* Multiplying the two tilted densities cancels  $G(e^{\beta f})(x)$  and produces the original joint law multiplied by  $e^{\beta f(y)}$  and its single total normalizer. Differentiation of the conditional normalizer gives the first two formulas just as in (I.5). Apply the ordinary second-order chain rule to (I.3). Its first term is the covariance of conditional means and its second is the mean of conditional covariances. The total covariance formula gives the final identity.  $\square$

The conditional covariance term is essential. Discarding it would treat a random observation as a deterministic projection. This connects the positive-kernel and source-response layers exactly, while keeping them distinct from Fisher information for a model parameter.

**I.2. A finite risk-sensitive recursion.** Consider a known prepared predictive model with finite action sets, state update  $F_t(s, a, y)$ , report kernel  $g_t(dy | s, a)$ , bounded observed stage cost  $c_t(s, a, y)$ , and bounded terminal cost  $v_N$ . The cost here is an optimization objective, not an undeclared memory resource. For  $\beta > 0$  define

$$v_t(s) = \min_{a \in A_t} \frac{1}{\beta} \log \int \exp\{\beta[c_t(s, a, y) + v_{t+1}(F_t(s, a, y))]\} g_t(dy | s, a). \quad (\text{I.6})$$

**Proposition I.3** (Finite-horizon exponential dynamic programming). *The value (I.6) is the least value of  $\beta^{-1} \log \mathbb{E} \exp(\beta[\sum_{j=t}^{N-1} c_j + v_N])$  over all causal strategies from  $s$ . A deterministic Borel minimizing policy exists with a fixed tie rule.*

*Proof.* Write  $w_t = e^{\beta v_t} > 0$ . Conditional expectation over the next report and then the future gives the multiplicative recursion  $w_t(s) = \min_a \int e^{\beta c_t(s, a, y)} w_{t+1}(F_t(s, a, y)) g_t(dy | s, a)$ . At the last step this is the exact conditional objective. Induct backwards; future randomized strategies cannot improve the already minimized continuation, and mixing current actions produces a convex

combination of their positive objective values, no less than their minimum. Finite Borel action comparisons give a Borel first minimizer. Taking logarithms recovers (I.6).  $\square$

This recursion needs a state sufficient for the future experiment and its cost. It cannot be applied to an arbitrary lossy statistic without proving closure. Compact infinite action sets would require appropriate existence and selection hypotheses; they are not needed for this proposition.

**I.3. Deterministic LDP contractions.** We record a conditional interface for later work. Suppose a sequence  $P_n$  on a Polish space has a good large-deviation principle with speed  $a_n \rightarrow \infty$  and rate  $I$ . If  $T$  is continuous into another Polish space, its pushforward has rate

$$I_T(y) = \inf_{T(x)=y} I(x). \quad (\text{I.7})$$

For an open or closed target set its inverse image has the same property, so the LDP lower and upper bounds give (I.7). Goodness follows because each finite-rate infimum is attained on compact sublevels and the sublevels of  $I_T$  are continuous images of those of  $I$ . This is the standard contraction principle [14]. It is not a consequence of the raw experiment axioms.

For a random channel one needs the joint law, a conditional rate, or a separate exponential approximation theorem. A small total-variation simulation defect does not supply these ingredients; an explicit example is given in Section J.

**I.4. Regular quadratic and labelled-phase interfaces.**

**Proposition I.4** (Quadratic constrained reduction). *If  $H > 0$  and  $A$  has full row rank, then*

$$\inf_{Ah=z} \frac{1}{2} h^\top H h = \frac{1}{2} z^\top (A H^{-1} A^\top)^{-1} z.$$

*Proof.* The Lagrange equations give  $h = H^{-1} A^\top (A H^{-1} A^\top)^{-1} z$ . The constraint matrix in parentheses is positive definite. Substitution gives the value, and strict convexity gives the unique minimizer.  $\square$

Only if a separate probabilistic theorem identifies  $H^{-1}$  with the appropriate Gaussian covariance does this become a covariance contraction statement. Existence of an LDP alone is not that theorem.

For finitely many labelled components with common speed  $a_n$ , suppose component rates are  $I_j$  and their positive weights satisfy  $-a_n^{-1} \log w_{n,j} \rightarrow c_j$ , with  $\min_j c_j = 0$ . The joint labelled rate is  $c_j + I_j$  and the unlabelled rate is  $\min_j (c_j + I_j(x))$ . To verify the bounds, bound a finite sum above by the number of summands times its largest summand and below by each summand; then apply the component LDPs. Compactness follows from the finite union of component compact sublevels. Dynamic prediction may still require the posterior over labels. A marginal log-sum is not automatically a Markov semigroup.

**I.5. Closure and finite-dimensional memory.** For a Markov kernel  $P$  and state map  $q$ , an exact reduced kernel  $\bar{P}$  satisfies

$$P(f \circ q) = (\bar{P}f) \circ q.$$

Iterating this identity proves every finite-time identity, including the corresponding exponential transforms. If it fails, the projected state need not be Markov. For a finite-dimensional linear evolution split as

$$\dot{u} = A_{PP}u + A_{PQ}v, \quad \dot{v} = A_{QP}u + A_{QQ}v,$$

variation of constants in the second equation gives exactly

$$\begin{aligned} \dot{u}(t) &= A_{PP}u(t) + A_{PQ}e^{tA_{QQ}}v(0) \\ &\quad + \int_0^t A_{PQ}e^{(t-s)A_{QQ}}A_{QP}u(s) ds. \end{aligned} \quad (\text{I.8})$$

The unresolved initial-state term is not optional. Unbounded generators, common form domains, and differentiable operator transports require separate analytic results; the finite-dimensional identity does not prove them. This is the precise boundary at which the history and memory program of [17] can be connected to the present foundations.

## APPENDIX J. BOUNDARY EXAMPLES AND NON-IMPLICATIONS

The following examples specify the distinctions used in the theorems. They are part of the mathematical scope, not objections to studying stronger results under stronger hypotheses.

**Example J.1** (Less statistical information, more exact prediction labels). Let  $Z \sim \text{Bernoulli}(1/2)$  and initially observe  $X = Z$ . The future target is again  $Z$ . Two labels retain the posterior mean exactly. Garble  $X$  into  $Y = X + \xi$  with independent  $\xi \sim N(0, 1)$ . Bayes' formula and the ratio of the two Gaussian densities give

$$p(Y) = \mathbb{P}(Z = 1 \mid Y) = \frac{1}{1 + e^{1/2 - Y}}.$$

This is a smooth strictly increasing bijection from  $\mathbb{R}$  to  $(0, 1)$ . The density of  $Y$  is positive everywhere, so the posterior law is continuous and has a positive density on every compact subinterval of  $(0, 1)$ . No finite set of centres can equal this posterior almost surely. More quantitatively, restrict to a compact interval where the density has a positive lower bound and apply Lemma E.3 in one dimension. Every fixed  $M$  has strictly positive optimal excess squared error. Both oracles are different: absolute decision risk obeys statistical data processing, but excess coding risk relative to each oracle need not.

**Example J.2** (The same mean map with two noise geometries). Let  $Z \sim N(\lambda, 1)$ . For known  $\epsilon > 0$ , observing  $Y_\epsilon = \epsilon Z$  is equivalent to observing  $Z$  by division, and its Fisher information for  $\lambda$  is one. Observing  $\tilde{Y}_\epsilon = \epsilon\lambda + \xi$  with independent  $\xi \sim N(0, 1)$  has Fisher information  $\epsilon^2$ . Both means equal  $\epsilon\lambda$ . Only the first covariance is scaled with the mean. Consequently a small singular value of a mean map does not determine its statistical information without the noise law.

**Example J.3** (Rank does not record global fibres or contact order). The derivative of  $x \mapsto x^2$  is nonzero at 1 and  $-1$ , yet their images coincide. For another distinction, let  $Y_i = \lambda^r + \xi_i$ ,  $\lambda \geq 0$ , with independent unit Gaussian noise and  $r \geq 2$ . Every mean map has zero derivative at zero, but its local amplitude scale is  $n^{-1/(2r)}$ . Indeed the divergence between  $\lambda = 0$  and a positive alternative is  $n\lambda^{2r}/2$ . Taking this bounded gives the lower resolution scale by two-point testing. For an upper bound use  $\hat{\lambda} = (\bar{Y}_+)^{1/r}$  and  $|x_+^{1/r} - b^{1/r}| \leq |x - b|^{1/r}$  for  $b \geq 0$ ; its mean squared error is  $O(n^{-1/r})$ . The order of contact, not just the rank, matters.

**Example J.4** (Reachability is not attained mass). A protocol may permit all commands in an interval while the exploration law always chooses a single command. The set of legally attainable predictions can be continuous even if that particular exploration law visits only one prediction. Another exploration on the same command interface can spread positive mass over a full interval. Their average quantization risks differ. Theorem G.1 avoids this ambiguity by specifying independent uniform commands and deriving their density.

**Example J.5** (Equal first two moments do not give equal laws). The two laws  $\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$  and  $\frac{1}{2}\delta_0 + \frac{1}{4}\delta_{-\sqrt{2}} + \frac{1}{4}\delta_{\sqrt{2}}$  have mean zero and variance one, but are different. Equality of a finite jet or covariance needs a model-specific identifiability theorem to imply anything stronger.

**Example J.6** (Fisher preservation need not be sufficiency). The Kagan–Shepp phenomenon, analyzed in [11], gives an insufficient statistic preserving Fisher information. We do not reproduce that external construction here or claim it as new. Its precise logical role is to prevent promotion of the equality case of Theorem D.3 into the parameter-uniform recovery assertion of Theorem D.2.

**Example J.7** (Small variation error can change the exponential rate). For  $a_n \rightarrow \infty$  and  $c > 0$ , let

$$P_n = \delta_0, \quad Q_n = (1 - e^{-a_n c})\delta_0 + e^{-a_n c}\delta_1.$$

Their total variation is  $e^{-a_n c} \rightarrow 0$ . At speed  $a_n$ , however,  $P_n$  gives rate  $+\infty$  to 1, while  $Q_n$  gives it rate  $c$ . Thus even exponentially small variation error need not preserve a complete rate function. Appropriate superexponential approximation hypotheses are substantially stronger than an ordinary vanishing simulation defect.

**Example J.8** (Linear-source pressure can convexify a nonconvex rate). Let  $P_n$  put probability  $1/2$  at each of  $-1$  and  $1$ , for all  $n$ . At speed  $n$  its rate vanishes at those two points and is infinite elsewhere. Its limiting pressure for linear source  $sx$  is  $|s|$ . The Legendre transform of  $|s|$  vanishes on the entire interval  $[-1, 1]$ . The full bounded-continuous-source Laplace functional and the restricted linear-source transform are not interchangeable.

**Example J.9** (Observation does not ensure Markov closure). Consider the deterministic chain  $a \mapsto a$ ,  $b \mapsto c$ ,  $c \mapsto c$ , with report  $q(a) = q(b) = 0$ ,  $q(c) = 1$ . The current report 0 does not specify the next report: the two compatible hidden states give different answers. Thus no single reduced kernel works for every preparation. Future-test equivalence distinguishes  $a$  and  $b$ ; merely keeping their current report does not.

**Example J.10** (Zero evidence has no canonical normalized limit). On a two-point space the positive measures  $(\delta, 0)$  and  $(0, \delta)$  both tend to the zero measure as  $\delta \downarrow 0$ . Their normalized probabilities tend to different point masses. A continuous normalized extension at zero cannot be assigned from the zero measure alone.

**Example J.11** (A simulator may need delay memory). Let the unknown parameter be  $\lambda \in \{0, 1\}$ . A source with no control actions reveals  $\lambda$  at time one and constants thereafter. A target reveals a constant at time one and  $\lambda$  at time two. Their terminal information is equivalent. However, a parameter-independent simulator with only one retained state after time one, and no other retained tape, has the same internal state and the same new inputs under both parameters. Its second output therefore has the same law under both parameters and cannot equal the required two different point masses. Two states suffice by retaining the bit. If a target controller also retains its own state, the direct implementation is a product. Terminal sufficiency does not erase this state cost.

**Example J.12** (A finite command alphabet is a different experiment). At a fixed finite time, finite actions and finite reports produce only finitely many histories. With enough labels they can be retained exactly. The positive lower bound of Theorem G.1 for all finite  $M$  uses the stipulated continuously distributed commands. Rounding the physical command alphabet itself changes that experiment. Approximating only the filter's arithmetic while observing the original experiment is instead the perturbation problem in Section F.

These examples rule out four unqualified shortcuts: deriving global identifiability from local rank, deriving memory from information order, deriving statistical rates from a noiseless feature map, and deriving exponential asymptotics from ordinary variation approximation. The positive results in this paper specify the extra structure needed in each case.

## APPENDIX K. INTERFACES WITH SINGULAR GEOMETRY AND DYNAMICS

The foundations established here do not require all experiments to be contractions of one physical path law or one rate function. They give a common language in which an actual map or causal simulator must be exhibited before a comparison is asserted.

**K.1. What is established in this installment.** The exact results are path construction, future-test minimality with stated realization conditions, charged simulation composition, posterior and entropy identities, the finite/dominated score identity, normalized response, product-observation fibres, checkpoint quantization, and exponential composition. The quantitative results are the feedback- and stopping-compatible finite-state emulation theorem and the uniform dynamic memory law for the refreshing model. Their proofs occur in this article. The LDP and operator discussions are conditional interfaces, with a standard contraction principle and a finite-dimensional memory identity; no general Sinai or kinetic limit theorem is asserted here.

The principal dynamic model verifies a compact shrinking domain, actual probability mass, and stable updates directly. It gives one time-uniform isotropic degeneration law. It does not classify all anisotropic collisions, all hidden Markov filters, all priors, or weak-refreshing limits. The bounded comparison constants are not exact optimizer transition constants.

**K.2. The role of the two geometric manuscripts.** The finite-horizon manuscript [15] provides finer anisotropic profiles for its positive monomial experiments and a separate bounded observation-algebra law. Those hypotheses should remain attached to their conclusions. The present dynamic result is not a replacement for their collision geometry, and their fixed-horizon constants are not assumed uniform as the horizon grows.

The contact manuscript [16] provides a different geometric realization of information loss. Theorem H.2 explains its linear-functional interface. A complete extension to singular statistical resolution still requires the actual noise metric, calibration, parameter nuisance, and admissible real domain. In particular independent measurement noise and correlated raw-data score noise can react differently to a small multiplication singular value. A complex determinantal component is not automatically a physically attainable real statistical submodel.

**K.3. A precise next target.** Let a parameter target be  $\tau(\lambda)$  and let an actual feature map be  $O(\lambda)$ . Its inverse modulus is

$$\omega(\delta) = \sup\{d(\tau(\lambda), \tau(\lambda')) : d_Z(O(\lambda), O(\lambda')) \leq \delta\}.$$

If a finite-memory procedure estimates that feature with a proved common error bound  $\delta$ , fitting the model on a compact parameter set gives an error at most  $\omega(2\delta)$ : the true parameter is an admissible fit, and the triangle inequality bounds the discrepancy of the two model features. A matched statistical lower bound requires actual distributional closeness. If  $\|P_{n,\lambda_0} - P_{n,\lambda_1}\|_{TV} \leq a < 1$ , any estimator has maximum squared target risk at least

$$\frac{1-a}{8} d(\tau(\lambda_0), \tau(\lambda_1))^2.$$

Indeed choose the nearer of the two target points as a test; on an incorrect choice the estimation error is at least half their separation. The two error probabilities sum to at least  $1 - a$ .

These two elementary bounds specify a route to a resource–statistical singularity theorem, but do not prove a universal matched modulus law. Covering, testing, attainable alternatives, and tail control must be supplied. The geometry-of-rates literature, including [12], is the appropriate comparison, not a claim that all minimax theory follows from one rank locus.

**K.4. The physical realization obligations remain.** For deterministic hyperbolic and particle systems, one must still establish the true experiment kernels, physical clock handling, tightness and recovery at exponential scale, and the domains of any response or memory operators. A pathwise finite-state prediction error does not provide a Fourier local limit theorem. A static covariance factorization does not construct a collision process. A projected state does not acquire a Markov semigroup merely because it is named a predictive coordinate.

The historical typed-contraction program [17] can therefore be reorganized as a family of verified realization theorems attached to these interfaces. Its model-specific proofs are not assumed

completed by the present definitions. In particular memory response in a varying Hilbert or form domain requires its own transport and domain arguments.

**K.5. Conclusion.** A common foundation is obtained by keeping the entire experiment ahead of its geometric representations. Positive preparation and future tests give prediction; a real acquisition law gives a distortion problem; stable updates give an online implementation; a causal simulator gives a comparison, with its state and error charged. In the refreshing model these steps yield a uniform finite-memory law through loss of the acquisition channel. The separate entropy, score, and exponential identities explain which forms of information transport are valid. Stronger singular and long-time results must enter through corresponding theorems, not through additional meanings assigned to the word “contraction”.

## APPENDIX L. MEASURABILITY, FINITE COVERS, AND A SCALAR BENCHMARK

**L.1. Finite-label implementation.** A finite codebook  $C = \{c_1, \dots, c_M\}$  in a metric space has a Borel nearest-centre selector whenever each distance  $s \mapsto d(s, c_j)$  is Borel. Define its label by the least index attaining the minimum. The set on which label  $j$  is selected is the intersection of the finitely many sets  $\{d(s, c_j) < d(s, c_i)\}$  for  $i < j$  and  $\{d(s, c_j) \leq d(s, c_i)\}$  for  $i > j$ . These sets are Borel. Thus the finite-label recursions used in this paper do not require an unproved measurable choice of a nearest point in an infinite set.

Codebook construction and codebook execution are different questions. Existence of finitely many representatives in nonempty cells is an ordinary finite choice. When an algorithmic construction is claimed, one must also supply a representation and a feasibility procedure for those cells. In Section G, rational calibration gives a rational polytope; finite rational linear feasibility supplies this additional construction. This does not charge an unbounded list of history-dependent centres to a read-only program.

**L.2. Conditional distributions and common versions.** The existence of regular conditional distributions on standard Borel spaces is a classical input [4]. Its relevant conclusion is existence of a kernel under a fixed joint law. It does not provide one canonical version on every null history, nor does it alone give a version continuous in a parameter or uniformly valid on an uncountable family of null sets. The statements in the main text accordingly take one of three forms: an explicit positive finite-report formula; a compact continuous realization as in Theorem B.3; or an already supplied jointly measurable predictive realization with its applicable quantifiers.

There is no hidden claim that all quotient images of a Borel space are standard Borel. The evaluation image into a countable product can be given its induced measurable structure, but to use ordinary kernel calculus on it one must verify that the image and descended maps have the required measurability. The compact theorem verifies precisely that fact in one useful class. Keeping the full posterior remains a valid alternative when compression of the state is not yet justified.

### L.3. Quantization after a readout change.

**Proposition L.1** (Bi-Lipschitz readout comparison). *Let  $S$  be a metric space with a probability  $\nu$ , and let  $p : S \rightarrow \mathbb{R}^q$  be an injective readout satisfying*

$$a d(s, t) \leq \|p(s) - p(t)\|_2 \leq b d(s, t), \quad a, b > 0.$$

*If  $S$  has an  $M$ -point cover of radius  $r$  with centres in  $S$ , the checkpoint prediction risk is at most  $b^2 r^2$ . If some measurable map  $T : \mathbb{R}^q \rightarrow \mathbb{R}^k$  has Lipschitz constant  $L > 0$ , and the law of  $Tp(s)$  has density at most  $A$ , that risk is at least*

$$\frac{1}{2L^2} (2Av_k M)^{-2/k}.$$

The second assertion does not require decoder centres to belong to  $p(S)$ .

*Proof.* For the upper bound map the cover centres through  $p$  and use the upper Lipschitz estimate. For any decoder centres  $z_j \in \mathbb{R}^q$ , the inequality  $\|p(s) - z_j\|_2 \geq L^{-1} \|Tp(s) - Tz_j\|_2$  reduces the lower bound to Lemma E.2. Take the infimum over the original centres. The constant  $a$  is useful for comparing intrinsic covers in both directions, but it is not needed in this particular density-based lower bound. In particular, the argument does not require inversion of a near-singular statistical covariance.  $\square$

A linear projection or bounded coordinate map can be used for  $T$ . A coordinate change whose Lipschitz constant tends to infinity changes the bound; it cannot be called a harmless normalization at a degeneracy.

**Proposition L.2** (Exact scalar uniform quantization). *For  $U$  uniform on an interval of length  $a > 0$ , the optimal squared  $M$ -centre quantization error is  $a^2/(12M^2)$ , for every integer  $M \geq 1$ .*

*Proof.* Order the centres. Their nearest-centre cells intersect the interval in intervals, allowing empty cells. If a cell has length  $l_j$ , its contribution is minimized by its midpoint and equals  $l_j^3/(12a)$ . Since  $\sum_j l_j = a$ , convexity gives  $\sum_j l_j^3 \geq a^3/M^2$ . Hence the total error is at least  $a^2/(12M^2)$ . Equal-length cells with their midpoints attain that value. Arbitrary randomized encoders cannot improve it by Theorem E.1.  $\square$

This exact formula is a benchmark for the meaning of label cardinality, not a claim that the dynamic posterior density in Section G is uniform. The latter is bounded by its actual Jacobian and can depend on the preceding history. Its two-sided order, rather than an exact optimizer constant, is what we prove.

**L.4. Predictable error budgets and random horizons.** The stopped version of Lemma C.2 is often more useful than the deterministic sum. In a comparison with common visible stopping rule  $\sigma$ , append an absorbing terminal symbol. The two conditional kernels coincide once the symbol has appeared. Their discrepancy at time  $t$  is therefore bounded by  $\mathbf{1}_{\{t < \sigma\}} e_t(H_t)$ , a measurable function of the common prefix. The telescoping proof integrates it against the actual prefix law. This gives the expectation of the active-step sum, without replacing a random horizon by an uncharged successful count.

If the active-step sum is bounded by a deterministic budget  $B$  on every path, its expectation is at most  $B$ . If only an expected budget is known, that is the exact kind of budget furnished by the theorem. A tail-budget statement requires an additional probability estimate. These are distinct resource certificates even though the same positive summands occur in all three.

## APPENDIX M. EXPLICIT FINITE CHECKS AND THE LIMITS OF COMPUTATION

**M.1. A two-state posterior calculation.** The smallest dynamic instance has  $d = 1$ ,  $\gamma = 4/5$ , and  $\epsilon = 1/4$ . Put  $v = \tanh u$  and select  $v = 1/2$ , which corresponds to the legal command  $u = \operatorname{artanh}(1/2) < 1$ . Starting from the uniform preparation, the pre-report belief is  $q = (1/2, 1/2)$ , and

$$\mathbb{P}(Y = +1) = \frac{17}{32}, \quad \mathbb{P}(Y = -1) = \frac{15}{32}.$$

The posterior probabilities of state 1 are respectively

$$p_1^+ = \frac{9}{17}, \quad p_1^- = \frac{7}{15}.$$

Thus the same current command produces two different posteriors. Discarding the negative report or assigning its conditional law mass one would alter the acquisition law used in the proof. With

the fixed terminal probe the corresponding success probabilities are

$$\frac{1}{2} + \frac{1}{4} \frac{9}{17} = \frac{43}{68}, \quad \frac{1}{2} + \frac{1}{4} \frac{7}{15} = \frac{37}{60}.$$

The mean posterior agrees with the pre-report probability:

$$\frac{17}{32} \frac{9}{17} + \frac{15}{32} \frac{7}{15} = \frac{1}{2}.$$

This is the posterior martingale property for the report step; the refresh step has its own affine prediction and is not being omitted.

The derivative with respect to  $v$ , rather than  $u$ , at this point for the positive report is

$$\frac{q_0 q_1 \epsilon}{D_+^2} = \frac{16}{289}.$$

Multiplying by  $dv/du = 1 - v^2 = 3/4$  gives  $12/289$ , exactly (G.13) in one dimension. These distinct derivatives are recorded because replacing a physical command by a nonlinear coordinate without its Jacobian would change the preparation density.

**M.2. A direct resource calculation.** Suppose a report converter has two persistent states (for example, it retains the previous binary report), the target predictor has  $M$  labels, and a feedback controller has  $D$  labels. The implemented composite uses at most  $2MD$  triples. On the next step it reads the old triple, the current input, and fresh independent randomness, and writes the next triple. The source history may be longer, but it is not consulted by this implementation. Any reduction below  $2MD$  needs a proved state identification; it is not obtained by calling the converter a sufficient statistic.

Conversely, if a converter is memoryless and deterministic, its state set has one element. In that special case it need not enlarge the target label count. This is the correct context for a nonincrease assertion. It does not imply the false general statement about exact predictive memory refuted in Example J.1.

**M.3. A finite precision requirement rather than a free real tape.** For  $d = 1$  the invariant domain is an interval inside the probability simplex. A codebook can be constructed using interval intersections, and its labels are ordinary finite integers. One evaluates the two rational Bayes fractions, with the command likelihood evaluated to a specified accuracy, then selects the nearest admissible representative. The perturbation form of Theorem F.1 accounts for the incurred state error. Since denominators are at least  $3/4$ , no division by a vanishing report probability occurs.

For general fixed  $d$ , the same procedure uses a finite polytope feasibility construction. It may be computationally expensive. Our cardinality theorem does not state an optimal running time or workspace bound. The program may store its predetermined codebook, as permitted by the read-only model; it may not put acquired commands or reports in that program. The current command must be discarded after updating the label, and the terminal decoder may not reread it as free side information.

**M.4. Role of the accompanying diagnostics.** The source package includes exact-rational checks of Bayes normalization, the odds inverse, the rank-one determinant identity, the refresh contraction inequality at a finite family of inputs, and the scalar quantization identity. It also checks a finite conditional-covariance identity, a feedback path-law comparison, and specified deliberately incorrect formulas. These computations are reproducibility aids for signs, constants, and implemented examples.

The universal statements in this article are proved analytically above. A finite test does not establish measurability, a global differential bound, a minimax lower bound, or uniformity over all times. In particular, no number of successful finite diagnostics certifies that every possible

hidden Markov model has the displayed memory law. Build receipts, source hashes, and diagnostic output belong to the accompanying delivery record rather than to the logical assumptions of any theorem.

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